The Witnessed Difference Programme - From Finite Distinction to Witnessed Measurement

Finite obstruction, information loss, and sufficient observation

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2026-10-16 · Version 2.4.5

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Abstract

The Witnessed Difference Programme studies what must be specified for a distinction to be retained, compared and reported without losing the question being asked. Its finite core begins with explicit comparison data, permitted changes and report maps. A report preserves a question exactly when the answer is constant on each report fiber. With a declared linear repair map, discrepancies are compared modulo its image; finite cochain complexes separate removable discrepancies, closed residual classes and failures of closure. Exact certificates can check proposed repairs and lower bounds on the remaining discrepancy.

A central result distinguishes defects that can occur from changes admitted as repairs. For finite rational linear data it gives sharp counts of additional rational linear observation rows needed to distinguish realizable defects, repair classes on the entire defect space, or both. Further results treat transport between presentations, retained histories and certified progress in finite decision problems. A paired count-guard family has the homotopy type of a real no-equal arrangement complement, giving an exact capacity threshold for coherence and a first nonzero homology obstruction.

The wider programme asks how these constructions can support richer mathematical and physical theories while making every further assumption explicit. Later parts develop additive and tensor constructions, conditional complex representations, finite quantum states and processes, and selected model-specific limits. Each depends on declared algebraic, operational, topological or analytic inputs. The present work does not derive those inputs, or quantum theory, from an unstructured mark. Written proofs, selected Lean formalizations, executable checks and external interpretations are recorded separately.

Mechanism inventory

The monograph is one **Paper** containing the following internal papers. They share the protected-question discipline, while retaining their separate input contracts and proof obligations.

Internal paper	Location and mechanism	Evidence boundary
Finite distinction and probability	Chapters 1–9	Finite carriers, declared weights and cochain identities; optional quantum structure is explicit
Observer and measurement	Chapters 10–15 and 23; Appendices T–V, X–Y	Protected questions, calibrated records and finite recovery; authority is a separate input

Internal paper	Location and mechanism	Evidence boundary
Histories and admissible operations	Chapters 16–17; Appendices I–L, N, P	Declared history relations, operation closure and finite schedule witnesses
Conditional quantum interfaces	Chapters 18–19A	Scalar, positivity, readout and composition premises are not consequences of report minimality
Limits and analytic extensions	Appendices M, O, W	Analytic hypotheses and external theorems remain separate from finite and Lean checks
Relative completeness and proof supplements	Chapters 20–22; Appendices A–H, Q–S	Question-relative theorems, source provenance and bounded mechanization scopes

Reader's mathematical entrance

The organizing question is simple: when a presentation changes, which specified questions remain answerable, which differences can the declared operations repair, and what additional observations suffice when it loses a needed distinction?

Let $p : X \to A$ be a report and $q : X \to Z$ the answer that must be retained. A decoder $\bar{q} : p(X) \to Z$ with $q = \bar{q} \circ p$ exists precisely when

$$p(x) = p(y) \Longrightarrow q(x) = q(y).$$

This standard factorization criterion gives a constructive decoder when it holds and a collision pair when it fails. For example, reporting the first bit of (a, b) cannot answer the question "what is b?" because $(0, 0)$ and $(0, 1)$ have the same report but different answers. Appendix P.1 supplies the full statement and its transport uses.

The first substantial finite theorem refines this question by separating realizable defects from admitted repairs. Given rational linear maps with $R = \operatorname{im} D_s$, $B = \operatorname{im} D_r$ and existing report kernel $K = \ker Q$, unrestricted additional rational linear observation rows require exactly $\dim(R \cap K)$ rows to distinguish individual realizable defects in R, and $\dim K - \dim(K \cap B)$ rows to distinguish repair classes **globally on** V, meaning $\ker(Q, E) \subseteq B$. Ensuring both of these properties requires $\dim K - \dim(K \cap B) + \dim(R \cap K \cap B)$ rows. If instead only candidates in R must be distinguished up to repair, using their actual augmented reports, the minimum is $\dim(R \cap K) - \dim(R \cap K \cap B)$. These are different questions. Appendix T supplies the matching lower bounds and constructions for the global theorem WDP-DIM-1; T.8 states and derives the realizable-domain corollary. Restricted detector menus, noise, conditioning and physical instruments are separate models, not consequences of this count.

A three-edge cycle gives the complementary repair example. A field assigning one to every oriented edge has circulation three, whereas every vertex-potential change has circulation zero. It cannot be erased by those permitted repairs; with the declared ℓ^1 norm, the exact residual cost is three. This is a finite algebraic obstruction, not yet a physical interpretation.

Theorem map: when a presentation preserves the question

The finite mathematical thread is whether a changed presentation retains the **specified question**. Fifteen results carry that thread. Each row names the assumptions that make its conclusion meaningful; none derives those assumptions from the initial mark.

Canonical bindings below abbreviate the fixed `WDP-` prefix before `CH`. Register bindings name the bridge register; R12/R13 bindings name the separate supplement table in the claim ledger. Row IDs are stable identifiers, not locations, and the binding does not expand the cited evidence scope.

Result	Premises and dependency	Exact conclusion and contribution	Binding
R1. Repair obstruction and certificate (§4)	A declared finite cochain window, admissible repairs, and an exact norm/linear programme.	`Phi([h-d0 f])=Phi([h])`: changing a representative cannot remove a nonzero quotient obstruction. A matched primal repair and bounded dual circulation certify the optimum. Quotient algebra is standard; the typed repair/certificate interface is this paper's construction.	CH04-P01; CH04-C01
R2. Protected-query factorization (Appendices J.2, P.1)	Actual maps `f:X->Y` and protected query `q:X->Z`.	A decoder on `f(X)` exists exactly when `q` is constant on every `f`-fiber; a collision pair is a negative witness. The set-theoretic lemma is standard; its use to audit reported loss and future questions is this paper's application.	GEN-WDP-03
R3. Behavioral quotient (CH03-O01, Appendix I.1)	Finite total labelled updates and exact readouts or validated report equivalence.	Partition refinement yields the greatest report-respecting update congruence and its minimal quotient. The algorithmic principle is standard; the declared observer/report interface fixes what must be preserved.	CH03-O01

Result	Premises and dependency	Exact conclusion and contribution	Binding
R4. Carrier transport and reflection (Appendix P.3; admission/transport supplement)	Supplied cochain maps, with a retraction or homotopy only when explicitly given.	Commuting squares transport repair classes; a down-map retraction reflects equality on the old image. A homotopy identifies closed classes but need not identify arbitrary repair classes. Standard cochain algebra is applied to concrete carrier-change and loss boundaries.	GEN-WDP-71
R5. Finite local/global test (CH11-O01, Appendix J.2)	A specified finite context site with coherent restrictions and a declared global report set.	A matching family glues uniquely exactly when its restriction fiber contains one global report. The sheaf criterion is standard; the constrained-assignment countermodel locates this paper's finite obstruction.	CH11-O01

Result	Premises and dependency	Exact conclusion and contribution	Binding
R6. Guarded five-edit model histories (Appendix P.13–P.21)	The actual five-edit raw semantics, tagged events, stable birth names when needed, and declared precedence/guards.	Legal schedules have an exact raw endpoint. Under all-order legality, ideal prices descend by zero diamond curls and swap proofs have a local presentation. P.18 shows that guarded legal schedules can be disconnected despite equal endpoints, and gives the exact componentwise price criterion. P.20 establishes that star-shaped legality makes every legal swap complex connected with trivial fundamental group, so parallel legal swap proofs are joined by legal local 3-witnesses, and that compatibly ordered native precedences give star-shaped legality. P.21 shows that, for at most three literal terms, count total at most four and two successful edits, preserving the joint mark, literal balance record, native offers, prices and continuations forces raw equality (29,771 classes) under either terminal convention, attained by the report of mark, faces and counts capped at two. These are selected finite source theorems, not a global source quotient functor.	GEN-WDP-40–45; GEN-WDP-69; GEN-WDP-73

Result	Premises and dependency	Exact conclusion and contribution	Binding
R7. Contextual histories and additive tensor completion (§17.7; derivation supplements)	An admitted typed generating signature, endpoint-typed protected questions, finite alternative formation, signed completion, and an explicitly required biadditive independent composition.	Two-sided contextual equivalence is the greatest congruence preserving every protected question in every compatible context. Its free additive envelope has the unique homomorphism factorization property of Grothendieck group completion, and the free signed group on generator pairs represents biadditive maps. This derives an integer additive tensor calculus while leaving dagger, positivity, complex scalars, Hilbert completion and Born readout open.	GEN-WDP-58–59; GEN-WDP-72
R8. Dagger and observer Gram radical (§17.8; derivation supplement)	A declared involution on generators, dagger-closed protected questions when descending through compression, and a supplied additive observer into a finite integer coordinate carrier.	Reversal extends uniquely to an involutive additive dagger and acts componentwise on the algebraic tensor. The observer induces a positive semidefinite Gram form whose radical is exactly its kernel, so equal reports are exactly null signed differences. Positivity is nonunique and rank one supplies no integer complex structure.	GEN-WDP-60–61

Result	Premises and dependency	Exact conclusion and contribution	Binding
R9. Finite Hermitianization (§17.9; derivation supplement)	The realized observer carrier is rationalized; a rational-linear map J is supplied with $J^2 = -I$ and is skew-adjoint for the observer Gram form.	$(a + ib)x = ax + bJx$ makes the carrier a $\mathbb{Q}(i)$-module, and $h(x, y) = g(x, y) + ig(Jx, y)$ is Hermitian, positive semidefinite and has exactly the real Gram radical. The same positive form admits opposite complex orientations, so this constructs but does not select the complex branch.	GEN-WDP-62–63
R10. Certified state/effect readout (§17.10; derivation supplement)	A nonzero Gaussian-rational Gram factor presents the state; complete effect factors satisfy $\sum_o C_o^\dagger C_o = I$; the trace pairing is explicitly admitted as the readout.	Frobenius-square certificates give exact nonnegative normalized rational probabilities and replay the trace formula. A depolarized affine family remains positive, normalized and phase invariant while differing on a sharp input, so these conditions do not select the undeformed Born readout.	GEN-WDP-64–65
R11. Certified process composition (§17.11; derivation supplement)	Gram-factor states and a finite Gaussian-rational operator family satisfying $\sum_o K_o^\dagger K_o = I$ are supplied; sequential or tensor carrier structure is explicitly admitted.	Branch factors give exact nonnegative probabilities whose masses sum to the input mass; sequential and tensor families compose and supplied unitaries are reversible. Equal channels can retain different instruments. Conventional Hadamard/T matrices expose a scalar boundary; R12 realizes the Hadamard ray action over $\mathbb{Q}(i)$ while retaining the T obstruction. No gate model capable of arbitrary quantum computation is selected.	GEN-WDP-66–68

Result	Premises and dependency	Exact conclusion and contribution	Binding
R12. From separation to transformation (Chapter 19A)	The R9 carrier has positive definite Hermitian form; R10's pure-state readout is supplied. A bijection of the entire ray space preserves orthogonality in both directions, with $d \geq 3$, or all transition probabilities when $d = 2$.	The map has a projectively unique semilinear similarity representative. Rational phases form an infinite group. Unitarily reachable rays are classified by the Gaussian norm class, while some even-dimensional similarities cross those classes. These standard algebraic ingredients give exact, original-task-bound certificates and sharpen the limits of report preservation.	WDP-R12-O01; WDP-R12-P02–P05; WDP-R12-O02
R13. Algebraic generators (§19A.6)	A K-linear derivation of $M_d(K)$ is supplied and obeys the declared adjoint law.	An explicit matrix-unit construction gives $\delta = -i[H_0, \cdot]$ with $H_0^\dagger = H_0$, unique modulo a rational scalar. The inner-derivation theorem is standard; the finite certificate checks its actual input table. No time evolution or physical Hamiltonian is selected.	WDP-R13-P01
R14. Question-preserving encoding (§10.8; encoding supplement)	A declared task and protected question, a question-faithful partial parser, a sound checker, and binding of the report to the claimed candidate.	Equal reports with unequal answers exclude a complete report-only decision. Parser fidelity transports checker acceptance to the original question. An exhaustive integer-quadratic experiment compares cochain reconstruction with both enumeration and direct reconstruction; it establishes no cochain-specific efficiency gain.	GEN-WDP-74; QPE-1–QPE-4

Result	Premises and dependency	Exact conclusion and contribution	Binding
R15. Sharp linear observation budgets (Appendix T)	Finite rational spaces, realizable image R, repair image B, existing kernel K, and unrestricted added rational linear rows.	WDP-DIM-1 gives sharp raw-on-R, quotient-on-V, and joint counts. Restricting the repair-class question to actual reports of candidates in R gives $\dim(R \cap K) - \dim(R \cap K \cap B)$ rows. Fixed menus and noisy reports require their own models.	WDP-DIM-1; WDP-DIM-R; GEN-WDP-75

Section 10.6 connects R1's repair classes to R2's report fibers through sharp observer completion. The raw-defect question (is a locally accepted configuration actually defective?) and the quotient question (does it lie in a nonzero repair class?) have different invisible subspaces and different measurement budgets. Checked repair retains the resulting certificates as evaluation evidence, with authorization as a separate operation.

Three forms of completion must be reported separately. A **written theorem** is complete when its finite or conditional proof, assumptions and countermodels are inspectable here. A **mechanized theorem** is complete only for the exact Lean statement and its audited axioms; an abstract model declaration does not verify the source program. **Concrete implementation verification** binds the actual Rust/Python inputs, transition semantics, checker and tested scope; a finite replay does not prove every run. Appendix D and the claim ledger record those evidence levels without interchanging them.

A finished *finite edition* can be defined by a declared collection of carriers, questions and theorems such as R1–R6, with every assertion in that collection proved or explicitly refused at its stated scope. The broader GEN-WDP-O01–O05 programme, physical reconstruction, continuum limits and higher coherence remain separately open until their own hypotheses and proofs are supplied. The edition boundary does not change their status or turn a checked finite example into an unrestricted theorem.

Publication Scope Boundaries and Open Reconstruction Obligations

To maintain rigorous claim hygiene and prevent over-extension of formal statements, the following explicit scope boundaries distinguish proven finite conditional theorems from open global reconstruction goals:

1. **Attainment and rational coefficients (§4.2, §10.2)**: Finite-dimensional real or complex normed repair spaces have compact bounded sublevel sets after the repair kernel is removed. Completeness of an arbitrary scalar field does not supply this compactness. General norms, including polyhedral norms with irrational coefficients, need not attain their infimum on rational repairs. A feasible bounded linear programme with rational data has a rational optimum; matching primal and dual certificates verify a proposed optimum, whereas primal feasibility alone does not.
2. **Graded Subspace Invariance (§16.1)**: Anticommutator zero-locus arguments require an explicitly declared graded domain preserved by d_s and d_h; general un-graded endomaps do not

automatically preserve zero loci.

3. **Specialization Inputs for W1–W6 (§18.1)**: Structural specializations W1–W6 represent admitted conditional inputs (reversible updates, Gaussian-integer linearizations, pairing/quadratic readouts) rather than derivations from primitive distinction alone.
4. **PSD Geometry vs. LP Polyhedra (§18.5)**: General Positive Semidefinite (PSD) membership must be kept distinct from finite linear programming (LP) polyhedra. For instance, the qubit state slice $\frac{1}{2}\begin{pmatrix} 1+z & x \\ x & 1-z \end{pmatrix}$ is PSD if and only if $x^2 + z^2 \leq 1$, defining a non-polyhedral disk that requires exact Hermitian/PSD certificates rather than finite LP polyhedra.

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- Chapter 4. Finite cochains, quotients, and certificates
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Part II — Order, quantity, and finite probability:

- Chapter 6. Recurrence, succession, and constitutive order
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Part V — Relative scope and the open programme:

- Chapter 20. Relative completeness for observer-indexed presentations
- Chapter 21. Falsifiers, nonclaims, and open obligations
- Chapter 22. Synthesis: what the finite calculus establishes
- Chapter 23. Observer completion on a finite 2D toric-code carrier

Appendices: A dependency and status table; B end-to-end example; C source integration map; D formalization status and evidence architecture; E glossary; F bibliography and citation policy; G finite decision procedures; H scoped completion results; I native integration; J expert derivations and native ownership; K source-bound observer adaptation; L separation of the archived source derivation from WDP; M Meron characters and finite shadows; N source form specifications; O presentation closure from finite shadows to completion; P the earlier component calculi as six reusable constructions, with five-edit model balance-edit, path, result-mark and guarded-history results; Q transfer of learned rejection rules; R a plain-language picture of observers and authority; S certified progress; T sharp completion with distinct images; U corrected toric instrument mathematics; V locality proofs; W Trotter proofs; X calibrated reports and observation cost; Y recovery costs from joint report supports.

How to read this monograph

There are three useful routes through the text.

1. A reader interested in the foundational chain can read Chapters 1–3, 6–9, 10–15, and 20–23. This route foregrounds what is admitted, what is constructed, and what is refused.
2. A reader interested in the finite mathematics can follow the central chain: **report preservation** (the entrance above and Appendix P.1) → **repair** (Chapter 4) → **sufficient observation** (§10.6 and Appendix T) → **observation cost** (§§14.6–14.7 and Appendices X–Y) → **lawful histories** (Appendices P.13–P.23). P.22 gives the guarded-schedule topology and coherence thresholds; P.23 gives integral chain transport and certificate bounds. Chapters 3–8, 16–19 and 19A supply further algebraic and quantum constructions with their own admitted inputs.
3. A reader interested in observers and measurement can begin with the typed glossary below, then read Chapters 10–15, Appendix T and Appendices X–Y. The shared-error example in Y.6 shows why retaining correlations can change recovery from obstructed to attainable. This route is independent of the optional quantum-carrier branch.

No reading route should treat chapter order as a metaphysical order. The order is a dependency order for the constructions presented here. A later formalism may admit a different starting package and still be mathematically coherent.

Appendix L follows the actual archived source derivation into WDP. Appendix M follows the Meron and finite-shadow branch toward continuum models, with separate proofs for cancellation, observable preservation and conditional numerical limits. Appendix N follows the source form specifications and proposed SQL profile; Appendix O follows metric presentation closure and actual Meron source limits. The spatial proof distinguishes signed finite generating functions from the constructed Gaussian random-distribution target.

Proof supplements and stable locations

Several full proofs extend beyond the summaries printed in this PDF. They are part of the accompanying source archive, under these relative paths:

- `docs/guarded_schedule_topology.md`: Appendix P.22, the native event complex, common-cover argument and its relationship to arrangement complements.
- `docs/sparse_guarded_chain_transport.md`: Appendix P.23, the minimum-sweep construction and fixed-degree computation and certificate-size bounds.

- `docs/derivations/README.md`: index of the seven full proof supplements for §§17.7–17.11 and Chapter 19A, including additive/tensor completion, Hermitianization, certified readout, process composition and preservers.
- `docs/derivation_review.md` and the supplements linked from Appendix J: detailed admission and transport arguments and their evidence boundaries.

The archive preserves these paths; `MANIFEST.sha256` binds their exact bytes. The repository snapshot provides an immutable location for the retained supplements. The present correction has no assigned version DOI; the older v0.5.0 DOI identifies that older edition. Appendix F includes the external bibliography in this PDF and preserves the citation keys used in the text.

Status grammar

Every major assertion belongs to one of the following classes.

- **Definition.** A stipulated mathematical object or predicate. A definition may be useful or useless, but is not a theorem just because it is precise.
- **Finite proposition.** A result proved in the text from stated finite hypotheses. A proof sketch is included when the argument is short enough to audit locally.
- **Typing proposition.** A claim about what the declared nominal interfaces do or do not coerce. It establishes a type-system fence, not every semantic nonimplication between possible interpretations of the types.
- **Conditional result.** A proposition whose conclusion is proved only after named imports or regularity assumptions are supplied.
- **Cited result.** A result attributed to an external source and used only within its declared scope. The primary-source registry records the intended source and consultation state.
- **Mechanized elsewhere.** A repository or external theorem is reported as machine checked outside this carrier. That report is not promoted here unless the exact source, statement, toolchain, and source reference are available.
- **Open claim.** A well-typed target without a completed proof.
- **Open interface.** An open claim whose missing object is a typed adapter or interface together with its preservation laws.
- **Conjecture.** An open claim for which examples or analogy suggest a direction.
- **Refuted draft.** A previously printed implication or classification for which a counterexample or typing failure is known.
- **Interpretive proposal.** A reading of the formalism, not a mathematical consequence.
- **Nonclaim.** A boundary the monograph expressly refuses to cross.
- **Synthesis conclusion.** A bounded summary of results already established in named rows; it introduces no additional theorem.

These labels describe **mathematical status**. The claim ledger records two additional, independent fields: **formalization scope** (an exact declaration, a narrower partial declaration, or no bound declaration) and **execution/reproduction status** (a dated paired replay or a pending replay). The strict paired reproduction gate still requires a successful, source-bound Lean elaboration and the bound Python check. Missing bindings, unavailable dependencies, or a check that has not been rerun are reproduction limitations; they do not make a written proof an open mathematical claim. A declaration name alone also does not establish a successful current build.

The alphanumeric claim identifiers and implementation names used below are repository lookup keys, not mathematical standards or additional assumptions. The definitions and cited source

statements, rather than those labels, determine the content of each result.

Identifiers of the form `WDP-CHxx-*` are the canonical row identifiers shared by this monograph and the unified claim ledger. The body supplies hypotheses and proof context, while the ledger is the authority for final evidence bindings and replay status.

Standing scope discipline

The base combinatorial and operational families are finite: token and event sets, graphs, covers, path families at a fixed bound, query families, and comparison tables. Coefficient and value spaces may be infinite when explicitly typed—for example $\mathbb{R}$, $\mathbb{C}$, real intervals, or a finite-dimensional real vector space. Finite cardinality and finite dimension are never interchangeable. The finite indexing makes many inspections exhaustive and avoids infinite path sums; it does not license continuum conclusions. "Canonical," "minimal," "unrestricted," "forced," "complete," and "equivalent" are used only with a stated category, order, optimization problem, or hypothesis class.

`WDP-CH00-N01` **Nonclaim.** The monograph does not derive all structure from Primitive Difference.

`WDP-CH00-N02` **Nonclaim.** The monograph does not identify formal witness with experience, consciousness, agency, or moral status.

`WDP-CH00-N03` **Nonclaim.** No local proof, test run, dashboard, archive, or status label confers authority.

Part I — Difference, witness, and finite obstruction

Chapter 1 — The admitted cut

1.1 The starting package

The programme begins not with a bare slogan but with an admitted finite package. Let X be a finite carrier of exhibited poles or tokens. A **cut** is a declared occurrence of apartness between roles in an exhibition. At minimum we require:

1. a way to distinguish the relevant roles in the present exhibition;
2. a structural carrier on which those roles can be compared;
3. a rule saying which comparisons count as sameness or repair for the present question.

We call the first ingredient **Primitive Difference** and the remaining organization **Structure**. They are co-inputs. The phrase "from difference" is therefore shorthand for "from an exhibited difference inside a declared finite structural package." We call the resulting finite-carrier setting the **finite obstruction calculus**: cochains, admissible repairs, quotient defects and their certificates over a declared finite carrier.

`WDP-CH01-D01` **Definition (admitted cut).** An admitted cut is a tuple $\mathfrak{C} = (X, E, \tau, \mathcal{K})$ where X is a finite set of exhibited tokens, E is a finite set of incidences or comparisons, τ assigns the roles needed to type those incidences, and $\mathcal{K}$ is the declared family of admissible comparisons. The tuple is data. No part of it is claimed to be selected from equality on X alone.

In the admission-versus-evaluation vocabulary of the earlier component papers (Root W vs. Root P), a **Root W realization** additionally supplies a relation and its protected irreflexivity law: $K = (X, R, \mathrm{Irr}(R))$, with witness $d = (a, b, w : R(a, b))$. The general admitted-cut tuple does

not itself entail these extra inputs. Once supplied, they type relational distinction; orientation still depends on the chosen relation, as in the symmetric and oriented examples of §1.4. Root P's cochain evaluation and repair quotient are further structures, not consequences of these initial assumptions.

The exact representation can vary. A graph is often convenient: vertices are tokens and oriented edges are typed comparisons. A hypergraph, relation, or small category may be more appropriate when comparisons have higher arity. What matters is that the representation be named before a theorem depends on it.

1.2 Difference and local sameness

Within a fixed comparison family, one may ask whether two presentations agree. Let P be a set of presentations and let

$$\mathrm{Same}_{\mathcal{K}} : P \times P \longrightarrow \{0, 1\}$$

be a declared predicate. A failed comparison is a residual at the level of predicates. If the presentations live in an abelian group or vector space, the failure may instead be encoded by a difference $p - q$. The algebraic encoding is stronger: subtraction, scalar structure, and a zero element have already been admitted.

`WDP-CH01-P01` **Finite proposition.** A failed declared sameness comparison exhibits that the two compared presentations are not interchangeable under that comparison.

This statement is intentionally local. It does not say the presentations are different under every possible comparison. It does not make the failed comparison intrinsic to the tokens. It does not supply an ordering of them.

1.3 Why Structure cannot be erased

Suppose only a finite set X with equality is given. For $|X| \geq 2$, every permutation of X is an automorphism. Any construction advertised as parameter-free and natural under all automorphisms must commute with the full symmetric group. A distinguished element, a direction, an ordered pair of distinct elements, or a nontrivial proper subset is generally moved by some permutation. Therefore it cannot be selected invariantly without additional data.

This elementary observation is the first discipline of the programme. It does not make structure illegitimate. It requires structure to be visible in the input ledger.

1.4 Worked example: a two-pole exhibition

Let $X = \{a, b\}$. Declaring just $a \neq b$ exhibits apartness but does not select which pole is source and which is target. The transposition exchanging a and b preserves equality and inequality. If we additionally declare an oriented edge $e : a \to b$, then orientation exists because it was supplied by e.

Three conclusions must remain distinct:

- the poles are apart in the exhibition;
- the edge orients them;

- an observer later may register that orientation.

The first does not imply the second, and neither implies the third.

The same separation holds one level down, for the relation that is supposed to exhibit apartness. A pointed relation (X, R, a, b) records a relation R on a finite carrier together with two marked poles.

WDP-CH01-P02 **Finite proposition (pointed relations and apartness).** For a pointed relation (X, R, a, b) on a finite carrier, $R(a, b)$ does not imply $a \neq b$; $a \neq b$ does not imply $R(a, b)$; and $R(a, b)$ together with irreflexivity of R implies $a \neq b$.

Proof. A one-point carrier with $R = \{(a, a)\}$ and $b = a$ has $R(a, b)$ and $a = b$. The empty relation on a two-point carrier has $a \neq b$ without $R(a, b)$. If R is irreflexive and $R(a, b)$, then $a = b$ would give $R(a, a)$. □

Irreflexivity is therefore the declared ingredient that lets a related pair witness apartness; it belongs to the admitted package and does not follow from relatedness. Symmetry is a second, independent choice. The symmetric two-point relation $\{(a, b), (b, a)\}$ admits the swap as an automorphism, and the oriented relation $\{(a, b)\}$ does not. Keeping both is nonselection in the sense of Chapter 3; choosing one is an entry in the injection ledger of §3.2.

1.5 Ground and spectral ground state

"Ground" in this chapter means the admitted starting package. A spectral ground state is instead a lowest-eigenvalue state of a declared operator on a declared inner-product space. The latter can be studied in the optional carrier branch, but it is not the foundation of the former and should not share an unqualified name.

WDP-CH01-N01 **Nonclaim.** No spectral, physical-vacuum, or cosmological ground state is derived from an admitted cut.

1.6 The Priced Grounding Span: From Raw Mention to Witnessed Verification

WDP-CH01-D02 **Definition (Priced Observer Boundary State).**
An observer boundary specification is an admitted triple (X_O, Y_O, π_O). For this profile take $X_O = (S_O, R, \mathrm{TL}_O^{\mathrm{priv}})$, with $R \in \mathbb{Q}_{\geq 0}$ and $\mathrm{TL}_O^{\mathrm{priv}} \in \mathbb{N}$. Choose an external-state report π_{ext}, a threshold $c_{\min}$ and a positive integer k. The public report is

$$\pi_O(S_O, R, t) = (\pi_{\mathrm{ext}}(S_O), \mathbf{1}_{\{R \geq c_{\min}\}}, \lfloor t/k \rfloor).$$

The Boolean flag may be labeled SOLVENT or EXHAUSTED. Fibers are inverse images $\pi_O^{-1}(y)$, not linear kernels: this mixed discrete/nonlinear carrier has no declared vector-space dimension. Coarsening a balance can hide its exact value at one instant. Persistent privacy requires equal reports under all admitted continuations, as in §20.14. Keeping exact balances in a separate host ledger is an architectural admission, not a consequence of noninjectivity.

WDP-CH01-D03 **Definition (Five-Gate Grounding Span).**

The grounding span for an ingress candidate is the sequence of five gated transitions:

$$\begin{aligned}
\text{External Supply} &\xrightarrow{g_{\text{sup}}} \text{Token-Erased Admission},\\
\text{Token-Erased Admission} &\xrightarrow{g_{\text{adm}}} \text{Asymmetry-Priced Retention},\\
\text{Asymmetry-Priced Retention} &\xrightarrow{g_{\text{ret}}} \text{Quotient-Defect Evaluation},\\
\text{Quotient-Defect Evaluation} &\xrightarrow{g_{\text{ev}}} \text{External Sovereign Receipt},\\
\text{External Sovereign Receipt} &\xrightarrow{g_{\text{auth}}} \text{Authorized transition}.
\end{aligned}$$

governed by explicit operational preconditions: 1. **External Supply (g_{sup}):** Ingress of external raw tokens. Operates as an identity transition on internal regime state (`supply_no_op`). 2. **Token-Erased Admission (g_{adm}):** Admitted if and only if form $f < |forms|$, argument arity matches, and argument data exist ($x < data$). Erases token spellings in favor of ordinal indices (NS-14), debiting ingress cost c_{adm} from ledger R. If $R < c_{\text{adm}}$, transitions to refusal `REFUSE_INSUFFICIENT_FUNDS`. 3. **Asymmetry-Priced Retention (g_{ret}):** Admits a retention policy across $\Delta t \leq \text{TL}_O$ and a separately chosen fee $c_{\text{ret}}\Delta t$. `WDP-CH03-P03` concerns the selected two-item no-form grammar; it does not derive this temporal fee or physical retention law. 4. **Quotient-Defect Evaluation (g_{ev}):** Evaluates discrepancy $\omega \in C^1$ via quotient distance $\Phi_1(\omega) = \inf_{\alpha \in C^0} \|\omega - d_0\alpha\|_1$. 5. **External Sovereign Receipt (g_{auth}):** Emits machine receipt $\text{Receipt}(\Phi_1, \varphi^*)$. Carries diagnostic evaluation only; confers zero sovereign standing (`ENC-0195`, `WDP-CH15-D01`).

`WDP-CH01-L01` **Lemma (Token-Erased Admission Obstruction).**
Let an unhosted syntactic generator operate with empty boundary presentation ($B_O = \emptyset$, $S_O = \emptyset$). Then gate g_{adm} cannot evaluate boundary consistency, returning typed refusal `REFUSE_EMPTY_BOUNDARY`.

`WDP-CH01-N02` **Nonclaim.**
The five-gate grounding span defines the formal operational requirements of the declared mathematical model; it is not asserted as an empirical model of human cognitive architecture.

Chapter 2 — Residuals and the typed witness taxonomy

2.1 Residual before witness

A **residual** is the part of a declared target that is not removed by an admissible repair. Let V be a finite-dimensional real vector space of defects, A a finite-dimensional space of repairs, $d : A \to V$ a linear repair map, and $\delta \in V$ an exhibited defect. For a declared seminorm or cost functional $J : V \to [0, \infty)$, define

$$\Phi_J([\delta]) = \inf_{\alpha \in A} J(\delta - d\alpha).$$

Here $[\delta]$ denotes the coset of δ modulo $\operatorname{im} d$ when that quotient is meaningful. The expression is not primitive: it imports a linear carrier, a repair map, and a cost.

In this lexicon, the exhibited defect δ, the repair kernel $\ker d$, and the quotient defect $\Phi_J([\delta])$ are classified strictly as **Root P**: the evaluation kernel and quotient defect structure on cochains. Root

P computes obstruction to global exactness and repair over an admitted complex. It presupposes Root W as underlying carrier and never substitutes for it.

WDP-CH02-D01 **Definition (residual locus witness).** A residual-locus witness is a typed locus L together with evidence that the restriction or component of the residual at L is nonzero according to the declared test.

The locus is a witness *to a proposition*. It is not thereby a memory, an observer, an agent, or a measurement apparatus.

2.2 Seven witness roles

The single word *witness* is insufficient. We use the following typed roles.

1. `ResidualLocusWitness[P]`: a locus at which residual proposition P is exhibited.
2. `RetentionWitness[P]`: a mark or state component enabling later re-access to evidence for P.
3. `RepairWitness[P]`: an admissible α exhibiting that P can be repaired, often by satisfying $d\alpha = \delta$.
4. `ContextWitness[P]`: a local assignment or section witnessing P inside a declared context.
5. `DualCertificateWitness[P]`: a functional separating a defect from the repair image.
6. `DescentReceipt[P]`: evidence that compatible local presentations glue or transport according to a declared descent rule.
7. `CounterexampleWitness[P]`: a finite inhabitant refuting a universally quantified P.

The general notation

$$\mathrm{Witness}_k[P](x, e)$$

means that evidence e makes subject x a witness of kind k for proposition P. The proposition and kind are part of the type.

WDP-CH02-P01 **Typing proposition.** In the nominal typed interface, no one of the seven role tags coerces to another unless an explicit adapter and its hypotheses are declared.

This is a syntactic fence, not a proof of all 42 directed semantic nonimplications. Simple models do separate several important cases: a nonzero residual component need not be stored; a stored mark need not repair anything; a repair element need not define a separating functional; a local section need not glue; and a counterexample need not be re-accessible.

2.3 A triangle example

Let G be an oriented triangle with vertices $0, 1, 2$ and edges $01, 12, 20$. Let $C^0(G; \mathbb{R}) = \mathbb{R}^3$ and $C^1(G; \mathbb{R}) = \mathbb{R}^3$. The coboundary is

$$(d_0 f)_{ij} = f_j - f_i.$$

Take $\delta = (1, 1, 1)$ in the edge order $(01, 12, 20)$. Its circulation is $1 + 1 + 1 = 3$. Every coboundary has zero circulation because the vertex terms cancel. Thus the cycle is a `DualCertificateWitness` against exact repair: the linear functional $\ell(x) = x_{01} + x_{12} + x_{20}$ vanishes on $\operatorname{im} d_0$ and satisfies $\ell(\delta) = 3$.

Each edge is also a possible residual locus if the local test reads its nonzero component. But no edge stores the result just by being nonzero, and the functional ℓ is not an observer.

2.4 Witness and evidence

Evidence is role-relative. A proof term can witness a proposition in a formal system. A data record can witness that a test returned a value. A residual can witness failure of a repair condition. These are not interchangeable. Whenever a chapter moves between them, it must provide an adapter—for example, a theorem connecting an executable output to a mathematical predicate, or a provenance verification record connecting a stored artifact to the source that generated it.

WDP-CH02-N01 **Nonclaim.** A typed witness does not acquire epistemic, phenomenal, legal, or institutional authority just by being called evidence.

Chapter 3 — Symmetry, nonselection, and the price of retention

3.1 The finite symmetry obstruction

Let X be a finite set, $|X| = n \geq 2$, carrying no structure beyond equality. The group S_X acts transitively on X. A parameter-free choice of a point would be an S_X-equivariant map from the one-point trivial S_X-set to X. Such a map selects $x \in X$ fixed by every permutation, but no point is fixed by all permutations when $n \geq 2$.

WDP-CH03-P01 **Finite proposition (no equivariant point selection).** There is no S_X-equivariant selection of a distinguished element of an unstructured finite set X with $|X| \geq 2$.

The same proof applies to an oriented edge or nontrivial proper subset whenever the full symmetry group acts without a global fixed target of the requested type.

This is a nonselection theorem, not an impossibility of choice. A label, base point, order, random seed, boundary condition, prior state, or external input can select. The theorem requires the asymmetry source to be named.

3.2 Injection ledger

For any proposed selection, record the first input that breaks the relevant symmetry. Typical entries are:

- a named element or base point;
- an orientation;
- an ordering or clock;
- a measure or weighting;
- a boundary condition;
- a representation or coordinate system;
- an observer profile;
- a stochastic device and its distribution;
- a physical law not present in the root package.

WDP-CH03-D01 **Definition (asymmetry-price frontier).** Relative to a specified implementation class and dependency preorder, the asymmetry-price frontier is the family of inclusion-minimal declared input sets whose removal restores a symmetry that moves the output. A scalar "price"

additionally requires a stated cost function; several frontier elements may tie or remain incomparable.

Minimality here is relative to the declared implementation class and order. No uniqueness or global-minimality theorem is claimed for the frontier in general; CH03-P03 below settles one declared instance, and a tie survives even there.

A small search makes the point concrete. A **declare-then-assert grammar** has a finite set D of data, a list of declared forms with positive arities, a set of asserted instances (a form with a tuple of data of its arity), and a set of forms carrying a rule that asserts every tuple of that form. A **context** is a declared form whose slots hold admitted data or holes, at least one of them a hole; a datum x satisfies the context when filling every hole with x gives an asserted instance or a tuple of a ruled form. Two data are **separated** when some context is satisfied by exactly one of them. A **response** is a list of items, each of which adds a datum, declares a form of positive arity, asserts one instance of an already declared form on admitted data, or admits the rule of an already declared form. An isomorphism of results is a pair of bijections, on data and on forms, preserving arities, instances, rules, and the required pair as an unordered pair. Fix a base with $a \neq b$ in D, no forms, no instances, no rules, and the single required pair $\{a, b\}$. A response is valid when every item applies and the result separates a from b. The first cost is the item count; the second refines it lexicographically by the sum of the arities of the newly declared forms.

`WDP-CH03-P03` **Finite proposition (separation price and the surviving tie).** Over this base:

1. no response of at most one item is valid, and declaring a unary form F and asserting $F(a)$ is valid;
2. for each $k \geq 1$, declaring a k-ary form and asserting it on $(a, \dots, a)$ is valid, and distinct k give non-isomorphic results, so the item count leaves infinitely many isomorphism classes of minimal responses;
3. under the lexicographic refinement, up to renaming the newly declared unary form, exactly two marked responses remain: the unary marks on a and on b. They form one isomorphism class, and the transposition $(a\,b)$, an automorphism of the base, exchanges them, so no $(a\,b)$-equivariant rule selects one of them.

Proof. (1) The base declares no form, so it has no context, and an instance or rule item cannot apply to it. A single new datum adds no context. A single new form adds contexts but asserts nothing, so every context fails at both a and b. For the unary mark, the context $F(_)$ holds at a and fails at b. (2) The all-hole context $F(_, \dots, _)$ holds at a and fails at b. The result has exactly one form, and an isomorphism preserves its arity. (3) By (1) a minimal response has two items. Its first item declares a form F: an instance or rule cannot apply to the base, and after a first datum the single remaining item cannot both declare and assert. Its second item is neither a datum nor a second form, which assert nothing, nor the rule of F, which makes every context hold at both a and b. It therefore asserts one instance of F, and the arity sum equals the arity of F, whose least value is 1. A unary form has the single context $F(_)$, which holds exactly at the asserted datum x, so separation forces $x \in \{a, b\}$. The transposition $(a\,b)$ fixes the base with its required pair and carries the mark on a to the mark on b. The two results are therefore isomorphic, and $H = \{1, (a\,b)\}$ acts on the two surviving responses without a fixed point. By CH03-P02 (stated in Chapter 5), no H-equivariant choice exists. □

Worked count. For a binary form, an asserted pair (x, y) separates a from b exactly when x or y lies in $\{a, b\}$: the contexts $F(_, _)$, $F(c, _)$, and $F(_, c)$ test the diagonal, the second slot, and the first slot. Up to isomorphism the separating pairs are (a, a) and (a, b) and, when D has a

third datum c, also (a, c) and (c, a). With at most two items and arity at most two, the minimal responses therefore fall into 3 classes when $D = \{a, b\}$ and 5 when $|D| \geq 3$, and into 1 class under the refinement. The bounded counts grow with the arity bound, as clause 2 requires.

The proposition is relative to this grammar, these two costs, and this isomorphism relation. It asserts no minimality over other formal languages, and the value two is a property of the declare-then-assert grammar. Choosing a over b is a further declared input of the kind listed in the injection ledger, not an output of the search. The absorbed archived working-paper source proved clause 1 for every base with no forms and no instances, the unary witnesses, clause 2 for two data, and the uniqueness half of clause 3 for two and three data in Lean at its recorded commit. Those modules are copies of an external derivation study, which the atlas rules exclude. They were not ported, and this row takes no Lean credit.

3.3 Retention is additional structure

A difference at one exhibition does not by itself remain available in another. Retention requires a carrier that persists, an update rule, and a criterion of re-identification. A generic retention packet is

$$\mathcal{R} = (M, u, r, \approx),$$

where M is a memory carrier, $u : M \times I \to M$ is an update under inputs I, $r : M \to O$ is a report map, and $\approx$ is the declared comparison used to recognize retained content.

This packet is a definition, not a derivation. Earlier drafts sought an order-free minimal retention constructor and component independence results. The declared finite behavioral classification below is now proved; Appendix J also classifies typed component choices and separates additional regularity laws from typing.

`WDP-CH03-001` **Finite proposition (resolved finite target).** For a finite nonempty carrier and finite input alphabet with total labelled deterministic updates and exact readouts or a validated report equivalence, descending partition refinement computes the greatest report-respecting update congruence, its minimal quotient, and behavioral classification by quotient isomorphism, without a global chronology. The proof, native ownership and scope are given in Appendix I.

`WDP-CH03-002` **Finite proposition (typed component independence).** For fixed finite nonempty carrier M, input alphabet I and nonempty report alphabet O, all typed retention extensions form the Cartesian product $M^{I \times M} \times O^M \times \mathcal{P}(O \times O)$. Named compatibility laws may couple its factors. For arbitrary report comparison the all-word relation is the greatest report-respecting forward-compatible relation; quotient and literal-readout descent require their separate stated conditions. Appendix J establishes this result and the complete declared predicate classifications.

3.4 Worked countermodel: witness without retention

At a single instant, let a detector bit be 1 exactly while an input contact is closed. If the bit resets to 0 when contact opens and no other state persists, then the bit is a `ResidualLocusWitness` for current contact but is not a `RetentionWitness` for past contact. Adding a latch creates retention. The latch is not contained in the original event; it is a new mechanism.

3.5 Relation to automata

Myhill–Nerode equivalence is a standard downstream framework for quotienting word histories by future distinguishability [REF-AUT-002], but its inputs are words and therefore already ordered. The native finite construction in Appendix I establishes its own minimal-quotient property for the declared total update packet; no global chronology or free selection of update structure is inferred. Bisimulation likewise presupposes a labelled transition system [REF-AUT-005]. Channel theory presupposes classifications and their connecting maps [REF-AUT-004]. These frameworks clarify what retention can mean after their input structures are admitted.

3.6 Component independence: retention countermodels CM1–CM5

To probe the independence structure of $\mathcal{R} = (M, u, r, \approx)$ under Claim CH03-O02, and to give a candidate branching-time / non-total-order behavioral relation for Claim CH03-O01, we give five concrete finite countermodels. They separate components in these instances. Appendix J supplies the full typed extension-space theorem and complete named predicate classifications; the finite behavioral classification is proved in Appendix I:

- **CM1 (Carrier Independence / Static Memory):** Let $M = \{m_0, m_1\}$ with $|M| \geq 2$, input alphabet $I = \{\text{tick}\}$, identity update $u(m, \text{tick}) = m$, injective report $r(m_i) = r_i$, and diagonal comparison $\approx = \{(r_0, r_0), (r_1, r_1)\}$. The memory carrier is non-trivial and observable, but updates perform no state transitions. Having a memory carrier M does not force active recording dynamics u.
- **CM2 (Update Independence / Readout Blindness):** Let $M = \{m_0, m_1\}$, $I = \{\text{toggle}\}$, update $u(m_0, \text{toggle}) = m_1$ and $u(m_1, \text{toggle}) = m_0$, but constant report $r(m) = o_0$ for all m. The internal state dynamics are invertible and non-trivial, but the readout is completely blind. Active updating u does not induce an informative report r without an explicit observability axiom.
- **CM3 (Report Independence / Coarse-Graining):** Let $M = \{m_0, m_1, m_2\}$, $I = \{\text{cycle}\}$, cyclic update $u(m_i, \text{cycle}) = m_{(i+1) \bmod 3}$, and non-injective report $r(m_0) = r(m_2) = \text{even}$, $r(m_1) = \text{odd}$. The dynamics are bijective, but report fibers merge states, showing that report equivalence $\ker(r)$ is strictly coarser than state equality, so a single report cannot reconstruct M. (The depth-1 relation below does separate m_0 from m_2: their successors m_1 and m_0 report differently.)
- **CM4 (Comparison Independence / Non-Trivial Tolerance):** Let $M = \{m_0, m_1\}$, $I = \{\text{tick}\}$, $u(m, \text{tick}) = m$, $O = M$, and $r = \text{id}_M$. Set $\approx = M \times M$, a non-trivial tolerance that is also an equivalence relation. The states and reports are distinct, yet declared recognition identifies them. The comparison relation $\approx$ is logically independent of (M, u, r).
- **CM5 (Lossy Update / Irreversible Reset):** Let $M = \{m_0, m_1\}$, $r = \text{id}_M$, and $I = \{\text{reset}, \text{keep}\}$ with $u(m, \text{reset}) = m_0$ and $u(m, \text{keep}) = m$ for all m. The readout is exactly faithful, but the reset input collapses all history. Faithful observation of current state does not ensure retention of past interactions.

Branching-Time / Non-Total-Order Behavioral Equivalence (CH03-O01): Standard automata equivalence uses words in I^* to compose admitted updates. Each word has a local order; these words do not supply a global chronology. The same behavior can be presented by taking input events organized along branching computation trees of depth k (where each branch carries local transition depth, but no global linear time axis is postulated across branches): two states $s \in M_1, s' \in M_2$ are k-equivalent if $r_1(s) \approx r_2(s')$ and for every declared input event $i \in I$, the

stepped states $u_1(s,i)$ and $u_2(s',i)$ are $(k-1)$-equivalent, with 0-equivalence given by $r_1(s) \approx r_2(s')$ alone. This gives a bounded-depth behavioral relation on finite branching structures without presupposing a global total order or linear clock; each branch is still a finite sequence of events. It is an equivalence relation when $\approx$ is one, as it is in CM4. For a reflexive symmetric comparison that is not transitive, the behavioral relation need not be transitive: use three states with identity reports and updates, with $a \approx b$ and $b \approx c$ but $a \not\approx c$.

`WDP-CH03-N01` **Nonclaim.** The symmetry obstruction does not establish that retention, dynamics, or observers are obstructed or unrealizable. It establishes that their selection is not free.

3.7 Presentation relabeling and invariant exports

`WDP-CH03-D02` **Definition (Presentation Relabeling and Invariant Passage Charge).** Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be presentations related by a declared invertible alphabet map σ. A protected export E is relabeling-invariant when $E(\sigma x) = E(x)$ on its declared domain. A passage cost, cochain action and induced quotient action are additional data. Zero cost does not by itself imply the identity action on $Q^1 = C^1/\operatorname{im}(d_0)$ or invariance of E.

`WDP-CH03-P04` **Proposition (limit of an invariant export).** If $E(\sigma x) = E(x)$ but a protected answer satisfies $q(\sigma x) \neq q(x)$, no decoder using only E recovers q on the whole domain. The proof is the same-report contradiction in §14.8. It requires an actual target-separating pair; a relabeling that preserves both E and q presents no such obstruction.

`WDP-CH03-N02` **Nonclaim.**
Proposition `WDP-CH03-P04` assumes invariance of a specified export. It does not establish invariance of arbitrary internal verdicts or natural-language tokenizers, and does not prohibit computing a defect from a supplied table.

Chapter 4 — Finite cochains, quotients, and certificates

4.1 The graph-cochain host

Let $G = (V, E)$ be a finite oriented graph. For a normed coefficient space A, define

$$C^0(G; A) = A^V, \qquad C^1(G; A) = A^E,$$

with coboundary

$$(d_0 f)(u, v) = f(v) - f(u).$$

An edge field $\delta \in C^1(G; A)$ is **exact** when $\delta = d_0 f$. The quotient

$$Q^1(G; A) = C^1(G; A)/\operatorname{im} d_0$$

records edge fields modulo vertex-potential repair. It is a quotient relative to the declared graph, coefficient space, and repair map. It is not the first cohomology group unless a next differential d_1 is supplied and $d_1 d_0 = 0$.

4.2 Quotient defect

For a norm J on $C^1(G;A)$, define

$$\Phi_J([\delta]) = \inf_{f \in C^0(G;A)} J(\delta - d_0 f).$$

WDP-CH04-P01 **Finite proposition (well-defined quotient defect).** Φ_J is independent of the representative of $[\delta]$.

Proof. If $\delta' = \delta + d_0 g$, then

$$\inf_f J(\delta' - d_0 f) = \inf_f J(\delta - d_0(f-g)) = \inf_h J(\delta - d_0 h).$$

The translation $h = f - g$ is a bijection on $C^0(G;A)$. □

WDP-CH04-P02 **Conditional result (attainment).** If the repair space is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$, and J is a norm, the infimum is attained after quotienting the kernel of d_0.

Proof sketch. Restrict to a complement of $\ker d_0$. On this complement, $f \mapsto J(\delta - d_0 f)$ is continuous and coercive because d_0 is injective and all finite-dimensional real norms are equivalent (view a complex space as a real space). Over $\mathbb{R}$, a continuous coercive function attains its minimum on compact sublevel sets. □

The finite-dimensional, real/complex scalar, and norm hypotheses matter. Finite dimension and a norm do **not** ensure attainment over general rational carriers $\mathbb{Q}$. For an explicit counterexample, take two parallel equally oriented edges between two vertices, fix the first potential to zero, and write $d_0 f = (x, x) \in \mathbb{Q}^2$ for the second potential $x \in \mathbb{Q}$. Let the edge field be $\delta = (1, 0)$. Equip $C^1(G;\mathbb{R})$ with the strictly convex weighted ℓ^4 norm $J(u_1, u_2) = (u_1^4 + 2u_2^4)^{1/4}$. The residual is $\delta - d_0 f = (1 - x, -x)$, with $J(\delta - d_0 f)^4 = (1-x)^4 + 2x^4$. The unique real minimizer is $x^* = 1/(1 + \sqrt[3]{2}) \notin \mathbb{Q}$. For all rational potentials $f \in C^0(G;\mathbb{Q})$, $J(\delta - d_0 f) > J(1 - x^*, -x^*)$, so the infimum over rational potentials is not attained. For scalar rational coefficients, positive rational weights and rational repair data, the weighted L^1 problem of §4.3 is a feasible bounded rational linear programme and has a rational optimizer. More generally the same conclusion holds for a norm given by finitely many rational linear forms and rational input data. Polyhedrality alone is insufficient: $J(u, v) = |u + \sqrt{2}v| + |v|$ is a norm with a polyhedral real unit ball, but on the same repairs its unique real minimizer is $x^* = 1/(1 + \sqrt{2})$. Rational repairs approach that value without attaining it.

4.3 Weighted L^1 geometry

Choose positive edge weights $w_e > 0$ and set

$$\|x\|_{1,w} = \sum_{e \in E} w_e \|x_e\|_A.$$

This norm is especially appropriate when disjoint positive witness masses must add exactly. Its dual is a weighted L^∞ constraint. In the real scalar case, finite-dimensional convex duality [REF-MATH-005] gives a circulation certificate of the form

$$\Phi_{1,w}([\delta]) = \max\{\langle y, \delta\rangle : d_0^* y = 0,\ |y_e| \le w_e\},$$

under the standard linear-programming hypotheses. The constraint $d_0^* y = 0$ says that y is a circulation annihilating coboundaries.

`WDP-CH04-C01` **Conditional result.** The displayed primal–dual identity is an imported finite-dimensional convex-duality result specialized to the declared weighted graph complex. It is not proved here for arbitrary coefficient spaces.

4.4 Closure requires another degree

Suppose a finite 2-cell structure supplies $d_1 : C^1 \to C^2$. A closure diagnostic may be defined by

$$\Gamma(\delta) = J_2(d_1\delta).$$

If J_2 is a norm, this descends to Q^1 if and only if $d_1 d_0 = 0$ on the admissible repairs. Necessity follows by comparing a repair with the zero representative; sufficiency follows from linearity. For a seminorm the exact condition is $\operatorname{im}(d_1 d_0) \subseteq \ker J_2$. The failure

$$K = d_1 d_0$$

is itself a typed curvature or composition defect. Calling Q^1 "cohomology" while $K \neq 0$ is an error. Section 16.7 computes K for declared restriction maps on a finite poset, where it is exactly the failure of the composition law.

4.5 Worked example: exact path and obstructed cycle

On a path $0 \to 1 \to 2$, let $\delta = (2, 3)$. Taking $f = (0, 2, 5)$ gives $d_0 f = \delta$, so $\Phi([\delta]) = 0$ for every norm.

The path is one case of a general fact. On a finite forest, $\dim \operatorname{im} d_0 = (|V| - c)\dim A = |E|\dim A$ for a finite-dimensional coefficient space A, where c is the number of components. For any vector coefficient space, d_0 is onto C^1 and every edge cochain is exact. A graph without cycles therefore has zero quotient for every coefficient choice, and the edgeless graph has no edge cochains at all. Neither carries a nontrivial obstruction. For the same reason every $U(1)$ connection of §5.2 on a forest is gauge-equivalent to the trivial connection: fix the gauge at one root per component and propagate along the unique paths. Its connection Laplacian is then unitarily equivalent to the ordinary weighted graph Laplacian.

On the oriented triangle, take $\delta = (1, 1, 1)$. As in Chapter 2, circulation is 3, whereas every coboundary has circulation 0. Hence δ is not exact. For uniform weights and scalar coefficients, symmetry shows that the best L^1 repair leaves total residual at least 3 under the chosen orientation; the dual circulation $y = (1, 1, 1)$ certifies the bound. The example illustrates a quotient obstruction. It says nothing about quantum contextuality until a measurement cover and empirical model are also supplied.

4.6 Replayed vector-valued three-cycle

The source corpus also supplied one numerical Python sanity check. On the oriented three-cycle it used the vector-valued edge defect

$$\delta = ((1,0),(0,1),(0,0)).$$

Its circulation is $c = (1,1)$. Every repaired residual $r = \delta - d_0\alpha$ has the same sum c, so the triangle inequality gives

$$\sum_e \|r_e\|_1 \geq \|c\|_1 = 2, \qquad \sum_e \|r_e\|_2 \geq \|c\|_2 = \sqrt{2}.$$

Both bounds are attained by $\alpha = ((0,0),(1,0),(1,1))$, which leaves residual $((0,0),(0,0),(1,1))$. The executable replacement in `code/witnessed_difference/vector_obstruction.py` therefore computes the exact certificate rather than depending on a continuous numerical optimizer. It also reproduces the historical display values `2.000000` and `1.414214`. This is a finite example under declared norms, not a general optimization oracle or a physical observer theorem.

4.7 Evaluation preorders and the degree-1 window

Sections 4.1–4.6 declared the cochain host directly. The primitive-axiomatics source (Appendix C) builds the same host from local comparisons, and its route is worth recording because every step is a choice.

Let V be a finite set of evaluation sites and $R \subseteq V \times V$ a declared relation of direct comparisons. Its reflexive–transitive closure R^* is a preorder: reflexivity is added by the closure, and concatenating two finite comparison chains gives a finite chain. Directed cycles are admitted. A cycle does not contradict the construction; it is a place where a residual can survive repair.

The preorder is not yet a cochain complex. A declared route

$$P \longrightarrow N(P) \longrightarrow \mathbb{F}[N(P)] \longrightarrow \text{normalized chains} \longrightarrow \text{dual cochains} \longrightarrow \text{2-truncation}$$

supplies $C^0 = \mathbb{F}^V$, $C^1 = \mathbb{F}^E$, and $C^2 = \mathbb{F}^{S_2}$ with simplicial coboundaries, where $\mathbb{F}$ is a declared coefficient field, E is a declared set of retained nondegenerate 1-simplices of the nerve (pairs $u \neq v$ with $(u,v) \in R^*$, oriented by the comparison), and S_2 is a declared set of retained nondegenerate 2-simplices, each of whose faces lies in E or is degenerate; degenerate faces vanish in normalized cochains. When R^* has directed cycles, the nerve has nondegenerate simplices in every degree, and the 2-truncation keeps only the declared E and S_2. The nerve $N(P)$ is a simplicial set, not a cochain complex and not itself a Dold–Kan object. A Dold–Kan comparison, if wanted, applies to the degree-reversed chain complex $A_2 = C^0$, $A_1 = C^1$, $A_0 = C^2$ and represents the declared truncation, not the nerve.

`WDP-CH04-D01` **Definition (evaluation preorder and degree-1 window).** The evaluation preorder of a declared finite direct-comparison relation R on V is its reflexive–transitive closure R^*. A degree-1 diagnostic window is a declared tuple $\mathcal{W} = (C^0, C^1, C^2, d_0, d_1)$ of finite-dimensional coefficient spaces and linear maps together with a certificate that $d_1 d_0 = 0$. The certificate is automatic on the simplicial route above and must be verified for any hand-declared pair (d_0, d_1).

When the certificate fails, $K = d_1 d_0$ is the composition defect of §4.4 and none of §§4.8–4.11 applies.

Orientation convention. Sections 2.3 and 4.5 orient the triangle cyclically, $(01, 12, 20)$. The simplicial route orients it by vertex order, $(01, 12, 02)$. Reversing the edge 20 to 02 negates that component, so the cyclic field $\delta = (1, 1, 1)$ is the simplicial field $(1, 1, -1)$; both have circulation 3. Every quantity below is invariant under reorientation once the cochain is transported with its edges.

4.8 Closure descent and the degree-1 exact sequence

`WDP-CH04-D02` **Definition (quotient residual, closure charge, and signature).** In a window $\mathcal{W}$ with a declared norm J on C^1, set $Q^1 = C^1/\operatorname{im} d_0$; define the descended closure map $\kappa^1 : Q^1 \to C^2$ by $\kappa^1([h]) = d_1 h$; call $H^1 := \ker \kappa^1$ the closed sector; and define the quotient residual mass $\Phi_1([h]) = \Phi_J([h])$ of §4.2, the closure mass $\Gamma_2([h]) = \|d_1 h\|_1$, and, when $\Phi_1([h]) > 0$, the closure ratio $R_{cl}([h]) = \Gamma_2([h])/\Phi_1([h])$. The diagnostic class of $[h]$ is `ExactClass` when $[h] = 0$, `ClosedResidual` when $[h] \neq 0$ and $\Gamma_2([h]) = 0$, and `ActiveClosureFailure` when $\Gamma_2([h]) > 0$. A window whose carrier, norm, closure data, or $d_1 d_0$ certificate is missing or ill typed receives a typed refusal (`NoRule`), not a class.

Because $d_1 d_0 = 0$ is certified, a class lies in $\ker \kappa^1$ exactly when one (equivalently, every) representative is closed. Thus H^1 is the usual Z^1/B^1 of the window. The name "cohomology" is earned by the certificate, in keeping with the migration rule of §21.12.

Norm hypothesis. Sections 4.9–4.11 take $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, or $\mathbb{F} = \mathbb{Q}$ with J the restriction of a norm on the real extension $C^1 \otimes \mathbb{R}$, such as a weighted L^1 or L^2 norm with rational weights. The restriction matters. The function $J(x) = |x_{01} + \sqrt{2}\, x_{12} + \sqrt{3}\, x_{02}|$ is a norm on $\mathbb{Q}^3$, because $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$; yet on the filled triangle of §4.11 the values $J(\delta - d_0 f)$ for $\delta = (1, 1, 0)$ accumulate at 0, so $\Phi_1([\delta]) = 0$ although $\Gamma_2([\delta]) = 2$. Under the hypothesis, Φ_1 of a rational class equals Φ_1 of the same class over $\mathbb{R}$, since rational repairs are dense among real ones and J is continuous; and a rational field lies in the real image of d_0 only if it lies in the rational image.

`WDP-CH04-P03` **Finite proposition (closure descent and exactness).** In a window with $d_1 d_0 = 0$, the map κ^1 is well defined on Q^1 and

$$0 \longrightarrow H^1 \longrightarrow Q^1 \xrightarrow{\kappa^1} \operatorname{im} d_1 \longrightarrow 0$$

is exact. The sequence is an extension. In finite dimension it splits, but a splitting $Q^1 \cong H^1 \oplus \operatorname{im} d_1$ requires a chosen section and is not canonical.

Proof. If $h' = h + d_0 \alpha$, then $d_1 h' = d_1 h + d_1 d_0 \alpha = d_1 h$, so κ^1 is constant on repair classes. The inclusion of $H^1 = \ker \kappa^1$ is injective with image $\ker \kappa^1$, which is exactness at H^1 and at Q^1. Every element $d_1 h$ of $\operatorname{im} d_1$ equals $\kappa^1([h])$, which is exactness at $\operatorname{im} d_1$. Closedness of a class is representative independent because κ^1 is. □

The Lean package states this on the literal quotient of C^1 by the repair relation $h \sim h + d_0 \alpha$ in an abstract additive setting; §D.1 records the declarations.

4.9 The obstruction signature

`WDP-CH04-P04` **Finite proposition (diagnostic trichotomy).** In a window with $d_1 d_0 = 0$ and a norm J on finite-dimensional C^1 satisfying the norm hypothesis, every class $[h] \in Q^1$ has exactly one diagnostic class, $\Phi_1([h]) = 0$ holds exactly for `ExactClass`, and every `ActiveClosureFailure` class has $\Phi_1([h]) > 0$.

Proof. The quotient seminorm is faithful: over $\mathbb{R}$ or $\mathbb{C}$ because $\operatorname{im} d_0$ is a closed subspace of a finite-dimensional space, and over $\mathbb{Q}$ by the comparison with the real extension under the norm hypothesis. So $\Phi_1([h]) = 0$ iff $[h] = 0$. If $[h] = 0$ then $\Gamma_2([h]) = \|d_1 0\|_1 = 0$. The three defining conditions—$[h] = 0$; $[h] \neq 0$ with $\Gamma_2 = 0$; $\Gamma_2 > 0$—are therefore pairwise disjoint and exhaustive. If $\Gamma_2([h]) > 0$ then $[h] \neq 0$, hence $\Phi_1([h]) > 0$. □

`WDP-CH04-P05` **Finite proposition (signature bounds).** Under the hypotheses of CH04-P04, write $\|d_1\|_{J\to 1} = \sup\{\|d_1 x\|_1 : J(x) \leq 1\}$, a maximum over $\mathbb{R}$ or $\mathbb{C}$ and finite over $\mathbb{Q}$ by the norm hypothesis. Then

1. $\Phi_1([h]) \geq 0$ and $\Gamma_2([h]) \geq 0$;
2. $\Phi_1([h]) = 0$ if and only if $[h] = 0$;
3. $\Gamma_2([h]) = 0$ if and only if $[h] \in H^1$;
4. $\Gamma_2([h]) \leq \|d_1\|_{J\to 1}\, \Phi_1([h])$;
5. if $H^1 \neq 0$, no constant $c > 0$ satisfies $\Gamma_2 \geq c\, \Phi_1$ on all of Q^1;
6. on Q^1/H^1, Γ_2 is a norm equivalent to the quotient norm.

Proof. Items 1–3 follow from the definitions and CH04-P04. For item 4, $d_1(h + d_0\alpha) = d_1 h$ for every repair α, so $\Gamma_2([h]) = \|d_1(h + d_0\alpha)\|_1 \leq \|d_1\|_{J\to 1} J(h + d_0\alpha)$; take the infimum over α. For item 5, a nonzero $[h] \in H^1$ has $\Phi_1 > 0$ and $\Gamma_2 = 0$. For item 6, κ^1 induces an injective linear map $Q^1/H^1 \to C^2$, and all norms on a finite-dimensional real or complex space are equivalent; over $\mathbb{Q}$, apply this on the real extension and restrict. □

For the unweighted L^1 norm, $\|d_1\|_{1\to 1}$ is the largest absolute column sum of the matrix of d_1.

`WDP-CH04-P06` **Finite proposition (closure-ratio bound).** Under the hypotheses of CH04-P04, if $\Phi_1([h]) > 0$, then $0 \leq R_{cl}([h]) \leq \|d_1\|_{J\to 1}$.

Proof. Divide item 4 of CH04-P05 by the positive number $\Phi_1([h])$. □

The normalized ratio $r_{cl} = R_{cl}/\|d_1\|_{J\to 1}$ is defined only when $d_1 \neq 0$. If $d_1 = 0$, every class is closed and r_{cl} is the undefined quotient $0/0$, not zero. Readings such as "R_{cl} near $\|d_1\|_{J\to 1}$ means the residual is dominated by closure failure" are interpretive summaries of the bound, not further theorems.

The unnormalized ratio depends on the declared norms. Rescaling J to cJ and the closure norm to $\lambda\|\cdot\|_1$ multiplies R_{cl} by λ/c, while r_{cl} is unchanged, because $\|d_1\|_{J\to 1}$ rescales in the same way.

The source also proposed a repair policy that accepts a transition only when Φ_1 decreases and R_{cl} stays bounded, with a termination argument for such runs. A policy of that kind belongs to a separately checked routing layer and is not absorbed here; see CH04-O01.

4.10 Dual certificates without importing strong duality

`WDP-CH04-P07` **Finite proposition (dual-certificate soundness).** Over $\mathbb{F} = \mathbb{Q}$, let $J = \|\cdot\|_{1,w}$ with positive rational weights w_e. If $y \in C^1$ satisfies $d_0^* y = 0$ and $|y_e| \leq w_e$ for every edge,

then $\langle y, \delta \rangle \le \Phi_{1,w}([\delta])$. Consequently, a repair f and such a y with $\|\delta - d_0 f\|_{1,w} = \langle y, \delta \rangle$ certify $\Phi_{1,w}([\delta]) = \langle y, \delta \rangle$ and show that f attains it.

Proof. For every f,

$$\langle y, \delta \rangle = \langle y, \delta - d_0 f \rangle + \langle d_0^* y, f \rangle = \langle y, \delta - d_0 f \rangle \le \sum_e |y_e| \, |(\delta - d_0 f)_e| \le \|\delta - d_0 f\|_{1,w}.$$

Taking the infimum over f gives the lower bound. A matching pair meets its own lower bound. □

The same argument holds verbatim over $\mathbb{R}$ with positive real weights; it uses the order of the field, so it does not apply over $\mathbb{C}$ or $\mathbb{F}_2$.

Here d_0^* is the transpose of d_0 for the standard pairing: $(d_0^* y)(v) = \sum_{e \to v} y_e - \sum_{v \to e} y_e$, the net inflow at v. The condition $d_0^* y = 0$ says that y is a circulation.

CH04-C01 asserts more: for every δ a matching pair exists. That existence statement is imported finite-dimensional convex duality and remains a conditional result. CH04-P07 needs no import. The executable package computes (f, y) with an exact two-phase simplex method using Bland's rule [REF-MATH-010] and then checks the pair independently against CH04-P07; a candidate pair that fails any check is refused, not reported. Soundness rests on that check, not on the solver.

`WDP-CH04-P10` **Finite proposition (repairable mass and certificate pairs).** Under the hypotheses of CH04-P07,

1. $0 \le \Phi_{1,w}([\delta]) \le \|\delta\|_{1,w}$, so the repairable mass $R_{\mathrm{rep}}(\delta) = \|\delta\|_{1,w} - \Phi_{1,w}([\delta])$ is nonnegative; $\|\delta\|_{1,w}$ and R_{rep} depend on the representative δ, while $\Phi_{1,w}$ depends only on its class;
2. for a repair f with residual $h = \delta - d_0 f$ and a circulation y with $|y_e| \le w_e$, the equality $\langle y, \delta \rangle = \|h\|_{1,w}$ holds exactly when $y_e = w_e \operatorname{sign}(h_e)$ wherever $h_e \neq 0$; in particular, if f attains $\Phi_{1,w}([\delta])$ and $\langle y, \delta \rangle = \Phi_{1,w}([\delta])$, then y saturates the weights on the support of h;
3. neither the minimizing residual nor the optimal circulation is unique in general.

Proof. The zero repair gives the upper bound in item 1, and CH04-P01 gives class dependence. For item 2, the chain in the proof of CH04-P07 reads $\langle y, \delta \rangle = \sum_e y_e h_e \le \sum_e |y_e||h_e| \le \sum_e w_e |h_e|$, and equality holds throughout exactly when $y_e h_e = w_e |h_e|$ for every e. For item 3, on the triangle of §4.5 with edges $(01, 12, 20)$, unit weights, and $\delta = (1, 0, 0)$, the repairs $f = (0, 0, 0)$, $(0, 1, 0)$, and $(0, \frac{2}{3}, \frac{1}{3})$ leave the residuals $(1, 0, 0)$, $(0, 1, 0)$, and $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, each of mass 1 and each met by $y = (1, 1, 1)$. On the theta graph (two vertices joined by three parallel edges $0 \to 1$) with $\delta = (1, 0, 0)$, the repair $f = 0$ is met by each of $y = (1, -1, 0)$, $(1, 0, -1)$, and $(1, -\frac{1}{2}, -\frac{1}{2})$; so $\Phi_1 = 1$ with three optimal circulations. □

On the path $a \to b \to c$, $\delta = (1, 0)$ is exact, so $\Phi_1 = 0$ and $R_{\mathrm{rep}} = 1$, while the zero representative of the same class has $R_{\mathrm{rep}} = 0$. On the lollipop, a triangle $0 \to 1 \to 2 \to 0$ with a pendant edge $2 \to 3$ and $\delta = (1, 0, 0, 1)$, the repair $f = (0, 0, 0, 1)$ and the circulation $y = (1, 1, 1, 0)$ meet at 1, so $\|\delta\|_1 = 2$, $\Phi_1 = 1$, and $R_{\mathrm{rep}} = 1$: the pendant unit is repairable and the cycle unit is not. Only the scalars are determined by the input. Two correct solvers may return different pairs; each pair is checked against CH04-P07, not against the other.

Declared selections. Over $\mathbb{R}$ the optimal residuals form a nonempty bounded convex polyhedron, and so do the optimal circulations once CH04-C01 supplies one. A declared strictly convex tie-break, such as least Euclidean norm in the declared edge coordinates, selects exactly one of each. The selection is noncanonical without those declared data. Once the coordinates and a unique tie-break

are fixed, the selected residual and dual are class-invariant: replacing δ by $\delta + d_0 g$ leaves $\{\delta - d_0 f : f \in C^0\}$ unchanged, and every feasible dual annihilates $d_0 g$, so its objective and optimal face are unchanged. The site potential itself need not be class-invariant. On the triangle, both $\delta = (1,0,0)$ and $\delta' = (2,-1,0) = \delta + d_0(0,1,0)$ select the same residual $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$ and dual $(1,1,1)$; on the theta graph it selects $(1,0,0)$ and $(1,-\frac{1}{2},-\frac{1}{2})$. The absorbed archived working-paper source 008 checked the lollipop pair and the selected triangle and theta pairs with exact rational code at its recorded commit. The current selection regression independently checks the two triangle representatives, their exact primal/dual witnesses and the squared-norm identity selecting the barycentre; it does not port the historical general selection code. On the triangle's optimal residual face, $r_i \geq 0$ and $\sum_i r_i = 1$, hence $\sum_i r_i^2 - \frac{1}{3} = \sum_i (r_i - \frac{1}{3})^2 \geq 0$, with equality only at the stated barycentre. This establishes its unique least-Euclidean-norm selection on the entire real optimal face.

Two routes between the same endpoints give a circulation, so a failed commuting diagram is a dual certificate.

WDP-CH04-P11 **Finite proposition (path discrepancy bounds the quotient defect).** Under the hypotheses of CH04-P07, let p and q be directed paths from s to t, read as edge-count vectors in $\mathbb{Q}^E$, and let $\gamma = p - q$. Then $d_0^* \gamma = 0$, the discrepancy $\Delta = \sum_{e \in p} \delta_e - \sum_{e \in q} \delta_e$ equals $\langle \gamma, \delta \rangle$ and does not change when δ is replaced by $\delta - d_0 f$, and if $\gamma \neq 0$,

$$|\Delta| \leq c_\gamma \, \Phi_{1,w}([\delta]), \qquad c_\gamma = \max_e \frac{|\gamma_e|}{w_e}.$$

In particular $\Delta \neq 0$ certifies $\Phi_{1,w}([\delta]) > 0$.

Proof. Traversing a directed path adds 1 to the net inflow at its end, -1 at its start, and 0 at every interior visit, so $d_0^* p = \mathbf{1}_t - \mathbf{1}_s = d_0^* q$ and $d_0^* \gamma = 0$. Hence $\Delta = \langle p, \delta \rangle - \langle q, \delta \rangle = \langle \gamma, \delta \rangle$ and $\langle \gamma, d_0 f \rangle = \langle d_0^* \gamma, f \rangle = 0$, which is the invariance of CH05-P04, item 2, for the 1-cycle γ. Both $\pm \gamma / c_\gamma$ are circulations with $|y_e| \leq w_e$, so CH04-P07 gives $\pm \langle \gamma, \delta \rangle \leq c_\gamma \Phi_{1,w}([\delta])$. □

For simple paths $\gamma_e \in \{-1, 0, 1\}$, so with unit weights the bound is $|\Delta| \leq \Phi_1$. On the triangle $A \to B$, $B \to C$, $A \to C$ with $\delta = (\frac{1}{10}, \frac{1}{5}, \frac{2}{5})$, the routes ABC and AC give $\Delta = -\frac{1}{10}$ and $\gamma = (1,1,-1)$. The repair $f = (0, \frac{1}{10}, \frac{3}{10})$ leaves $\frac{1}{10}$ on AC and meets $y = (-1,-1,1)$, so $\Phi_1 = \frac{1}{10} = |\Delta|$, while $\|\delta\|_1 = \frac{7}{10}$ and $R_{\mathrm{rep}} = \frac{3}{5}$: the edge-mass bound $|\Delta| \leq \sum_e |\delta_e|$ is seven times weaker. With $\delta = (\frac{1}{10}, \frac{1}{5}, -\frac{2}{5})$ instead, $\Delta = \Phi_1 = \|\delta\|_1 = \frac{7}{10}$. With weights $(\frac{1}{2}, 1, 1)$ and $\delta = (1,0,0)$, $c_\gamma = 2$ and $f = 0$ meets $y = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$, so $\Phi_{1,w} = \frac{1}{2}$ and $|\Delta| = 1 = c_\gamma \Phi_{1,w}$.

The discrepancy locates repair-invariant mass on the cycle γ; it does not say which edge is wrong. A version with declared transports into a common comparison space and a declared gauge per edge reduces to the triangle inequality under its hypotheses and is not carried; with identity restrictions a section-induced field $\delta = d_0 s$ is exact and every discrepancy vanishes. The absorbed source that stated the path bound had no Lean and no executable check of it.

4.11 Worked example: a filled triangle with one non-closing edge field

WDP-CH04-P08 **Finite proposition (filled-triangle signature).** Take vertices $\{0,1,2\}$, edges $(01, 12, 02)$, and one 2-simplex $(0,1,2)$, with

$$d_0 f = (f_1 - f_0,\ f_2 - f_1,\ f_2 - f_0), \qquad d_1 x = x_{01} + x_{12} - x_{02}.$$

For $\delta = (1,1,0)$ over $\mathbb{Q}$ with unit weights: $d_1 d_0 = 0$; $\Phi_1([\delta]) = 2$, attained at $f = (0,1,0)$ and certified by the circulation $y = (1,1,-1)$; $\Gamma_2([\delta]) = 2$; $\|d_1\|_{1\to 1} = 1$; and $R_{cl}([\delta]) = 1$, which saturates CH04-P06.

Proof. $d_1 d_0 f = (f_1 - f_0) + (f_2 - f_1) - (f_2 - f_0) = 0$. For $f = (0,1,0)$, $d_0 f = (1,-1,0)$ and $\delta - d_0 f = (0,2,0)$, which has L^1 mass 2. The vector $y = (1,1,-1)$ has $|y_e| \le 1$, and its net inflow is $0 - (1-1) = 0$ at vertex 0, $1 - 1 = 0$ at vertex 1, and $(1-1) - 0 = 0$ at vertex 2; equivalently $\langle y, d_0 f\rangle = (f_1 - f_0) + (f_2 - f_1) - (f_2 - f_0) = 0$ for every f, so $d_0^* y = 0$. Since $\langle y, \delta\rangle = 2$, CH04-P07 gives $\Phi_1 = 2$. Next, $d_1 \delta = 2$, so $\Gamma_2 = 2$. The matrix of d_1 is the row $(1,1,-1)$, whose largest absolute column sum is 1. Finally $R_{cl} = 2/2 = 1$. □

The class is `ActiveClosureFailure`. Compare the hollow triangle of §4.5: with no 2-simplex, $C^2 = 0$, so every class is closed; $\delta = (1,1,1)$ has $\Phi_1 = 3$, certified by $y = (1,1,1)$, and $\Gamma_2 = 0$. That class is `ClosedResidual` and witnesses item 5 of CH04-P05: filling the triangle converts a closed residual into closure charge.

Growing a complex is a natural candidate for a repair mechanism. It is not one.

WDP-CH04-P09 **Finite proposition (growth does not lower Φ_1).** Let K be a subcomplex of a finite complex K', with weights on K restricted from K', and let h be the restriction of an edge field h' on K'. Then $\Phi_1(h) \le \Phi_1(h')$. Attaching 2-simplices leaves Φ_1 unchanged, and can raise Γ_2. Subdividing an edge through a new vertex of degree two, with the old value split into two summands and both new edges carrying the old weight, leaves Φ_1 unchanged.

Proof. For any repair α' on K', dropping the edges outside K from the weighted sum gives $\|h' - d_0\alpha'\|_{K'} \ge \|h - d_0(\alpha'|_K)\|_K \ge \Phi_1(h)$; take the infimum over α'. Attaching 2-simplices changes neither C^0, C^1, nor d_0. For a subdivided edge $u \to m \to v$ with values $x_1 + x_2 = x$ and weight w, $w|x_1 - (f_m - f_u)| + w|x_2 - (f_v - f_m)| \ge w|x - (f_v - f_u)|$ by the triangle inequality, with equality at $f_m = f_u + x_1$; so every repair of the subdivided complex has a repair of the original of no greater cost, and conversely. □

On the hollow triangle with the cyclic field of §4.5, $\Phi_1 = 3$ and $\Gamma_2 = 0$; after filling, $\Phi_1 = 3$ and $\Gamma_2 = 3$. Splitting an edge value as $(\frac{1}{2}, \frac{1}{2})$ or $(5,-4)$ leaves $\Phi_1 = 3$. None of these growth moves lowers Φ_1. A decrease requires deleting cells, changing declared weights (including unequal weights on a subdivided edge), or changing existing edge values: deleting edge 20 leaves a tree with $\Phi_1 = 0$, and subdividing edge 01 with new weights $(\frac{1}{2}, \frac{1}{2})$ lowers Φ_1 to $\frac{3}{2}$.

Restricted repair sets. The same argument covers a declared set $A \subseteq C^0(K)$ of admissible repairs, with $\Phi_A(h) = \inf_{\alpha \in A} \|h - d_0\alpha\|_{1,w}$. If K' arises from K by the growth moves of CH04-P09, h' extends h as there, and the declared repair set $A' \subseteq C^0(K')$ restricts into A on the old vertices, then $\Phi_{A'}(h') \ge \Phi_A(h)$: for $\alpha' \in A'$ the proof of CH04-P09 gives $\|h' - d_0\alpha'\|_{K'} \ge \|h - d_0(\alpha'|_K)\|_K \ge \Phi_A(h)$. A decrease needs a repair set whose restrictions leave A. On one edge $0 \to 1$ with value 1 and $A = \{0\}$, $\Phi_A = 1$. Subdividing through m with values $(1,0)$ and $A' = \{\alpha' : \alpha'(0) = \alpha'(1) = 0\}$ keeps $\Phi_{A'} = \min_t(|1-t| + |t|) = 1$. Taking $A' = C^0(K')$ gives 0, but only because its restrictions include repairs with $\alpha(1) \ne \alpha(0)$. That is a change of declared repair data, not growth.

4.12 What the signature does not do

WDP-CH04-N01 **Nonclaim.** A diagnostic class, signature value, or certificate is not a routing terminal, repair action, stabilization verification record, or authority. In particular, `ExactClass` is not `ExactRepair`, and `ClosedResidual` is not `StabilizedSurvivor`.

Routing a class to a terminal requires a separately checked policy that consumes the operation, witness, budget, and stabilization evidence. The source's terminal enumerations, lift specifications, guarded repair policy, and reflective completion of contextual systems are not automatically carried by a diagnostic. Chapter 21 and Appendix J give the declared finite routing interface and its limits as CH04-O01.

`WDP-CH04-N02` **Nonclaim.** Growth of the complex does not repair a quotient residual: passing to a larger complex, face attachment, and equal-weight subdivision never lower Φ_1; a decrease requires deleting cells, changing declared weights (including unequal subdivision weights), or changing existing edge values; and survival of a class to the limit page of a filtration spectral sequence records cohomology of the final complex rather than positivity of Φ_1.

Projection is not repair either. On the filled triangle with $\delta = (1, 1, 0)$, orthogonal projection onto $\ker d_1$ subtracts $(2/3, 2/3, -2/3) \in \operatorname{im} d_1^*$ and returns $(1/3, 1/3, 2/3)$, which is closed. But $\operatorname{im} d_1^*$ meets $\operatorname{im} d_0$ only in 0, so the difference is not a repair and Φ_1 of the original class is still 2. The projection replaces the datum; it does not diagnose it.

The window supports one sharp falsifier. Two implementations given the same declared window, coefficient field, weights, and cochain must return the same signature. A divergence refutes an implementation or exposes an undeclared assumption. Certificate pairs are not compared this way: by CH04-P10 two correct implementations may return different residuals and circulations, and each pair is checked against CH04-P07 on its own.

Interpretive remark (two opposite misreadings of the quotient). The quotient $C^1/\operatorname{im} d_0$ blocks two errors. The first treats every visible difference as an obstruction, although it may be a coordinate choice, a unit mismatch, a calibration offset or a naming convention, all of which are coboundaries once the repair data declare them. The second treats every difference as removable redescription; the quotient refutes it whenever a class survives all declared repairs. Under a declared repair space, the coboundaries are the differences that redescription removes, and the nonzero class is what survives it. A difference in formatting between two generated answers is of the first kind under a formatting repair. A lost boundary condition, proof obligation or database constraint is of the second. Whether a difference can be removed is a question about the declared repair data, never about the appearance of the difference.

Chapter 5 — Optional quantum-capable carriers

5.1 Branch declaration

This chapter begins a branch, not the trunk. We now **declare** complex coefficients, phases, inner products, and matrix representations. Those imports are recorded because later quantum comparisons require them. Nothing in Chapters 1–4 selects them uniquely.

`WDP-CH05-D01` **Definition (quantum-capable carrier package).** A finite quantum-capable carrier is

$$\mathfrak{Q} = (G, w, \mathbb{C}^N, U, \nabla, \langle -, - \rangle),$$

where G is a finite graph or cell complex, w is a positive weighting, $\mathbb{C}^N$ is a declared complex fiber, U assigns unitary transport to oriented edges, ∇ is the associated covariant difference, and $\langle -, - \rangle$

is a declared Hermitian inner product.

The adjective "quantum-capable" means only that familiar finite quantum structures can be represented. It does not mean that a physical quantum theory has been derived.

5.2 $U(1)$ connection and holonomy

For a line fiber, assign $U_{uv} \in U(1)$ with $U_{vu} = U_{uv}^{-1}$. For a vertex field $\psi : V \to \mathbb{C}$, set

$$(\nabla_U \psi)_{uv} = \psi(v) - U_{uv}\psi(u).$$

A gauge transformation $g : V \to U(1)$ acts by

$$\psi(u) \mapsto g(u)\psi(u), \qquad U_{uv} \mapsto g(v)U_{uv}g(u)^{-1}.$$

WDP-CH05-P01 **Finite proposition (gauge covariance).** Under the displayed action,

$$(\nabla_{U^g}\psi^g)_{uv} = g(v)(\nabla_U \psi)_{uv}.$$

The proof is a direct substitution. Consequently, any norm invariant under vertexwise phase multiplication gives a gauge-invariant defect magnitude.

For an oriented loop $\gamma = e_1 \cdots e_k$, the holonomy is

$$\mathrm{Hol}_U(\gamma) = U_{e_k} \cdots U_{e_1}.$$

In the abelian $U(1)$ case it is gauge invariant. In a nonabelian unitary group it transforms by conjugation, so conjugacy-invariant functions such as trace or spectrum are gauge invariant.

The same product is defined on any walk that may traverse an edge against its declared orientation: such a step contributes $U_{vu} = U_{uv}^{-1}$, which is the complex conjugate in $U(1)$. **Finite proposition (walk transport).** For a walk γ from u to v, the transport $T_U(\gamma)$ satisfies $T_{U^g}(\gamma) = g(v)\, T_U(\gamma)\, g(u)^{-1}$. The gauge factors of each interior vertex cancel in adjacent steps. On a closed walk ($u = v$) the $U(1)$ transport is therefore gauge invariant. On an open walk it is only covariant at its two endpoints. An open-path phase is not an invariant observable of the connection until its endpoint frames are fixed by a further declaration.

5.3 Connection Laplacian

Let W be the positive diagonal edge-weight operator. The finite connection Laplacian is

$$L_U = \nabla_U^* W \nabla_U.$$

WDP-CH05-P02 **Finite proposition.** L_U is Hermitian and positive semidefinite.

Proof. Hermiticity follows from the adjoint construction. For every ψ,

$$\langle \psi, L_U \psi \rangle = \langle \nabla_U \psi, W \nabla_U \psi \rangle \geq 0$$

because W is positive. □

The kernel of such an operator is exactly the joint kernel of its pieces, but only when the weights are definite.

WDP-CH05-P07 **Finite proposition (kernel of a weighted sum of forms).** For finitely many linear maps A_a out of a finite-dimensional inner-product space over $\mathbb{Q}$, $\mathbb{Q}(i)$, $\mathbb{R}$, or $\mathbb{C}$, and positive-definite weights W_a, the operator $H = \sum_a A_a^* W_a A_a$ is Hermitian and positive semidefinite, and $H\psi = 0$ exactly when $A_a\psi = 0$ for every a. The equivalence can fail when some W_a is only positive semidefinite.

Proof. $\langle \psi, H\psi \rangle = \sum_a \langle A_a\psi, W_a A_a\psi \rangle \geq 0$. If $H\psi = 0$, every summand vanishes, and definiteness gives $A_a\psi = 0$; the converse is immediate. With $W = 0$ and $A \neq 0$, $H = 0$ while $A\psi \neq 0$. □

The connection Laplacian of CH05-P02 is the case of a single covariant difference, so its kernel is exactly the set of covariantly constant fields.

Gauge-related connections yield unitarily conjugate Laplacians, hence identical spectra. This is a representation theorem inside the declared package, not a selection theorem for the package.

5.4 Real cochains and phases

Given a real edge cochain a and a declared scale $\alpha \in \mathbb{R}$, one may set

$$U_e = \exp(i\alpha a_e).$$

If a changes by a real coboundary $d_0 f$, then U changes by the $U(1)$ gauge transformation $g(v) = \exp(i\alpha f(v))$. Thus exponentiation descends from real cochain classes to gauge classes once α is fixed.

The phase scale is not selected by the real quotient. Distinct real cochains can also exponentiate to the same phase because of 2π periodicity. The map is therefore structured and potentially lossy.

The scale must also be fixed before the cochain is seen. Choosing $\alpha = 1/I_1(a)$, so that each cochain is normalized by its own raw ℓ^1 mass, does not define a phase convention. It is not linear in a. It is not constant on cochain classes either, because I_1 depends on the representative and gauge-equivalent representatives can then receive different scales. A fixed α, declared as observer data, is what makes the descent statement above meaningful. Neither I_1 nor Φ_1 selects it.

5.5 Matrix fibers

For $N \geq 2$, a generalized Gell-Mann basis gives a real basis of traceless Hermitian $N \times N$ matrices. Together with the identity, it coordinatizes Hermitian matrices. The basis is useful for calculations but depends on a declared complex fiber and basis convention. It does not derive $SU(N)$ from finite difference.

Finite proposition (generalized Gell-Mann basis). With matrix units E_{jk}, take $E_{jk} + E_{kj}$ and $-iE_{jk} + iE_{kj}$ for $j < k$, and, for $1 \leq \ell \leq N-1$, $\sqrt{2/(\ell(\ell+1))}\,\mathrm{diag}(1, \ldots, 1, -\ell, 0, \ldots, 0)$ with ℓ leading ones. These $N^2 - 1$ matrices λ_a are Hermitian, traceless and satisfy $\mathrm{Tr}(\lambda_a \lambda_b) = 2\delta_{ab}$, so they form a real basis of the traceless Hermitian matrices.

Proof. There are $N(N-1)/2$ symmetric, the same number of antisymmetric and $N-1$ diagonal elements, N^2-1 in all. Each is Hermitian and traceless by inspection. Off-diagonal elements with different supports have zero trace product. The symmetric and antisymmetric elements on one support have trace product $\mathrm{Tr}((E_{jk}+E_{kj})(-iE_{jk}+iE_{kj})) = -i+i = 0$. The diagonal elements are orthogonal by direct summation, and the displayed scale makes every squared Frobenius norm 2. Orthogonal nonzero elements are linearly independent, and their number equals the real dimension N^2-1. □

With $T_a = \lambda_a/2$, the commutators define structure constants by $[T_a, T_b] = i\sum_c f_{abc}T_c$. They are real, because $-i[T_a, T_b]$ is Hermitian and traceless and the T_c are a real basis. They are totally antisymmetric, because $f_{abc} = -2i\,\mathrm{Tr}([T_a, T_b]T_c)$ and the trace is cyclic. None of this structure comes from a graph. The label space and its basis may be attached to vertices or edges only by a separate modeling choice. The exact companion checks of these identities, together with walk transport, the connection Laplacian and cochain exponentiation, are the `omega_quantum_physics` integration tests `tests/foundations/quantum_carrier_primitive_difference.rs`.

5.6 Worked example: phase around a triangle

Let each oriented edge of a triangle carry the same real value $a_e = \theta$ and take $\alpha = 1$. The loop holonomy is $e^{3i\theta}$. If $\theta = 2\pi/3$, the holonomy is 1 even though each edge phase is nontrivial. If $\theta = \pi/3$, the holonomy is -1. The example shows why local phase and global loop residue differ. It does not establish a particle, an amplitude rule, or an experiment.

`WDP-CH05-N01` **Nonclaim.** This chapter does not derive Hilbert space, unitarity as dynamics, tensor composition, entanglement, observables, the Born rule, collapse, or measurement.

`WDP-CH05-N02` **Nonclaim.** The existence of a connection Laplacian does not identify its lowest eigenvector with a physical vacuum or foundational ground.

5.7 Nonabelian transport and its exact scope

For a declared fiber $\mathbb{C}^N$, edge transport can take values in $U(N)$. With the convention

$$(\nabla_U \psi)_{uv} = \psi(v) - U_{uv}\psi(u),$$

a vertexwise gauge $g : V \to U(N)$ acts by

$$\psi(v) \mapsto g(v)\psi(v), \qquad U_{uv} \mapsto g(v)U_{uv}g(u)^{-1}.$$

The same substitution as in the line-bundle case gives covariance at the target vertex. Loop holonomy based at v transforms as $H_\gamma \mapsto g(v)H_\gamma g(v)^{-1}$. Thus its conjugacy class, trace, and spectrum are invariant. A chosen matrix representative is not invariant.

This distinction is important when a source speaks of "the holonomy." The gauge-invariant object is generally a conjugacy class or class function, not a specific matrix. Likewise, a basis of $\mathfrak{su}(N)$ is computational data; the Lie algebra does not privilege one generalized Gell-Mann basis.

5.8 Density matrices are another import

Given the complex Hilbert carrier, one may declare a density operator $\rho \succeq 0$ with $\operatorname{tr}\rho = 1$. Unitary conjugation preserves positivity and trace. A completely positive trace-preserving map can then be studied, and trace norm has useful contractivity properties under appropriate hypotheses. Every noun in this sentence—density operator, positivity cone, trace, complete positivity, tensor extension—is additional structure.

The monograph therefore uses the following import ladder:

$$\begin{gathered}\text{real graph cochains}\\ \downarrow \text{ declare phase scale and complex line fiber}\\ U(1) \text{ transport}\\ \downarrow \text{ declare } \mathbb{C}^N \text{ and Hermitian product}\\ U(N) \text{ transport and connection Laplacian}\\ \downarrow \text{ declare state/effect cones and normalization}\\ \text{finite operational quantum carrier}\\ \downarrow \text{ declare instruments/composition/dynamics}\\ \text{candidate quantum theory.}\end{gathered}$$

Each arrow is an interface. None is licensed by similarity of notation.

5.9 Stability under perturbation

Finite-dimensionality gives a useful but limited stability statement. If two connection matrices differ slightly in an operator norm, their Laplacians also differ continuously; standard finite-matrix perturbation bounds can then control eigenvalue changes. To use such a bound one must specify the norm, weights, graph, and convention for comparing connections. Gauge-equivalent connections need no perturbation estimate because they are exactly unitarily conjugate.

`WDP-CH05-O01` **Conditional result (resolved finite target).** For two unweighted rational U(1) connection Laplacians on the same ordered finite carrier, the exact maximum absolute-real-plus-imaginary row sum of their difference bounds every ordered eigenvalue displacement; a separately valid gap bound greater than twice this value ensures gap survival. The proof, native ownership and scope are given in Appendix I.

5.10 Higher-degree windows on finite simplicial complexes

Sections 5.10 and 5.12 use real or rational cochains, and §5.11 compares rational with mod-2 coefficients; they sit in this branch because they extend the degree-1 window of §§4.7–4.11 rather than the trunk. Let K be a finite abstract simplicial complex on a totally ordered vertex set, and let $C^k(K;\mathbb{F}) = \mathbb{F}^{\Sigma_k}$ with

$$(\delta^k\varphi)([v_0,\dots,v_{k+1}]) = \sum_{i=0}^{k+1}(-1)^i\varphi([v_0,\dots,\widehat{v_i},\dots,v_{k+1}]).$$

`WDP-CH05-P03` **Finite proposition (degree-k windows).** On a finite oriented simplicial complex, $\delta^{k+1}\delta^k = 0$ for every k. Hence each triple (C^{k-1}, C^k, C^{k+1}) with a norm J on C^k satisfying the

norm hypothesis of §4.8 is a window in the sense of CH04-D01: $\kappa^k([h]) = \delta^k h$ is well defined on $Q^k = C^k / \operatorname{im} \delta^{k-1}$, the sequence $0 \to H^k \to Q^k \to \operatorname{im} \delta^k \to 0$ is exact, and $\Gamma_{k+1} \le \|\delta^k\|_{J\to 1} \Phi_k$.

Proof. In $\delta^{k+1}\delta^k\varphi$, each (k)-face of a $(k+2)$-simplex arises by deleting two vertices in either order, with opposite signs [REF-MATH-002]. The remaining statements are CH04-P03 and CH04-P05 applied to the degree-k window. □

This is a finite simplicial extension. It says nothing about composition-level gluing of evaluator defects, cup products, or continuum differential forms.

With the standard inner products, the combinatorial Laplacian $\Delta_k = \delta^{k*}\delta^k + \delta^{k-1}\delta^{k-1*}$ has kernel isomorphic to H^k over $\mathbb{Q}$ or $\mathbb{R}$. This is elementary finite linear algebra: $\ker \Delta_k = \ker \delta^k \cap \ker \delta^{k-1*}$, the orthogonal complement of $\operatorname{im} \delta^{k-1}$ inside $\ker \delta^k$. [REF-MATH-001] develops the real Whitney setting in which it is usually met. The obstruction signature does not use it.

5.11 Coefficients matter: a torsion witness

The coefficient field is part of the declared window, not a detail.

WDP-CH05-P06 **Finite proposition (a rational solver misses 2-torsion).** For the six-vertex, ten-triangle triangulation K of the real projective plane, $H^2(K;\mathbb{Q}) = 0$ while $H^2(K;\mathbb{F}_2) \cong \mathbb{F}_2$.

Proof. K has 6 vertices, 15 edges, and 10 triangles. Exact row reduction gives $\operatorname{rank}_{\mathbb{Q}} \delta^1 = 10$ and $\operatorname{rank}_{\mathbb{F}_2} \delta^1 = 9$. Since $C^3 = 0$, $\dim H^2 = 10 - \operatorname{rank} \delta^1$, which is 0 over $\mathbb{Q}$ and 1 over $\mathbb{F}_2$. This agrees with $H^2(\mathbb{RP}^2;\mathbb{Z}) \cong \mathbb{Z}/2$ and the algebraic homology coefficient theorem [REF-MATH-002]. □

A window declared over $\mathbb{Q}$ therefore reports no degree-2 class where a mod-2 window exposes one. Neither answer is wrong; they answer different declared questions. The nontrivial mod-2 class is represented by the indicator of a single triangle; the all-ones 2-cochain is a mod-2 coboundary, because the ten triangles pair off. Over $\mathbb{Q}$ the single-triangle indicator is exact ($\Phi_2 = 0$), and every rational repair of it has non-integral values: that is the concrete sense in which a rational solver misses the torsion.

The algebra behind it is short: multiplication by 2 on $\mathbb{Z}$ has cokernel $\mathbb{Z}/2$, but becomes invertible after tensoring with $\mathbb{Q}$. Rational coefficients are flat and see no torsion.

5.12 Balance identities on finite graphs

WDP-CH05-P04 **Finite proposition (combinatorial balance and invariant pairings).** Items 1 and 2 hold on every finite oriented graph; item 3 concerns a finite simplicial complex:

1. $\sum_v (d_0^* x)(v) = 0$ for every $x \in C^1$, so $d_0^* x = \rho$ is solvable only if $\sum_v \rho(v) = 0$;
2. a functional F on C^1 with $F(A + d_0\lambda) = F(A)$ for every λ descends uniquely to Q^1; in particular, the pairing $\langle A, \gamma\rangle$ with a 1-cycle γ ($\partial\gamma = 0$) is invariant;
3. on a finite simplicial complex with 3-simplices, $\delta^2\delta^1 = 0$.

Proof. (1) $\sum_v (d_0^* x)(v) = \langle \mathbf{1}, d_0^* x\rangle = \langle d_0 \mathbf{1}, x\rangle = 0$ because d_0 kills constants; equivalently, each edge contributes $+x_e$ at its target and $-x_e$ at its source. (2) $\langle A + d_0\lambda, \gamma\rangle = \langle A, \gamma\rangle + \langle \lambda, \partial\gamma\rangle$. (3) is CH05-P03 at $k = 1$. □

Item 1 needs no closedness, connectedness, or boundary condition. The condition $\sum_v \rho(v) = 0$ is necessary, not sufficient: on a disconnected graph each component must be neutral separately. The source stated item 1 only "on a closed complex" and attributed a boundary leak to graphs with

sinks. On a finite graph there is no such leak; it appears only if some edge is allowed an endpoint outside the vertex set, which is not a graph in this sense.

WDP-CH05-N03 **Nonclaim.** Reading d_0^* as divergence, $d_1 A$ as flux or curvature, item 1 as charge conservation, $\delta^2\delta^1 = 0$ as a Bianchi identity or an absence of monopoles, or averaging over a finite symmetry group as a Noether law is interpretive vocabulary. No Maxwell equation, Lorentz structure, physical charge, monopole physics, or physical conservation law is derived.

5.13 A projective phase obstruction

Declare a finite group G and unitary matrices $U(g)$ with $U(g)U(h) = \omega(g,h)U(gh)$ for phases $\omega(g,h) \in U(1)$. Associativity of matrix multiplication forces the 2-cocycle identity $\omega(g,h)\,\omega(gh,k) = \omega(g,hk)\,\omega(h,k)$. Rephasing $U'(g) = \lambda(g)U(g)$ multiplies ω by the coboundary $\lambda(g)\lambda(h)/\lambda(gh)$, so only the class $[\omega] \in H^2(G;U(1))$ is invariant [REF-MATH-011].

WDP-CH05-P05 **Finite proposition (Klein-four Pauli obstruction).** Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{e, x, z, xz\}$ with $U(e) = I$, $U(x) = \sigma_x$, $U(z) = \sigma_z$, and $U(xz) = \sigma_y$. Then U is a projective representation, its phase function satisfies the 2-cocycle identity, and the ratio $\omega(x,z)/\omega(z,x) = -1$ is unchanged by every rephasing. Hence $[\omega] \neq 0$: no rephasing makes U a linear representation.

Proof. $\sigma_x\sigma_z = -i\sigma_y = -i\,U(xz)$ and $\sigma_z\sigma_x = i\sigma_y = i\,U(xz)$, so $\omega(x,z) = -i$, $\omega(z,x) = i$, and the ratio is -1. Under rephasing, the coboundary factors at (x,z) and (z,x) are equal because G is abelian, so they cancel in the ratio. A linear representation would have $\omega \equiv 1$ and ratio 1. □

The source wrote $\sigma_x\sigma_z = -U(xz)$; the correct factor is $-i$. Its conclusion survives because only the ratio is invariant. The result is finite group cohomology of a declared complex representation. It derives no spin, qubit, or quantum theory from difference; CH05-N01 still applies.

The Klein-four class is the case $N = 2$ of a family, and the family has an additive counterpart whose nonzero form no finite-dimensional carrier over a characteristic-zero field supports. Positive-characteristic exceptions are part of CH05-P08 below.

WDP-CH05-P08 **Finite proposition (trace obstruction to a finite commutation relation).** Let R be a commutative ring, $N \geq 1$, and $c \in R$ with $Nc \neq 0$. No $A, B \in M_N(R)$ satisfy $AB - BA = cI_N$. Over a field K, some $A, B \in M_N(K)$ satisfy $AB - BA = cI_N$ for some $c \neq 0$ exactly when K has characteristic $p > 0$ and $p \mid N$. The hypothesis $Nc \neq 0$ cannot be weakened to "R has characteristic zero": $R = \mathbb{Z} \times \mathbb{F}_2$ has characteristic zero, $c = (0,1)$ is nonzero with $2c = 0$, and $A = (0, D)$, $B = (0, X)$ with

$$D = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

satisfy $AB - BA = cI_2$.

Proof. In a commutative ring $\mathrm{tr}(AB) = \sum_{i,j} A_{ij}B_{ji} = \mathrm{tr}(BA)$, so $\mathrm{tr}(AB - BA) = 0$, while $\mathrm{tr}(cI_N) = Nc \neq 0$. Over a field, $Nc = 0$ with $c \neq 0$ forces $N = 0$ in K, that is, characteristic $p > 0$ with $p \mid N$. Conversely, on $\mathbb{F}_p[x]/(x^p)$ with basis $1, x, \ldots, x^{p-1}$ let D differentiate and X multiply by x. Then $[D,X]x^k = (k+1)x^k - kx^k = x^k$ for $k < p-1$ and $[D,X]x^{p-1} = -(p-1)x^{p-1} = x^{p-1}$, so $[D,X] = I_p$; for $p \mid N$, block-diagonal copies of cD and X give cI_N. For $p = 2$ these are the displayed matrices, and $M_2(\mathbb{Z} \times \mathbb{F}_2) = M_2(\mathbb{Z}) \times M_2(\mathbb{F}_2)$ componentwise. □

In particular no complex carrier, and no exact carrier over $\mathbb{Q}(i)$ or $\mathbb{Q}(\zeta_N)$, supports $AB - BA = cI$ with $c \neq 0$. The absorbed archived working-paper source 004 stated the obstruction for every commutative ring of characteristic zero; the ring above refutes that form. Its cited external Lean establishes the case of integer matrices only; it was read, not ported, and gives no Lean credit. Nothing here represents a position, a momentum, or Planck's constant; CH05-N01 applies.

WDP-CH05-P09 **Finite proposition (the clock-and-shift cocycle).** Let $N \geq 2$, let ω be a primitive N-th root of unity, and on $\mathbb{C}^N$ with indices modulo N set $Xe_j = e_{j+1}$ and $Ze_j = \omega^j e_j$. Then X and Z are unitary with entries in $\mathbb{Z}[\omega]$, $X^N = Z^N = I$, and $XZ = \omega^{-1} ZX$. On $G = \mathbb{Z}_N \times \mathbb{Z}_N$ the assignment $U(a,b) = X^a Z^b$ is a projective representation with $U(a,b)U(a',b') = \omega^{ba'} U(a+a', b+b')$. For $g = (a,b)$ and $h = (a',b')$ its commutator ratio $c(g,h)/c(h,g) = \omega^{ba'-b'a}$ is unchanged by every rephasing and equals ω^{-1} at $g = (1,0)$, $h = (0,1)$, so the class $[c] \in H^2(G; U(1))$ has order exactly N. For $N = 2$, U is the assignment of CH05-P05 up to the rephasing $U(1,1) = XZ = -i\sigma_y$.

Proof. $XZe_j = \omega^j e_{j+1}$ and $ZXe_j = \omega^{j+1} e_{j+1}$, so $Z^b X^{a'} = \omega^{ba'} X^{a'} Z^b$, which gives the product rule; exponents may be read modulo N because $X^N = Z^N = I$. As earlier in this section, a rephasing multiplies c by $\lambda(g)\lambda(h)/\lambda(g+h)$, which is symmetric in g and h because G is abelian, so the ratio is unchanged. The cocycle c^k has ratio $\omega^{-k} \neq 1$ at $(1,0), (0,1)$ for $0 < k < N$, so it is not a coboundary, while $c^N \equiv 1$. □

The two propositions answer one question in two ways. The additive relation $AB - BA = cI$ has no finite solution when $Nc \neq 0$; the multiplicative relation $XZ = \omega^{-1} ZX$ holds in every dimension as a projective class of order N. The absorbed source proposed the second as the finite substitute for the first and did not prove it. Neither derives quantum kinematics; CH05-N01 applies.

5.14 Trace-norm certificates and channel contractivity

With the imports of §5.8 in place, two standard finite-dimensional results are used inside this branch and nowhere upstream.

WDP-CH05-C01 **Conditional result (cited result).** On $M_d(\mathbb{C})$ with the pairing $\operatorname{Re}\operatorname{tr}(W^\dagger X)$, for a subspace V and an operator L,

$$\min_{X \in V} \|L - X\|_1 = \max\{\operatorname{Re}\operatorname{tr}(W^\dagger L) : W \in V^\perp,\ \|W\|_\infty \leq 1\},$$

and every completely positive trace-preserving map $\mathcal{E}$ satisfies $\|\mathcal{E}(\rho) - \mathcal{E}(\sigma)\|_1 \leq \|\rho - \sigma\|_1$ for density operators ρ, σ [REF-QM-020; REF-QM-021; REF-MATH-005].

The first identity is the Schatten-norm analogue of §4.10; its weak direction is the same one-line argument with the trace pairing. The second is the operator analogue of observer passivity (§10.5). Neither makes a channel an observer or derives measurement. The source checked these results numerically with floating-point eigensolvers and a declared tolerance; this monograph imports them as theorems and does not treat a numerical check as a certificate.

Two lifts of real edge fields into this branch show what the first identity does and does not add to §4.10.

Diagonal lift. For real $x \in \mathbb{R}^E$, $\|\operatorname{diag}(x)\|_1 = \|x\|_1$, because the singular values of a real diagonal matrix are the $|x_e|$. Take $L = \operatorname{diag}(\delta)$ and the real subspace $V = \{\operatorname{diag}(d_0 f) : f \in \mathbb{R}^V\}$. The primal side is then the unit-weight $\Phi_1([\delta])$. On the dual side only the real parts $y_e = \operatorname{Re} W_{ee}$ of the diagonal of W enter the pairing with L and the condition $W \in V^\perp$; they form a circulation, and

$|y_e| \le |W_{ee}| \le \|W\|_\infty \le 1$. Conversely $\operatorname{diag}(y)$ is feasible for every such circulation. So the §4.10 problem is the diagonal instance of the first identity, whose weak direction there is CH04-P07. For $h = (1, -2, \frac{1}{2})$, $\|\operatorname{diag}(h)\|_1 = \frac{7}{2} = \|h\|_1$. The lift represents a chosen field: $\operatorname{diag}(d_0 f) \neq 0$ whenever $d_0 f \neq 0$, so it does not quotient repairs.

Density lift. A nonzero real field h also has a declared rank-one lift, $\psi(h)_e = \operatorname{sign}(h_e)\sqrt{|h_e|/\|h\|_1}$ and $\rho(h) = \psi(h)\psi(h)^\top$, a symmetric projector of trace one. The real, phase-free section is a choice: phases $e^{i\theta_e}$ give other lifts with the same moduli. The lift remembers exactly the line through h: $\rho(h) = \rho(k)$ if and only if $k = ch$ for some real $c \neq 0$, because $\rho(h)$ determines $\psi(h)$ up to sign and $h_e/\|h\|_1 = \operatorname{sign}(\psi_e)\psi_e^2$. For unit vectors ψ, ψ' the difference $D = \psi\psi^\top - \psi'\psi'^\top$ has trace 0, rank at most 2, and $\operatorname{tr} D^2 = 2 - 2\langle\psi, \psi'\rangle^2$, so two lifts have trace-norm separation $2\sqrt{1 - \langle\psi, \psi'\rangle^2}$. Every functional of lifts is blind to magnitude. Replacing δ by $\lambda\delta$ with $\lambda \neq 0$ multiplies Φ_1 by $|\lambda|$ and the least-Euclidean-norm residual and dual selections of §4.10 by λ and $\operatorname{sign}\lambda$, which leaves their lifts unchanged. This scaling statement does not extend to an arbitrary strictly convex tie-break, such as a norm centered at a fixed nonzero target. On the theta graph the selected residual $(1, 0, 0)$ and circulation $(1, -\frac{1}{2}, -\frac{1}{2})$ have $\langle\psi, \psi'\rangle^2 = \frac{1}{2}$, so their lifts have trace-norm separation $\sqrt{2}$. On the triangle the selections $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $(1, 1, 1)$ have equal lifts, at distance 0, although $\Phi_1 = 1$. A distance between lifts is not an obstruction magnitude and is not used as one. The lift is a density operator on the declared Hilbert carrier. It is not thereby identified with a physical state; unit trace alone supplies no empirical probability interpretation (CH05-N01).

5.15 Two finite reconstruction frames

With the imports of §5.8, an operator on $\mathbb{C}^d$ can be recovered exactly from finitely many numbers, and for the two frames below those numbers lie in a cyclotomic field when the operator's entries do. Both frames are declared structure on the complex carrier; the calculus selects neither.

Two bases are *unbiased* when $|\langle u, v\rangle|^2 = \langle u, u\rangle\langle v, v\rangle/d$ for every u in one and v in the other. In dimension 4 take the standard basis and

Basis	Vectors in $\mathbb{Z}[i]^4$
B_1	$(1,-1,1,-1)$, $(1,1,-1,-1)$, $(1,-1,-1,1)$, $(1,1,1,1)$
B_2	$(1,-i,i,1)$, $(1,i,-i,1)$, $(1,-i,-i,-1)$, $(1,i,i,-1)$
B_3	$(1,-1,-i,-i)$, $(1,1,-i,i)$, $(1,1,i,-i)$, $(1,-1,i,i)$
B_4	$(1,-i,-1,-i)$, $(1,i,-1,i)$, $(1,i,1,-i)$, $(1,-i,1,i)$

WDP-CH05-P10 **Finite proposition (complete unbiased bases in dimensions 3 and 4).** In dimension 3, with ω a primitive cube root of unity, the standard basis and the three bases $\{v_{r,k}\}_{k=0,1,2}$ with $v_{r,k} = (\omega^{2rj^2+jk})_{j=0,1,2}$, $r = 0, 1, 2$, are pairwise unbiased. In dimension 4, the standard basis and $B_1, \dots, B_4$ are pairwise unbiased. For the $d(d+1)$ rank-one projectors $\Pi = uu^*/\langle u, u\rangle$ of either family, every complex $d \times d$ matrix ρ satisfies

$$\sum_{\Pi} \operatorname{tr}(\rho\Pi)\,\Pi = \rho + \operatorname{tr}(\rho)\, I_d.$$

With any one of the $d + 1$ bases removed, the identity fails.

Proof. Unbiasedness is direct computation over $\mathbb{Z}[\omega]$ and $\mathbb{Z}[i]$: in dimension 3, $\langle v_{r,k}, v_{r,k'}\rangle = 3\delta_{kk'}$, $|\langle e_j, v_{r,k}\rangle|^2 = 1$, and $|\langle v_{r,k}, v_{r',k'}\rangle|^2 = 3$ for $r \neq r'$; in dimension 4, each B_m is orthogonal with

$\langle u,u\rangle = 4$, $|\langle e_j,u\rangle|^2 = 1$, and $|\langle u,v\rangle|^2 = 4$ between different B_m. For the identity, let V_m be the real span of the traceless Hermitian matrices $\Pi - I/d$ for Π in basis m; it has dimension $d-1$. Across bases $\operatorname{tr}(\Pi\Pi') = 1/d$, so $\operatorname{tr}\left((\Pi - \frac{I}{d})(\Pi' - \frac{I}{d})\right) = 0$: the $d+1$ spaces are orthogonal for the positive-definite trace form, and since $(d+1)(d-1) = d^2-1$ they fill the traceless Hermitian matrices. Let $T(\rho) = \sum_\Pi \operatorname{tr}(\rho\Pi)\Pi$. An element $H = \sum_j h_j \Pi_j^{(n)}$ of V_n has $\operatorname{tr}(H\Pi_j^{(n)}) = h_j$ and $\operatorname{tr}(H\Pi) = \operatorname{tr}(H)/d = 0$ for Π in another basis, so $T(H) = H$; and $T(I) = (d+1)I$. Hence $T(\rho) = \rho + \operatorname{tr}(\rho)I$ on Hermitian matrices, and on all matrices by linearity. Without basis n, a nonzero $H \in V_n$ has $T(H) = 0 \neq H$. □

When ρ is a density operator the numbers $\operatorname{tr}(\rho\Pi)$ are the $d(d+1)$ outcome probabilities of the $d+1$ bases, and in the language of CH14-P02 the map $\rho \mapsto (\operatorname{tr}(\rho\Pi))_\Pi$ is a linear presentation that is injective on all of $M_d(\mathbb{C})$, with the explicit inverse above. The projectors have entries in $\mathbb{Q}(\omega)$ or $\mathbb{Q}(i)$, so for ρ with entries there the numbers and the reconstruction stay there. The absorbed archived working-paper source 004 normalized the dimension-3 vectors by $1/\sqrt{3}$, which does not lie in $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, and did not list its dimension-4 bases. It cited an external Lean check of orthogonality and unbiasedness for the unnormalized vectors used here; that check was read, not built or ported, and gives no Lean credit, and the reconstruction identity had only an executable test there. Informational completeness is a property of the declared bases; it derives no measurement postulate (CH05-N01, CH19-O04).

Let N be odd, let ω, X, and Z be as in CH05-P09, and let $Pe_j = e_{-j}$ with indices modulo N. Then $P^2 = I$ and $\operatorname{tr} P = 1$, because $j \equiv -j$ only for $j = 0$ when N is odd; for even N the trace is 2.

`WDP-CH05-P11` **Finite proposition (the phase-point frame).** For odd N and $(q,k) \in \mathbb{Z}_N^2$ put $A(q,k) = X^q Z^k P Z^{-k} X^{-q}$. Then $A(q,k)e_j = \omega^{2k(q-j)} e_{2q-j}$; each $A(q,k)$ is Hermitian with $\operatorname{tr} A(q,k) = 1$; $\sum_{q,k} A(q,k) = NI_N$; and $\operatorname{tr}\left(A(q,k)A(q',k')\right) = N\delta_{qq'}\delta_{kk'}$. Consequently, for every complex $N \times N$ matrix ρ the function $W_\rho(q,k) = \operatorname{tr}(\rho A(q,k))/N$ satisfies

$$\rho = \sum_{q,k} W_\rho(q,k)\, A(q,k), \qquad \sum_{q,k} W_\rho(q,k) = \operatorname{tr}\rho,$$

W_ρ is real when ρ is Hermitian, and W_ρ takes values in $\mathbb{Q}(\omega)$ when the entries of ρ do.

Proof. $X^{-q}e_j = e_{j-q}$; Z^{-k} multiplies it by $\omega^{-k(j-q)}$; P sends it to e_{q-j}; Z^k multiplies by $\omega^{k(q-j)}$; and X^q sends it to e_{2q-j}. This gives the formula. A diagonal entry at j needs $2(j-q) \equiv 0$, that is $j = q$ since N is odd, and equals 1. The entry at $(j, 2q-j)$ is the conjugate of the entry at $(2q-j, j)$, so A is Hermitian. For fixed q and j, $\sum_k \omega^{2k(q-j)}$ is N when $j = q$ and 0 otherwise, because then $\omega^{2(q-j)} \neq 1$ is an N-th root of unity; hence $\sum_{q,k} A(q,k) = NI$. The product $A(q,k)A(q',k')$ sends e_j to a multiple of $e_{2q-2q'+j}$, so its trace vanishes unless $q = q'$, and then it is $\sum_j \omega^{2(k-k')(j-q)} = N\delta_{kk'}$. The N^2 Hermitian operators are therefore orthogonal for the pairing $\operatorname{tr}(A^*B)$ and form a basis of $M_N(\mathbb{C})$; the coefficient of $A(q,k)$ in ρ is $\operatorname{tr}(\rho A(q,k))/N$, and summing the coefficients gives $\operatorname{tr}\rho$. For Hermitian ρ and A, $\operatorname{tr}(\rho A)$ is real. □

This frame is the standard odd-dimensional discrete phase space; it is proved here directly, and no external theorem about it is used. The phase factor that the absorbed archived working-paper source 005 placed in its displacement operators cancels in $A(q,k)$. That source established the frame for $N \in \{3,5\}$ only by executed tests in an external repository; the Lean it cites establishes $P^2 = I$ and $\operatorname{tr} P = 1$ for $N \in \{3,5\}$. It was read, not ported, and gives no Lean credit.

WDP-CH05-P12 **Finite proposition (a signed phase-point valuation).** Let $N \geq 3$ be odd and J the all-ones $N \times N$ matrix. The density operators $\rho_0 = e_0 e_0^{\mathsf{T}}$ and $\rho_+ = J/N$ have $W_{\rho_0}(q,k) = \delta_{q,0}/N$ and $W_{\rho_+}(q,k) = \delta_{k,0}/N$, both nonnegative. For $m = e_1 - e_{N-1}$, the density operator $\rho_M = \frac{1}{2} m m^{\mathsf{T}}$, whose entries are 0 and $\pm\frac{1}{2}$, has $W_{\rho_M}(0,0) = -1/N$.

Proof. $W_{\rho_0}(q,k) = A(q,k)_{00}/N$, and the $(0,0)$ entry of $A(q,k)$ is nonzero only when $2q \equiv 0$, that is $q = 0$, where it is 1. Next, $W_{\rho_+}(q,k) = N^{-2} \sum_{i,j} A(q,k)_{ij} = N^{-2} \sum_j \omega^{2k(q-j)} = \delta_{k,0}/N$. At the origin $A(0,0) = P$, and $Pm = e_{N-1} - e_1 = -m$, so $\operatorname{tr}(\rho_M P) = \frac{1}{2} m^{\mathsf{T}} P m = -\frac{1}{2} m^{\mathsf{T}} m = -1$. □

For $N = 3$ the other eight values of W_{ρ_M} equal $\frac{1}{6}$, and $-\frac{1}{3} + 8 \cdot \frac{1}{6} = 1$. For $N = 5$ the values lie in $\mathbb{Q}(\sqrt{5})$: on the line $q = 0$ they are $-\frac{1}{5}$, $\frac{1+\sqrt{5}}{20}$, $\frac{1-\sqrt{5}}{20}$, $\frac{1-\sqrt{5}}{20}$, $\frac{1+\sqrt{5}}{20}$ for $k = 0, \ldots, 4$, on the lines $q = 1$ and $q = 4$ they are $\frac{1}{10}$, and on $q = 2, 3$ they are 0; three are negative and they sum to 1. So the frame represents ρ_M by real weights on the finite set $\mathbb{Z}_N^2$ with total 1 and a negative entry: the kind of signed valuation that the remark after the representation theorem of §18.2 separates from a probability state. The Lean that the absorbed source cites establishes $Pm = -m$, $\langle m, m \rangle = 2$, and $\langle m, Pm \rangle = -2$ for $N \in \{3, 5\}$ and defines the value $-1/N$ as a constant, as that source recorded; the connecting step is the computation above. It was read, not ported, and gives no Lean credit.

The two frames of this section place the negativity differently. The numbers $\operatorname{tr}(\rho\Pi)$ of CH05-P10 are nonnegative for every density operator, but that reconstruction subtracts $\operatorname{tr}(\rho) I$; the phase-point reconstruction is a plain sum, but its weights can be negative. This compares two declared frames and is not a theorem about all frames. A negative phase-point value is not claimed to certify nonclassicality, contextuality, or a computational resource; CH05-N01 and CH19-N02 apply.

Technical note for Chapters 1–5: morphisms and verification records

The foundational constructions become composable only after morphisms are fixed. For admitted cuts, a morphism may consist of maps on tokens and incidences preserving role types and the selected comparison family. For graph cochains, a graph map can induce pullback or pushforward only under the usual variance and coefficient hypotheses. For observer-facing use, the morphism must also preserve the proposition witnessed; a map that forgets the relevant edge can destroy a residual witness.

A minimal verification record for any construction in Part I should contain:

1. the finite carriers and their explicit enumeration or source hash;
2. the role and comparison declarations;
3. the construction performed;
4. the proposition claimed of its output;
5. the verifier or proof and its assumptions;
6. the morphisms under which the proposition is invariant;
7. the known countermodel class;
8. an explicit statement of what the verification record does not establish.

This verification record schema prevents a common migration error. Suppose a cycle certificate is valid on graph G, and a later document replaces G by a quotient graph G'. The old certificate does not automatically migrate. One must show that the relevant circulation functional descends, that the target cochain is transported, and that the repair image maps into the new repair image. A matching filename or informal diagram is not a proof of that transport.

A useful, non-exhaustive checklist names what a presentation change can alter: the repair space, the

closure map, the carrier and its maps, the coefficient field, the declared norms, and the differential itself (a perturbation, as in §16.5). Each needs its own transport statement; a change on one axis does not license a claim on another.

The same point sharpens the symmetry theorem. For an H-set Y, an H-equivariant map from the trivial one-point H-set to Y exists exactly when Y has an H-fixed point. Thus every equivariant selection problem can be reduced to a fixed-point question in the target action. The full symmetric group on an unstructured carrier is one instance. If the input includes a partition, the automorphism group shrinks and fixed targets may appear. The new output is then relative to the partition.

WDP-CH03-P02 **Finite proposition (fixed-target criterion).** Let a finite group H act on a finite set Y. A parameter-free H-equivariant choice of a point of Y exists if and only if Y^H, the set of global fixed points, is nonempty.

This proposition gives an exact audit recipe: identify the automorphism group of the admitted input, identify its action on the proposed outputs, and compute the fixed locus. Claims that a target is "forced" should specify this action and show that the fixed locus has the required uniqueness property.

Typed witnesses also need morphisms. An adapter from a residual-locus witness to a retention witness consists of a write map, a persistence law, and a readback proof. An adapter from a repair witness to a dual statement may not exist: a repair certifies membership in an image, whereas a separating functional establishes nonmembership. They face opposite directions. An adapter from a data record to a proposition requires a semantics and a validity predicate. These obligations are small enough to state explicitly and large enough to block invalid prose inferences.

Finally, the quotient defect admits a stability estimate. By the triangle inequality,

$$|\Phi_J([\delta]) - \Phi_J([\delta'])| \le J(\delta - \delta')$$

whenever J is a norm and both defects live in the same quotient problem. Indeed, any repair candidate for one defect supplies an upper bound for the other after adding the norm of their difference; exchange the defects for the reverse inequality. This one-Lipschitz property is useful when Chapter 13 adds measurement uncertainty. It does not compare different graphs, repair spaces, or gauges without transport data.

Part II — Order, quantity, and finite probability

Chapter 6 — Recurrence, succession, and constitutive order

6.1 Recurrence is not yet succession

If a retained mark can be accessed in two distinguishable exhibitions, one has a recurrence pattern. Recurrence alone does not say which exhibition is earlier. A pair of records can be compared symmetrically; order appears only when an asymmetric relation is declared or constructed from additional data.

Let S be a finite set of retained states or events. A **succession candidate** is a binary relation $\prec \subseteq S \times S$. To function as a strict partial order it must be irreflexive and transitive. Local transition

data may instead give a directed relation $\rightsquigarrow$ that is not transitive; its transitive closure can define $\prec$ when cycles are absent.

WDP-CH06-D01 **Definition (constitutive succession).** A constitutive succession packet is

$$\mathfrak{S} = (S, \rightsquigarrow, \prec, \pi),$$

where $\rightsquigarrow$ is an exhibited immediate-transition relation, $\prec$ is a declared or proved strict partial order containing $\rightsquigarrow$, and π records the assumptions under which that containment and order are valid.

This definition forces an often-hidden choice into view: whether order is primitive, generated by acyclic reachability, or inferred by an observer from records. Those are different models.

6.2 Immediate succession and cycles

If $G = (S, E)$ is a finite directed acyclic graph, define $x \prec y$ when there is a nonempty directed path from x to y. Then $\prec$ is irreflexive and transitive. Conversely, if G contains a directed cycle, reachability is not irreflexive.

WDP-CH06-P01 **Finite proposition.** Nonempty reachability on a finite directed graph is a strict partial order exactly when the graph is acyclic.

Proof sketch. Transitivity follows by concatenating paths. A directed cycle gives $x \prec x$. If $x \prec x$, the witnessing nonempty path contains a directed cycle. □

The proposition is graph theory, not a derivation of physical time. It says what an admitted transition graph must satisfy to support a strict order.

6.3 No external clock in the definition

A transition record can be indexed for storage without treating the index as physical duration. If a computation enumerates states $s_0, s_1, ...$, the enumeration order is part of the implementation. A foundational claim must not silently transfer that order to the represented system.

The programme therefore distinguishes:

- **execution order**, the order in which an algorithm processes data;
- **record order**, an order stored in a trace;
- **constitutive order**, the relation asserted of the modeled events;
- **coordinate time**, a numeric label assigned after a clock convention;
- **duration**, an additive or otherwise composed magnitude on intervals.

Any bridge among them is a theorem or a declaration, not a synonym.

6.4 Worked example: fork without total order

Let $S = \{a, b, c\}$ with edges $a \rightsquigarrow b$ and $a \rightsquigarrow c$. Reachability gives $a \prec b$ and $a \prec c$, while b and c are incomparable. The packet has succession but no total order. A file format may list b before c; that list does not change the constitutive relation.

Adding an edge $b \rightsquigarrow c$ makes one branch later than the other. Adding both $b \rightsquigarrow c$ and $c \rightsquigarrow b$ destroys strictness by creating a cycle.

6.5 The finite kernel and its boundary

The finite order facts above are elementary and amenable to proof assistants. The stronger programme targets are not automatic:

- constructing the event carrier without presupposing ordered inputs;
- proving that a retained-difference mechanism selects one transition relation;
- connecting the order to physical causal structure;
- recovering continuum topology or metric;
- explaining an experienced passage of time.

WDP-CH06-O01 **Finite proposition (relational retention and succession).** A declared finite nonempty labelled transition system with exact reports or a validated report equivalence admits greatest strong bisimilarity and its minimal bounded quotient. Its raw directed relation has a strict transitive edge extension exactly when acyclic. Cycles obstruct such an extension; SCC condensation changes the carrier. A nonempty strict order cannot be chosen invariantly from a fully symmetric carrier. Directional transition or symmetry-breaking order data must therefore be declared. See Appendix J.

Equivariance Analysis and Mechanized Separation.

1. **Order Data Necessity via** $\mathrm{Sym}(M)$**-Equivariance:** Let M be a bare finite memory carrier with $|M| \geq 2$, constant report, and no distinguished relational or ordering data. Its symmetry group is the full symmetric group $G = \mathrm{Sym}(M)$. Any strict partial order $\prec \subset M \times M$ must be irreflexive and asymmetric ($a \prec b \implies \neg(b \prec a)$). Any parameter-free construction from order-free retention must be G-equivariant. On the set of distinct pairs $M \times M \setminus \Delta$, the action of G is transitive, because G acts 2-transitively on M. The only G-invariant subsets of distinct pairs are $\emptyset$ and $M \times M \setminus \Delta$. The empty relation $\emptyset$ is formally a trivial strict partial order, but contains no pairs and therefore provides no succession, while the full set of pairs is symmetric, violating asymmetry. Hence no nonempty (or nontrivial) G-equivariant strict partial order exists on an order-free carrier M: to obtain any actual succession, additional order data (or symmetry-breaking selection) is mathematically necessary. While this symmetry argument establishes the necessity of order data in the parameter-free G-equivariant setting, Appendix J also classifies declared finite relational retention and strict succession extensions. It does not select an undeclared transition relation.
2. **Constitutive succession and immediate transitions:** On the three-cut carrier `Chain3` with immediate edges $\alpha \rightsquigarrow \beta \rightsquigarrow \gamma$, nonempty reachability relates α to γ, although there is no immediate edge $\alpha \rightsquigarrow \gamma$. Thus reachability and immediate transition differ on the same ordered pair (α, γ). Appendix D records the separate formalization evidence for `lean4/WitnessedDifference/Succession.lean`.

WDP-CH06-N01 **Nonclaim.** A finite acyclic relation is not a continuum spacetime, a causal-set model of our universe [REF-TIME-001; REF-TIME-002], or an account of temporal experience.

Chapter 7 — Duration, exactness, periods, and clocks

This chapter merges the two duration drafts. Succession and duration are kept separate: order answers *which intervals compose*; duration assigns a magnitude to those intervals.

7.1 Edge valuation

Let $G = (S, E)$ be a finite directed acyclic graph or, more generally, a directed graph on which a chosen path family is defined. A duration gauge is a map

$$\lambda : E \longrightarrow M$$

into a commutative additive monoid $(M, +, 0)$. For a path $p = e_1 \cdots e_k$, define

$$D_\lambda(p) = \sum_{j=1}^{k} \lambda(e_j).$$

WDP-CH07-D01 **Definition (finite duration gauge).** The pair of a path system and an additive edge valuation is a finite duration gauge. Positivity, Archimedean structure, units, or real values are additional hypotheses.

WDP-CH07-P01 **Finite proposition (concatenation additivity).** Whenever path concatenation pq is defined,

$$D_\lambda(pq) = D_\lambda(p) + D_\lambda(q).$$

The proof is reassociation of a finite sum. This is the strongest unconditional duration theorem available at this level.

7.2 Exact, closed nonexact, active failure, undetermined

When edge durations lie in an abelian group, ask whether $\lambda = d_0 t$ for a vertex potential t. Four statuses must not be collapsed.

1. **Exact.** A displayed t satisfies $d_0 t = \lambda$.
2. **Closed nonexact.** A declared next differential gives $d_1 \lambda = 0$, while a period or dual certificate establishes $\lambda \notin \operatorname{im} d_0$.
3. **Active closure failure.** $d_1 \lambda \neq 0$ on an exhibited 2-cell.
4. **Exactness undetermined.** The available structure or search did not decide the preceding cases.

"Closed nonexact" is unavailable when no valid d_1 exists. On a graph without declared 2-cells, loop periods can still refute exactness, but the word *closed* should be used only relative to a named complex.

7.3 Path independence

WDP-CH07-P02 **Finite proposition.** If $\lambda = d_0 t$, then all directed paths from a to b have the same duration $t(b) - t(a)$.

Proof. The edge differences telescope along each path. □

Conversely, suppose a finite directed graph has a base vertex o from which every vertex is reachable, and sums agree along all directed paths from o to each vertex. Define $t(v)$ by this common sum. Appending any edge $u \to v$ to an o-to-u path gives $t(v) = t(u) + \lambda(u, v)$, so $d_0 t = \lambda$. The converse

needs this reachability and closure under edge extension; connectedness of the underlying undirected graph or an arbitrary selected path family is insufficient.

7.4 Worked examples

Chain. On $a \to b \to c$ let $\lambda = (2, 5)$. With $t(a) = 0, t(b) = 2, t(c) = 7$, the valuation is exact and $D(a \to b \to c) = 7$.

Diamond. Let $a \to b \to d$ and $a \to c \to d$. Give the upper path weights $2, 5$ and lower path weights $3, 4$. Both total 7; taking $t = (0, 2, 3, 7)$ shows exactness. Changing the last lower edge from 4 to 6 makes the lower path total 9. The discrepancy 2 is a period witness around the undirected cycle and refutes exactness. With unit weights it also bounds the quotient defect from below (CH04-P11), and here $\Phi_1 = 2$ exactly: the potential $t = (0, 2, 3, 7)$ leaves the single residual 2 on the last lower edge.

Unit tally. Assign every edge value 1. Path duration becomes edge count. This is a valid declared gauge, but two refinements of the same modeled process can change the count. Unit tally therefore requires a granularity convention. It is additive under concatenation, yet it is not exact on a graph with a directed cycle: a directed cycle of length $n \geq 1$ has unit-tally circulation $n \neq 0$, so no vertex potential produces it.

Commuting square. Let two routes of two steps each join the same endpoints, one carrying values $(1, 1)$ and the other $(1, 0)$. The valuation is additive along each route, but the circulation around the square is 1, so it is not exact. Counting steps cannot tell the routes apart, since both have two items.

7.5 Clock packets

A clock is not just a duration function. It is a comparator between a target process and a repeatable reference process.

`WDP-CH07-D02` **Definition (finite clock packet).** A clock packet is

$$\mathfrak{T} = (C, \rightsquigarrow_C, \lambda_C, r_C, \kappa, \varepsilon),$$

where C is a reference-state carrier, $\rightsquigarrow_C$ its transition relation, λ_C its duration gauge, r_C a readout, κ a declared coupling or comparison with target intervals, and ε an error or tolerance model.

A clock is **adequate for a target family** only relative to stated criteria: coverage, stability, resolution, monotonicity, and error bounds. No clock is adequate just because it ticks.

7.6 Calibration joins duration to measurement

Calibration supplies a relation between an instrument readout and a reference quantity. In this monograph, duration becomes a measurement quantity only when a clock packet is attached to the measurement architecture of Chapter 13. The edge valuation alone is a mathematical magnitude; the calibrated result requires a measurand, procedure, unit, and uncertainty statement.

`WDP-CH07-O01` **Finite proposition (resolved finite target).** For a positive calibrated finite occurrence chain whose edge residuals are bounded by nonnegative declared budgets, every interval

residual is bounded by the sum of those budgets, and endpoint-preserving block-sum refinement preserves the bound. The proof, native ownership and scope are given in Appendix I.

WDP-CH07-N01 **Nonclaim.** Additive finite duration does not select a global unrestricted unit, a continuum parameter, Lorentzian geometry, or a physical arrow of time.

Chapter 8 — Multiplicity, additive weight, and finite probability

8.1 Matching before number

Let A and B be finite exhibited families. A **matching** is a finite set of pairing incidences satisfying:

1. no endpoint participates in two pairings;
2. every endpoint in both families participates in a pairing;
3. the pairing preserves the comparisons declared relevant.

These conditions describe a bijective correspondence without taking a number object as primitive. They still presuppose the exhibited families and a frozen comparison scope.

WDP-CH08-D01 **Definition (comparison-relative congruence).** A and B are congruent relative to $\mathcal{K}$ when a matching preserving $\mathcal{K}$ is exhibited.

This construction does not show that plurality itself is derived from one cut. It describes how finite multiplicity can be compared after plural exhibited tokens are available.

8.2 Witness mass

Let $W = \{w_1, \dots, w_n\}$ be a finite family of typed witnesses and let $m : W \to \mathbb{R}_{\geq 0}$ be a declared mass. For $A \subseteq W$, define

$$\mu(A) = \sum_{w \in A} m(w).$$

For disjoint A, B,

$$\mu(A \sqcup B) = \mu(A) + \mu(B).$$

This is finite additivity by construction. If $m(w) = 1$, $\mu(A)$ is a tally. If witnesses are indistinguishable under a declared transitive symmetry and m is required to be equivariant, equal weights follow on that orbit. The equivariance requirement is the load-bearing assumption; indifference is not obtained from ignorance alone.

8.3 Normalized valuation

Let $\mathcal{P} = \{A_1, \dots, A_k\}$ be a declared finite partition of W and assume $\mu(W) > 0$. Define

$$p(A_i) = \frac{\mu(A_i)}{\mu(W)}.$$

WDP-CH08-P01 **Finite proposition.** The values $p(A_i)$ are nonnegative and sum to 1.

The proof is finite arithmetic. The result is a normalized additive valuation on a declared finite partition. Calling it probability is mathematically standard in a finite sample model, but its physical interpretation remains open.

8.4 Refinement

Suppose a cell A_i is refined into disjoint cells $B_{i1}, \dots, B_{ir}$. Finite additivity gives

$$p(A_i) = \sum_j p(B_{ij}).$$

If the mass assigned to a witness changes under refinement, this equality can fail because a new model has been introduced. A refinement theorem therefore requires a transport rule preserving total mass.

`WDP-CH08-001` **Finite proposition (resolved finite target).** Every nonnegative additive valuation on a finite Boolean event algebra has unique algebra-atom masses and measurable-map pushforward; nonnegative masses on individual witnesses extending it are unique exactly when every nonsingleton algebra atom has mass zero. The proof, native ownership and scope are given in Appendix I.

This conditional result is load-bearing for any assertion that L^1 aggregation is forced rather than just chosen.

8.5 Why L^1 appears—and what it does not exclude

For nonnegative vectors with disjoint support,

$$\|x + y\|_1 = \|x\|_1 + \|y\|_1.$$

For $p > 1$, $\|x + y\|_p$ is generally not the sum of the norms. Thus L^1 is the natural norm when the magnitude itself must be degree-one homogeneous and exactly additive over disjoint positive sectors.

This does not ban L^2. Squared L^2 magnitude is additive on orthogonal components:

$$\|x \oplus y\|_2^2 = \|x\|_2^2 + \|y\|_2^2.$$

The two expressions answer different questions. Moving from witness mass to amplitude geometry changes the object and its rules.

The primitive-axiomatics source isolates the hypotheses under which an aggregator of edge magnitudes must be L^1. Let $\mathcal{A}$ send finite families $(a_e)_{e \in E}$ of nonnegative edge magnitudes to $[0, \infty)$.

With additivity over disjoint edge sets assumed outright, the conclusion is finite. If $\mathcal{A}$ is edge-decomposable, additive over disjoint edge sets, positively homogeneous on each edge, invariant under relabeling edges, and faithful ($\mathcal{A}_{\{e\}}(1) > 0$), then additivity over the singleton decomposition gives $\mathcal{A}((a_e)) = \sum_e \mathcal{A}_{\{e\}}(a_e)$, homogeneity gives $\mathcal{A}_{\{e\}}(a_e) = a_e \mathcal{A}_{\{e\}}(1)$, relabeling invariance makes $\mathcal{A}_{\{e\}}(1) = c$ independent of e, and faithfulness gives $c > 0$. So $\mathcal{A}((a_e)) = c \sum_e a_e$. The conditional part is the replacement of additivity by the weaker replica extensivity.

WDP-CH08-C01 **Conditional result (L^1 aggregation from replica extensivity).** For every finite list a of nonnegative reals, let $\mathcal{A}(a) \geq 0$. Assume $\mathcal{A}(\emptyset) = 0$, $\mathcal{A}((t)) = ct$ for a fixed $c > 0$, and a continuous associative operation $F : [0,\infty)^2 \to [0,\infty)$ with identity 0 such that

$$\mathcal{A}(a \sqcup b) = F(\mathcal{A}(a), \mathcal{A}(b)).$$

Assume replica extensivity: $\mathcal{A}(a^{\sqcup N}) = N\mathcal{A}(a)$ for positive integers N. Then $F(x,y) = x + y$ and $\mathcal{A}((a_e)) = c\sum_e a_e$.

Proof. For $t \geq 0$, a singleton of value t/c has aggregate t. Replicas and the composition law give $F(mt, nt) = (m+n)t$ for nonnegative integers m, n; the cases with a zero argument use the identity. For any nonnegative rational x, y, choose a common denominator N and put $t = 1/N$ to obtain $F(x,y) = x + y$. Continuity and density of rational pairs extend this equality to all nonnegative real x, y. The composition law then gives the claimed sum by induction on list length. □

The singleton condition follows from positive homogeneity, relabeling invariance and faithfulness on singleton inputs. The explicit composition law is the meaning of edge decomposability used here. This finite argument does not require Aczél's representation theorem [REF-MATH-006].

The hypotheses are not idle. Bottleneck (max), density (mean), saturating (capped-sum), and L^p aggregators with $p > 1$ fail replica extensivity, as do typical aggregators with cross-edge coupling. The sum of squares $\sum_e a_e^2$ fails only homogeneity; a weighted sum $\sum_e c_e a_e$ with unequal positive c_e fails only relabeling invariance; and the zero aggregator fails only faithfulness. Edge-decomposability is the format in which the others are stated. The result selects L^1 for evaluators that require independent defects to add. It does not make L^1 globally unrestricted, and the classification of gauges by edge symmetry monoids belongs to a separate source that is not absorbed here.

Identifying this nonnegative-magnitude aggregator with a norm on signed cochains additionally assumes dependence only on coordinate magnitudes (independent sign invariance), or additivity on every disjoint-support pair. Additivity restricted to the positive cone alone does not classify signed norms.

8.6 Worked example: a declared partition

Let $W = \{w_1, w_2, w_3, w_4\}$ with masses $(1, 1, 2, 4)$. Partition it into $A = \{w_1, w_2\}$ and $B = \{w_3, w_4\}$. Then

$$\mu(W) = 8, \qquad p(A) = \frac{1}{4}, \qquad p(B) = \frac{3}{4}.$$

Refining B gives probabilities $1/4$ and $1/2$, which sum to $3/4$. Nothing in this calculation says the witnesses were random trials or that the numbers predict frequencies. Those readings require a sampling or physical model.

Chapter 9 — Probability scope fences

9.1 Five distinct notions

The following must not be collapsed:

1. a finite normalized valuation on an exhibited partition;
2. a limiting frequency in a repeatable trial model;
3. a propensity or physical chance;
4. a Bayesian degree of belief;
5. a countably additive measure on a sigma-algebra.

Chapter 8 constructs only the first. A finite valuation can be embedded into some of the others after additional data are supplied, but there is no unique embedding in general.

9.2 Normalization does not select the Born rule

Let a nonzero finite amplitude vector be $z = (z_1, \dots, z_n)$. For any $q > 0$, define

$$p_i^{(q)} = \frac{|z_i|^q}{\sum_j |z_j|^q}.$$

Then $p_i^{(q)} \geq 0$ and $\sum_i p_i^{(q)} = 1$. Normalization alone therefore does not select $q = 2$.

`WDP-CH09-P01` **Finite proposition (normalization nonselection).** On any amplitude vector with at least two unequal nonzero magnitudes, the normalized vectors for exponents $q, r > 0$ with $q \neq r$ are distinct, while each is a probability vector.

Proof. For two nonzero coordinates of unequal magnitude, their normalized coordinate ratio is $(|z_i|/|z_j|)^q$. Because the positive base differs from 1, equality of the ratios forces $q = r$. □

For example, for magnitudes $(1, 2)$, $q = 1$ gives $(1/3, 2/3)$ and $q = 2$ gives $(1/5, 4/5)$. Any derivation of the Born rule must use structure beyond positivity and normalization—for example, noncontextuality and Hilbert-space assumptions of a Gleason-type theorem [REF-PROB-002], or operational axioms in a reconstruction [REF-QM-003; REF-QM-004; REF-QM-006].

The reverse direction loses information too. **Finite proposition (mass-to-amplitude embedding is many-to-one).** Given nonnegative masses μ_j, not all zero, and separately declared phases φ_j, put $p_j = \mu_j / \sum_k \mu_k$ and $\psi_j = \sqrt{p_j}\, e^{i\varphi_j}$. Then $\sum_j |\psi_j|^2 = 1$. The map $(\mu, \varphi) \mapsto \psi$ forgets the overall scale of μ, and the phases are inputs rather than outputs. If the masses come from a signed cochain through absolute values, the signs are lost as well. *Proof.* Unit norm is the normalization; μ and $c\mu$ for $c > 0$ give the same ψ, and changing φ changes ψ without changing μ. □ Reading $|\psi_j|^2$ back as a probability returns the declared p_j by construction. That is a consistency check of the embedding, not a derivation of the Born rule.

9.3 Finite additivity is not countable additivity

On a finite algebra, finite and countable additivity coincide trivially because only finitely many disjoint nonempty cells exist. This does not provide a sigma-algebra over an infinite outcome space, regularity, convergence, or a continuum measure. Those are separate imports.

9.4 Probability and observers

A partition may be selected by an observer profile: which distinctions the observer can report, which outcomes are grouped, and which events are outside scope. The resulting valuation is then profile-relative. This does not make it subjective in a psychological sense; it makes the dependence on a partition map explicit.

Later chapters will require an observer to publish the partition and value map used in a measurement. Without that declaration, two normalized tables may disagree because they answer different questions.

WDP-CH09-N01 **Nonclaim.** The finite witness valuation is not a derivation of chance, frequency convergence, exchangeability, a continuum probability space, or the Born rule.

WDP-CH09-N02 **Nonclaim.** Equal weights under a declared symmetry do not establish an unrestricted principle of indifference.

9.5 Order-preserving maps and temporal coarse-graining

Two succession packets can describe the same finite events at different resolutions. Let $(S, \prec_S)$ and $(T, \prec_T)$ be strict partial orders. A map $f : S \to T$ is monotone when $x \prec_S y$ implies either $f(x) = f(y)$ or $f(x) \prec_T f(y)$, depending on whether the target convention permits collapsed intervals. If equality is forbidden, use strict monotonicity.

Collapsing a chain segment can preserve causal order while changing unit-tally duration. For example, $a \prec b \prec c$ may map to $f(a) = u, f(b) = v, f(c) = v$. The image remembers that u precedes v but erases the second step. A duration transport theorem must therefore say whether collapsed edges contribute zero, are summed into a coarse edge, or make the map inadmissible.

WDP-CH06-D02 **Definition (duration-respecting coarse-graining).** Given duration packets (S, λ_S) and (T, λ_T), a monotone map f is duration respecting relative to an edge/path assignment F when every source path p satisfies

$$D_{\lambda_T}(F(p)) = D_{\lambda_S}(p)$$

or lies within a separately stated error bound. Such preservation is extra data; order preservation alone does not imply it.

9.6 Duration uncertainty

Suppose edge durations are not exact values but intervals $I_e = [\ell_e, u_e]$. For a finite path p, interval addition yields

$$I_p = \left[\sum_{e \in p} \ell_e, \sum_{e \in p} u_e \right].$$

This is a conservative bound when edge values can vary independently inside their stated hard intervals. Correlation constraints can only preserve or narrow that Minkowski bound because they select a subset of the Cartesian product. A wider set would require changing the interval semantics—for example, treating the inputs as marginal confidence sets rather than hard bounds—and must be represented as a different model. If two paths from a to b have disjoint duration intervals, exact path independence is refuted robustly. If their intervals overlap, exactness is not established; the data are just inconclusive at that resolution.

A clock comparison adds its own uncertainty. If a target interval maps to k reference ticks, and each tick has interval I, naive interval addition gives kI. Drift, shared calibration error, and

quantization may be correlated and cannot always be added as independent contributions. Chapter 13's uncertainty packet is the correct place to state those rules.

WDP-CH07-P03 **Finite proposition (interval accumulation).** Under the model that each edge value can be selected independently from its declared interval, the set of possible finite path sums is contained in the Minkowski sum of the edge intervals, which for real intervals is the displayed endpoint sum.

9.7 Orbit weights

Let a finite group H act on a witness family W. If a mass function $m : W \to \mathbb{R}_{\geq 0}$ is H-invariant, then it is constant on each orbit. The converse also holds: assigning one nonnegative mass to each orbit defines an invariant mass.

WDP-CH08-P02 **Finite proposition (invariant mass classification).** H-invariant nonnegative mass functions on W are in bijection with nonnegative functions on the orbit set W/H.

Proof. Invariance forces equal values on orbit-related witnesses. Conversely, pulling a function back along the orbit projection is invariant. □

This theorem states exactly what symmetry buys. It does not select the mass of one orbit relative to another. If the action is transitive and total mass is fixed, equal atomic mass follows uniquely. If there are several orbits, their relative weights remain free. A statement of "equal probability by symmetry" must therefore name the group, action, orbit structure, and total-mass normalization.

9.8 Coarse partitions and sufficient reports

An observer often sees only a partition of W. Let $q : W \to P$ assign each witness to a report class. The pushforward mass is

$$(q_*\mu)(p) = \sum_{w : q(w) = p} m(w).$$

Normalization commutes with this finite pushforward when total mass is preserved. If two fine-grained models have the same pushforward, no evaluator using only P can distinguish them. This is the probability analogue of the presentation blind spot in Chapter 14.

A report is sufficient for a particular query when the query factors through q. Sufficiency is query-relative. A partition sufficient for predicting one outcome can erase information needed for another.

9.9 Frequencies, exchangeability, and updating

Suppose a protocol repeats a finite measurement N times and records outcome counts n_i. The empirical frequency n_i/N is a finite normalized valuation on the observed trial multiset. It is a fact about those records. A law of large numbers relates such frequencies to a probability parameter only after a stochastic model—often independence or a weaker condition—is supplied.

Exchangeability is another substantive assumption: the joint assignment is invariant under permutation of trial labels. Finite exchangeability by itself does not provide an infinite de Finetti representation [REF-PROB-005], and an observer's choice to ignore order does not prove the physical process exchangeable.

Bayesian updating likewise begins with a prior and likelihood model. The finite obstruction calculus can record those objects and audit their transformations; it does not select a uniquely rational prior from symmetry alone. A maximum- entropy rule [REF-PROB-006] adds a constraint set, reference measure, and optimization principle.

9.10 Probability-to-measurement interface checklist

Before a finite valuation is used as a measurement probability, record:

1. the trial or instrument protocol;
2. the outcome partition and its version;
3. whether weights are counts, modeled chances, likelihoods, or decision weights;
4. the sampling or exchangeability assumptions;
5. missing-data and stopping rules;
6. calibration and detection-efficiency treatment;
7. the uncertainty statement;
8. the inference rule and its scope.

Without these fields, a normalized table remains valid arithmetic but has no unique empirical reading.

Part III — Observer, observation, and measurement

Chapter 10 — A layered observer architecture

10.1 Why one tuple is not enough

The source papers contain a quotient observer, a nine-component evaluator, a six-component source packet, a five-component operational packet, and a self-witnessing extension. These objects overlap, but they are not definitionally equal. Attempting to compress them into one long tuple hides which components are mathematical presentation, temporal memory, evaluation, policy, or institutional authority.

We instead use four layers.

`WDP-CH10-D01` **Definition (observer presentation).** An observer presentation is

$$\mathsf{Pres}_{\mathcal{O}} = (X_{\mathcal{O}}, P_{\mathcal{O}}, q_{\mathcal{O}}, \mathcal{C}_{\mathcal{O}}),$$

where $X_{\mathcal{O}}$ is the admitted input carrier, $P_{\mathcal{O}}$ is the presentation or report carrier, $q_{\mathcal{O}} : X_{\mathcal{O}} \to P_{\mathcal{O}}$ is a presentation map, and $\mathcal{C}_{\mathcal{O}}$ is a declared context/profile family.

`WDP-CH10-D02` **Definition (observer memory).** An observer memory is a retention packet

$$\mathsf{Mem}_{\mathcal{O}} = (M_{\mathcal{O}}, u_{\mathcal{O}}, r_{\mathcal{O}}, \approx_{\mathcal{O}})$$

as in Chapter 3, with a typed interface from presentations into memory and from memory into reports.

`WDP-CH10-D03` **Definition (observer evaluator).** An observer evaluator is

$$\mathsf{Eval}_{\mathcal{O}} = (D_{\mathcal{O}}, A_{\mathcal{O}}, d_{\mathcal{O}}, J_{\mathcal{O}}, \Gamma_{\mathcal{O}}, T_{\mathcal{O}}),$$

consisting of a defect carrier, admissible repair carrier and map, declared cost, any well-typed closure tests, and a terminalizer returning a status in a declared finite status set.

`WDP-CH10-D04` **Definition (observer policy).** An observer policy specifies what the system does with evaluator outputs: store, request more data, emit a report, refuse an ill-typed query, or take a domain-specific action. Policy is not authority.

The layered observer is

$$\mathcal{O} = (\mathsf{Pres}_{\mathcal{O}}, \mathsf{Mem}_{\mathcal{O}}, \mathsf{Eval}_{\mathcal{O}}, \mathsf{Policy}_{\mathcal{O}})$$

together with explicit adapters among layers.

Interpretive remark (what an observer is, and what a perspective is). In this monograph an observer is not a person, a sensor or a conscious state. It is the declared packet that makes a claim evaluable: what is compared, what counts as defect, which repairs are admissible, how the residual is measured, and which terminal statuses exist. Two such packets applied to the same objects can return different verdicts without contradiction, because they evaluate different declared questions. A *perspective*, in the same vocabulary, is available only once such a packet is declared. A viewpoint, camera angle or opinion becomes a perspective when it can be terminalized by a stated evaluator. A human or a test harness can operate the packet; neither is the packet. One reading of this picture must be corrected. It is sometimes said that the model proposes and the evaluator authorizes. Under CH10-D04 and CH12-P01 the evaluator returns a status, and a policy acts on it. Authorization is a separate record supplied by an authority packet (Chapter 15), so the correct slogan is that the model proposes, the evaluator evaluates, and an independently declared regime may authorize.

10.2 Quotient evaluators

The quotient construction of Chapter 4 gives a useful evaluator. Fix its repair image $B_{\mathcal{O}} = \operatorname{im} d_{\mathcal{O}}$ and let the presentation expose $[\delta] \in Q^1_{\mathcal{O}} = C^1/B_{\mathcal{O}}$. The evaluator computes

$$\Phi_{\mathcal{O}}([\delta]) = \inf_{\alpha \in A_{\mathcal{O}}} J_{\mathcal{O}}(\delta - d_{\mathcal{O}}\alpha).$$

If the presentation instead retains an existing quotient C^1/B, this formula descends to that quotient exactly when $B \subseteq B_{\mathcal{O}}$ in the faithful finite-dimensional norm setting. Sufficiency follows by translating the repair variable; necessity follows by comparing the zero representative with each $b \in B$, since zero distance to the closed repair image means membership. Equality of repair images retains exactly the same zero classes. A strict inclusion gives a seminorm on the old quotient, with kernel $B_{\mathcal{O}}/B$; those extra zero classes must be declared.

For example, take $C^1 = \mathbb{R}^2$, $B = \operatorname{span}(e_0)$ and $B_{\mathcal{O}} = \operatorname{span}(e_1)$ with the ℓ^1 norm. The old quotient identifies 0 and e_0, but the new formula gives values 0 and 1. This mismatched profile is refused before numerical evaluation; it does not define an evaluator on the old classes. The bound profile evaluator checks the exact repair-image inclusion before invoking the rational solver.

Under the finite-dimensional real or complex normed-space hypotheses of Chapter 4, the infimum is attained. Rational input data and a rational polyhedral norm likewise give a rational optimizer through a feasible bounded rational linear programme, as for the weighted L^1 problem of §4.3. Attainment over rational repairs can fail for other norms, including polyhedral norms with irrational coefficients (§4.2). A terminalizer may return `Accept` when the value is below a declared tolerance, `Reject` when a certified lower bound exceeds it, and `Undetermined` otherwise. The tolerance, repair family, and cost are observer-profile data.

This is evaluation, not perception in the phenomenological sense. It is also not proposal generation: the evaluator compares a supplied candidate against a declared profile.

`WDP-CH10-N01` **Nonclaim.** A tolerance status belongs to the evaluator, not to the class. A class whose quotient residual lies below a declared tolerance is not thereby exact, absent, infinite, or continuous, and changing the tolerance changes the status while leaving the class, its residual, and its certificate unchanged.

For the cyclic field $\delta = (2, 2, 2)$ on the hollow triangle, $\Phi_1 = 6$ with repair residual $(0, 0, 6)$ and circulation $y = (1, 1, 1)$. A terminalizer with tolerance 10 returns `Accept`. With tolerance 1 the certified lower bound 6 exceeds the tolerance and it returns `Reject`. An exact class is accepted at every positive tolerance, so reading "below tolerance" as "infinite" or "continuous" would give that name to every exact class. In the terms of §21.9, moving the tolerance is a profile change, not an evidential update.

10.3 Adapters for legacy packets

An adapter from a legacy packet L to the layered observer is a partial map

$$\mathrm{adapt}_L : L \rightharpoonup \mathcal{O}$$

with a domain predicate listing the required fields and laws. Typical mappings are:

- a quotient observer supplies `ObserverPresentation`; an evaluator requires its own test or certificate;
- a report map and context family map into `ObserverPresentation`;
- update and re-access maps map into `ObserverMemory`;
- a mathematical terminal status belongs to `ObserverEvaluator`; a separately declared status-to-action map supplies `ObserverPolicy`;
- an authorization field does **not** map into any of these layers and must be carried separately.

`WDP-CH10-O01` **Conditional result (source-bound partial adapter completeness).** Every identified observer family in the selected source corpus admits an explicit typed partial adapter preserving its declared maps, relations, undefined fields and separate grammar/terminal interfaces. For surjective sort maps, the target interpretation exists uniquely exactly when mapped operation results and predicate truth are constant on input fibers. Missing semantics are recoverable exactly when their own visible-fiber criterion holds. Appendix K supplies the complete criterion, source-specific adapters and independent necessary-hypothesis countermodels. All 44 original texts and all 529 absorbed source occurrences are verified. Arbitrary history functions require their actual declared semantics; finite transducer and rational certificate verification retain their explicit executable domains.

10.4 Worked observer

Let $X = \mathbb{R}^3$ represent three sensor readings. Let the presentation be $q(x_1, x_2, x_3) = (x_1 - x_2, x_2 - x_3)$. Let memory retain the previous presented pair. Write the current presented pair as (u, v). With diagonal repair $d(a) = (a, a)$ and L^1 cost, the evaluator computes $\inf_a(|u - a| + |v - a|) = |u - v|$. The value is zero exactly when the two reported differences are equal. Subtracting a common offset cannot change their discrepancy. A policy may emit `stable`, `unstable`, or `insufficient-history`.

Every component is visible. Nothing about this construction authorizes a decision outside the policy's declared domain, and nothing makes the system conscious.

10.5 The quotient observer as an adapter

The primitive-axiomatics source defines an observer as a repair-compatible quotient of Q^1. That legacy packet fits the layered architecture through one adapter, and the fit is exact enough to state.

`WDP-CH10-D06` **Definition (quotient-observer adapter).** A quotient observer on a window (§4.7) is a linear map $\mathcal{O} : Q^1 \to Q^1_{\mathcal{O}}$, equivalently a linear report map $R : C^1 \to \mathbb{F}^k$ with $Rd_0 = 0$. It enters the layered observer only as a presentation report map and as an evaluator cost, the induced norm on the **realized report space** $\operatorname{im} R$, $\Phi_{\mathcal{O}}(y) = \inf\{\Phi_1(r) : r \in Q^1,\ \mathcal{O}(r) = y\}$. It supplies no memory, policy, or authority layer, and requests for those layers are refused.

The ambient codomain may contain unrealizable report values. For $R = 0$ and $y = 1$ the fiber is empty, so the displayed expression is not a finite norm value there. The evaluator refuses reports outside $\operatorname{im} R$ instead of issuing an ordinary numerical acceptance. On the realized space, the real-extension norm hypothesis gives the stated norm and passivity properties. Attainment holds over the finite-dimensional real or complex carrier, and for rational queries with rational polyhedral norms; a general rational norm restriction need not attain its infimum (§4.2).

Because $\Phi_1(r)$ is itself an infimum over representatives, $\Phi_{\mathcal{O}}(y) = \inf\{J(h) : h \in C^1,\ Rh = y\}$. For weighted L^1 this is a finite linear program, which the executable package solves exactly.

`WDP-CH10-P01` **Finite proposition (observer passivity).** For a quotient observer, $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \le \Phi_1(q)$ for every $q \in Q^1$. Hence an observer cannot report positive obstruction for an exact class, and states identified by the core profile remain identified by every quotient observer.

Proof. The class q lies in the fiber over $\mathcal{O}(q)$, and an infimum is at most any member of its set. If $\Phi_1(q) = 0$, then $0 \le \Phi_{\mathcal{O}}(\mathcal{O}(q)) \le 0$. If a core profile p has $p(x) = p(y)$, then $\mathcal{O}(p(x)) = \mathcal{O}(p(y))$ because $\mathcal{O}$ is a function; linearity is not needed. □

Two consequences are worth naming. If an observer sends an exposed context residual (§11.5) to zero, it carries no context witness for that anomaly. Two reports q_A, q_B from observers $\mathcal{O}_A, \mathcal{O}_B$ share a core preimage exactly when $\mathcal{O}_A^{-1}(q_A) \cap \mathcal{O}_B^{-1}(q_B)$ is nonempty; an empty fiber means the reports cannot describe one core class. This adapter covers the repair-compatible quotient family. Appendix K supplies the other source-specific partial adapters and completes CH10-O01.

`WDP-CH10-P02` **Finite proposition (declared observer families).** For a declared finite family of report maps, set $\mathcal{O} \preceq \mathcal{P}$ when $\ker \mathcal{O} \subseteq \ker \mathcal{P}$, that is, when $\mathcal{P}$ identifies whatever $\mathcal{O}$ identifies. This relation is a preorder; modulo equal kernels it is a partial order; the identity observer, when present, is finest, and the zero observer, when present, is coarsest.

Proof. Kernel inclusion is reflexive and transitive, and mutual inclusion is equality of kernels. Every kernel contains $\{0\}$ and is contained in the whole space. □

Over an infinite field the set of all quotient observers on a nonzero Q^1 is infinite, and when $\dim Q^1 \geq 2$ even their kernels form an infinite family. Finiteness belongs to the declared family.

WDP-CH10-P03 **Finite proposition (grammar resolution bound).** A deterministic report map r from a declared finite set of core classes into a finite grammar $\mathcal{G}$ exposes at most $|\operatorname{im} r| \leq |\mathcal{G}|$ visible classes. If the core set has more than $|\mathcal{G}|$ elements, two distinct core classes share a report.

Proof. Visible classes are the fibers of r, and there is one fiber per element of $\operatorname{im} r \subseteq \mathcal{G}$. The pigeonhole principle gives the collision. □

For reports of length at most L over an alphabet Σ, $|\mathcal{G}| \leq \sum_{\ell=0}^{L} |\Sigma|^{\ell}$. Grammar design is therefore a resolution constraint on the observer, not a presentation detail.

Two consequences follow. A quotient observer separates all of Q^1 exactly when $\ker R = \operatorname{im} d_0$, which needs at least $\dim Q^1$ report coordinates. And a reporter that reads windows of radius d over k letters has a grammar of at most k^{2d+1} reports, so a longer run of distinct local situations must repeat a report.

10.6 Sharp observer completion: raw defects and repair classes

GEN-WDP-46 (finite rational observer-completion theorem). Supply rational matrices $D : \mathbb{R}^p \to V$ and $Q : V \to W$, interpreted over their real extensions, and put

$$B = \operatorname{im} D, \qquad K = \ker Q, \qquad H = B \cap K, \qquad k = \dim K, \quad r = \dim H.$$

Here B is both the image of realizable raw defects and the space of allowed repair changes. This is an explicit hypothesis: distinct realization and repair operators require distinct subspaces. Give V, W, and each augmented report space faithful rational polyhedral norms. For an arbitrary cochain $v \in V$, define

$$I(b) = \|b\|_V \quad (b \in B), \qquad \Phi(v) = \operatorname{dist}_V(v, B), \qquad \Phi_Q(Qv) = \operatorname{dist}_W(Qv, Q(B)).$$

Required property	Exact criterion	Minimum additional rational scalar measurements
Raw: $Qb = 0 \Rightarrow b = 0$ for every $b \in B$	$H = 0$	r
Quotient: $\Phi_Q(Qv) = 0 \Rightarrow \Phi(v) = 0$ for every $v \in V$	$K \subseteq B$	$k - r$
Both	$K = 0$	k

These are unrestricted linear measurement counts. A restriction to local sensors, coordinate reads, finite precision, or a fixed sensor menu requires its own attainability theorem. The augmented norm is supplied separately; minimum row count alone does not minimize the recovery constant.

Proof. Raw invisibility is precisely H. For the quotient, $Q^{-1}(Q(B)) = B + K$, so observed zero implies original zero exactly when $K \subseteq B$. The two criteria together require $K = 0$. The canonical exact sequence

$$0 \longrightarrow H \longrightarrow K \longrightarrow (K + B)/B \longrightarrow 0$$

separates the two invisible dimensions. For an augmentation $A = (Q, R)$, raw soundness requires $R|_H$ injective, quotient soundness requires $\ker(R|_K) \subseteq H$, and both require $R|_K$ injective. Rank-nullity gives the three lower bounds. Extend a rational basis of H first to K, then by the remaining directions of B, and finally to V. Its dual coordinates on the first r, the next $k - r$, or all first k directions attain the respective bounds. □

For a fixed sound augmentation A, the least recovery constants in

$$\|b\|_V \le C_{\mathrm{raw}} \|Ab\|_U, \qquad \Phi(v) \le C_{\mathrm{quo}} \operatorname{dist}_U(Av, A(B))$$

are finite and rational under the declared polyhedral norms. Exhaustive rational vertex enumeration and matching primal/dual distance certificates prove sharpness. A nonzero invisible vector gives an infinite constant for the corresponding failed property. The constant is zero when its recovery domain is zero: $B = 0$ for raw recovery, or $V/B = 0$ for quotient recovery. A zero report space with a nonzero recovery domain gives infinity. The full derivation and certificate specification give the norm hypotheses, complete finite constructions, counterexamples, and deterministic error bounds. No efficiency claim is made.

The source three-patch carrier has $(\dim B, k, r) = (15, 6, 3)$, giving minimum counts $(3, 3, 6)$. Its realized family satisfies $I(Ds) = 6|\lambda|$ and $QDs = 0$, while $\Phi(Ds) = 0$ because every section defect is exact. A separate invisible nonexact cochain witnesses quotient failure. Three cycle reads restore quotient recovery with sharp ℓ^1 constant 1; six missing coordinate reads restore both properties with constants 1. The coupled carriers have $(\dim B, k, r) = (2, 2, 0)$, giving counts $(0, 2, 2)$: raw soundness already holds, but quotient soundness fails. Their source report coordinate conventions differ and are checked separately. The original coupled raw constant is 2 for stacked ℓ^1 reports and 4 for the maximum individual score.

This theorem joins the raw false-acceptance test (a locally accepted but actually defective configuration) to WDP's quotient-observer test without conflating them. In §10.5's convention $QD = 0$, quotient soundness specializes to $K = B$; raw soundness is deliberately lost when $B \neq 0$. Projection Obstruction's physical-residual-to-cochain adapter must still be supplied separately. A checked-repair evaluator may evaluate the retained exact certificates; neither a repaired observer nor a recovery certificate grants authority to install it or execute a policy. The reference package records written proofs, source-pinned replay, 15 adversarial controls and 27 scoped Python tests. It supplies no new Lean formalization or verification of the native Rust owners. This bridge claim supplements the canonical claim table.

10.7 Sufficient observation under physical locality

The finite completion theorem of §10.6 permits arbitrary declared linear observation rows. A physical instrument must additionally realize its rows with the declared spatial support, communication and precision constraints. These are separate obligations; algebraic completion does not supply an instrument.

For density matrices rho and sigma with equal **joint** marginals on a region R, let O_R be supported there and let ||O-O_R|| <= beta. Then

$$|\operatorname{Tr}((\rho - \sigma)O)| \le \|\rho - \sigma\|_1 \, \beta \le 2\beta.$$

Subtract the zero local readout Tr((rho-sigma)O_R), then apply trace/operator norm duality. A supplied Lieb–Robinson estimate and a specified conditional expectation can provide beta after the spatial tail is summed. Equal single-site marginals alone do not establish the joint-marginal premise.

For a physical-to-target map F defined on **states**, the separation is between F(rho) and F(sigma), not F(rho-sigma), which is generally outside its domain. A reconstruction map on readouts with Lipschitz constant kappa and endpoint errors at most alpha satisfies delta <= 2 alpha + kappa eta, where eta is the actual readout separation. When eta <= 2 beta and delta > 2 alpha, stable reconstruction requires the corresponding conditioning price. This does not turn an error upper bound into a device error floor.

Readout noise balls overlap if the separation is at most twice their radius. At a common report one reconstruction error is at least half the target separation. No exact waiting threshold T >= r/v follows from an exponential upper envelope without its constants, desired precision and lower evidence.

The corrected derivation supplies the full finite recovery/completion theorem, locality and conditioning proofs, actual CAR exchange signs and executable counterexamples. GEN-WDP-56 in §20.11 binds its written, formal and executable scopes separately. This section supersedes the duplicate locality bridge imported at remote-main commit 9b55e6c1; the original §10.6 and its canonical claims are retained.

10.8 Question-preserving encodings and a direct checker

The encoding bridge and experiment apply the existing report-fiber criterion to a bounded symbolic task. For a fixed question q and encoding E, equal reports with unequal answers exclude a decoder that correctly decides both. A checker may still soundly refuse both; the obstruction does not require false acceptance.

For the positive bridge, let P be an explicit partial parser and assume $P(E(c))$ is defined with $q_Z(P(E(c))) = q_C(c)$. If a checker K obeys $K(z, w) = \text{accept} \Rightarrow q_Z(z) = \text{true}$, then accepting $K(P(E(c)), w)$ implies $q_C(c) = \text{true}$. This follows by substitution of the two premises. Checker soundness and the binding of a received report to an original candidate remain explicit obligations; cochain closure does not supply either premise. This specializes the faithful-representation soundness interface of §20.6 to an explicit parser and bounded experiment.

The finite implementation represents integer quadratics with coefficients in $[-2, 2]$. Its protected question is an exact derivative identity together with a specified value at zero. The edge differences around a triangle lose the common coefficient offset: $(0, 0, 0)$ and $(1, 1, 1)$ have the same report but different answers to the zero task. Adding one coefficient as an anchor gives an explicit inverse parser. The parser calls the canonical closure functional and rejects malformed reports; the direct polynomial checker then evaluates the original question.

Across 225 tasks and 125 candidates, enumeration uses 20,375 checker calls. Both direct algebraic reconstruction and anchored cochain reconstruction use 125. The latter adds a scalar and a closure test for each successful proposal. Thus this declared experiment demonstrates no cochain-specific search or diagnostic efficiency gain. All 28,000 incorrect task/candidate pairs nevertheless have closed encoded cochains. Correctness rests on the checker; representation consistency, source/world

interpretation and action authorization remain distinct. The supplement records full proofs, cost scope, failure controls and native replay bindings.

10.9 The Priced Observer Boundary State and Referential Binding Protocol $\mathcal{G}_{\text{bind}}$

WDP-CH10-D07 **Definition (Refined Observer Boundary State).**
The observer boundary state is formally specified as the projection pair:

$$\pi_O : X_O \to Y_O, \quad X_O = (S_O, R, \mathrm{TL}_O^{\text{priv}}), \quad Y_O = (B_O, R_{\text{pub}}, \mathrm{TL}_O^{\text{pub}})$$

where $R_{\text{pub}} = \mathbf{1}_{\{R \geq c_{\min}\}}$ projects a coarsened operational solvency predicate, while the exact quantitative ledger $R \in \mathbb{Q}_{\geq 0}$ is arbitrated out-of-band by the external host runtime (ENC-0195) and contained within the private projection fiber $\mathrm{Fib}_O := \ker \pi_O$ satisfying $\dim(\mathrm{Fib}_O) > 0$.

WDP-CH10-P04 **Proposition (Private Fiber Invariance in Boundary Observation).**
For all internal perturbations $v \in \mathrm{Fib}_O$ and all external observation projections π_O, $\pi_O(x + v) = \pi_O(x)$. Public observation is invariant under internal fiber shifts.

WDP-CH10-D08 **Definition (Referential Binding Protocol $\mathcal{G}_{\text{bind}}$).**
The protocol $\mathcal{G}_{\text{bind}}$ coordinates reference across the observer boundary via four discrete steps: 1. **Ingress:** Receiving candidate token stream under Gate 1; 2. **Admission:** Validating form and data preconditions under Gate 2, debiting ledger R; 3. **Retention:** Evaluating structural asymmetry cost under Gate 3; 4. **Attestation:** Computing quotient defect Φ_1 under Gate 4 and emitting receipt under Gate 5.

Chapter 11 — Intersubjective presentation, descent, and redundancy

11.1 Many observers, one carrier

Let observers $\mathcal{O}_i$ have presentation maps $q_i : X \to P_i$. A comparison map $r_{ij} : P_i \rightharpoonup P_j$ may allow one presentation to be transported into another. These maps are data and may be partial.

For a family of overlapping contexts U_i, local reports s_i are compatible when their restrictions agree after declared transport on overlaps. A descent problem asks whether there exists a global report s whose restrictions recover the s_i.

WDP-CH11-D01 **Definition (observer descent receipt).** A descent receipt is a tuple containing the local reports, overlap maps, compatibility checks, a global candidate when one exists, and verification that its restrictions agree.

The definition resembles sheaf descent, but a sheaf is not obtained until a site or cover, restriction maps, and gluing laws are supplied.

11.2 Local satisfiability is not global satisfiability

Every proper subfamily of a finite constraint family can be jointly satisfiable while the whole family is inconsistent. For example, each context in a three-cycle can admit assignments satisfying its own parity constraint, and each proper pair of contexts can admit a compatible assignment, although no single assignment satisfies all three parity constraints. This is not a compatible family of fixed local sections: if fixed sections actually agree on all overlaps, the ordinary gluing problem has already been solved at the level of variable assignments. Constraint-cycle examples motivate obstruction diagnostics, but contextuality in the quantum-foundational sense additionally requires a measurement scenario and empirical model.

The same gap has an exact quotient form. Let G be the oriented m-cycle with vertices $0, \dots, m-1$, edges $e_k = (k, k+1)$ with indices mod m, unit weights, and an edge field δ over $\mathbb{Q}$ with circulation $c = \sum_k \delta_{e_k}$. For each j let $P_j = E \setminus \{e_j\}$ be the path patch that omits e_j, and set $J_{\max}(x) = \max_j \sum_{e \in P_j} |x_e|$.

WDP-CH11-P02 **Finite proposition (three readings of one cycle class).** For $m \geq 3$: (1) the restriction of δ to each patch P_j is exact, so each patch reports $\Phi_1 = 0$; (2) the unit L^1 quotient defect is $\Phi_1([\delta]) = |c|$; (3) $J_{\max}$ is a norm on $C^1(G;\mathbb{Q})$ and $\Phi_{J_{\max}}([\delta]) = \frac{m-1}{m}|c|$, attained by a rational repair.

Proof. (1) A path is a tree: fix the potential at one end and add the edge values along the path. (2) With the cyclic orientation every coboundary sums to 0 around the cycle, so every residual $r = \delta - d_0 f$ has $\sum_e r_e = c$ and $\sum_e |r_e| \geq |c|$. The repair built along P_j leaves c on e_j alone, and $y = \pm(1, \dots, 1)$ is a circulation with $|y_e| \leq 1$ and $\langle y, \delta \rangle = |c|$, so CH04-P07 certifies the value. (3) Each edge lies in $m - 1 \geq 2$ patches, so $J_{\max}(x) = 0$ forces $x = 0$, and a maximum of seminorms is a seminorm. For every repair,

$$\max_j \sum_{e \in P_j} |r_e| \;\geq\; \frac{1}{m} \sum_j \sum_{e \in P_j} |r_e| \;=\; \frac{m-1}{m} \sum_e |r_e| \;\geq\; \frac{m-1}{m}|c|.$$

The image of d_0 is the sum-zero hyperplane, because the cycle is connected and d_0 has rank $m-1$. So the residual with every entry c/m equals $\delta - d_0 f$ for a rational f, and then every patch reads $\frac{m-1}{m}|c|$. □

For the cyclic field $\delta = (2, 2, 2)$ on the triangle the three readings are 0, 6, and 4. Each patch alone is repaired exactly. The union of edges carries the whole circulation: $f = (0, 2, 4)$ leaves $(0, 0, 6)$, and $y = (1, 1, 1)$ certifies 6. One shared repair read patch by patch through the maximum leaves 4, attained at $f = 0$ with residual $(2, 2, 2)$. On the triangle each patch repair $f^{(j)}$ of P_j is unique up to an additive constant, since each patch meets all three vertices. Two patch repairs differ by a constant on the endpoints of their shared edge, and the three constants $f^{(0)} - f^{(1)}$, $f^{(1)} - f^{(2)}$, $f^{(2)} - f^{(0)}$ always sum to c, whatever constants are chosen. So the local repairs glue exactly when $c = 0$: the family is compatible up to gauge, and the gauge constants carry the obstruction. The three numbers are values of three declared evaluator costs on one class. None of them is the size of the class, and the proposition concerns a single cycle with unit weights.

Provenance: the absorbed archived working-paper source 011 proved in Lean, over the integers at its recorded commit, the triangle case of (1) and the non-exactness of $(2, 2, 2)$. It checked the three numbers by exact rational Python. Its Lean evaluates one candidate for (3) and does not prove the minimum. Not ported; no Lean credit.

WDP-CH11-O01 **Finite proposition (explicit observer descent).** On the finite inclusion site of intersection-closed contexts with union covers, checked report sets and coherent restriction maps form a presheaf. It is a sheaf exactly when every matching-family fiber contains one global report. Constrained assignment reports admit a glue exactly when their pointwise union is admitted; the equality-constraint countermodel refutes unrestricted local-to-global gluing. Appendix J defines the category and coverage and establishes these criteria.

The same gap appears for composed procedures. If a step γ_A passes a local evaluator E_A and a step γ_B passes E_B, the composite $\gamma_B \circ \gamma_A$ need not pass a global evaluator E_G. CH11-P02 is an instance: every proper patch passes and the cycle fails. Two local passes compose to a global pass

only when the boundary between them preserves what E_G inspects. A tool or subagent that passes its own check is therefore evidence-bearing only for the question that check declares.

11.3 Redundancy

Suppose several conditionally independent channels carry presentations of one source. Redundancy can improve robustness: a report recoverable from any k of n channels survives some channel failures. A finite redundancy theorem must state the channel model, failure family, decoder, and target predicate.

Quantum Darwinism uses environmental fragments, decoherence, and preferred observables to study a related but stronger physical setting [REF-OBS-001; REF-OBS-002]. The monograph may borrow the structural question—when do independent channels support stable agreement?—but not the physical mechanism or its conclusions.

11.4 Agreement and authority

Even unanimous observer reports do not create authority. Agreement can be a premise in an external decision rule, but the rule and its jurisdiction must be declared. Conversely, an authorized institution can issue a decision that is not an observer-consensus theorem. Keeping these types separate prevents a descent result from becoming a governance claim.

11.5 Contextual congruence and witness rank

The primitive-axiomatics source supplies a finite notion of context-relative failure that complements descent. A finite contextual system S consists of a finite state set X, a finite context family C acting by maps $\rho(c) : X \to X$, and a profile map $p : X \to Q^1$ into the quotient of a window (§4.8).

`WDP-CH11-D02` **Definition (contextual congruence and context witness).** Set $x \sim y$ when $p(x) = p(y)$. The relation $\sim$ is a contextual congruence when $x \sim y$ implies $\rho(c)x \sim \rho(c)y$ for every context c. A critical context witness is a triple (x, y, c) with $p(x) = p(y)$ and $p(\rho(c)x) \neq p(\rho(c)y)$. Its exposed residual is $\nu(x, y, c) = p(\rho(c)x) - p(\rho(c)y) \in Q^1$. Such a triple is a `ContextWitness` in the sense of Chapter 2.

The source's slogan, "obstruction alone does not force refinement; context-visible non-congruence does," is a design principle for when to enlarge a presentation. It is not a theorem about which enlargement to choose.

`WDP-CH11-P01` **Finite proposition (witness-rank invariance).** The witness rank $\mathrm{wrank}(S) = \dim \mathrm{span}\{\nu(w)\}$, taken over all critical context witnesses w, is invariant under invertible linear coordinate changes of Q^1 and under isomorphisms of contextual systems, meaning bijections of states and contexts that intertwine the actions, composed with a linear isomorphism of profile spaces.

Proof. A linear isomorphism preserves linear independence, hence the dimension of a span. An isomorphism of systems carries critical witnesses bijectively to critical witnesses and transforms their exposed residuals by the profile isomorphism. □

Paths of contexts refine the picture. For context paths a, b with common endpoints that are declared homotopic, the path defect $\Theta(x; a, b) = p(\rho(a)x) - p(\rho(b)x)$ lies in Q^1. It is closed and non-exact when $\Theta \in H^1 \setminus \{0\}$ and open when $\kappa^1(\Theta) \neq 0$. Homotopy fillers are added only by declaration; none arises from the nerve of the context family alone.

The source also stated a reflective completion of contextual systems, a left adjoint to the inclusion of congruence-complete systems, as a conjecture that depends on an unproved initiality lemma for minimal refinements. It is not absorbed here; see CH04-O01.

11.6 The Inter-Observer Descent Operad $\mathcal{O}_{\text{Bridge}}$, Relational Correspondence Span, and Represented Bridge Quotient Class $Q^1_{\mathcal{B}_{12}}$

WDP-CH11-D03 **Definition (Coloured Planar Operad $\mathcal{O}_{\text{Bridge}}$).**
The inter-observer descent operad is a coloured planar operad $\mathcal{O}_{\text{Bridge}} = (\mathcal{C}, \Sigma, \circ, \mathcal{I})$ with colour palette:

$$\mathcal{C} = \{\mathsf{World}_{\mathcal{O}_i}, \mathsf{Fact}_{\mathcal{O}_i}, \mathsf{Export}_{\mathcal{O}_i}, \mathsf{Bridge}_{\mathcal{O}_1 \to \mathcal{O}_2}, \mathsf{Defect}_1, \mathsf{Receipt}_{\text{joint}}\}$$

where colour $\mathsf{Fact}_{\mathcal{O}_i} := (R_i, \sim_i, G_i, \rho_i)$ encapsulates the quotient reference carrier R_i together with the admissible internal gauge group action representation (G_i, ρ_i). Generator slot arities are strictly linearized: - $g_{\text{fact},i} \in \mathcal{O}(\mathsf{World}_{\mathcal{O}_i}; \mathsf{Fact}_{\mathcal{O}_i})$, arity 1; - $g_{\text{defect}} \in \mathcal{O}(\mathsf{Fact}_{\mathcal{O}_1}, \mathsf{Bridge}_{\mathcal{O}_1 \to \mathcal{O}_2}, \mathsf{Fact}_{\mathcal{O}_2}; \mathsf{Defect}_1)$, arity 3, evaluating $\alpha_{\text{designation}}$ over encapsulated (G_1, G_2) representations without auxiliary wire inputs; - $g_{\text{receipt}} \in \mathcal{O}(\mathsf{Defect}_1; \mathsf{Receipt}_{\text{joint}})$, arity 1; yielding overall planar composition tree $\tau = g_{\text{receipt}} \circ_1 (g_{\text{defect}} \circ (g_{\text{fact},1}, 1_{\mathsf{Bridge}}, g_{\text{fact},2}))$ of arity 3 with zero wire crossings and fan-out 1.

WDP-CH11-D04 **Definition (Relational Correspondence Span).**
Given two observers $\mathcal{O}_1, \mathcal{O}_2$ with respective report quotient spaces R_1, R_2, the inter-observer comparison is formalized by the relational correspondence span:

$$R_1 \xleftarrow{\pi_1^B} K_{12} \xrightarrow{\pi_2^B} R_2$$

where $K_{12} \subseteq R_1 \times R_2$ is a closed sub-relation replacing the unconstructed categorical pullback apex $\mathcal{O}_1 \wedge \mathcal{O}_2$.

WDP-CH11-D05 **Definition (Four Congruence Annihilators).**
Admissibility of inter-observer transport requires satisfaction of four congruence annihilators evaluated by g_{defect}: 1. α_{core}: verifies support compatibility on span apex K_{12}; 2. α_{equiv}: verifies quotient descent preservation ($k \sim_1 k' \iff k \sim_2 k'$); 3. $\alpha_{\text{designation}}$: verifies gauge invariance under internal presentation relabeling across encapsulated gauge orbits:

$$\forall (g_1, g_2) \in G_1 \times G_2,\ \forall k \in K_{12},\ \pi_1^B(\rho_1(g_1) \cdot k) \sim_1 \pi_1^B(k) \wedge \pi_2^B(\rho_2(g_2) \cdot k) \sim_2 \pi_2^B(k);$$

4. α_{defect}: verifies coboundary exactness $\Phi_1(\omega_{12}) = \inf_\alpha \|\omega_{12} - d_0^{12}\alpha\|_1 = 0$.

WDP-CH11-T02 **Theorem (Represented Bridge Quotient Class and LP Dual Circulation Witness).**
Let $\mathcal{B}_{12}$ be an inter-observer bridge complex terminating at degree 1 ($C^2_{\mathcal{B}_{12}} = \emptyset$). Then: 1. The 1-cocycle condition is vacuous ($Z^1_{\mathcal{B}_{12}} = C^1_{\mathcal{B}_{12}}$), and the degree-1 obstruction class is represented by the quotient class:

$$q_{\mathcal{B}_{12}} = [\omega_{12}] \in Q^1_{\mathcal{B}_{12}} := C^1_{\mathcal{B}_{12}} / \operatorname{im}(d_0^{12})$$

2. The dual space is canonically isomorphic to the adjoint kernel $(Q^1_{\mathcal{B}_{12}})^* \cong \ker((d_0^{12})^*)$; 3. By finite-dimensional linear programming duality, there exists an optimal dual witness $\varphi^* \in \ker((d_0^{12})^*)$ with $\|\varphi^*\|_\infty \leq 1$ such that:

$$\langle \varphi^*, \omega_{12} \rangle = \|q_{\mathcal{B}_{12}}\|_{Q^1_{\mathcal{B}_{12}}}$$

attaining zero rational duality gap.

WDP-CH11-R01 **Remark (Connection to Trace-Zero Monograph Theorem 3.1).**
When potential repairs $\alpha \in C^0_{\mathcal{B}_{12}}$ are unrestricted, the dual witness condition is $(d_0^{12})^*\varphi^* = 0$ (exact vanishing divergence, matching Problem (8)). When potential repairs are constrained by trace-zero or mean-zero conditions across observer frames, the dual witness condition relaxes to $(d_0^{12})^*\varphi^* \in \mathbb{R}I$ (scalar divergence, matching Problem (7)).

Chapter 12 — The observation ladder

12.1 Five levels

The word *observation* often compresses several events. We separate them.

1. **Interaction.** A system and target participate in a declared coupling.
2. **Registration.** The coupling changes a record carrier.
3. **Typed witness.** The record is linked by a declared evidence relation to a proposition about the interaction.
4. **Re-comparison.** The record can be accessed and compared under the same or a transported profile.
5. **Evaluator acceptance.** A declared evaluator returns an admissible status for the typed proposition.

WDP-CH12-D01 **Definition (observation record and validated observation).** An observation record is

$$\mathsf{Obs} = (i, g, w, m, e)$$

whose fields record interaction i, registration g, typed witness w, memory/re-comparison m, and evaluator result e. A **validated observation** is a well-typed observation record for which the first four level predicates verify and $e = \mathsf{Accept}$.

A receipt for a failed registration, invalid witness, missing re-access, refusal, rejection, or undetermined result records an observation attempt but does not inhabit the validated observation type.

12.2 Failure modes

The ladder makes failure localizable.

- **No coupling:** the proposed interaction did not occur under the model.
- **No registration:** coupling occurred but left no admissible record.
- **Misregistration:** the record does not satisfy the witness relation.
- **No retention:** the record existed but cannot be re-accessed.
- **Profile drift:** the later comparison uses an incompatible context.
- **Evaluator refusal:** the query is ill typed or outside the evaluator's domain.
- **Evaluator rejection:** a well-typed claim fails its declared criterion.
- **Undetermined:** available evidence does not settle the criterion.

These statuses should not be replaced by a single Boolean unless the loss of information is deliberate.

12.3 Observation is not authorization

`WDP-CH12-P01` **Typing proposition.** `ObservationAccept` does not imply `AuthStanding` or `Promote` in the layered architecture, because those predicates belong to a separate authority object and no coercion between the types is defined.

This proposition is almost syntactic, but it repairs a consequential ambiguity in the measurement source. An evaluator may accept that a thermometer read 20.1 °C under its procedure. That does not authorize opening a facility, publishing a medical conclusion, or changing a controlled standard.

12.4 Worked ladder

A finite switch closes, changing a voltage at a sensor. An analog-to-digital converter stores the integer `813`. A calibration table associates the code with a voltage interval. Memory stores the code, calibration version, and time index. An evaluator checks checksum, range, and calibration validity.

- Interaction: switch/sensor coupling.
- Registration: stored code `813`.
- Witness: code and calibration receipt support a voltage proposition.
- Re-comparison: the stored record can be checked later.
- Evaluator acceptance: validity predicates pass.

This is already a validated observation. It becomes a measurement only after the separate measurand specification and result carrier, value map, unit, calibration, uncertainty/error semantics, metrological-traceability evidence when claimed, and provenance/re-access fields of Chapter 13 are supplied.

Chapter 13 — Measurement as a typed extension of observation

13.1 Metrological repair

The earlier measurement draft correctly recognized that measurement requires a scale or partition and re-access. It was incomplete in four ways: it used authority where evaluator standing was required; it did not distinguish measurand from indication; it lacked calibration and uncertainty; and it lacked traceability and instrument-response semantics. We repair those omissions here without claiming a full physical metrology theory.

`WDP-CH13-D01` **Definition (finite measurement packet).** A finite measurement packet in this WDP-specific interface is

$$\mathfrak{M} = (\mathsf{Obs}, Q, Y, h, u, \mathcal{S}, \mathsf{Cal}, \mathsf{Unc}, \mathsf{Err}, \mathsf{Tr}_{\mathrm{met}}, \mathsf{Prov}),$$

where:

- Obs is a validated observation record, a stronger WDP-specific prerequisite rather than part of the VIM definition of measurement [REF-MET-001];
- Q specifies the measurand: the quantity intended to be measured, including the relevant system, quantity kind, and stated conditions;
- Y is the distinct carrier for reported quantity values or results;

- h maps admissible indications/records to reported values;
- u is a unit or scale convention when the value type requires one;
- $\mathcal{S}$ is the permitted value/partition structure;
- Cal is a versioned calibration record relating reference quantity values and their uncertainties, indications and their uncertainties, and the conditions and relation used to obtain a result;
- Unc is the measurement-uncertainty model and budget, when that metrological claim is made;
- Err separately records any correction, measurement-error estimate, worst-case error bound, confidence set, or result enclosure, with its own semantics;
- $\mathsf{Tr}_{\mathrm{met}}$ records any claim of metrological traceability as a property of the measurement result established through a documented unbroken calibration chain, each calibration contributing to the stated measurement uncertainty;
- Prov is the separate provenance and re-access record linking the result to the procedure, calibration version, evaluator, and raw record.

The packet is **valid relative to a profile** only when each component is typed, the evaluator accepts its declared checks, and the result lies within the procedure's domain.

13.2 Measurand, indication, result

The measurand specification Q is not a value carrier and neither is the raw record. An instrument produces an indication z; a value map and calibration yield an estimate $\hat{y} = h_{\mathsf{Cal}}(z)$ in Y. A measurement result should pair that estimate with an uncertainty description rather than present an exact numeral when the procedure does not justify one.

A finite interval $E(z) = [\hat{y} - \epsilon_-, \hat{y} + \epsilon_+]$ can be declared as a worst-case result enclosure or error bound, and a confidence set can be declared under a specified sampling model. Neither object is automatically a VIM/GUM measurement uncertainty [REF-MET-001; REF-MET-002; REF-MET-003]. A claim of measurement uncertainty must state its uncertainty components, evaluation and propagation method, coverage convention when used, and relationship to the reported result. Probabilistic uncertainty requires a probability model and must not be inferred just from finite witness weights.

When serialized for machine audit and custody ledgers, a validated observation or measurement packet could be recorded in a 12-field evidential receipt such as:

$$R = (\text{claim_id}, \text{theory_profile}, \text{inputs}, \text{operation}, \text{result}, \text{witness_role}, \\ \text{assumptions}, \text{provenance}, \text{uncertainty}, \text{falsifier}, \text{tool_version}, \text{dependencies}).$$

Such a record would keep raw indication, calibration lineage, and uncertainty model separate. The schema is a proposal of this text, not a package interface: the executable `Receipt` in `code/witnessed_difference/receipts.py` has six fields (subject, input and output digests, claim IDs, implementation version, and receipt digest).

13.3 Calibration

Let reference inputs $y_1, \ldots, y_k$ have accepted reference quantity values with stated uncertainties and conditions, and let the instrument produce indications $z_1, \ldots, z_k$ with their indication uncertainties. A calibration record relates those standards and indications and records how that relation is used to obtain a result and its uncertainty. A fitted affine calibration curve,

$$\hat{y} = az + b,$$

is one component rather than the whole calibration. The parameters a, b are fitted or otherwise certified from the reference data; reference-value, indication, model, and fit-residual contributions enter the appropriate uncertainty or error accounts. Extrapolation outside the calibrated range must be separately flagged.

`WDP-CH13-C01` **Conditional result.** Write $I \prec J$ for compact indication intervals with $\sup I < \inf J$. Let F be the complete declared interval map, including calibration and every uncertainty/error enlargement used in the report. If F is order-preserving on the admitted range in the exact sense $I \prec J \Rightarrow F(I) \prec F(J)$, then ordered separated indication intervals yield ordered separated reported-value intervals. A monotone point calibration alone is insufficient: the interval hypothesis must be verified, for example by an explicit gap-versus-enlargement bound.

13.4 Metrological traceability, provenance, and re-access

Metrological traceability is a property of a measurement result, not a synonym for a hash chain or data lineage. A claim of it requires a documented unbroken chain of calibrations to a stated reference, with every calibration contributing to the measurement uncertainty. The chain may terminate in an external standard. This monograph can type and verify supplied finite records and their internal links; it does not certify an external institution, reference, or calibration event.

Provenance records where the data came from and which transformations were applied. Re-access is the stronger WDP-specific ability of a later evaluator to retrieve the raw indication, calibration version, transformation, uncertainty inputs, and profile used for the report. Neither property alone establishes metrological traceability. A number without the provenance/re-access packet is an assertion, not a replayable measurement result in this architecture.

13.5 Worked measurement

Consider a four-level finite sensor with raw codes $Z = \{0, 1, 2, 3\}$. A calibration table maps them to intervals in volts:

$$0 \mapsto [0.0, 0.4], \quad 1 \mapsto [0.4, 0.9], \quad 2 \mapsto [0.9, 1.5], \quad 3 \mapsto [1.5, 2.0].$$

An interaction registers code 2. The checksum and range tests pass; memory stores the raw code and calibration hash. The measurand is terminal voltage under the declared loading condition. The reported result may be the interval $[0.9, 1.5]\,\mathrm{V}$, or a representative value with an explicitly derived uncertainty. It is not justified to report "1.2 V exactly."

If a new calibration changes code 2 to $[1.0, 1.6]$ V, the old result remains interpretable because its provenance/re-access packet identifies the old version. A policy may request re-evaluation, but the new table does not retroactively alter the stored raw code. Any claim that either result is metrologically traceable must separately verify its calibration hierarchy and uncertainty contributions.

13.6 Measurement and probability

Repeated measurement can produce a finite table of outcomes. Chapter 8 can normalize their declared weights. To interpret that table as sampling frequencies or a stochastic instrument, one must

add a trial protocol, exchangeability or independence conditions as appropriate, and a probability model. A single measurement packet does not contain those assumptions.

WDP-CH13-O01 **Conditional result (resolved finite target).** For sequential affine calibration stages with declared gain, offset and residual error bounds, the radius recurrence (abs(a)+alpha)r+alpha abs(x)+beta+eta contains every admissible realization without independence assumptions; nominal calibration updates have exact interval sup error abs(da x+db)+abs(da)r. The proof, native ownership and scope are given in Appendix I.

WDP-CH13-N01 **Nonclaim.** Measurement validity does not imply authority, causal explanation, physical completeness, or observer independence.

Chapter 14 — Self-reference and the awareness boundary

14.1 Self-presentation

An observer can receive as input a representation of some of its own state. Let $\sigma : \mathcal{O} \to X_{\mathcal{O}}$ encode the accessible self-state and let $q_{\mathcal{O}}\sigma$ be its presentation. A self-report is then an ordinary report with a reflexive provenance path.

WDP-CH14-D01 **Definition (self-witnessing event).** A self-witnessing event is a validated observation whose target proposition concerns a typed component of the same observer packet and whose provenance/re-access record exhibits that reflexive target.

The definition requires no phenomenological vocabulary. A program checking its own version hash is a simple instance. It witnesses a proposition about an accessible representation, not every fact about itself.

14.2 The refuted implication ladder

An earlier draft printed a hierarchy roughly of the form

$$O_3 \Rightarrow O_2 \Rightarrow O_1 \Rightarrow O_0.$$

As stated, the levels mixed incompatible context actions and made O_0 and O_1 contrary in some models. The full carrier ladder is therefore a `RefutedDraft`, not a theorem.

The repair is to use independent predicates and explicit bridges:

- R_1: a self-report is linked to an accessible self-state;
- R_2: the report remains stable under a declared context transport;
- R_3: an evaluator can compare the report with later behavior or state.

Appendix J supplies the complete finite classifications under explicit bridge data. Comparison availability and calibration success are separate predicates; partial maps cannot be completed by defaults.

14.3 Blind spots

If the presentation map $q_{\mathcal{O}}$ is not injective, there are distinct input states with the same presentation. No evaluator operating only on that presentation can distinguish them.

WDP-CH14-P01 **Finite proposition (presentation blind spot).** If $x \neq y$ but $q_{\mathcal{O}}(x) = q_{\mathcal{O}}(y)$, then every function $F : P_{\mathcal{O}} \to Z$ assigns the same output to the presentations of x and y.

This trivial factorization fact is a precise limit on self-knowledge in the model. It is not the philosophical hard problem and does not imply that a human or machine lacks awareness.

14.4 Capacity and diagonal questions

A finite observer with a finite report alphabet cannot injectively encode an arbitrarily large state space. Pigeonhole arguments yield blind spots relative to the selected encoding. Stronger diagonal or self-reference results require a language capable of naming its own reports, an interpretation map, and a logic. Those structures must be listed before invoking Gödelian language.

Frauchiger–Renner-style scenarios can serve as adversarial tests for context mixing in a quantum host [REF-OBS-009]. They are not evidence for a general awareness theorem, and no protocol or proof from that work is imported here.

Linear presentations make the capacity bound quantitative. Let $q : V \to \mathbb{F}^M$ be a linear presentation map on a finite-dimensional space and let $Y \subseteq V$ be a declared target subspace. Here Y is a declared target, not a defect: in Chapter 4 the image of d_0 is the repair image, and a quotient observer is built to be blind to it (CH10-D06).

WDP-CH14-P02 **Finite proposition (injectivity on a declared subspace).** A linear presentation map q on a finite-dimensional space is injective on a declared subspace Y exactly when Y meets $\ker q$ only in 0. This requires at least $\dim Y$ report coordinates, and with fewer, exact elimination over the coefficient field exhibits a nonzero $y \in Y$ with $q(y) = 0$.

Proof. Injectivity on Y means $q(y) = q(y')$ implies $y = y'$ for $y, y' \in Y$, that is, $q(y - y') = 0$ implies $y - y' = 0$. By rank–nullity, $\dim(Y \cap \ker q) \geq \dim Y - M$, and Gaussian elimination on the matrix of q restricted to a basis of Y produces a nonzero kernel vector when $M < \dim Y$. □

WDP-CH14-P03 **Finite proposition (recovery bound).** If a linear presentation map q is injective on a finite-dimensional declared subspace Y, and norms are declared on Y and on reports, over $\mathbb{R}$ or $\mathbb{C}$ or, over $\mathbb{Q}$, as restrictions of norms on the real extensions, then some $c > 0$ satisfies $\|y\| \leq c\|q(y)\|$ for every $y \in Y$, and a report perturbed by η with $\|\eta\| \leq \varepsilon$ still gives $\|y\| \leq c(\|q(y) + \eta\| + \varepsilon)$.

Proof. The restriction of q to Y is an injective linear map between finite-dimensional spaces, so it has a bounded inverse on its image, because all norms on a finite-dimensional real or complex space are equivalent; over $\mathbb{Q}$ the constant for the real extension serves. The hypothesis is needed: with $Y = \mathbb{Q}^2$, q the identity, the L^1 norm on Y, and the report norm $|r_1 + \sqrt{2}\, r_2|$, no constant exists. The noisy bound follows from $\|q(y)\| \leq \|q(y) + \eta\| + \|\eta\|$. □

For rational matrices and rational polyhedral norms, §10.6 and its full certificate construction replace this existential bound by the **least** raw or quotient recovery constant. Complete vertex coverage supplies a global upper bound and a saturating witness supplies the matching lower bound. Measurement units, the report norm and the noise bound are part of the instance. Rescaling a measurement row while leaving its physical noise or units unstated is not an informational improvement: a change of units must transport the norm and the error bound as well. The integer coupled fixture, for example, has sharp raw constant 2 for stacked ℓ^1 reports and 4 for the maximum individual score.

For Euclidean norms on a nonzero real space Y, assuming q is injective on Y, let V be a basis

matrix with Gram matrix $G = V^T V$. Then the best constant is $c = 1/\sqrt{t}$ for the largest t with $V^T q^T q V - tG \succeq 0$. An exact LDL^T factorization certifies a rational lower bound on t. The exact Rayleigh quotient of a rational vector bounds t only from above, so it bounds the best constant from below and cannot certify a recovery constant. For $Y = \{0\}$ the optimal nonnegative constant is 0 and there is no largest finite t. Over complex coordinates, replace every transpose here by the conjugate transpose. In the coupled example $Y = \{(s_0, s_1, s_0, s_1)\} \subseteq \mathbb{Q}^4$ with $q(y) = (y_0, y_3)$, each of the two coordinate reads has a nontrivial kernel on its own edge, yet $\|q(y)\|_2^2 = \frac{1}{2}\|y\|_2^2$ on Y, so $c = \sqrt{2}$. If every probe contributes one report coordinate at a declared positive cost, a suite that is injective on Y costs at least the sum of the $\dim Y$ cheapest probes.

`WDP-CH14-N02` **Nonclaim.** Injectivity of a presentation map on a declared subspace does not detect inputs outside that subspace, does not certify detection of unmodeled or adversarial behavior, and does not select, price, or authorize a sensor suite.

14.5 The awareness firewall

The monograph reserves **awareness** for a bridge not supplied by the present finite architecture. Functional self-report, global availability, integration, or higher-order representation may be compared with theories of consciousness [REF-OBS-004; REF-OBS-005; REF-OBS-006; REF-OBS-007; REF-OBS-008], but no equivalence is asserted.

`WDP-CH14-N01` **Nonclaim.** Self-witnessing, self-modeling, evaluator closure, or report integration does not establish phenomenal consciousness.

`WDP-CH14-O01` **Finite proposition (declared functional bridges).** For independently declared finite presentation, report, context and calibration maps, factorization and stability have the complete implication classification of Appendix J. Explicit accessible self-state, reflexive provenance, later-state and partial comparison maps separately classify linkage, stability, comparison availability and comparison success; later-target and successful linked-report existence are exact fiber-constancy questions. No phenomenology premise occurs.

Interpretive remark (reports, not experience). The firewall concerns which part of an experience report can enter a shared evaluation, not whether the experience exists. Two kinds of report differ here. An *adjectival* report attaches a quality to an object ("the apple appears red"). It has external anchors: the object, other reporters, instruments and a shared colour space. An *adverbial* report describes a manner of undergoing ("I am remembering vividly"). Its anchors are temporal profile, salience and later reports by the same subject, so it is harder to place in a declared packet. Neither report is nothing, and neither governs a shared verdict by being felt. Each enters a shared evaluation only through a declared report, probe and comparison structure. No claim about consciousness follows.

14.6 Minimum-cost sufficient observation

The unrestricted rational row counts in §10.6 answer a dimension question. A supplied instrument catalogue with prices asks a different finite optimization question. Let `X` and `I` be finite, with existing report `p:X->P`, protected question `q:X->A`, measurements `m_i:X->M_i` and positive rational prices `c_i`. Define the unresolved unordered pairs

$$U = \{\{x, y\} : p(x) = p(y),\ q(x) \neq q(y)\}, \qquad E_u = \{i : m_i(x) \neq m_i(y)\} \quad (u = \{x, y\}).$$

Finite priced-observation theorem (GEN-WDP-47). A fixed suite `S` is sufficient exactly when it meets every `E_u`. Equivalently its distinction sets cover `U`. Its minimum cost is the exact weighted hitting-set optimum, or infinity if some `E_u` is empty. A feasible suite gives an upper bound; optimality additionally requires a replayed lower bound covering every cheaper alternative. An LP bound alone may be strictly smaller: for the seven-point Fano incidence instance, the fractional optimum is `7/3`, while the integer optimum is `3`. The full finite proof and certificates give the explicit table, primal/dual fractional witnesses and integer counting argument.

For adaptive observation, let `C` be the states remaining after the free report and previous measurements. Put `V(C)=0` if `q` is constant on `C`, including the empty set. Otherwise, with a minimum over splitting measurements only,

$$V(C) = \min_{i:\,|m_i(C)|>1} \left(c_i + \max_{a \in m_i(C)} V(C \cap m_i^{-1}(a)) \right).$$

An empty minimum is infinity. Each child has smaller cardinality, so exact dynamic programming terminates. Induction establishes the Bellman lower bound for every strategy; choosing a minimizing measurement at each finite-valued node realizes it. Since `p` is already observed, the initial worst-case price is `max_{z in p(X)} V(p^{-1}(z))`, with zero for empty `X`, not `V(X)` in general. It is at most the optimal fixed-suite price. Asking the full shelf establishes feasibility precisely when there is no differently answered pair indistinguishable under `p` and every measurement. Such a pair defeats every adaptive strategy because both states follow the same branch at every step.

For a concrete strict gap, use four states with distinct answers and constant `p`. Let `A=(0,0,1,1)`, `B=(0,1,0,0)`, `C=(0,0,0,1)` with prices `1,3,3`. The pairs `(1,3)`, `(1,2)` and `(3,4)` force `A`, `B` and `C`, respectively, so every sufficient fixed suite costs `7`. Asking `A` first and then `B` or `C` costs `4`; the path through state `1` must ask both `A` and `B`, proving that lower bound. A refinement of the free report can only decrease either optimum; coarsening can only increase it.

The supplied derivation is preserved with its correction and source record. In particular its original eight-state adaptive table was infeasible, its report-monotonicity implication was reversed, and a cover witness was incorrectly listed as a lower bound. The corrected finite checker binds the original instance and replays submitted optimality data. This is exact bounded computation, not a polynomial-time algorithm, an empirical speedup, instrument physics or permission to buy or use the selected observations. No canonical `CH14` identifier is reused or silently strengthened.

14.7 Realizable observation and the cost of preserving future operations

The measurements of §14.6 leave the measured state unchanged. A measurement that changes state needs an additional preservation test. Let $k : X \to K$ be a finite quotient preserving the protected answers, target truth, enabled future actions, their prices and their successor-block sets. Supply actual measurement transitions, post-state reports, admissibility conditions and positive rational costs.

Track the nonempty relation $z \subseteq X \times X$ of possible original and current states. For a measurement i with successors $T_i(y)$ and post-state report λ_i, the observed update is

$$z_{i,o} = \{(x, y') : \exists y \ ((x, y) \in z, \ y' \in T_i(y), \ \lambda_i(y') = o)\}.$$

Finite observation-realization theorem. A completed measurement procedure identifies the original future class and preserves that class at handoff exactly when every terminal relation satisfies $\exists C \ \forall (x, y) \in z : \ k(x) = C = k(y)$. The least worst-case cost is the finite reachability/minimax optimum of this relation arena. An initial-containing loss trap certifies nonrealizability; a terminating policy and all-action lower inequalities certify an attained optimum. Initial free-report fibers are treated separately. The full argument, constructive decoder and certificate conditions are in the derivation.

This is more than a refinement of a present-answer partition. In the specified eight-state example, the least fixed nondisturbing measurement family costs 6, adaptive nondisturbing observation costs 4, and temporary disturbance followed by restoration costs 15/4. Only identifying the original class costs 7/2 but leaves the wrong future operation enabled. Removing restoration makes the selected disturbing measurement family insufficient. If preservation is required at every prefix, the disturbing action is excluded and the optimum is 4.

The finite replay contains five instances and twelve hostile controls, checked by a separate construction verifier and the existing finite-game certificate checker. Five core-only Lean declarations check relation composition, retention of the original coordinate, decoder uniqueness and the current-state collision; they use no axioms. These are distinct from the complete written cost proof and from physical realization of a measurement apparatus.

14.8 Reasoning under a declared information boundary

`WDP-CH14-T01` **Theorem (target-separating report obstruction).** Let $E : W \to R$ encode all information available at the decision, and let $q : W \to Q$ be the protected answer. If there are admitted worlds w, w' with $E(w) = E(w')$ and $q(w) \neq q(w')$, there is no function $f : R \to Q$ such that $f(E(x)) = q(x)$ for every $x \in W$.

Proof. Such an f would give $q(w) = f(E(w)) = f(E(w')) = q(w')$, a contradiction. Noninjectivity of E alone is insufficient: a constant question is recoverable from a constant report.

`WDP-CH14-C01` **Corollary (post-processing preserves the collision).** For any deterministic computation c on the available report, $c(E(w)) = c(E(w'))$. Thus additional internal computation alone cannot settle this target on both worlds. For interaction, the premise must cover all admitted observation/action histories. A one-step blind spot can disappear after an intervention; swapping two coordinates exposes a previously hidden coordinate.

`WDP-CH14-N03` **Nonclaim.** The theorem concerns a specified information map and target, not every autonomous system or language model. Internal computation can prove consequences of supplied tables and assumptions. Physical actuators or external verifiers change the available information and require a new check of the theorem's premises; they do not supply an automatic exemption or guarantee. An invariant export cannot distinguish inputs that it identifies, whereas an arbitrary internal verdict need not be invariant. Zero passage charge alone does not establish the required collision.

This replaces the earlier three-axis formulation. The unsourced Lemma D and retired NB-9 do not support it. Formal self-reference limits require their own coding, expressiveness and soundness hypotheses; an institutional grant still requires a separately declared authority rule. The corrected

argument agrees with the epistemic boundary analysis. The complete inference calculus, source-task interpretation and formal evidence are given in §20.14 and its derivation supplement.

Chapter 15 — Authority is a separate layer

15.1 Typed authority

Authority concerns who or what is permitted to issue, accept, promote, or act on a statement within a jurisdiction. It is not an observer component.

WDP-CH15-D01 **Definition (authority packet).** An authority packet is

$$\mathfrak{A} = (R, J, \mathsf{Perm}, \mathsf{Proc}, \mathsf{Appeal}),$$

where R is a set of roles, J a jurisdiction, Perm a typed permission relation, Proc a procedure for exercising permissions, and Appeal any review or revocation mechanism.

The packet is external to the layered observer. A bridge from measurement or proof status into an authority procedure is a policy choice in a jurisdiction, not a theorem of obstruction calculus.

15.2 Four separations

The following implications are invalid without explicit bridge rules:

$$\begin{aligned} \text{witness exists} &\not\Rightarrow \text{observer accepts},\\ \text{observer accepts} &\not\Rightarrow \text{measurement valid},\\ \text{measurement valid} &\not\Rightarrow \text{authority standing},\\ \text{authority standing} &\not\Rightarrow \text{claim true}. \end{aligned}$$

An authorized role can make an incorrect statement; a correct measurement can be outside a role's jurisdiction; an evaluator can accept an internally valid record that lacks metrological fields; and a residual can exist without being registered.

15.3 Repository status

A green test run says that specified commands returned expected results in a specified environment. A formal proof says that a term inhabits a proposition under the proof assistant's logic and imports. A release tag identifies bytes. None of these confers institutional, scientific, or moral authority. They may be inputs to an authority process, but only through a declared rule.

WDP-CH15-N01 **Nonclaim.** This monograph assigns no authorization and changes no jurisdiction. It carries no operational authority or institutional standing.

15.4 Observer morphisms

To compare observers, define the morphisms rather than comparing tuple lengths. A morphism $F : \mathcal{O} \to \mathcal{O}'$ can contain maps

$$F_X : X_{\mathcal{O}} \to X_{\mathcal{O}'}, \quad F_P : P_{\mathcal{O}} \to P_{\mathcal{O}'}, \quad F_M : M_{\mathcal{O}} \to M_{\mathcal{O}'}$$

that commute with selected presentation, update, and report maps. Evaluator transport additionally requires a map on defects and repairs preserving the repair action and controlling the cost. A strong isometry condition may require exact equality of costs; a simulation condition may require only a one-sided bound.

There is no single correct morphism class for every purpose. If the question is whether one observer can reproduce another's reports, behavioral simulation is appropriate. If the question is whether obstruction magnitudes agree, gauge and cost preservation are needed. If the question is whether authority transfers, no observer morphism is enough: an authority-packet morphism must be defined separately.

WDP-CH10-D05 **Definition (report simulation).** Observer $\mathcal{O}'$ report- simulates $\mathcal{O}$ on input family D when there is an input adapter and a report decoder making the declared report diagram commute on D. This does not imply memory-state isomorphism or evaluator equivalence.

15.5 A finite descent example

Take three contexts $U_{12} = \{x_1, x_2\}$, $U_{23} = \{x_2, x_3\}$, and $U_{31} = \{x_3, x_1\}$ with Boolean values. Impose local parity constraints

$$x_1 \oplus x_2 = 0, \qquad x_2 \oplus x_3 = 0, \qquad x_3 \oplus x_1 = 1.$$

Each context has local assignments. Any two constraints can be satisfied together, but all three cannot: adding the left sides modulo two gives 0, while adding the right sides gives 1. The summed parity is a finite counterexample witness to global assignment.

This example has the *shape* of a contextual obstruction. To call it an empirical contextuality model, one must additionally say what the measurements and outcomes are, provide compatible distributions on contexts as required, and connect the logical support to those distributions. The arithmetic contradiction alone does not supply an experiment.

15.6 Observation receipt invariants

An observation receipt should survive benign representation changes. Suppose a raw record is reencoded by a bijection $c : R \to R'$. Registration and readback maps can be transported through c; the witnessed proposition should remain the same if its semantics is transported as well. A checksum over raw bytes, however, changes unless recomputed and provenance-linked. Thus semantic invariance and byte identity are different predicates.

For each observation, retain at least:

- interaction identifier and target role;
- registration source and raw record digest;
- witness kind and proposition identifier;
- context/profile identifier;
- memory/re-access pointer;
- evaluator version, inputs, output, and any certificate;
- transformations applied to the raw record;
- explicit exclusions.

This schema is deliberately more demanding than storing the final status. It allows later reviewers to distinguish a changed evaluator from changed data.

15.7 Measurement equations and uncertainty propagation

Let a reported scalar be $y = h(z, c)$, where z is an indication and c is a calibration parameter vector. A first-order uncertainty approximation around (z_0, c_0) is

$$\mathrm{Var}(y) \approx \nabla h^{\mathsf{T}} \Sigma \nabla h,$$

where Σ is the joint covariance of indication and calibration inputs. This familiar formula is conditional on a probabilistic model, differentiability, and the validity of local linearization. It is not part of the finite root.

For purely bounded errors, one can instead use interval or Lipschitz transport. If h is L-Lipschitz on the admitted domain and $\|x - x_0\| \le \epsilon$, then

$$|h(x) - h(x_0)| \le L\epsilon.$$

This deterministic statement often matches the finite calculus better. A measurement packet should declare which uncertainty semantics it uses. Mixing a worst-case bound with a standard deviation without an adapter is a type error.

15.8 Repeated instruments and state change

Repeatability has at least three meanings:

1. repeated readout returns the same code;
2. repeated measurement returns compatible value intervals;
3. the first measurement prepares a state that, under declared assumptions, ensures the second result.

The third belongs to an operational state-transition theory and, in quantum models, to instrument structure. The finite measurement packet by itself supports only the first two when suitable stability predicates are provided. Davies–Lewis and Ozawa supply primary quantum-instrument comparisons [REF-MET-006; REF-MET-007]; their results should not be back-projected into the general definition.

15.9 Self-model calibration

A self-report can be calibrated against externally accessible behavior without being exhaustive. Let an observer emit a finite report $r \in R$ and let a test procedure observe a behavior $b \in B$. A calibration relation $K \subseteq R \times B$ records which pairs count as agreement. Repeated tests can reveal systematic mismatches. The result concerns K and the sampled behavior family, not the total internal state.

This gives a precise repaired hierarchy:

- **self-presentation:** a report is generated from an encoded self-state;
- **self-retention:** the report is available later;

- **self-comparison:** a later procedure compares it with a typed target;
- **self-calibration:** the comparison satisfies a declared relation on a test family;
- **cross-context robustness:** the relation is preserved under named context transports.

No implication should be asserted in reverse, and forward implications require the adapter laws. A stable report can be consistently wrong; a calibrated report can cover only a narrow test family; a broad functional self-model still does not settle phenomenology.

15.10 Authority bridge example

Suppose a laboratory procedure says: "A calibration-valid measurement below threshold θ, signed by role `Operator`, permits action A." This rule is a bridge

$$\mathsf{MeasurementReceipt} \times \mathsf{RoleCredential} \longrightarrow \mathsf{PermissionDecision}.$$

The measurement supplies a value and uncertainty; the credential supplies standing; the procedure supplies the jurisdictional inference. If any input is missing, the bridge refuses. If a later calibration changes the value interval, the procedure determines whether prior decisions are reviewed. The obstruction calculus can audit the inputs and rule. It does not create the rule's legitimacy.

15.11 Part III interface theorem schema

The most useful theorem schema for this part is compositional and conditional. If:

1. an interaction-to-record map satisfies a declared fidelity relation;
2. the record-to-proposition semantics is valid on a stated domain;
3. memory preserves the required fields;
4. the evaluator is sound for its status predicates;
5. calibration and uncertainty transport are valid;
6. provenance links all transformations and any metrological-traceability claim satisfies the documented calibration-chain and uncertainty requirements;

then the final measurement result is valid **relative to that package**. Each premise has its own falsifier. The conclusion does not include authorization or truth outside the measurand/profile domain.

`WDP-CH13-002` **Finite proposition (resolved finite target).** The declared finite registration/witness/measurement instance replays the retained indication, composed affine map, correction and uncertainty enclosure against separate re-access data; altered records, understated uncertainty and rehashed false enclosures are rejected. The proof, native ownership and scope are given in Appendix I.

15.12 Causal representations are proposals

Structural causal models [REF-CAUSE-001; REF-CAUSE-002] express interventional and counterfactual queries. This section places such queries in the separations above. It adds no theorem about causal inference itself.

Definition (causal proposal). A causal proposal is a represented interventional or counterfactual claim in a declared causal representation M, for example "under $\mathrm{do}(A = a)$ the outcome Y changes by Δ" or "had A not occurred, Y would have differed".

Definition (causal overclaim). A causal overclaim passes from "M can represent the intervention or counterfactual" to a standing such as "the system is trustworthy" or "acting on the answer is authorized" without the evaluation and authority packets that standing requires.

Finite proposition (a model and query do not fix a verdict). Let a domain of packets $(M, q, \mathcal{O})$ carry a protected status $s(M, q, \mathcal{O})$. If two packets share (M, q) and differ in s, then no function of (M, q) alone equals s on that domain.

Proof. This is the fiber criterion of P.1 and L.2 with report (M, q). A function of the report takes one value on the shared fiber, and s takes two. □

The hypothesis is met by a two-variable instance. Let M be the binary model $Y := X$ with $X \in \{0, 1\}$, and let q state that $\mathbb{E}[Y \mid \mathrm{do}(X = 1)] - \mathbb{E}[Y \mid \mathrm{do}(X = 0)] = 1$, which is true in M. An evaluator $\mathcal{O}_{\text{in}}$ declares the target population to be the one M represents and supplies the support repair, so the declared support defect is exact and $\Phi = 0$. An evaluator $\mathcal{O}_{\text{out}}$ declares a target outside the represented support and admits no transport repair; its one-coordinate defect has $\Phi = 1$. With the same threshold rule (accept when $\Phi \leq \frac{1}{2}$), the first returns accept and the second refuse, although M and q are unchanged. The defect values here are declared by the two evaluators, not computed from data; the instance shows only that the hypothesis is satisfiable. Even the accept status is an evaluation, not an authorization. By §15.2 and P.19, a regime must still supply the grant.

An earlier informal statement of this point said that there is no observer-independent map from queries to authority. As stated it is false: a constant assignment is such a map. The fiber statement above is the corrected form, and it concerns agreement with a declared status on a declared domain. Likewise, descent through an evaluator yields evaluable evidence. It does not yield authority.

An interventional proposal enters an evaluation only after the following are declared, each with its own falsifier: the unit or population carrier; the admissible variables; the operational meaning of the intervention; the comparison made; the identification assumptions; the data support in the intervention region; any remaining confounding, selection or transport defect; the residual after admissible repair; and the terminal statuses. A counterfactual proposal additionally needs the factual state, the structural equations or update rule, the evidence (abduction) regime, the prediction semantics, and a ledger of where the answer extrapolates beyond support. Without these, counterfactual language is a richer proposal syntax. Because the evaluator and the regime remain separate, this section asserts no causal claim about any system.

15.13 Separation V: Authority Isolation from Prose Assertions (`AuthorityEffect: None` and Executable Witness Dominance)

`WDP-CH15-D02` **Definition (Executable Machine Receipt).**
An executable machine receipt is a typed record Receipt $= (c, \Delta, \varphi^*, \text{status})$ emitted by a sovereign test harness or compiled kernel, where c is the command, Δ the diagnostic vector, φ^* the rational dual witness, and status $\in \{\mathsf{PASS}, \mathsf{FAIL}, \mathsf{REFUSE}\}$.

`WDP-CH15-P04` **Proposition (Prose Authority Impotence).**
Markdown declarations, natural language assertions, and cited rules carry AuthorityEffect = None. An autonomous agent citing prose rules does not constitute verification. Compliance is established solely by the generation of a passing machine receipt.

`WDP-CH15-N02` **Nonclaim.**

Proposition WDP-CH15-P04 defines repository verification governance; it does not model legal or social authority structures.

Part IV — Histories and the quantum boundary

Chapter 16 — Spatial/history diagnostics and the curved-complex fork

16.1 Two directions

Suppose a model has a spatial or contextual degree p and a historical degree q. Let

$$C^{p,q}$$

be a finite bigraded family with candidate maps

$$d_s : C^{p,q} \to C^{p+1,q}, \qquad d_h : C^{p,q} \to C^{p,q+1}.$$

A genuine anticommuting double complex requires

$$d_s^2 = 0, \qquad d_h^2 = 0, \qquad d_s d_h + d_h d_s = 0.$$

Earlier drafts treated these identities as likely constructions while also recording defects that can make them false. Here the identities are gates, not labels.

WDP-CH16-D01 **Definition (curvature defects).** Define

$$K_s = d_s^2, \qquad K_h = d_h^2, \qquad \Lambda = d_s d_h + d_h d_s.$$

The vanishing locus (or flat locus) Z is the subfamily on which all three defects vanish. With this anticommuting convention the unsigned total operator $D = d_s + d_h$ satisfies $D^2 = K_s + \Lambda + K_h$, so $D^2 x = 0$ for each $x \in Z$. However, the vanishing locus Z is **not** automatically a subcomplex: forming a genuine chain complex $(Z, D|_Z)$ requires the additional hypothesis that Z is preserved by the differentials ($d_s(Z) \subseteq Z$ and $d_h(Z) \subseteq Z$).

A finite countermodel demonstrates that differential preservation does not follow from pointwise vanishing: let $C^{0,0} = \mathrm{span}(x)$, $C^{1,0} = \mathrm{span}(y)$, $C^{0,1} = \mathrm{span}(u)$, $C^{1,1} = \mathrm{span}(z)$, and $C^{1,2} = \mathrm{span}(w)$, with all other components zero. Set $d_s(x) = y$, $d_s(u) = -z$, and $d_s = 0$ elsewhere; set $d_h(x) = u$, $d_h(y) = z$, $d_h(z) = w$, and $d_h = 0$ elsewhere. On the element $x \in C^{0,0}$, all three defects vanish: $K_s(x) = d_s(y) = 0$, $K_h(x) = d_h(u) = 0$, and $\Lambda(x) = d_s(u) + d_h(y) = -z + z = 0$, so $x \in Z$. Yet for its differential image $d_s(x) = y$, the historical curvature is $K_h(d_s x) = d_h^2(y) = d_h(z) = w \neq 0$. Thus $d_s(x) \notin Z$, so $d_s(Z) \not\subseteq Z$. Without higher Bianchi coherence ensuring differential stability, the zero-defect locus is just a subset, not a chain subcomplex. The alternative signed convention $D = d_s + (-1)^p d_h$ instead pairs with commuting bidegree maps; the two conventions must not be mixed.

16.2 The three outcomes

There are three honest outcomes.

1. **Flat complex.** Prove all square-zero and mixed identities on a declared domain. Cohomology and spectral-sequence language then becomes available [REF-HOM-001].
2. **Curved system.** Retain the nonzero defects as typed curvature and use a curved differential formalism whose axioms are separately checked [REF-HOM-002].
3. **Higher coherent repair.** Supply explicit homotopies or higher operations satisfying the relevant coherence identities [REF-HOM-003; REF-HOM-004].

Just renaming a defect "higher structure" is not Outcome 3. Higher cells are new data and their identities are new obligations.

16.3 Projection-induced memory

Let a full state evolve by $T : X \to X$, while an observer sees a surjective presentation $q : X \to P$ (or replace P by $\operatorname{im} q$). If a section $i : P \to X$ is chosen, a reduced update can be written $S = qTi$. Descent of the full dynamics through q instead requires

$$q(x) = q(y) \Longrightarrow q(Tx) = q(Ty).$$

Without descent, the next presentation can depend on information discarded by q. To retain a single-valued deterministic update on the unchanged presentation carrier, one must add enough state or history to distinguish the relevant fibers. Alternatives include refining P or adopting relation-valued, set-valued, or stochastic reduced dynamics.

`WDP-CH16-P01` **Finite proposition (descent criterion).** For a surjective $q : X \to P$, a function $\bar{T} : P \to P$ with $\bar{T}q = qT$ exists exactly when $q(x) = q(y)$ implies $q(Tx) = q(Ty)$.

Proof. Necessity follows by applying $\bar{T}$ to equal presentations. For sufficiency, define $\bar{T}(p) = q(Tx)$ for any x with $q(x) = p$; the condition makes this well defined. □

Selecting $S = qTi$ without descent gives one reduced dynamics, but it depends on the section and does not establish quotient dynamics.

The same criterion governs any target, not only dynamics. For a surjective presentation $q : X \to P$ and a map F on X, call the pairs with equal presentation and unequal target the obstruction set of F.

`WDP-CH16-P02` **Finite proposition (presentation descent and sections).** For a surjective presentation q from X onto P and a target F on X, the obstruction set is empty exactly when F factors through q; the factor is then unique; and $F = F \circ i \circ q$ for one section i of q exactly when the obstruction set is empty, in which case the equality holds for every section.

Proof. Factorization is the fiber-constancy criterion behind CH16-P01, applied to F in place of $q \circ T$: F factors through the surjection q exactly when F is constant on the fibers of q, which is emptiness of the obstruction set. Uniqueness uses surjectivity: two factors agree on $q(x)$ for every x. If $F = F \circ i \circ q$, then $q(x) = q(y)$ gives $F(x) = F(i(q(x))) = F(i(q(y))) = F(y)$. Conversely, an empty obstruction set gives $F(i(q(x))) = F(x)$ for every section, since $q(i(q(x))) = q(x)$. □

Without surjectivity, uniqueness of the factor off the image of q is not guaranteed, and no section exists at all. For a singleton answer target, there is still exactly one factor on the entire report target; with two answer values, an unconstrained report point can instead admit distinct extensions.

16.4 Worked example: forgotten bit

Let $X = \{0,1\}^2$, $q(x_1, x_2) = x_1$, and $T(x_1, x_2) = (x_2, x_1)$. States $(0,0)$ and $(0,1)$ have the same presentation, but their next presentations are 0 and 1. Hence T does not descend through q. Remembering the discarded second bit repairs prediction. The example gives a precise sense in which projection can induce memory; it is not by itself a Mori–Zwanzig kernel theorem [REF-HOM-005].

`WDP-CH16-N01` **Nonclaim.** No cohomology, spectral sequence, or homotopy- coherent descent is available until its defining identities and data are supplied.

Even when a filtration spectral sequence is available, it is built from the filtered complex alone and does not see Φ_1 (compare CH04-N02).

16.5 Chain contractions and perturbation

Presentation changes between cochain complexes are often contractions: a smaller complex that retains the cohomology, with explicit comparison maps. This section fixes the convention and records what those maps do and do not preserve.

`WDP-CH16-D02` **Definition (chain contraction and retraction defect).** A contraction of a finite cochain complex C onto D over a declared field consists of chain maps $i : D \to C$ and $r : C \to D$ with $ri = 1_D$, together with a map h of degree -1 satisfying $1_C - ir = dh + hd$. It is normalized when $hi = 0$, $rh = 0$, and $hh = 0$. Its retraction defect is $\varepsilon = 1 - ir$.

`WDP-CH16-P03` **Finite proposition (retraction defect).** For chain maps i and r with $ri = 1$ on a finite complex over a field, ir is idempotent, $\varepsilon = 1 - ir$ commutes with d and so is closed in the degree-zero window of the endomorphism complex, a homotopy h with $\varepsilon = dh + hd$ exists exactly when the class of ε vanishes, which holds exactly when $\ker r$ is acyclic, and any such h can be modified to satisfy the normalization conditions; the retraction class is therefore `ExactClass` or `ClosedResidual` and never `ActiveClosureFailure`.

Proof. $(ir)(ir) = i(ri)r = ir$. Since i and r are chain maps, $d\varepsilon = d - idr = \varepsilon d$. In the endomorphism complex, the differential of a degree-zero map f is $df - fd$, so ε is a cocycle, and it is a coboundary exactly when some h gives $\varepsilon = dh + hd$. Over a field $C = \operatorname{im} i \oplus \ker r$ as complexes, and ε is the projection onto $\ker r$; it is nullhomotopic exactly when $\ker r$ is acyclic. Given any h, set $\pi = 1 - ir$, $h' = \pi h \pi$, and $h'' = h' d h'$; these satisfy the homotopy equation together with the three normalization conditions. □

`WDP-CH16-C01` **Conditional result (perturbation lemma, imported standard).** With the convention $1 - ir = dh + hd$, if (i, r, h) is a normalized contraction, δ has degree one with $(d + \delta)^2 = 0$, and $1 + \delta h$ is invertible, then with $A = (1 + \delta h)^{-1}\delta$ the maps $d_D + rAi$, $i - hAi$, $r - rAh$, and $h - hAh$ form a normalized contraction of $(C, d + \delta)$ onto D [REF-HOM-007].

The signs depend on the convention. The source that proposed this layer used the formulas with $(1 - \delta h)^{-1}$ that belong to the opposite convention $ir - 1 = dh + hd$; with its own convention they fail to give a differential on D. The executable package checks every perturbed contraction it produces rather than trusting the formula. The same source listed side conditions for a transfer step that

omitted the homotopy equation itself; with $C = \mathbb{Q}^2$, $D = \mathbb{Q}$, $d = 0$, and $h = 0$ every listed condition holds while the cohomology dimensions differ.

16.6 Truncation leakage

Restricting an operator to a subspace is a presentation change of a different kind. Let A act on a finite-dimensional inner-product space and let P be an orthogonal projector.

WDP-CH16-P04 **Finite proposition (truncation leakage).** For a linear operator A on a finite-dimensional inner-product space over $\mathbb{Q}$, $\mathbb{Q}(i)$, $\mathbb{R}$, or $\mathbb{C}$ and an orthogonal projector P, $PA^*AP = (PAP)^*(PAP) + L_P$ with $L_P = ((1-P)AP)^*((1-P)AP)$ positive semidefinite; $L_P = 0$ exactly when the range of P is A-invariant; and a vector ψ in the range of P with $PAP\psi = 0$ satisfies $A\psi = 0$ exactly when $\langle \psi, L_P \psi \rangle = 0$, so $L_P = 0$ is sufficient but not necessary for a reduced zero to be genuine.

Proof. Insert $1 = P + (1-P)$ between A^* and A and use $P^2 = P = P^*$ and $(1-P)^2 = 1-P$. The range of P is invariant exactly when $(1-P)AP = 0$, which is exactly when $L_P = 0$. For $\psi = P\psi$ with $PAP\psi = 0$, $A\psi = (1-P)AP\psi$, whose squared norm is $\langle \psi, L_P \psi \rangle$. □

On $\mathbb{Q}(i)^3$ with $P = \mathrm{diag}(1,1,0)$ and $A = e_3 e_2^T$, $L_P = \mathrm{diag}(0,1,0) \neq 0$, yet e_1 is a genuine zero; e_2 is a reduced zero that is not genuine. Leakage-free is also not the same as descent through P: for nonsymmetric A, $(1-P)AP = 0$ does not force $PA(1-P) = 0$.

WDP-CH16-N02 **Nonclaim.** The leakage term does not classify vectors: a vector in the range of P can satisfy $A\psi = 0$ while L_P has arbitrarily large norm, and no Hamiltonian, energy, zero-mode, probability-leak, or superspace interpretation is derived from the identity.

16.7 Composition curvature of a restriction system

Restriction data often come without a composition law. Let $(P, \leq)$ be a finite poset. For each $x \in P$ declare a vector space $F(x)$ over a declared field, and for each strict comparable pair $a < c$ a linear map $r_{c,a} : F(c) \to F(a)$. Nothing is assumed about how these maps compose. Index cochains by strict chains, valued in the fiber at the bottom:

$$C^0 = \prod_x F(x), \quad C^1_< = \prod_{a<c} F(a), \quad C^2_< = \prod_{a<b<c} F(a), \quad C^3_< = \prod_{a<b<c<d} F(a),$$

with

$$(d_0 s)_{c,a} = r_{c,a} s(c) - s(a), \qquad (d_1 g)_{c,b,a} = r_{b,a} g_{c,b} - g_{c,a} + g_{b,a},$$

$$(d_2 h)_{d,c,b,a} = r_{b,a} h_{d,c,b} - h_{d,c,a} + h_{d,b,a} - h_{c,b,a}.$$

WDP-CH16-P05 **Finite proposition (composition curvature).** Put $K_{c,b,a} = r_{b,a} r_{c,b} - r_{c,a}$ for every strict triple $a < b < c$. For every $s \in C^0$, every $g \in C^1_<$, and every strict chain $a < b < c < d$,

$$(d_1 d_0 s)_{c,b,a} = K_{c,b,a}\, s(c), \qquad (d_2 d_1 g)_{d,c,b,a} = K_{c,b,a}\, g_{d,c}.$$

Hence $d_1 d_0 = 0$ exactly when $K = 0$, which holds exactly when $r_{b,a} r_{c,b} = r_{c,a}$ on every strict triple; and then $d_2 d_1 = 0$ as well.

Proof. Using linearity of $r_{b,a}$, $(d_1 d_0 s)_{c,b,a} = r_{b,a}(r_{c,b} s(c) - s(b)) - (r_{c,a} s(c) - s(a)) + (r_{b,a} s(b) - s(a)) = (r_{b,a} r_{c,b} - r_{c,a}) s(c)$. Since C^0 is the full product, $s(c)$ ranges over all of $F(c)$, so $d_1 d_0 = 0$ forces every $K_{c,b,a} = 0$. For the second identity, expand d_2 on the four components of $d_1 g$: the terms in $g_{d,b}$, $g_{d,a}$, $g_{c,b}$, $g_{c,a}$, and $g_{b,a}$ cancel in pairs and leave $(r_{b,a} r_{c,b} - r_{c,a}) g_{d,c}$. □

On a chain $a < b < c$ with one-dimensional fibers, $r_{c,b} = 2$, $r_{b,a} = 3$, and $r_{c,a} = 5$, the curvature is $K = 3 \cdot 2 - 5 = 1$. The section $(s(a), s(b), s(c)) = (0, 0, 1)$ has $(d_0 s)_{c,b} = 2$, $(d_0 s)_{b,a} = 0$, and $(d_0 s)_{c,a} = 5$, so $(d_1 d_0 s)_{c,b,a} = 3 \cdot 2 - 5 + 0 = 1 = K\, s(c)$. Declaring $r_{c,a} = 6$ instead makes the triple flat. A diagnostic that reads only the cover pairs $a \prec b \prec c$ never sees $r_{c,a}$ and cannot detect K.

This is Outcome 2 of §16.2 in its smallest form: K is typed curvature, and $d_1 d_0 = K$ is exact bookkeeping, not a cohomology class. On the flat locus, with identities adjoined by definition ($r_{a,a} = 1$), the declared maps satisfy the identity and composition laws, so they form a functor on $(P, \leq)^{\mathrm{op}}$ [REF-MATH-004], and the differentials square to zero through degree three. The identities use only additivity of $r_{b,a}$, so abelian groups may replace vector spaces. No norm is used, and no H^2 class, no descent of d_1 to a quotient of $C^1_{\leq}$, and no higher-coherent repair is claimed; CH16-N01 still applies. The absorbed archived working-paper source 009 checked the first identity and the flat equivalence in Lean for a single triple of maps on one carrier type, and the case $g = d_0 s$ of the second identity for a single quadruple; that module is not ported and earns no Lean binding here.

Chapter 17 — Finite weighted histories

17.1 Path carrier

Let $G = (V, E)$ be a finite directed graph with a declared admissibility predicate on paths. For vertices a, b, let $\Omega^{(n)}_{a,b}$ be the finite set of admissible paths of length n from a to b. A history is a path, not an instantaneous state.

A length cutoff $n \leq N$ creates a finite set, but the union of length-bounded paths is not generally closed under concatenation: two paths of length at most N can concatenate to a path longer than N. Thus it is not a category unless the composition law is restricted or the graph is suitably acyclic.

17.2 Semiring weights

Let $(R, \oplus, \otimes, 0, 1)$ be a declared semiring and assign edge weights $w : E \to R$. The weight of a path $p = e_1 \cdots e_n$ is

$$w(p) = w(e_1) \otimes \cdots \otimes w(e_n),$$

with the empty path assigned 1. The finite length-n history aggregate is

$$Z^{(n)}_{a,b} = \bigoplus_{p \in \Omega^{(n)}_{a,b}} w(p).$$

WDP-CH17-P01 **Finite proposition (matrix path sum).** For the full family of directed graph paths of length n in the stated graph, with weighted adjacency matrix M and conventional ordered multiplication matching path composition,

$$(M^n)_{a,b} = Z^{(n)}_{a,b}.$$

Proof sketch. Induct on n. The step expands $(M^{n+1})_{a,b} = \bigoplus_c (M^n)_{a,c} \otimes M_{c,b}$ and distributes over the last edge of every length-$(n+1)$ path. □

For an admissibility predicate compiled into a finite augmented-state graph, let λ_a be the declared row of admissible augmented start states over original vertex a and ρ_b the declared column of accepting augmented end states over original vertex b, including any start/end weights. If augmented paths are in weight-preserving bijection with admissible original paths, then the correct aggregate is $\lambda_a M^n \rho_b$, generally a weighted block sum, not a single entry indexed by the original vertices. An arbitrary path-global predicate is not counted by a matrix power unless such finite-state compilation and endpoint maps are supplied.

The theorem is finite. An all-length sum $I \oplus M \oplus M^2 \oplus \cdots$ requires nilpotence or a finite acyclic truncation, a complete or $*$-continuous semiring, or an analytic convergence structure. A bare Conway/Kleene star operation can define an algebraic M^* without identifying it with that countable indexed sum, and a bare semiring does not evaluate the infinite series.

17.3 Families of meaning

The same path formula has different interpretations for different semirings.

- Boolean: reachability.
- Natural numbers: path count.
- Nonnegative reals: positive path weight.
- Tropical min-plus: optimal cost.
- Complex numbers: amplitude-like cancellation, after complex structure is declared.

Changing the semiring changes the theory. A proof over nonnegative reals cannot be advertised as a theorem about complex interference without a transport argument.

17.4 Cancellation and zerosumfree carriers

A semiring is zerosumfree when $a \oplus b = 0$ implies $a = b = 0$. In such a carrier, nonzero path contributions cannot cancel to zero just through addition.

WDP-CH17-P02 **Conditional result.** In a zerosumfree semiring, a finite sum is zero only if every summand is zero. The binary property extends by induction on the number of summands, using associativity.

Complex numbers are not zerosumfree: $1 + (-1) = 0$. That cancellation is a necessary ingredient for some amplitude phenomena, but it is not sufficient for quantum theory. Quadratic readout, tensor composition, contextual structure, and dynamics remain separate.

17.5 Obstruction action and history dependence

Suppose a local action assigns $A(e) \in \mathbb{R}$ and define

$$A(p) = \sum_{e \in p} A(e), \qquad w_\beta(p) = e^{-\beta A(p)}.$$

This is a declared positive-real weighting. If $A = d_0 f$ on edges, then $A(p) = f(b) - f(a)$ depends only on endpoints, so all a-to-b paths have equal weight. Nonzero periods can make the weighting genuinely history-dependent.

The converse requires care: equal weights at one chosen β may hide action differences if the exponential is noninjective in another coefficient system. Over real exponentials with $\beta \neq 0$, equality of weights implies equality of actions.

17.6 Worked semiring example

Take two paths from a to b: p with edge weights $2, 3$ and q with one edge weight 5.

- Over $(\mathbb{N}, +, \cdot)$, the total is $2 \cdot 3 + 5 = 11$.
- Over min-plus, interpreting the listed numbers as costs, the total is $\min(2 + 3, 5) = 5$.
- Over complex numbers, choosing path weights 1 and -1 gives total 0.

The graph is unchanged; the aggregate's meaning is not.

17.7 Contextual histories, free differences, and algebraic tensor composition

The path carrier above presupposes a graph and a weight semiring. The more primitive construction begins with an admitted typed generating signature $\Sigma = (O, E, s, t)$ and freely forms well-typed histories. Given an endpoint-indexed family of protected questions, define two parallel histories to be equivalent when every compatible prefix and suffix gives the same answer to every protected question. This is the greatest two-sided congruence with that preservation property. Equal current reports alone are insufficient: an allowed continuation can distinguish them. The full construction and the counterexample are proved in `generative_histories_contextual_compression_derivation.md`.

For a declared generator carrier X, finite alternative formation is the free commutative monoid $\mathbb{N}^{(X)}$. Its signed completion is the free abelian group $\mathbb{Z}^{(X)}$. The canonical map $\mathbb{N}^{(X)} \to \mathbb{Z}^{(X)}$ has the Grothendieck extension property: every additive map from positive alternatives into an abelian group extends uniquely to signed differences. Applying this construction to every contextual-history hom-set and extending composition bilinearly gives the free preadditive envelope of the contextual history category.

`GEN-WDP-58` **Finite algebraic proposition (free signed history completion).** For every admitted generator type X, the alternative carrier and signed carrier satisfy the free commutative-monoid and free abelian-group initial factorisation properties. Every signed class is a difference of two positive finite alternatives, and every alternative-monoid map into an abelian group has a unique group-homomorphic extension.

For two free signed carriers, explicitly requiring independent composition to be additive in each argument yields

$$\mathbb{Z}^{(X)} \otimes_{\mathbb{Z}} \mathbb{Z}^{(Y)} \cong \mathbb{Z}^{(X \times Y)}.$$

`GEN-WDP-59` **Finite algebraic proposition (biadditive tensor factorization).** Every biadditive map $B : \mathbb{Z}^{(X)} \times \mathbb{Z}^{(Y)} \to G$ into an abelian group factors through a unique group homomorphism

$\overline{B} : \mathbb{Z}^{(X\times Y)} \to G$, with $\overline{B}(a \otimes b) = B(a, b)$. The proof defines the factor on pair generators and applies the free-group uniqueness theorem successively in the right and left inputs.

These results derive the algebraic tensor object only after biadditivity has been declared. They do not derive a Hilbert tensor product or a physical independence principle. A determinant-one 2×2 integer tensor is already non-pure, but algebraic non-factorability is not yet physical entanglement. Complex scalars, a dagger, positivity, norm completion, normalized states and Born readout remain separate interfaces. Full proofs, exact information-loss boundaries, core-only Lean, and replay evidence are in `additive_tensor_completion_derivation.md`.

17.8 Dagger extension and the observer Gram radical

A dagger is additional reversal data. On a typed generating signature it sends every generator $e : x \to y$ to an involutive reverse $e^\dagger : y \to x$ and reverses the order of a history. It descends through the protected contextual quotient if and only if its equivalence relation is invariant under reversal. Closure of the protected questions under reversal is a sufficient condition for this invariance. Once supplied on a generator carrier, free-group uniqueness extends it to an additive involution on signed differences.

`GEN-WDP-60` **Finite algebraic proposition (dagger extension).** A declared generator involution has a unique additive extension D to the free signed carrier, with $D^2 = I$. For componentwise involution on generator pairs,

$$(a \otimes b)^\dagger = a^\dagger \otimes b^\dagger.$$

The tensor equation follows because both sides are biadditive and agree on every pair of generator units. It does not assert that a native guarded edit has an executable reverse.

Now supply an additive observer $L : \mathbb{Z}^{(X)} \to \mathbb{Z}^m$. Its coordinate Gram form is

$$g_L(a, b) = L(a)^T L(b).$$

`GEN-WDP-61` **Finite algebraic proposition (observer Gram radical).** The form g_L is symmetric, biadditive and positive semidefinite, and

$$\mathrm{Rad}(g_L) = \ker L, \qquad L(u) = L(v) \iff g_L(u - v, u - v) = 0.$$

The proof is the exact sum-of-squares identity $g_L(a, a) = \sum_i L(a)_i^2$. Quotienting by the radical produces the realized image geometry and removes exactly the observer-blind signed differences. This is relative to the supplied observer. Distinct positive forms already exist in rank one, while rank one over the integers admits no endomorphism squaring to $-I$. Thus dagger plus positivity does not select complex structure, Hilbert completion or probability readout. The full derivation, Lean scope and replay are in `dagger_observer_pairing_derivation.md`.

17.9 Finite Hermitianization from a checked complex structure

Let the realized observer carrier be rationalized to a finite rational vector space V with Gram form g. Complex structure is a new admitted datum: a rational-linear map $J : V \to V$ satisfying

$$J^2 = -I, \qquad g(Jx, y) = -g(x, Jy).$$

The second law is equivalent to J preserving g. In coordinates the checker recomputes the observer Gram matrix $G = L^T L$ and verifies $J^2 = -I$, $J^T G = -GJ$, and $J^T G J = G$.

GEN-WDP-62 **Finite conditional theorem (Gaussian-rational carrier).** Define

$$(a + ib)x = ax + bJx.$$

Then V is a vector space over $\mathbb{Q}(i)$. Scalar associativity follows exactly from rational linearity and $J^2 = -I$; no analytic completion is used.

GEN-WDP-63 **Finite conditional theorem (sharp Hermitianization).** Define

$$h(x, y) = g(x, y) + ig(Jx, y).$$

With the first slot conjugate linear, h is a positive-semidefinite Hermitian form and

$$\mathrm{Rad}(h) = \mathrm{Rad}(g).$$

For an observer Gram form, $h(x, x) = 0$ exactly when $Lx = 0$. Thus the complex extension preserves the observer's distinction boundary. On the rational plane, $J(x_0, x_1) = (-x_1, x_0)$ gives the standard Gaussian-rational pairing. The equally admissible orientation $-J$ has the same positive g but conjugates the off-diagonal Hermitian values, proving that positivity does not select J. The full proof, refusal example, Lean formalization and source-bound replay are in finite_hermitianization_derivation.md.

17.10 Certified finite states, effects and readout

Hermitianization supplies a carrier and pairing, but no normalized state or effect cone. Assume a supplied standard Hermitian coordinate system over $\mathbb{Q}(i)$; this need not follow from Hermitianization over that same field. For the explicit rectangular Gram-factor class, let $B \neq 0$ present

$$\rho_B = \frac{B^\dagger B}{\|B\|_F^2},$$

and let a binary effect be presented by factors C, D satisfying $C^\dagger C + D^\dagger D = I$.

GEN-WDP-64 **Finite conditional theorem (certified state/effect readout).** The admitted trace readout has the exact factor form

$$\mathrm{tr}(\rho_B C^\dagger C) = \frac{\|CB^\dagger\|_F^2}{\|B\|_F^2}.$$

The accepted and rejected values are nonnegative rationals summing to one. The proof is finite trace cyclicity plus the submitted completeness equation; the checker reconstructs every Gram

matrix and refuses a zero state or incomplete effect. The certificate language covers matrices with supplied $\mathbb{Q}(i)$ Gram factors, not every positive matrix over a larger scalar field.

GEN-WDP-65 **Finite countermodel (readout nonselection).** For $0 \le \lambda \le 1$,

$$p_\lambda(\rho, E) = \lambda \operatorname{tr}(\rho E) + (1-\lambda)\frac{\operatorname{tr} E}{n}$$

is positive, normalized on every finite POVM, phase invariant and affine for $n \ge 1$. For $n \ge 2$, a sharp state/effect pair gives $p_\lambda = \lambda + (1-\lambda)/n$, which differs from the undeformed value 1 when $\lambda \ne 1$. At $n = 1$, every normalized state is $[1]$ and an effect $[e]$ gives $p_\lambda = e$ for every λ. Thus the theorem constructs an exact Born-form interface after that readout is admitted; positivity and normalization do not uniquely select it. Full proof, Lean scope, refusal controls and replay are in finite_state_effect_readout_derivation.md.

17.11 Certified finite processes and composition

For an ensemble of state factors B_s, let

$$T = \sum_s \|B_s\|_F^2 > 0, \qquad \rho = T^{-1}\sum_s B_s^\dagger B_s.$$

A finite process certificate is a labelled family (K_o) satisfying $\sum_o K_o^\dagger K_o = I$. Its branch factors are $B'_{s,o} = B_s K_o^\dagger$.

GEN-WDP-66 **Finite conditional theorem (certified process composition).** Every branch mass

$$T_o = \sum_s \|B_s K_o^\dagger\|_F^2$$

is nonnegative and $\sum_o T_o = T$. The unconditioned output is $\sum_o K_o \rho K_o^\dagger$. Applying (K_o) and then (L_r) has composite factors $L_r K_o$; complete families on admitted independent carriers tensor; and a supplied unitary factor preserves mass and is reversed by its adjoint. These are exact algebraic statements over $\mathbb{Q}(i)$.

GEN-WDP-67 **Finite countermodel (channel/instrument nonselection).** The one-outcome instrument $\{I\}$ and the two-outcome instrument $\{(3/5)I, (4/5)I\}$ induce the same identity channel. Their outcome reports are respectively (1) and $(9/25, 16/25)$. A presentation retaining only the unconditioned channel therefore loses future conditioning and branch questions.

GEN-WDP-68 **Exact scalar obstruction.** No rational r satisfies $r^2 = 1/2$. The standard Hadamard matrix and the usual T phase consequently do not lie in the current exact $\mathbb{Q}(i)$ matrix language. This does not rule out approximate synthesis or a different rational gate set; those require a topology, error metric, alphabet and compilation theorem. Full proofs, exact replay, refusal controls and Lean scope are in finite_process_composition_derivation.md. Chapter 19A distinguishes a conventional matrix from its ray action: the Gaussian Hadamard there realizes that ray action exactly, while the T eigenvalue ratio and the order-three Fourier requirements remain genuine field obstructions.

WDP-CH17-001 **Finite proposition (resolved finite target).** For a finite rational row-source transition matrix and a nonempty block partition, representative-independent endpoint-block path sums

at every finite length exist exactly when outgoing block sums are constant within each source block, equivalently M H = H B; B is unique and finite partition enumeration is exhaustive. The proof, native ownership and scope are given in Appendix I.

Partial Characterization and Mechanized Path Algebra.

1. **Semiring Lumpability Condition:** Let $G = (V, E, W)$ be a finite directed graph with transition matrix $M \in \mathrm{Mat}_{|V|\times|V|}(R)$ over a semiring $(R, \oplus, \otimes, 0, 1)$. Let $\mathcal{P} = \{V_1, \ldots, V_k\}$ be a partition of V with block projection matrix $P \in \mathrm{Mat}_{k\times|V|}(\{0,1\})$ (where $P_{I,u} = 1 \iff u \in V_I$) and transposed membership matrix $P^T \in \mathrm{Mat}_{|V|\times k}(\{0,1\})$. The representative-independent outgoing-block condition requires that for each source vertex $u \in V_I$ and target block V_J, the aggregate transition weight $\bigoplus_{v\in V_J} M_{u,v} = \bar{M}_{I,J}$ is independent of the choice of representative $u \in V_I$. In matrix form, evaluating the (u, J)-entry shows this corresponds to the intertwining relation $MP^T = P^T\bar{M}$. (The dual relation $PM = \bar{M}P$ corresponds to pushforward of distribution column vectors $\bar{x} = Px$ under the stated matrix convention, commuting via $P(Mx) = \bar{M}(Px)$.) Zerosumfreeness prevents nontrivial cancellation to zero ($a \oplus b = 0 \implies a = 0 \wedge b = 0$), precluding paths with nonzero weights from destructively canceling to the zero element, though it does not in general force equality between arbitrary sums to be termwise rigid without additional cancellative or idempotence assumptions. A comprehensive classification of coarse-grainings preserving arbitrary path aggregates across general semirings remains a broader research problem. Claim CH17-O01 selects the rational endpoint-block aggregate, whose complete finite criterion and partition classification are proved in Appendix I.
2. **Mechanization of Finite Path Distributivity and Coboundary Holonomy:** In `lean4/WitnessedDifference/SemiringTransport.lean`, we formally verify under `FWSemiring` that finite sums distribute on the right: `WitnessedDifference.lsum_mul_right` establishes $\mathrm{mul}(\mathrm{lsum}\, xs, y) = \mathrm{lsum}(xs.\,\mathrm{map}(\lambda x \implies \mathrm{mul}\, xy))$. Furthermore, `coboundary_trivial_holonomy` demonstrates that coboundary connection potentials telescope along walks and evaluate to 0 around any closed walk. Neither declaration involves a partition or a coarse-graining; the lumpability condition of item 1 is not mechanized.

`WDP-CH17-N01` **Nonclaim.** A finite semiring path sum is not a continuum path integral, a measure over all trajectories, or the Feynman amplitude rule [REF-AUT-007].

Chapter 18 — Quantum reconstruction as an audit

18.1 The benchmark question and candidate finite operational axiom package

The correct question is not whether the notation resembles quantum mechanics. It is whether a stated set of operational principles singles out complex finite-dimensional quantum theory from a comparison class that includes classical and post-quantum models [REF-QM-003; REF-QM-004; REF-QM-005; REF-QM-006].

To make this audit mathematically precise without smuggled assumptions, we formulate a candidate finite operational axiom package. Axioms W1–W3 abstract the finite witness relation, presented report carrier, and rational mass structures introduced in Chapters 1–13, while Axioms W4–W6 are additional specialization hypotheses introduced to probe the boundary with complex quantum mechanics. Earlier packets in Chapters 1–13 do not require reversible dynamics (W4) or quadratic readout (W6); in general, observer updates may be non-injective descents and readouts may be affine or report-based.

1. **Axiom W1 (Primitive Difference Carrier):** An admitted finite irreflexive witness relation $K = (X, R, \operatorname{Irr} R)$ with witness $d = (a, b, w : Rab)$.
2. **Axiom W2 (Presented Report Carrier):** A finite carrier presentation $C = (X, Q, \mathrm{pres} : X \to Q)$ where two points share a report $\mathrm{sameReport}(x, y)$ if and only if $\mathrm{pres}(x) = \mathrm{pres}(y)$.
3. **Axiom W3 (Integer/Rational Weights):** A mass assignment $w : Q \to \mathbb{Z}$ (or $\mathbb{Q}$) with strictly positive total mass $W = \sum_{q \in Q} w(q) > 0$.
4. **Axiom W4 (Reversible Discrete Dynamics, Specialization):** State updates are bijections $\sigma \in S_Q$ on the report carrier preserving total mass.
5. **Axiom W5 (Free Linearization & Admitted Sesquilinear Form, Specialization):** The free module $\mathrm{Vec}_K(Q)$ over $\mathbb{Z}[i]$ (or $\mathbb{Z}$) equipped with the standard positive-definite inner pairing $\langle u, v\rangle = \sum_{q \in Q} \bar{u}_q v_q$.
6. **Axiom W6 (Phase-Blind Quadratic Readout, Specialization):** The coordinatewise readout map $\pi : \mathrm{Vec}_K(Q) \to \mathbb{Z}_{\geq 0}^{Q}$ given by $\pi(v)_q = |v_q|^2$.

18.2 The classical-simplex fork and classical representation theorem

Finite nonnegative normalized witness weights form a simplex when all such weights and their convex mixtures are admitted. Normalization alone does not imply nonnegativity: W3 also permits the signed weights $(2, -1)$. A simplex has unique decompositions into extreme points under the standard finite-dimensional model. The full density-operator state space of complex quantum theory is nonsimplicial in Hilbert dimension at least two; the one-dimensional case is a singleton simplex, and commutative classical subtheories can also be simplicial.

Nonsimpliciality is not a quantum signature by itself.

`WDP-CH18-P01` **Finite proposition (a finite-vertex non-simplex polytope).** The convex polytope with four extreme points $v_1 = (1, 0)$, $v_2 = (0, 1)$, $v_3 = (-1, 0)$, and $v_4 = (0, -1)$ (whose convex hull is a continuum of states generated by finitely many vertices) is not a simplex: its centre equals both $\frac{1}{2}v_1 + \frac{1}{2}v_3$ and $\frac{1}{2}v_2 + \frac{1}{2}v_4$. A nonsimplicial finite-dimensional state polytope therefore exists outside quantum theory, and nonsimpliciality alone is not a quantum discriminator.

Proof. Both mixtures equal $(0, 0)$, and the two decompositions use disjoint sets of extreme points. The square, with affine effects, is the state space of the gbit of generalized probabilistic theories [REF-QM-005], which is not a quantum state space. □

The centre has a one-parameter family of decompositions, $t(v_1 + v_3) + (\frac{1}{2} - t)(v_2 + v_4)$, while points on an edge still decompose uniquely.

Under Axioms W1–W6, the reachable representation theorem establishes that finite reversible discrete dynamics admits a faithful representation within an integer permutation-isometry envelope:

Theorem (Representation in Permutation-Isometry Envelope). Let (X, Q, pres) be a finite presented carrier satisfying Axioms W1–W4 with integer weights $w : Q \to \mathbb{Z}$ and total mass $W = \sum_{q \in Q} w(q) > 0$. Every reversible report update $\sigma \in S_Q$ admits a faithful representation as an isometric automorphism $P_\sigma \in \mathrm{GL}_{|Q|}(\mathbb{Z})$ preserving the standard Euclidean form $\langle u, v\rangle = u^T v$ with two-sided inverse $P_{\sigma^{-1}} = P_\sigma^T$, and identically preserving total mass $\sum_{q \in Q} (P_\sigma w)_q = W$.

Proof. Let $|Q| = n$ with standard basis vectors e_q for $q \in Q$. For any $\sigma \in S_Q$, define the linear operator P_σ by $P_\sigma e_q = e_{\sigma(q)}$. Since σ is a bijection, P_σ is a permutation matrix with entries in $\{0, 1\} \subset \mathbb{Z}$. For any vectors $u, v \in \mathbb{Z}^n$, $\langle P_\sigma u, P_\sigma v\rangle = \sum_q u_{\sigma(q)} v_{\sigma(q)} = \sum_{q'} u_{q'} v_{q'} = \langle u, v\rangle$, so P_σ is an exact isometry with $P_\sigma^T P_\sigma = I_n$. Furthermore, $\sum_{q \in Q} (P_\sigma w)_q = \sum_{q \in Q} w_{\sigma^{-1}(q)} = \sum_{q' \in Q} w_{q'} = W$.

Distinct permutations act differently on some basis vector, so the action of S_Q on the report carrier Q embeds faithfully into the integer permutation-isometry envelope, preserving total mass. This does not make the presentation $\mathrm{pres} : X \to Q$ injective. □

Remark on Stochastic vs. Signed Valuations. Axiom W3 permits signed integer weights with positive total ($W > 0$) without requiring pointwise nonnegativity. For example, the weight vector $w = (2, -1)$ satisfies $\sum_q w(q) = 1 > 0$ but contains a negative weight, so it is a signed discrete valuation rather than a probability state on a simplex. Consequently, the theorem establishes a permutation-isometry representation over $\mathbb{Z}$, not a classical stochastic or probabilistic realization; a stochastic realization requires the additional standing hypothesis of pointwise nonnegativity $w(q) \geq 0$.

The classical corner. The theorem above assumes Axiom W4, that reversible updates are bijections of Q. The next four propositions derive that assumption from different premises, show which premise does the work, and locate where the package stops being classical. They are mechanized in `lean4/WitnessedDifference/ClassicalCorner.lean`, a Mathlib-backed module. Other modules also import Mathlib; a standalone core-only replay therefore does not establish compilation of this result or of the full package.

`WDP-CH18-P02` **Finite proposition (nonnegative weights force permutations).** Let Q be a finite report carrier and $W > 0$ a rational number, and call $w : Q \to \mathbb{Q}$ a *state of mass* W when $w(q) \geq 0$ for every q and $\sum_q w(q) = W$. A linear update T sends every state of mass W to a state of mass W if and only if T is column-stochastic. It has an inverse S with the same property ($ST = I$) if and only if $T = P_\sigma$ for some $\sigma \in S_Q$. Once W3 carries pointwise nonnegativity, the content of W4 is a theorem for linear updates that are reversible within the states.

Proof. Applying T to We_j shows that column j of T is nonnegative with sum 1; conversely, a column-stochastic matrix preserves nonnegativity and total mass. Let $ST = I$ with both matrices column-stochastic. For $i \neq j$ the entry $(ST)_{ij} = \sum_k S_{ik}T_{kj} = 0$ is a sum of nonnegative terms, so $S_{ik}T_{kj} = 0$. Column j of T has an entry $T_{kj} > 0$; column k of S then vanishes off row j and, summing to 1, equals e_j. Since T is square, $ST = I$ gives $TS = I$, and reading $TS = I$ at column k gives $Te_j = e_k$. Two columns of T cannot use the same k, since column k of S would then equal two different basis vectors, so $j \mapsto k$ is an injection of the finite set Q into itself, hence a permutation. Every permutation matrix is column-stochastic with a column-stochastic inverse. □

The same module restates the preservation clauses of the theorem above for every finite carrier: a permutation matrix over $\mathbb{Z}$ preserves the standard form, the total mass, and the ℓ^1 norm. `QuantumImports.lean` establishes them only for $Q = \mathrm{Bool}$.

`WDP-CH18-P03` **Finite proposition (signed weights do not force permutations).** With the signed weights that W3 admits, total mass and invertibility do not force a permutation. The matrices $T = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ and $T^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$ both preserve $\sum_q w(q)$, T is not a permutation matrix, and T sends the basis weight $(1, 0)$ to the signed weight $(2, -1)$ of the remark above.

Proof. Both matrices have unit column sums, $T^{-1}T = I$, and $T_{00} = 2 \notin \{0, 1\}$. □

`WDP-CH18-P04` **Finite proposition (unitaries over $\mathbb{Z}[i]$ are monomial).** Let U be a $Q \times Q$ matrix over the Gaussian integers with $U^*U = I$. Then there are $\sigma \in S_Q$ and units $u_q \in \{\pm 1, \pm i\}$ with $Ue_q = u_q e_{\sigma(q)}$. Consequently the phase-blind readout of W6 moves by the same permutation, $\pi(Uv)_{\sigma(q)} = \pi(v)_q$, and no such U separates two vectors with equal readout.

Proof. The diagonal of $U^*U = I$ gives $\sum_k N(U_{kj}) = 1$, where $N(a + bi) = a^2 + b^2$ is a nonnegative

integer. Exactly one entry of column j therefore has norm 1, which makes it a unit, and the others vanish. If two columns $j \neq j'$ had their units in the same row k, the entry $(U^*U)_{jj'} = \overline{U_{kj}}\,U_{kj'}$ would be a product of units and hence nonzero. So the rows are distinct and $j \mapsto k$ is a permutation. Then $(Uv)_{\sigma(q)} = u_q v_q$, and N is multiplicative with $N(u_q) = 1$. □

Along any finite run of such updates, the readout is the initial readout relabelled by one permutation (Lean: `gaussian_run_readout`; not a ledger row). The package is therefore classical under either of two strengthened readings: W3 with pointwise nonnegativity and linear updates reversible within the states (CH18-P02), or dynamics that preserve the form of W5 over $\mathbb{Z}[i]$, read through W6 (CH18-P04). Under either reading every readout the package predicts comes from a deterministic reversible model on Q, and the phase fiber of principle P2 in §18.3 is never observed. For W3 as stated, with signed weights, CH18-P03 shows that the first route fails. Over $\mathbb{Z}$ the same argument, with $N(a) = a^2$, gives the signed permutation matrices (not mechanized).

WDP-CH18-P05 **Finite proposition (where the classical corner ends).** The unnormalized Hadamard matrix $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ used for interference in §18.3 satisfies $H^*H = 2I$. It separates the readout fiber of $(1,1)$ and $(1,-1)$, which CH18-P04 shows no unitary over $\mathbb{Z}[i]$ can do, so it is not a unitary over $\mathbb{Z}[i]$. Unitaries that separate readout fibers exist over finite localizations of the scalars. The rotation $R = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$ over $\mathbb{Z}[\frac{1}{5}]$ sends the readouts of $(1,1)$ and $(1,-1)$, both $(1,1)$, to $(\frac{1}{25}, \frac{49}{25})$ and $(\frac{49}{25}, \frac{1}{25})$. The matrix $M = \frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$ over $\mathbb{Z}[i][\frac{1}{2}]$ sends the readouts of $(1,1)$ and $(1,i)$, both $(1,1)$, to $(1,1)$ and $(2,0)$, and M^2 is the toggle $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Extending the scalars extends the readout of W6 to rational values.

Proof. Direct computation over $\mathbb{Z}[i]$, $\mathbb{Q}$, and $\mathbb{Q}(i)$. □

So unitary interference needs neither the continuum field (import 1) nor the circle group (import 2). It needs a non-monomial unitary: none exists over $\mathbb{Z}[i]$ (CH18-P04), and a finite localization of $\mathbb{Z}$ or $\mathbb{Z}[i]$ already has one. The similitude H leaves the classical corner by another route, since it is not an isometry. The square root M of the toggle also shows that the missing half-step of principle P11 is missing only among report permutations. This does not reconstruct quantum theory: CH18-N01 and the eight interfaces of §18.5 stand.

The strengthened package S. Call S the package W1, W2, W3$^+$, W4′, W6′ and W7, where W3$^+$ is W3 with pointwise nonnegative weights, W4′ says that an update is a linear map that sends states to states (no reversibility is required), W6′ says that a measurement is a *classical instrument*, a family $(T_o)_{o \in O}$ of nonnegative updates whose sum is column-stochastic, and W7 says that the empirical model of the measurement cover has a nonnegative global section, which by CH19-P05 means it is the image of a table in which every context reads the same runs. S is not a strengthening of W1–W6: it omits W5, replaces W6 by W6′, and drops the reversibility of W4. It is a package of hypotheses, not a theorem about the packets of Chapters 10–13: with CH19-P04's free tuple-formation and repetition rule, those admit tables with contextual images, so W7 is a hypothesis on the recorded runs and does not follow from the packet types.

WDP-CH18-P07 **Finite proposition (stochastic updates).** Over a finite report carrier, the updates of W4′ are the column-stochastic matrices. They are closed under composition and contain the

identity. A stochastic update whose entries are all 0 or 1 is exactly the update M_f induced by a function f of reports, and so is every presentation map. Every stochastic update is a mixture of function updates, $T = \sum_f w_f M_f$ with $w_f = \prod_j T_{f(j)j} \geq 0$ and $\sum_f w_f = 1$. Its units are the permutations (CH18-P02).

Proof. A product of column-stochastic matrices is column-stochastic. A column-stochastic $\{0,1\}$-matrix has exactly one 1 in each column, which is the matrix of a function. Expanding $\prod_j \sum_i T_{ij} = 1$ gives $\sum_f w_f = 1$, and fixing $f(k) = i$ in the same expansion gives $\sum_{f:f(k)=i} w_f = T_{ik}$ because the other columns still sum to one; this is $T = \sum_f w_f M_f$. □

`WDP-CH18-P08` **Finite proposition (classical facts about S, and their sharpness).** Let $(T_o)_{o \in O}$ be a classical instrument on $\mathbb{Q}^Q$ and p a state. The law of an outcome word $o_1 \dots o_n$ is the mass of $T_{o_n} \cdots T_{o_1} p$. It is nonnegative; the laws of the words of length n add up to the mass of p; and the law of a word is the sum of the laws of its one-letter extensions. Under W7 the empirical model of the measurement cover has a nonnegative global section by definition, and every such model on the CHSH cover has CHSH ≤ 2. Both hypotheses matter to these conclusions. Without W3$^+$ the signed shear of CH18-P03 has a negative entry, so it is not stochastic; it preserves mass and has a mass-preserving inverse but is not a permutation. Without W7 the PR box of CH19-P05 is an empirical model with CHSH $= 4$ and no nonnegative global section.

Proof. Nonnegativity follows by induction on the word, for a state $p \geq 0$ and finite O. Because $\sum_o T_o$ is column-stochastic, $\sum_o \sum_i (T_o q)_i = \sum_i q_i$ for every q, which gives consistency and, by induction on n, normalization. The CHSH bound and the PR-box values are those of CH19-P05; its outcome-availability and free table-formation hypotheses are needed only for the equivalence with global tables. □

Read in the vocabulary of §19.10, under S a state is a distribution on Q, an update is a mixture of functions (CH18-P07), an instrument outcome law is a hidden-Markov law, and a noncontextual model has a hidden-variable representation. This reading is an interpretation of CH18-P07 and CH18-P08, and no unrestricted structure-preserving representation of the packets of Chapters 10–13 into a classical joint-run model is asserted here; Appendix J constructs the exact admitted report-instrument quotients.

`WDP-CH18-O01` **Conditional result (reconstruction classification and obstruction).** Signed report weights have a classical state representation exactly when every observable block mass is nonnegative; signed instrument updates admit a classical report-instrument quotient exactly under the stated fiber-intertwining and stochastic-quotient conditions. The unrestricted witness/observer/measurement class admits accepted PR tables, refuting reconstruction that preserves their declared probabilities together with classical joint runs or bipartite local Hilbert measurement composition. Classical and quantum alternatives require the explicit additional packages in Appendix J; neither is forced by the original packets.

The original two alternatives were:

- a representation theorem into classical finite stochastic theory; or
- a reconstruction theorem with all additional quantum assumptions exposed.

Outcome (a) is proved for two strengthened readings of the package W1–W6 of §18.1: W3 with pointwise nonnegativity and linear updates reversible within the states (CH18-P02), and dynamics that preserve the form of W5 over $\mathbb{Z}[i]$ (CH18-P04). CH18-P07 and CH18-P08 prove classical facts for the package S, which drops reversibility and adds classical instruments and joint runs; without

W3$^+$ or W7 those conclusions fail. CH18-P03 shows that mass and invertibility alone do not force permutations, and CH18-P04 shows that the preserved form over $\mathbb{Z}[i]$ does. Appendix J resolves the unrestricted classical reading negatively: full signed indicators can be negative, and accepted PR tables admit no nonnegative joint representation. They also violate the imported Hilbert CHSH bound, so the original packet class cannot universally reconstruct standard quantum theory while preserving fixed local factor measurements. The constructive report-partition and instrument-factorization criteria identify the precise classical subclasses. A separately declared Hilbert/matrix-unit package gives a conditional quantum composition theorem, with the probability readout still exposed as an assumption. CH18-P05 remains an earlier algebraic milestone.

18.3 Discriminator suite and audit table

`WDP-CH18-D01` **Definition (reconstruction audit).** A reconstruction audit is a table whose rows are candidate principles and whose columns record:

1. the candidate principle;
2. its status in the finite obstruction calculus;
3. its formal statement and proved theorem or defect locator;
4. classical models satisfying it;
5. quantum models satisfying it;
6. post-quantum countermodels and conclusion.

Note on claim hierarchy and internal anchors: The principles P1–P12 and internal identifiers (`H006`, `H013`, `H021`, etc.) refer to exploratory computational and formal test fixtures in `code/witnessed_difference/quantum.py` and `lean4/WitnessedDifference/QuantumImports.lean`. They do not constitute standalone canonical claim rows in the canonical ledger. The canonical rows governing this chapter are CH18-D01, CH18-P01 to CH18-P08, the conditional reconstruction classification CH18-O01, and the nonclaim CH18-N01.

The table below audits the 12 candidate principles across six questions: nonclassical geometry, contextuality, composition, probability readout, continuity and reversibility, and purification or dilation. A row labelled "derived" means a consequence of its stated specialization assumptions, including W4–W6 or an additional operator when that row requires one. It does not mean that the primitive distinction alone supplies those assumptions. "Refuted" applies to the specified implication or host model, not to quantum mechanics or to every possible reconstruction route.

Candidate Principle	Status in the finite obstruction calculus	Proved Theorem / Defect Locator	Classical Model	Quantum Model	Post-Quantum / Conclusion
P1: Permutation Invariance	**DERIVED**	Form-preserving isometry with two-sided inverse (`H006.permutation_isometry`)	Symmetric group S_n	Permutation matrices $\subset U(n)$	Isometry envelope over $\mathbb{Z}[i]^n$ is the monomial matrices with entries in $\{\pm 1, \pm i\}$ (CH18-P04 gives one inclusion; the other is immediate); strictly smaller than $U(n)$.
P2: Coordinate-readout fiber	**CONSEQUENCE OF W5–W6**	For the admitted quadratic readout $\pi(v)_q = \lvert v_q\rvert^2$, equal reports can leave sign or phase freedom (`H009`).	A declared readout can identify distinct underlying representatives.	On normalized complex vectors with fixed coordinate probabilities and support of size r, the fiber modulo common phase is $(S^1)^{r-1}$; $\mathbb{C}P^{n-1}$ is the full pure-state ray space, not this single fiber.	A separately supplied mixer can distinguish some equal-readout vectors. This does not select complex scalars or the quadratic readout.
P3: Interference Non-Descent	**DERIVED (only with H admitted)**	Hadamard mixing separates equal-weight fiber states; no descent through fiber (`H010.int_no_descent`). $H^*H = 2I$, so H lies outside W4, and no unitary over $\mathbb{Z}[i]$ separates a readout fiber (CH18-P04)	Trivial descent ($f(\pi) = \pi'$)	Unitary basis rotation $U\psi$	Runs on $\mathbb{Z}$ with ± 1 once H is admitted; an isometric separator is unrealizable over $\mathbb{Z}[i]$ (CH18-P04) and exists over $\mathbb{Z}[\frac{1}{5}]$ (CH18-P05); complex field $\mathbb{C}$ is NOT selected.

Candidate Principle	Status in the finite obstruction calculus	Proved Theorem / Defect Locator	Classical Model	Quantum Model	Post-Quantum / Conclusion
P4: Phase Group Selection	**REFUTED (derived); IMPORTED**	Sign group $\{\pm 1\}$ suffices; circle group $U(1)$ is an explicit import (`H011Circle`)	Trivial phase $\{1\}$	Continuous circle $U(1)$	Interference requires only signs; $U(1)$ requires explicit continuity import.
P5: Quadratic Born Scaling	**DERIVED (discrete; uses H)**	On uniform superpositions $v = (1, 1)$, Hadamard norm ratio $\|Hv\|_p^p / \|v\|_p^p = 2$ holds identically iff $p = 2$; vanishing third-order interference $I_3 \equiv 0$ follows in the declared additive-amplitude/quadratic-measure model (Sorkin 1994 [REF-QM-018]; `H021`)	Linear L^1 norm	L^2 Hilbert norm $\langle\psi\|\psi\rangle$	$p = 2$ uniquely preserves Hadamard scaling on uniform superposition, while $I_3 \equiv 0$ is a structural consequence of additive amplitudes and quadratic probabilities.
P6: Gleason Uniqueness	**REFUTED (host); IMPORTED ($n \geq 3$)**	For state $(1, 0)$, the standard frame and $((3, 4)/5, (-4, 3)/5)$ give fourth-power sums 1 and 337/625, while both squared-coordinate sums are 1 (`H013Contract`). This counterexample is not Gleason uniqueness [REF-QM-017]	Simplex valuation	Spectral measure P_i on continuum lattice	Gleason's theorem applies to normalized weight-one frame functions on the complete projection structure [REF-QM-017]. These finite real-ray reports do not determine a complex Hermitian operator or certify its positivity.

Candidate Principle	Status in the finite obstruction calculus	Proved Theorem / Defect Locator	Classical Model	Quantum Model	Post-Quantum / Conclusion
P7: Algebraic Non-Factorable Witness	**DERIVED (algebraic)**	Vector $\Phi = (1,0,0,1)^T$ on $Q \times Q$ has nonfactorable coefficient matrix ($\det = 1 \neq 0$), proving tensor rank ≥ 2 in $\mathbb{Z}[i]^{Q \times Q}$ (`H014_bell_not_product`)	Factored distributions	Entangled bipartite states $\sum c_{ij}\|ij\rangle$	Algebraic non-product vectors exist in free modules over Cartesian products; physical quantum entanglement requires composite state tensor structure and LOCC constraints, which remain unpaid (§19.2).
P8: Tensor Uniqueness	**REFUTED (forced); IMPORTED**	Weaker combining interface admits `dumpRight` $\neq$ `tensorComb` (`H014Interface`)	Cartesian product $X \times Y$	Tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$	Injective composite tagging excludes degenerate maps like `dumpRight`, but does not uniquely force the Hilbert tensor product or its categorical factorisation property.
P9: Non-Contextuality (KS)	**REFUTED (classical); OB-STRUCTED**	Cabello 18-ray in $\{-1,0,1\}^4$ with 9 tetrads yields mod-2 parity contradiction $9 \not\equiv 2k$ (CH18-P06; [REF-QM-014]; `H020` checks the supplied rays, projective distinctness, all 54 orthogonality products and exact two-incidence before parity)	Non-contextual $\{0,1\}$ valuations	Kochen-Specker state space	Refutes non-contextual binary valuations on this ray configuration. No classification of its fractional frame-polytope vertices is supplied.

Candidate Principle	Status in the finite obstruction calculus	Proved Theorem / Defect Locator	Classical Model	Quantum Model	Post-Quantum / Conclusion
P10: Reversible Dilation	**DERIVED (census)**	Census on 2-point carrier with 1-bit prepared ancilla ($d_A = 2, d_E = 2$): exactly $2 \cdot 2 = 4$ reversible permutations dilate collapse: the two prepared states must go to the two states with system value 0, and the other two to the rest (`H015CollapseDilations` enumerates every permutation of the supplied composite carrier)	Markov embedding	Stinespring dilation ($d_E \leq d_A d_B$) [REF-QM-016]	Collapse has no same-basis isometry (`H015_collapse_no_isometry` checks its Gram matrix); while finite channels admit finite ancillary dilations ($r \leq d_A d_B$ [REF-QM-016]), quantum dilation requires importing CPTP maps, complex Hilbert spaces, and unitary rotations.

Candidate Principle	Status in the finite obstruction calculus	Proved Theorem / Defect Locator	Classical Model	Quantum Model	Post-Quantum / Conclusion
P11: Continuous Dynamics	**REFUTED (host); REFUSED**	Toggle permutation on Bool has no report square root (`import7_unforced` in `QuantumImports.lean`)	Continuous Markov flow	Stone's theorem e^{-iHt} [REF-QM-019]	In topology, $\mathrm{SO}(n,\mathbb{Q})$ is dense in $\mathrm{SO}(n,\mathbb{R})$, not discrete; operational discreteness consists in restricting evolution to discrete integer time steps $\tau \in \mathbb{Z}$ without Lie infinitesimal generators. The missing half-step is missing only among report permutations: the unitary M of CH18-P05 over $\mathbb{Z}[i][\frac{1}{2}]$ squares to the toggle.
P12: Finite Matrix Recovery	**DERIVED ($\mathbb{Z}[i]$)**	$\mathrm{End}_{\mathbb{Z}[i]}(\mathbb{Z}[i]^Q) \cong M_n(\mathbb{Z}[i])$ with conjugate-transpose adjoint (standard linear-algebra theorem; `H017Fin` checks supplied finite basis images, matrix, adjoint and composition)	Stochastic matrices	Matrix algebra $M_n(\mathbb{C})$	The code checks declared Gaussian-integer linear instances; it does not establish the unrestricted isomorphism by sampling or recover continuous $M_n(\mathbb{C})$.

18.4 Contextuality in exact arithmetic

The Mermin–Peres square [REF-QM-015], the Peres 33-ray configuration in $\mathbb{Z}[\zeta_8]^3$, and the Cabello 18-ray configuration in $\{-1,0,1\}^4$ [REF-QM-014] establish contextuality obstructions in exact arithmetic without continuous angles. The eighteen-ray case is a finite parity count on the following table of nine contexts.

Context	Vectors in $\mathbb{Q}^4$
C_1	$(0,0,0,1)$, $(0,0,1,0)$, $(1,1,0,0)$, $(1,-1,0,0)$
C_2	$(0,0,0,1)$, $(0,1,0,0)$, $(1,0,1,0)$, $(1,0,-1,0)$
C_3	$(1,-1,1,-1)$, $(1,-1,-1,1)$, $(1,1,0,0)$, $(0,0,1,1)$
C_4	$(1,-1,1,-1)$, $(1,1,1,1)$, $(1,0,-1,0)$, $(0,1,0,-1)$
C_5	$(0,0,1,0)$, $(0,1,0,0)$, $(1,0,0,1)$, $(1,0,0,-1)$
C_6	$(1,-1,-1,1)$, $(1,1,1,1)$, $(1,0,0,-1)$, $(0,1,-1,0)$
C_7	$(1,1,-1,1)$, $(1,1,1,-1)$, $(1,-1,0,0)$, $(0,0,1,1)$
C_8	$(1,1,-1,1)$, $(1,-1,-1,-1)$, $(1,0,1,0)$, $(0,1,0,-1)$
C_9	$(1,1,1,-1)$, $(1,-1,-1,-1)$, $(1,0,0,1)$, $(0,1,-1,0)$

`WDP-CH18-P06` **Finite proposition (an eighteen-ray parity obstruction).** In the table, each context consists of four pairwise orthogonal vectors of $\mathbb{Q}^4$, no two of the eighteen vectors are parallel, and every vector lies in exactly two contexts. No map v from the eighteen vectors to $\mathbb{F}_2$ satisfies $\sum_{r\in C_k} v(r) = 1$ for $k = 1,\dots,9$; in particular no $\{0,1\}$-valuation puts exactly one 1 in every context.

Proof. The 54 inner products within contexts are 0 by direct integer computation, no two vectors are proportional, and each vector occurs in exactly two rows. Adding the nine equations gives $\sum_k \sum_{r\in C_k} v(r) = \sum_r 2v(r) = 0$ in $\mathbb{F}_2$ on the left and $9 = 1$ on the right. A solution with values in $\{0,1\}$ or in $\mathbb{Z}$ would reduce modulo 2 to one over $\mathbb{F}_2$. □

The proposition concerns a declared finite configuration. It becomes the Kochen–Specker obstruction for $\mathbb{C}^4$ [REF-QM-008; REF-QM-014] only after the vectors are read as rank-one projectors of the Hilbert carrier of §5.8 and a noncontextual valuation is defined on them, and it yields no Bell violation, no correlation bound, and no reconstruction (CH18-N01, CH19-N02). CH19-P05 applies the same count to the four CHSH contexts. The absorbed archived working-paper source 004 printed a different eighteen-vector table in which seven of the nine contexts contain non-orthogonal pairs, and read the contradiction as a nonzero class in the first cohomology of the context hypergraph with $\mathbb{Z}_2$ coefficients. That reading is not carried: no cochain complex was declared, and the falsifier of §21.6 applies. The source cited an external Lean check of a correct table, equivalent to this one under a signed permutation of coordinates; it was read, not built or ported, and gives no Lean credit.

The finite frame equations constrain ray probabilities. Such a probability vector is not a matrix and has no eigenvalues without a specified reconstruction. The listed rays are real, so their expectations do not determine a complex Hermitian matrix. For example,

$$H_t = \tfrac{1}{4} I_4 + t\, i(E_{12} - E_{21})$$

has expectation $1/4$ on every real unit ray. Its eigenvalues are $1/4 + t, 1/4 - t, 1/4, 1/4$; for $t > 1/4$ it is not positive semidefinite, yet its ray report is also the report of $I_4/4$. Thus these finite tests do not certify positivity of a submitted operator. This example does not prove that the frame polytope has a nonquantum vertex. The image of the density-matrix set under the ray-readout map is a spectrahedral shadow; neither equality nor strict inclusion in this particular frame polytope is established here.

Gleason's theorem [REF-QM-017], on a real or complex separable Hilbert space of dimension at least three, concerns the complete projection structure: a nonnegative, normalized weight-one frame

function is represented by a unique density operator. A finite list of rays does not supply those hypotheses.

WDP-CH18-N01 **Nonclaim.** The primitive finite calculus alone is not shown to select the quantum set of correlations, a complex Hilbert representation, or quantum dynamics. Under separately supplied finite Hilbert-space, tensor and trace assumptions, CH19-O02 establishes the CHSH bound $S^2 \leq 8$. That conditional result does not derive those assumptions from the primitive calculus.

18.5 The eight external interfaces exposed by this reconstruction route

To reach textbook continuous quantum mechanics $M_n(\mathbb{C})$, this reconstruction route exposes eight external mathematical interfaces along its boundary. We describe them as interfaces exposed by this specific constructive progression, rather than asserting that all eight are logically independent necessities for every conceivable axiomatic route:

1. **Import 1: The Continuous Field $\mathbb{C}$.** Cauchy completeness, archimedean topology, and algebraic closure (refused on discrete host; host uses $\mathbb{Z}[i]$).
2. **Import 2: The Continuous Circle Group $U(1)$.** Connected Lie group phase symmetries (refused on host; discrete interference selects only $\{\pm 1\}$).
3. **Import 3: The Spectral Theorem for Self-Adjoint Operators.** Orthonormal eigenbases (refused on host; Hermitian matrices in $M_n(\mathbb{Z}[i])$ need not have eigenvalues in $\mathbb{Z}[i]$).
4. **Import 4: The complete projection structure in dimension at least three.** Gleason's theorem represents normalized weight-one frame functions by unique density operators under its full real or complex Hilbert-space hypotheses. The finite real-ray example above does not provide those hypotheses or a classification of all probability vectors satisfying its frame equations.
5. **Import 5: Unique Hilbert Tensor Product.** Postulating the Hilbert tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ and its initial bilinear factorisation property; injective composite report tagging excludes degenerate combining maps such as `dumpRight`, but does not uniquely force the Hilbert tensor product structure.
6. **Import 6: CPTP Channels & Unitary Stinespring Dilation.** Completely positive trace-preserving (CPTP) channel structure and complex unitary rotations on an enlarged system (finite-dimensional quantum channels admit finite ancillary constructions with environment dimension $r \leq d_A d_B$ [REF-QM-016]; the missing structures are CPTP condition, complex Hilbert inner products, and continuous unitary groups, whereas the discrete carrier provides only a finite permutation census).
7. **Import 7: Continuous 1-Parameter Unitary Groups / Stone's Theorem.** Continuous time evolution $t \in \mathbb{R}$ generated by self-adjoint Hamiltonians e^{-iHt} [REF-QM-019] (refused on discrete host; $\mathrm{SO}(n, \mathbb{Q})$ is topologically dense in $\mathrm{SO}(n, \mathbb{R})$, but discrete dynamics is restricted to integer steps $\tau \in \mathbb{Z}$).
8. **Import 8: The positive-semidefinite state cone.** Require state matrices to satisfy $\rho \succeq 0$. A fixed finite collection of ray inequalities gives polyhedral constraints and, in general, does not characterize this entire cone. For a supplied exact Hermitian matrix, positivity can instead be checked by valid principal-minor, Gram-factor or congruence certificates; semidefinite programming is another method for problems with matrix positivity constraints. These methods check the supplied state condition; they do not derive its operational interpretation.

The route also has steps that are not among these interfaces. CH18-P05 shows that a finite localization of the scalars, such as $\mathbb{Z}[\frac{1}{5}]$ or $\mathbb{Z}[i][\frac{1}{2}]$, already admits unitaries that separate readout fibers.

18.6 Audit verdict

The defensible verdict is: the finite obstruction calculus, under the stated specialization hypotheses, establishes an integer permutation-isometry envelope over $\mathbb{Z}$, a phase-blind superposition fiber, an interference non-descent obstruction over $\mathbb{Z}$ once the similitude H is admitted, an algebraic non-factorable tensor-rank witness, and the finite matrix algebra $M_n(\mathbb{Z}[i])$. For the continuous circle group, Hilbert tensor uniqueness, Gleason uniqueness at dimension ≥ 3, CPTP dilations, Lie algebras, and continuous unitary dynamics it gives one small finite witness each of why the discrete host does not supply them; the reconstruction route in §18.5 exposes eight admitted interfaces. This is not a theorem that every reconstruction must use those interfaces, or that they are independent. Under the two strengthened readings of CH18-P02 and CH18-P04 the package is classical, which is outcome (a) of CH18-O01 for those readings; CH18-P07 and CH18-P08 add classical facts for the package S, which has stochastic updates, classical instruments and joint runs.

This verdict can be stated with exact claim discipline across the four-tier foundational ambition taxonomy:

1. **Tier 1 (Syntactic Representation):** Established for the declared finite encodings. Parse/print and token encode/decode round-trip laws concern their specified carriers; selected implementation replays have the separate scopes recorded in Appendix D.
2. **Tier 2 (Faithful Operational Representation):** Established for the stated finite models and adapter laws. Integer permutation actions and cochain transports preserve the specified operations and questions when the corresponding commutation conditions hold. No unrestricted theory-profile adapter follows from an encoding alone (§20.4).
3. **Tier 3 (Reconstruction Under Explicit Principles):** Characterized with open interfaces. This reconstruction route exposes eight admitted interfaces; under the strengthened readings of CH18-P02 and CH18-P04 the package stays inside the monomial $\mathbb{Z}[i]$ envelope, which is classical. The similitude H (P3) and a finite localization of the scalars (CH18-P05) each leave that envelope, and neither is one of the eight interfaces.
4. **Tier 4 (Unique Selection From Primitive Kernel):** Explicitly refused. The monograph does not claim that textbook continuous quantum mechanics $M_n(\mathbb{C})$ is forced or uniquely selected by primitive difference, and it establishes no theorem that it cannot be (§18.5, CH19-N01).

Chapter 19 — The unpaid quantum interfaces

19.1 Context and empirical content

A family of contexts is not selected by the existence of cochains. To reach an empirical contextuality model one needs measurements, outcome sets, compatible probability distributions, and a cover. Chapter 13 provides a general finite measurement packet, but a quantum scenario requires a specialized adapter into effects and instruments.

`WDP-CH19-O01` **Conditional result (arbitrary-cover repair and sampling).** For any declared finite cover and nonempty rational run tables, the minimax context L^1 distance to the compatible polytope is the optimum of the exact standard-form LP in Appendix J, with independently verified primal/dual certificates. Under declared within-context iid categorical trials and compatible population marginals, the trial counts give explicit finite-sample pair tolerances and a union-bound failure probability. The empirical image retains only the previously stated outcome information.

CH19-P02 to CH19-P05 below construct this map for tables whose frequencies agree exactly on every overlap of contexts, and state what survives it. CH19-P06 and CH19-P07 add a tolerant adapter for tables whose context frequencies agree only up to a declared tolerance. Appendix J completes arbitrary-context exact repair and derives finite-sample tolerance under the explicitly declared trial protocol and probability model of §13.6. The specialization to effects and instruments is CH19-O04.

The adapter uses the packet and a declared profile. A declared cover consists of a finite set X of measurements, a finite outcome set O, and contexts $C_i \subseteq X$. A profile assigns each measurement x a measurand and a partition of calibrated results into outcomes. The partition stands in for the structure $\mathcal{S}$ of CH13-D01; it is declared per measurement rather than carried by the packet. A packet may fill slot x only when its measurand is the one declared for x, and it reports the bin of its calibrated value. A run in context C_i holds one such packet for each measurement of C_i, and a table lists the runs recorded in each context. The frequency $e_i(s)$ of a section $s : C_i \to O$ is the normalized multiplicity (Chapter 8) of the runs that report s. Weights are rational throughout. An empirical model [REF-QM-001] is a probability distribution with rational weights on the sections of each context whose marginals agree on every overlap $C_i \cap C_j$. It is noncontextual when a nonnegative rational distribution on the global assignments $X \to O$ restricts to it, and strongly contextual when no global assignment restricts, in every context, to a section of positive weight. For a model with rational weights a real nonnegative global section yields a rational one, because the constraints are rational linear inequalities; that step is not mechanized.

`WDP-CH19-P02` **Finite proposition (the packet adapter).** On a declared cover, the adapter sends a table of measurement packets to its frequencies. It refuses the table when some context has no runs, naming that context, and when the frequencies of two contexts disagree on their overlap, naming the two contexts and the overlap section at which they differ. Otherwise it returns an empirical model. It returns the model m exactly when no context is empty and m is the table of frequencies.

Proof. Frequencies are nonnegative, and in a context with at least one run they sum to one. Agreement on overlaps is the compatibility condition of an empirical model. □

`WDP-CH19-P03` **Finite proposition (what survives the adapter).** A section has positive weight in the image exactly when a recorded run reports it. The image depends only on the list of reported outcomes in each context, and two nonempty tables have the same frequencies in a context exactly when their outcome counts there are proportional. So a packet field reaches the image only through the outcome it reports. The world, the presentation map, the indication and the calibrated value reach it that way. The trace, the evaluator, the calibration chain and its uncertainty contributions, the unit and scale, the provenance record, the order of runs, and the absolute counts do not affect it. Replacing the calibration chain of every packet, for example, changes the documented traceability and leaves the image unchanged. Two packets that differ only in their evaluators' verdicts on reports that were not recorded have the same measurand and readout, so exchanging them leaves the image unchanged. Every packet of a run does carry the declared measurand, an accepted evaluator status, a report that the observer's memory returns, and a provenance record that replays the indication and the calibrated value. It carries them by its type (CH13-D01), in every table, and none of them can be read back from the image. The answer to the second half of CH19-O01 is therefore that no observer or evaluator field survives the map except through the outcome it reports. Acceptance survives only as the condition for a run to be counted.

Proof. A count is positive exactly when some run reports the section. Counts are functions of

the outcome list, and $c/n = c'/n'$ exactly when $cn' = c'n$. The readout reads the report, the indication, the calibrated value, and the profile's bin, and nothing else. □

WDP-CH19-P04 **Finite proposition (the image of freely formed tables is every rational model).** Suppose every outcome of every measurement is reported by some admissible packet, context runs may be formed from arbitrary tuples of those packets, and each run may be repeated any finite number of times. Then every empirical model on the cover with rational weights is the image of a finite table. Under this table-formation rule, the packet type alone neither forces nor excludes contextuality.

Proof. For each section s of context i, choose a packet reporting each component of s and form their tuple. Let N be a common denominator of the weights in that context, and record $Ne_i(s)$ copies of the tuple. Their frequencies are exactly e_i. □

This is a formal table-realization theorem. Individual outcome availability does not imply joint physical availability: an operational theory may forbid some tuples or constrain which runs can coexist. Such restrictions must be declared separately; the construction does not realize a PR box as a physical experiment.

WDP-CH19-P05 **Finite proposition (contextuality as a missing global table).** Call a table global when each of its runs reads every measurement at once and every context reads the same runs. Assume the freely formed, repeatable tables of CH19-P04, with the same tuple-formation rule for all measurements at once. An empirical model with rational weights on any declared cover is noncontextual exactly when some global table has it as its image. On the CHSH cover, with Alice's settings a_0, a_1, Bob's settings b_0, b_1, contexts $\{a_x, b_y\}$ and outcomes in $\{0, 1\}$, let E_{xy} be the expected value of $+1$ for agreeing outcomes and -1 for differing ones. Every noncontextual model, and so every global table, has $E_{00} + E_{01} + E_{10} - E_{11} \leq 2$. The PR box [REF-QM-009] gives probability $\frac{1}{2}$ to each outcome pair with $a_x \oplus b_y = xy$. It is an empirical model, its CHSH value is 4, and it is strongly contextual. It is the image of a finite table of accepted measurement packets, and no global table has it as its image. It has the signed global section $\frac{1}{2}(\delta_{0000} - \delta_{0010} + \delta_{0110} + \delta_{1011})$ on (a_0, a_1, b_0, b_1), and no nonnegative one.

Proof. A nonnegative global section with rational weights is the frequency table of a finite list of global assignments, by freely forming the global tuples and clearing denominators as in CH19-P04. If a rational empirical model has a real nonnegative global section, the finite rational linear system specifying that section has a rational feasible point: restrict to the support of a real solution, use rational Gaussian elimination and approximate its free parameters closely enough to keep all positive coordinates positive. Thus the rational-weight restriction loses no noncontextual rational models. Conversely, the frequencies of the global assignments in a global table are a nonnegative global section. One global assignment has CHSH value at most 2, and so does any mixture of them. In the PR box the four context conditions add up to 1 modulo 2, while each measurement occurs in two contexts, so no global assignment meets all four. This is the mod-2 count of CH18-P06 (§18.4), applied to the four CHSH contexts instead of the nine contexts there. The signed section is checked by direct computation. □

So, under the stated free table-formation rule, a rational model is contextual exactly when no single table of global runs reproduces it. The packet type alone imposes no Tsirelson bound, since the PR box is an image, and CH18-N01 stands.

WDP-CH19-P06 **Finite proposition (the tolerant adapter).** For contexts i, j let D_{ij} be the ℓ^1 distance between the marginals of the frequency tables e_i, e_j on the overlap $C_i \cap C_j$. For $\varepsilon \geq 0$ the

tolerant adapter accepts a table exactly when no context is empty and every $D_{ij} \le \varepsilon$. It returns the frequencies together with a certificate that every $D_{ij} \le \varepsilon$, and otherwise names the empty context or a pair with $D_{ij} > \varepsilon$. At $\varepsilon = 0$ it accepts exactly the tables of CH19-P02 and returns the same frequencies, and acceptance is monotone in ε. If every e_i lies within ε in ℓ^1 of the i-th distribution of a compatible model, then every $D_{ij} \le 2\varepsilon$.

Proof. At $\varepsilon = 0$ the condition is agreement of the overlap marginals, which is compatibility. Marginalization does not increase ℓ^1 distance and the compatible model has equal overlap marginals, so the triangle inequality gives $D_{ij} \le \varepsilon + \varepsilon$. □

What the tolerant adapter returns is not an empirical model, since its overlap marginals may differ by up to ε; it is a table with a certified signalling defect.

`WDP-CH19-P07` **Finite proposition (exact repair on two contexts).** On a cover with two contexts and a nonempty outcome set, let D be the defect of the two frequency tables. Replace each overlap marginal by the average ν of the two, rescale the conditional weights over each overlap section, and, where a context has no weight over a section with $\nu > 0$, place ν on one fixed extension of that section. The result is an empirical model, and it lies at ℓ^1 distance exactly $D/2$ from each table (the distance is taken context by context). Hence a compatible model within ε of both tables exists exactly when $D \le 2\varepsilon$.

Proof. Over an overlap section t with positive weight $\mu_i(t)$, rescaling moves $\mu_i(t)$ to $\nu(t)$ and changes the weights over t by $|\mu_i(t) - \nu(t)|$ in total. Where $\mu_i(t) = 0$ the added mass is $\nu(t) = |\mu_i(t) - \nu(t)|$. Summing over t gives $\sum_t |\mu_i(t) - \nu(t)| = D/2$, because $\nu - \mu_i = \pm(\mu_1 - \mu_2)/2$. The converse is CH19-P06. □

The closeness hypothesis of CH19-P06 is a declaration. Appendix J derives a sufficient finite-sample tolerance from actual trial counts under explicit within-context iid and compatible-population assumptions, and solves the exact compatible-polytope distance for arbitrary finite covers. The three-point-mass example shows why half the pairwise defect is insufficient for three contexts. These assumptions cannot be inferred from the table (§9.9).

19.2 Composition

Two systems require a rule for states, effects, and transformations of the composite. Cartesian product, tensor product, minimal and maximal ordered tensor products, and constrained subspaces yield different theories. Primitive difference does not choose among them.

`WDP-CH19-O02` **Conditional result (paid quantum composition).** Declaring positive finite Hilbert density states, local Hermitian dichotomic observables and trace pairing on the tensor composite gives the exact CHSH bound $S^2 \le 8$, excluding PR. Commuting complete local matrix units yield the tensor-factor representation up to a multiplicity space, removed by the stated dimension condition. The native instrument and observer adapters exhibit consistency with the finite architecture. These assumptions are inputs; CP alone does not supply them.

CH19-P08 to CH19-P10 state a composition principle that needs no import and prove what its clauses do and do not decide. The principle CP has three clauses: (T) injective tagging, so that the composite report determines both local reports; (NS) no-signalling, so that a local instrument on one side never changes the marginal of the other, which on the CHSH product cover is the compatibility of an empirical model on overlaps; and (B) bilinear positivity, so that a composite state is a bilinear form on product effects that is nonnegative and normalized. Clause (B) is

formulated here for classical carriers only.

WDP-CH19-P08 **Finite proposition (injective tagging).** For nonempty finite carriers A, B and a composite tagging $c : A \times B \to C$, there are decoders d_A, d_B with $d_A(c(a,b)) = a$ and $d_B(c(a,b)) = b$ if and only if c is injective. The tagging $\mathrm{dumpRight}(a,b) = b$ is not injective when A has two elements, so it has no decoder for the left factor.

Proof. If decoders exist, $c(a,b) = c(a',b')$ gives $a = a'$ and $b = b'$. If c is injective, decode with any left inverse. □

WDP-CH19-P09 **Finite proposition (no-signalling and presentation maps).** On the Cartesian composite $A \times B$ a joint state is a nonnegative weight $p(a,b)$, and a classical instrument (K_o) on B acts as $1 \otimes K_o$. For every a, $\sum_o \sum_{b'} ((1 \otimes K_o)p)(a,b') = \sum_b p(a,b)$, and the same holds with the factors exchanged. So the marginal of one factor is the same for any two instruments on the other. A presentation map $\pi : A \to Q$ lifts to $\pi \circ \mathrm{fst}$ on $A \times B$, and pushing forward the A-marginal gives the Q-marginal of the lifted map.

Proof. Summing over the outcomes and the post-states uses only that $\sum_o K_o$ is column-stochastic. The lift is a reordering of a finite sum. □

WDP-CH19-P10 **Finite proposition (classical composites are Cartesian).** Let A, B be finite carriers and ω a bilinear form on effect vectors of A and B that is nonnegative on every pair of effects (values in $[0,1]$) and is 1 on the pair of constant effects. Then there is a unique joint distribution p on $A \times B$ with $\omega(e,f) = \sum_{a,b} e_a f_b p(a,b)$, namely $p(a,b) = \omega(1_a, 1_b)$; conversely every joint distribution defines such a form. On the CHSH product cover, whose contexts $\{a_x, b_y\}$ are the products of two two-setting scenarios, the PR box of CH19-P05 is compatible on every overlap, so it satisfies (NS), and yet it is not noncontextual and has CHSH = 4.

Proof. Evaluating ω on indicator effects gives p and its nonnegativity, and bilinearity gives the formula. The second part is CH19-P05. □

By its clauses CP excludes taggings that lose a local report and composites that signal. For classical carriers (B) forces every composite state to be a joint distribution on $A \times B$; it does not force every joint distribution to be a state. Clauses (T) and (NS) do not exclude the PR box, and (B) is not formulated here for its nonclassical factors, so its status under CP is undecided. On the Cartesian composite of classical carriers, local presentations and instruments act on one factor, do not signal, and the composite report determines both local reports (CH19-P08, CH19-P09); this is the sense in which CP is consistent with the finite observer architecture here, and no more is proved. Nothing here shows that CP selects the Hilbert tensor product. Appendix J adds a sufficient finite Hilbert composition package that excludes PR and conditionally yields tensor factors. Its matrix algebra, positivity and trace assumptions pay import 5 of §18.5 explicitly; they are not consequences of CP.

19.3 Continuity and reversible dynamics

Finite-dimensional real or complex vector spaces can support continuous parameter groups once their topology and transformations are specified. Finite dimension is different from finite cardinality. Every finite Hausdorff space is discrete, although finite spaces with other topologies exist. Consequently, a continuous action of the connected group $\mathbb{R}$ on a finite discrete state set is trivial. Nontrivial continuous reversible dynamics requires additional structure beyond finitely many witness types.

19.4 Phase and interference

The complex semiring permits cancellation. A $U(1)$ connection permits loop phase. Neither supplies the operational rule connecting amplitude to observed frequency. Conversely, positive finite probability does not supply phase. A valid bridge must specify:

1. the amplitude carrier;
2. the composition/addition of alternatives;
3. the transformation law;
4. the measurement/effect pairing;
5. the probability readout;
6. the empirical interpretation.

WDP-CH19-O03 **Conditional result (phase and operational readout bridge).** A declared continuous character $\chi_\theta(t) = e^{i\theta t}$ maps real cochains into complex path amplitudes and descends under gauge to phase-blind operational readouts; its cycle-period lattice and anchor dependence are explicit. An independent exact rational-turn/cyclotomic branch is implemented. Declared effect/instrument pairing supplies a conditional operational model, while normalized quadratic and quartic readouts prove nonselection by phase invariance. No Born rule or faithful real phase embedding is inferred.

CH19-P11 and CH19-P12 declare the first three slots of this bridge and prove what the last three can and cannot depend on. The amplitude carrier is $\mathbb{Q}(i)$. Alternatives add in $\mathbb{Q}(i)$. The transformation law multiplies by the unit-norm phase $u^{a(v,w)}$ along each step $v \to w$ of a walk, for an antisymmetric integer edge cochain a and the fixed rational phase $u = (3+4i)/5$. The effect pairing and the probability readout stay parameters.

WDP-CH19-P11 **Finite proposition (gauge descent of phase-blind readouts).** Let $z_1 \neq 0$. Two pairs of amplitudes (z_1, z_2) and (w_1, w_2) in $\mathbb{Q}(i)$ differ by one common unit-norm phase exactly when $N(z_1) = N(w_1)$, $N(z_2) = N(w_2)$ and $z_1\bar{z}_2 = w_1\bar{w}_2$. So a readout of two alternatives that a common phase does not change is a function of $|z_1|^2$, $|z_2|^2$ and the cross term $z_1\bar{z}_2$. For two walks with common endpoints and magnitudes m_1, m_2, $z_1\bar{z}_2 = m_1 m_2$ times the ratio of the two walk amplitudes, which for an antisymmetric cochain is the holonomy of the loop formed by the first walk and the reverse of the second (the mechanized statement is about the ratio and does not use antisymmetry). If the cochain changes by a coboundary $d_0 f$, each walk amplitude is multiplied by the same phase $u^{f(\text{end})-f(\text{start})}$, so the holonomy and every phase-blind readout are unchanged: they depend on the cochain only through its class modulo coboundaries. The quadratic readout $N(z_1+z_2)$ and the quartic readout $N(z_1+z_2)^2$ are both phase-blind and differ (4 against 16 for $z_1 = z_2 = 1$), so gauge invariance does not select the quadratic readout.

Proof. If $w = gz$ with $N(g) = 1$, the three invariants agree because N is multiplicative and $g\bar{g} = 1$. Conversely put $g = w_1\bar{z}_1/N(z_1)$. Then $gz_1 = w_1$ and $N(g) = 1$, and the equality $z_2\bar{z}_1 = w_2\bar{w}_1$, the conjugate of the third condition, gives $gz_2 = w_2$. By induction along a walk the amplitude of $a + d_0 f$ is that of a times $u^{f(\text{end})-f(\text{start})}$; the two walks share this factor, which cancels in their holonomy. □

WDP-CH19-P12 **Finite proposition (the exponentiation obstruction).** The phase $u = (3+4i)/5$ has no square root among the unit-norm elements of $\mathbb{Q}(i)$, so no group homomorphism from $(\mathbb{Q}, +)$ to those elements sends 1 to u. Integer exponents compose, $u^{m+n} = u^m u^n$, and $u^n = 1$ only for $n = 0$. The map $a \mapsto u^a$ sends coboundaries to gauge transformations (CH19-P11). So the exponentiation $U_e = \exp(i\alpha a_e)$ of §5.4 is not available on rational cochains with this scale inside

$\mathbb{Q}(i)$; it needs an extension of the scalars that contains roots of u of every order (an algebraic closure suffices, and import 1 of §18.5 supplies one), or a restriction to integer cochains. The obstruction is to a homomorphism with values in the unit-norm elements of $\mathbb{Q}(i)$, not a claim that no exponentiation exists.

Proof. A square root $x + yi$ would satisfy $x^2 + y^2 = 1$ and $x^2 - y^2 = 3/5$, hence $x^2 = 4/5$, and $20 = (5x)^2$ is not the square of a rational number. A homomorphism φ with $\varphi(1) = u$ would give $\varphi(\frac{1}{2})^2 = u$. For the order of u, write $u = (2+i)/(2-i)$. If $u^n = 1$ with $n > 0$, then $(2+i)^n = (2-i)^n$, so the imaginary part b_n of $(2 + i)^n = a_n + b_n i$ vanishes. Modulo 5, $a_n + 3b_n \equiv 0$ for $n \geq 1$ and $a_n - 3b_n \equiv (-1)^n$, so $b_n = 0$ gives $0 \equiv \pm 1$. Negative n follow by inversion. □

Appendix J supplies the real-character bridge, its period lattice and the exact rational-turn alternative. Probability readout and empirical interpretation are explicitly declared operational parameters. CH19-P11 and the quadratic/quartic countermodel establish that gauge invariance cannot select them (CH19-N01).

19.5 Born uniqueness

Chapter 9 establishes only nonselection by normalization. Gleason-type theorems [REF-PROB-002] begin with a Hilbert space and noncontextual probability assignments of a specified kind. Operational reconstructions begin with other strong axioms. Those are legitimate routes when their assumptions are visible. They cannot be retroactively described as consequences of finite witness mass.

`WDP-CH19-N01` **Nonclaim.** Born-quadratic uniqueness is not proved.

`WDP-CH19-N02` **Nonclaim.** Complex phases, contextual obstructions, or destructive cancellation separately do not characterize quantum theory.

19.6 A small curved-bicomplex example

Let $C^{0,0} = C^{1,0} = C^{0,1} = C^{1,1} = \mathbb{R}$ and define scalar maps

$$d_s^{0,0}(x) = ax, \qquad d_h^{0,0}(x) = bx,$$

with the maps into $C^{1,1}$ given by multiplication by c from $C^{1,0}$ and by d from $C^{0,1}$. The mixed defect on x is

$$\Lambda(x) = (db + ca)x$$

for the chosen sign convention. Even in this one-dimensional toy model, anticommutation is a nontrivial algebraic condition $db + ca = 0$. If it fails, calling the square a double complex is false.

One can add a homotopy h only after specifying its degree and an identity such as $\Lambda = dh + hd$ in a surrounding complex. The existence of the scalar $db + ca$ does not manufacture that homotopy. This example is deliberately simple: the typing obstruction is already visible before geometry or physics enters.

19.7 Coarse-graining histories

Let $\pi : \Omega \to \bar{\Omega}$ group fine histories into coarse classes. A natural coarse weight is

$$\bar{w}(\bar{p}) = \bigoplus_{p:\pi(p)=\bar{p}} w(p).$$

This definition preserves the total finite aggregate by associativity and commutativity of $\oplus$ when the fibers are finite. It may fail to preserve composition: two fine histories in the same coarse class can compose with a third history into different coarse classes. For a compositional coarse model, π must be a congruence relative to the path product, or the coarse carrier must retain additional memory.

`WDP-CH17-P03` **Finite proposition (aggregate-preserving pushforward).** For a finite history family and a commutative additive operation,

$$\bigoplus_{\bar{p}\in\bar{\Omega}} \bar{w}(\bar{p}) = \bigoplus_{p\in\Omega} w(p).$$

The theorem is a finite reindexing. It does not prove Markovianity, decoherence, or consistent histories.

19.8 Destructive cancellation versus operational interference

Suppose two complex-weighted paths contribute z_1 and z_2. Their aggregate amplitude is $z_1 + z_2$, and a separately declared quadratic readout gives

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1\overline{z_2}).$$

The cross term is absent if one first converts each path to a positive weight and then adds. This elementary identity explains why positive witness mass and complex amplitude are different layers. The identity itself does not say which alternatives are physically coherent, which phases occur, or why the quadratic readout is a probability.

The history chapter should therefore use "cancellation" for an algebraic sum that vanishes and reserve "interference" for a model containing alternatives, phase-sensitive addition, and an operational readout. This vocabulary repair prevents the zerosumfree theorem from masquerading as a quantum theorem.

The higher-order version of the same bookkeeping is an identity about declared readouts. For alternatives with amplitudes a_j and a subset S, let $A(S) = \sum_{j\in S} a_j$, and define the alternating sum $I_k(S) = \sum_{T\subseteq S}(-1)^{|S|-|T|}P(T)$ for $|S| = k$.

`WDP-CH19-P01` **Finite proposition (Sorkin sums under a quadratic readout).** If the amplitudes of finitely many alternatives add and the readout is $P(S) = B(A(S), A(S))$ for a declared biadditive B, then the alternating sum I_k vanishes for every set of $k \geq 3$ alternatives, while I_2 equals $B(a, b) + B(b, a)$ and can be nonzero. Vanishing higher-order interference is therefore a consequence of the declared quadratic readout and not a derivation of it.

Proof. Expanding by biadditivity, $P(T) = \sum_{i,j \in T} B(a_i, a_j)$. The coefficient of $B(a_i, a_j)$ in $I_k(S)$ is $\sum_{\{i,j\} \subseteq T \subseteq S} (-1)^{|S|-|T|}$, which is zero whenever $S \setminus \{i, j\}$ is nonempty, that is, whenever $k \geq 3$. For $k = 2$ only the cross terms survive. □

With $a = 1$, $b = \frac{3}{5} + \frac{4}{5}i$, and $B(x, y) = x\overline{y}$, the readouts are $P(a) = P(b) = 1$ and $P(a + b) = \frac{16}{5}$, so $I_2 = \frac{6}{5}$; the value exceeds 1 because the readout is unnormalized intensity, not probability. A quartic readout $|A|^4$ gives $I_3 = 36$ when all three amplitudes equal 1, and an additive readout gives $I_2 = 0$. Commutativity of the coefficients is not used: the identity holds for matrix amplitudes with the conjugate-transpose readout as well. The condition $I_3 = 0$ does not distinguish quantum from classical models, since the additive readout satisfies it too [REF-QM-018].

19.9 Reconstruction audit matrix

The following matrix states the present programme boundary. `Declared` means the structure can be added consistently in an optional host; it does not mean the root selects it.

Feature	Root finite calculus	Optional carrier	Reconstruction obligation
finite state/evidence sets	available	available	specify operational interpretation
convex mixtures	not intrinsic	declarable	justify mixing and normalization
non-simplex state space	absent	declarable	prove from operational principles
effects/instruments	absent	declarable	connect to measurement packets (one finite qubit adapter: CH19-P13, CH19-P14)
complex Hilbert space	absent	declared in Ch. 5	show why real/classical alternatives fail
$U(1)$ phase	absent	declared	connect phase to operational alternatives (integer cochains and rational phases: CH19-P11, CH19-P12)
tensor composition	absent	not fixed	state product rule and locality axiom (CP of §19.2 forces classical composite states to be joint distributions)
contextual empirical model	absent	possible	supply cover, outcomes, distributions
continuous reversible dynamics	absent	possible	supply topology/group and derive relevance
purification	absent	possible	state exact axiom and countermodels
Born rule	absent	not selected	prove under named assumptions

Feature	Root finite calculus	Optional carrier	Reconstruction obligation
CHSH Tsirelson bound	not derived from the primitive calculus	$S^2 \leq 8$ under CH19-O02's supplied finite Hilbert, local-observable, tensor and trace assumptions	deriving or selecting those assumptions remains a reconstruction obligation

This table makes a classical no-go verdict meaningful. If every structure actually constructed before the optional branch has a faithful simplex model, then no quantum conclusion follows at that stage. The verdict is not that quantum theory is false; it is that the current premises underdetermine it.

Under the four-tier ambition taxonomy, this audit matrix establishes that:

- **Tiers 1 & 2 (Syntactic & Faithful Operational Representation)** are satisfied for discrete carrier models (such as permutation isometries and integer cochain lattices over $\mathbb{Z}$).
- **Tier 3 (Reconstruction Under Explicit Principles)** is the exact function of the table above: cataloging which physical structures remain unforced and must be adjoined as explicit operational axioms.
- **Tier 4 (Unique Selection From Primitive Kernel)** is refused as an ambition for quantum theory: the monograph does not claim that primitive difference forces complex Hilbert spaces, Gleason frame uniqueness, or the Born rule, and the finite import witnesses of Appendix D are not proofs that no derivation could (CH19-N01).

19.10 A classical comparison model

Take a finite set Λ of hidden states. Let normalized states be probability distributions $p \in \Delta(\Lambda)$, effects be functions $e : \Lambda \to [0, 1]$, and transformations be stochastic matrices. Measurement probability is

$$\Pr(e \mid p) = \sum_{\lambda \in \Lambda} e(\lambda) p(\lambda).$$

This model supports finite probability, observers with coarse presentation maps, memory, stochastic instruments, and measurement verification records. Its state space is a simplex. Any axiom package satisfied by this model cannot by itself distinguish quantum theory from classical theory.

Thus a proposed quantum principle should be tested on this model first. If it passes, either the principle is too weak or it contributes only to a shared operational foundation. That shared contribution may still be valuable, but it should not be labelled a quantum discriminator.

No-signalling is one such shared principle. With the Cartesian composite outcome set and product effects, summing any local partition of unity on one side returns the other side's marginal statistics. Because the PR box of §19.11 is also nonsignalling, no-signalling alone is not a discriminator either.

19.11 A post-quantum comparison

A nonsignalling correlation such as a PR-box fixture can exceed classical and quantum CHSH bounds while respecting nonsignalling [REF-QM-009; REF-QM-010]. Therefore nonlocality or contextuality beyond a classical model does not suffice to identify quantum theory. A reconstruction

needs principles that exclude both classical and post-quantum alternatives in the intended domain. A rational violation of the classical bound is cheap: with $P(a,b \mid x,y) = \frac{1}{4}(1 + ab\,E_{xy})$ on outcomes ± 1 and $(E_{00}, E_{01}, E_{10}, E_{11}) = (\frac{4}{5}, \frac{4}{5}, \frac{4}{5}, -\frac{2}{5})$ the model is nonsignalling with CHSH value $\frac{14}{5}$, and it is exactly $\frac{3}{5}$ of the PR box plus $\frac{1}{5}$ of the shared-bit model $a = b$ plus $\frac{1}{5}$ of uniform noise. A value below $2\sqrt{2}$ does not by itself place a model in the quantum set.

The fixture should be used operationally: list inputs, outputs, conditional probabilities, normalization, nonsignalling equalities, and the correlation value. Do not refer to "strong contextuality" without specifying which formal definition and support structure are being used. CH19-P05 is such a fixture: it lists the four contexts, the conditional probabilities, normalization, agreement on overlaps, the CHSH value 4, and strong contextuality in the sense of §19.1. It also establishes the classical bound: every noncontextual model on the CHSH cover has CHSH value at most 2. CH19-O02 and Appendix J.4 prove the quantum bound under their separately supplied finite Hilbert-space, local-observable, tensor-product and trace assumptions. CH18-N01 retains the distinction between that conditional theorem and reconstruction from the primitive calculus.

19.12 Quantum measurement adapter

If the optional Hilbert carrier is adopted, a finite quantum instrument can be represented by completely positive maps indexed by outcomes whose sum is trace preserving [REF-MET-006; REF-MET-007]. A measurement packet can then be populated as follows:

- interaction: system–apparatus channel or dilation, when supplied;
- registration: outcome label and apparatus record;
- witness: instrument outcome linked to the declared effect;
- value map: outcome-to-reported-value map;
- state change: normalized post-measurement state when defined;
- calibration/uncertainty: empirical characterization of the instrument;
- metrological-traceability evidence: the calibration hierarchy and uncertainty contributions, when such traceability is claimed;
- provenance/re-access: implementation, raw-record, transformation, evaluator, and version records.

This is an adapter specification, not a derivation of instruments. It also shows why a POVM alone is not the whole measurement packet: a POVM gives outcome probabilities but not in general the conditional state change, calibration history, uncertainty budget, or authority rule.

`WDP-CH19-O04` **Finite proposition (complete finite instrument replay).** Grouped finite Gaussian-rational Kraus instruments preserve their declared probabilities, post-states and canonical dilation. On a common input/output carrier, outcome repeatability is equivalent to $K_{jb}K_{oa} = 0$ for every $j \neq o, a, b$. Independent replay checks the complete registration, witness, accepted observation, indication, calibration enclosure and declared documentary hierarchy against actual inputs. Documentary traceability does not authenticate a physical standard or select the Born readout.

CH19-P13 and CH19-P14 give one finite adapter, for a qubit over $\mathbb{Q}(i)$ in Kraus form, with the least structure that states it. A state is a finite mixture $\rho = \sum_j w_j |v_j\rangle\langle v_j|$ with rational $w_j \geq 0$ and $\sum_j w_j \langle v_j, v_j\rangle = 1$; positivity is part of the declaration, so no spectral theorem (import 3) is used. An instrument is a family of 2×2 matrices K_o over $\mathbb{Q}(i)$ with $\sum_o K_o^\dagger K_o = 1$, the trace-preserving condition; no Stinespring dilation and no Choi matrix is used.

WDP-CH19-P13 **Finite proposition (probabilities and state change of the qubit adapter).** The adapter declares the outcome probabilities $p_o := \sum_j w_j \langle K_o v_j, K_o v_j \rangle$ for a state given as a list of weights and vectors. Then $p_o = \sum_j w_j \langle v_j, E_o v_j \rangle$, the trace $\mathrm{tr}(E_o \rho)$ of the effect $E_o = K_o^\dagger K_o$ against the mixture, the p_o are nonnegative, and $\sum_o p_o = 1$. For $p_o > 0$ the post-measurement state $K_o \rho K_o^\dagger / p_o$, the mixture $\{(w_j/p_o, K_o v_j)\}$, is again a state of the same form. The probabilities are an empirical model on the one-measurement cover, so by CH19-P04 the table of declared probabilities is the image of a finite table of accepted measurement packets whenever every outcome can be reported. Appendix J now supplies independent native replay through the full declared measurement packet; the historical Lean qubit theorem below does not formalize that additional replay. The rule $p_o = \mathrm{tr}(E_o \rho)$ is the declared readout of the adapter, not a derived one (CH19-N01). That ρ does not depend on the decomposition of the mixture is not used.

Proof. The Gram identity $\langle Kv, Kv \rangle = \langle v, K^\dagger K v \rangle$ and completeness give $\sum_o \langle K_o v, K_o v \rangle = \langle v, v \rangle$, so $\sum_o p_o = \sum_j w_j \langle v_j, v_j \rangle = 1$. Nonnegativity is that of $\langle x, x \rangle$ over $\mathbb{Q}(i)$, and the trace of the post-state is $\sum_j w_j \langle K_o v_j, K_o v_j \rangle / p_o = 1$. □

WDP-CH19-P14 **Finite proposition (a POVM does not determine the instrument).** The Lüders instrument $K_o = |o\rangle\langle o|$ and the measure-and-prepare instrument $M_o = |w\rangle\langle o|$ with $w = (3/5, 4/5)$ are both complete and have the same effects $E_o = |o\rangle\langle o|$, so they give the same outcome probability on every state. On the input $(1, 0)$ the outcome 0 leaves $(1, 0)$ under one instrument and $(3/5, 4/5)$ under the other, and the two vectors are not parallel.

Proof. Direct computation over $\mathbb{Q}(i)$: $M_o^\dagger M_o = \langle w, w \rangle |o\rangle\langle o| = |o\rangle\langle o|$. If $(1, 0) = c(3/5, 4/5)$ then $c \cdot 4/5 = 0$, so $c = 0$ and $1 = 0$. □

So a POVM is not the whole measurement packet already at this size, as the adapter specification above says. Calibration, uncertainty and authority are carried by neither instrument, and CH19-P03 shows that the calibration chain of a packet does not reach the image. Appendix I extends the native adapter to arbitrary finite input/output dimensions and grouped Kraus maps, with a canonical outcome-resolved isometric dilation and independent verification record replay. Appendix J.6 supplies the repeatability criterion and describes replay of the complete declared packet, including calibration arithmetic and documentary traceability. These are the finite scopes recorded in CH19-O04; documentary checks do not authenticate physical standards or identify an actual apparatus. Unitary completion, general dilation-independence and selection of the Born rule remain outside the stated result (CH19-N01, imports 4 and 6).

Chapter 19A — From certified observation to admissible transformation

19A.1 What observation determines

This chapter starts after R8–R10. Declare $K = \mathbb{Q}(i)$, a carrier $V = K^d$, $d \geq 1$, and $h(x, y) = x^* H y$ with $H = H^* > 0$. The preceding observer construction supplies positive semidefiniteness; requiring zero radical then supplies positive definiteness. Nondegeneracy without that positivity premise would be insufficient. The field and form are admitted data, and the probability rule remains R10's declared readout.

For $f, y \neq 0$, write $r_f = h(f,f)$ and

$$\tau_f([y]) = \frac{N(h(f,y))}{r_f r_y}, \qquad N(z) = \bar{z}z.$$

For $d \geq 2$, $\mathbb{P}(K^d)$ is countably infinite; for $d = 1$, it has one ray. On the infinite carrier, a finite example cannot certify an arbitrary black-box permutation.

WDP-R12-P01 — Finite-dimensional proposition (pure-ray separation). The table is independent of both representatives, lies in $\mathbb{Q} \cap [0,1]$, and $\tau_f = \tau_g$ exactly when $[f] = [g]$. Indeed, with $\alpha = h(f,y)/r_f$,

$$r_f h(y - \alpha f, y - \alpha f) = r_f r_y - N(h(f,y)).$$

Positivity gives the bounds and makes equality at one equivalent to collinearity. Equal tables agree at the ray $[f]$; representative independence gives the converse. The d^2 probes $e_j, e_j + e_k, e_j + ie_k$, $j < k$, also suffice by polarization of $y^* H f f^* H y / r_f$. This is a sufficient tomography menu, not a minimal pure-state measurement claim.

19A.2 Preservers, branches and projective uniqueness

WDP-R12-O01 — Conditional result (global preserver classification). For $d \geq 3$, a bijection of the entire ray space preserves orthogonality in both directions exactly when it has the form

$$T[f] = [M\sigma(f)], \qquad M^* H M = c\,\sigma(H), \qquad c \in \mathbb{Q}_{>0}, \quad \sigma \in \{\mathrm{id}, \mathrm{conj}\}.$$

The branch and matrix modulo nonzero scalar are unique, and the resulting map preserves all transition probabilities.

The proof reconstructs projective lines by double perpendiculars, then normalizes a coordinate frame. Preservation of affine lines forces a single coordinate map to preserve addition and multiplication. A field automorphism fixes $\mathbb{Q}$ and sends i to i or $-i$. Equality of the zero kernels of the transported and original forms forces one positive rational multiplier. Basis, sum and mixed rays prove uniqueness. The full coordinate proof is in the preserver supplement, §3. The related algebraic literature is recorded as REF-PRES-001; the written proof here does not rely on an unread classification theorem.

The projective linear-similarity group sits in a split extension by C_2. A splitting is obtained after choosing an h-orthogonal K-basis, where the Gram matrix is rational diagonal. Bare coordinate conjugation in an arbitrary basis need not preserve H. The conjugate branch is an antilinear map on states; it is not the operator adjoint. Nor is a similarity automatically an R11 unitary.

19A.3 Phases and the exact reachability obstruction

WDP-R12-P02 — Finite-dimensional proposition (corrected phase group). Replacing M by aM changes its multiplier to $N(a)c$. Equal multipliers leave

$$K^1 = \{a : N(a) = 1\} = \{-1\} \cup \left\{ \frac{1 - t^2 + 2it}{1 + t^2} : t \in \mathbb{Q} \right\}.$$

The inverse parameter is $\mathrm{Im}\, a/(1 + \mathrm{Re}\, a)$. This group is infinite: $(3 + 4i)/5$ is already outside the four Gaussian-integer units. Only the torsion subgroup is μ_4. Thus a chosen unitary lift can have

cocycles in K^1 outside μ_4; Chapter 5's cubic phases retain their separate cyclotomic carrier, and Chapter 18's four-unit theorem retains its integral hypothesis.

WDP-R12-P03 — Conditional arithmetic result (norm realization). The integer two-squares theorem [REF-NORM-001, Theorem 9.10], applied to a reduced numerator and denominator, gives

$$r \in N(K^\times) \iff v_p(r) \in 2\mathbb{Z} \text{ for every prime } p \equiv 3 \pmod 4,$$
$$\mathbb{Q}_{>0}/N(K^\times) \simeq \bigoplus_{p\equiv 3 \pmod 4} \mathbb{Z}/2.$$

Denominator valuations are included. A positive witness exhibits two rational coordinates; a negative witness exhibits a prime and its exact odd valuation. A resource limit is neither witness. The Gaussian Hadamard

$$B = \frac{1+i}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad B^*B = I, \quad B^2 = iI$$

implements the conventional Hadamard ray action. This corrects any reading of §17.11 that rules out that action just because the real normalization $1/\sqrt{2}$ is absent. Order-three Fourier normalization fails the norm test; its cubic root of unity is a second, distinct missing scalar. The conventional T eigenvalue ratio also remains outside K.

WDP-R12-O02 — Conditional result (complete unitary ray criterion). A supplied positive form admits a unitary taking $[f]$ to $[g]$ exactly when $r_f/r_g \in N(K^\times)$. Antiunitaries do not enlarge these orbits. For sufficiency choose a with that norm, put $v = ag$, and, unless $f = v$, set

$$w = f - v, \qquad \beta = h(w, f), \qquad Ux = x - w\, h(w, x)/\beta.$$

Here $h(w, w) = \beta + \bar{\beta}$; positivity ensures $\beta \neq 0$. Expansion establishes $h(Ux, Uy) = h(x, y)$ and $Uf = v$. The inverse replaces β by $\bar{\beta}$. This construction uses no square root or assumed matching of complementary Gram frames.

A similarity normalizes to a unitary precisely when its multiplier is a Gaussian norm. Its determinant gives $N(\det M) = c^d$, so this holds in odd dimension. In dimension four the supplement's exact matrix satisfies $M^*M = 3I$, preserves every probability, and crosses unitary norm classes. The whole similarity group therefore cannot be identified with the reversible unitary operations. Classifying the multiplier subgroup for every supplied form is a separate question.

19A.4 Postselection and the dimension-two boundary

WDP-R12-P04 — Finite-dimensional proposition (the specified Lüders branch). The operator $\Pi_g = gg^*H/r_g$ is an h-selfadjoint idempotent. The instrument $\{\Pi_g, I - \Pi_g\}$ is complete, and

$$\Pr(\Pi_g \mid f) = h(\Pi_g f, \Pi_g f)/r_f = \tau_f([g]).$$

Its conditional ray is $[g]$ only at positive probability; a zero branch has no normalized conditional state. This probability is not an optimal operating cost. For $f = e_1, g = (0, 1, 1, 1)$, the direct branch vanishes and the norm class obstructs unitarity, yet projections through $f + g$ reach g with probability 3/16. Admissible operations matter.

CM-W1. In dimension two, orthogonality is a perfect matching. Swapping the axes and exchanging the orthogonal pair $\{[1, 1], [1, -1]\}$ with $\{[1, 2], [2, -1]\}$ in both directions, while fixing every other ray, preserves orthogonality. It changes a transition probability from 1/2 to 4/5 and is not a

semisimilarity. The six-ray pocket is closed under orthogonal complement, which is what justifies this extension to the entire carrier.

WDP-R12-P05 — Finite-dimensional proposition (full-probability qubits). In dimension two the full transition table does force a projectively unique semisimilarity. Normalize the images of two orthogonal axes and the plus ray. The imaginary plus ray then has image with coefficient i or $-i$. After fixing that branch, the four reference readouts recover the norm, real part and imaginary part of every ray coordinate. The supplement gives the argument for every positive Hermitian form. This settles the proposed qubit conjecture at its declared field.

19A.5 Stabilizers and the price of sufficient observation

WDP-R12-O03 — Conditional result (three distinct consolidation laws). For named readout functions E and a declared transformation group G, let G_E fix every named function pointwise and let $\mathrm{Fix}(G)$ be the functions fixed by every group element. Then

$$E \subseteq \mathrm{Fix}(G) \iff G \subseteq G_E.$$

This antitone Galois pair concerns the actual labelled observations. For a unital h-dagger-closed K-operator algebra A, one also has $A'' = A$ and the commutant of the unitaries of A' equals A. Invariant orthogonal projections on the amplified carrier prove the bicommutant identity; rational Cayley transforms recover the algebra from its unitaries. These are algebraic statements without topological closure.

Finally, for an explicitly supplied finite rational linear defect space R and linear observation map Q_E, arbitrary additional rational rows achieve injectivity with the sharp count

$$k = \dim_{\mathbb{Q}}(R \cap \ker Q_E).$$

Each row removes at most one kernel dimension; extended duals of a kernel basis attain the bound. This is the injectivity case of Appendix T. Restricted instrument shelves, original-carrier realization, nonlinear transformation sets and operation costs retain their separate obligations.

For $d \geq 2$, an injective menu has trivial projective pointwise stabilizer; the converse is false. In dimension one, the identity and conjugate semilinear classes act identically on the single ray, so faithfulness requires quotienting this action kernel. The supplement supplies two exact three-dimensional effects with trivial stabilizer and two distinct rays reporting $1/2$ for both. A residual conjugation branch is not automatic: an imaginary cross-term probe detects it.

19A.6 From a supplied derivation to its generator

WDP-R13-P01 — Finite-dimensional proposition (inner compatible generators). Let $\delta : M_d(K) \to M_d(K)$ be K-linear and satisfy Leibniz. For matrix units define

$$S = \sum_j \delta(E_{j1}) E_{1j}.$$

Differentiating $E_{kl}E_{j1} = \delta_{lj}E_{k1}$ gives $[S, E_{kl}] = \delta(E_{kl})$, hence $\delta = [S, \cdot]$. If also $\delta(A^\dagger) = \delta(A)^\dagger$, then $S + S^\dagger$ commutes with all matrix units and is a rational scalar matrix. Replace S by $(S - S^\dagger)/2$, and put $H_0 = iS$. Thus

$$\delta = -i[H_0, \cdot], \qquad H_0^\dagger = H_0,$$

with H_0 unique modulo a rational scalar. The supplied finite matrix-unit table can be checked independently for linear extension, every Leibniz identity, the dagger law, and the submitted generator. This is an inner-derivation result, not Stone's theorem. No time parameter, exponential flow, physical Hamiltonian or authorization follows.

19A.7 Dependencies and evidence scope

R12 uses the admitted R8 involution, R9 positive form and R10 readout. It compares its preservers with R11's operations. Chapter 5's phases and Chapter 18's integral carrier are comparisons with different coefficients, not hypotheses that turn K^1 into μ_4. Only the linear completion clause uses Appendix T. R13 uses a declared matrix algebra, adjoint and derivation; it does not assume a selected physical dynamics or need the preserver classification as an algebraic premise.

The full proof supplement, supplement evidence rows, and evidence guide separate written proofs, the Lean model, concrete Rust certificates, and Python exploration. These supplement IDs do not alter the historical 154-row canonical inventory or its meaning of status coverage. Physical reconstruction, branch selection, global synthesis, source refinement and continuum completion retain their existing open status.

Part V — Relative scope and the open programme

Chapter 20 — Relative completeness for observer-indexed presentations

20.1 From "cosmogenesis" to a defensible target

The earlier title "witness-relative cosmogenesis" promised more than its formal content supplied. The actual construction concerns the status of a query relative to a finite observer profile. We therefore treat the chapter as a calculus of **relative completeness for observer-indexed presentations**.

Let an observer profile Π contain a grammar of admissible queries, an input domain, a presentation map, an evaluator, a resource bound, and a policy. For a query q, define a status only after checking that q is well typed and within profile.

20.2 Status algebra

A useful finite status set is:

$$\mathsf{Status}_\Pi = \{\mathsf{Proved}, \mathsf{Refuted}, \mathsf{Undetermined}, \mathsf{OutsideProfile}, \mathsf{Refused}\}.$$

The terms require disjoint typing rules.

- `Proved`: a declared proof/evidence predicate is satisfied.
- `Refuted`: a declared counterevidence predicate is satisfied.
- `Undetermined`: the query is in profile and well typed, but neither terminal predicate is established within the declared resources.
- `OutsideProfile`: the query is meaningful in the ambient language but is not represented by this profile.

- `Refused`: processing is prohibited or unsafe under a separate policy despite the query being syntactically processable.

An ill-typed string should be handled by a parse/type error before this status set, rather than simultaneously being `OutsideProfile` and `Refused`.

`WDP-CH20-D01` **Definition (profile-relative completeness).** An evaluator is complete for a finite query family Q relative to profile Π when every $q \in Q$ receives exactly one terminal status according to the declared status rules, and the evidence attached to `Proved` and `Refuted` passes its verifier.

This is a finite coverage property. It says nothing about all mathematical truths, all meaningful questions, or reality independent of the profile.

20.3 Exclusivity and coverage

To prove that statuses are exclusive, the rules must exclude overlap. In particular, consistency conditions should prevent both `Proved` and `Refuted` for one query under one fixed logic and evidence base. To prove coverage, every in-profile query must either terminate or be assigned `Undetermined` under a bounded procedure.

`WDP-CH20-O01` **Finite proposition (resolved finite target).** A declared finite Boolean semantic model with admission, policy and bounded independent evidence replay has five exclusive and exhaustive statuses; injective value-preserving profile adapters transport verified evidence and rerun target evaluation rather than assuming status invariance under changed policy or resources. The proof, native ownership and scope are given in Appendix I.

20.4 Observer-relative does not mean arbitrary

Profiles can be compared. If Π_1 embeds faithfully into Π_2, one can ask whether proofs transport, whether counterexamples remain valid, and whether an `OutsideProfile` query becomes expressible. These are naturality questions. A status is relative to a profile, but constrained by its evidence and transport laws.

Formally, a theory profile transport specification from a target theory T to a host substrate G uses an encoder E_T and a left-inverse decoder D_T, satisfying four core commutation properties. The decoder is inverse on the encoded image; no surjectivity onto all host states is required:

1. **Source round trip:**
$$D_T(E_T(s)) = s \quad \text{for all } s \in \mathrm{State}_T\,.$$
2. **Checker fidelity:**
$$\mathrm{Check}_T(\pi, \varphi) \iff \mathrm{Check}_{G,T}(E_T\pi, E_T\varphi).$$
3. **Operational step commutation:**
$$E_T(\mathrm{step}_T(s, a)) \simeq \mathrm{step}_{G,T}(E_T s, E_T a).$$
4. **Observational commutation:**
$$\mathrm{Obs}_G(E_T(s)) = E_O(\mathrm{Obs}_T(s)).$$

Across profile boundaries, the following cumulative specification ladder adds obligations at eight levels:

$$\text{Syntactic} \longrightarrow \text{Judgmental} \longrightarrow \text{Operational} \longrightarrow \text{Observational}$$
$$\longrightarrow \text{Probabilistic} \longrightarrow \text{Causal} \longrightarrow \text{Higher Coherence} \longrightarrow \text{Semantic Adequacy}.$$

Each level retains the earlier obligations and adds its own commutation conditions and falsifiers. The arrows describe that cumulative specification, not implications between independently stated properties. Proving round-trip syntactic encoding (Level 1) does not imply operational step commutation (Level 3) or semantic truth transport (Level 8).

20.5 No root observer

The calculus contains no theorem selecting a privileged observer whose profile captures every other profile. A candidate root observer would require a initial factorisation property in a specified category and a proof that the necessary morphisms exist uniquely in the relevant sense. Neither is present.

`WDP-CH20-N01` **Nonclaim.** Relative completeness is not absolute omniscience, a truth-decider, a Gödel escape, world creation, or cosmogenesis.

20.6 Faithful representation and certificate completeness

GEN-WDP-49 (conditional theorem). Fix an effectively encoded original problem P, sound initial witness decoder, and constructive coverage of its solutions. Each admitted representation change supplies a backward map preserving accepted witnesses, a forward map preserving solution coverage, and equality of their decoded original answers. Both laws compose along every admitted finite history. Therefore every accepted terminal witness decodes to a solution of P, and

$$\mathrm{Sol}(P) \neq \emptyset \iff \exists \tau : \mathrm{Check}(\mathrm{Encode}(P), \tau) = \mathsf{Solved}.$$

Here `Solved` is a verified positive existential claim, represented by `Proved`. Soundness, coverage and the existence of enough admitted resources are explicit hypotheses. The theorem does not reinterpret the status-coverage definition of §20.2. In particular, a sound backward adapter can discard every source solution; only forward coverage permits negative-result transfer. The complete derivation establishes composition, original-answer soundness, certificate existence and fair search separately. The new core-only Lean module establishes those laws without assuming the combined end-to-end conclusion.

20.7 A constructive finite constraint foundation

GEN-WDP-50 (finite theorem). Given finite variable domains and finite tables of allowed local assignments, add full-domain unary tables at uncovered variables. Choose one permitted row from each table. Compare occurrences of each variable using integer indicator differences against its first occurrence. Zero raw defect is equivalent to agreement on every overlap. Gluing a compatible family gives a unique original assignment satisfying every table; restriction gives the inverse map. Thus original solutions and compatible local families correspond bijectively, including the empty-domain, empty-table, nullary and empty-variable cases.

The encoder reads supplied local tables and domains; it does not consume global solution bits. Bijective renaming, exact joins, checked lossless decomposition and total deterministic auxiliary

variables supply constructive forward/backward witness maps. The local defect domain is a constrained finite set, not automatically a vector space. A compressed zero detector must be adequate on this actual set. Choosing the whole ambient space as formal repairs cannot manufacture an admissible solving endpoint.

The finite replay compares all 4,096 Boolean triangle relation systems with direct original-assignment enumeration, covering 32,768 local families. Both the consistent and globally inconsistent cycles have nonempty local relations. These finite checks, the written all-finite proof, and the new Lean gluing/indicator/product theorems retain their separate evidence scopes.

20.8 Recursive preservation, search and stopping

GEN-WDP-51 (conditional finite-history theorem). Re-entry retaining the original problem, assumptions, authorization requirements and witness interpretation preserves that anchor under finite composition. If each counted operation consumes one unit of a shared fixed ceiling B, a history of n operations has spent count s_0+n at most B. Re-entry cannot reset that count. Explicit new budget grants are separate from re-entry; records of requirements or grants do not establish their institutional legitimacy.

A fair enumeration of a declared effective certificate language, with terminating checks, eventually finds an admitted finite solving witness when one exists and admission/resources persist. For a finite local product of size M, canonical enumeration with at most one optional proposal before each canonical candidate needs at most 2M candidate checks, plus the admitted transport checks. This is a search-unit bound, not a wall-clock bound. An exhaustive negative result requires retained coverage; budget exhaustion and interruption remain `Undetermined`. Rejected proposals do not change an accepted mathematical state. Completeness does not follow just from strict residual descent or an arbitrary continuation policy.

The reference checker independently replays supplied witnesses, transport parents, enumeration and fuel. General source refinement, future-operation congruence for new compressed states, guarded-history coherence and physical reconstruction remain separate obligations. The full statement, constructive proofs and replay instructions are in the supplement.

20.9 Certified progress, dynamic compression and resource claims

GEN-WDP-52 (finite theorem). From a declared finite transition/report model, compute reachable nonempty beliefs and their universally enabled actions. The target attractor is exactly the winning region. A natural-ranked strategy certifies guaranteed finite success; an initial-containing, target-disjoint region with an actual internal successor for every enabled action certifies loss. Its dead ends remain in that region. With positive rational prices, finite-horizon minimax constructs an exact optimum and its own strategy rank. Independent upper/lower certificate replay establishes that optimum without rerunning the producer. Initial membership, full winning partition, original model targets and zero terminal values are required boundaries.

GEN-WDP-53 (finite theorem). A partition of this belief arena preserving target truth, enabled actions, exact costs and successor-block sets preserves winning and every achievable finite budget under its two-sided quotient map. Iterative signature splitting computes the coarsest such refinement of a supplied partition. Congruence is sufficient; accidental agreement of a single current

answer does not establish it. Quotient strategies lift, and class-constant fine strategies descend. The mathematical quotient does not create a physical observer realizing its distinctions.

GEN-WDP-54 (conditional resource and guard result). Optional operations preserving old outcomes and prices cannot worsen the optimum. A restrictive guard can raise finite cost or destroy winning, but a failed rejection-rule transfer alone establishes neither loss nor original-problem impossibility. An optimum above budget establishes budget exhaustion; an expensive feasible strategy is only an upper bound. Reports and certificate refusal preserve their distinct meanings. Original-answer soundness and negative-result coverage still require §20.6's actual maps; neither a loss verification record nor a supplied authorization flag supplies them.

The full written proofs and retained counterexamples are in Appendix S below; the proof/evidence map binds 39 new selected Lean declarations and independent Python replay to their separate scopes. The local-acceptance analysis supplies the concrete local/global problem and overlap question; Projection Obstruction/WDP identifies the information lost by a report; checked repair verifies the proposed witness and its bound task; the admission and continuation protocol and the discrete deformation steps must still supply admissible realized changes and continuation. A change of report cannot itself repair a nonzero original obstruction. This connects the owners through preserved questions and checked maps, without equating their different diagnostics or granting authority.

20.10 Exact progress quotients and checked task binding

GEN-WDP-55 (finite structural theorem). An exact operation-preserving quotient of the belief arena exists if and only if presentation fibers agree in terminal truth and retained reports, enabled action sets, positive rational prices, and successor class sets. Necessity is for this structural definition; accidental equality of one task's optimum is weaker. An explicit representative section constructs the quotient. All achievable budgets, winning and optimal cost are preserved and reflected. Retained resolution IDs need actual common task-witness binding; equal target booleans alone do not supply it.

The self-contained sharp theorem reproves the attractor, certificates, finite optimal-cost horizon and coarsest refinement, separating standard game theory from the WDP evidence specifications. The bounded calculus adapter replays the actual archived antiderivative through its supplied task checker. It resolves at operating cost three; a forever-unhelpful proposal-only menu loses without making the integral unrealizable. No missing proposal-protocol stage or native review binding is asserted. Source-sound rejection-rule transfer, budget cutoff and optional shortcut setup costs retain their separate rejection tests.

20.11 Sufficient observation and conditional physical locality

GEN-WDP-56 (finite and conditional theorem family). For admitted binary errors E and repair subspace B in a syndrome kernel, syndrome-only recovery exists exactly when each admitted syndrome fiber lies in one B-class. Added linear observations must separate every forbidden same-syndrome difference; the minimum is a finite hitting-suite problem. If E is a linear admitted-error subspace, $B \subseteq E \cap \ker Q$, and arbitrary additional linear rows are available, the sharp count is $\dim(\ker(Q|_E)/B)$. A supplied physical quasi-local readout with tail norm beta distinguishes equal joint-marginal states by at most 2 beta. Reconstruction error alpha and conditioning kappa obey delta <= 2 alpha + kappa eta, where eta is the actual readout separation.

The complete corrected locality derivation establishes these statements and retains Bell-marginal,

gauge-cycle, CAR exchange and false error-floor controls. A locality upper bound does not give a physical error floor. A finite recovery theorem does not manufacture a device adapter. The analytic Lieb–Robinson hypothesis is imported separately from the new finite observation/conditioning deductions.

20.12 Trotter smearing, time consistency and cochain descent

GEN-WDP-57 (finite-carrier time theorem and exact adapter theorem). For the supplied full sixteen-state, four-site CAR trace, response matrices have size 8M and retain their declared 1/(L(2M)^2) quadratic normalization. Adjacent temporal pairing bounds the staggered average by D/(4M). The all-M rank-one compressed contribution tends to zero under a source-derived linear growth bound. The complete quadratic has a conservative explicit A/M error envelope against the fixed four-site continuous-time covariance. That covariance is positive for the declared real sources by its finite thermal spectral representation; a strictly positive supplied margin gives eventual strict positivity with floor(A/m)+1, not an unqualified ceiling.

The complete corrected Trotter/cochain derivation provides the analytic proof, a source-derived constant-profile margin, four independently checked finite inertia witnesses, and exact rational schedules. A cochain adapter descends iff it kills the declared repair subspace. A low-pass projection alone does not ensure this; weighted subdivision requires its explicit norm convention. The staggered cochain class is not automatically the physical CAR repair relation. Neither four spectra nor a fixed spatial carrier's time limit closes all-M inertia or the spatial continuum programme.

Written analytic proofs, core-only Lean certificate consequences and executable exact witnesses are bound separately in the derivation evidence map. The new bridge IDs leave the 154 canonical rows and GEN-WDP-O01–O05 at their existing scopes.

20.13 Bridge Operad Equivariance and Relative Completeness Across Observers

`WDP-CH20-T02` **Theorem (Bridge Operad Descent Equivariance).**
Let $\mathcal{O}_1, \mathcal{O}_2$ be two bounded observers with relational correspondence span K_{12}. If the four congruence annihilators $(\alpha_{\text{core}}, \alpha_{\text{equiv}}, \alpha_{\text{designation}}, \alpha_{\text{defect}})$ evaluate to zero on candidate cochain ω_{12}, then the induced quotient representations descend equivariantly:

$$q_1 \sim_{\mathcal{B}_{12}} q_2 \iff [R_1] \cong_{K_{12}} [R_2]$$

without requiring the existence of a global root observer.

`WDP-CH20-C02` **Corollary (Multi-Observer Consistency Without Root Observer).**
Intersubjective agreement across a network of observers is certified pair-by-pair through bridge operad compositions τ_{ij}, establishing profile-relative consistency while preserving `WDP-CH20-N01` (refutation of global root observer).

20.14 From an admitted distinction to witnessed reasoning

The presentation <> acquires inferential meaning through an explicitly admitted carrier, facts, rules and interpretation. With no facts or rules, the everywhere false model excludes every derivation. Admitting an atom as a fact changes that result. The glyph therefore does not uniquely select the resulting logic.

For the first reasoning profile, choose a finite definite Horn theory $T = (A, \Gamma, R)$. A rule has a finite list of premises and one conclusion. The judgment $T \vdash a$ is generated by facts in Γ and rule

firing from already derived premises. A model makes the facts true and preserves truth under every admitted rule. The following claims have separate bindings in the ledger's reasoning supplement.

- `WDP-RSN-01`: every derivation is true in every model, by induction on the proof.
- `WDP-RSN-02`: the derivable atoms themselves form the least model, hence derivability equals truth in all models. A countermodel proves non-entailment.
- `WDP-RSN-03`: atom substitutions preserving facts and rule consequences transport proofs; adding an already derived fact is conservative by applying soundness to the old derivability model. This is the reuse/cut rule.
- `WDP-RSN-04`: every checked trace entry has a derivation. Finite saturation inserts at most $|A|$ atoms; it returns a proof or a closed-set countermodel. A smaller exhausted budget yields no negative conclusion.
- `WDP-RSN-05`: encoding a directed edge $u \to v$ as $u \Rightarrow v$ and the start vertex as a fact makes proofs correspond to paths. Induction on proofs appends edges; induction on paths fires rules. A separate graph checker checks the decoded original-task witness.
- `WDP-RSN-06`: a same-report, different-answer pair prevents a decoder, even after deterministic post-processing, as in §14.8.
- `WDP-RSN-07`: for total deterministic dynamics the same argument applies to the whole function from finite action words to reports. An instantaneous collision alone does not establish this persistent limit.
- `WDP-RSN-08`: if the fact and rule semantics are valid in every admitted original world, every checked derivation has a true decoded interpretation. World adequacy is an explicit premise, not supplied by the inference checker.

The full derivation gives definitions, proofs, certificate formats, source bindings and falsifiers. The first fifteen root-library Lean declarations verify the abstract inference and report arguments. Section 20.15 adds executable checker refinement and formal graph transport. Finite saturation retains its written termination proof; whole-program Python/Rust refinement remains open. The historical monograph-wide Lean census is unchanged.

This establishes a conditional route from admitted distinctions to checkable reasoning and its information limits. It neither derives empirical premises nor authorizes action. Probabilistic or defeasible reasoning, full executable refinement, general source adapters and physical/institutional interpretation retain their named residual obligations. Finite Horn entailment is decidable; stronger self-reference limits cannot be inferred for it without changing the formal system and stating the additional hypotheses.

20.15 Executable refinement and kernel replay

The abstract trace now has a concrete executable realization in the root Lean library. A finite theory contains explicit atom names, fact indices and ordered rules. A certificate contains its source theory, goal and steps with indices into earlier conclusions. The executable checker validates the carrier, source and query, matches every referenced premise in order, and requires the final conclusion to equal the requested goal.

`WDP-RSN-09` **Theorem (Concrete proof refinement).** For every finite input, an accepted concrete step is abstractly justified. Successful replay refines the abstract checked trace, reversing the chronological list to match its prepend constructor. Hence an accepted proof certificate yields $T \vdash a$. The proof follows the executable recursion and establishes its invariant at every step, including empty and repeated premise lists.

WDP-RSN-10 **Theorem (Concrete countermodel refinement).** A model certificate checks every fact, every rule, vector length, source/query equality and a false value for the goal. These checks construct a model of T omitting a, so soundness gives $T \nvdash a$. Positive and negative certificates cannot both be accepted for the same task and query. A rejected or exhausted replay provides neither judgment.

WDP-RSN-11 **Theorem (Original graph transport).** For the edge-to-rule encoding of §20.14, derivability is equivalent to the independent inductive relation of directed reachability, including the zero-edge path. The two inductions are now formalized for arbitrary finite edge lists. Composing this equivalence with the concrete checkers turns accepted graph certificates into reachability or non-reachability theorems for the supplied graph.

WDP-RSN-12 **Construction (Foreign certificate replay).** Python and Rust export concrete task and certificate data to closed Lean proof obligations. The kernel checks each instance by reduction of the verified Boolean checker using `by decide`; an externally supplied success flag supplies no premise. Unicode scalar constructors keep names as data. The live Python adapter returns a logical judgment only after kernel acceptance. Its source hash and query bind the reported result to the generated obligation.

Seventeen new Lean declarations establish the executable and graph results. The scoped replay checks 2,048 two-atom Horn queries, 64 two-vertex graph queries, malformed Python obligations and freshly emitted native Rust cases. The full derivation and verification boundary separates the general Lean theorems from these bounded interoperability checks. Whole-program foreign-language equivalence, serialization correctness, compiler trust, production deployment, empirical adequacy and formal saturation bounds retain explicit obligations. Neither a closed proof nor a countermodel grants institutional authorization or supplies missing observations.

Chapter 21 — Falsifiers, nonclaims, and open obligations

21.1 Why falsifiers belong in the monograph

A finite programme can be unusually explicit about failure. Each theorem family should name a small countermodel that would refute it, and each interface should name the missing type that prevents composition. This is more informative than a list of aspirations.

21.2 Foundational falsifiers

- The full-symmetry nonselection theorem fails if a parameter-free equivariant selector exists for a moved target under the stated group action.
- A minimality or uniqueness claim for a separating response fails if its grammar, cost, and isomorphism relation are not all stated: under item count the grammar of CH03-P03 has infinitely many minimal classes, and the refinement that leaves one class still leaves a swap pair with no equivariant choice.
- A retention theorem fails if its proposed memory packet cannot distinguish a displayed pair of histories that the theorem says it retains.
- A "difference alone" theorem fails if its construction uses an unrecorded label, order, measure, topology, or observer profile.

21.3 Cochain falsifiers

- Quotient well-definedness fails if changing a representative changes the claimed value.

- Cohomology language fails if $d_1 d_0 \neq 0$ on the declared domain.
- A dual certificate fails if it does not annihilate every admissible repair or violates its norm constraint.
- Gauge invariance fails if gauge-related inputs give nonconjugate or otherwise inequivalent outputs under the stated invariant.
- A reported Φ_1 value for a weighted L^1 norm is uncertified unless a repair and a bounded circulation with equal values accompany it (CH04-P07).
- A path-discrepancy bound fails if two directed paths with common endpoints give $|\Delta| > c_\gamma \Phi_{1,w}([\delta])$ (CH04-P11).
- The signature bound fails if a class is exhibited with $\Gamma_2 > \|d_1\|_{J\to 1}\Phi_1$ on a window whose $d_1 d_0 = 0$ certificate verifies.
- Two implementations given one declared window, coefficient field, weights, and cochain must return one signature; a divergence refutes an implementation or exposes an undeclared assumption.
- A coefficient-free cohomology claim fails when changing the declared field changes the answer, as on the projective plane (CH05-P06).
- A claimed repair fails when it subtracts an element outside $\operatorname{im} d_0$, such as an orthogonal projection onto $\ker d_1$; the result lies in a different class, and the original class and its Φ_1 are not repaired.
- A claim that passing to a larger complex, attaching faces, or equal-weight subdivision lowers Φ_1 fails (CH04-P09); only deleting cells, changing declared weights, or changing existing edge values can lower it.
- A restriction system reported flat fails if some strict triple has $r_{b,a} r_{c,b} \neq r_{c,a}$, equivalently if $d_1 d_0 s \neq 0$ for some section (CH16-P05).

21.4 Time and probability falsifiers

- Succession as strict order fails on a directed cycle.
- Exact duration fails when two paths with the same endpoints have unequal declared sums.
- Clock adequacy fails under an admitted target interval that has no comparable reference readout or exceeds the stated error bound.
- Refinement invariance fails if transported atomic mass is not conserved.
- Born selection from normalization is refuted by the positive-power family in Chapter 9.

21.5 Observer and measurement falsifiers

- A legacy observer adapter fails when it silently drops a field used by a theorem or invents memory not present in the source packet.
- Observation validity fails when a record cannot be connected to its target proposition.
- Measurement validity fails when the measurand specification, result carrier and value map, calibration, declared uncertainty/error semantics, or required provenance/re-access record is absent. If metrological traceability is claimed, absence of the documented calibration hierarchy or any calibration's uncertainty contribution refutes that claim.
- Any implication from evaluator acceptance to authority is ill typed unless an authority bridge is declared.
- A self-knowledge completeness claim fails whenever the presentation map has a displayed nontrivial fiber.

21.6 History and quantum falsifiers

- A length-bounded history category fails when two admissible paths compose to a path beyond the bound.
- A star-sum theorem over a bare semiring fails when the infinite sum has no declared semantics.
- A zerosumfree noncancellation theorem fails in a carrier with nonzero additive inverses.
- A quantum characterization fails on any nonquantum GPT satisfying its proposed discriminator set.
- A contextuality/cohomology claim fails when only non-coboundary, not cocycle, has been shown.
- An eighteen-ray parity claim fails when some context contains two non-orthogonal vectors or some vector does not lie in exactly two contexts; an absorbed source's printed table failed the first test.
- A trace obstruction stated for every ring of characteristic zero fails over $\mathbb{Z} \times \mathbb{F}_2$; the hypothesis is $Nc \neq 0$.

21.7 Consolidated nonclaims

The monograph does **not** claim:

1. that Primitive Difference alone produces structure;
2. that witness implies memory, observer, or experience;
3. that retention uniquely produces time;
4. that finite duration is physical or continuum time;
5. that finite normalized weight is physical probability;
6. that normalization selects the Born rule;
7. that a quotient evaluator is a conscious observer;
8. that observer acceptance confers authority;
9. that a measurement packet is observer-independent;
10. that finite path sums are continuum or quantum path integrals;
11. that a cochain obstruction alone is quantum contextuality;
12. that declared complex carriers are derived from difference;
13. that contextuality is sufficient for quantum theory;
14. that the calculus reconstructs quantum mechanics;
15. that self-reference establishes awareness;
16. that profile-relative completeness is absolute completeness;
17. that any chapter creates worlds or establishes cosmogenesis;
18. that repository tests, formal elaboration, or publication status decides the interpretation of the mathematics;
19. that an obstruction signature or certificate selects a repair, stabilizes a residual, or routes a claim;
20. that combinatorial balance identities on finite complexes are physical conservation laws;
21. that a residual below an evaluator's tolerance is exact, infinite, or continuous, or that the tolerance status belongs to the class;
22. that a negative value of a declared phase-point frame, or the failure of $AB - BA = cI$ on a finite carrier, is a quantum signature, a computational resource, or a statement about physical position and momentum.
23. that a positive product-projection residual (Paper 000) represents a computational or logical failure; such residuals are expected and necessary during entangling gate operations, and a raw residual bound does not constitute an obstruction modulo repairs without a declared adapter

and repair space;

24. that nonzero physical realization error by itself prevents quantum-computing scalability; fault-tolerance threshold theorems show that scalable computation is achievable under local architectures whenever physical error rates fall below threshold.

21.8 Completed derivations and formal coverage

The nineteen registered open targets now have finite or conditional resolutions in Appendices I–K. Their original identifiers are retained so that the proof or counterexample can be followed from the earlier target. The canonical registry contains no open claim or open interface.

This statement concerns the 154 canonical rows at their declared finite or conditional scopes. The enlarged source-derivation obligations in Appendix L and the analytic and physical-model obligations in Appendix M remain open. Their separate register prevents a resolved finite target from being read as a completed derivation from the primitive mark to a field theory.

The results cover retention and succession classification, clock adequacy, finite valuations, observer adapters and descent, uncertainty and measurement replay, self-reference, semiring coarse-graining, reconstruction countermodels, paid composition and phase bridges, and profile-relative evaluation. Each construction states the maps, policies or physical assumptions it uses.

Further formalization can extend the partial Lean bindings for the complete prose results. The real-character, matrix-analysis, probability and optimization arguments are not all mechanized. Physical calibration, Born selection, unrestricted quantum reconstruction and external source programmes retain their explicit nonclaim or conditional scope; the finite adapter result does not turn them into conclusions of Primitive Difference.

`WDP-CH04-001` **Conditional result (routing and contextual reflection).** Under a declared terminal, lift and guarded-change policy, independently verified quotient/closure diagnostics feed checked finite routing verification records. Context-equivariant profile-preserving systems admit the least same-carrier congruent coarsening, initial in the specified comma category; a profile-reflecting completion of a noncongruent source has no solution under stated axioms. Appendix J supplies the construction and refutation. The diagnostic does not select its policy.

The primitive-axiomatics source records an unconstrained initiality target and its failure. Appendix J resolves the declared finite profile-preserving coarsening category and refutes the incompatible profile-reflecting demand; it does not identify these different categories.

One consequence bounds any such layer in advance. Growth by vertices, edges, faces, or equal-weight subdivision, with a repair set that restricts into the old one, never lowers Φ_1 (CH04-P09 and the remark after it). A lift-admission policy over those moves therefore never admits a candidate, and refusal is its only outcome. A policy with content must declare a change of weights, a deletion, an edit of existing values, or an enlarged repair set.

21.9 Evidence monotonicity and revision

Statuses can change when evidence or profiles change, but not every change is a monotone upgrade. Adding a proof can move an in-profile query from `Undetermined` to `Proved`. Adding a counterexample can move it to `Refuted`. Changing definitions may instead create a different query. Expanding a profile can move a query from `OutsideProfile` to an evaluable status. A policy change can add or remove `Refused` without altering mathematical evidence.

Every status transition should therefore record one of four causes:

1. **evidence addition or correction** under the same proposition and profile;
2. **profile change** altering admissible inputs or resources;
3. **statement change** producing a new proposition version;
4. **policy change** altering processing permission.

Only the first is ordinarily an evidential update. If the proposition changes, the old status must remain attached to the old proposition rather than being silently overwritten.

A status of `Undetermined` can hide two causes that this list separates. If a named requirement is absent—an input, a calibration record, a proof obligation—the status can move by evidence addition. If a declared resource bound was consumed, only a profile change that raises the bound can move it. Recording which of the two holds says which cause could change the status; the bare label does not.

21.10 Claim identity

A claim identifier should bind to normalized statement text, scope, hypotheses, and version. Reusing one ID after strengthening a quantifier, deleting a hypothesis, or changing the carrier produces false continuity. Conversely, typographic repair that preserves the formal statement need not mint a new claim, but the changelog should say why the identity was preserved.

A useful claim record has fields:

```
claim_id
statement
status
hypotheses
scope
dependencies
proof_or_counterexample
formal_anchor
executable_anchor
source_revision
nonclaims
last_checked
```

The narrative anchor IDs in this monograph are intentionally lighter. They locate prose; they do not replace these records.

21.11 Dependency-aware status

If claim C uses claims A and B, its status cannot exceed the weakest load-bearing dependency unless C carries an independent proof. An imported standard can support a conditional result only when the source theorem's hypotheses are matched. A mechanized special case cannot support an unrestricted prose theorem. A historical verification record cannot support a new revision without a transport or rerun.

This yields a simple audit algorithm:

1. parse the claim's explicit dependencies;
2. verify each dependency's exact statement and current evidence;

3. check hypothesis discharge at the use site;
4. search for hidden imports in definitions and examples;
5. downgrade or split the claim when its supported scope is narrower;
6. preserve the stronger target as an open row rather than deleting it.

21.12 Canonical terminology migrations

The following migrations should be applied consistently in future source consolidation:

Legacy phrase	Canonical phrase	Reason
"witness" without modifier	typed witness kind	removes seven-way polysemy
"observer tuple"	named observer layer or full layered observer	avoids false equivalence of tuples
"observer standing"	evaluator standing	separates evaluation from authority
"authority-standing observation"	validated observation with evaluator acceptance	repairs the measurement conflict
"probability from difference"	finite normalized valuation after mass and partition inputs	exposes imports
"ground state" at root	admitted ground/cut	avoids spectral homonym
"history integral"	finite semiring history aggregate	avoids continuum implication
"interference" for any cancellation	algebraic cancellation unless operational readout is present	preserves quantum boundary
"cosmogenesis"	profile-relative completeness	matches actual content
"cohomology class" without d_1	non-coboundary or quotient class	respects typing
"curvature" for $\operatorname{im} d_1$	closure charge	no connection or geometry is declared
"physical gauge orbit space" for Q^1	quotient of edge fields by vertex repairs	removes a physics import
"magnetic flux mass," "absence of monopoles"	closure mass Γ_2; $\delta^2\delta^1 = 0$	interpretive vocabulary (CH05-N03)
`ExactClass` as `ExactRepair`	diagnostic class, routed only by a checked policy	separates diagnosis from routing (CH04-N01)
"observer triple $(\mathcal{A}, \mathcal{M}, \mathcal{J})$"	evaluator cost with declared aggregation and symmetry data	an evaluator layer, not a full observer
"observer" for a bare linear map	presentation map $q_{\mathcal{O}}$	a single layer (CH10-D01)
"sound observer"	injective on a declared subspace	avoids collision with evaluator soundness
"fake zero-mode"	reduced zero not certified by the leakage test	a typed truncation blind spot
"Hodge repair"	orthogonal projection (not a repair)	projection can change the repair class
"finite/infinite regime" of a residual against a floor	evaluator tolerance status (`Accept`, `Reject`, `Undetermined`)	the status belongs to the evaluator, not the class (CH10-N01)

21.13 Editorial tests for the monograph

The manuscript itself should pass finite editorial checks:

- every restricted word is locally scoped;
- every theorem-like sentence has a status;
- every external result has a registry key and consultation state;
- no internal file path is treated as an external citation;
- no optional quantum object appears upstream of its branch declaration;
- no observation or measurement statement implies authority;
- every use of "witness" is typed by context;
- duration is defined once, with the former duplicate mapped to provenance;
- open claims remain visible rather than being narrated as completed results;
- nonclaims survive summary and conclusion sections.

These checks are necessary for a reliable release but do not prove the mathematics.

21.14 Worked status transition

Suppose the claim "all accepted readings lie within 0.1 V of the reference" is `Undetermined` because only three of four sensor codes have calibration data. A fourth reference record is added and a finite exhaustive check passes. Under the unchanged claim, calibration model, and profile, the status can move to `Proved` relative to those four codes. It must not become "proved for every voltage" because the domain is finite and declared.

If instead the calibration model changes from interval lookup to affine interpolation, that is a statement/profile change. The old proof remains valid for the old model; the new claim needs a new check. If a safety policy refuses code `3` regardless of calibration, the mathematical bound and the policy status coexist: the reading can be calibration-valid and policy-refused.

21.15 Research claims versus release claims

A research claim answers "what has been established under these hypotheses?" A release claim answers "what exact artifact contains the statement and evidence, and can another reader recover it?" Both are needed for publication, but they are not the same. A manuscript can contain a sound proof with poor artifact custody; a well-hashed archive can contain a false theorem. The canonical package should report both dimensions without substituting one for the other.

21.16 The sharp open research program on observer completion and quantum viability

The integration across FSP, Majorana, MUB6, Paper 000, and WDP turns the question of physical realizability into a sharply answerable research program:

> **Can the available observations and recovery dynamics resolve the error classes produced by the noise, within the available time and resources, while preserving the logical quantum state?**

This formulation unites the four research directions under the finite obstruction calculus:

- **FSP:** Does the chosen fermionic representation remove sign holonomy on configuration space while preserving the requested physical observables?
- **Majorana:** Does the syndrome observer Q distinguish the admitted error set $\mathcal{E}$ up to stabilizer-trivial changes ($(\mathcal{E} - \mathcal{E}) \cap \ker Q \subseteq B$), preserving the logical Pauli/error class quotient $H^1 = \ker Q/B$ on the binary Pauli carrier (rather than the complex Hilbert codespace itself)?

- **MUB6:** Does the regional cocycle circulation obstruction $\Phi_1 > 0$ extend to an exhaustive coverage certificate across the full complex Hadamard moduli space?
  The proposed Matolcsi–Matszangosz–Varga–Weiner (2025) Conjecture 2 conditions are $G_H(\pi(\gamma_1)) = 0$ and $G_H(\pi(\gamma_2))G_{H^*}(\pi(\gamma_2)) = 0$ for the twenty sign permutations. The retained evidence establishes the integer parity identity $1 + (-1)^{6m-9} = 0$ for $m \in \{0, 1, 2, 3\}$ and checks explicitly declared tensor/two-circulant families. It does not establish the proposed conditions for every MUB transition or classify the full complex Hadamard moduli space. In particular, the R6 record is explicitly marked `NOT_A_CERTIFICATE`: its Lean moduli reduction decides an admitted row-count table, not a theorem about all matrices. The former promotion from that lookup to confinement in $F(x, y) \cup B_6(\theta)$ is withdrawn; a matrix-level necessity and classification proof would be needed. The Tao matrix supplies a separate exact negative control: $G_{S_6}(\pi(\gamma_1)) = 9 \neq 0$ for all twenty permutations, as recorded in the G1 artifact. Thus the single-matrix Hadamard equations alone do not imply that vanishing condition. This says neither that Tao is a counterexample to the conjecture's MUB premise nor that the remaining matrices belong to two named families. Connecting local circulation or Positivstellensatz certificates to an exhaustive, hypothesis-correct MUB6 cover remains open.
- **Paper 000 / WDP:** Does the discarded correlation residual lie in an unrepairable class modulo stabilizers under local Lieb–Robinson dynamics and bounded readout precision?

Chapter 22 — Synthesis: what the finite calculus establishes

22.1 The trunk

The defensible trunk of the programme is modest and coherent.

An admitted cut contains exhibited difference and the structure needed to type it. A declared comparison can fail, leaving a residual. Typed witnesses can locate, retain, repair, separate, or refute claims about that residual, but those roles require explicit adapters. Full symmetry blocks a deterministic equivariant point choice of a moved target. It does not block invariant probability laws; realizing a point from such a law adds a random outcome or seed. Every successful deterministic point selection must depend on a declared symmetry-breaking input. When the candidate input family is finite and removing the full declared dependency set restores a symmetry that moves the output, that dependency set contains at least one inclusion-minimal frontier element; a particular implementation may also carry redundant inputs. On a declared finite graph, cochains and quotient defects provide exact mathematical tools; finite-dimensional optimization yields bounded propositions and certificates.

Retention plus an admitted transition relation supports finite succession. Succession plus an additive valuation supports duration. Exhibited finite witness families plus declared masses and partitions support normalized finite probability. Each layer has a clear input ledger and none supplies its usual physical interpretation for free.

22.2 The observer/measurement waist

The central architectural result of the synthesis is the separation

$$\text{witness} \longrightarrow \text{observer presentation/memory/evaluation} \longrightarrow \text{observation} \longrightarrow \text{measurement},$$

with authority outside the chain.

A witness is typed evidence for a proposition. An observer packages a presentation boundary, memory, evaluator, and policy. An observation is a registered, witnessed, re-accessible interaction accepted by an evaluator. In the WDP interface, a measurement adds a measurand specification, a distinct reported-value carrier and map, scale/unit, calibration, uncertainty, any claimed metrological-traceability evidence, and separate provenance/re-access. This "waist" lets duration and probability meet operational records without claiming that either was derived as physical law.

22.3 The branches

Histories form one branch. Projection can fail to descend; if a single-valued deterministic update on the unchanged presentation carrier is retained, extra state or history is then required. Refinement or relation-valued, set-valued, or stochastic dynamics are alternatives. Finite path aggregates over semirings are well defined at fixed length and have a matrix representation under the stated admissibility hypothesis. Infinite sums and quantum readings need more.

Quantum-capable carriers form another branch. Declared complex fibers, unitary connections, holonomy, and connection Laplacians support rigorous finite geometry. They remain assumption-bearing. A reconstruction audit shows the additional assumptions: empirical contexts, composition, continuity, purification or an alternative discriminator, and a probability rule stronger than normalization.

Relative completeness forms a third branch. A finite evaluator can cover a declared query family with exclusive statuses if its rules are proved correct. That is an engineering and logical property of a profile, not an origin story for a universe.

22.4 The four-tier ambition hierarchy

The entire programme is organized across four distinct levels of foundational ambition:

1. **Tier 1 — Syntactic Representation:** Demonstrated for the declared finite encodings. The round-trip claims concern their specified source and image carriers; selected implementation checks retain their recorded scope. Encoding a packet does not establish its operational interpretation.
2. **Tier 2 — Faithful Operational Representation:** Demonstrated for the stated constructions and adapter laws. Finite transition quotients preserve their declared reports and updates; permutation actions and cochain maps preserve their specified algebra. Formation, checker fidelity, operational steps and observational answers remain separate obligations for each theory-profile adapter (§20.4).
3. **Tier 3 — Reconstruction Under Explicit Principles:** Characterized as an audit framework. Domain structures such as empirical contextuality covers, operator-algebraic states, and continuous measurement uncertainties are reconstructed only by adjoining explicit operational axioms and cataloging their external interfaces.
4. **Tier 4 — Unique Selection From Primitive Kernel:** Explicitly refused. The programme does not claim that textbook continuous spacetime, the complex field $\mathbb{C}$, the continuous unitary circle group $U(1)$, Gleason frame uniqueness, or the Born probability rule is forced or uniquely selected by primitive difference. It establishes no theorem that they cannot be; the nonclaims and the finite witnesses of §18.5 and Appendix D mark where the programme stops.

Maintaining strict boundaries between these four tiers prevents confusing a verified syntactic embedding (Tier 1) or finite operational model (Tier 2) with unrestricted ontological necessity (Tier 4).

22.5 Strongest defensible conclusion

WDP-CH22-C01 **Synthesis conclusion.** Finite witnessed-difference systems can be organized into an explicit obstruction calculus in which every transition from residual to retention, order, magnitude, probability, observer, observation, measurement, history, or quantum-capable representation is exposed as a typed construction with a visible assumption cost.

This conclusion is architectural. It does not claim that all listed open constructions exist or are unique. Its value is that failed derivations become local: one can point to the missing order, map, law, calibration, or theorem rather than covering it with vocabulary.

22.6 Closing boundary

The monograph ends where its evidence ends. It supplies finite definitions, written proofs of the scoped finite and conditional theorem families, selected formalizations, bounded computations, and an explicit map of remaining obligations. Each kind of support retains its stated scope and reproduction status. It does not turn those materials into a physical theory by declaration. A future result may close one of the interfaces; if so, the new hypothesis and proof should be attached at that interface, and the nonclaims elsewhere should remain intact.

Chapter 23 — Observer completion on a finite 2D toric-code carrier

Active exposition status, 2026-10-07. The active exposition of this chapter develops the corrected mathematics of finite toric observer completion. The historical draft text originally imported with withdrawn global-kernel and restricted-error detector-rank/adaptivity arguments has been retired to docs/source_archive/toric_code_chapter_23_import.md. The original draft identifiers (WDP-CH23-D01 through WDP-CH23-P20) are mapped to their corrected statements (WDP-CH23-P01C through WDP-CH23-P18C) in the crosswalk register of §23.8 and Appendix U.

What this chapter does. It instantiates the operational question of the previous session — *given what the observer already sees, which further observations are necessary and sufficient, and what is the cheapest way to obtain them?* — on a declared finite carrier with actual instruments and prices. The finite procedures return a sufficient observation set with a checked lower bound on its cost. Adaptivity depends on the declared candidate set and instruments: P17C gives a worst-case adaptive cost of four against a fixed-suite cost of six. The chapter is a **conditional result**: it presupposes a declared noise model, a declared schedule, and a declared precision, and it says nothing about any physical device.

23.1 The declaration packet

WDP-CH23-D01 **Definition (toric-code completion packet).** A *completion packet* is a tuple

$$\mathfrak{T} = (\Lambda,\ \mathcal{S},\ \mathcal{D},\ \mathcal{T},\ \mathcal{P},\ \mathcal{E},\ q,\ B)$$

with the following fields, all finite and all declared:

1. **Carrier.** $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ with $L \geq 3$, horizontal qubits $h_{i,j}$ joining (i,j) to $(i,j+1)$ and vertical qubits $v_{i,j}$ joining (i,j) to $(i+1,j)$ (indices modulo L), $n = 2L^2$ data qubits, and the Pauli carrier $\mathcal{P} = \mathbb{F}_2^n \oplus \mathbb{F}_2^n$ with $e = (e_X, e_Z)$ and weight convention
$$\mathrm{wt}(e) = |\mathrm{supp} e_X| + |\mathrm{supp} e_Z|,$$
so a single Y contributes 2.
2. **Stabilizers.** $S = \langle \sigma_v^\star,\ \sigma_f^\square \rangle$, $\sigma_v^\star$ the X-star of site v, $\sigma_f^\square$ the Z-plaquette of face f.
3. **Detector locations.** A *detector* is a pair $d = (\sigma, k)$ with σ a stabilizer generator and k a round index. A schedule D is a finite subset of $\mathcal{S} \times [R]$; the *dark windows* of D are the maximal open intervals of $[R]$ containing no round whose full stabilizer set is in D.
4. **Classical communication.** Per round, a measurement duration τ_m and a link latency τ_c; the syndrome of round k is available to the policy at time $t_k = k(\tau_m + \tau_c) + \tau_c$.
5. **Precision.** Each detector returns one bit (exact readout), or — in the noisy branch of §23.6 — one bit subject to a declared independent flip with rational parameter ε, or no value at all (that is, the location is dark).
6. **Admissible errors.** $\mathcal{E} \subseteq \mathcal{P}$, declared; the standard instance is $\mathcal{E}_t = \{e : \mathrm{wt}(e) \leq t\}$.
7. **Protected question.** A group homomorphism $q : \mathcal{P} \to A$ with $B = \ker q$ and $S \subseteq B$. The standard instance is $q(e) = [e]$ in $\mathcal{P}/S$, with $B = S$: *the recovery left a stabilizer, so the logical state is the prepared one.*
8. **Recovery.** A policy $\rho : \mathrm{im}\, Q_D \to \mathcal{P}$ applied to the accumulated detector record.

`WDP-CH23-D02` **Definition (measured history record).** For a schedule $D \subseteq \mathcal{S} \times [R]$, put $D_{\leq k} = \{(\sigma, j) \in D : j \leq k\}$. Its record is
$$Q_{D,k}(e^{(1)}, \ldots, e^{(k)}) = \big(\mathrm{syn}_\sigma(e^{(j)})\big)_{(\sigma,j) \in D_{\leq k}} \in \mathbb{F}_2^{D_{\leq k}}.$$

Each coordinate records the error at its actual measurement time. A previous measurement does not reveal a later syndrome. With no measurement error, one complete round observes
$$Q(x \mid z) = (Ax, A_v z),$$
where $A \in \mathbb{F}_2^{L^2 \times n}$ is the face–qubit incidence matrix and $A_v \in \mathbb{F}_2^{L^2 \times n}$ is the site–qubit incidence matrix. The rank, separation and observation-cost results below concern this single-round map, or repeated tests of one fixed error. They do not solve recovery for an evolving error history.

23.2 Exact structure of the commutation kernel: $S^\perp$, not S

Let A be the $L^2 \times n$ face-incidence matrix and A_v the star-incidence matrix over $\mathbb{F}_2$. Each edge lies in two faces, so the sum of all rows of A is zero. The dual graph is the connected torus grid, so the left kernel is the constant vectors and $\mathrm{rank}\, A = L^2 - 1$. Similarly, $\mathrm{rank}\, A_v = L^2 - 1$.

On the Pauli space of dimension $4L^2$, the block commutation syndrome $Q(x \mid z) = (Ax, A_v z)$ has rank
$$\mathrm{rank}\, Q = (L^2 - 1) + (L^2 - 1) = 2L^2 - 2,$$
so its kernel has dimension
$$\dim \ker Q = 4L^2 - (2L^2 - 2) = 2L^2 + 2.$$

The stabilizer space S is spanned by the star rows in the X block and face rows in the Z block. Commutation relations give $AA_v^{\mathsf{T}} = 0$, so $S \subseteq \ker Q$. Since the two blocks are independent, $\dim S = 2L^2 - 2$. Hence

$$\dim(\ker Q/S) = (2L^2 + 2) - (2L^2 - 2) = 4.$$

WDP-CH23-P01C **Finite proposition (commutation kernel structure).** On this carrier, with this weight convention, the full stabilizer commutation syndrome has kernel $S^\perp \neq S$, and $\dim(S^\perp/S) = 4$. The kernel of Q is the set of Pauli vectors commuting with every generator of S (the symplectic orthogonal $S^\perp$), which properly contains S. An exhibiting vector is the weight-L support Z_1 (horizontal edges $h_{i,0}$ for $i = 0, \ldots, L-1$) placed in the X block with Z block zero: it lies in $\ker A$, outside the star span, and outside S.

23.3 Distance under declared weight and coarser-question thresholds

Let $x \in \ker A$. Let $\alpha_i(x)$ be the sum of horizontal edges in lattice row i, and $\beta_j(x)$ the sum of vertical edges in column j. Summing the face constraints along a row gives $\alpha_i = \alpha_{i+1}$; summing them along a column gives $\beta_j = \beta_{j+1}$. Both fluxes vanish on the star span, so (α, β) descends to an isomorphism $\ker A/\operatorname{rowspan}(A_v) \cong \mathbb{F}_2^2$, as proved in Appendix U.2.

If $\alpha = 1$, each row contributes at least one edge, giving weight $\geq L$. The same bound holds in each sector.

WDP-CH23-P02C **Finite proposition (code distance).** $\min\{\mathrm{wt}(e) : e \in \ker Q \setminus S\} = L$, and a weight-$L$ vector exists in each sector.

A class nontrivial in both sectors has weight at least $2L$ under the declared additive weight.

WDP-CH23-P07C **Finite proposition (diagonal distance and coarser-question threshold).** Under $\mathrm{wt}(e) = |\operatorname{supp} e_X| + |\operatorname{supp} e_Z|$, the diagonal class has distance $2L$, not $2L - 1$. However, this distance does not establish the fault-tolerance threshold $t \leq L - 1$ for the coarser question $B = S + \langle u + v \rangle$. The relevant separation distance is

$$d_B = \min\{\mathrm{wt}(z) : z \in \ker Q \setminus B\},$$

measuring forbidden differences. Because the diagonal class $u + v$ belongs to the allowed subspace B, its weight of $2L$ does not govern d_B. The forbidden difference set $\ker Q \setminus B$ still contains the pure single-sector logical classes u and v, each of weight L. Thus $d_B = L$, retaining the single-sector threshold $2t < L$. The inference that $2t < 2L$ tolerates $t \leq L - 1$ fails: at $L = 3, t = 2$, the errors $e_1 = h_{0,0}$ and $e_2 = \{h_{1,0}, h_{2,0}\}$ have weights one and two, identical full syndromes, and difference $u \notin B$, producing an undetectable collision.

WDP-CH23-P08C **Withdrawal of halving slogans.** The imported slogans that diagonal coarsening halves the required lattice and is factorially cheaper are withdrawn. No cost or lattice-size identity of that form is proved. The coarser question retains the threshold requirement $2t < L$.

23.4 The restricted fiber criterion and failure certificates

Let E_{adm} be a declared finite set of Pauli vectors, Q_D a declared detector map, and q a protected question with $B = \ker q$.

WDP-CH23-P03C **Finite proposition (restricted fiber criterion).** For every admitted pair $e, e' \in E_{\mathrm{adm}}$,

$$Q_D(e) = Q_D(e') \implies q(e) = q(e')$$

if and only if

$$(E_{\mathrm{adm}} - E_{\mathrm{adm}}) \cap \ker Q_D \subseteq B.$$

WDP-CH23-P04C **Corollary (primal recovery table).** When the inclusion holds, choosing one admitted representative e_s in each nonempty syndrome fiber gives an exact recovery policy $\rho(s) = e_s$ such that $e + \rho(Q_D e) \in B$ for all $e \in E_{\mathrm{adm}}$. Both construction and verification terminate by finite enumeration.

WDP-CH23-P05C **Corollary (dual failure certificate).** When the inclusion fails, finite search returns an admitted pair $e_1, e_2 \in E_{\mathrm{adm}}$ such that $Q_D e_1 = Q_D e_2$ but $e_1 - e_2 \notin B$, providing an independently verifiable failure witness.

WDP-CH23-P10C **Finite proposition (global rank vs. admitted collisions).** A global rank defect is not an admitted collision. Dropping one independent check raises $\dim(\ker Q_D/S)$ from 4 to 5. At $t = 0$, $E_{\mathrm{adm}} = \{0\}$, so $(E_{\mathrm{adm}} - E_{\mathrm{adm}}) = \{0\} \subseteq B$. The empty purchase is sound at $t = 0$. Whether an enlarged kernel contains an admitted collision is determined by searching $E_{\mathrm{adm}} - E_{\mathrm{adm}}$, not by quoting the global quotient dimension.

23.5 Full rank sufficiency, $t = 0$ purchase, and the matroid boundary

Let $\mathcal{E}_t = \{e \in \mathcal{P} : \mathrm{wt}(e) \le t\}$ and $B = S$.

WDP-CH23-P06C **Finite proposition (sufficiency of full rank).** For $1 \le t$ and $2t < L$ with $B = S$, a passive stabilizer-check set of rank $2L^2 - 2$ separates $\mathcal{E}_t$ up to S. Sufficiency follows from $d = L$: any vector of weight $\le 2t < L$ in $\ker Q = S^{\perp}$ already belongs to S. Full rank is not necessary for separation: at $L = 4$, twenty-nine independent checks separate $\mathcal{E}_1$ ($t = 1, 2t = 2 < 4$), dropping one check without producing an admitted collision.

WDP-CH23-P11C **Finite proposition (uniform cost minimum and zero purchase).** At $t = 0$, the empty detector set is sufficient and has cost 0. For $1 \le t$ and $2t < L$ with $B = S$, rank $2L^2 - 2$ is sufficient but need not be the minimum independent count. The strict saving stated here is the $L = 4, t = 1$ example; it is not asserted for every L, t.

WDP-CH23-P12C **Restriction of the matroid shortcut.** Greedy selection in the stabilizer-check matroid returns a minimum-cost basis only when sufficiency is proved equivalent to full rank. For general E_{adm}, sufficiency is the hitting condition of P03C, requiring exact enumeration under positive rational costs.

WDP-CH23-P13C **Finite proposition (uniform-cost check bounds).** The uniform-cost minimum on a declared finite shelf is returned by exhaustive enumeration. Every detector in a minimal sufficient suite is essential for that suite, but need not span the full check space.

23.6 Adaptivity, schedules, and spacetime consistency

WDP-CH23-P17C **Finite proposition (refutation of unrestricted adaptive lower bounds).** The imported unrestricted claim that adaptivity buys nothing is false. On a four-hypothesis toy model with tests of cost 2, nonadaptive separation requires three tests (cost 6), while an adaptive policy requires two tests (cost 4). The claimed lower bound of 6 is refuted.

WDP-CH23-P18C **Refutation of global purchase bounds.** Every unrestricted detector-purchase bound requiring full check rank along every branch is withdrawn. The $t = 0$ family and the $L = 4$ rank-29 suite are explicit counterexamples.

WDP-CH23-P15C **Conditional status of causal-cone bounds.** Comparing the cone radius $\lceil v_{\mathrm{LR}} \Delta t \rceil$ with $\lfloor L/2 \rfloor$ is a geometric comparison, not a proof of schedule failure. A failure certificate requires an explicit dual pair in $(E^R - E^R) \cap \ker Q_{D,R} \setminus B^R$.

WDP-CH23-P16C **Spacetime consistency and passive readout.** The spacetime chain complex, Bernoulli readout model, and repetition threshold $R \geq L$ remain conditional on an explicit joint error model and state-evolution map. Detectors are passive queries; a classical bit table does not prove physical non-disturbance.

23.7 Terminating observer-completion procedure

WDP-CH23-D16 **Definition (`ToricObserverCompletion`).** Given $(\Lambda, \mathcal{S}, D, \mathcal{E}, B)$, the procedure: 1. enumerates Pauli coordinates of the declared carrier; 2. constructs H_D; 3. computes $\ker H_D$; 4. enumerates $\mathcal{E} - \mathcal{E}$; 5. tests $(\mathcal{E} - \mathcal{E}) \cap \ker H_D \subseteq B$; 6. if true, outputs primal recovery dictionary $\rho(\sigma) = e_\sigma$; 7. if false, returns dual failure pair e_1, e_2.

WDP-CH23-P19 **Finite proposition (termination and correctness).** The procedure terminates for every finite instance and outputs either an exact primal recovery dictionary or an exact dual failure pair.

WDP-CH23-P20 **Finite proposition (complexity).** For $d = 2L^2$, $r = |D|$, and $m = |\mathcal{E}|$, Gaussian elimination uses $O(d^3 + d^2 r)$ operations over $\mathbb{F}_2$. Testing membership in B is polynomial in d, and difference enumeration is polynomial in m.

23.8 Identifier crosswalk and archival register

The original draft text of Chapter 23 is archived in docs/source_archive/toric_code_chapter_23_import.md. The crosswalk below maps each imported draft identifier to its canonical disposition and corrected identifier:

Draft ID	Topic	Draft Claim	Canonical Disposition	Corrected ID
D01	Packet	Toric completion packet	Proved definition	WDP-CH23-D01
D02	Detection	Cumulative syndrome map	Proved definition	WDP-CH23-D02

Draft ID	Topic	Draft Claim	Canonical Disposition	Corrected ID
P01	Kernel	$\ker Q = S$, quotient dim 2	Corrected: $\ker Q = S^{\perp} \neq S$, quotient dim 4	WDP-CH23-P01C
P02	Distance	Distance L	Corrected: distance L via winding fluxes	WDP-CH23-P02C
P03	Fiber	$(E-E) \cap \ker Q \subseteq B$	Corrected: sound equivalence proof	WDP-CH23-P03C
P04	Recovery	Primal recovery dictionary	Proved	WDP-CH23-P04C
P05	Dual	Dual failure pair	Proved with explicit search	WDP-CH23-P05C
P06	Sufficiency	$2t < L$ iff full rank	Corrected: full rank sufficient, not necessary	WDP-CH23-P06C
P07	Diagonal	Diagonal distance $2L-1$, $t \leq L-1$	Corrected: distance $2L$, but $d_B = L$ retaining $2t < L$	WDP-CH23-P07C
P08	Slogans	Halves lattice, factorially cheaper	Refuted and withdrawn	WDP-CH23-P08C
P09	Terminal	Terminal error convention	Proved for stated convention	WDP-CH23-P09
P10	Omission	Dropped check gives low-weight collision	Corrected: rank defect is not admitted collision	WDP-CH23-P10C
P11	Check count	Full rank necessary for $2t < L$	Corrected: false at $t = 0$ and for $L = 4$ rank-29	WDP-CH23-P11C
P12	Matroid	Greedy matroid basis	Corrected: restricted to verified full-rank cases	WDP-CH23-P12C
P13	Uniform cost	Minimum count $2L^2 - 2$	Corrected: enumeration minimum, not rank bound	WDP-CH23-P13C
P14	Dichotomy	Primal/dual certificate dichotomy	Proved	WDP-CH23-P14
P15	Causal cone	Causal cone failure threshold	Conditional on error propagation model	WDP-CH23-P15C
P16	Spacetime	Spacetime consistency chain	Conditional on joint error model	WDP-CH23-P16C
P17	Adaptivity	Worst-case adaptivity buys nothing	Refuted as unrestricted claim; toy countermodel	WDP-CH23-P17C
P18	Cost	Adaptive cost $\geq 2L^2 - 2$	Refuted as unrestricted claim	WDP-CH23-P18C

Draft ID	Topic	Draft Claim	Canonical Disposition	Corrected ID
P19	Procedure	Algorithm termination	Proved	WDP-CH23-P19
P20	Complexity	Bit-operation bounds	Proved	WDP-CH23-P20

23.9 Claim hygiene and boundaries

1. What this chapter supports

- The full commutation syndrome has kernel $S^{\perp} \neq S$ with logical quotient dimension 4.
- The exact observer completion condition is $(E_{\mathrm{adm}} - E_{\mathrm{adm}}) \cap \ker Q_D \subseteq B$.
- For $\mathcal{E}_t$, $B = S$, and $2t < L$, full independent stabilizer rank $2L^2 - 2$ is sufficient, but not necessary.
- For coarser diagonal questions $B = S + \langle u + v \rangle$, the forbidden distance is $d_B = \min\{\mathrm{wt}(z) : z \in \ker Q \setminus B\} = L$, retaining the single-sector threshold $2t < L$.
- At $t = 0$, the empty detector set is sufficient and costs 0.
- Every finite instance admits either an exact primal recovery dictionary or an exact dual failure pair.
- Unrestricted claims that adaptivity never reduces cost or that full rank is required along every branch are refuted.

2. What this chapter does not support

- A physical toric-code quantum device, its hardware, calibration, or noise spectra.
- Fault tolerance under undeclared stochastic, correlated, or adversarial noise models.
- Increase of fault tolerance beyond $2t < L$ for diagonal-stabilizer quotients.
- Universality of matroid greedy selection for general error-separation tasks.
- Non-disturbance of quantum states by passive classical readouts.
- Computational scalability for unbounded lattice sizes or spacetime histories.
- Any claim about human cognition, general intelligence, or consciousness.

3. Narrowest defensible claim

Under the declared finite 2D toric carrier and Pauli error family, observer completion is an unresolved quotient-homology problem governed by $(E_{\mathrm{adm}} - E_{\mathrm{adm}}) \cap \ker Q_D \subseteq B$. For the CSS commutation syndrome, the kernel is $S^{\perp}$ with quotient dimension 4; full rank $2L^2 - 2$ is sufficient for $1 \leq t, 2t < L, B = S$, but not necessary; coarser diagonal questions retain $d_B = L$ and threshold $2t < L$; and every finite instance terminates with an exact primal recovery table or dual failure witness.

4. Next test needed

Calibrated execution pairing measured lattice error statistics with the finite kernel solver, verifying that the predicted primal dictionary or dual failure pair matches observed error-correction performance on physical benchmark data.

Appendix A — Dependency and status table

Construction	Required inputs	Output	Present status
Admitted cut	finite exhibition + Structure	typed apartness	Definition
Residual locus	comparison + failed repair	local residual witness	Definition / finite checks
Symmetry no-selection	full group action + equivariance	obstruction to invariant choice	Finite proposition
Separation price	declare-then-assert grammar + item cost + isomorphism	minimal responses and the surviving tie	Finite proposition
Retention	memory carrier + update + report + comparison	re-accessible mark	Finite deterministic quotient; broader relational classification open
Cochain quotient	graph + coefficient group + d_0	defect modulo repair	Definition; bounded propositions
Closure/cohomology	C^2, d_1, $d_1 d_0 = 0$	cocycles/classes	Conditional; often absent
Evaluation preorder and window	comparison relation + nerve route or certified (d_0, d_1)	degree-1 window	Definition
Obstruction signature	window + norm J + closure declaration	$(\Phi_1, \Gamma_2, R_{cl})$ and diagnostic class	Finite propositions; routing interface open
Exact certificate	weighted L^1 + repair + bounded circulation	certified Φ_1	Finite propositions (soundness; repairable mass and non-unique pairs); existence imported
Path discrepancy	two same-endpoint paths + weighted L^1	lower bound on Φ_1	Finite proposition
Quotient observer	window + report map with $Rd_0 = 0$	presentation map + evaluator cost	Definition; passivity proved
Context witness	states + contexts + profile into Q^1	exposed residuals and witness rank	Definition; rank invariance proved
Patch cover of a cycle	oriented cycle + path patches + declared aggregation	independent, shared-maximum, and union readings of one class	Finite proposition
Degree-k and torsion windows	finite simplicial complex + coefficient field	Q^k, H^k, coefficient-dependent classes	Finite propositions (branch)
Projective phase class	finite group + declared unitary matrices	$[\omega] \in H^2(G; U(1))$	Finite proposition (branch)
Finite commutation	matrices over a commutative ring; clock and shift over $\mathbb{Z}[\zeta_N]$	trace obstruction; Weyl class of order N	Finite propositions (branch)

Construction	Required inputs	Output	Present status
Finite reconstruction frames	complex carrier + complete unbiased bases ($d = 3, 4$) or phase-point operators (odd N)	exact linear reconstruction of ρ; signed valuation W_ρ	Finite propositions (branch)
Complex growth	larger complex, face attachment, or equal-weight subdivision	comparison of Φ_1	Finite proposition (growth never lowers Φ_1)
Presentation descent	surjective presentation + target	factor or obstruction set	Finite proposition
Chain contraction	chain maps + homotopy + convention	retraction defect; perturbed contraction	Definition; proposition; imported lemma
Restriction curvature	finite poset + declared comparable-pair maps	K, with $d_1 d_0 = K$ and $d_2 d_1 = K$ on the top edge	Finite proposition
Truncation leakage	operator + orthogonal projector	leakage form L_P	Finite proposition
Linear recovery	presentation map + declared subspace + norms	recovery constant or kernel witness	Finite propositions
Non-simplex state space	extreme points + affine effects	two decompositions of the centre	Finite proposition (branch)
Sorkin sums	additive amplitudes + declared biadditive readout	vanishing $I_{k\geq 3}$	Finite proposition (branch)
Succession	events + directed acyclic relation	strict partial order	Finite proposition after inputs
Duration	path system + additive valuation	path magnitude	Definition; exactness propositions
Clock	reference process + coupling + error rule	calibrated comparison	Definition; adequacy open
Finite probability	witness family + mass + partition	normalized valuation	Finite proposition; atomicity open
Observer	presentation + memory + evaluator + policy	profile-relative reports/status	Definition; adapters open
Observation	interaction + registration + witness + re-access + evaluation	verification record; validated observation when status is `Accept`	Definition
Measurement	accepted observation + measurand specification + result carrier, map, unit, and calibration + uncertainty and error semantics + optional metrological-traceability evidence + provenance/re-access	measurement result	Repaired WDP definition; calculus open
Authority	role + jurisdiction + permission + procedure	authorized act/status	Separate declared layer
History sum	path carrier + semiring weights	finite aggregate	Finite proposition at fixed length

Construction	Required inputs	Output	Present status
Quantum carrier	complex/unitary imports	quantum-capable geometry	Conditional branch
Certified preserver (R12)	R8 involution, R9 positive definite form, R10 readout, global ray bijection and dimension hypothesis	unique projective semisimilarity; rational phase group; typed norm obstruction	Native written proof; selected Lean theorems; concrete submitted-witness checks
Observation stabilization (R12-O03)	named readouts, declared group/algebra, linear defect carrier; Appendix T for the count	Galois closure, dagger bicommutant, sharp linear observation count	Conditional algebra; no nonlinear or restricted-menu optimality inferred
Compatible generator (R13)	matrix algebra, K-linear derivation and declared adjoint law	$\delta = -i[H_0, \cdot]$, unique modulo rational scalars	Written finite-dimensional theorem; Python basis-table checker; no supplied flow
Kochen–Specker parity	declared rays in $\mathbb{Q}^4$ + orthogonal contexts	no $\mathbb{F}_2$ valuation with sum 1 per context	Finite proposition (branch)
Classical package S	nonnegative weights + stochastic updates + classical instruments + joint runs	mixtures of function updates; instrument laws; CHSH bound; sharpness	Finite propositions (classical corner extended); no packet representation
Tolerant packet adapter	frequency tables + declared tolerance	table with certified signalling defect; exact repair on two contexts	Finite propositions and finite-cover/i.i.d. concentration scope; general non-i.i.d. sampling open
Composition principle CP	injective tagging + no-signalling + bilinear positivity (classical carriers)	composite states of classical carriers are joint distributions; PR box not excluded	Finite propositions; tensor product open
Amplitude bridge	integer cochain + rational phase u + two walks	gauge-class dependence of phase-blind readouts; no rational exponentiation	Finite propositions; readout open
Qubit instrument adapter	Kraus family over $\mathbb{Q}(i)$ + mixture state	probabilities, post-state, POVM underdetermination	Finite propositions and finite-dimensional instrument laws (Appendix J); infinite-dimensional case open
Quantum reconstruction	operational axioms + comparison class	representation/no-go	Open audit
Relative completeness	finite grammar, evaluator and status rules; sound decoding and constructive coverage for solving claims	finite status coverage; original-goal witness and search preservation under declared adapters	Finite coverage and §§20.6–20.8 solving theorems; arbitrary profile fidelity remains open

Construction	Required inputs	Output	Present status
Toric-code observer completion	finite carrier $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ + stabilizers + schedule D + admissible errors $\mathcal{E}$ + protected subgroup B	exact primal recovery table ρ or dual failure pair (e_1, e_2)	Finite admitted-fiber criterion and corrected Chapter 23/Appendix U bounds; joint spacetime/readout matrices outside supplied family open

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Appendix B — A compact end-to-end finite example

This example illustrates composition without claiming physical completeness.

Let a device have states $S = \{s_0, s_1, s_2\}$ and transitions $s_0 \to s_1 \to s_2$. A binary contact changes from open at s_0 to closed at s_1 and remains closed at s_2.

1. **Cut.** Open/closed roles are declared distinct.
2. **Residual witness.** At s_1, a sensor record differs from the open baseline.
3. **Retention.** A latch stores the change bit through s_2.
4. **Succession.** The transition graph is acyclic, so reachability is a strict order.
5. **Duration.** Edge weights $(2, 3)$ give duration 5 from s_0 to s_2 in a declared unit.
6. **Observer presentation.** The observer exposes `(contact, latch)` and hides other device state.
7. **Observation.** Interaction, ADC registration, typed contact witness, retained record, and an evaluator range check that returns `Accept` form an validated observation.
8. **Measurement.** The measurand Q is terminal voltage under the declared loading condition; Y is the ordered carrier of closed voltage intervals. A calibration table maps ADC code into Y, attaches volts, a separately typed worst-case error bound, and a calibration-version verification record. No metrological-traceability claim is made without an external calibration chain and uncertainty contributions.
9. **Probability.** Repeated, separately declared trials may be tallied and normalized; the resulting finite table has no stochastic interpretation until a trial model is added.
10. **History.** Positive-real edge weights can be multiplied and alternative path weights summed. With only one path, no interference question arises.
11. **Authority.** Whether the result permits an external action is determined by a separate jurisdictional rule.

At each step, deleting the new input removes the corresponding conclusion. The example is therefore a dependency demonstration, not a derivation from one bit of difference.

Appendix C — Source integration map

The canonical chapters consolidate the source folders as follows.

Source folder	Canonical destination
022_ground_state	Chapters 1–2; spectral appendix separated into Chapter 5 scope
001_from_primitive_difference_to_cochain_systems	Chapters 3, 10–11, 16, and open programme
000_primitive_difference_for_quantum_carriers	Chapters 4–5
003_witnessed_succession_and_duration	Chapters 6–7
024_duration	Merged into Chapter 7; no parallel live duration theory
023_probability	Chapters 8–9
004_calculus_of_observers_and_observation	Chapters 10–12 and 15
025_measurement	Repaired and expanded in Chapter 13
002_self_witnessing_and_the_awareness_boundary	Chapter 14 with refuted ladder fenced
005_finite_witness_history_integral	Chapter 17, dependent on Chapter 16
006_quantum_reconstruction_or_classical_no_go	Chapters 18–19
007_witness_relative_cosmogenesis	Retitled and narrowed into Chapter 20
../006_primitive_axiomatics (v2.10.0)	§§4.7–4.12, §§5.10–5.14, §8.5, §10.5, §11.5, and CH04-O01 in §21.8
../../000_genesis_alpha_program/012_primitive_difference_residual_quotients_and_simplex_growth	CH04-P09 and CH04-N02, which replace its growth rule
../../000_genesis_alpha_program/015_primitive_projection_obstruction_descent_and_retraction_calculus	§§16.3 and 16.5–16.6; the §4.12 projection remark; the Chapter 1–5 technical note
../../000_genesis_alpha_program/003_operational_probabilistic_state_spaces	CH18-P01, CH19-P01, and the §19.10 no-signalling sentence
../../000_genesis_alpha_program/035_cost_of_closing_observer_blind_spots	CH14-P02, CH14-P03, and CH14-N02
../../000_genesis_alpha_program/020_finite_hilbert_lift_adequacy	CH05-P07 and CH16-P04
../../000_genesis_alpha_program/007_what_the_basis_forces_minima_difference_cost	CH01-P02 and merged examples in §3.2 and §7.4
../../000_genesis_alpha_program/004_nonlocality_contextuality_finite_commutator_obstructions	CH18-P06 (§18.4); CH05-P08 and CH05-P09 (§5.13); CH05-P10 (§5.15); the PR-box mixture sentence in §19.11
../../000_genesis_alpha_program/005_discrete_phase_spaces_stabilizer_grounded_semantics	CH05-P11 and CH05-P12 (§5.15)
../../000_genesis_alpha_program/006_no_free_conclusions_minimal_calculus	CH03-P03 and its worked count in §3.2; a §21.2 falsifier; an Appendix A row
../../000_genesis_alpha_program/008_l1_coboundary_certificates_and_lift_refusal	CH04-P10 and CH04-P11 in §4.10; the restricted-repair remark after CH04-P09; sentences in §4.12 and §7.4; a §21.3 falsifier; the lift-admission paragraph after CH04-O01 in §21.8
../../000_genesis_alpha_program/009_representation_lifts_and_coherence_defects	CH16-P05 in the new §16.7 with a pointer in §4.4; the diagonal-lift and density-lift remarks in §5.14; the Euclidean selections after CH04-P10; a §21.3 falsifier
../../000_genesis_alpha_program/010_reasoning_and_intelligence_as_containers	Not carried: an interpretive essay whose separations are already §10.2, CH12-P01, §§15.2–15.3, §20.2, §20.5, and CH00-N03

Source folder	Canonical destination
`../../000_genesis_alpha_program/011_finite_and_infinite_as_observer_relative_regimes_of_obstruction`	CH11-P02 in §11.2 and CH10-N01 in §10.2; §21.7 item 21, a §21.12 migration, and an Appendix A row
`../../000_genesis_alpha_program/022_finite_presentation_calculus_for_computation`	No rows: its descent criterion is CH16-P02; the awaiting/exhausted split of `Undetermined` is merged into §21.9

Merged-source papers under the observer folder remain provenance sources. They do not become additional canonical chapters just because they contain useful definitions or examples.

The primitive-axiomatics source is a sibling package in this series with its own deposit record. It was absorbed on 2026-09-29 as a source input: its bytes are bound in the `absorbed_snapshots` block of `source_provenance.json`, and its claim IDs appear in the historical crosswalk of the ledger without inheriting status. Its trace-instrumentation bridge, guarded repair policy, terminal and lift enumerations, reflective completion, adjudication protocol, and edge-symmetry gauge classification were not carried. Six corrections to the source were made on absorption: the balance identity needs no closedness hypothesis (§5.12); the Pauli product carries a factor $-i$ (§5.13); the mod-2 class on the projective plane is represented by a single-triangle indicator, not the all-ones cochain (§5.11); the source's brute-force Φ_1, which searched a small box of integer potentials, and its Bland's-rule test were replaced in the executable package; and a mis-cited identifier for the trace-distance contraction reference was replaced (REF-QM-020). Its Lean anchors for the exact sequence, ratio bound, passivity, and triangle were restated over $\mathbb{Q}$ and the literal quotient (Appendix D).

Six archived working papers were absorbed on the same date, each bound by its own `absorbed_snapshots` block. Audit found much of their stated content false, packaged, or out of scope, and none of it enters here without a new proof or a correction. Not carried: 012's claim that complex growth repairs residuals (it cannot; CH04-P09), its spectral-sequence reading of Φ_1, and its physics appendix; 015's two-category claim, routing alphabet, hash tape, and superspace counterexample, and its perturbation formulas as printed (sign error; CH16-C01 gives the corrected form); 003's cyclotomic census and mechanization claims; 035's greedy optimality, its floating-point "dual witness," its ℓ^1 modulus bound (false), and its physics and audit readings; 020's claim that vanishing leakage is necessary for a genuine reduced zero (false; CH16-P04) and its Born-rule reading; and 007's framing that depends on an external derivation study, and its vendored source code, which the atlas rules exclude. Each paper's claim IDs appear in the ledger's historical crosswalk without inherited status.

The remaining eight archived working papers (004, 005, 006, 008, 009, 010, 011, and 022) were absorbed on 2026-09-30 under the same discipline, each bound by its own `absorbed_snapshots` block. Every surviving statement was corrected, generalized, or reproved before it entered. Not carried: 004's rational CHSH model (its correlators are not those of its vectors, and the model is a PR-box mixture), its printed ray table (seven of nine contexts not orthogonal) and its $H^1(\mathbb{Z}_2)$ reading, its characteristic-zero form of the trace obstruction (false for rings), its microcausality reading, and its $1/\sqrt{3}$ normalization; 005's stabilizer codes (correct, outside scope), its Heyting semantics, empirical-reference definition, panel result, and firewall, its truth-maintenance system and the source-withdrawn AGM claim, and its pin to an external commit other than the required tag; 006's regime framing that depends on an external derivation study, and its vendored source code, which the atlas rules exclude, its number-theory catalogue (correct, but outside scope), and its two-

fermion exchange bridge (a physics reading whose cited Lean names are absent from its package); 008's lift-admission and refusal calculus, net-utility and rollback rules, and typed terminals (routing layer, CH04-O01), its LP complexity and coordinate-descent material, its SAT and belief-revision readings, and its claim that only a repair column equal to the residual lowers Φ_1 (false); 009's tie-break functional Ξ with its evaluator hierarchy and transition gate, its quarantined Hilbert-lift conjecture, and its sheaf predictive-coding note (circular where not withdrawn); 010 entirely, an interpretive essay whose measurements and verification records refer to paths absent from this repository; 011's finite/infinite vocabulary, which labels every exact class infinite, its reading of the quotient class as a continuum, its Riemann-hypothesis springboard, the quantum bound it cites, and its refusal flags, which are policy booleans; and 022's machine proposal and open theorem queue, and its productivity remark, which fails for runs from one stage of a deterministic machine with finitely many events. No absorbed Lean or code was ported, and no new row takes Lean or executable credit. Every absorbed archived working-paper directory has now been removed from the tree; the bytes remain at the commits recorded in `source_provenance.json`.

Appendix D — Formalization status and executable evidence architecture

This appendix records the evidence architecture and mechanization status of the monograph. Selected finite discrete statements are formalized in Lean 4 (`lean4/WitnessedDifference/`) and backed by executable Python test suites (`code/witnessed_difference/`). This appendix documents the evidence binding discipline, distinguishing proved formal theorems, executable tests, and open interfaces without letting one artifact stand for another.

D.1 Mechanized finite core

The source inventory at remote main `096551f4f98d9e801d95acfa84fe85deaf309b70`, counted on 2026-10-11, is **102** `.lean` **files under** `lean4/`: the lakefile, the historical root module, and **100 modules** in `lean4/WitnessedDifference/`. This is a source-file count, excluding build outputs and the separate reference sources under `code/`. It is not a count of compiled modules, audited declarations, or fully formalized claims.

The dated evidence surfaces are distinct:

Audit date	Recorded source scope	Reports	Interpretation
2026-09-30	Historical Mathlib-backed root: then 29 source files and 341 theorem/lemma declarations	202	Historical root audit, retained in `lean4/VERIFICATION.md`; not a current full-package build.
2026-10-03	Standalone core spine: 29 supplement modules	408	Extends the earlier 28-module, 393-report tensor checkpoint with `DyadicPresentation`.

Audit date	Recorded source scope	Reports	Interpretation
2026-10-08	Standalone core spine: 50 modules listed in `scripts/verify_finite_derivations.py`	691	Later audit of the expanded spine, with source-bound reports in `lean4/*_axioms.txt`; supersedes the 29/408 scope for this entry point.
2026-10-09	`GuardedScheduleTopology`	44	37 theorems, 6 definitions and 1 inductive declaration; the arrangement homotopy theorem remains a written proof.
2026-10-09	`GuardedChainTransport` and `SparseChainTransportBounds`	143	78 and 65 explicit declarations, respectively, including 99 theorems; conditional certificate and arithmetic results, not a formalization of the entire cellular construction.

The last two entries have their own source hashes and scope exclusions in `manifests/guarded_schedule_topology_lean.json`, `manifests/guarded_chain_transport_lean.json` and `manifests/sparse_chain_transport_lean.json`. They do not silently extend the 50-module entry point. The further module compilations recorded in the 2026-10-08 review overlap existing sets and must not be summed as disjoint coverage. Dates above identify the original stored audits; their receipts bind source revisions by SHA-256. The 2026-10-11 editorial pass additionally replayed the unchanged 50-module, 691-declaration core spine. That replay does not renew the unrelated audits or the paired canonical reproduction gate.

The standalone replay uses Lean 4.33.1 without Mathlib and checks only `{propext, Classical.choice, Quot.sound}`, with no admitted proofs: `scripts/verify_finite_derivations.py --lean /path/to/lean-4.33.1/bin/lean`. The default `lake build` and historical `AxiomReport.lean` have a separate Mathlib-backed import graph, pinned in `lean4/lakefile.lean`. Neither source counts nor selected declaration counts measure whole-row formalization. For the canonical claims, the following exact bindings and the ledger's separate formalization/reproduction fields state the relevant scope.

In accordance with the canonical claim-to-declaration map (`lean4/claim_map.md`), the formal package establishes exact bindings for twenty-nine canonical theorems and one definition:

- **Exact theorem bindings:**
  - CH03-P01: No point of a finite set with at least two elements is fixed by every permutation (`WitnessedDifference.Symmetry.no_fully_invariant_point`);
  - CH07-P01: Exact path duration additivity under concatenation (`WitnessedDifference.FinitePath.duration_append`);

– CH07-P02: Path independence of exact duration potentials (`WitnessedDifference.FinitePath.exact_duration_path_independent`, with telescoping lemma `duration_potential`);
– CH14-P01: Presentation blind spots under noninjective observation maps (`WitnessedDifference.presentation_blind_spot`);
– CH16-P01: A map descends through a surjective presentation exactly when it is constant on fibers (`WitnessedDifference.Descent.function_descends_iff_fiber_constant`);
– CH04-P07: Dual-certificate soundness over $\mathbb{Q}$ on a finite edge list (`dual_certificate_lower_bound`, with `matching_certificate_is_optimal`, in `WitnessedDifference.Obstruction`);
– CH04-P08: The filled-triangle signature over $\mathbb{Q}$ (`WitnessedDifference.Obstruction.filled_triangle_signature`), whose lower bound is derived from the certificate lemma;
– CH10-P02: The kernel order on declared observer families (`declared_family_order`, in `WitnessedDifference.ObserverResolution`);
– CH10-P03: The grammar resolution bound (`grammar_resolution_bound`, in the same namespace);
– CH01-P02: Pointed relations and apartness (`pointed_relation_independence_fin` in `WitnessedDifference.PointedRelation`);
– CH16-P02: Presentation descent and sections (`presentation_descent_signature` in `WitnessedDifference.Descent`);
– CH18-P02: Nonnegative weights force permutations (`nonnegative_weights_force_permutations` in `WitnessedDifference.ClassicalCorner`);
– CH18-P03: Signed weights do not (`signed_weights_break_the_corner`, same namespace);
– CH18-P04: Unitaries over $\mathbb{Z}[i]$ are monomial with entries ± 1 or $\pm i$ and move the readout by the same permutation (`unitary_package_is_classical`, same namespace);
– CH18-P05: Where the classical corner ends (`classical_corner_ends`, same namespace);
– CH19-P02: The packet adapter (`measurement_packet_adapter` in `WitnessedDifference.PacketAdapter`);
– CH19-P03: What survives the adapter (`what_survives_the_adapter`, same namespace);
– CH19-P04: Every rational model is an image (`every_model_is_a_packet_image`, same namespace);
– CH19-P05: Contextuality as a missing global table (`contextuality_is_a_missing_global_table`, same namespace), with the general equivalence `noncontextual_iff_global_table` in `WitnessedDifference.Empirical`;
– CH18-P07: Stochastic updates are mixtures of function updates (`stochastic_updates_are_mixtures_of_functions` in `WitnessedDifference.ClassicalPackage`);
– CH18-P08: Classical instruments and the sharpness of the package S (`classical_package_and_sharpness`, same namespace);
– CH19-P06: The tolerant adapter (`tolerant_adapter_signature` in `WitnessedDifference.ToleranceAdapter`);
– CH19-P07: Exact repair on two contexts (`two_context_tolerance_iff`, same namespace, with `exists_repair`);
– CH19-P08: Injective tagging (`locallyRecoverable_iff_injective` and `dumpRight_not_recoverable` in `WitnessedDifference.Composition`);
– CH19-P09: No-signalling of local instruments and observer compatibility (`noSignalling_right`, `noSignalling_left`, `marginal_independent_of_choice`, `presentation_commutes_with_marginal`, same namespace);
– CH19-P10: Classical composites are Cartesian, and the PR box on the product cover (`composite_state_iff_joint_distribution`, `joint_unique`, `no_signalling_does_not_force_the_lo-`

cal_composite, same namespace);
  - CH19-P11: Gauge descent of phase-blind readouts (common_phase_iff, phaseBlind_readout_factors, holonomy_gauge_invariant, cross_eq_holonomy, readout_depends_on_cochain_class, gauge_invariance_does_not_select_the_readout in WitnessedDifference.AmplitudeBridge);
  - CH19-P12: The exponentiation obstruction (exponentiation_obstruction, with phaseU_no_square_root, no_rational_exponentiation and phaseU_zpow_ne_one, same namespace);
  - CH19-P14: A POVM does not determine the instrument (povm_does_not_determine_the_instrument in WitnessedDifference.InstrumentAdapter).
- **Exact definition binding:**
  - CH07-D01: Finite path structure and duration valuation (WitnessedDifference.FinitePath, FinitePath.duration).

In addition to these thirty exact bindings, auxiliary modules mechanize focused finite milestones:

- constitutive order vs. immediate succession separation on M_3 posets (Succession.lean);
- transitivity of constitutive order, and irreflexivity and asymmetry under the M_3 hypotheses (Succession.lean); the reachability criterion of CH06-P01 is prose only;
- nonzero natural multiplicity normalization (Probability.lean);
- typed observation verification records, status refusal, and authority non-entailment (Observer.lean);
- nonempty calibration hierarchies and non-negative uncertainty fields (Measurement.lean);
- fixed-length semiring path-sum right-distributivity and coboundary holonomy (SemiringTransport.lean);
- one small finite witness for each of the eight external imports of §18.5, such as the absence of an integer square root of -1 and of a Boolean square root of the toggle; they illustrate why the discrete host does not supply each import and are not independence proofs (QuantumImports.lean);
- well-definedness and set-level exactness of the closure map on the quotient by the repair relation for CH04-P03, the closure-ratio bound for $\mathbb{Q}$-valued costs for CH04-P06, the operator bound of item 4 of CH04-P05 from the infimum property, and a packaged form of observer passivity whose fiber lower bound is a hypothesis (Obstruction.lean; each narrower than its row);
- the probabilities, the trace formula and the post-measurement state of the qubit adapter of CH19-P13, and their form as an empirical model on the one-measurement cover (sum_prob, prob_eq_effect, prob_nonneg, post, instrumentModel in InstrumentAdapter.lean); the application of CH19-P04 to that model and the replay of the verification records through the packet of CH13-D01 are prose, so CH19-P13 is narrower than its row;
- restriction and one-edge subdivision for CH04-P09 from the infimum halves (ComplexGrowth.lean), the Sorkin identity for two and three alternatives under any biadditive readout for CH19-P01 (Interference.lean), and the commuting part of the retraction defect for CH16-P03 (Descent.lean), each narrower than its row.

The Lean anchors inherited from the primitive-axiomatics source were restated rather than copied. The source's exact-sequence file proved kernel/range exactness for an arbitrary function, its ratio bound used natural-number floor division, its passivity file assumed the inequality it named, and its triangle replay ranged over integer potentials; lean4/PROVENANCE.md records each replacement.

The remaining 124 canonical claim rows have partial or absent whole-row Lean bindings, including definitions and nonclaims, as recorded in lean4/claim_map.md. A prose result can be proved without being mechanized. The nineteen historical targets now have bounded finite or conditional resolutions in Appendices I–K; that status does not upgrade their partial Lean bindings or resolve the five separate source-derivation obligations of Appendix L.

D.2 Executable test suites and property testing

The Python test suite across `code/unit_tests/` and `code/tests/python/` (`integration_tests/`, `injection_tests/`, and `e2e_tests/`) provides executable evidence across tests covering:

- enumeration of the full symmetric groups on small sets, confirming the full-group case of the fixed-point criterion (no invariant point, oriented edge, or strict total order);
- enumeration of small graphs testing exactness, loop periods, and duration statuses;
- observer and measurement packets with deliberately missing fields asserting typed refusal;
- verification that calibration-version updates preserve raw-record custody and provenance;
- bounded path enumeration over exact rationals with cancellation reports, together with signed rational coarse-graining and length-one intertwining tests; the general-semiring implication has its separately scoped Lean proof;
- classical permutation-isometry representation and quantum audit fixtures (P1–P12). The current `H015_collapse_no_isometry` computes the supplied basis-map Gram matrix; `H015CollapseDilations` enumerates dilations on at most eight composite states; `H016_no_report_half_step` enumerates endomap roots on at most six report states; `H017Fin` checks a supplied Gaussian-integer linear instance; and `H020` verifies the full supplied eighteen-ray incidence and orthogonality witness before using parity. Malformed inputs and census limits refuse. The input mutation tests establish these executable scopes, separately from the written carrier-wide results and historical Lean audits;
- an exact two-phase simplex method with Bland's rule, whose every reported Φ_1 is re-checked by the dual-certificate test of CH04-P07, with tamper tests on each certificate field and Beale's cycling example started from its classical basis;
- degree-k simplicial windows, the six-vertex projective-plane torsion witness, balance identities, the Klein-four projective phase class, quotient-observer passivity by exact linear programming, observer-family kernel order, grammar collisions, and witness-rank invariance;
- exhaustive pointed-relation censuses, growth verification records for restriction, face attachment, and subdivision, exact congruence certificates for positive semidefiniteness and recovery constants, truncation leakage over $\mathbb{Q}(i)$, presentation descent, chain contractions with the sign-corrected perturbation lemma re-validated on every output, the square state space, and Sorkin sums under declared readouts.

D.3 Evidence separation

For every formalized claim, distinct pointers are preserved for:

1. prose statement;
2. claim-ledger row;
3. implementation;
4. tests;
5. formal theorem;
6. axiom/import report;
7. source revision and hash;
8. release artifact, if any.

A missing pointer is `unknown` or `not present`, not green by inheritance.

Appendix E — Glossary

Notation K. Appendix T uses $K = \ker Q$; CH01-D01 uses $\mathcal{K}$ for the admitted comparison family; Chapter 16 uses $K_{c,b,a} = r_{b,a}r_{c,b} - r_{c,a}$ for the restriction-composition defect.

Admitted cut. A finite exhibition of difference together with the structure needed to type and compare it.

Asymmetry price. The declared input whose removal restores a symmetry that moves the construction's output.

Authority. A role/jurisdiction/permission relation external to observer and measurement structure.

Calibration. A versioned relation between reference quantity values and their uncertainties and instrument indications and their uncertainties, including the conditions and use under which a result and its uncertainty are obtained.

Closure defect. A nonzero value of a proposed next differential, or the failure of a differential composition to vanish.

Descent. The existence of a well-defined global or quotient-level object compatible with declared local/restriction data.

Duration gauge. An additive valuation on composable finite intervals or edges.

Observation verification record. A record of an interaction, registration, typed witness, recomparison state, and evaluator result; a refusal or rejection verification record is not an accepted observation.

validated observation. An observation verification record whose first four ladder predicates verify and whose evaluator result is `Accept`.

Evaluator. A declared defect/repair/cost/status mechanism. It does not propose truth or confer authority.

Measurement. In the WDP interface, a validated observation extended by a measurand specification, a distinct result/value carrier and value map, scale/unit, calibration, separate uncertainty and error semantics, any claimed metrological-traceability evidence, and separate provenance/re-access. The validated observation prerequisite is WDP-specific.

Metrological traceability. A property of a measurement result supported by a documented unbroken calibration chain to a stated reference, each calibration contributing to measurement uncertainty; it is not ordinary data provenance.

Observer. A layered presentation, memory, evaluator, and policy with typed adapters.

Primitive Difference. The admitted role of apartness in a finite exhibition, not a complete structure generator.

Profile. The contexts, grammar, inputs, comparisons, resources, and terminal rules under which an observer operates.

Residual. The portion of a declared target not removed by admissible repair.

Retention. A mechanism supporting later re-access and comparison of a typed mark.

Provenance/re-access. A replayable chain linking a result to its raw record, procedure, calibration version, evaluator, and transformations. It does not by itself establish metrological traceability.

Typed witness. Evidence of a declared kind for a declared proposition; never the unqualified source of observer, measurement, or authority structure.

Appendix F — Bibliography and citation policy

All external references are registered in references.md. A registry entry records the support role and affected chapters. An entry marked `UNCONSULTED` or `UNRESOLVED` may identify a research target but may not support a theorem statement. Secondary expositions can orient the reader, but claims about an original theorem should be anchored to a primary source when one is available.

Internal source manuscripts are provenance inputs, not external validation. Their release labels, claim counts, and artifact references require independent custody checks. The unified claim ledger, not this narrative, is the definitive location for evidence status.

F.1 Bibliography

The following records resolve every `REF-*` citation in the main text. They reproduce bibliographic records already held in the supporting registry; inclusion does not extend the recorded consultation or theorem-support scope. Panov and Tril [REF-MATH-013], Construction 4.5 and Theorem 4.6, supply the prior task-complex/arrangement connection used in Appendix P.22.

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Appendix G — Finite decision procedures and proof obligations

The programme repeatedly appeals to finiteness. This appendix states what that finiteness actually buys. A finite carrier permits exhaustive algorithms for many local questions, but decidability of an encoded instance is not the same as a conceptual derivation, an efficient algorithm, or a theorem about an unbounded family.

G.1 Symmetry audit

For a finite input structure A and finite target set Y:

1. compute or enumerate the automorphism group $\mathrm{Aut}(A)$;
2. compute its induced action on Y;
3. intersect the fixed sets of the generators;
4. report the global fixed locus $Y^{\mathrm{Aut}(A)}$;
5. if a construction returns $y \notin Y^{\mathrm{Aut}(A)}$, identify the input field or implementation convention that broke symmetry.

If the fixed locus is empty, invariant point selection is unrealizable. If it has one element, that element is unique relative to the stated action. If it has several elements, symmetry alone underdetermines the choice. This procedure does not decide which input structure is metaphysically correct; it audits one finite presentation.

G.2 Exactness and certificate audit

Choose bases for finite-dimensional C^0 and C^1, represent d_0 by a matrix D, and represent the defect by a vector δ. Exactness is the linear feasibility problem

$$Df = \delta.$$

Gaussian elimination over an exact field can either return f or show inconsistency. A separating row vector y satisfying

$$yD = 0, \qquad y\delta \neq 0$$

is a dual nonmembership certificate. Over integers, modular rings, or floating- point approximations, the algorithm and certificate semantics differ; the coefficient domain must be recorded.

For weighted L^1 residual minimization, introduce nonnegative slack variables and solve a finite linear programme. A primal repair and dual feasible circulation with matching objective values certify optimality under exact arithmetic. A floating-point solver output is evidence only after tolerances, rounding, and certificate checking are specified.

G.3 Duration classifier

Given a finite directed graph and exact rational edge valuation:

1. test the declared 2-cell closure equations, if a valid d_1 exists;
2. solve $d_0 t = \lambda$;
3. if solvable, return `Exact` with potential t;
4. if not solvable, seek a cycle-period or dual certificate;
5. return `ClosedNonExact` only when closure and nonexactness are both certified;
6. return `ActiveClosureFailure` with an exhibited cell when closure fails;
7. return `ExactnessUndetermined` when the structural or arithmetic inputs do not justify one of the prior statuses.

The four statuses are evidence-bearing outputs, not labels inferred from solver failure alone. A timeout or missing 2-cell structure yields `Undetermined`, not `ClosedNonExact`.

G.4 Observer-packet validator

A finite schema validator can check whether an observer packet contains the required fields and whether table-valued maps are total on their declared domains. Deeper checks include:

- presentation commutation under a proposed adapter;
- update/report consistency for every finite memory state and input;
- representative independence of quotient evaluation;
- soundness of terminal statuses on an exhaustive fixture family;
- preservation of raw-record and calibration identifiers through policy output.

Exhaustive success establishes only the finite instance or bounded family actually enumerated. A generic theorem requires symbolic proof or a verified reduction from arbitrary instances to the enumeration.

G.5 Observation and measurement validator

For a candidate measurement verification record, validate in dependency order:

1. interaction and registration identifiers are present;
2. the raw record hash matches the stored bytes;
3. the witness semantics names a target proposition and context;
4. re-access recovers the same record and profile version;

5. the first four observation-ladder predicates reproduce, the evaluator inputs and result reproduce, and the result is exactly `Accept`; otherwise the object is an attempt verification record rather than the validated observation required by a measurement packet;
6. the measurand specification is distinct from the result carrier, and the indication and result types match the value map;
7. unit and scale are valid for the reported value type;
8. calibration version covers the indication range;
9. uncertainty and error/enclosure procedures are separately named and their inputs are present;
10. the provenance/re-access chain has no missing internal link;
11. when metrological traceability is claimed, the documented unbroken calibration hierarchy, stated reference, and every calibration's uncertainty contribution verify;
12. any authority decision is stored separately with its role and procedure.

The validator should return a structured list of failures. Collapsing missing calibration, rejected range, and absent authority into one `false` value makes repair and audit harder.

G.6 Finite probability audit

For a witness family W, mass table m, and partition $\mathcal{P}$:

1. verify that every mass is in the declared nonnegative carrier;
2. verify that partition cells are disjoint and cover W;
3. compute total mass and reject normalization when it is zero;
4. compute normalized cell values using exact arithmetic when possible;
5. verify that they sum to one in the chosen carrier;
6. if symmetry is claimed, enumerate group generators and check mass invariance;
7. for a refinement, verify the transport rule and mass conservation.

This procedure certifies a finite valuation. A stochastic interpretation needs the separate protocol fields listed in Chapter 9.

G.7 History dynamic programme

For a finite graph and semiring with executable operations, the aggregate over all graph paths of a fixed length can be computed iteratively:

$$Z^{(0)} = I, \qquad Z^{(n+1)} = Z^{(n)} M.$$

The recurrence avoids enumerating every path explicitly, though complexity still depends on graph size, length bound, and semiring operations. For a finite-state compilation of admissibility, the reported aggregate must use the declared endpoint selectors $\lambda_a Z^{(n)} \rho_b$ and a weight-preserving bijection to the original admissible paths. To attach a proof verification record, record the orientation/product convention and verify the matrix recurrence on small explicitly enumerated fixtures. An all-length algorithm must state why it terminates or how the star is evaluated.

Coarse-graining can be tested by computing fine and pushed-forward aggregates and comparing them. Composition preservation requires a separate congruence check; equality of one total aggregate is not enough.

G.8 Reconstruction countermodel loop

For each proposed quantum principle:

1. formalize it independently of quantum terminology where possible;
2. test it on finite classical simplex models;
3. test it on the intended finite quantum models;
4. test it on post-quantum nonsignalling fixtures when well typed;
5. record which model families it excludes;
6. search for conjunctions whose discriminating power exceeds the individual principles;
7. only then attempt a representation theorem.

A principle satisfied by all comparison families may be foundational but is not a discriminator. A principle satisfied by the quantum examples but also by an unexamined post-quantum family is not yet sufficient. Countermodel search is not a proof of completeness, but it is an efficient guard against premature reconstruction claims.

G.9 Profile-status verifier

For finite query family Q, enumerate each query and check:

- parsing and typing;
- profile membership;
- policy refusal predicates;
- proof and refutation predicates;
- exclusivity of terminal conditions;
- evidence attachment and verifier success.

Coverage holds for Q when each member receives exactly one allowed status. If both proof and refutation predicates hold, the evaluator is inconsistent on that query or the predicates refer to different proposition versions. The verifier should expose the conflict rather than choose by priority.

G.10 What algorithms do not settle

All procedures above consume a representation. They do not select the representation, the relevant automorphism group, the semantics of a record, the physical measurand, the observer profile, the semiring, or the reconstruction axioms. Those choices remain in the assumption ledger. Finiteness turns many well-typed questions into checkable ones; it does not erase the need to type the question.

This is the operational form of the monograph's conclusion: every verified output should be accompanied by the finite input, declared semantics, algorithm or proof, and scope boundary that made verification possible.

Appendix H. Scoped completion results and obstruction review

This supplement adds finite results under explicit hypotheses. It does not replace the original canonical obligations by broader conclusions. The executable witnesses use exact rational arithmetic; finite examples validate implementations, while the arguments below state the general finite results.

H.1 Chain-clock adequacy and refinement

Declare a finite occurrence chain with positive edge durations d_i, clock readings r_i, calibration $k > 0$, and nonnegative error budgets b_i. Assume $|k(r_{i+1} - r_i) - d_i| \le b_i$ on every edge. For every interval $[a, c]$,

$$\left| k(r_c - r_a) - \sum_{i=a}^{c-1} d_i \right| \le \sum_{i=a}^{c-1} b_i.$$

Proof: sum the edge residuals, telescope the readings, and apply the triangle inequality. Restricting the readings to partition endpoints and summing the durations and budgets inside each block preserves these inequalities. Conversely, every endpoint-preserving refinement whose fine block sums recover the declared coarse durations and budgets induces the same coarse bounds. This is adequacy relative to the declared reference duration and coupling, not physical calibration evidence. An endpoint reading alone does not supply the edgewise hypotheses. The implementation is `ChainClock` in `code/witnessed_difference/finite_adequacy.py`. The native owner is `omega_measurement::finite_chain_clock`; WDP independently replays named clock adequacy and selected block/refinement verification records. Given the admitted durations, calibration and starting reading, the exact construction $r_{i+1} = r_i + d_i/k$ has zero residual budgets and commutes with block sums. This conditional construction does not select a physical clock.

H.2 Atomic representation and stable pushforward

For a finite set X with its full Boolean event algebra, a nonnegative, finitely additive valuation v, with $v(\emptyset) = 0$, has the unique representation $v(A) = \sum_{x \in A} m_x$, where $m_x = v(\{x\})$. Proof: each event is the disjoint union of its singleton atoms. Conversely, any nonnegative singleton masses define such a valuation. On a proper Boolean subalgebra, the same argument gives masses of the algebra's atoms; individual witnesses inside one atom are not uniquely distinguishable.

A declared map $p : X \to Y$ transports masses by $m'_y = \sum_{p(x)=y} m_x$, so $v'(B) = v(p^{-1}B)$. For two mass assignments, put $\delta_x = m_x - n_x$. Their exact event distance is

$$\max_{A \subseteq X} |v(A) - w(A)| = \max\left(\sum_{\delta_x > 0} \delta_x, -\sum_{\delta_x < 0} \delta_x \right).$$

The upper bound follows by separating positive and negative terms; the sets of positive or negative terms attain it. Pushforward is contractive because coarse events pull back to a subset of the fine events. The half-ℓ^1 formula additionally requires equal total mass. No probability readout or Born rule is selected by this theorem. See `AtomicValuation` and the exact valuation routines in `finite_adequacy.py`.

H.3 A specified finite function sheaf

On the inclusion category of subsets of a finite set X, use union covers and reports $F(U) = Y^U$. Restriction is selection of the values on a subset; identity and composition laws follow from function restriction. Compatible local reports glue by assigning to each point the value of any patch containing it. Compatibility makes this well defined; coverage establishes both existence and

uniqueness, including the empty carrier/empty-cover case. The checked Rust adapter is `omega_wdp::presheaf`, including restriction, cover validation and independent replay of the candidate global section.

This theorem does not hold for arbitrary report presheaves. On $X = \{a, b\}$, allow arbitrary Boolean singleton reports but require global reports to have equal values. Restrictions form a presheaf, yet singleton reports $a \mapsto 0$, $b \mapsto 1$ agree on the empty overlap and have no global section. Observer categories and their report constraints must therefore be specified before a general descent claim is made.

H.4 Conditional factorization and the reflective-completion boundary

For a surjection $\eta : A \to B$, a map $f : A \to M$ factors uniquely through η exactly when $\eta(x) = \eta(y)$ implies $f(x) = f(y)$. Necessity follows from applying the factor map to an equality. For sufficiency, choose one preimage of each b and define $g(b) = f(x_b)$; kernel compatibility makes the choice immaterial, and surjectivity establishes uniqueness. An explicit section makes the construction independent of classical choice. Collapsing the two Boolean points cannot factor their identity map.

For contextual systems, suppose a morphism η preserves and reflects the profile relation, and intertwines every context action. If its target profile is a congruence, its source profile is already a congruence: transport a profile-related pair to the target, apply the target context congruence, intertwine, then reflect the resulting relation. These assumptions therefore cannot complete a genuinely noncongruent source. A one-step refinement also cannot be placed in the category of congruent systems until membership in that category is established. This is a set-level obstruction, not a HoTT identification or a general reflector theorem.

All seven supporting theorems compile independently with Lean 4.33.1 in `lean4/WitnessedDifference/RefinementFactorization.lean`. Their separate axiom report is `lean4/refinement_factorization_axioms.txt`; its union is `Classical.choice` and `Quot.sound`, with no admitted proofs. This supplement is not registered in the historical 202-declaration axiom census.

H.5 Exact finite quantum diagnostics

For real rational qubit density matrices, their difference is $\Delta = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$, and $\Delta^2 = (a^2 + b^2)I$.

Thus trace distance satisfies $D^2 = a^2 + b^2$ exactly, even when D is irrational. The diagonal-shadow total variation is $|a|$, so its squared contraction slack is exactly b^2. Returning a rounded square root as an exact rational distance is invalid. The Rust adapter now retains the squared certificate and refuses an irrational rational-root request.

For two declared unweighted finite $U(1)$ connection Laplacians on the same ordered carrier, let $H = L' - L$ and $B = \max_i \sum_j (|\operatorname{Re} H_{ij}| + |\operatorname{Im} H_{ij}|)$. Hermiticity implies $\|H\|_2 \le B$, because the absolute row and column sums coincide. Weyl's inequality then bounds each ordered eigenvalue displacement by B, and each adjacent gap shrinkage by $2B$. A valid independently supplied gap lower bound $g > 2B$ certifies survival of that gap. The computation does not certify a caller's spectral-gap premise. `omega_wdp::connection` delegates the actual Laplacians to `omega_holonomy`; no numerical eigensolver is substituted for that premise. General weighted vector-bundle uncertainty remains separate.

H.6 A finite profile evaluator

Declare a finite Boolean semantic table, an admitted query subset, a refusal subset, a replay budget and candidate evidence. Recheck candidates against the table independently of their claimed result. On admitted, allowed queries, the verified true and false predicates are disjoint because each query has one Boolean value. Their negation gives Undetermined; the two remaining cases are OutsideProfile and Refused. The five guards are exclusive and exhaustive. This is a conditional finite model, not an oracle for arbitrary statements.

An injective, total, table-value-preserving query encoding transports verified evidence when the source and target both replay it successfully. A change in admission, policy or budget can change the status, even with preserved semantic values. The implementation in `code/finite_profile_evaluator.py` checks both model preservation and replay; it does not equate successful transport with status invariance under arbitrary resource changes.

H.7 Outstanding requirements and native decomposition

The full row-by-row review is in `docs/derivation_review.md`. No unconditional quantum reconstruction follows: contextual tables need not have joint runs, no-signalling permits a PR box, gauge invariance permits distinct readouts, and Gaussian rational phases do not admit the required rational-cochain extension. Trial-derived statistical tolerances, physical traceability, nonclassical composition axioms and a well-typed refinement category remain prerequisites, rather than conclusions of the finite witnesses above.

Native ownership stays with the existing math crates: cochain boundaries and Betti numbers in `omega_homology`, exact image/rank and PSD checks in `omega_linear_algebra`, quotient defects and primal-dual certificates in `omega_functional`, and transport/Laplacian calculations in `omega_holonomy`. WDP retains packet declarations, adapters and independently checked verification records. An abstract simplex extension and candidate-face membership check are not a Kan completion of a simplicial set; the historical adapter names are explicitly scoped accordingly.

Validation of this supplement: 42 targeted Python tests passed and the seven new Lean theorems compiled. Six function-sheaf module tests passed in an isolated source harness. During the first pass, production Rust integration was blocked by a reviewed specification pin mismatch. Appendix I records its explicit current-state review and the subsequent production validation; the verification code was preserved. The first pass also ran five quantum/transport tests in an actual-source harness. The subsequent integration validates the production dependencies.

Appendix I. Native integration and discharged finite targets

The original target identifiers are retained for custody. Eight former open targets acquired precisely stated finite propositions or conditional results; the eleven targets left at that stage are resolved in Appendices J–K. The local research report on the source derivation and its primitives inform the packet vocabulary. Access to the external source repository is not a prerequisite for these derivations.

I.1 Greatest behavioral congruence and minimal quotient

For a finite nonempty carrier M, total updates $u_i : M \to M$ and exact readout r, put $R_0 = \ker r$ and

$$R_{k+1} = R_k \cap \bigcap_i (u_i \times u_i)^{-1} R_k.$$

Every R_k is an equivalence relation by induction. A strict refinement increases the number of classes, so stabilization occurs after at most $|M| - |M/R_0|$ strict refinements. The fixed point R_* is an update congruence. If B is any report-respecting invariant relation, induction gives $B \subseteq R_k$ for every k; conversely R_* itself has those properties. It is therefore the greatest such relation. Induction also identifies R_k with agreement after all generated input words of length at most k. Words here are finite compositions of the admitted updates, not an external clock or global ordering of events.

Reports and updates descend to M/R_*. For every surjective report/update morphism $f : M \to N$, its kernel lies in R_*, and $g(f(m)) = [m]$ is well defined and unique. Hence $|M/R_*| \leq |N|$. Two packets with the same declared input labels have the same behavior set exactly when every behavior class in their disjoint union meets both carriers; then their minimal quotients are label/readout-preserving isomorphic. A declared report equivalence is handled by readout into its equivalence classes. Nontransitive tolerances are rejected rather than silently closed.

The reusable algorithm and checked refinement/factorization verification records belong to `omega_category_theory::finite_moore`; WDP translates named packets into that owner. Seventeen new Lean theorems establish the semantic congruence, quotient and comparison properties. The finite termination bound above is proved in prose and implemented in the checked native refinement routine.

I.2 Boolean atoms and individual-witness uniqueness

For any finite Boolean event algebra, points with identical membership in all events form an atom. Each atom is the finite intersection of its signature's events and complements, hence is measurable. The atoms partition the carrier, and every measurable event is their disjoint union. Disjoint additivity gives its unique sum of atom masses. This establishes the representation on a coarse algebra as well as on a full powerset.

For nonnegative extensions to individual witnesses, a zero-mass atom forces all its point masses to vanish. A singleton atom fixes its point's mass. A positive nonsingleton atom admits distinct first-point and last-point concentrations with identical measurable restrictions. Thus singleton-level extension is unique exactly when every nonsingleton atom has zero mass. For a declared measurable map f, pushforward is $v'(B) = v(f^{-1}B)$, preserving disjoint additivity and total mass. Coarsening restricts the event algebra. None of these operations selects a canonical point inside a positive indistinguishable atom.

`omega_finite_probability::finite_valuation` owns the checked algebra and exact mass operations. WDP retains named witness/event declarations.

I.3 Coarse-graining classified by intertwining

Select rational weights, allowing signed values, with row-source matrix M. For a nonempty block partition, let $H_{x,j} = 1$ when x lies in block j. A quotient matrix B preserves representative-independent endpoint-block path sums at every finite length exactly when each outgoing sum $\sum_{y \in B_j} M_{xy}$ is constant for x in a fixed source block. Those constants uniquely define B, and the criterion is $MH = HB$. Necessity follows at length one. Sufficiency follows by induction:

$$M^0H = H = HB^0, \qquad M^{n+1}H = M(M^nH) = M(HB^n) = (MH)B^n = HB^{n+1}.$$

The argument uses semiring laws, not positivity or absence of cancellation. Finite enumeration of all set partitions gives a complete classification for the selected input matrix. Equal unrestricted totals do not imply this criterion, and endpoint-block preservation does not imply preservation of individual histories; signed countermodels distinguish those predicates. The generic matrix routine lives in the linear-algebra owner; WDP supplies the rational adapter. The new Lean module establishes the power-intertwining implication and its length-one converse under explicit associative/unital laws.

I.4 Sequential calibration and replayed measurement

Assume an input enclosure $|t - x| \le r$, nominal calibration $at + b$, gain error $|\delta a| \le \alpha$, offset error $|\delta b| \le \beta$, and residual $|e| \le \eta$. Subtracting the nominal center gives $a(t-x)+\delta a\, x+\delta a(t-x)+\delta b+e$, so the next radius is

$$r' = (|a| + \alpha)r + \alpha|x| + \beta + \eta.$$

Induction propagates every admissible chained realization, including adversarially correlated errors. Exact nominal composition follows $(A, B) \mapsto (aA, aB + b)$. A nominal update $(a, b) \mapsto (a', b')$ changes the output over the input interval by exact supremum $|(a'-a)x+(b'-b)|+|a'-a|r$, attained at an endpoint. This is deterministic uncertainty calculus under declared coefficient bounds, not a statistical confidence interval or authentic calibration traceability.

The paper's end-to-end instance uses its actual registration, witness, observation and measurement APIs. It rechecks a separately re-accessed mark, the composed nominal map, exact calibration correction and declared enclosure. False records and freshly rehashed false enclosures fail algebraic replay. For the explicit instance, the retained indication has center 2 and radius 1/10. Stage one has $(a, b, \alpha, \beta, \eta) = (2, 1, 1/10, 1/20, 0)$, so its center is 5 and radius 23/50. Stage two has $(-3, 2, 0, 0, 1/50)$, giving center -13 and radius 7/5. The composed nominal map is $t \mapsto -6t - 1$. The declared correction $+1$ and documentary additive hierarchy bound 1/10 give corrected center -12 and radius 3/2. Replay binds these values to the actual registration, witness and measurement packet; it does not infer physical traceability from the documentary fields. The native measurement owner uses checked kernel intervals; WDP binds named calibration stages, input and recomputed output in a replayable verification record.

I.5 Finite quantum instruments and canonical dilation

For grouped rectangular Gaussian-rational Kraus operators K_{oi} with $\sum_{o,i} K_{oi}^* K_{oi} = I$, define $\mathcal{M}_o(\rho) = \sum_i K_{oi}\rho K_{oi}^*$ and $E_o = \sum_i K_{oi}^* K_{oi}$. Congruence preserves positivity, and cyclicity of trace gives $p_o = \mathrm{tr}(E_o\rho)$, $p_o \ge 0$ and $\sum_o p_o = 1$. When $p_o > 0$, the normalized post-state is $\mathcal{M}_o(\rho)/p_o$ and stays Gaussian rational. Zero-probability post-state normalization is refused.

The declared dilation $V\psi = \sum_{o,i} K_{oi}\psi \otimes |o, i\rangle$ obeys $V^*V = I$. Projecting onto an outcome's environment coordinates and tracing out the environment recovers each $\mathcal{M}_o(\rho)$, not only their sum. The exact channel and Hermitian owners supply this algebra. The new instrument owner supplies grouping and normalized post-states; WDP independently replays verification records against the actual input and ordered instrument presentation. A shared POVM need not determine the

post-state, as the dimension-three countermodel verifies. Unitary completion, Born-rule selection, apparatus identification and metrological traceability remain outside this result.

I.6 Native profile evaluator and integration repairs

The finite Boolean evaluator of Appendix H now has a checked native adapter. It replays an ordered evidence prefix under the declared budget, respects admission/refusal guards, binds a verification record to the profile and candidates, and recomputes it during verification. Profile transport checks a total injective value-preserving encoding and evaluates transported evidence in the target's own policy and resource model. It never copies a status as evidence.

Production integration required two narrow repairs. The quotient-transport projection received a new explicit review against the actual committed current source/projection, with the prior row archived verbatim. Verification code and specification laws were unchanged. The quantum owner also had two incompatible `PauliString` definitions; the phase-carrying type was renamed `PhasedPauliString`, preserving the consumed legacy API and both test families. These repairs enabled tests against the real production dependencies.

The native ownership map and exact validation results are recorded in `docs/derivation_review.md`. The historical package axiom census remains unchanged; each new core-only Lean module has its own source-bound report.

Appendix J — Expert derivations, exact interfaces, and native ownership

This supplement answers ten of the eleven then-open finite targets by constructive theorems, exact obstruction criteria or explicit countermodels. CH10-O01 is completed in Appendix K after exact original source recovery. All 154 original identifiers remain unique. No completion selects physical laws or authority. Full proofs and implementation evidence are in the linked expert notes and `docs/derivation_review.md`; new Lean milestones have separate source-bound axiom verification records, without altering the historical 202-declaration census.

J.1 Retention components and succession

Fix finite typed sets of sizes $n = |M| > 0$, $k = |I|$, $m = |O| > 0$. Retention extensions are precisely independent choices of $u : I \times M \to M$, $r : M \to O$ and a subset of $O \times O$, hence their count is $n^{nk} m^n 2^{m^2}$. Projection and assembly are inverse: typing supplies no cross-component law. Restricting updates to permutations gives $(n!)^k$; restricting readouts to injections gives $m!/(m-n)!$ when $n \leq m$, and none otherwise. Reflexive symmetric comparisons number $2^{m(m-1)/2}$; equivalence comparisons number the Bell number B_m. A congruence condition mentioning both updates and reports is an additional coupling law. Identity updates and diagonal comparison are canonical choices; a readout cannot be selected invariantly under a fixed-point-free independent output symmetry.

For arbitrary comparison C, define $W(x, y)$ by $C(r(u_w x), r(u_w y))$ for every finite input word w. The empty word gives report respect; prepending a label gives forward compatibility. Induction shows any compatible relation lies in W. Deleting immediately compared pairs whose successors fail computes W on a finite carrier. Reflexivity and symmetry are inherited from C, but transitivity need not be. A state quotient requires W to be an equivalence; for reflexive symmetric comparison this is exactly the additional transitivity check. A literal descended readout additionally requires $W(x, y) \Rightarrow r(x) = r(y)$. Neither condition is supplied by tolerance.

Under total deterministic updates and validated report equivalence, the complete six-predicate retention classification has 19 signatures: nontrivial carrier, reversible updates, raw injectivity, realized comparison identification, behavioral minimality and active update. The only constraints are the singleton signature, immediate recognized injectivity implying minimality, and inactive updates being reversible with minimality exactly when the recognized report is injective. Cycles and delayed-report chains construct every permitted signature on at most four states. These invariants supplement the structural-data theorem, rather than replace its vocabulary.

For finite labelled relational updates with exact reports or validated report equivalence, unions of strong bisimulations give greatest bisimilarity; inverse and composite bisimulations prove equivalence. The bounded quotient satisfies both forward and back clauses and is minimal. Its finite refinement and quotient-isomorphism classification need no global clock. A raw directed relation embeds in a strict transitive relation iff it has no directed cycle: transitive closure establishes sufficiency, and a cycle contradicts irreflexivity. SCC condensation always gives an order on the changed carrier. Swapping a selected ordered pair in a fully symmetric carrier establishes that a nonempty invariant strict order cannot be selected.

Full proofs: typed choices, predicate classification, and relational retention.

J.2 Routing, reflection and report descent

Fix a context alphabet. In the category of profile equivalences with total context actions, arrows preserve equivalence and intertwine every action. The least action-stable equivalence containing the source profile exists by intersection. The identity function is its unit. Any equivariant profile-preserving map into a congruent target respects that closure by induction on its generators and factors uniquely through the unit. The construction is functorial and the unit is natural. The unit is initial in the comma category $S \downarrow \mathrm{Cong}$, with an initial same-carrier **coarsening**. If the unit must also reflect profile equality, equivariance and target congruence force source congruence; a noncongruent three-state source therefore has no such completion. This refutes the stronger initiality proposal without losing the paid reflector.

The typed native routing adapters first replay the actual quotient and closure certificates, then apply a separately declared partial three-class terminal table, coarsening/value-edit permissions and requests. Missing policy is refused. A cochain lift supplies checked maps L_0, L_1, L_2 with $L_1 d_0 = d_0' L_0$ and $L_2 d_1 = d_1' L_1$, actual coordinate inclusion and field transport. It induces a quotient map and preserves closedness; it need not preserve defect norms. Guarded edits keep the window and weights fixed, strictly decrease replayed Φ_1, and bound the actual Γ_2 damage by the declared nonnegative budget. A stabilized terminal additionally requires an actual fixed point of a complete declared deterministic update table; equal diagnostic values alone do not suffice. Finite checked sequences obey the summed damage bound, without proving an infinite limit or physical stability. Every verification record is bound to its actual problem. Neither a defect norm nor a reflective unit chooses a routing policy.

For sheaf semantics use the finite inclusion category of an intersection-closed context family, with families covering by union. Report sets and coherent total restriction maps define a FinSet-valued presheaf. The category of elements gives actual observer/report presentations. For a cover, the map from global reports to compatible local families is bijective iff every fiber has one element; fiber sizes zero and greater than one are respectively obstruction and ambiguity. For constrained assignment charts, matching gives a unique pointwise union; it is a permitted global report iff the global constraint admits that union. On $\{x, y\}$ with the equality constraint, singleton reports 0 and

1 match on the empty overlap but do not glue. Of four local families, two glue and two obstruct. Adding the two missing global assignments restores descent. No additive cohomology conclusion follows from these set-valued data.

Full proofs: reflection and routing and observer descent.

J.3 Functional self-reference and archival observers

Declare independent maps $q : X \to P$, $s : X \to R$, $t : X \to X$ and $b : X \to R$. Let F mean s factors through q, P presentation stability, S report stability, C equality of report and target, and B a presentation collision. Exactly 20 signatures occur: $\neg B \Rightarrow F$ and $F \wedge P \Rightarrow S$ are the only Boolean constraints. Necessity is fiber constancy and composition; the explicit constructors on at most three states prove sufficiency. Choosing $b = s$ or changing one target independently toggles calibration success.

The original R1/R2/R3 vocabulary additionally needs accessible state, reflexive provenance, later-state replay and a possibly partial comparison protocol. These are now explicit checked maps. Linked-report generation is replayed against captured and re-accessed self-state and observer identity. Report stability compares actual transported reports. Comparison availability requires both reports/targets and protocol admission; success additionally requires their declared comparison to pass. A rejected comparison is available. Later targets factor through a map iff constant on its fibers. Successful linked-report plans therefore require actual generation, comparison-domain and later-target compatibility, without silently completing partial tables. Presentation blind spots constrain presentation-only functions; richer accessible state can distinguish them. No phenomenology conclusion is used.

The legacy inventory now reproduces all 44 original source texts and every referenced blob in15 absorbed snapshots (529 occurrences). Appendix K adds the complete typed adapter criterion and original-source constructions. The original six-component observer tuple, ten-component operational observer, five-component world-observer tuple, five-component policy tuple, nine-component normalized observer tuple and general report/sensing/operator forms have distinct semantics and explicit adapter domains.

The finite layers below retain actual maps and verified verification records. Linear report/repair semantics belong to the functional owner using the matrix owner; quantum operators remain full payloads rather than lossy diagonal reports. Memory, policy, optional closure and source authority metadata stay separate. See original-source completion and the recovered-family adapters.

J.4 Reconstruction obstruction, composition and phases

A signed normalized weight vector represents a classical distribution on a declared report partition iff every block sum is nonnegative. Representing every individual indicator is possible iff every individual weight is nonnegative. An outcome-resolved signed update has a stochastic quotient iff its block aggregation factors through the report map and the resulting outcome matrices are nonnegative with stochastic total. Fiber constancy establishes necessity and constructs the unique quotient on realized report classes.

For four binary measurements let $w(\lambda) = (1 + \mathrm{CHSH}(\lambda))/16$. Eight entries are $3/16$ and eight are $-1/16$; all four pair marginals are nonnegative PR distributions. Its negativity is $1/2$. For negativity N, the total absolute mass is $1 + 2N$, so $|S| \leq 2(1 + 2N)$; PR saturates the resulting $N \geq 1/2$ lower bound. Accepted measurement tables can realize these observable marginals,

while no positive joint classical model can. They also obstruct standard quantum reconstruction preserving fixed bipartite local measurements, since the additional quantum package below excludes PR. A centralized classical machine depending on the measurement settings can reproduce PR; context-indexed global POVMs $E_{ab}^{xy} = p_{\mathrm{PR}}(a,b|x,y)I$ can do so even on a one-dimensional Hilbert space. These encodings do not preserve the required joint-run or local-factor structure.

Declare finite complex Hilbert spaces, density positivity, local Hermitian involutions, tensor composition and trace pairing. The CHSH operator obeys $\mathsf{S}^2 = 4I - [A_0, A_1] \otimes [B_0, B_1] \preceq 8I$; positivity and trace Cauchy-Schwarz give $S^2 \leq 8$. Native exact Hermitian and positive-semidefinite audits verify the stated matrix bound. Commuting complete matrix units identify the composite as $\mathbb{C}^m \otimes \mathbb{C}^n \otimes \mathbb{C}^r$: simultaneous diagonal blocks have a common multiplicity, and off-diagonal units identify them. The dimension condition $\dim H = mn$ forces $r = 1$. This paid package supplies the intended composition while excluding PR and hidden multiplicity under the additional dimension clause; CP and primitive difference do not select it.

For a real cochain declare $\chi_\theta(a) = e^{i\theta a}$. A path amplitude multiplies by the character of its edge sum. Gauge change $a \mapsto a + df$ telescopes to $e^{i\theta(f(v)-f(u))}$, so common-endpoint phase-blind readouts descend. For $\theta \neq 0$, cycle periods are retained modulo $(2\pi/\theta)\mathbb{Z}$, not faithfully as real cochain classes. A chosen logarithm anchors $\chi_\theta(1) = u$; continuous real characters are periodic. The exact rational-turn/cyclotomic branch is a distinct scale, not an extension of the fixed infinite-order Gaussian-rational phase. Declared effects and Kraus maps provide a covariant operational interpretation. For amplitudes $(1, 2)$, normalized quadratic readout gives $(1/5, 4/5)$ and quartic gives $(1/17, 16/17)$; both are phase invariant. Neither normalization nor gauge invariance selects Born.

Full proofs: reconstruction, tensor factors and real/rational phases.

J.5 Arbitrary-context repair and conditional sampling

For context frequencies f_i, minimize t over compatible probabilities $p_i \geq 0$, coordinate envelopes $d \geq |p - f|$ and context sums $\sum_s d_{i,s} \leq t$. Equal overlap marginals and normalization are linear equalities. Nonnegative slacks give a rational standard-form LP. Every feasible LP bounds the desired maximum context distance; every compatible family extends with $d = |p - f|$, proving equality of the optima. Uniform global marginals ensure feasibility. A primal/dual certificate $Ax = b$, $x \geq 0$, $A^T y \leq c$, $c \cdot x = b \cdot y$ establishes global optimality by weak duality against every feasible candidate. Replay reconstructs the actual problem.

Three point-mass distributions on a three-outcome context require the uniform repair with radius $4/3$: distances are $2(1 - p_i)$ and $\min_i p_i \leq 1/3$. All pair defects are 2, so half the maximum is insufficient. PR data are already compatible and require radius zero, without a positive global joint distribution.

For n_i iid categorical trials on m_i sections, $\mathbb{E}\|f_i - p_i\|_2^2 = (1 - \sum_s p_i(s)^2)/n_i$ and Cauchy-Schwarz give $\mathbb{E}\|f_i - p_i\|_1^2 \leq (m_i - 1)/n_i$. Markov gives failure at most $\min(1, (m_i - 1)/(n_i r_i^2))$ for a declared $r_i > 0$; for $r_i \geq 2$ the strict tail event is empty. Union bounds give simultaneous confidence without cross-context independence. If the population marginals are compatible, marginal L1 contraction and triangle inequality bound pair defects by $r_i + r_j$ on the good event. The data do not certify the population or iid premises. Full proof: repair and trial protocol.

J.6 Repeatability, full verification records and geometric soundness

For two uses of the same complete instrument and distinct outcomes $j \neq o$, the outcome-pair switching probability is $\sum_{a,b} \operatorname{tr}(K_{jb} K_{oa} \rho K_{oa}^* K_{jb}^*)$ (and the total switching probability sums over all unequal outcome pairs $j \neq o$). All cross-outcome products ($j \neq o$) vanishing establishes repeatability. Conversely test the full-rank state I/d: the switching probability is a sum of nonnegative squared matrix entries divided by d, forcing every product to vanish. Repeatability need not force projective effects or idempotent state update.

The native adapter checks actual instrument/state/event, external retained-mark re-access, interaction/evaluator acceptance, indication, composed calibration and enclosure, units and the declared traceability DAG. Rehashing false records cannot satisfy the arithmetic replay. Documentary hierarchy checks do not authenticate physical standards. Full proof: measurement and repeatability.

Legacy geometry bridges are also corrected. A zero period for one cochain does not prove path homotopy; a nonzero period obstructs it only after closedness on all declared faces is checked. Horn diagnostics check existing faces and their typed boundaries and cannot fabricate Kan fillers. Simplicial conversion refuses oriented-cell collapse or missing face boundaries. Path kernels check composability, closed cycles and nonnegative zero-absorbing mass traces. No Kan fibrancy or HoTT equivalence is inferred from these finite diagnostics.

Appendix K — Complete source-bound observer adaptation

This appendix completes CH10-O01 at its typed partial-adapter scope. All nineteen registered targets now have finite or conditional resolutions. The original 44 files are recovered byte for byte from the recorded source commit and retained in `docs/source_archive/coda_original_source_snapshot/`. The 529 absorbed occurrences are independently checked at their recorded Git commits. The inventory indexes all 573 occurrences and reviews all 79 definition-cue records, including every one of the 53 semantic candidates. Custody identifies the actual statements; the following proofs establish the adapter semantics.

K.1 Necessary and sufficient adapter criterion

Fix actual source carriers and a typed signature of partial operations and relations. Declare a surjection on each sort. A unique target interpretation preserving these mapped semantics exists exactly when each mapped operation result is constant on its input fibers and each relation's truth value is constant on its input fibers. Treat undefined as a distinct result: both definedness and undefinedness are retained. Numerical or terminal values that must remain literal use identity output maps.

Necessity evaluates the same target input twice. For sufficiency choose a source preimage of each target input and use its mapped result. Fiber constancy makes the choice independent; surjectivity supplies every preimage and makes the target unique. Apply this to every symbol. Products of surjections are surjective, including the one empty tuple of a nullary operation. Empty input sorts have no rows. Identity sort maps give a faithful adapter for every well-typed declared structure.

A required missing semantic field is recoverable from an onto visible projection exactly when it is constant on the visible fibers. Two admitted completions with identical visible data and different field values refute recovery. The actual joint image of visible and required fields is the coarsest faithful completion retaining both; any other onto faithful completion has a unique comparison

to it. Unused target inputs destroy whole-target uniqueness, so an image restriction or explicit extensions are required. These are complete criteria, not a default completion policy.

The category owner decides the finite table case, returning the first conflicting representatives. The WDP wrapper supplies 13 typed operations and three predicates. Independent countermodels cover every operation, every predicate, definedness loss and an omitted layer. The standalone Lean result establishes the same algebraic iff statements on arbitrary actual carriers; a finite table does not certify an arbitrary program or a real vector space.

K.2 The original six-component observer tuple, ten-component operational observer and five-component world-observer tuple

The original source packet is `(M,U,p,rho,G,T)`, not the later education six-tuple. Its context acts on all finite histories. Put `q(w)=U*(m0,w)`. A unique context on reached memory states exists iff `q(w)=q(v)` implies `q(rho(w))=q(rho(v))`. Necessity substitutes into the context; sufficiency chooses a representative history, with uniqueness on the actual reachable image. Conversely a state context lifts to histories iff its images of reached states remain reachable; choose a representative of each such image.

For a checked finite transducer, a product vertex records original memory, transducer state and memory after its emitted word. Induction on word length shows the initial vertex and each input edge represent exactly the source history. Applying the final emission gives `q(rho(w))`. Every reachable vertex has a representative word, and the graph is closed under all input symbols. Checking its fibers therefore decides the criterion for every finite history, with at most the transducer-state count times the square of the memory-state count vertices. A failing fiber gives two actual words. This is all-word verification, not a length cutoff.

The original O1 criterion is nonconstant induced profile. O2 requires two equal-profile histories separated by a context. The product graph also decides these criteria, grouping by source profile for O2. A context changing one profile is insufficient: parity flip changes every profile while preserving profile equality. Grammar and terminal routing do not occur in O1/O2, so changing only G/T cannot change them; the source's six-field independence proposal is refuted. An input-label-independent toggle retains length parity, also refuting its proposed O0 corollary. A profile-to-grammar encoder remains an additional explicit map.

The ten-component operational observer has an input-dependent Mealy report and an independent profile. Both are retained. The five-component world-observer tuple `(image p,C,H,W,G)` instead supplies a history presentation with actual contexts, witnesses and grammar. Contexts need not descend to profile equality. Its facts require an explicit grammar interpretation and forward stability: membership in a subset must survive every declared context. The finite world adapter checks the actual image, history closure, witnesses and this condition; it adds no update or evaluator.

K.3 Compressed policy, normalized evaluator and report families

The five-component policy tuple `(C,D,G,R,T)` uses G for transformations and R for repairs. It is distinct from the source grammar G and profile codomain R. Actual comparison defects are computed from the carrier coordinates and restriction maps. A scalar norm is separately declared. In the checked rational specialization, simultaneous finite signed permutations act on states, comparisons, defects and repairs; actual intertwining and norm invariance are checked. Reindexing repairs then preserves the quotient infimum. A verified primal/dual verification record must match the actual

defect, repair matrix and weights; a valid certificate for another problem is refused. The declared terminalizer is replayed separately.

The nine-component normalized observer tuple retains its carrier/stalk data, comparison, actual defect, repair, optional closure, gauge, transformation class and terminalizer. A finite interaction carrier does not make its coefficient spaces finite. Linear and operator instances therefore use their actual symbolic matrices/certificates; finite verification record domains use checked tables. Optional closure descends iff it annihilates the repair image. An undefined closure ratio or omitted control budget is not supplied by a default.

Aggregation/transformation/gauge triples need independent carrier, defect, repair and terminal declarations. General report maps, token predicates, invariant classifiers and finite-capacity reports retain their actual codomains without acquiring other layers. General linear sensing may expose repairs: it becomes a quotient report iff it annihilates the repair image. A supplied blindspot witness is checked by applying both actual matrices. Memory uses its exact directed comparison, rather than a symmetric or reflexive closure. Measurement and self-target extensions retain their separately checked bridges.

K.4 Operator source and necessary numerical hypotheses

For a finite-dimensional Hermitian repair space W and the declared weighted half-trace norm, quotient optimization has an attained minimum. A Jordan decomposition of the residual and a polar-bounded linear functional annihilating W give matching primal and dual costs. The polar is taken for the declared ambient norm: on trace-zero Hermitian inputs a functional is represented modulo scalar identities, so the dual bound cannot be copied unchanged from the full Hermitian space while also retaining the old divergence constraint. Weak duality establishes sufficiency; finite norm attainment and a supporting quotient functional prove necessity. The native rational-complex certificate checker replays every equation, positivity condition, bound and actual-input binding. Some exact spectral witnesses are irrational; they remain covered by the mathematical theorem and outside that rational certificate representation.

A comparison transport descends to quotients iff it sends the actual source repair image into the target repair image. Positive trace-preserving maps contract trace norm on Hermitian inputs by transporting their positive and negative parts. Together with calibrated weights and repair inclusion this bounds the transported quotient obstruction. Shape or CPTP contraction alone does not supply the repair-inclusion hypothesis.

For state-generated source defects, a compatible trace-one reference section sigma yields the traceless repair `rho-sigma` and `delta(rho)=d0(rho-sigma)`. If this repair is admitted, the obstruction vanishes. An arbitrary free-edge circulation cannot replace those data. Incompatible reset comparisons instead give a genuine nonzero state-generated fixture. The source's dg Lie enrichment is also preserved separately: degree-zero nilpotence does not imply nilpotent action on degree one, so its finite exponential requires the representation-level condition.

K.5 Source and implementation evidence

The completion proof and family crosswalk bind every family to its actual source and declared domain. Detailed proofs: unrestricted adapters and sensing, original histories, world facts, policy evaluator, and operator/vector geometry. The recovery report records byte custody.

Reusable partial-structure and history algorithms belong to the category owner; linear sensing

belongs to the functional owner using the matrix owner; trace certificates belong to the quantum owner. WDP assembles their source packets and independently replays verification records. Lean establishes the complete stated algebraic and all-history criteria, with source-bound axiom reports. The canonical row retains partial Lean coverage because its full source reconciliation and all matrix/analysis arguments are not one mechanized corpus theorem.

Appendix L. Separating the archived source derivation from WDP, and replay bridges

The programme is oriented from the archived source derivation and its ongoing continuation through WDP to mathematical realizations and checked reasoning protocols. Its source origins and remaining obligations are recorded in the dependency programme and its separate bridge register. These additional results do not enlarge the scope of the earlier 154 canonical rows.

L.1 Primitive provenance

The source seed treats `<>` as one opaque datum. Its source establishes inertia and the distinction between proposal, supply and admission. The relation interpretation of `<arb>`, the protected prohibition represented by `<ara>`, and the resources for witnessing them require explicit admission and law provenance. Once R(a,b) and irreflexivity are given, a=b would imply R(a,a), so the distinction and identification obstruction follow by substitution. The source's provenance proof also shows why those inputs do not follow from a single opaque inscription alone. The continuation's distinct seeds have distinct commitments and are kept as named source contexts.

L.2 Evaluation, permission and exact information

For a report E:X->Reports and a protected decision A:X->Decisions, a decoder on the actual report image exists iff E(x)=E(y) implies A(x)=A(y). Necessity follows by composing with the decoder. Sufficiency assigns each occupied report its representative's decision; the fiber condition makes that choice well-defined. Equal successful evaluations with opposite permissions supply a finite countermodel to recovering independent permission from evidence. No deterministic postprocessing of that same report removes the collision.

The joint signature (E,A) identifies exactly agreement of these two meanings. Its projections recover both, and every report sufficient for both maps to it. This is the coarsest protected-question signature; it need not identify full source states. A declared policy can make permission a function of evidence. Such a policy and its recognized standing are additional inputs.

One program may implement proposal, evaluation and admission as distinct typed functions. Three programs may share a false grant flag. The semantic separation criterion therefore does not prove a required physical actor count. `ProposalAdmissionBridge.lean` establishes the abstract image criterion, its countermodel, joint projections and equality characterization. Native and Python adapters reuse the category owner's finite descent checks.

L.3 Replay and certification

For a declared deterministic step, replay is an ordered fold and obeys replay(s,first++second)=replay(replay(s,first),second). Induction on the first log establishes this law. Induction on a declared run establishes equality to replay; conversely replay supplies a

declared run. In the guarded admission model, invalid or unpermitted proposals leave admitted state unchanged. These laws are mechanized in the same standalone Lean module.

A record of histories reconstructs protected semantics iff those semantics are constant on its fibers. Multisets can lose event order, and final states can lose occurrence count. Newly hashed false records remain false. The executable verification record check binds independently supplied initial state, ordered events, declaration, scope and actual transitions; it does not mint authority. Faithful future execution additionally needs extension congruence and preservation of protected future observations.

One retained proposal-protocol calculus episode now connects an actual model proposal to host audit, extraction and Stage 5. Its unchanged seed asks for the real integral of `f(x)=x` from `0` to `10`. A fresh Space Bunny response supplied `F(x)=x^2/2` and `50`; the host checked the derivative and endpoint difference using the original calculus checker. The control `F(x)=5x` has the same endpoint difference but is refused because `F'=5` is not `f(x)=x`. The saved packet, raw response, candidate identity, source hashes and host verification record support this one execution. Audit and extraction here are host operations, not additional model calls. Provider-reported cost is zero; this is not an independent account-balance reconciliation or authorization grant.

L.4 Directed source quotients and scalar realizations

The fixed-carrier continuation bridge preserves all n^2 integer assertion coordinates, including loops and independent reverse arcs. Raising supplies d0g(x,y)=g(x)-g(y). Marked cyclic diagnostics cancel these repairs exactly, so D_F d0=0. Rational repair equality reflects integer equality: fixing a root o in a nonempty carrier yields integer h(x)=e(x,o) and d0h=e whenever e=d0g for rational g. The empty carrier has the trivial equality. These are repair-equality results, not equality of discrete change and norm costs.

A filled mark need not balance, and balancing marked triples need not detect loops or all other defects. Antisymmetric simplicial interpretation requires its extra hypotheses. Source term-addition countermodels prevent inferring a global dynamic quotient update. The adapter uses the existing matrix and functional owners, with no loop-dropping graph surrogate.

The separate finite quantum realification theorem starts with declared complex Hilbert states, effects and instruments. The block map R preserves multiplication and adjoint, has real trace 2Re of the complex trace, and represents states as R(rho)/2 while effects and Kraus operators remain unscaled. It preserves outcome probabilities, post-states and sequential composition in the real submodel carrying the chosen complex structure. Local composition now has an explicit balancing map T from the ordinary 4mn-dimensional real tensor to the 2mn-dimensional real view of the declared complex tensor. Its kernel is exactly the complex balancing relation, and P=T^t T/2 is the balanced projector. Operators compress with a factor 1/2; products of individually normalized realified densities do not. Arbitrary complex composites have the normalized lift T^t R(rho)T/4, preserving local effects and declared CP instruments. General written finite-matrix proofs, selected integer coefficient proofs in Lean and exact Q(i) native checks are separated in the tensor supplement. This is representation equivalence. It does not select complex scalars, Hilbert geometry, Born readout or unrestricted real/tensor composition from an inscription. The proof and exact owner checks are recorded in the quantum bridge.

L.5 Complete raising and marked results

The two-relation model supplement defines the carrier and source correspondence used here. For that model, fixed-root normalization N retains the complete directed integer table, including loops. At a fixed writing, N(e)=N(f) iff f-e is an integer gradient. Minimizing N after each common rewriting gives the complete source result key. A sorted collection of loop, pair and cycle values can lose incidence and therefore need not determine this key. The source's nothing result is characterized by zero loops, two-way sums and ascending triangle sums. These criteria have carrier-wide written proofs and selected standalone Lean proofs; their concrete programs have separate finite original/native comparisons.

For the five-edit model, the complete mark is the jointly minimized pair (N(e),F), where the same writing transports both the table and the oriented fillings F. Separate minima can lose their joint incidence. At a fixed writing, mark equality is exactly equality of fillings together with a gradient difference. Marked cycle sums are invariant under that gradient. Raw canonicalization and alignment retain the original assertion multiplicities needed to replay the declared costs.

The retained original-function verification records compare 779 two-relation model observations, 16,292 bounded bulk raisings, 3,643 rewritings and 779 alignments, followed by 2,314 five-edit model observations, 17,489 raisings, 12,907 rewritings and 2,314 alignments. These are finite differential evidence. The sixteen and nineteen selected Lean declarations for these two supplements prove their stated algebraic laws and countermodels; they do not formally verify the concrete Python or Rust implementations. AddTerm and Identify can separate equal result marks, and identification can destroy marked balance. Ordered raw intermediates remain necessary for the actual source execution.

Full statements, source pins and evidence: raising results and marked results.

L.6 Larger frontier

The remaining foundational questions concern the primitive admission floor, end-to-end source/run refinements, dynamic carrier/context coherence, operational probability/scalar reconstruction, and recognized governance and execution maps. Jurisdiction, appeal, facts and agreement each require their own declared meaning and soundness. Checked repair, the admission, adapter-soundness and continuation protocol, and the discrete deformation steps must bind to actual carriers, detector adequacy, priced changes, scoped terminal policies and separately recognized permission. Existing finite witnesses and proofs support these constructions at stated scopes; none silently licenses the entire enlarged programme.

The L.3 calculus episode closes one concrete model-to-host path. It does not establish the entire staged demo campaign, every recursive re-entry, or the general source-refinement obligation. The current evidence index separates this execution from older reported runs and from written or mechanized mathematical results.

Appendix M. Meron characters, finite shadows and conditional limits

This appendix integrates the finite-to-continuum source merged in PR #401, at revision `3bec3e2dd5659b53c33889d967fcd2b09689043c`. The source review and replay boundaries identify the actual functions, assumptions and unfinished targets. These additional bridge results use the separate source-bridge register; the canonical 154 rows retain their original scopes. Operational authority remains null.

M.1 Meron cancellation and the observables it preserves

Declare a complete orientation orbit U=(Z/2Z)^k, a constant nonnegative weight A, a baseline sign s in {+1,-1}, and a sign character chi_m(u)=(-1)^(m dot u), for m in U. Its signed partition contribution is

$$Z_U = As\sum_{u\in U}\chi_m(u) = \begin{cases} As\,2^k, & m=0,\\ 0, & m\neq 0. \end{cases}$$

For m nonzero choose a coordinate j with m_j=1. The involution u -> u+e_j pairs equal weights with opposite signs, proving cancellation. For m=0 every character value is one. A positive surviving sector further requires s=+1. Completeness of the orbit, invariant weights and the character law are explicit hypotheses to be established for the selected model.

For a declared real or complex observable O, put

$$\widehat{O}(m) = 2^{-k}\sum_{u\in U}\chi_m(u)O(u). \qquad N_U = As\,2^k\widehat{O}(m).$$

For A>0, deleting a nonzero-meron orbit preserves its observable numerator iff its coefficient at m vanishes. Orbit-constant O is sufficient, since its coefficient is then proportional to the cancelled character sum; it is not necessary. With k=1, A=1, s=1 and O(u)=(-1)^u, the partition contribution is zero but the numerator is two. For m=0 the retained quantity is the orbit average of O. One orientation represents it only under an additional constancy condition. Normalized expectations also require a nonzero total Z.

This supplies an interface between WDP and the shadow-to-completion bridge: a compressed report determines a protected query on its actual image exactly when that query is constant on report fibers. If partition-only reports lose an observable coefficient, retain that coefficient or its separately verified improved estimator. Cancellation alone does not preserve arbitrary queries, paid histories or future source transitions. The image-factor criterion of Appendix K applies to the chosen report; the character calculation supplies its model-specific query, rather than supplying an institutional terminal verdict.

M.2 Finite characters and the declared circle carrier

For a positive integer N and a declared complex root zeta of exact order N, the characters of Z/NZ are chi_j(n)=zeta^(jn). For a nontrivial character, its sum S obeys (1-zeta^j)S=1-(zeta^j)^N=0, so S=0. The trivial sum is N. Products give the corresponding finite-torus statement. The finite dual has N characters, or the product of the declared finite orders, rather than a free circle. Meron orientation characters use only the subgroup {+1,-1}.

After admitting the unbounded discrete group Z and the topological group U(1), every character is uniquely n -> z^n for its value z at one. Evaluation at one is a bijection onto U(1). For the compact-open topology it is a homeomorphism: compact subsets of discrete Z are finite, and finite families of powers vary continuously with z. The same argument identifies the dual of Z^d with U(1)^d. These are conditional representation theorems once the unbounded group, circle and topologies have been supplied.

They do not construct a lattice limit from a single finite period. A refinement programme still needs compatible groups/maps and a chosen limit topology. Continuous translations R^d require another declared carrier and the appropriate continuous-character theorem. Radian coordinates and Haar measure normalization are separately specified; rational interval enclosures for pi do not make pi an exact rational coefficient or select the circle from an L1 obstruction. The source's staged declarations record these changes of scope. Its orthogonality status functions encode the theoretical expected delta; they do not independently evaluate root-of-unity matrices. This describes the PR 401 source reviewed on 2026-10-02. The later PR 407 finite presentation also sums rational character intervals and checks containment of its assigned expected delta. The new exact native evaluation in the dyadic supplement instead actually sums cyclotomic roots; these are different evidence scopes.

M.3 What a continuum enclosure assumes

Let a>0, C>=0 and p>0. Let I_a=[l_a,u_a] contain the actual discrete value d_a, and suppose a model-specific proof supplies a common real target c with |d_a-c| <= C a^p. Then

$$c \in [l_a - Ca^p, u_a + Ca^p].$$

Indeed d_a-Ca^p <= c <= d_a+Ca^p, and the input enclosure extends both endpoints. For several mesh directions replace C a^p by the sum of their proved nonnegative bounds. Intersecting such enclosures remains sound. If their widths tend to zero they contain at most one common target, since any two common targets have distance bounded by every width. Existence from nested nonempty bounded closed intervals additionally uses completeness of the declared real carrier.

The merged defect carriers compute these expanded intervals and record overlap and a contraction flag. They accept the constants and orders as inputs; overlap does not prove a derivative bound, a PDE estimate or a common physical target. A smaller declared error term need not shrink the total width if the discrete input uncertainty grows. Compatible field, action, state and observable maps require their own preservation and convergence arguments.

M.4 Finite-size extrapolation and its missing remainder

The actual occupation-basis checkerboard transfer replay gives Z(4)=912, Z(6)=33012 and Z(8)=973824 for repetitions=2 and q=3. This function traces over all 2^L occupation states; it performs no fixed-particle-number projection. Define the dimensionless log-partition density g(L)=ln(Z(L))/L. Interpreting it as a thermodynamic free energy requires the separately specified temperature, sign and normalization conventions.

For x_i=1/L_i^2, the source's Neville tableau encloses the value at zero of the interpolating quadratic. Its exact weights for L=(4,6,8) are

$$(w_4, w_6, w_8) = (4/15, -81/35, 64/21), \qquad \hat{f} = \sum_i w_i g(L_i).$$

The weights obey sum w_i=1, sum w_i x_i=0 and sum w_i x_i^2=0. Thus under the exact quadratic law g(L)=f+c_1/L^{2+c_2/L}4 the result encloses f. More generally, if a proof binds the same f and coefficients to all selected values with remainders |r_i| <= b_i, then

$$f \in [\hat{f}_{lo} - B, \hat{f}_{hi} + B], \qquad B = \sum_i |w_i| b_i.$$

This follows by cancelling the first two correction terms and bounding sum w_i r_i by triangle inequality. The required b_i are model-error bounds, distinct from the arithmetic intervals for the logarithms.

Three exact values alone cannot supply them. Set H(x)=(x-1/16)(x-1/36)(x-1/64). The zero function and H agree at all three sample nodes; their limits at zero are respectively zero and -1/36864. Both are polynomials with expansions in powers of 1/L^2. Adding any multiple of H to a fitted quadratic preserves the samples while changing the limit. The original extrapolator returns [0,0] on those three zero inputs. This countermodel refutes the claim that exact arithmetic or a finite polynomial fit alone bounds an unrestricted thermodynamic limit.

For the actual three Meron values, the computed extrapolation is near 1.6941183355 with arithmetic width below 10^-19. Its current verification record carries no independently justified scaling remainder. That width therefore measures the enclosed interpolation calculation. Appendix O separately establishes time/volume and selected spatial-source limits using the actual Hamiltonian and explicit bounds. Those proofs do not validate this three-point quadratic fit or supply its missing model remainder.

M.5 The field-theory frontier

At the PR 401 source pin reviewed on 2026-10-02, the five higher-dimensional targets provide a useful order of work: an actual 2D Meron ensemble, metric/deficit-angle data, 3D winding/linking, 4D simplicial transport and action, and a field-theoretic topological sector. Their verification records at that pin have different scopes. T1 reuses the 1D value 912 without constructing its labelled 16-site torus model. T2 encloses the declared angle pi/3 for a five-triangle toy geometry. T3 records proposed winding/linking data; T4 supplies zero simplicity and Bianchi defects as constants; T5 supplies charge and sign fields rather than deriving them from a field configuration and a fermion determinant.

In particular, the declared sign (-1)^(2Q) equals -1 at both Q=1/2 and Q=-1/2. Charge reversal alone therefore cannot justify the opposite sign assigned in T5. An actual sign-changing involution, weight invariance and observable transport need independent proofs. Wire syntax and verification record hashes certify their respective artifacts, without establishing these physical laws.

Later merges add exact Regge code and further 2D/3D/4D Meron implementations. The preceding classifications describe the pinned source audit; those later implementations require their own source correspondence and model-error proofs.

A completed route to the selected field theory must bind actual carrier refinements, fields and action, model-derived error bounds, a limit construction, and preserved observables and dynamics. Meron cancellation can reduce a sign obstruction within such a route. Finite shadow and refinement interfaces can expose its missing assumptions. The current additions advance these interfaces while retaining the enlarged O03/O04 obligations.

M.6 The actual dyadic refinement and its topology choice

The later PR 407 presentation declares C_n=Z/(2^n) with k -> 2k. The coordinates k/2^n are unchanged under inclusion; equality at a common later level is exactly equality modulo integers. Thus the algebraic direct limit is D=Z[1/2]/Z. It is not already the full circle.

With the discrete topology on D, compatible finite character indices a_(n+1) mod 2^n=a_n identify its continuous dual with Z_2. With the circle-subspace metric topology, D is dense in the supplied circle and its metric completion is that circle. Continuous characters in this second branch extend to the circle and are indexed by ordinary integers. The refinement supplement establishes the common-level initial property, inverse-family classification and completion/character arguments. Algebraic refinement selects neither topology.

Each finite character has exactly two adjacent extensions, a and a+2^n. Fifteen selected Lean residue laws prove refinement, restriction and this two-element fiber at every natural level. They do not formalize the infinite groups or topology. Eight math cases and one WDP case actually evaluate finite roots, retain checked prefixes and compare 441 original source observations under explicit resource limits.

After supplying circle length measure and a continuous periodic F, its dyadic average differs from its integral by at most omega_F(2^-n). For an L-Lipschitz function the bound is L/2^(n+1), by integrating the cellwise linear error. This conditional limit argument requires the declared carrier, measure and regularity; it supplies no determinant, action or thermodynamic remainder for a physical model.

The original enclosure flag checks adjacent overlap and nonincreasing widths. It accepts [0,1], [1/2,3/2], [1,2], [3/2,5/2] at declared rate 1/2; the original mark lift consequently sets its witness flag true. Yet the first and last intervals are disjoint and all widths stay one. The retained original-function output therefore refutes using that flag as a common-target or contraction certificate. Even a sound finite nested prefix still needs an infinite refinement law and model-specific common-target bounds. GEN-WDP-20 through GEN-WDP-22 record these results without changing the canonical 154 rows or closing the five enlarged obligations.

Appendix N. Source form specifications and native mathematical ownership

The source-derivation course already contributes the opaque admission boundary, the separation of source/object/apparatus declarations, ordered replay, protected reports and actual source raising/marked results to Appendix L. Its 19 family labels and level 00–100 correspondence are educational indexes, not evidence that all these structures are forced by the initial inscription. The source-graded education crosswalk makes the relationship to Chapters 1–4, 6, 8–17, 20–21 and Appendices K–M explicit.

N.1 What the 14 form specifications reconstruct

The existing source YAML specifications declare their relational-profile qualification gap. Apartness constructs an extensional quotient under its supplied laws; isomorphism canonicalization recovers labeled data only with a transport; primitive integer rays recover original weights only with their scale; a final-state digest does not reconstruct an omitted history. These different reconstruction targets must not be conflated with a relational lossless project-join theorem or a declaration-level decision procedure.

This distinction matters for continuation. The five-edit model counterexample of Appendix L.5 already shows why a current canonical mark cannot replace the complete raw representative needed for future edits. Protected current reports and protected future operations impose different preservation obligations. An index entry or an `executable` metadata flag does not discharge either one.

N.2 One implementation per native owner

The native decomposition moves exact matrix path kernels into `omega_epk`, replayed diagnostic/proposal routing into `omega_coda`, and finite lift/closure/terminal policies into `omega_prime_algos`. Existing WDP paths remain compatibility surfaces. The owners use the existing graph, matrix, certificate and cochain owners and do not depend back on WDP. Source custody and paper-specific presentation remain in WDP.

These are explicitly scoped mathematical profiles. The checked-repair profile supplies no operational authority grant, and the admission-protocol profile does not replace the nine-conjunct operational continuation gate. Agreement on a supplied overlap list is not a general sheaf gluing theorem; a declared contracting sequence is not proof that a source algorithm produced it. Those boundaries travel with the extracted implementations.

N.3 The shared language as a declared interface

The repository's schema grammar and typed intermediate representation are software interfaces for versioned carriers, maps, normal forms, protected questions and evidence. The language boundary proposes one SQL profile as an initial executable target, with explicit set/bag, NULL, numeric, dependency and reconstruction semantics. This source increment does not claim that SQL lowering, a HoTT kernel or an all-encompassing mathematical language has already been implemented.

Finite restriction/composition laws and extension fibers can be checked by the current report-descent owners; exact cochains and primal/dual certificates by their current algebraic owners. Higher identity and transport laws require their separately declared formal backend. A hash or supplied proof flag refers to evidence rather than checking it. A vanishing class in an incomplete obstruction detector does not establish a global extension. The canonical claim ledger and the five open source-derivation obligations retain their existing statuses; this appendix supplies integration and ownership, not an unearned foundational completion.

Appendix O. Presentation closure from finite shadows to completion

O.1 Relative continuity closes the realized image

The presentation-completion supplement combines factorization on the realized image with uniform extension. For p:X→P into a metric space and q:X→Y into a complete metric space, require

$$\forall \varepsilon > 0 \; \exists \delta > 0 : \quad d_P(p(x), p(x')) < \delta \implies d_Y(q(x), q(x')) < \varepsilon.$$

This condition is equivalent to a uniformly continuous image factor g:p(X)→Y. Its unique extension to the completion of p(X) gives the completed protected query. Metric separation supplies exact fibre constancy at equal profiles. The extension applies to the actual image, not unused labels. If p is also uniformly continuous on a metric source, its completed transport commutes with the completed query by uniqueness.

The separate metric Lean module now establishes this equivalence and commuting square using actual Mathlib metric completion, with fifteen selected declarations audited by its own axiom report. The dated 2026-10-08 record in the completion/duality supplement reports the scoped Lake build on Lean 4.33.1 and the pinned Mathlib revision. A separate 2026-10-13 compatibility replay compiled the unchanged completion module and its fifteen-declaration report with Lean 4.34.0 and Mathlib `5ed2965256430c3649e86755f9576b54eca72435`; its reported axiom union was `propext`, `Quot.sound` and `Classical.choice`, without `sorryAx` or a custom axiom. This is not a fresh build of the full pinned 4.33.1 package: the current 4.33.1 cache lacked compiled Mathlib completion modules. This classical existence result supplies no executable modulus. The character, Haar and physical-model statements below have separate scopes; the historical formal censuses retain their original meanings.

Source Lipschitz control alone is insufficient. On discrete X={0,1,2,…}, p(0)=0, p(n)=1/n, q(0)=0 and q(n)=1 give injective p and a 1-Lipschitz q, yet the image query is discontinuous at zero. Every finite factorization can pass while the all-level relative continuity condition fails.

O.2 Character closure selects the circle branch

For dyadic torsion D with the circle-subspace topology, every continuous character extends to T. The character group is Z, and its compact-open topology is discrete: K={0}∪{1/2^n:n≥1} is compact and detects every nonzero integer character at value −1. Taking the character dual again gives T, with evaluation D→T equal to its dense completion embedding. The proof uses uniform extension and explicit dual topologies; D in this topology is not locally compact, so general LCA reflexivity is not applied to it. Discrete D instead has dual Z_2 and bidual discrete D.

Thus a coherent character tower requires the selected continuity condition. The alternating-bit 2-adic family is coherent but sends inputs 1/4^m→0 to character turns tending to 1/3. It is excluded from the circle branch. Neither topology is forced by the primitive mark.

The same proof works for each supplied finite product D^d: coordinate characters give discrete Z^d and character bidual T^d, with dense evaluation embedding. Coordinatewise Lipschitz bounds give product-grid sampling error at most Σ_i L_i/(2N_i). This supplies the selected 2-, 3- and 4-torus branch; it does not identify the physical dynamics on those spaces.

O.3 Observable closure and its remaining model premises

Band-limited trigonometric integrals descend exactly to sufficiently fine finite grids. General L-Lipschitz integrals do not: zero and distance to the grid have identical samples and different integrals. Averages of a single admitted observable recover its circle integral with error L/(2N). Arbitrary measurable observables do not share this sampling theorem.

The incoming Haar source returns a recovery flag on a prefix below its threshold, and a direct shrinking-overlap flag need not bind a common target. The supplement retains actual calls and distinguishes the wrapper that detects the illustrated empty intersection. Its positive example computes actual samples of one piecewise-linear circle observable.

Gaussian-rational scalars with the Archimedean metric also have the standard coordinate completion R^2, identified with C; extending multiplication requires bounded-set control. Fixed-size matrix sums, products and traces then extend entrywise. Infinite-dimensional or unbounded-operator limits, state selection and physical field models need further premises.

These results provide a specific conditional presentation closure and an observable transport criterion. A Meron route must still bind its ensemble, sign/weight identity, protected numerator, topology and model-specific time and volume errors. Numeric convergence does not extend a discontinuous Boolean threshold or supply institutional authorization. The five enlarged source-derivation obligations remain open; the canonical 154-row ledger is unchanged.

O.4 A common target for the actual finite Meron chain

The time/volume supplement binds the original full-Fock transfer to the admitted spinless U=t Hamiltonian. Its local block is I+(q-1)P_x for an actual CAR projection. At M repetitions the physical temperature parameter is theta=M log(q); fixed q with growing M changes temperature. The removed scalar subtracts M log(q)/4 from the unscaled log partition density.

The positive similar product and matrix-logarithm bounds give time error at most M(log q)^2 q^L. For rational q=1+r/M and a common parameter in [r-rho,r+rho], the corrected density has error at most r^2 $q^{L/M+(3/4)(r}$2/(2M)+rho) against the exact finite Hamiltonian. Open-chain splitting changes one bond of norm 3/4, so its log partitions are near-additive. The resulting thermodynamic target f(theta) satisfies |ell_L(theta)-f(theta)|<=3 theta/(2L) for periodic chains.

Thus the corrected actual source interval, expanded by r^2 $q^{L/M+(3/4)(r}$2/(2M)+rho)+3(r+rho)/(2L), encloses one target. For bounded r and M>=L, this radius tends to zero as even L and M grow and rho tends to zero. This establishes a selected joint time/volume limit without assuming a quadratic finite-size scaling polynomial. The target is 3/4-Lipschitz in theta, so its rational-parameter presentation closes to the admitted nonnegative real parameter line by PC-1.

The analytic model, near-additivity and limit proofs are written. The new native quantum owner reuses the existing CAR/matrix owners; WDP retains the declared plan and actual trace. Nineteen Python and five native checks cover their separate finite scopes. A spatial-mesh field theory, a complete infinite operator algebra, all observable numerators and general sign-problem complexity are not established by this partition-density closure.

O.5 Spin-sector and actual observable transport

The spin/observable supplement derives the periodic fermion boundary in the actual occupation carrier. A staggered diagonal unitary turns the bulk into the antiferromagnetic Heisenberg chain. Odd parity uses periodic spin coupling; even parity uses a pi twist. Both matching restricted sectors are needed for full-Fock readouts. The source and unrestricted periodic spin chain have the same thermodynamic log density at fixed theta by a one-bond norm bound, while their finite spectral readouts can differ.

Expanding the actual local breakups gives a graph/orientation-to-history bijection and binds the SQL staggered observable to two insertion derivatives of the transfer trace. In the canonical fixture, the raw squared numerator is 36864 with partition 912. Its time-average square is 48/19; the equal-time square is 149/57. Retaining the query therefore matters even after the Hamiltonian has been identified.

A three-block matrix-jet lift establishes the normalized time-average query converges at fixed volume to the finite Hamiltonian's imaginary-time Kubo correlation. Its explicit bound preserves parameter uncertainty and gives derivative convergence directly. It does not establish the growing-volume correlation or spatial scaling limit. The general analytic proofs remain written; generic

exact derivative multiplication belongs to linear algebra, the model to quantum physics, and WDP retains the actual plan/readout.

O.6 A checked finite language path

The normal-form presentation closes one existing repository schema grammar specification profile through actual typed intermediate representation, native FD checking, SQL storage and independent reconstruction. On legal set relations, the binary projection has natural join as a left inverse. The written closure and two-tuple countermodel proofs establish the schema criterion independently of the current instance. BCNF components and lossless join remain distinct from dependency preservation; the existing S,J,T loss counterexample is replayed.

The category module supplies generic finite functional-dependency mathematics. The schema grammar supplies profile admission, real typed intermediate representation construction, SQL emission and whole-artifact replay. WDP retains the declared instance and recovery target. Eight new native and twenty-two actual compiler/SQLite checks pass. The profile binds original inputs and the executing native binary and refuses substituted certificates, illegal rows and incompatible recovery targets. It is an immutable text-set snapshot path; later component edits, richer dependency theories, a second domain and an all-encompassing mathematical language require further evidence.

A worked relational instance: point coincidence. In general relativity the hole argument and the point-coincidence reply [REF-REL-001; REF-REL-002] lead to labelling events by coincidences of invariant fields rather than by coordinates; Komar's four scalar invariants are the standard example [REF-REL-003; REF-REL-004]. Read as a finite relation, a table of events with attributes $\varphi_0, \dots, \varphi_3$ (and further fields) satisfies point coincidence exactly when equal values of $\Lambda = \{\varphi_0, \dots, \varphi_3\}$ identify the row, that is, when Λ is a **superkey** of the instance. This is the factorization criterion of CH16-P02 (and R2) with the row identity as the target: the identity factors through the projection onto Λ exactly when rows with equal projections are equal, so a superkey is injectivity of the projection on the instance (compare CH14-P02). Calling Λ a *candidate* key adds minimality, which is a separate claim about the instance and its declared dependencies, not a consequence of four dimensions. The archived source paper made that conflation; on its own three sample rows the minimal keys are {kretschmann}, {matter density}, $\{\varphi_0\}$, $\{\varphi_1, \varphi_2\}$ and $\{\varphi_1, \varphi_3\}$, so Λ is a superkey that is not minimal. A constructed five-row instance has Λ as its only candidate key, so minimality is instance-level. Heath's decomposition then applies as in the normal-form presentation: if $X \to Y$ holds, $R = R[X \cup Y] \bowtie R[X \cup Z]$. That is a statement about one snapshot. It does not establish that updates are anomaly-free, and the S,J,T countermodel above shows locally legal component edits whose rejoin violates a dependency. The relational coincidence checks replay these statements exactly (28 tests, including lossy-join negative controls). No physical claim about spacetime follows; the instance shows only that point coincidence has a precise relational reading.

O.7 Actual source pressure and response closure

The source-field supplement establishes a common thermodynamic pressure for the declared full-Fock U=t chain with staggered source lambda. Even-block near-additivity gives field-uniform volume error 3theta/(2L). A positive similar transfer product gives an explicit joint time/volume radius; for bounded target parameters it vanishes when M/L grows, L grows and supplied parameter radii shrink. The original SQL histories, weighted by v^Q, equal the actual nonzero-source transfer.

Convexity and uniform pressure convergence give a quantitative weak limit of source-response measures. Three pressure enclosures also close the triangularly averaged response at every fixed

field resolution epsilon, with error amplified by epsilon^-2. This is a protected response with its resolution retained. A global uniform-convergence countermodel shows why pointwise susceptibility cannot be inferred by twice differentiating the pressure bound. The later O.8 supplies uniform derivative control in an explicit high-temperature region; outside that region and for spatial field scaling, the remaining derivative, sector and temperature-scaling obligations stay open.

The written analytic results are separate from twenty-one new Python and seven new native checks. Thirteen affected Python and eleven affected native cases are replayed. Quantum physics owns the model/field plan, measurement owns generic response interval propagation, and WDP retains the declared plan and trace. These results add no Lean coverage or operational authority.

O.8 Pointwise response in a proved analytic region

The analytic-source supplement uses even CAR support factorization and two exact polymer expansions to prove a volume- and time-uniform complex-source bound for 0<=theta<=1/192, |z|<=1/64. It applies the Kotecky–Preiss criterion with all model premises derived explicitly, including the periodic wrapping bond and both parity sectors. Cauchy estimates then supply the derivative control missing from pressure convergence alone.

For the actual source curve the intensive query $<Q^{2>/((2M)}2L)$ converges to the common thermodynamic Kubo response. Its exact error majorant is 4B/epsilon^2 + $64epsilon^{2/(19R}4)$, with B the joint pressure error and R=1/64. A polynomial schedule L=4k^4, M=L^2, epsilon=R/(4k) gives O(k^-2). The written proof is separate from twelve new Python and five new native checks, with thirty-four/fifteen affected earlier cases replayed.

The quantum owner retains the admitted domain and bound; WDP retains its actual intensive readout. The q=3,M=2 fixture lies outside this region and receives no analytic certificate. All-temperature susceptibility and interacting spacetime/critical mesh/coupling scaling remain open. O.9 closes a specific static spatial source-functional profile under declared fluctuation scaling. No new Lean coverage, general sign-problem algorithm or authority is claimed.

O.9 Spatial Gaussian closure of the actual source functional

The spatial-source supplement keeps the same full periodic Meron chain and original signed time histories. A weighted polymer expansion bounds the local source-response kernel by C_K exp(-distance/4), uniformly in volume and time count. For one fixed Lipschitz circle profile F, this identifies the smeared quadratic query $<Q_F^{2>/((2M)}2L)$ with chi(theta) integral F^2, with explicit response and spatial-variation errors.

At mesh a=1/L and source amplitude sqrt(a), Cauchy estimates and particle-hole symmetry make the higher source terms vanish. The actual normalized source functional converges to exp[u^2 chi(theta) integral F^2/2] along the existing polynomial common-temperature schedule. Integer circle characters construct its static Gaussian distribution-valued target. Its variance is nonzero at theta=0 and on the explicit sufficient range theta<=2^-20. The general source-functional theorem retains theta<=1/192.

The generic functional owner supplies exact periodic test profiles and derived bounds; quantum physics supplies model bounds and weighted moments; WDP retains those objects and existing bounded readings. Twenty new Python and six new native checks pass, with forty-six/twenty-two affected cases replayed. Four new comparisons use original SQL histories. These finite checks do not formalize the written infinite-family proof in Lean.

The finite signed generating functions are not assumed to be probability characteristic functions. The limiting Gaussian field is constructed explicitly; no finite-measure tightness, real-time CCR dynamics, physical coupling renormalization or interacting quantum field theory is claimed. The selected circle geometry, mesh and fluctuation scaling remain premises.

O.10 Time-labelled sources and higher connected responses

The time-labelled source derivation keeps the full periodic CAR transfer and inserts an independent source at each site and half-step. Its occupation-basis expansion retains the periodic hop signs and the factor $q^{-LM/4}$. The original SQL rows do not carry these extra time labels, so the identity is an exact transfer extension, not a claim of new SQL execution. At zero coupling it reduces exactly to $\prod_x 2\cosh((4M)^{-1}\sum_j z_{j,x})$, fixing the time-source normalization and showing that nonzero temporal modes require interaction.

At $q = 1+r/M$, $0 \le r \le 1/192$, the LM bond and $2ML$ source factors have exponent norm at most $1/(256M)$. Their physical even-CAR support graph has degree at most $7M-1$, independently of the source values. The rooted polymer majorant is at most $1/(57M)$; its incompatibility sum is at most $7|C|/57 < |C|$. Consequently the imported polymer criterion gives, for every $|z_{j,x}| \le R = 1/64$, a zero-free polydisk and the uniform bound

$$\left|\log X_{L,M}(\mathbf{z}) - L\log 2\right| \le L/19. \tag{O-10}$$

For a bounded spacetime array Ψ, let $\Phi_{L,M}(\Psi) = (2M\sqrt{L})^{-1}\sum_{j,x}\Psi_{j,x}a_x(s_j)$. Its connected responses are derivatives of the logarithm of the *signed* normalized source sum. Cauchy's estimate on O-10 gives

$$|\kappa_n(\Phi_{L,M}(\Psi))| \le \frac{2n!}{19}\left(\frac{\|\Psi\|_\infty}{R}\right)^n L^{1-n/2}. \tag{O-11}$$

The bound is uniform in the time count. Particle-hole symmetry kills odd orders; every connected order $n \ge 3$ vanishes under the displayed fluctuation scaling. For fixed u, the nonquadratic source-logarithm remainder is $O(L^{-1})$, uniformly in M. This closes the higher-cumulant subproblem for bounded time-labelled profiles. It does **not** identify the non-equal-time covariance, prove a joint spacetime limit or positivity of a finite signed-history measure, or select a spacetime field topology. Those are separate open obligations.

O.11 Positive finite-volume Euclidean covariance and temporal modes

The Euclidean covariance derivation uses the exact Gibbs trace of the same full periodic CAR Hamiltonian. Its occupation-basis matrix is real symmetric, including the periodic sign, and the staggered site operators are real diagonal. In a real energy eigenbasis, pair the two orientations of every unequal-energy transition. For integer time frequency m, each such pair contributes a nonnegative multiple of the spatial outer product $v_{\alpha\beta}(x)v_{\alpha\beta}(y)$, with coefficient

$$\frac{2(p_\alpha - p_\beta)\theta(E_\beta - E_\alpha)}{[\theta(E_\beta - E_\alpha)]^2 + (2\pi m)^2} \quad (E_\alpha < E_\beta). \tag{O-12}$$

Equal-energy terms contribute only at zero frequency. Particle-hole symmetry makes every site mean zero. Thus every finite-volume Fourier coefficient matrix of the exact time-ordered covariance is positive semidefinite. The staggered spatial row sum is the $A_L = \sum_x a_x$ covariance divided by L. Because $[H_L/t, A_L] \neq 0$ already on the one-particle sector, its Fourier coefficient is strictly positive at every nonzero time mode for every even $L \geq 4$ and $\theta > 0$. At $\theta = 0$ the row-sum kernel is exactly $1/4$ and these modes vanish. The zero mode is exactly $p_L''(\theta, 0)$ by the Duhamel formula, so the earlier static susceptibility limit is the selected zero-frequency limit.

The local double commutator obeys $0 \leq D_{L,\theta} := \operatorname{Tr} \rho[A_L, [H_L/t, A_L]] \leq 2L$: its operator is twice the sum of nearest-neighbor hopping terms, each of norm one. The spectral coefficients therefore also give, uniformly in even $L \geq 4$,

$$\operatorname{Lip}_{\mathbb{T}}(k_{L,\theta}) \leq \theta, \qquad 0 < \hat{k}_{L,\theta}(m) \leq \frac{\theta}{2\pi^2 m^2} \quad (m \neq 0), \qquad |k_{L,\theta}(u) - p_L''(\theta, 0)| \leq \theta/6. \tag{O-13}$$

In the declared analytic region, the source-disk bound controls p_L'' uniformly. Hence the row-sum time kernels are uniformly bounded and equicontinuous. They have uniformly convergent subsequences with nonnegative Fourier coefficients and the proved static zero mode.

This closes finite-volume Hamiltonian positivity and temporal nontriviality plus a uniform temporal upper bound. It does not prove uniqueness of that subsequential limit, a uniform positive lower bound for nonzero modes as $L \to \infty$, spatial summability of the non-equal-time kernel, or comparison with the time-labelled discrete transfer. Consequently the spacetime field topology and joint continuum schedule remain open. The finite signed SQL histories are not declared probability measures.

O.12 The unique Euclidean thermodynamic limit of the row-sum covariance

The thermodynamic-limit derivation uses the same object as O.11. Take even $L \geq 4$, the full 2^L CAR space with the physical periodic hopping sign, P_x, K_L, $a_x = (-1)^x(n_x - 1/2)$ and $\rho_{L,\theta}$. Let $k_{L,\theta}(u) = \sum_y C_{0y}^L(u)$. It uses the polymer representation of the analytic response together with the Kotecký–Preiss criterion that representation imports, together with finite-dimensional spectral theory, matrix trace inequalities, Cauchy estimates and standard Fourier/measure representation results. The model-specific estimates are proved in the linked derivation; the named standard results remain mathematical inputs.

At fixed u, the two Euclidean segments of length $1-u$ and u give a gas of bond and source labels. Its label norm is $1/128$ and its degree is 7, so $S_1 = 3/43$ and $(\Delta+1)S_1 = 24/43 < 1$. Every cluster that contains both sources contains a chain of bonds joining them. Cauchy and Schwarz estimates then give, uniformly in even L and u on $|\theta| \leq 1/96$,

$$|C_{0y}^L(u)| \leq M_1(96|\theta|)^{d_L(0,y)}, \qquad M_1 = \frac{24576}{43}, \qquad \sup_L \sup_u \sum_y e^{\mu d_L(0,y)} |C_{0y}^L(u)| \leq M_1 \frac{1+q}{1-q} \tag{O-14}$$

for $\mu \geq 0$ and $q = 96|\theta|e^{\mu} < 1$. The absolute value is essential: at $\theta = -1/96$, $\mu = 0$, a signed ratio would give a zero upper bound, although $C_{00}^L(0) = 1/4$ because $a_0^2 = I/4$. The corrected

ratio is $q = 1$, outside the weighted bound's domain. This does not affect the separate positive-temperature theorem. The open-chain functions on any two intervals containing $[-R, R]$ differ by at most $2M_1(96|\theta|)^{R+1}$. So the line two-point function C^∞ exists, does not depend on the exhaustion, is translation invariant, and inherits (O-14). The wrapping term is the CAR hopping between modes $L-1$ and 0. A cluster with fewer than L bonds therefore lives on an arc, which is the same CAR system as a segment of the line. The same holds with the wrap sign erased, which is the spin-periodic chain. The bounds above use only five premises: each bond term is four times a projector; every word on pairwise disjoint components has a factorizing normalized trace; bond terms and sources commute with every local operator off their supports; the staggered source has vanishing mean; and, on the physical ring, bond terms and sources are even. They are CAR statements for the physical ring and spin-language statements for the erased one. The word-level factorization follows from the support and evenness premises (the locality lemma of TL-1). They are checked exactly on $L = 4, 6$ for every set of disjoint bond terms and sources. The check also confirms that the Hamiltonian is built from those bond terms, and it rejects four mutations that pass weaker versions of it: a nonlocal bond term, a source dressed by the total parity, an odd bond term, and an extra nonlocal term added to the Hamiltonian. An exact per-site comparison on $L = 4, 6$ shows the two kernels agree below order L and first differ at L. The ring kernel minus its wrapped line images vanishes below order L, and Schwarz on $|\theta| \le 1/128$ gives, for $0 \le \theta < 1/128$,

$$\begin{aligned} \sup_u |k_{L,\theta}(u) - k_\theta(u)| &\le 4M_1 L(128\theta)^L \to 0, \\ \|E_L - C^\infty\|_{\ell^1_\mu(\mathbb{Z};C[0,1])} &\le M_1 q^{L/2}\Big(4L + \frac{4}{1-q}\Big). \end{aligned} \tag{O-15}$$

Here $k_\theta = \sum_n C^\infty_{0n}$. The second bound is for $0 \le \theta \le 1/192$, $\mu \ge 0$, and $q = 96|\theta|e^\mu < 1$, where E_L is the ring kernel placed on the window $-L/2 < n \le L/2$. The limit is unique. All of O.11's subsequential limits equal k_θ, and the rate is uniform on $0 \le \theta \le 1/192$.

The limit has the following properties.

- Its zero mode is the static susceptibility $\chi(\theta)$.
- At $\theta = 0$, $k_0 = 1/4$.
- $k_\theta(u) = k_\theta(1-u)$.
- $k_\theta(u) > 0$ for $\theta < 43/56631360$.
- Every Fourier mode is nonnegative, as a uniform limit of O.11.

Strict positivity is a separate statement. Integrating by parts twice over the time circle gives

$$(2\pi m)^2 \hat{k}_\theta(m) = 2\theta F(\theta) - \theta^2 \hat{g}(m).$$

Here F is the bond energy and g is a sum of local Euclidean correlations. The constants are $\|B_0\|^2 \le 3$, certified exactly, $|F - \theta/4| \le M_F(96\theta)^2$ and $|g - 1/2| \le 288G\theta$. This yields, for every $m \ne 0$,

$$\hat{k}_\theta(m) \ge \frac{\theta^2}{16\pi^2 m^2} > 0 \qquad (0 < \theta \le \theta_* = \frac{2537}{100281465600}). \tag{O-16}$$

The intermediate bound $\theta^2(1/2 - c\theta)$ is strictly positive for $\theta < 1/(2c) = 2\theta_*$, so strict positivity holds on $0 < \theta < 2\theta_*$.

Evidence grade:

- The analytic estimates are written proofs, with the polymer criterion imported.
- Exact rational CAR series on rings $L = 4, 6, 8$ and open chains check the vanishing orders, the lift, the zero mode, the integration-by-parts identity and the B_0 certificate.
- Mutations of the normalization, the staggering, the P scale and the wrap sign each move an exact coefficient.
- `code/tests/test_meron_thermodynamic_limit.py` runs these checks.
- The Lean module `MeronThermoLimitConstants.lean` states only the rational constants. It compiles standalone with Lean 4.33.1, and `#print axioms` reports no axiom for its 18 theorems; the receipt `_intelligence/research/briefs/meron_thermo_lean_receipt_20261007.json` records the file hash and exit codes. It is not imported by a build root; the full package build was not run.
- Floating-point values are diagnostics only.

The later window argument in the thermodynamic-limit supplement shows in this model $\hat{k}_\theta(m) \geq \theta^2/(36\pi^2 m^2) > 0$ for $0 < \theta \leq 1/192$ and nonzero integer m, under its imported polymer criterion. The earlier small-temperature estimate is no longer the full positivity range proved in this package.

Not proved:

- anything for $\theta > 1/128$;
- real-time or spectral-density statements;
- the X_{LM} comparison;
- the joint spacetime schedule and the distribution topology.

The finite signed SQL histories are still not declared probability measures.

Appendix P — The earlier component calculi decomposed into reusable WDP mathematics

The earlier nine-layer stack, its eight component papers and its layered-authority monograph are consolidated here by mathematical dependency. Their nine runtime stages apply six constructions; they do not supply nine independent foundational calculi. The source and implementation correspondence retains every primary text, its extra assumptions and its native owner. This appendix gives written proofs. It adds no claim of a fresh all-estate Lean build, general Master Normal Form or operational authority.

P.1 Reports, queries and one factorization criterion

Let `p:X→A` be a report and `q:X→B` a protected query. Put `P=p(X)`. There is a unique `q̄:P→B` with `q=q̄∘p` exactly when

$$p(x) = p(y) \implies q(x) = q(y).$$

Necessity follows by applying `q̄` to the equal reports. For sufficiency, define `q̄(p(x))=q(x)`; fiber constancy makes this independent of the representative. Each realized report has a representative, giving uniqueness. Extending a decoder to all of `A` requires additional values outside `P`; the theorem does not supply them. The finite-shadow and shadow-to-completion constructions use this same criterion.

If p=h∘p_fine, equality of fine reports implies equality of coarse reports. Thus fiber constancy for the coarse report implies it for the fine report. The reverse fails for X={0,1}^{2}, p(a,b)=a and q(a,b)=b: the complete pair report decodes q, while the first-coordinate report does not. Post-processing p cannot recover the lost second coordinate.

In a pseudometric output space, a decoder satisfying d(q(x),D(p(x)))≤ε implies, on each report fiber,

$$d(q(x), q(y)) \leq 2\varepsilon.$$

This is the triangle inequality through the common decoded value. It is a necessary bound, not a converse in an arbitrary metric space. Three points at mutual distance 2, with outputs taking all three values and epsilon 1, satisfy that bound but have no radius-1 center in that three-point space. At epsilon zero, output equality additionally needs metric separation: the indiscrete pseudometric has zero distances between distinct values. An input Lipschitz estimate is another premise, not this decoder theorem.

Standing is itself a protected query when it is to be decoded from a report. Two states with equal defect numbers and different grant outcomes violate its fiber criterion. This explains the collision between descent standing and layered standing without making authorization a consequence of observation.

P.1.a Decoders and collisions (DP-1, audited 2026-10-06) Statement. Let X, A and B be finite sets. Let p:X→A and q:X→B be total tables, and put P=p(X). That each value of q lies in B is a typing premise of the table; it is not a further hypothesis to be tested. Exactly one of the following holds.

- **(F)** There is a decoder D:P→B with D(p(x))=q(x) for every x∈X.
- **(K)** There is a collision: a pair x,y∈X with p(x)=p(y) and q(x)≠q(y).

In case (F) the decoder is unique as a function on P. The statement includes X=∅. Then P=∅, and the empty function is the unique decoder, also when B=∅.

Proof. Suppose (K) holds and D is a decoder. Then q(x)=D(p(x))=D(p(y))=q(y), which contradicts q(x)≠q(y). So (F) and (K) exclude each other.

Suppose (K) fails. Then q is constant on every fiber p^{-1}(a) with a∈P. The construction above sets D(a)=q(x) for any x with p(x)=a. Every a∈P has such an x, so two decoders agree at every point of P.

For X=∅, the decoder equation is vacuous. A function ∅→B exists and is unique for every B.

A finite scan decides which case holds. For each realized report it records the first query value seen, and it stops at the first disagreement. It returns either the table of D or a colliding pair, and each step compares two entries of the supplied tables.

DP-1′ (scope of uniqueness). DP-1′ supersedes the candidate statement "Factorable: a unique table D:P→B satisfies D(p(x))=q(x) for every x∈X" in one respect. Uniqueness is asserted for functions on P=p(X). Nothing is asserted, neither existence nor uniqueness, for functions on all of A. Two countermodels show why.

1. Let `X={x}`, `A=B={0,1}` and `p(x)=q(x)=0`. The decoder on `P={0}` is unique. Both total extensions, with value `0` or `1` at the unrealized report `1`, satisfy `D(p(x))=q(x)`.
2. Let `X=B=∅` and `A={a}`. The empty decoder on `P=∅` exists, but no total function `A→B` exists.

Refinement. Suppose `p=h∘p_f` with `h:A_f→A`. Then (F) for `(p,q)` implies (F) for `(p_f,q)`, with decoder `D_f=D∘h` restricted to `p_f(X)`, because `D(h(p_f(x)))=D(p(x))=q(x)`. The converse fails in the example above: `X={0,1}`^{2}, `p_f` the identity, `h(a,b)=a` and `q(a,b)=b`. There `(0,0)` and `(0,1)` collide for the coarse report.

Worked cases. Take `X={x0,x1,x2}`, `p=(0,0,1)` and `q=(1/3,1/3,2/3)`. The decoder is `D(0)=1/3`, `D(1)=2/3`. Changing `q(x1)` to `2/3` produces the collision `(x0,x1)`.

The test reads only the tables `p` and `q`. Profile metadata attached to a report is neither sufficient nor necessary for factorization. A constant report with `q` the identity on `{0,1}` collides at `(0,1)`, whatever metadata accompanies the report.

Evidence grade.

- *Written proof:* the argument above.
- *Exact finite replay* (2026-10-06 audit):
  - an exhaustive enumeration over `|X|`$\leq$`3` and `|A|,|B|`$\in${`2,3`}, 1423 table pairs, each in exactly one of (F) and (K) and never with two decoders (`dp1_exhaustive_factorable_xor_collision_sizes_le3`, `dp1_exhaustive_case_count`);
  - `dp1_positive_fixture_decoder`, `dp1_mutated_decoder_rejected`;
  - `dp1_reverse_refinement_fine_factorable`, `dp1_reverse_refinement_coarse_collides`, `dp1_refinement_transfer_exhaustive_violations`;
  - `dp1_metadata_not_sufficient_collision`;
  - `dp1_decoder_on_all_A_not_unique`, `dp1_empty_image_but_no_function_A_to_B`, `dp1_empty_X_empty_B_unique_empty_decoder`.
- *Lean kernel:* the following, stated for arbitrary types, which contain the finite case. Codomain membership is carried by the Lean typing, matching the typing premise above.
  - `WitnessedDifference.ComponentReports.decoder_exists_iff_fiber_constant`, `WitnessedDifference.ComponentReports.decoder_unique`, `WitnessedDifference.ComponentReports.not_fiber_constant_iff_collision`, `WitnessedDifference.ComponentReports.decoder_xor_collision`;
  - `WitnessedDifference.ComponentReports.empty_source_has_decoder`;
  - `WitnessedDifference.ComponentReports.refinement_transfers`, `WitnessedDifference.ComponentReports.refinement_converse_fails`;
  - `WitnessedDifference.ComponentReports.fixture_factorable`, `WitnessedDifference.ComponentReports.constant_report_collides`;
  - `WitnessedDifference.ComponentReports.decoder_off_image_not_unique`.

  The axiom report is `lean4/component_reports_axioms.txt`. The module was checked standalone with Lean 4.33.1; the rooted `lake build` was not run.
- *Native exact implementation:* exact rational tables with typed refusals.
  - `omega_category_theory::finite_report_query` decides on total exact tables, with decisions `Factorable` and `Collision`. It refuses duplicate identities, duplicate, missing or extraneous assignments, and values outside the codomain.
  - `omega_coda Layer3Cs::evaluate_report_query` and `omega_wdp PresentationMap::decide_query` route through it. The legacy `Layer3Cs::evaluate` used to skip unmapped tokens and accept an extraneous row; it now refuses both. `PresentationMap::new` likewise used to accept keys outside the carrier and now refuses them. Each defect matched a typing gap named by

the DP-1′ correction.

- *Unresolved residual:* countermodel 2 (no total function A→B) is replayed but not stated in Lean. The termination and cost of the scan rest on the written argument.
  - In the native decision, membership in the codomain is a typing premise. The decision applies only to total exact tables.

P.1.b Separation of declared output classes (DP-2, audited 2026-10-06) Statement. Let S be a finite set of section records and d an exact rational pseudometric on S. Put Z={(s,t):d(s,t)=0}, and let E be an equivalence relation on S.

1. d_E([s],[t]):=d(s,t) is a well-defined pseudometric on S/E iff E⊆Z.
2. It is a metric iff E=Z.
3. Suppose E=Z and S/E has at least two classes. Then Δ=min{d(s,t): not sEt} is attained by some pair, and Δ>0.
4. Let p:X→A be a report and q:X→S a query into the declared family, so that q(X)⊆S. Let D:p(X)→T be a decoder into an admitted pseudometric space T⊇S that extends d. Let $\varepsilon \geq 0$ with d(q(x),D(p(x)))≤ε for every x. If E=Z, at least two classes exist and 2ε<Δ, then the class-valued query x↦[q(x)] factors through p.

In the weighted coordinate instance, every record is a total table v_s:Y→Q on one declared common coordinate set Y. The distance is d(s,t)=Σ_{y∈Y}w_y|v_s(y)−v_t(y)|, with rational w_y>0. Then Z is equality of the represented functions.

Proof.

Weighted coordinate instance. Each summand is nonnegative and symmetric. The scalar triangle inequality, multiplied by w_y>0 and summed, gives the triangle inequality. With all weights positive, a zero sum forces every coordinate difference to vanish.

Item 1. If s E s′, t E t′ and E⊆Z, then d(s,t)≤d(s,s′)+d(s′,t′)+d(t′,t)=d(s′,t′). Interchanging the primes gives the reverse inequality. The laws of a pseudometric then descend to S/E. Conversely, if d_E is well defined and s E s′, then d(s,s′)=d_E([s],[s])=0.

Item 2. Separation of d_E says exactly d(s,t)=0 ⇒ sEt, that is Z⊆E.

Item 3. By item 2, every interclass distance is positive. There are finitely many interclass pairs and at least one, so the minimum is one of them.

Item 4. If p(x)=p(y)=a, then d(q(x),q(y))≤d(q(x),D(a))+d(D(a),q(y))≤2ε<Δ. Since q(x),q(y)∈S, item 3 forces q(x) E q(y), and P.1 gives the factorization. At ε=0 the same chain gives d(q(x),q(y))=0, which yields class equality whenever E=Z, with no positive gap needed. Literal equality of outputs needs E to be literal equality.

Worked example. Take Y={u,v}, weights (1,2) and records a=(0,0), a′=(0,0), b=(1/3,1/2) and c=(1,0), with classes {a,a′}, {b} and {c}. Then:

- d(a,a′)=0;
- d(a,b)=1/3+2·1/2=4/3;
- d(a,c)=1;
- d(b,c)=2/3+1=5/3.

So E=Z and Δ=1, attained at (a,c). A decoder with ε=1/3 has fiber diameter at most 2/3<1, so the queried class is constant on report fibers. If instead E were identity on record names, (a,a′) would

be a separation collision, even though the least nonzero distance is still 1. A minimum taken after skipping zero pairs therefore certifies nothing.

Countermodels to dropped hypotheses.

- *Indiscrete pseudometric.* Every distance is zero. A constant report, q(0)=0, q(1)=1, decoder 0 and ε=0 pass every error check, yet the outputs differ.
- *Strict threshold.* In Q with $|\cdot|$, the outputs {0,1} have Δ=1. The decoder 1/2 gives ε=1/2, so 2ε=Δ, and the outputs still differ.
- *Declared domains.* Records s:{x}→Q and t:{y}→Q, both with value 0, have equal length and entries but different domains. They are not a zero-distance pair of one comparison problem.
- *Partial pair checks.* Take values 0,0,1 with literal E. Checking only the pairs that involve the third record misses the zero collision of the first two.
- *Non-equivalence.* Declaring aEb and bEc without aEc gives no quotient. The declaration is refused rather than silently closed.
- *Zero weight.* w=0 makes 0 and 1 zero-distance on one coordinate.
- *Triangle law.* Symmetric distances 3,1,1 on {a,b,c} fail the triangle law. The decoder c, with errors 1 and 1, does not bound d(a,b)=3 by 2.
- *Symmetry.* A directed table with d(a,c)=d(b,c)=0 and every other off-diagonal entry 1 passes the directed triangles. It has zero errors toward c, yet d(a,b)=1.
- *Fiber bound without a center.* The three-point space of P.1 shows that a fiber bound alone is not sufficient.

Evidence grade.

- *Written proof:* the argument above.
- *Exact finite replay:* dp2_distance_table, dp2_weighted_is_pseudometric, dp2_E_equals_Z_and_gap, dp2_two_eps_below_gap, dp2_identity_E_has_zero_collision, dp2_indiscrete_pseudometric_errors_zero_values_differ, dp2_strict_threshold_control, dp2_fiber_bound_not_sufficient, dp2_triangle_control_refused, dp2_symmetry_control.
- *Lean kernel:* axiom report lean4/component_reports_axioms.txt.
  - WitnessedDifference.ComponentReports.Pseudometric.zero_equivalence, WitnessedDifference.ComponentReports.Pseudometric.descends_of_zero;
  - WitnessedDifference.ComponentReports.Pseudometric.well_defined_iff (item 1), WitnessedDifference.ComponentReports.Pseudometric.quotDist_mk;
  - WitnessedDifference.ComponentReports.Pseudometric.separated_iff (item 2);
  - WitnessedDifference.ComponentReports.fiber_diameter_le, WitnessedDifference.ComponentReports.strict_gap_forces_equivalence (item 4);
  - WitnessedDifference.ComponentReports.equality_threshold_countermodel, WitnessedDifference.ComponentReports.indiscrete_quotient_singleton, WitnessedDifference.ComponentReports.worked_example_distances.
- *Native exact implementation:*
  - omega_category_theory::finite_section_separation (decide_separation, verify_separation, epsilon_decoder_class_factorization) returns either a zero-distance collision or a separation with an attained gap and its attaining pair. It accepts at most 512 points.
  - The certified path in omega_coda is Layer4CsQ::evaluate_declared_separation.
- *Unresolved residual:* the attained minimum of item 3 is a written proof with finite replay. The declared-domain discipline is a specification on inputs, and it is not formalised.
  - The legacy Layer4CsQ::evaluate remains a screen labelled "not a separation certificate", and it is not a certificate. It still accepts a pair flagged equivalent at positive distance, and it

succeeds when nothing is compared.

P.2 Repair quotients, closure and exact certificates

Admit finite-dimensional real spaces, linear maps `d0:C0→C1`, `d1:C1→C2`, `a2:U2→C2`, the law `d1 d0=0`, and strictly positive coordinate weights `w1,w2`. Write `R1=im d0`, `R2=im a2` and

$$\Phi(x) = \min_f \|x + d_0 f\|_{1,w_1}, \qquad \Gamma(x) = \min_u \|d_1 x + a_2 u\|_{1,w_2}.$$

These are quotient norms on `Q1=C1/R1` and `Q2=C2/R2`, respectively, with Γ evaluated after the induced map `D:Q1→Q2`. This map is well-defined because `d1` kills `R1`. A minimizing residual exists: the affine repair coset is closed, and its intersection with a sufficiently large norm ball is compact. Hence `Φ=0` iff `x∈R1`, and `Γ=0` iff `d1x∈R2`. **The latter is closure modulo the second repair space.** Taking `R2={0}` recovers raw closure `d1x=0`. More generally, the two zero tests agree for every `x∈C1` if and only if `im d1 ∩ R2={0}`: only vectors in that intersection can be both nonzero raw defects and zero quotient defects. The re-presentation papers' raw closed/open square labels must retain this distinction from the general trichotomy of the finite obstruction calculus.

Since `Φ=0` forces `d1x=0`, there are exactly three numerical regimes: `Φ=0,Γ=0`; `Φ>0,Γ=0`; `Φ>0,Γ>0`. This classification does not prove that a coherent residual survives every later allowed correction. Such survival quantifies over a declared move system. Strictly positive weights change prices but not either nullity condition; zero weights can destroy that statement. Block-diagonal independent systems have additive Φ and Γ, because their variables and coordinate objectives separate.

For a general repair matrix `A`, finite weighted L1 duality is

$$\min_f \|x + Af\|_{1,w} = \max\{z^T x : A^T z = 0,\ |z_i| \le w_i\}.$$

To derive the dual, minimize `Σ_i w_i t_i` subject to `−t≤x+Af≤t`. The two nonnegative multiplier vectors `α,β` satisfy `α+β=w` and `A^⊤(α−β)=0`; setting `z=α−β` gives precisely the stated capacities and annihilator. Conversely `α=(w+z)/2`, `β=(w−z)/2` recover feasible multipliers. Finite linear-programming strong duality applies to this feasible bounded-below primal. For rational input and weights, rational LP gives rational optimal certificates; compactness of a rational ball is not used.

A replayable optimum needs primal feasibility, **all** annihilator equations, all capacities and equality of the two objectives, bound to the actual input. Weak duality establishes the equality certifies optimality: `z^⊤x=z^⊤(x+Af)≤Σ_i w_i|x_i+(Af)_i|`. One selected annihilator vector is only a lower bound. For `A=(1,0,0)^⊤`, `x=(0,0,1)^⊤`, `z=(0,1,0)^⊤` is feasible but pairs to zero while the optimum is one.

Compatible transport maps satisfy `T1 d0=d0′ T0`, `T2 d1=d1′ T1`, `T2 a2=a2′ S2`. Transport a minimizing repair to obtain

$$\Phi'(T_1 x) \le \|T_1\|\Phi(x), \qquad \Gamma'(T_1 x) \le \|T_2\|\Gamma(x).$$

For weighted L1, `‖T‖=max_j (Σ_i w′_i|T_ij|)/w_j`. Equality or reversibility requires stronger hypotheses. The finite Python and primary Rust checked-repair pipelines implement an unweighted

first quotient and a raw next defect norm; they do not implement the whole weighted two-repair theorem.

P.2.a Raw and quotient second defects (DP-3, audited 2026-10-06) Statement.

Setting. Write `D` for the rational matrix of `d1:C1→C2` and `A` for that of `a2:U2→C2`. Let `w=w2` be strictly positive, with `m` coordinates. Define

$$\Gamma_{\mathrm{raw}}(x) = \|Dx\|_{1,w}, \qquad \Gamma_{\mathrm{quot}}(x) = \Gamma(x) = \min_{u} \|Dx + Au\|_{1,w}.$$

Let `B_w={z:|z_i|≤w_i}` be the capacity box, and let `K_w={z∈B_w:A^⊤z=0}`.

Claims.

1. `Γquot(x)≤Γraw(x)` for every `x`.
2. The following are equivalent.
   - **(G)** `Γquot(x)=Γraw(x)` for every `x`.
   - **(P)** `D^⊤K_w=D^⊤B_w`.
   - **(GV)** For every sign vector `σ∈{−1,1}^m` there is `z∈K_w` with `D^⊤z=D^⊤(w∘σ)`. Here `w∘σ` is the coordinatewise product.
3. The two zero tests agree for every `x` iff `im D∩R2={0}`, as in P.2. Claim (G) implies this agreement. The converse fails.
4. For every vector `y`, `(min_i w_i)‖y‖_1≤‖y‖_{1,w}≤(max_i w_i)‖y‖_1`. This bounds Γ_{raw} and Γ_{quot} by their unweighted values times the minimum and maximum entries of w_2. For Φ use its own weight vector w_1 and the corresponding extrema. On an empty coordinate space all these values are zero; no minimum or maximum of an empty weight vector is used.

Proof.

Claim 1. Take `u=0`.

Claim 2. By the P.2 duality, `Γquot(x)=max{(D^⊤z)^⊤x : z∈K_w}`. This is the support function of the polytope `D^⊤K_w`. Putting `A=0` gives `Γraw(x)=max{(D^⊤z)^⊤x : z∈B_w}`, the support function of `D^⊤B_w`. Both sets are nonempty, compact and convex, and `D^⊤K_w⊆D^⊤B_w`.

- (G) ⇒ (P). Suppose `y∈D^⊤B_w\D^⊤K_w`. Exact separation of a point from a rational polytope gives a rational `x` with `x^⊤y>max_{y′∈D^⊤K_w}x^⊤y′`. Hence `Γraw(x)≥x^⊤y>Γquot(x)`, and (G) fails.
- (P) ⇒ (G). Equal sets have equal support functions.
- (P) ⇔ (GV). Positive weights make `B_w` a full box whose vertices are the points `w∘σ`. Every `b∈B_w` is the convex combination `b=Σ_σ c_σ(w∘σ)`, with `c_σ=∏_i(1+σ_ib_i/w_i)/2≥0`. These are the product weights of independent signs with means `b_i/w_i`, so `Σ_σc_σ=1` and `Σ_σc_σσ_iw_i=b_i`. Therefore `D^⊤B_w=conv{D^⊤(w∘σ)}`. The convex set `D^⊤K_w` equals it exactly when it contains each `D^⊤(w∘σ)`, which is (GV).

A vertex that fails (GV) has a replayable negative certificate. The rational system `|z_i|≤w_i`, `A^⊤z=0`, `D^⊤z=D^⊤(w∘σ)` is infeasible exactly when Fourier–Motzkin elimination returns a multiplier vector `η≥0`. Writing the system as `Mz≤b`, the vector satisfies `η^⊤M=0` and `η^⊤b<0`. A checker replays these equations without trusting the solver.

Claim 3. `Γraw(x)=0` iff `Dx=0`, and `Γquot(x)=0` iff `Dx∈R2`. (G) equates the two values for every `x`, so it equates their zero sets.

Claim 4. Compare coordinatewise.

Hypotheses are needed.

- *Zero weight.* One coordinate with `D=1`, `A=0` and `w=0` gives Γ`=0` at `x=1` although `Dx`$\neq$`0`.
- *Negative weight.* This can make the quotient objective unbounded below: with one coordinate, `D=A=1` and `w=-1`, it is `-|x+u|`. It need not do so: `Dx=(x,x)`, `Au=(u,u)` and `w=(-1,2)` give `-|x+u|+2|x+u|=|x+u|`, with minimum zero at `u=-x`.
- *Missing annihilator.* With `A=1`, `D=1` and `w=1`, the vector `z=1` meets the capacity bound and pairs to `1` with `Dx` at `x=1`. Yet Γ`quot(1)=min_u|1+u|=0`. Every annihilator equation must be checked.

Example 1 (zero tests agree, values differ). Take `Dx=(x,x)`, `Au=(u,0)` and `w=(1,1)`.

- Γ`raw(x)=2|x|`, while Γ`quot(x)=min_u(|x+u|+|x|)=|x|`, attained at `u=`$-$`x`.
- `K_w={(0,z_2):|z_2|`$\leq$`1}`, so `D`$^\top$`K_w=[`$-$`1,1]`, whereas `D`$^\top$`B_w=[`$-$`2,2]`.
- The vertices `(1,1)` and `(`$-$`1,`$-$`1)` fail (GV): any `z`$\in$`K_w` has `D`$^\top$`z=z_2`$\leq$`1<2`. The separator `x=1` gives raw `2` and quotient `1`, with dual `z=(0,1)`.
- The zero tests agree, since `im D`$\cap$`R2={(x,x)}`$\cap${(u,0)}={0}`.
- With weights `(2,3)`, Γ`raw(1)=5` and Γ`quot(1)=3`. The quotient value is attained at `u=`$-$`1` and certified by `z=(0,3)`. The ratio is `5/3`.

Example 2 and DP-3′. Keep `Dx=(x,x)` and set `Au=(u,`$-$`u)`.

- With unit weights, Γ`quot(x)=min_u(|x+u|+|x`$-$`u|)=2|x|=`Γ`raw(x)`.
- For each σ, the vector `z_`σ`=((`σ`_1+`σ`_2)/2,(`σ`_1+`σ`_2)/2)` lies in `K_w` and has `D`$^\top$`z_`σ`=`σ`_1+`σ`_2`.
- So (G) holds although `R2=span(1,`$-$`1)`$\neq${0}`.

DP-3′ supersedes the candidate wording "demonstrating that `R2={0}` is sufficient but not necessary for value agreement". The corrected wording reads: `R2={0}` implies (G), since then `K_w=B_w`, and with unit weights `w=(1,1)` this example has (G) with `R2`$\neq${0}`. The example is weight-specific. With weights `(1,3)` and the same `D` and `A`, (GV) fails at $\sigma$`=(1,1)`. There `D`$^\top$`(w`$\circ$`)=4`, while every `z`$\in$`K_w` has `z_1=z_2` and `|z_1|`$\leq$`1`, so `D`$^\top$`z`$\leq$`2`. Directly, $\Gamma$`raw(1)=4` and $\Gamma$`quot(1)=min_u(|1+u|+3|1`$-$`u|)=2`, at `u=1`.

Evidence grade.

- *Written proof:* claims 1–4 and the certificate replay.
- *Exact finite replay:*
  - `dp3_ex1_raw`, `dp3_ex1_quot_lp`, `dp3_ex1_dual_certificate`, `dp3_ex1_zero_tests_agree_on_samples`, `dp3_ex1_weights_23_values`, `dp3_ex1_weight_bounds`, `dp3_ex1_gv_vertex_status`, `dp3_ex1_farkas_and_separator`;
  - `dp3_farkas_replay_rejects_tampered_certificates`;
  - `dp3_ex2_four_vertex_certificate`, `dp3_ex2_values_equal`;
  - `dp3_new_ex2_weights_13_vertex_fails`, `dp3_new_ex2_weights_13_values`;
  - `dp3_convex_coefficients`;
  - `dp3_control_mixed_weights`, `dp3_control_zero_weight_nonzero_defect`, `dp3_control_missing_annihilator_false_certificate`, `dp3_control_negative_weight_unbounded`.
- *Lean kernel:* these cover claim 1 and Example 1 only. The axiom report is `lean4/component_repair_axioms.txt`.
  - `WitnessedDifference.ComponentRepair.lower_bound_le_raw`, `WitnessedDifference.ComponentRepair.quot_le_raw`;
  - `WitnessedDifference.ComponentRepair.countermodel_raw`, `WitnessedDifference.ComponentRepair.countermodel_quot`, `WitnessedDifference.ComponentRepair.countermodel_`-

dual, WitnessedDifference.ComponentRepair.countermodel_zero_tests, WitnessedDifference.ComponentRepair.countermodel_values_differ;
  - WitnessedDifference.ComponentRepair.weighted_countermodel.
- *Native exact implementation:*
  - omega_functional::gamma_reconciliation provides the raw defect and the certificate verifiers for the quotient value, the vertex table, the vertex Farkas certificate, the value gap and the zero-test gap.
  - omega_solvers::gamma_reconciliation_search provides the quotient defect and the agreement decisions.
  - Replays: Example 1 has raw 2 and quotient 1 at unit weights, and raw 5 and quotient 3 at weights (2,3). Example 2 at unit weights extends all four vertices, and so agrees.
- *Unresolved residual:*
  - Native vertex enumeration is capped at 16 rows. Above that it returns a typed refusal, not a verdict.
  - The CODA first layer computes the raw Γ only and takes no repair map, so the quotient Γ is not yet wired into it.
  - The general equivalence (G)⇔(P)⇔(GV) is a written proof, exercised on the examples above; it is not stated in Lean.
  - Zero-test agreement in Example 1 is replayed on samples. Its general form is the P.2 argument.
  - Example 2 and the weights-(1,3) failure are finite replay only.

P.3 Lawful transport and an exact carrier-change criterion

For linear repair spaces R⊆V, R′⊆V′, a linear raw operation T:V→V′ induces V/R→V′/R′ iff T(R)⊆R′. Sufficiency follows by comparing representatives; necessity follows by applying the induced map to [r]=[0]. This is P.1 specialized to quotient reports.

For general reports p:X→A, p′:X′→A′, a lawful transport is the square p′F=Fp. Two such squares paste by substitution: p′′GF=Gp′F=GFp. Equal source reports remain equal after transport. A noninjective collapse has no global left inverse: any two distinct source values in one fiber would both equal the same recovered value. Pasting provides neither target standing nor bicategorical coherence.

Here is a complete criterion for the **actual** fiber-sum operation on directed integer pair tables. For a total finite map f:X→Y, retain all declared target vertices, including empty fibers. Set

$$m_a = |f^{-1}(a)|, \quad s_a = \sum_{f(x)=a} g_x, \quad (T_f e)_{ab} = \sum_{f(x)=a,\ f(y)=b} e_{xy}.$$

For the gradient (dg)xy=gx−gy, direct expansion gives

$$T_f(dg)_{ab} = m_b s_a - m_a s_b.$$

Its triangle sum is

$$(m_b - m_c)s_a + (m_c - m_a)s_b + (m_a - m_b)s_c.$$

Finite carrier theorem. `T_f` carries every integer gradient to a gradient iff at least one of these conditions holds: `|Y|`$\leq$`2`; at most one target fiber is occupied; all target fiber sizes are equal.

For at most two vertices, every antisymmetric zero-loop table is a gradient. With at most one occupied fiber the displayed transported gradient is zero. With common fiber size `k`, it is `d(ks)`. For necessity suppose there are at least three target vertices, at least two occupied fibers and unequal sizes. Some occupied `a` has two other vertices `b,c` with `m_b`$\neq$`m_c`. Otherwise all sizes away from each occupied vertex agree; using two occupied vertices and a third vertex forces all sizes to agree, a contradiction. Put a unit potential at one source vertex over `a`. Then `s_a=1` and all other sums vanish, so the triangle is the nonzero integer `m_b`$-$`m_c`. No gradient has that cycle. This also constructs a counterwitness for independent raw transport replay.

On a surjective map to at least three vertices, the criterion reduces to equal fiber sizes. Empty fibers cannot be discarded to manufacture a pass. The term-addition operation has fiber sizes `(1,…,1,0)` and descends exactly when the old carrier has at most one vertex. Identify has one size-2 fiber and the remaining size-1 fibers and descends exactly when the old carrier has at most three vertices. With map `(0,0,1,2)` and potential `(0,0,1,0)`, the target cycle on `(0,1,2)` is -1.

Raw pair transports compose by regrouping finite sums. If both preserve gradients, their composite does too. A root-normalized target potential is `h_a=T_f(dg)_{a0}`; it is well-defined on the potential modulo constants, with the empty potential used for an empty target, and successive normalization agrees with the composite. In particular, finite carriers with these lawful total maps form a subcategory of finite sets: identities have size-1 fibers and composition preserves gradients. The assignment `X` $\mapsto$ `Z^(X×X)/{dg}` with arrows induced by `T_f` is a functor to abelian groups, since identity and composition hold already on raw pair tables. This is a category for the selected quotient profile, rather than a claim that every actual source edit is an arrow in it. The native owner replays these identities exactly. This result covers fixed-writing gradient quotients; raw counts, oriented marks, edit prices, future enabled moves and all source-specific result equivalences require their own laws.

P.4 A residual section is a chosen splitting

Let `C:V`$\to$`V` be linear with `C`^{2}`=C`, and put `L=I`$-$`C`. Then `L`^{2}`=L`, `CL=LC=0`, `ker L=im C`, and `im L=ker C`. Direct expansion establishes the first identities; `Cx=x` characterizes membership in `im C`, and `Ly=y` characterizes `ker C`. Consequently

$$V/\operatorname{im} C \;\simeq\; \ker C, \qquad [x] \longmapsto Lx.$$

Well-definedness uses `LC=0`; injectivity uses `ker L=im C`; surjectivity uses `Ly=y` on `ker C`. If `im C=R1`, this represents the obstruction quotient, and Φ`(x)>0` iff `Lx`$\neq$`0`. Without that image equality, the link fails: take `V=Q`, `C=I`, `d0=0`, `x=1`. Then `Lx=0` but Φ`(x)=1`.

If `TC=C′T`, expansion gives `TL=L′T`, and residual sections respect transport. An arbitrary repair or optimizer need not satisfy that commutation law. Choosing `C` is extra structure; an L1 quotient norm does not choose a canonical projection. Nor does this vector-space splitting establish A's broader ontology or persistence policy.

The dressing-field method [REF-REL-005] is the group-action analogue of a chosen splitting. Let a finite group `G` act on configurations by $\varphi\mapsto$`g`$\cdot\varphi$. A dressing field `u` is **equivariant** when `u(g`$\cdot\varphi$`)=g`$\cdot$`u(`φ`)`, and the dressed configuration is φ`^u=u(`φ`)^{-1}`$\cdot\varphi$. Then `(g`$\cdot\varphi$`)^u=u(`φ`)^{-1}g^{-1}g`$\cdot\varphi=\varphi$`^u`, so the

dressed configuration is invariant, and u(φ^u)=u(φ)^{-1}u(φ)=e by equivariance, so dressing is idempotent. Both hold for nonabelian G. Two printed conventions in the archived general-relativity source fail. Dressing by ρ(u)φ without the inverse is not invariant even for the abelian group Z3. A proposition requiring u(g·φ)=g^{-1}u(φ) together with that dressing has no solution in S3 (it forces (g_1g_2)^{-1}=g_1^{-1}g_2^{-1}). Its proof step moves ρ(g^{-1}) past ρ(u), which needs commutativity. The exact checks enumerate all 6^6 maps from the free S3 orbit to S3: exactly six are equivariant, each is invariant and idempotent, and the printed conventions fail at stated configurations. As with C above, a dressing field is chosen extra structure. On a finite set an equivariant field exists exactly when the action is free: g·φ=φ would give u(φ)=g·u(φ) and so g=e, while for a free action one representative per orbit defines u(g·r)=g. Nothing here concerns continuous gauge groups, where a global choice can also fail for topological reasons.

P.4.a Contractive splittings in weighted L1 (DP-4, audited 2026-10-06) Setting. Let V=Q^n carry the norm ‖x‖=Σ_iw_i|x_i|, with rational w_i>0. Let R⊆V be a subspace with basis matrix B (n×r). Choose N (n×(n−r)) so that [B N] is invertible, and write its inverse in block rows [U;Q]. Then BU+NQ=I, UB=I, QN=I, UN=0 and QB=0. Put Φ_R(x)=min_{y∈R}‖x−y‖, the quotient norm of P.2 with R1=R. For a matrix M, the induced norm is

$$\|M\| = \max_j \frac{\sum_i w_i |M_{ij}|}{w_j}.$$

Statement.

1. The idempotents C with im C=R are exactly C_H=B(U+HQ), for H ranging over the rational r×(n−r) matrices. For each, L_H=I−C_H=(N−BH)Q and ker C_H=im(N−BH).
2. For each such C, with L=I−C, every x satisfies Φ_R(x)≤‖Lx‖≤‖L‖Φ_R(x).
3. If R≠V, then ‖L‖≥1. The identity ‖Lx‖=Φ_R(x) for every x holds iff ‖L‖=1; call C *contractive* in that case. If R=V, then L=0 and Φ_R=0; this is a separate branch.
4. A contractive C exists iff the following rational linear program in (H,t) is feasible: t_ij≥(L_H)_ij, t_ij≥−(L_H)_ij and Σ_iw_it_ij≤w_j.
5. In Q³ with unit weights and R=span(e), e=(1,1,1), every projection onto R has ‖L‖≥4/3. Equality holds for a=(1/3,1/3,1/3), so no contractive complement exists.
6. If R has codimension one, a contractive C exists, for every positive weight vector. In particular this holds for every line in weighted ℓ_1^2.
7. Let T be rational linear, and let C and C′ be projections. Then TC=C′T iff T(im C)⊆im C′ and T(ker C)⊆ker C′. When these hold, TL=L′T.

Proof.

Item 1. Expanding with UB=I and QB=0 gives C_H^2=C_H, and im C_H=R. Conversely, let C be idempotent with im C=R. Then C=BM for some M, and C^2=C with B injective gives MB=I. Hence M=M(BU+NQ)=U+(MN)Q, which is C_H with H=MN. The other two formulas follow by expansion.

Item 2. Cx∈R gives Φ_R(x)≤‖x−Cx‖. For y∈R, Ly=0 and so ‖Lx‖=‖L(x−y)‖≤‖L‖‖x−y‖; minimize over y. The column formula is the induced norm: ‖Mx‖≤Σ_j|x_j|Σ_iw_i|M_ij|, with equality at the maximizing e_j.

Item 3. A nonzero idempotent satisfies ‖L‖=‖L^2‖≤‖L‖^2. If ‖L‖=1, item 2 gives equality. Conversely, if ‖Lx‖=Φ_R(x)≤‖x‖ for every x, then ‖L‖≤1.

Item 4. L_H is affine in H.

Item 5. Every projection onto R has the form C=ea$^\top$ with a_1+a_2+a_3=1. Column j of L has norm c_j=|1−a_j|+2|a_j|. Since |1−a_j|≥1−a_j and |a_j|≥a_j, we get c_j≥1+a_j. So c_1+c_2+c_3≥4 and max_jc_j≥4/3.

A replayable infeasibility certificate for the LP of item 4 has thirteen rows:

- −a_j−t_jj≤−1 for each j (3 rows);
- a_j−t_ij≤0 for i≠j (6 rows);
- Σ_it_ij≤1 for each j (3 rows);
- −Σ_ja_j≤−1 (1 row).

Adding all thirteen gives 0≤−1. At a=(1/3,1/3,1/3) every c_j=4/3. For x=e1, Lx=(2/3,−1/3,−1/3) has norm 4/3. But Φ_R(e1)=1, attained at y=0 and certified by the dual z=(1,−1,0), which has z$^\top$e=0 and |z_i|≤1.

Item 6. Write R=ker ψ for a nonzero rational functional. Rescale ψ so that max_i|ψ_i|/w_i=1, and let i attain the maximum. Put v=sign(ψ_i)e_i/w_i. Then ‖v‖=1 and ψ(v)=1. Set L=vψ$^\top$ and C=I−L.

- L^2=L, ker L=R, ψ(Cx)=0, and Cx=x on R, so im C=R.
- ‖Lx‖=|ψ(x)|‖v‖≤Σ_i|ψ_i||x_i|≤‖x‖, so ‖L‖=1.

The 4/3 obstruction of item 5 therefore needs codimension at least two, and so n≥3.

Item 7. (⇒) For x∈im C, Tx=TCx=C′Tx∈im C′. For x∈ker C, C′Tx=TCx=0. (⇐) Write x=Cx+Lx. Then C′TCx=TCx and C′TLx=0, so C′Tx=TCx. In the control below, C=[[1,0],[0,0]] and C′=[[1,1],[0,0]] share the image span(e1). Their kernels span(e2) and span(1,−1) differ, T=I fails the condition, and Le2=(0,1) while L′e2=(−1,1).

Worked examples. In Q^2 with unit weights and R=span(1,1):

- C=(1/2) [[1,1],[1,1]] gives L=(1/2) [[1,−1],[−1,1]], with ‖L‖=1.
- C0=[[0,1],[0,1]] is also idempotent with image R. It gives L0=[[1,−1],[0,0]], with ‖L0‖=1.
- For x=(3,1), Φ_R(x)=2, certified by z=(1,−1). Then Lx=(1,−1) and L0x=(2,0), both of norm 2.

So two contractive sections represent one quotient class by different residuals, and the choice of C is extra structure. The scalar C=2 on Q is not idempotent and is refused.

Evidence grade.

- *Written proof:* items 1–7. Item 6 was added in this audit pass, and it resolves the audit's open question about lines in weighted $\ell 1^2$.
- *Exact finite replay:*
  - dp4_diag_line_lp4_infeasible_H_param, dp4_thirteen_row_certificate, dp4_diag_line_attaining_norm, dp4_diag_line_Lx_and_phi;
  - dp4_2d_C_idempotent_norm_Lx, dp4_2d_C0_alternative, dp4_2d_phi_x;
  - dp4_new_l1_squared_lines_contractive (four sampled lines in weighted $\ell 1^2$, each feasible);
  - dp4_zero_quotient_norm_zero, dp4_R_zero_norm_one;
  - dp4_transport_control, dp4_control_C2_not_idempotent.
- *Lean kernel:* these cover item 5 and the planar example only. The axiom report is lean4/component_repair_axioms.txt.
  - WitnessedDifference.ComponentRepair.colNorm_ge, WitnessedDifference.ComponentRepair.some_column_ge_four_thirds, WitnessedDifference.ComponentRepair.uniform_column_-

`norm`, `WitnessedDifference.ComponentRepair.repaired_example_norm`, `WitnessedDifference.ComponentRepair.example_distance_to_R`, `WitnessedDifference.ComponentRepair.example_ratio`;
  - `WitnessedDifference.ComponentRepair.planar_control`.
- *Native exact implementation:*
  - `omega_functional::residual_splitting` provides the parametrized splitting, the weighted operator norm, and the splitting and transport verifiers.
  - `omega_solvers::rational_feasibility` decides the LP of item 4 and replays Farkas certificates.
  - `omega_solvers::contractive_complement` decides the distortion bound and whether a contractive complement exists. Pivot-budget exhaustion is a refusal, not a verdict.
  - Replays:
    * the diagonal line in $\ell 1^3$ has minimum distortion `4/3`, attained;
    * the bound `1333/1000` is infeasible;
    * the thirteen-row certificate replays, and a copy with its last multiplier zeroed is rejected;
    * the planar diagonal has a contractive complement.
- *Unresolved residual:*
  - The `omega_wdp` projection bounds do not yet call the contractive complement decision.
  - The Lean module takes the induced norm to be the largest column norm without formalising that identity.
  - It treats only rank-one projections onto the diagonal line.
  - Items 1–4, 6 and 7 are not in Lean.
  - Item 6 has no exact replay beyond the four samples.

P.5 Overlaps, comparison spans and retained histories

Pairs `(a,b)` and `(b′,c)` glue to a triple with both projections iff `b=b′`; when they do, `(a,b,c)` is the unique triple. This is the smallest fiber-product argument. For spans `X←M→Y` and `Y←N→Z`, composition uses

$$M \times_Y N = \{(m, n) : \mathrm{right}(m) = \mathrm{left}(n)\}.$$

Matching pairs have the evident outer endpoints. An incompatible overlap supplies no member. Iterated fiber products have the canonical tuple rebracketing bijection; a general enriched comparison category still needs its declared morphisms, identities and coherence data. The inherited faithful point-span groupoid embedding claim is not reinstated: a one-point span loses distinct arrows with the same endpoints.

The layered-authority source defines each composite comparison price to be the sum of its two declared prices. Associativity of that sum follows from associativity of addition. It is not a theorem about an optimized minimum price. Erasing a trace can preserve endpoints and span composition while losing steps, work and the intermediate raw states. Distinct histories can have the same terminal state and different future permissions or accumulated cost. Thus those protected queries must be tested against P.1 before erasure.

An empty support sieve, a missing terminalizer, an invalid leg or a mismatched target blocks the standing predicate that requires those fields. This is a consequence of a declared predicate. Finite overlap checks and these non-creation results do not construct a general sheaf, global section, omniversal comparison object or higher cohomological optimizer.

P.5.a Finite function gluing and the presheaf boundary (DP-6, audited 2026-10-06)
Statement. Let X be a finite set and (U_i)_{i∈I} a finite family of declared subsets of X. Let B be a finite output alphabet, and let s_i:U_i→B be total function tables. Put U=⋃_iU_i. The following are equivalent:

1. *(compatibility)* s_i(x)=s_j(x) for every i,j and every x∈U_i∩U_j;
2. *(unique glue)* there is exactly one s:U→B with s|U_i=s_i for every i.

A global report on a declared target X additionally requires U=X. When U≠X, the missing points are reported as IncompleteCoverage, and gluing on the smaller union is a separate problem. The case I=∅ is included: then U=∅, and the unique glue is the empty function.

For a general finite presheaf, the question is different. Let F be a presheaf on a finite site, validated for totality, identities and composition. Fix a cover of X, let M be its set of matching families, and let r:F(X)→M be the restriction map. The extensions of a family m∈M are exactly the fiber $r^{-1}(m)$. That fiber can have zero, one or more elements, and pairwise compatibility does not decide which.

Proof.

(1)⇒(2). Assume compatibility, and let G=⋃_i{(x,s_i(x)):x∈U_i}. The domain of G is U.

- G is a graph of a function. Suppose (x,b)∈G and (x,c)∈G. Then there are i,j with x∈U_i∩U_j, b=s_i(x) and c=s_j(x), and compatibility gives b=c. So G is the graph of some s:U→B.
- Restriction recovers every table. For x∈U_i, G contains (x,s_i(x)), so s(x)=s_i(x).
- The glue is unique. Let t:U→B restrict to every s_i. Each x∈U lies in some U_i, and then t(x)=s_i(x)=s(x).

(2)⇒(1). At x∈U_i∩U_j, s_i(x)=s(x)=s_j(x). This direction uses existence only. No step selects an element, so the empty union is covered.

Presheaf fibers. By functoriality, every global restricts to a matching family, so r is well defined. From the definitions of fiber, onto and one-to-one:

- every matching family extends iff r is onto;
- extensions are unique whenever they exist iff r is one-to-one;
- the sheaf condition for this cover is both.

Worked example. Take X={a,b,c}, U_0={a,b}, U_1={b,c} and B={0,1/2,1}. Set s_0(a)=0, s_0(b)=1/2, s_1(b)=1/2 and s_1(c)=1. The only overlap comparison is 1/2=1/2. The glue is {(a,0),(b,1/2),(c,1)}, and every global extending both tables takes these values.

Mutating s_1(b) to 1 gives the incompatibility witness (b,1/2,1). Its discrepancy is 1−1/2=1/2.

Countermodels for the dropped hypotheses.

- *Coverage omitted.* Let X={a,b} with only U_0={a}, s_0(a)=0 and B={0,1}. The glue on {a} is unique, but (a↦0,b↦0) and (a↦0,b↦1) both extend it on X.
- *Local totality omitted.* Let X=U_0={a} with an empty table. Both a↦0 and a↦1 complete it, so no local function on U_0 is specified.
- *Existence fails for a presheaf.* Use the category of the four subsets of {a,b} under inclusion.
  - Put F(∅)={*} and F(V)={0,1} for every nonempty V.
  - Restrictions between nonempty sets are identities, and restrictions to ∅ are constant.
  - Every functor law holds by direct evaluation.
  - Cover X by {a} and {b}, with locals 0 and 1. They match at ∅.

  - The globals restrict to `(0,0)` and `(1,1)`, neither of which is `(0,1)`. So the fiber is empty.
- *Uniqueness fails for a presheaf.* On the same category, put `F(X)={g0,g1}` and `F(V)={*}` for every proper `V`. The family `(*,*)` is matching and has the two extensions `g0` and `g1`.
- *The empty intersection is skipped.* Put `F(V)={0,1}` for all four subsets, $\emptyset$ included, with identity restrictions. The disjoint locals `0` and `1` restrict to unequal values at $\emptyset$, so they do not match. A check that skips empty overlaps would accept them. For function tables this cannot happen, since there is exactly one function on $\emptyset$.
- *Erased histories.* Take histories `h0,h1` with the same endpoint `p(h0)=p(h1)`.
  - Their costs differ: `c(h0)=0` and `c(h1)=1`.
  - Their future permissions differ: `u(h0)=false` and `u(h1)=true`.
  - By P.1 each query has a collision, so neither factors through the endpoint.

  Erasure is valid for a specified query exactly when that query is constant on endpoint fibers.

Evidence grade.

- *Written proof:* the argument above.
- *Exact finite replay:*
  - `dp6_function_glue`, `dp6_function_incompatible`;
  - `dp6_empty_cover_empty_function`;
  - `dp6_existence_presheaf_functor_laws`, `dp6_existence_presheaf_fibers`;
  - `dp6_uniqueness_presheaf_functor_laws`, `dp6_uniqueness_presheaf_fibers`;
  - `dp6_empty_intersection_family_01`;
  - `dp6_history_query_collisions`.
- *Lean kernel:* the axiom report is `lean4/component_reports_axioms.txt`.
  - `WitnessedDifference.ComponentReports.glue_exists_unique`, `WitnessedDifference.ComponentReports.compatible_of_restricts`, `WitnessedDifference.ComponentReports.not_compatible_iff_witness`, `WitnessedDifference.ComponentReports.empty_cover_glues`;
  - `WitnessedDifference.ComponentReports.CoverPresheaf.restriction_matches`, `WitnessedDifference.ComponentReports.CoverPresheaf.every_matching_extends_iff`;
  - `WitnessedDifference.ComponentReports.existence_failure_zero_extensions`, `WitnessedDifference.ComponentReports.uniqueness_failure_two_extensions`.
- *Native exact implementation:*
  - `omega_sheaf_theory::finite_report_gluing` glues on the union and on the declared target, verifies function glues, and classifies presheaf gluing as a unique glue, no extension, or ambiguous. It also provides the history-erasure decision.
  - The `omega_wdp` presheaf descent wraps the existing fiber enumeration. It replays fibers of size `0`, `1` and more than one.
- *Unresolved residual:*
  - The existing general sheaf equalizer routine still skips empty intersections. A test documents this, and the routine is unchanged.
  - The native presheaf carrier is capped at `64` points. The theorem is not a claim that every finite size is admitted.
  - The history countermodel is finite replay and written proof only.

P.6 Completion, common targets and independent standing

Let `P=p(X)` carry a declared metric, and let `B` be complete. After P.1, the image query $\bar{q}$`:P`$\to$`B` has a unique uniformly continuous extension to the metric completion exactly when it is uniformly continuous on `P`. Equivalently, for every positive epsilon there is a positive delta such that

d(p(x),p(y))<delta implies d(q(x),q(y))<epsilon, for all source states. Extend by limits of Cauchy representatives; uniform continuity makes the image sequence Cauchy and the limit independent of the representative. Restriction establishes necessity and density establishes uniqueness.

This is the actual CS-to-Q interface. Finite factorization alone is insufficient, and source continuity for an unrelated metric does not replace relative continuity. Appendix O.1 supplies the formal selected metric criterion; O.2 retains the separate circle and discrete dyadic dual topologies. O.4–O.9 bind model-specific common targets and error moduli. No primitive mark chooses those metrics, physical models or limits.

If every interval [l_n,u_n] contains the same real b and its width tends to zero, its midpoint error is at most (u_n−l_n)/2. Interval order and shrinking widths without common-target membership do not give that conclusion. Q's rational interval stage is not a construction of an unknown enclosed point. Completing a field of scalars likewise supplies neither an infinite operator domain nor interacting field dynamics.

Finally, operational descent standing consists of an actual terminal t, a pipeline equality to that terminal and its seven evidence-derived guards. Regime standing has the different form

$$\mathrm{Standing}_R(g,p) \iff (\exists e.\, \mathrm{verifies}_R(p,e)) \ \wedge \ \mathrm{recordedGrant}_R(g,p).$$

Assume verification implies truth and a recorded grant implies entitlement. Applying those two assumptions to the standing witnesses establishes truth and entitlement. This is a sufficient soundness theorem, not a proof of those assumptions or an equivalence recovering them. Constant-success verification and grant predicates with false truth give an immediate countermodel without the assumptions. An interpretation relating operational and regime standing is another obligation.

Refusal propagation and terminal-history reconstruction follow by unfolding the partial pipeline; disjoint terminal constructors follow by case analysis. These elementary laws are useful but do not prove general Kleisli factorization, conservation of a tower or institutional authorization. Python's finite application verifies a configured one-use HMAC grant; the primary Rust descent stage currently refuses supplied grant strings because its external verifier is unavailable. Neither result selects the grantor's institutional standing. Proposal, mathematical evaluation, evidence and authorization therefore retain distinct interfaces throughout the compression.

P.6.a Finite enclosure verification records and an exact square-root enclosure (DP-5, audited 2026-10-06) Correction. The candidate asked for "a nested-without-common-target countermodel". The audit read it as: a nested family of nonempty, closed, bounded real intervals whose intersection is empty. No such family exists, so that reading of the request is refuted. **DP-5′ supersedes the candidate statement.** It keeps the theorem below and replaces the requested example with controls that each change exactly one premise or the ambient field.

Statement.

1. *(Finite midpoint bound.)* Let [l_j,u_j], 0≤j≤N, be a nonempty prefix with rational endpoints. Let b be a declared target with replayable evidence l_j≤b≤u_j for every submitted j. Then |b−(l_j+u_j)/2|≤(u_j−l_j)/2 for every j≤N. If u_j−l_j≤2ε, the midpoint has error at most ε. Nesting is not needed here.
2. *(Common point.)* Every nested sequence of nonempty, closed, bounded real intervals has a common point. If the widths tend to zero, that point is unique.

3. *(Finite intersection.)* A nonempty finite family of rational intervals has a common point iff max_jl_j≤min_ju_j. This finds an unnamed common point. It does not identify a separately designated target.
4. *(Exact square-root enclosure.)* Let a=p/q≥0 be reduced with q>0, and let δ>0 be rational.
   - If a=0, return [0,0].
   - Otherwise put l_0=0 and u_0=H=max(1,a). Bisect at m=(l+u)/2: return [m,m] if m^2=a; replace l by m if m^2<a; replace u by m if m^2>a.
   - Each comparison is an integer comparison: for m=s/t, compare s^2q with pt^2.
   - Every stage satisfies 0≤l_n≤u_n, $l_n^2 \le a \le u_n^2$, [l_{n+1},u_{n+1}]⊆[l_n,u_n] and u_n−l_n≤H/2^n.
   - The procedure reaches width ≤δ within the least n with 2^nδ≥H.
   - Taking δ=2ε gives a midpoint with error at most ε from the nonnegative root.
5. *(Prefix limit.)* A checked finite prefix does not certify a modulus n≥M(ε)⇒u_n−l_n≤2ε for later indices. It also does not certify membership of a target that was never witnessed.

Proof.

Item 1. l≤b≤u and m=(l+u)/2 give −(u−l)/2≤b−m≤(u−l)/2.

Item 2. Let b=sup_nl_n. This exists, since u_0 bounds every l_n. Fix j. For n≥j, nesting gives l_n≤u_n≤u_j. For n<j, l_n≤l_j≤u_j. So b≤u_j, and l_j≤b by the definition of b. Two common points are at most every width apart, so if the widths tend to zero they coincide.

Item 3. A common point lies in [max l,min u]. Conversely, if max l≤min u, then max l is a common point.

Item 4. - *Base case.* If a≤1, then u_0=1 and 1≥a. If a≥1, then a^2≥a. Either way l_0^2=0≤a≤u_0^2. - *Step.* The step keeps whichever half preserves $l^2 \le a \le u^2$, and it halves the width. Order is preserved because l≤m≤u. - *Common root.* Squaring is increasing on nonnegative numbers. So each interval contains the nonnegative root, and only that root. - *Termination.* Repeated integer doubling of δ finds the least n with 2^nδ≥H.

Nonnegativity is needed. With a=1 and l=u=−1, $l^2 \le a \le u^2$ holds, but the interval excludes the positive root.

Item 5. After any prefix [0,2^{−j}], j≤N, extend in two ways:

- continue the same rule, which gives widths 2^{−n} and target 0;
- set every later interval to [2,3].

The two extensions agree on the inspected prefix. The second fails both the target obligation and the modulus obligation.

Worked table (a=2).

n	l_n	u_n	width	decisive comparison
0	0	2	2	0≤2≤4
1	1	2	1	1<2
2	1	3/2	1/2	9/4>2
3	5/4	3/2	1/4	25/16<2
4	11/8	3/2	1/8	121/64<2
5	11/8	23/16	1/16	529/256>2

n	l_n	u_n	width	decisive comparison
6	45/32	23/16	1/32	2025/1024<2048/1024

The last midpoint is 91/64, with error at most 1/64.

- *Tamper.* Replacing the final lower endpoint by 23/16 is rejected, because 529/256>2.
- *Exact root.* For a=9/16, the second midpoint is 3/4, and the procedure returns [3/4,3/4].
- *Zero.* a=0 returns [0,0].

Controls for DP-5′. Each changes one premise of item 2 or item 1.

- *Incomplete field.* The table above, read in Q, consists of nonempty closed sets [l_n,u_n]∩Q that are nested and shrinking. Their intersection is empty.
  - A common rational q would satisfy $l_n^2 \le q^2 \le u_n^2$ for every n. Also $u_n^2 - l_n^2 = (u_n+l_n)(u_n-l_n) \le 4H/2^n$ tends to zero (the Archimedean step). So $q^2=2$.
  - For reduced q=m/n, $m^2=2n^2$ makes m even, then n even, a contradiction.
- *Closedness dropped.* (0,1/(n+1)) has empty intersection.
- *Target membership dropped.* [1,1+2^{−n}] is closed, nested and shrinking, but it misses the designated b=0, and every midpoint error is above 1.
- *Nesting dropped.* [n,n+1/(n+1)] has no common point.
- *Shrinking dropped.* [0,2] at every index keeps midpoint error 1.
- *Boundedness dropped.* [n,∞) is nested, closed and nonempty, with empty intersection.

Evidence grade.

- *Written proof:* items 1–5 and every control.
- *Exact finite replay:*
  - dp5_sqrt2_table, dp5_midpoint_and_error, dp5_invariants_width_nesting, dp5_long_run_invariant_a2_delta_2_pow_minus40;
  - dp5_termination_bound_a1/2_d1/100, dp5_termination_bound_a2_d1/32, dp5_termination_bound_a7_d1/10;
  - dp5_exact_root_9_16, dp5_zero_radicand;
  - dp5_tamper_lower_endpoint_rejected, dp5_negative_endpoint_control;
  - dp5_no_rational_sqrt2_small_denominators;
  - dp5_finite_intersection_max_lo_le_min_hi;
  - dp5_prefix_cannot_separate_tails.
- *Lean kernel:* the axiom report is lean4/component_completion_axioms.txt.
  - WitnessedDifference.ComponentCompletion.inv_start, WitnessedDifference.ComponentCompletion.inv_step, WitnessedDifference.ComponentCompletion.inv_iter;
  - WitnessedDifference.ComponentCompletion.width_step, WitnessedDifference.ComponentCompletion.step_nested, WitnessedDifference.ComponentCompletion.width_iter;
  - WitnessedDifference.ComponentCompletion.root_in_bracket, WitnessedDifference.ComponentCompletion.mid_close, WitnessedDifference.ComponentCompletion.negative_endpoint_control;
  - WitnessedDifference.ComponentCompletion.start_two, WitnessedDifference.ComponentCompletion.table_row1 through WitnessedDifference.ComponentCompletion.table_row6, and WitnessedDifference.ComponentCompletion.table_iter6;

  - `WitnessedDifference.ComponentCompletion.sq_mod_two`, `WitnessedDifference.ComponentCompletion.nat_sq_two_descent`, `WitnessedDifference.ComponentCompletion.no_rational_sqrt_two`.
- *Native exact implementation:*
  - `omega_kernel::square_root_enclosure` provides the bisection, trace replay, the step count for a width, finite-prefix verification and the finite common point, with a designated target.
  - At its resource limit it returns the last certified enclosure, never the requested precision.
- *Unresolved residual:*
  - The Archimedean step of the incomplete-field control is a written proof and is not formalised.
  - Item 2 is the real-analysis argument above. It is not stated in Lean.
  - A modulus for an arbitrary generator must come with its own proof. The square-root generator is the only one supplied here.

P.7 Complete balance-preservation rules for the five permitted edits

The model in P.7–P.21 has a finite term set, two nonnegative integer relation-count tables and oriented filled triples. Its five permitted edits identify two terms, remove one assertion, fill a triple, add a term, or add one assertion. This is the model specified in the twenty-fifth source derivation and implemented in `twentyfifth/state.py`. The source number is only an archive index. The orientation convention, identification of filled triples and ordering of offered edits are explicitly chosen readings in that implementation, not consequences of the initial inscription or external standards. The statements below use those stated choices as hypotheses.

This is a fixed-context bounded statement about the actual directed five-edit model transition system. Fix its ordered carrier `0,…,n−1` and its sorted set `F` of canonical cyclic triples. Both directions and loops remain independent entries. For integer excess `e=r−s`, put `C_(a,b,c)(e)=e_ab+e_bc+e_ca`. Write `D_F` for the matrix with these cycle rows; balance means `D_F e=0`. Every integer table is realizable by nonnegative raw counts, using its positive and negative parts. Adding the same positive offset to both tables leaves excess unchanged and enables any selected removal. A marking records an equation to check; the source kernel permits an unbalanced marking and does not solve it during `Fill`.

Call an enabled event **universally balance-preserving in this context** if every balanced raw state with this carrier and face set, on which the same event is enabled, has a balanced actual output. This quantifier differs from balance of the particular supplied input/output and from descent of all gradient classes.

Actual edit	Necessary and sufficient condition
`AddTerm`	Always, for the existing carried face set.
`AddAssertion` or `Remove` at directed `(x,y)`, in either relation	No face of `F` uses that directed arc.
`Identify(x,y)` with `x<y`	No surviving output face contains the merged vertex `x`.
`Fill(t)` with `t` absent from `F`	The row `C_t` belongs to the rational row span of `D_F`.

Proof for AddTerm. The actual operation preserves the old table entries and old faces, pads incident entries with zero and adds no face. Thus every old cycle sum is unchanged. This holds

even when P.3 says that the same operation fails to descend on the full gradient quotient: the new carrier has additional coordinates whose quotient semantics differ from the old face equations.

Proof for assertion changes. The excess changes by δ=±E_xy: plus for adding R or removing S, minus for adding S or removing R. Each old cycle changes by C_t(δ). All such changes vanish exactly when the directed arc occurs in no filled face. Necessity is witnessed by zero excess, realized with common count one on every coordinate so every removal is enabled. A face using the arc then has output value +1 or −1. The reverse arc is a different coordinate and loops are absent from triples, so neither is silently identified with the tested arc.

Proof for Identify. The actual endpoint map is f(y)=x and f(v)=v—[v>y] otherwise. It aggregates both count tables over endpoint fibers, carries faces by cyclic normalization, drops degeneracies and merges duplicates. If no surviving face meets x, each endpoint of every surviving face has a singleton preimage. Its cycle therefore equals that of an old filled face and remains zero. Conversely choose a surviving cyclic face (a,b,c) meeting x and rotate so a=x. Its three fiber sizes are (2,1,1). Let the old potential be one at the unique preimage of b and zero elsewhere, and let e=dg. Every old cycle vanishes. P.3's actual aggregate formula gives cycle (m_b s_a—m_a s_b)+(m_c s_b—m_b s_c)+(m_a s_c—m_c s_a)=—1. Positive/negative parts with common count one give a legal balanced raw source and an actual unbalanced identified output. This argument does not require any antisymmetry law for arbitrary source tables.

Proof for Fill. The operation keeps both tables and adds just the new diagnostic row. If C_t=c$^\top$D_F, every old balanced table satisfies the new equation. If no such rational coefficient vector exists, exact linear separation of D_F$^\top$c=C_t$^\top$ gives a rational vector w with D_F w=0 and C_t w≠0. Clear all denominators by their positive product to obtain an integer table with the same properties. Realize it as raw counts. The selected filling is enabled, its old faces are balanced, and its newly filled face is unbalanced. Thus rational row-span membership is equivalent even when all source excess tables are integral. Clearing denominators is exact and need not be minimal.

A directed tetrahedron falsifier. With old faces 013,023,123, set only e_20=1. Every old cycle is zero, but filling 012 gives value one. The familiar three-face implication for antisymmetric cochains does not apply to this raw table. With both orientations retained, the actual identity is

$$C_{012} - C_{021} - C_{013} + C_{031} + C_{023} - C_{032} - C_{123} + C_{132} = 0.$$

Hence seven directed tetrahedron face equations imply the eighth. This identity is checked coordinate by coordinate; it does not add an antisymmetry assumption.

The generic native owner linear_algebra::row_constraint_implication delegates elimination to the existing exact-field solver and independently replays either c$^\top$D_F=C_t or D_Fw=0, C_tw≠0. The WDP adapter genesis_balance_transition::Source25BalanceProfile retains the actual source, event and target, reports their separate balance values, and supplies a complete checked raw counterstate for a negative carrier-wide verdict. It never substitutes that counterstate for the supplied input. Dense Fill allocation/solver budgets refuse explicitly; arbitrary integer bit lengths and total source execution cost are outside those budgets.

The written classification above covers all five cases and arbitrary finite carriers. The new Lean module checks selected cycle, update and directed tetrahedron laws; it does not formalize the complete all-case necessity theorem or verify the Rust compiler. Exhaustive native kernel replay covers all 263 marked contexts through four terms and their 19,388 enabled edits. That finite evidence

is separate from the written carrier-wide proof. These rules close one dynamic balance question. General higher fillers, global site/HoTT semantics, physical limits and institutional soundness in GEN-WDP-O01–O05 remain open.

P.8 Affine balance paths and their exact coherence criterion

P.7 classifies one enabled edit at a time. A path needs a second question: whether its **endpoint** is balanced for every balanced input, even when an intermediate state is not. The following result answers that question at the integer-excess layer without identifying a raw history with its endpoint.

For a declared five-edit model context `C=(X,F)`, let `E_C=Z^(X×X)` retain every directed coordinate and loop, let `D_C` be its integer marked-cycle matrix, and put `K_C=ker_Z D_C`. A context change has an integer affine excess expression `F(e)=Ae+b`, where `A:E_C→E_C'` is an integer matrix. The actual five edit forms have exactly these expressions: AddTerm is zero padding, Identify is the pair-fiber sum of P.3, Fill leaves `e` fixed, and AddAssertion/Remove have `A=I` and `b=±E_xy` with the signs in P.7. The expression is a statement about excess; the original raw counts, edit enablement, occurrences, price and ordered history remain separately typed.

Affine balance theorem. The expression sends every balanced integer excess table to a balanced target table, `F(K_C)⊆K_C'`, if and only if

$$D_{C'}b = 0 \quad \text{and} \quad D_{C'}A = \Lambda D_C \quad \text{for some rational matrix } \Lambda.$$

The displayed pair is a finite positive certificate. The constant condition is necessary by applying `F` to zero. After it is imposed, preservation says that each row of `D_C'A` annihilates `ker_Z D_C`. The integer kernel spans `ker_Q D_C`: every rational kernel vector becomes integral after clearing denominators. Over `Q`, the row space lies in the annihilator of the kernel; both have dimension `rank D_C`, so they coincide and give `Λ`. The converse follows by substitution. If the constant condition fails, `e=0` is a negative witness. If the row relation fails, exact rational elimination gives `w∈ker_Q D_C` with `D_C'Aw≠0`; clearing denominators gives a replayable integer witness. Its positive and negative parts are legal nonnegative assertion tables. For an individual actual edit, adding a common sufficiently large count to both relations also enables a selected Remove without changing `e`; therefore P.7's all-states quantifier over enabled one-step raw states really ranges over all of `K_C`. This row-space test certifies a full-carrier *homogeneous implication*. It is not a certificate that an unrelated integer equation `Mx=b` has a solution; that question needs an integer solution or Smith divisibility evidence.

The affine expressions compose exactly:

$$(A_2, b_2) \circ (A_1, b_1) = (A_2A_1,\ A_2b_1 + b_2).$$

Identity has `(I,0)`. If the two steps have positive certificates `D_1A_1=Λ_1D_0` and `D_2A_2=Λ_2D_1`, then the composite has `D_2A_2A_1=(Λ_2Λ_1)D_0` and `D_2(A_2b_1+b_2)=0`. Thus finite contexts and their balance-preserving affine maps form a category at this selected excess level; certificates admit maps but are not part of arrow equality. A composite can pass the endpoint criterion when an intermediate step fails it: on a context with marked face `012`, AddAssertion in R at `(0,1)` followed by Remove of that new assertion returns the complete raw tables to their starting state. The first step can make the face sum one, but the composite is identity with `b=0` and row certificate `Λ=I`. Consequently checking each step is sufficient, not necessary, for endpoint balance. The converse failure is concrete rather than a claim about policy.

There is also an exact *path comparison* test. For two affine paths with the same source and target coordinate lists, `(A,b)` and `(B,c)` agree on every balanced integer excess table if and only if

$$b = c \quad \text{and} \quad A - B = HD_C \quad \text{for some rational matrix } H.$$

Apply equality at zero, then use the same integer-kernel/row-space argument for every row of `A-B`; the displayed equations also prove sufficiency. Unequal constants or an integer kernel vector on which `A-B` is nonzero are exact negative witnesses. In particular, a proposed restriction/edit square commutes on balanced excess precisely when its two route expressions pass this comparison test. Equalities of these extensional maps paste under composition when the adjacent maps preserve their stated kernels, giving coherent finite diamonds **for this declared layer**; they do not provide fillers between arbitrary source histories or a Kan structure.

Path-saturation lemma for the five permitted edits. Fix an initial context and a finite word of the five actual edit descriptors. Assume the word has at least one legal raw execution. Its intermediate contexts are determined by that context and the word; only Remove's positive-count test depends on the tables. Every raw prefix has a nonnegative integer linear part: AddTerm pads zeros, Identify aggregates fibers, Fill is the identity, and assertion edits add constants. Adding any common nonnegative table to both initial relations therefore cannot disable a previously legal execution. For an arbitrary integer excess `e`, start with `(e_+,e_−)` and add a common scalar `t` to **every** initial pair coordinate. At each Remove the tested count is α`t+`β`(e)` with integer $\alpha \geq$`0`. If α`=0`, the count has no initial-table dependence and equals its positive value in the known legal execution. If α`>0`, a sufficiently large `t` enables that Remove. One finite `t` handles the whole word. Consequently the excess projection of its legal initial states is the entire `E_C`, and the projection of legal **balanced** states is exactly `K_C`. If two words have one common legal initial state, the same argument handles both words. Thus the all-kernel endpoint criterion above is necessary as well as sufficient for an actual nonempty five-edit model path, not just for its affine extension. This lemma assumes unbounded nonnegative integer multiplicities; a native allocation or bit-budget refusal is a separate execution outcome.

For a different guarded move system, let `S`$\subseteq$`K_C` be the excess projection of its legal whole-path inputs. When nonempty, choose `e_0`$\in$`S` and put `L_S=span_Q{e-e_0 : e`$\in$`S}`. The exact guarded criterion is `D_C'F(e_0)=0` and `D_C'A(L_S)=0`; for two paths replace `D_C'F` by their difference on their **common** legal domain. A supplied finite rational basis of `L_S` makes this a finite check. Without that basis, a tested sample cannot certify the full-domain guarded statement. An empty common domain is reported as no enabled comparison, rather than as successful agreement.

Endpoint **raw-count** equality has a stronger exact test. Fix the same ordered target carrier and face set for two raw affine path expressions. Write the difference between their complete output count pairs as `U r+V s+c`, with `r,s`$\in$`N^(X`$\times$`X)` and source balance `D_C(r−s)=0`. It vanishes on every balanced raw input if and only if

$$c = 0, \qquad U + V = 0, \qquad U = HD_C \quad \text{for some rational matrix } H.$$

Zero raw counts force `c=0`. Taking `r=s=t` for each nonnegative unit table forces `U+V=0`. The difference is then `U(r−s)`. Every integer `e`$\in$`K_C` occurs as `r−s` by its positive/negative parts, so the earlier row-space argument forces `U=H D_C`; these equations conversely make the difference zero. A failed equation has, respectively, the zero state, a diagonal unit-count state, or an integer-kernel counterstate as

its witness. For two five-edit model words with a nonempty common legal domain, the displayed test is also necessary and sufficient **on that domain**. The saturation lemma supplies a legal representative of every balanced `e`. Moreover, adding a diagonal unit count to both tables at one initial coordinate cannot disable either word, so equality on the legal domain forces `U+V=0` one coordinate at a time. A legal representative with `e=0` then forces `c=0`, and the representatives of all `e∈K_C` force `U=H D_C`. Failed conditions have legal witnesses after common-offset padding; no arbitrary invalid raw candidate is counted as a counterexample. For other guarded move systems, their actual common domain must be checked separately. If the five-edit model raw test passes and both paths end in the same declared context, all deterministic future *state* questions agree, including source-enabled moves and plain-state price. It does not equate the paths' occurrences or accumulated costs. Identity and the path that adds one R and one S assertion at the same arc have identical endpoint excess on every input but different raw counts; at the zero state, only the latter enables their removal. P.3's AddTerm example likewise shows that balance preservation alone need not descend to the gradient-result quotient. These meanings require the matching raw, quotient, history and permission interfaces rather than an excess-only label.

P.9 Whole-word descent to fixed-writing result classes

P.8 concerns balance and equality of raw or excess endpoints. The actual five-edit model result mark asks a different question. At one fixed ordered writing and fixed oriented face set `F`, the source result relation is

$$e \sim_X e' \quad \Longleftrightarrow \quad e - e' \in G_X, \qquad G_X = \operatorname{im}_{\mathbb{Z}} d_X, \quad (d_X g)_{xy} = g_x - g_y.$$

The source-specific completeness proof in the marked-result supplement identifies this relation with equality of the normalized table component when the same `F` is retained. It does not identify raw counts or ordered histories. Write `Q_(X,F)=(E_X/G_X,F)` for this **fixed-writing** class carrier; the fully permutation-minimized `result_mark` has an additional orbit relation.

Fix an initial context `(X,F)` and a finite word `w` of the five actual five-edit model edit descriptors with at least one legal execution. Follow each original vertex through AddTerm inclusions and Identify mergers to obtain a total map `f_w:X→Y` to the final ordered carrier. The final face set `F_w` depends on `(X,F)` and `w`, but not on the initial count tables. Pair-table aggregation satisfies `T_gT_f=T_(g∘f)` by regrouping finite sums. AddTerm is aggregation along an inclusion with an empty new fiber; Identify is aggregation along its merger; Fill and assertion edits have the identity linear part. Induction over the word therefore gives the **exact** expression

$$e_w = T_{f_w} e + b_w, \qquad b_w = \sum_{j \text{ assertion edit}} T_{j \rightsquigarrow w}\, \delta_j,$$

where δ`_j=±E_xy` is the excess change at an AddAssertion or Remove edit and `T_(j⇝w)` is aggregation through the remaining carrier changes. Remove has the corresponding opposite sign from AddAssertion in the same relation. The formula does not reconstruct raw counts, costs or occurrences; those fields must remain available through their separate interfaces.

Fixed-writing whole-word descent theorem. The word defines a total, representative-independent map `[e]↦[T_(f_w)e+b_w]` from `Q_(X,F)` to `Q_(Y,F_w)` by choosing any *legal* raw representative of each source class and applying the word if and only if

$$T_{f_w}(G_X) \subseteq G_Y.$$

Sufficiency is immediate: two equivalent source excesses differ by `d_Xg`, and their output difference is `T_(f_w)d_Xg`; the translation `b_w` cancels. For necessity, if one integer gradient fails the inclusion, P.8's path saturation supplies legal raw representatives with excesses `0` and `d_Xg` and the same initial `F`. They have one source class and two target classes. The legal domain therefore cannot hide a failed gradient. The source and target face sets in this theorem are fixed; a word whose descriptors are invalid in the chosen context has no induced map by vacuous truth.

P.3 supplies the complete carrier test for this actual `T_f`. The inclusion holds **exactly** when `|Y|`$\leq$`2`, at most one fiber of `f_w` is occupied, or all target fiber sizes are equal. Empty target fibers, including new surviving terms, count. A positive integral certificate is a matrix `H` of integer potentials satisfying `T_(f_w)d_X=d_YH`, one column per source unit potential. For `Y`$\neq\emptyset$, each proposed target column can be checked directly: all loops vanish, reverse directed entries are negatives, and every distinct-vertex triangle sum vanishes. Set the root potential to zero and `h_a=t_(a,0)`; the triangle through the root then establishes `t_(a,b)=h_a-h_b` with **integer** `h`. For `Y`$=\emptyset$, the pair table is zero. If the fiber-size test fails, P.3 constructs a source unit potential whose transported gradient has a nonzero target triangle; this is an exact negative witness using two legal raw representatives after common-offset padding. Rational span membership alone would not certify an integer lift for a general lattice; this particular full directed-pair gradient has the explicit integer root construction.

The result-class and balance requirements are independent. A word induces a map on **balanced** fixed-writing classes `(K_(X,F)/G_X)`$\to$`(K_(Y,F_w)/G_Y)` precisely when its fiber-size test passes and the P.8 endpoint equations also hold:

$$D_{F_w} b_w = 0, \qquad D_{F_w} T_{f_w} = \Lambda D_F \quad \text{for some rational } \Lambda.$$

Every gradient lies in the appropriate balance kernel, so both quotients are well-defined. AddTerm from two old terms to three with no faces preserves balance vacuously but has fiber sizes `(1,1,0)` and fails result descent. Conversely, adding one R assertion on a directed arc of a marked triangle has `f_w=id` and passes result descent, but `D_(F_w)b_w=1` and fails balance. These are countermodels to replacing the two tests with one status flag.

Two words with the same initial context and the **same** final ordered carrier and face set, each passing descent and having a nonempty common legal domain, induce the same map on every fixed-writing result class iff

$$b_w - b_v \in G_Y \quad \text{and} \quad (T_{f_w} - T_{f_v})(E_X) \subseteq G_Y.$$

Evaluate at zero and at every source unit table for necessity; those excesses have common legal raw representatives by P.8. The same equations prove sufficiency by additivity. Exact integer potential witnesses for the constant and each matrix column certify the condition; a failed column plus a unit source table is a counterwitness. If both words also pass the P.8 balance test, comparison restricted to balanced classes replaces the unit-column test by an **integer** generating set for `K_(X,F)`, with the constant still tested separately. A rational kernel basis alone does not certify integral membership in a target gradient lattice. Different final face sets fail fixed-writing equality before any table comparison.

For example, on X={0,1}, the potential g=(1,0) gives e_01=1 and e_10=-1. AddTerm produces Y={0,1,2} with zero new incident entries, so the target triangle (0,1,2) has sum one: descent fails even for this legal representative. Follow AddTerm instead by Identify of the new term with old term 0. The complete word has final fiber sizes (1,1) and acts as the identity on every initial pair coordinate, including loops. It descends although its first prefix does not. It still incurs two occurrences and a term-addition price. The theorem classifies fixed canonical writings and complete word endpoints; it does not make a word equivariant under arbitrary source renamings, equate raw histories, recover permission, or construct the missing global site/higher fillers in O02/O03.

P.10 Full result-mark descent on one source face orbit

P.9 fixes a writing. The five-edit model's complete result_mark also identifies the simultaneous renamings of the excess table and oriented fillings. Here is an exact test for **one fixed literal edit word** on the orbit of one initial face context. It does not silently rename the edit descriptors.

Write E_X=Z^(X×X), G_X=im d_X, Q_X=E_X/G_X, and let P_p transport pair coordinates under a permutation p∈S_X. The same p carries each oriented filling through cyclic canonicalization; write its result as pF. Let w have one legal execution on (X,F), with the P.9 excess expression Ae+b, where A=T_(f_w), and let F_1=w(F) be its final face set. For every p∈S_X, run only the context skeleton of the same word on (X,pF). Let Adm_w(F) contain the p for which all descriptors are structurally valid, and write F_p=w(pF) for their final face sets. The identity belongs to Adm_w(F). The carrier maps and excess translations of the five edits do not depend on fillings, so every such writing has the **same** A,b, applied to its own input table. The known legal base count run is also a legal count run in each structurally valid face context; P.8 saturation then supplies a legal raw representative of every integer excess there. Invalid contexts are excluded, not counted as successful executions.

Full-mark word theorem. Applying w to any legal raw representative defines a total map from the full five-edit model result classes with face set in the orbit of F to full target result classes if and only if:

1. A(G_X)⊆G_Y;
2. for every p∈Adm_w(F), some *single* q∈S_Y satisfies qF_1=F_p and both integer-gradient conditions

$$b - P_q b \in G_Y, \qquad (AP_p - P_q A)(E_X) \subseteq G_Y.$$

The word *single* means that q works for every source excess table for that p; the witness q may differ between source permutations. These conditions give a finite, exact certificate: check the final oriented face lists, then supply integer target potentials for the constant and each source-unit column. Condition 1 is the P.3 fiber-size criterion. The certificate asserts a map on result classes, not equality of raw histories or a group homomorphism p↦q. Enumerating source and target permutations and the negative grid can be factorial or exponential; no polynomial-time or implemented decision procedure is asserted.

For sufficiency, condition 1 makes the output independent of a source gradient within any admitted writing. Given another source representative with table P_p e+g and faces pF, where g∈G_X, condition 2 gives [A(P_p e+g)+b]=[P_q(Ae+b)] in Q_Y and carries F_1 to F_p. Thus the two outputs have one full joint mark. Every source class has a legal representative in the base writing by saturation, so the induced map is total on the stated face orbit.

For necessity of condition 1, take g∈G_X. All tables kg, for integer k≥0, have the same source class as zero and have legal representatives. Their target classes are [b]+k[Ag]. The target permutation orbit

of `([b],F_1)` is finite. The lattice `Q_Y` is torsion-free: when `Y`$\neq\emptyset$, subtract the integer gradient with potential `h_a=e_(a,0)` for `a`$\neq$`0` and `h_0=0`; this splits `E_Y` into gradients and tables whose nonroot entries in the root column vanish. Hence `[Ag]`$\neq$`0` would produce infinitely many distinct target classes in a finite orbit. In fact, one of `k=1,…,|S_Y|` is already a legal counterwitness. Therefore every source gradient must map into `G_Y`.

For necessity of condition 2, fix an admitted `p`. Full-mark equality for every `e`$\in$`E_X` says that for each `e`, at least one target permutation `q` carrying `F_1` to `F_p` makes the affine difference

$$[AP_p e + b - P_q(Ae + b)] \in Q_Y$$

zero. There are finitely many candidate `q`. If none made this difference identically zero, choose a nonzero integer affine coordinate of the difference for each candidate, using the preceding lattice splitting. Their product is a nonzero polynomial over `Q`, which cannot vanish on all of `Z^(X×X)`: induction on the number of variables establishes that the integer lattice is Zariski dense. Thus one `e` would defeat every `q`, contradicting full-mark equality. An identically zero candidate gives exactly the two displayed gradient conditions, by evaluating at zero and each source-unit table. If there are `m` face-compatible candidate permutations but none passes, the same polynomial argument finds a countertable in the explicit grid `{0,…,m}^(X×X)`; its degree in each variable is at most `m`. If `m=0`, zero already witnesses the face mismatch. P.8 provides legal raw representatives for the base and renamed contexts. This finite-union step uses the torsion-free target gradient quotient; it would not hold for an arbitrary finite torsion report, where proper congruence classes can cover the input lattice.

The distinction is concrete. On three unmarked terms, `Identify(0,1)` passes P.9 because the target has two terms. Take only diagonal excesses `(e_00,e_11,e_22)=(0,1,2)`, with all other entries zero. The identified output has diagonal `(1,2)`. Rename the source by swapping terms `0` and `2`, then apply the **same** `Identify(0,1)`: its output diagonal is `(3,0)`. Gradients preserve loops, and no target permutation changes the unordered loop multiset `{1,2}` into `{0,3}`. Both raw inputs are legal and have the same source result mark, so the fixed literal word does **not** descend to that full mark. Conversely, AddTerm followed by Identify of the new last term with old term `0` returns every original raw pair coordinate and filling on any nonempty carrier. That word acts as identity on full result marks despite its non-descending AddTerm prefix when at least two old terms are present. The complete criterion remains finite and source-specific. P.11 constructs the separate raw-layer transport of **renamed descriptors**; neither result supplies cross-system refinements, a global site or higher fillers.

P.11 Transported five-edit model words commute with renaming

P.10 holds the edit descriptors literally fixed while renaming the input. For a different, useful square, rename the descriptors *with* the input. Let `C=(X,r,s,F)` be an actual five-edit model count/filling state on the ordered carrier `X={0,…,n−1}`. A permutation `p`$\in$`S_X` acts by simultaneous transport of both directed count tables and every oriented filling; it regenerates the source's canonical labels `t0,…,t(n−1)` rather than permuting those literal strings. Write the result `pC`. The following table defines the transported edit `a^p` and a target-carrier permutation `p_a` for a well-formed edit descriptor `a`, including one whose count or face guard later refuses.

Edit `a`	Transported descriptor `a^p`	Target permutation `p_a`
`AddTerm`	`AddTerm` with the same new last term	Extend `p` by fixing that new term.
`AddAssertion(R/S,x,y)`	Same relation at `(p x,p y)`	`p`
`Remove(R/S,x,y)`	Same relation at `(p x,p y)`	`p`
`Fill(t)`	`Fill(cyc(p t))`, rotating the least vertex first without reversing orientation	`p`
`Identify(x,y)`, `x<y`	`Identify(u,v)`, where `u=min(p x,p y)` and `v=max(p x,p y)`	The unique `p_a` satisfying `p_a f_(x,y)=f_(u,v)p`.

Here `f_(x,y)` is the actual canonical quotient map that identifies `y` with `x` and shifts labels above `y` down by one. For Identify, the displayed commuting equation defines a **bijection** on target carriers: `p` carries exactly the merged fiber `{x,y}` to `{u,v}` and singleton fibers to singleton fibers, so `f_(u,v)p` is constant on each fiber of the surjective `f_(x,y)`. The induced map is well-defined and bijective, and surjectivity makes it unique. The order of the two renamed endpoints only chooses the source's canonical surviving label; it does not erase the target permutation.

One-step naturality. On the raw pair/filling layer, admission and the output commute exactly:

$$a \text{ enabled on } C \iff a^p \text{ enabled on } pC, \qquad a^p(pC) = p_a(aC).$$

AddTerm zero-pads both tables and leaves old faces, and its new vertex is fixed by the extended permutation. Assertion edits transport their unit coordinate because `P_p E_(x,y)=E_(p x,p y)`; a Remove's positive count guard is the same integer at the transported coordinate. Fill's absent-face guard and oriented cyclic canonicalization commute with renaming. For Identify, pair aggregation obeys `T_(f_(u,v)) P_p=P_(p_a) T_(f_(x,y))` by reindexing each fiber sum. The same commuting square carries filled triples: distinctness, degeneracy after merging, cyclic orientation and duplicate removal are preserved by the bijective `p_a`. These checks use both count tables; an excess-only argument would not prove Remove admission.

For a legal word `w=a_k⋯a_1`, transport each descriptor using the permutation at its **current** carrier, starting with `p_0=p`. The table produces `a_i^(p_(i−1))` and `p_i`; call the resulting word `w^p`. Induction over the one-step square establishes

$$w^p(pC) = p_k(wC).$$

The transported word is legal on `pC`; transporting it back by $\mathtt{p}^{-1}$ recovers `w`, so the same statement holds in both directions. A refused one-step guard has the corresponding refusal under renaming, but an invalid descriptor does not acquire an output by this theorem.

The rule is coherent under both compositions. For consecutive words `w` then `v`, transport of the concatenation is `w^p` followed by `v^(p_k)`. For initial renamings `p` then `q`, transporting first by `p` and then by `q` gives the same literal transported descriptors as transporting once by `qp`, and the terminal permutations compose. At Identify this follows from the uniqueness of the induced target permutation; at the other edits it is coordinate transport or the extension that fixes the new

vertex. Thus identities and finite word composition give strict, replayable renaming squares on this **selected raw source layer**.

The raw equality implies corresponding affine excess conjugacy, `A_(w^p)P_p=P_(p_k)A_w` and `b_-(w^p)=P_(p_k)b_w`, and transports marked cycle values with their oriented faces. It therefore carries a passing P.8 balance test or P.9 gradient-descent test to the transported word in the renamed context. It does not make a **fixed literal** word act on every full result mark: the three-term Identify failure in P.10 keeps `Identify(0,1)` fixed, whereas its transported descriptor after swapping source terms `0` and `2` is `Identify(1,2)`. Neither exact renaming square supplies a gradient quotient map when P.3 fails, nor does it supply institutional permission or standing. P.12 checks the selected source prices and complete ordered runs separately. Cross-system source refinements, covering restrictions and higher fillers remain O02/O03 obligations.

P.12 Five-edit model prices and complete runs under descriptor transport

The declared five-edit model native execution adapter retains two count tables, canonical term labels, all intermediate raw states and the ordered edit occurrences. Its two state quantities and its run price have the exact definitions

$$S(C) = n + \sum_{x,y}(r_{xy} + s_{xy}), \qquad P(C) = n + \sum_{x,y}|r_{xy} - s_{xy}|,$$

$$A(R) = (\#\,\text{AddTerm}, \#\,\text{AddAssertion}).$$

These are respectively `source_size`, `plain_state_cost` and the two-component `addition_price`. `Fill`, `Identify` and `Remove` each contribute `(0,0)` to the latter, although each remains an occurrence. For a legal run `R=(C_0,a_1,C_1,…,a_k,C_k)`, use P.11 to transport every state and its selected descriptor with successive permutations `p_i`. The result `R^p` is a legal run on `pC_0`, and

$$|R^p| = |R| = k, \qquad A(R^p) = A(R), \qquad S(p_iC_i) = S(C_i), \quad P(p_iC_i) = P(C_i)\ (0 \le i \le k).$$

The proof is coordinatewise: each `p_i` bijects the directed pair coordinates and preserves the carrier size; hence both sums are unchanged. The transported descriptor has the same event kind and relation, so its addition-price increment is unchanged. P.11 gives the exact equality of every transported intermediate raw state. Transport by the inverse initial permutation recovers the original run, with all occurrences in the original order. Thus transport is a bijection between **complete legal runs** starting at `C_0` and those starting at `pC_0`; it does not collapse different runs that happen to have equal endpoints or result marks.

The native run splice `R.then(T)` requires complete equality of the joining raw state and adds the two price pairs. Since renaming is bijective on complete states, a defined splice transports exactly as `(R.then(T))^p=R^p.then(T^(p_k))`; a mismatched join remains a mismatch. At one state, the enabled **set** also transports exactly:

$$\text{Enabled}(pC) = \{a^p : a \in \text{Enabled}(C)\}.$$

This is P.11's admission equivalence in both directions. It does not preserve the source's lexicographic **list order** of enabled events: swapping terms `1` and `2` on a three-term state exchanges the order of the distinct assertion descriptors `(0,1)` and `(0,2)`. Selected run order is preserved because its occurrences are transported positionwise, not regenerated from that enumeration list.

The limits are witnessed on the same carrier. On two terms, zero excess and the gradient with `e_01=1`, `e_10=-1` have the same fixed-writing result class and no fillings, but their plain-state costs are respectively `2` and `4`; positive/negative parts give legal raw count tables. Thus `P` does not descend to that result class. On any nonempty carrier, identity replay and the legal word AddTerm followed by Identify of the new term with old term `0` have the same complete raw endpoint, but addition prices `(0,0)` and `(1,0)` and different occurrence counts. Price and history are not functions of the endpoint even when its raw tables agree.

These equalities close price and history transport for this declared renaming profile. They do not make `P(C)` constant on gradient-result classes, make a fixed literal word equivariant, prove equivalence to every original executable source run, or transport institutional authorization. Those are separate O02/O03/O05 obligations.

P.13 Permutation cubes for initially addressed permitted edits

P.11 transports one word under a renaming; P.12 retains that word's ordered history. Here the question is different: when may a fixed collection of source edits be scheduled in different orders? The result concerns the declared native five-edit model transition semantics on constructible finite carriers. It does not treat an edit addressed to a term created *during* the schedule.

Let `C=(X,r,s,F)` be one valid raw state, with `X={0,...,n-1}`, two nonnegative directed count tables, and a set of oriented cyclic faces. Let `M` be a finite **tagged multiset** of well-formed descriptors, each addressed to this initial `X`: `AddTerm`, `AddAssertion`, `Remove`, `Fill`, or `Identify`. Tags distinguish planned occurrences of an identical descriptor; they do not change the native event type. On any prefix, residualize each still-pending descriptor by sending its original vertices through the prefix's current quotient map. An Identify uses the smaller of the two current labels as its survivor; a nondegenerate Fill is cyclically rotated to put its least vertex first, without reversing orientation. `AddTerm` appends a fresh zero row and column in each table. No residual may silently become a no-op: a collapsed Identify, degenerate Fill, already-filled face, or empty Remove is refused.

Write `I` for the multigraph on `X` whose tagged edges are the Identify occurrences. Let `~` be the equivalence relation generated by those edges, and `q:X -> X/~` the quotient with classes labelled in increasing order of their least original vertex. For a face `t`, write `q#t` for its cyclically canonical image if its three images are distinct, and for `none` otherwise; `q#F` discards `none` and deduplicates. If `a_R(x,y)` and `m_R(x,y)` count AddAssertion and Remove occurrences at an original coordinate in relation `R`, define analogous counts for `S`.

Raw endpoint theorem. Any two *legal complete* orderings of `M` have the same final five-edit model state and the same addition-price pair, although their ordered runs remain distinct. The final carrier has `|X/~| + #AddTerm` canonical labels. On its original-class block, its tables are

$$r_{\mathrm{fin}} = q_* r + \sum_{x,y} (a_R(x,y) - m_R(x,y)) E_{q(x),q(y)},$$
$$s_{\mathrm{fin}} = q_* s + \sum_{x,y} (a_S(x,y) - m_S(x,y)) E_{q(x),q(y)}.$$

All coordinates incident to newly added terms are zero. Its face set is `q#(F union {t : Fill(t) occurs in M})`, and its addition price is `(#AddTerm, #AddAssertion)`. This statement does not assert that any complete ordering exists. It also does not identify the two run objects: their edit occurrences and intermediate raw states retain their actual orders.

All-order legality and cube criterion. Every permutation of the tagged occurrences is a legal complete run if and only if all three conditions hold:

1. `I` is a forest as a multigraph; a repeated Identify edge is a two-edge cycle.
2. At every original directed coordinate and separately in each relation, the initial count is at least its total number of planned Remove occurrences: `r_xy >= m_R(x,y)` and `s_xy >= m_S(x,y)`.
3. Every planned Fill has a nondegenerate `q#t`; these final oriented faces are pairwise distinct and none belongs to `q#F`.

When these conditions hold, every subset `S` of the tagged occurrences has a well-defined raw endpoint `C_S` independent of its order. For each pending tag `a`, its residual gives an enabled edge `C_S -> C_(S union {a})`. Consequently the subset lattice carries an **endpoint-coherent Boolean-cube skeleton** of actual raw states and edits: each square and every higher-dimensional boundary reaches the same terminal raw state. The diagram records the boundary routes as distinct runs. Endpoint equality supplies replayable boundary data; it does not by itself define a 2-cell between runs, a 3-cell between squares, or a higher-category filler. Those structures still require a declared comparison relation and composition laws.

Proof. For any finite vertex map `f`, pair-table pushforward is linear and `(g f)_*=g_* f_*`. Each Identify step aggregates both tables by this pushforward; each AddTerm uses the inclusion pushforward; an assertion edit adds or subtracts one basis table at its residual coordinate. Sending that unit through every later quotient yields `E_(q(x),q(y))`. Thus the displayed table formulas depend only on `M`, not its order. The source's Identify operation removes the higher current label, so induction shows that current classes are labelled by increasing minimum original vertex. Fresh AddTerm vertices stay after the original classes because no descriptor in `M` names one. Oriented-face transport satisfies `(g f)#=g# f#`: once a face becomes degenerate it cannot become distinct again, and cyclic rotation commutes with taking the nondegenerate image. Fill adds a singleton face, while Identify transports and deduplicates the set. This establishes the face formula even when a Fill later collapses or merges with another face. The native addition price counts kinds, so it is also order-independent.

For necessity of the three conditions, schedule all Identifies first. A cycle's last edge has endpoints already joined and is refused. Schedule all Removes before any Identify or AddAssertion: repeated subtraction at each original coordinate succeeds only with the stated initial supply. Schedule all Identifies before all Fills: each Fill must survive `q`, avoid `q#F`, and avoid every earlier Fill image, in any ordering. For sufficiency, a forest edge always joins distinct components of every preceding edge subset. At any intermediate quotient coordinate, the remaining count is the sum of original supplies in that fiber, plus preceding additions, minus preceding removals; the per-coordinate supply inequalities make the next Remove positive. A face that is nondegenerate and distinct after the *full* quotient was nondegenerate and distinct after every partial quotient; otherwise later identifications could not separate it. Hence every Fill guard succeeds. AddTerm and AddAssertion are unguarded on their well-formed residual addresses. The same argument applies to each subset and establishes the cubical assertion. This is an arbitrary-finite proof; bounded replay is only a check of its implementation-facing cases.

The obstructions are actual, not inferred from an endpoint quotient. Three Identifies on the edges

of a triangle are individually enabled but no complete ordering can execute the last edge. On one term with `r_00=1`, two tagged `Remove(R,0,0)` occurrences exhaust their supply after the first. On three terms, `Fill(0,1,2)` followed by `Identify(0,1)` is legal, but reversing them makes the Fill degenerate. On four terms with `F={cyc(0,2,3)}`, `Fill(1,2,3)` is initially enabled; after `Identify(0,1)`, its image collides with the existing face and is refused. The two cyclic orientations remain distinct unless an actual quotient identifies their oriented images.

For a two-occurrence fork, the criterion specializes to exact strict-swap guards: an Identify/Fill pair swaps precisely when the filled face survives the identification and avoids the transported old faces; two Removes at one relation/coordinate require count at least two; two identical Fills or Identifies do not swap. Other pairs of individually enabled initial-carrier descriptors satisfy the criterion. This includes overlapping but distinct Identify edges: their two-edge graph is still a forest. The theorem itself does not cover a later edit that first names an AddTerm-created vertex; P.14 supplies an additional named-token presentation for that case. Selected gradient-result quotient descent, arbitrary restrictions/covers, permission and institutional standing remain separate. Resource and allocation refusals are outside these mathematical guards.

P.14 Causal names and exact guards for dynamically created terms

P.13 compares edits addressed only to the starting carrier. A literal numeric address in this model is not a stable name for a newly created term when two `AddTerm` occurrences change order. Supply that missing comparison datum explicitly, without modifying the native five-edit kernel.

Fix an initial valid raw state `C=(X,r,s,F)`. Let `E` be a finite set of tagged edit occurrences with an acyclic precedence relation. Each `AddTerm` occurrence births one unique token; put `U=X disjoint-union B` for the initial and birth tokens. Every other descriptor names tokens in `U`, not current numeric labels. `Identify` names an unordered token pair; `Fill` names an oriented cyclic token triple. Require the birth of each named token to precede its use. Additional declared precedence edges are allowed. A linear extension is executed natively by maintaining a map from every born token to its current five-edit model label: `AddTerm` appends its token, `Identify` sends the lesser and greater current labels as native `(x,y)` with `x<y` and canonically renumbers, and `Fill` cyclically rotates its three current labels so the least is first. `Remove` and `AddAssertion` use their named tokens' current labels in order. A token remains a valid name after its class is merged. A reference before birth is an invalid presentation, not a negative mathematical instance.

For an order ideal `J` of the event poset, let `U_J` be `X` plus the tokens born in `J`. Let `I_J` be its Identify-edge multigraph and `q_J` the quotient of `U_J` by those edges. Use a fixed total order on token names to label its classes by their least token; this is a **symbolic** writing, which may differ from the native writing reached by a chosen linear extension. Define two formal count tables and one oriented face set on these classes:

$$R_J = (q_J)_* r + \sum_{\substack{a\in J \\ a=\mathrm{AddAssertion}(R,u,v)}} E_{q_J u, q_J v} - \sum_{\substack{a\in J \\ a=\mathrm{Remove}(R,u,v)}} E_{q_J u, q_J v},$$

$$S_J = (q_J)_* s + \sum_{\substack{a\in J \\ a=\mathrm{AddAssertion}(S,u,v)}} E_{q_J u, q_J v} - \sum_{\substack{a\in J \\ a=\mathrm{Remove}(S,u,v)}} E_{q_J u, q_J v},$$

$$F_J = (q_J)_{\#} \left(F \cup \{t : \mathrm{Fill}(t) \in J\}\right).$$

The initial tables are extended by zero on birth-token coordinates before pushforward. `q#` cyclically normalizes nondegenerate oriented triples and discards degenerate ones. These formulas can have negative entries for an unreachable ideal; calling them a formal profile does not admit an invalid five-edit model state.

Causal endpoint theorem. Every *legal* linear extension of `E` ends in the symbolic profile `(R_E,S_E,F_E)` up to the unique permutation that maps each final native label to its token-equivalence class. Thus any two legal linear extensions have isomorphic complete raw endpoints. Their occurrence counts and AddTerm/AddAssertion addition-price pairs agree; P.12 then gives equal endpoint source size and plain-state cost and transports their future enabled **sets**. Their ordered histories, intermediate prices, and lexicographic enabled-event lists need not agree. The theorem does not assert that a legal linear extension exists.

All-linear-extension criterion. Assume the Identify-edge multigraph on all of `U` is a forest. Then *every* linear extension is legal if and only if, for every order ideal `J` and every occurrence `a` whose predecessors lie in `J` but which is not in `J`, the following applicable guard holds in the formal profile of `J`:

Pending kind	Exact ideal-local guard
`Remove(R/S,u,v)`	`R_J(q_Ju,q_Jv)>0` or `S_J(q_Ju,q_Jv)>0`, in the named relation.
`Fill(u,v,w)`	The three `q_J` images are distinct and their oriented cyclic face is absent from `F_J`.
`Identify(u,v)`	The two images differ; this follows from the forest assumption.
`AddTerm`, `AddAssertion`	Enabled once the declared birth precedences make their names available.

The test ranges over the finite lattice of order ideals, rather than a sample of linear extensions. It is a finite exact decision procedure: enumerate ideals, construct their quotient and the displayed integer tables/faces, and replay the applicable guard. It may be exponential in `|E|`; no claim of an efficient generic solver is made. If the forest condition fails, no complete native execution can perform all its Identify occurrences. If a guard fails, choose a smallest failing ideal. Its proper ideal prefixes pass, so a linear extension reaching that ideal gives a concrete native refusal at the named pending occurrence. Refusal is never counted as a successful no-op.

Proof. In any legal prefix, the token-to-native-label map has fibers exactly the connected components of the performed Identify edges. Indeed, an `AddTerm` creates a singleton fiber, and an enabled Identify unions two different fibers; no other edit changes them. It therefore induces a unique bijection from the symbolic `q_J` classes to current native labels. By P.11's raw transport, the two writings have the same guards and results under that bijection. Pair-table pushforward is functorial, and an assertion update at tokens `(u,v)` contributes exactly one signed unit at their current class pair, then at their final class pair under later identifications. Fill and Identify act on oriented faces by functorial nondegenerate image and set union. Induction on the prefix establishes the three symbolic formulas for every legal prefix, hence the endpoint and price statements.

Every order ideal is the prefix set of some linear extension of a finite poset. If all complete extensions are legal, its actual prefix realizes the symbolic profile, and every currently available next occurrence must pass the displayed guard. A cycle of Identify edges has a last edge in every linear extension;

its endpoints are already joined, so a legal complete extension is unrealizable. Conversely assume the forest and all ideal-local guards. Starting at the empty ideal, induction makes every available next event legal and identifies its native output with the symbolic profile of the enlarged ideal. Thus every linear extension succeeds. A smallest bad ideal in a failed test has all proper prefixes admitted, giving the claimed replayable negative witness. Adjacent swaps of incomparable events connect any two linear extensions; the same ideal endpoints make their squares commute on raw states up to the uniquely induced token permutation. This is endpoint coherence, not an unprovided 2-cell between the run objects.

The added names matter. Start with one term and zero tables, birth tokens `b` and `c` in either order, then add one `R` assertion at `(b,c)`. Both orders are legal and their final tables agree after aligning the token classes; the correct native descriptor is `(1,2)` in one order and `(2,1)` in the other. Holding one literal numeric descriptor fixed would ask a different question. A guard obstruction is equally concrete: from one initial term and zero tables, birth `b`, then plan AddAssertion and Remove in `R` at `(0,b)` with no precedence between those two. The order birth, Remove, AddAssertion refuses. Adding AddAssertion-before-Remove precedence admits its linear extension, and the ideal-local table test sees both cases. P.13 is recovered when no later descriptor names a birth token and the event poset is discrete: its forest, initial Remove-supply and final Fill-survival conditions are the simplified all-ideal guards.

Stable token names and event precedence are **additional presentation data**. Any one native run can be annotated noncanonically by tagging its births and choosing token representatives of the current classes, but the native enum does not itself assert that two runs share one such presentation. P.14 therefore advances the selected O03 dynamic carrier/guard question without proving a global source quotient functor, compatible restrictions over covers, higher cells on histories, or institutional authorization.

The retained combined execution verifier now checks a bounded supplied P.14 presentation against the actual `omega_wdp` five-edit model transitions through a thin Rust JSON adapter. It binds the initial raw tables and faces, occurrence tags, birth tokens, precedence, all replayed native states, token-class alignments and protected prices. Admission is capped at eight events and 10,000 linear extensions. A family is accepted only after every extension actually executes. The retained fixtures include all five edit kinds across 24 legal schedules; a valid poset with an obstructed `Remove` is refused. Binary and source hashes are checked before and after replay. This is native execution evidence for the supplied finite family, not a formal verification of the native implementation or an exhaustive test proof.

P.15 Exact price descent on the causal ideal lattice

P.14 fixes the raw state of each order ideal, up to its token-class writing. It does **not** make every price accumulated along a schedule independent of that schedule. Here is an exact test for an additive price. Fix one P.14 family whose *every* linear extension is legal, and write `L=Idl(E)` for its finite lattice of event order ideals. For each cover `J -> J+e` choose a value `w(J,e)` in an abelian group `A`. An actual source-derived price must first be invariant under the token-to-native relabeling of P.14, so that it really is a function of the ideal and event, not of an unrecorded prefix history. The theorem below is algebraic for any `A`; the retained checker handles integer-vector prices `A=Z^d` only.

For two distinct events `e,f` both available after `J`, all four edges of their diamond exist. Define its **price curl**

$$\kappa_w(J; e, f) = w(J, e) + w(J + e, f) - w(J, f) - w(J + f, e).$$

Exact price theorem. The following conditions are equivalent:

1. Every two complete linear extensions from the empty ideal to `E` have the same accumulated `w` price.
2. Every elementary diamond has `kappa_w(J;e,f)=0`.
3. There is a unique `Phi:L -> A` with `Phi(empty)=0` and `w(J,e)=Phi(J+e)-Phi(J)` on every cover.

When they hold, *every* pair of saturated routes from any `J` to any containing `K` have the same price `Phi(K)-Phi(J)`. Thus the degree-one square-closed cochains on this declared ideal lattice are exact. This is a finite schedule statement, not a cohomology calculation for a proposed global source site.

Proof. A potential telescopes along every route, proving (3) implies (1) and (2). To obtain (2) from (1), choose any legal prefix reaching `J` and a common continuation from `J+e+f` to `E`. The two full runs differing only by `e,f` versus `f,e` have price difference exactly `kappa_w(J;e,f)`; by (1) it is zero. For (2) implying (3), any two linear extensions of an ideal are connected by adjacent swaps of incomparable events. To see this, move the first event of one extension leftward in the other. Each event it passes is incomparable with it, because both orders are linear extensions; then induct on the suffix. A swap changes the path price by precisely one diamond curl. Zero curls therefore make the sum from the empty ideal to `J` independent of the extension. Define that sum as `Phi(J)`. Appending one cover gives its required difference. Every ideal is reachable, so the root normalization makes `Phi` unique. The same swap argument works between any two fixed ideals.

For the native five-edit model addition-price pair, `w(J,AddTerm)=(1,0)`, `w(J,AddAssertion)=(0,1)`, and the other three kinds have `(0,0)`. The curls vanish; `Phi(J)` is the pair of event-kind counts in `J`. Likewise the *difference* `source_size(C_(J+e))-source_size(C_J)` is exact with potential `source_size(C_J)-source_size(C_empty)`, since P.12 makes the state quantity invariant under relabeling. These are different price definitions: an arbitrary fee evaluated *at* a state need not be a state difference.

For a concrete obstruction, start with one five-edit model term, zero counts and no fillings. Let `t=AddTerm` and `a=AddAssertion(R,0,0)` be incomparable. Both orders are legal and have the same raw endpoint. Charge zero for `t`; charge `source_size(C_J)` for `a` when executed after ideal `J`. The route `t,a` charges `2`, while `a,t` charges `1`, so `kappa_w(empty;t,a)=1`. The two complete orders are an exact negative certificate for schedule-independent price. Endpoint equality cannot silently replace this protected accumulated-cost question.

The integer ideal-price checker enumerates all ideals and their covers, checks every diamond in `Z^d`, and returns either the root-normalized potential or the first nonzero curl together with two complete topological orders and their different exact totals. Missing edges, dimension mismatches and cyclic precedence are malformed input, not negative mathematical instances. An independent schedule-enumeration test compares its decisions with all complete orders on 256 deterministic four-event DAG/price cases, plus the two displayed positive and negative controls. The checker does not itself verify P.14's native admissibility premise or source-price binding.

The combined P.14 verifier supplies those premises for its admitted bounded profile. It derives the two native addition prices, source-size difference and plain-state-cost difference from actual replay, verifies their ideal bindings, and invokes the price checker on those exact values. Just submitting an independently valid potential is insufficient.

If a curl is nonzero, choosing one path to each ideal still defines a candidate potential and its exact edge differences. Replacing `w` by those differences is a **changed price specification**, not a proof that the original price descends. Evaluation of the change and any authorization to use it remain separate. Arbitrary history-sensitive functionals, guarded subgraphs missing ideal diamonds, and higher cells between raw histories are outside this theorem; GEN-WDP-O03 remains open at its full scope.

P.16 Causal swap certificates and a chosen thin history 2-category

P.14 gives endpoint-coherent diamonds; it does not assert that a five-edit model history has a native 2-morphism. Here is one explicit, scoped construction which retains the ordered 1-morphisms and shows exactly where a new higher identification is chosen. Fix the same finite tagged event poset `E`, stable birth tokens, and a P.14 certificate that **every** linear extension is legal. For each ideal `J`, use its token-class normal form `C_J` and keep the ideal label even if two different ideals happen to have identical packets.

On a cover `J -> J+e`, translate `e` to the current native descriptor, execute that five-edit model step, and record the unique *token-induced* bijection from its native output labels to the normal writing `C_(J+e)`. The bijection is unique because it must send each native label to its named token class; it is not a claim that the underlying packet has no other automorphisms. P.11 transports the next descriptor through this bijection. A **framed history** from `J` to `K` is a saturated path of these actual one-step executions with all intermediate packets, occurrence tags and alignment maps retained. Concatenation splices at the common framed boundary and concatenates lists. P.12's exact-state splice and P.11's renaming law make this associative, with the empty list as identity. The resulting 1-category has distinct arrows for distinct event orders, even when their endpoints and addition prices agree. An alignment is comparison data, **not** a sixth native edit: transporting later descriptors back through the accumulated alignments recovers one literal native run, while the framed form keeps the same boundaries for composition.

For two different events `e,f` available after `J`, define an **elementary swap certificate** with the two native paths `e;f` and `f;e`, their four ideal-labelled boundary packets, the translated descriptors and the two token-induced maps to `C_(J+e+f)`. P.14 establishes both paths legal and both endpoint maps correct. Thus the verification record has two genuinely different ordered 1-boundaries and one checked common framed endpoint. Prefix and suffix framed histories can be pasted onto it; a list of such pasted verification records is a proof-relevant swap chain. Reversing the list reverses the comparison, and concatenating two lists composes comparisons vertically. No equality of distinct proof-relevant swap chains is inferred from their endpoint alone.

Connectivity and optimal verification record theorem. Let `h,h'` be any two framed histories from `J` to `K`. Count unordered pairs of occurrences in `K\J` whose order is opposite in `h` and `h'`; call this integer `inv(h,h')`. Comparable events cannot contribute. There is a swap chain from `h` to `h'` of exactly `inv(h,h')` elementary verification records, and no shorter chain. To construct it, scan `h'` from left to right. Move its next wanted event leftward through the current word until it reaches the wanted position. Each crossed event lies after it in `h'`, hence is incomparable with it; the two are simultaneously available after their common prefix ideal. Their P.14 diamond supplies the required actual native swap certificate. Each swap fixes exactly one inverted pair and no other pair order. Conversely an adjacent swap changes only one pair order, so every chain needs at least `inv(h,h')` steps. This also establishes that the verification record graph of histories between any fixed pair of ideals is connected.

The minimum-swap checker validates two linear extensions and emits precisely those prefix ideals, event pairs and positions. An independent BFS test replays every emitted diamond and compares its length with shortest-path search for two selected order pairs in each of 64 four-event forward DAGs; it also checks a nonempty start ideal and invalid schedules. The checker verifies the poset and swap layer only. It does not execute native five-edit model packets; the actual source-step square is supplied by P.14's separate admission and transport proof.

There is a **declared thin completion**: for any two framed histories with the same ideal-labelled source and target, identify *all* swap chains with those boundaries into one invertible 2-cell. Such a cell exists by the connectivity theorem. The hom-category is then the indiscrete groupoid on its distinct history 1-arrows. Horizontal composition is induced by concatenation and whiskering, and vertical composition is the unique cell between its boundaries. Identity, associativity and middle-four interchange hold strictly because both sides are parallel cells in a thin hom-category. This constructs a strict 2-category for the selected causal family. Thinness is an **extra quotient of verification records**, not a property of native five-edit model or a proof that distinct histories are equal as 1-arrows.

P.15 gives the exact price obstruction to this completion. Send every object to the one object of the discrete 2-category whose 1-arrows are an abelian price group `A` under addition, and send a history to its accumulated ideal-local price. This extends to a strict 2-functor precisely when every diamond curl vanishes: the two boundaries of each elementary 2-cell must have the same target price, and P.15 establishes that condition sufficient for all histories. Native five-edit model addition prices satisfy it. The legal AddTerm/assertion square charged by current source size in P.15 has curl one, so no price-preserving 2-functor sends that square to an identity 2-cell. The unpriced swap certificate still exists; an evaluator protecting that fee must refuse the proposed price-preserving comparison.

For three independent events, the six orders form a swap hexagon. Its native endpoint alignments agree by P.14, and its two parallel swap chains become equal **in the declared thin completion**. They remain different proof-relevant verification records before that quotient. No nontrivial 3-cell between those verification records, Kan filler, compatible restriction over arbitrary source covers, result-mark quotient functor, or institutionally authorized history identification follows. GEN-WDP-O03 therefore remains open beyond this selected source-specific 2-category.

Within the bounded combined profile, swap and local-relation replay now binds each swap to the two actual native histories, their common prefix, framed two-step boundary, token alignments and protected prices. The older poset-only checker remains combinatorial; native evidence is credited only to this combined path.

P.17 Local 3-witnesses between causal swap proofs

P.16 made all parallel swap chains equal by a declared thin quotient. For the same **selected** P.14 all-order-legal tagged event poset, that equality has a finite local presentation. Retain each framed actual run as a distinct 1-arrow and each elementary swap certificate as a generating 2-comparison. A swap proof from one history `h` to another is a word in adjacent-position swaps `s_i`, with every intermediate event order a linear extension. Its 2-boundary is the pair of framed histories, not just their shared packet.

Adjoin the following invertible **local 3-witnesses**, with arbitrary admissible swap-proof prefixes and suffixes whiskered onto either side:

1. A backtrack `s_i s_i <-> id`, where the two events at positions `i,i+1` are incomparable at the insertion site. Deletion from an already valid proof automatically satisfies this condition.
2. A distant square `s_i s_j <-> s_j s_i` for `|i-j|>1`. These swaps touch disjoint positions and retain the same two incomparable event pairs.
3. A braid hexagon `s_i s_(i+1) s_i <-> s_(i+1) s_i s_(i+1)`. A valid side swaps each of the three pairs among its three tagged events. All three pairs must therefore be incomparable, so the other side is valid too.

Each boundary order of a local witness has its P.16 framed native run and token-induced alignment. The square and hexagon are comparisons of **swap proofs**, not new native permitted edits. In particular, a square here is not the ideal-local price diamond of P.15, though its constituent swaps have such diamonds as verification records.

Local coherence theorem. Any two valid swap proofs with the same framed source history and target history are joined by a finite sequence of these local 3-witnesses. Number the event positions in the common source order. A swap proof gives a word in the adjacent generators of the symmetric group and acts on that order to produce its target; two proofs with the same boundary therefore represent the same permutation. The Coxeter word property, Theorem 2.2, reduces each word using braid moves and deletion of `s_i s_i`, and connects any two reduced expressions by braid moves. In this symmetric group the braid moves are exactly the distant square and the three-term hexagon above. The restriction to legal event orders does not break the argument: deleting a valid backtrack leaves a valid path; exchanging disjoint swaps preserves both incomparable pairs; and the three-term move is valid on its alternate route because its original route already swapped all three pairs. Thus both reductions, their reduced-word connection, and the reverse of the second reduction remain entirely inside the graph of linear extensions. P.14 supplies actual native execution and endpoint alignment at each intermediate order. The result includes the empty and single-event intervals, where the swap proof is necessarily empty.

A **3-certificate** is the finite list of such local replacements, naming the swap-word offset and, for an inserted backtrack, its adjacent position. Replay checks every intermediate linear extension and the fixed two boundary histories. This makes the P.16 thin 2-cell the quotient of proof-relevant swap chains by explicitly witnessed local relations, rather than an unexplained identification. For a three-event antichain, the two three-swap routes from one order to its reverse are distinct 2-proofs and the braid hexagon is their first nontrivial 3-witness. P.15 zero diamond curls remain exactly the condition for an ideal-local additive price to respect the generating 2-comparisons; the 3-witnesses introduce no new price equality between different boundaries.

The executable 3-certificate checker checks backtrack, distant-square and braid replacements, including whiskered replacements at a declared swap-word offset. Every intermediate order must be legal in the bound native family and both framed boundary histories must remain fixed. Retained tests refuse a comparable-event braid, an invalid offset, changed boundaries and corrupted intermediate states. Distinct replacement lists retain distinct certificate identities. The replay closes the selected certificate problem; the general existence argument above remains a written theorem, not a consequence of enumeration.

These 3-witnesses are **chosen presentation data**, not canonical native five-edit model operations. The theorem says parallel 2-proofs are connected; it does not identify different 3-certificates, furnish 4-cells or a coherent tricategory, define restrictions over arbitrary covers, extend to histories with dynamic carrier changes outside P.14, or grant authority. GEN-WDP-O03 remains open at those broader scopes.

P.18 Guarded histories and the componentwise price criterion

P.14–P.17 assumed that **every** linear extension of the declared causal event family is legal. Drop that assumption, but retain stable event tokens, birth-before-use precedences, the actual five-edit model guards, and at least one legal complete execution. P.14's endpoint formula still identifies the complete raw endpoint of *each legal* execution up to token-class writing. It does not say that those executions can be connected by legal swaps.

Let $\mathcal{H}$ be the finite set of legal complete linear extensions. Build a graph Γ on $\mathcal{H}$: an edge exchanges two adjacent, incomparable tagged events when both resulting complete executions are legal. Such an edge has a common prefix ideal J, the two orders e, f and f, e, and a common suffix. Both two-step prefixes end at the same formal state of $J + e + f$ by P.14, so the suffix has the same guard outcomes after token alignment. A proposed swap for which one two-step prefix refuses is not an edge.

Let A be an abelian group and let $w(J, e) \in A$ be an ideal-local price on each legal cover $J \to J + e$, invariant under the token-to-native relabeling as in P.15. The total $W(h)$ is its sum along a legal complete history. For a realized edge of Γ, put

$$\kappa_w(J; e, f) = w(J, e) + w(J + e, f) - w(J, f) - w(J + f, e).$$

Guarded componentwise price theorem. The totals $W(h)$ are constant on each connected component of Γ if and only if the price curl vanishes on every realized edge. They are constant on *all* legal complete histories if and only if those curls vanish **and** one representative total from every connected component has the same value. In particular, zero local curls suffice without the second check when Γ is connected. P.15 is the all-order-legal special case, where adjacent swaps connect every pair of linear extensions. This theorem does not claim a potential on the entire ideal lattice: some ideals may have no legal prefix, and the legal-prefix graph need not be the full ideal lattice.

Proof. For one edge, the common prefix and suffix cancel in the difference of the two totals, leaving exactly $\kappa_w(J; e, f)$. Therefore constancy on a component forces every edge curl to vanish. Conversely, zero edge curls make the total unchanged along every edge path. Comparing one total per component is then both necessary and sufficient for global constancy. All statements use actual legal full histories, so a diamond that cannot be continued to a complete legal execution is not silently promoted to a complete-run price condition.

Actual five-edit-kernel counterexample to connectivity. Start with one term, zero `R` and `S` count tables, and no filled faces. Use only the native `AddAssertion(R,0,0)` and `Remove(R,0,0)` kinds, with five distinct tagged occurrences:

Tag	Descriptor	Declared predecessor
`p`	`AddAssertion(R,0,0)`	none
`x`	`Remove(R,0,0)`	none
`a`	`AddAssertion(R,0,0)`	`x`
`y`	`Remove(R,0,0)`	none
`b`	`AddAssertion(R,0,0)`	`y`

The edges `x<a` and `y<b` are permitted additional causal precedences in P.14. Initially `x` and `y` fail the source's positive-count guard; `a` and `b` are unavailable by precedence. Hence `p` must occur first,

leaving count one. If `x` occurs next, the count becomes zero, so `y` cannot occur before `a`; precedence forces the unique continuation `a,y,b`. The symmetric choice `y` forces `b,x,a`. Thus the **only** legal complete histories are

$$h_x = (p, x, a, y, b), \qquad h_y = (p, y, b, x, a).$$

Both reach the same raw one-term state with `R(0,0)=1`, zero `S`, no faces, and native addition-price pair `(0,3)`. Yet neither has a legal adjacent swap: swapping `p` with its following Remove fails the initial count guard; swapping a Remove with its paired Add violates precedence; swapping the middle Add with the other Remove attempts a Remove at count zero. Therefore Γ consists of two isolated vertices. Equal raw endpoints do **not** imply legal-swap connectivity.

To see why the price theorem needs its component comparison, set `w(J,e)=1` only for `J={p}, e=x`, and zero on every other legal cover. This is an ideal-local integer price. There is no realized legal swap diamond, so every local curl condition holds vacuously, but `W(h_x)=1` and `W(h_y)=0`. This does not contradict the native addition price `(0,3)`, which is equal on both histories.

The counterexample settles guarded **connectivity** negatively and classifies price descent componentwise. It does not decide **coherence** of parallel swap proofs within a connected component. P.17's Coxeter reductions used paths through all linear extensions; with illegal schedules removed, each backtrack, distant square and braid hexagon must be checked for legality before it is a 3-witness. Connectivity alone does not supply those checks or identify distinct 3-certificates. P.20 gives a sufficient guard condition under which every component is coherent, and states what the bounded native search found when the condition fails. GEN-WDP-O03 remains open at its broader source/site scope.

P.19 Admission, transport, and finite continuation

Checked repair and the admission, adapter-soundness and continuation protocol have different jobs in the same declared execution. Checked repair tests the proposed mathematical witness against its carrier and task. The protocol separates admission from evaluation, requires the supplied representation change to meet its transport obligations, and accounts for continuation. An authorization requirement remains independently supplied. The consolidation map locates the component sources and their original proof and executable evidence inside this package.

Admission precedence. Let $A_P(w)$ be the admission predicate for a candidate w on the original task P. Define an evaluation operation only on pairs equipped with evidence of $A_P(w)$. Then every successful evaluation carries an admitted candidate. This is immediate by projection from that dependent input. In the retained Lean model of the protocol, the input is instead a packet with a Boolean `admitted` field and a partial `measure` operation. Its theorem `Prime.admission_precedes_measurement` establishes the corresponding Boolean gating law. Neither formulation derives the admission predicate, its checker soundness, or authority from packet syntax.

Two independent transport obligations. Suppose an admitted change sends P to P', and let $\mathrm{Sol}(P)$ retain the original solution relation. A backward map $b : \mathrm{Sol}(P') \to \mathrm{Sol}(P)$ supplies sound decoding. A forward map $f : \mathrm{Sol}(P) \to \mathrm{Sol}(P')$ supplies solution coverage. Backward maps compose contravariantly and forward maps compose covariantly along a finite history, by ordinary function composition. The laws needed for original-witness recovery must be stated separately; these two maps alone need not be inverses. Soundness can hold vacuously when the target has no

solutions, so it cannot replace coverage. This is the same distinction used in §§20.6–20.8 and the finite constraint representation.

Continuation conjunction. Fix the nine declared continuation conditions $c_1, \dots, c_9$: an obligation, its witness, a declared next carrier, adapter soundness, detector adequacy, minimal lift, skeleton preservation, remaining budget, and absence of the declared terminal refusal. Set

$$S = \bigwedge_{i=1}^{7} c_i, \qquad B = c_8, \qquad N = c_9.$$

Then $\bigwedge_{i=1}^{9} c_i \iff S \wedge B \wedge N$, by associativity of conjunction. The retained theorem `Prime.cc9_implies_cc3` establishes one direction for this explicit grouping. An implementation whose three fields omit an original conjunct has only a weaker condition; matching field counts does not establish equivalence. Verifying the supplied predicates is also different from just receiving nine `true` values.

Finite operational stopping. Assume a shared initial budget $B_0 \geq 0$, positive cost floor $\delta > 0$, and executed operation costs $c_i \geq \delta$. Require every admitted prefix to satisfy $\sum_{i=1}^{n} c_i \leq B_0$, including operations executed after recursive re-entry. Then

$$n\delta \leq \sum_{i=1}^{n} c_i \leq B_0, \qquad n \leq \lfloor B_0/\delta \rfloor.$$

For positive rational costs, multiply by a common denominator to obtain the integer inequality. The retained `Prime.ops_le_floor_div` mechanizes its natural-number arithmetic form under the supplied cumulative bound. It does not establish that the concrete runtime charges every operation. A re-entry that resets the budget violates the prefix premise and escapes the theorem. Uncharged proposal rounds likewise require their own finite bound or charge.

Stopping at this bound yields resource exhaustion when the task has not been resolved. It supplies no impossibility theorem and no zero-residual witness. For example, a controller with budget two and unit-cost operations stops before a declared three-operation solution even though the task is solvable. Finite impossibility requires the exhaustive coverage certificate of §20.8; guaranteed resolution and its optimal cost require the additional arena and certificate hypotheses of Appendix S.

The common finite specification. A checked resolution therefore binds the original question, its admitted representation history, the submitted witness, the independent checker result, and the shared cost ledger. Local satisfaction may still leave an overlap disagreement; a quotient-zero diagnostic may still lack a realizable solving operation; finite protocol stopping may still leave an undetermined question. These are different obligations in one account, and none can discharge the others by renaming a terminal status. The finite Python application retains its narrower specification: its local authorization variant requires a caller-configured, task-bound grant and is not a proof of institutional standing or native Rust execution.

P.19.a Rational instruments, no-signalling and an algebraic CHSH certificate (DP-9, audited 2026-10-06) Statement. Let `X,Y,Z` be finite nonempty carriers, and let `A,B` be finite nonempty outcome sets.

- A *classical rational instrument* K:X→Y is a family of matrices K_a∈Q_{≥0}^{Y×X} with Σ_{a,y}K_a(y,x)=1 for every x. Write U(K)=Σ_aK_a for its outcome-forgetting map.
- For a second instrument L:Y→Z, put (L∘K)_{(a,b)}=L_bK_a.

Then:

1. L∘K is an instrument.
2. Composition is associative, up to reassociation of outcome tuples.
3. Forgetting is a homomorphism: U(L∘K)=U(L)U(K).
4. For total maps f:A→A′ and g:B→B′, coarse-graining the composite gives (Σ_{g(b)=b′}L_b)(Σ_{f(a)=a′}K_a) at (a′,b′). This needs L to be fixed independently of the first outcome. An adaptive continuation L^a forgets to Σ_{a,b}L^a_bK_a instead.
5. For a joint table p(a,b) and a complete local instrument K on B, put q(o,a,b′)=Σ_bK_o(b′,b)p(a,b). Summing over all o and b′ returns the A marginal of p.
6. The PR box p(a,b|x,y)=1/2 when a⊕b=xy, and 0 otherwise, satisfies every no-signalling equation. It has no global assignment table.
7. Let r be the algebraic number with r²=2 and 1<r<3/2. For every finite bipartite density and all Hermitian involutions A_x, B_y acting on separate tensor factors, the CHSH value S=E_00+E_01+E_10−E_11 satisfies −2r≤S≤2r. The bound is attained.
8. Every Hermitian expectation over Q(i) is rational. So no example over Q(i) attains S²=8.

Proof.

Items 1–4. Every entry of a product is a finite sum of nonnegative rationals. For fixed x,

$$\sum_{a,b,z,y} L_b(z,y)K_a(y,x) = \sum_{a,y} K_a(y,x)\Big[\sum_{b,z} L_b(z,y)\Big] = 1.$$

For a third instrument M, both bracketings give the (a,b,c) entry Σ_{z,y}M_c(w,z)L_b(z,y)K_a(y,x) at (w,x). Forgetting and coarse-graining interchange the same finite sums.

Item 5. Interchange sums: Σ_{o,b′}q=Σ_bp(a,b)Σ_{o,b′}K_o(b′,b)=Σ_bp(a,b).

Item 6. Each local marginal is 1/2, so the no-signalling equations hold. A global bit assignment (a0,a1,b0,b1) ∈ {0,1}⁴ would need a0⊕b0=0, a0⊕b1=0, a1⊕b0=0 and a1⊕b1=1. Their xor gives 0=1.

A nonnegative global distribution that reproduces the zero cells is supported on such assignments. So no such distribution exists. Converting bits a_x,b_y ∈ {0,1} to signs A_x=(-1)^{a_x} and B_y=(-1)^{b_y} in {-1,+1}, a deterministic assignment has CHSH value A0(B0+B1)+A1(B0−B1)=±2, and averaging keeps |S|≤2. The PR box has S=4.

Item 7. Let B=A0⊗(B0+B1)+A1⊗(B0−B1). Put F0=A0⊗I−I⊗(B0+B1)/r and F1=A1⊗I−I⊗(B0−B1)/r. Expand, using:

- A_x²=B_y²=I;
- the fact that operators on different factors commute;
- r²=2.

The two mixed terms ±I⊗(B0B1+B1B0)/2 cancel. This yields

$$F_0^*F_0 + F_1^*F_1 = 4I - rB, \qquad 2rI - B = \tfrac{1}{r}(F_0^*F_0 + F_1^*F_1) \succeq 0.$$

Reversing both inner signs gives 2rI+B≥0. For ρ=Σ_jd_jv_jv_j^* with d_j≥0, Tr(ρF^*F)=Σ_jd_j‖Fv_j‖²≥0. So |Tr(ρB)|≤2r.

The cancellation does not use B0B1=B1B0. The identity therefore holds for any Hermitian involutions on the second factor. The audit checked it with B0=Z and B1=[[3/5,4/5],[4/5,−3/5]].

Attainment. Take Z=diag(1,−1) and X=[[0,1],[1,0]]. Put A0=Z, A1=X, B0=(Z+X)/r and B1=(Z−X)/r. Let ρ=vv^⊤/2 with v=(1,0,0,1)^⊤. - ZX+XZ=0, so each B_y²=I. - Tr(ρZ⊗Z)=Tr(ρX⊗X)=1 and Tr(ρZ⊗X)=Tr(ρX⊗Z)=0. - So the four correlators are (1/r,1/r,1/r,−1/r). - Hence S=4/r=2r and S²=8. The bracket 2<S<3 selects the positive root.

In this instance F0=Z⊗I−I⊗Z and F1=X⊗I−I⊗X, and F0²+F1²=4I−2(Z⊗Z+X⊗X).

Arithmetic is exact in pairs (u,v) representing u+vr, with (u,v)(s,t)=(us+2vt,ut+vs).

Item 8. The trace of a matrix over Q(i) lies in Q(i). Hermiticity makes the imaginary part zero, so the expectation is rational. A rational S with S²=8 would give a rational square root of 2, and the parity argument of P.6.a rules that out.

Worked instrument. Take K0=diag(1/2,1/3), K1=diag(1/2,2/3), L0=[[1/3,1/2],[0,0]] and L1=[[0,0],[2/3,1/2]]. Then

- L0K0=[[1/6,1/6],[0,0]] and L1K0=[[0,0],[1/3,1/6]];
- L0K1=[[1/6,1/3],[0,0]] and L1K1=[[0,0],[1/3,1/3]].

Their sum is U(L)U(K)=[[1/3,1/2],[2/3,1/2]]. On p=(2/5,3/5)^⊤ the four masses are 1/6, 7/30, 4/15 and 1/3, summing to 1. The forgotten state is (13/30,17/30)^⊤.

Correction to the dropped-nonnegativity control. The audit wrote: "If nonnegativity is dropped, one-outcome [[2,0],[-1,1]] has each column sum one but sends (1,0)^T to a vector with negative mass." The image is (2,−1)^⊤, whose total mass is 1. **DP-9′ supersedes that sentence:** the image has a *negative entry*, −1, with total mass 1. The control still refutes nonnegativity, but by a negative cell, not by negative mass.

Further controls.

- *Adaptive continuation.* K0=diag(1,0), K1=diag(0,1), L^0=I and L^1=[[0,1],[1,0]] forget to [[1,1],[0,0]]. The unconditioned average gives [[1/2,1/2],[1/2,1/2]] instead.
- *Postselection.* For p=[[1/3,1/6],[1/6,1/3]], the A marginal is (1/2,1/2). Keeping only B=0 and renormalizing gives (2/3,1/3).
- *Normalization dropped.* The scalar [2] doubles mass.
- *Unnormalized state.* 2ρ gives 4r.
- *Non-involutions.* A0=A1=2I with B0=B1=I gives S=4.
- *Non-Hermitian involutions.* A0=A1=[[1,4],[0,−1]], with B0=B1=I on the plus state, gives S=4.
- *No tensor separation.* On one qubit, take A0=Z, A1=−X, B0=Z and B1=X. Then B=2I+2ZX, which is not Hermitian. Its expectation on (I+Y)/2 is 2+2i. With A1=X instead the operator is 0.

Evidence grade.

- *Written proof:* items 1–8, the correction and the controls.
- *Exact finite replay:*
  - dp9_instrument_products, dp9_masses, dp9_forgotten_state, dp9_associativity, dp9_forgetting_homomorphism;
  - dp9_adaptive_vs_average, dp9_postselection, dp9_local_marginals_zero, dp9_pr_box;

  - dp9_chsh_correlators_and_value, dp9_root_selector, dp9_sos_identity_upper_and_lower, dp9_new_sos_generic_involutions;
  - dp9_hermiticity_control, dp9_nonlocal_control_c1_is_zero, dp9_nonlocal_control_c2;
  - dp9_line779_negative_entry_not_negative_mass.
- *Lean kernel:* the axiom report is lean4/component_quantum_axioms.txt.
  - WitnessedDifference.ComponentQuantum.pauli_relations;
  - WitnessedDifference.ComponentQuantum.sos_identity, WitnessedDifference.ComponentQuantum.sos_factors_symmetric;
  - WitnessedDifference.ComponentQuantum.chshBs_table, WitnessedDifference.ComponentQuantum.bs_quad_le, WitnessedDifference.ComponentQuantum.bs_quad_bell;
  - WitnessedDifference.ComponentQuantum.deterministic_chsh, WitnessedDifference.ComponentQuantum.pr_box_not_deterministic;
  - WitnessedDifference.ComponentQuantum.single_qubit_minus_x, WitnessedDifference.ComponentQuantum.single_qubit_plus_x, WitnessedDifference.ComponentQuantum.single_qubit_minus_x_not_symmetric.
- *Native exact implementation:*
  - omega_finite_probability::classical_instrument provides fixed and adaptive composition, forgetting, the PR box, signalling witnesses, local-model verification and CHSH. It finds that the PR box has no local table by exhaustive search over the 16 deterministic assignments.
  - omega_quantum_physics::finite_chsh_correlations admits the extreme value only for parties embedded in separate tensor factors. It takes the sign from an exact ordered test, not from s^2.
  - Both single-carrier controls are refused. With A1=X, the Bell operator is zero. With A1=−X, it is 2I+2ZX.
- *Unresolved residual:*
  - The Lean module states the bound in the rescaled form for Z⊗Z+X⊗X, which carries no r. It records non-symmetry of the one-qubit operator rather than the value 2+2i.
  - Items 1–5 are written proof with finite replay. They are not stated in Lean.

P.19.b Exact density and channel representations (DP-8, audited 2026-10-06) Setting. Fix positive finite dimensions and an ordered basis. Take F=Q, with transpose as the adjoint, or F=Q(i), with conjugate transpose.

- A *density* is a Hermitian ρ over F with trace one and nonnegative quadratic form.
- A linear map Φ is given on the matrix units E_ij. Its input-first Choi matrix is J_Φ=Σ_{ij}E_ij⊗Φ(E_ij).

Statement.

1. *(PSD decision.)* For a Hermitian H over F, the following are equivalent:
   - H≥0;
   - every principal minor is nonnegative;
   - there is an exact factorization $PHP^* = LDL^*$, where P is a permutation matrix, L is unit lower triangular, and D is diagonal with nonnegative entries.

   When H is not PSD, the procedure returns an exact vector v with v^*Hv<0. Leading principal minors alone do not suffice.
2. *(Choi.)* Φ is completely positive iff J_Φ≥0. It is trace preserving iff Tr_outJ_Φ=I_in.
3. *(Weighted Kraus.)* When J_Φ≥0, there are rational weights d_s≥0 and operators K_s over F with Φ(X)=Σ_sd_sK_sXK_s^*. When Φ is also trace preserving, Σ_sd_sK_s^*K_s=I. A unit-weight family

over F always exists: a weight p/q becomes pq copies of K/q.

4. *(Realification.)* For M=A+iB, put R(M)=[[A,−B],[B,A]]. Then:
   - R(MN)=R(M)R(N), R(M^*)=R(M)$^\top$ and Tr R(M)=2 Re Tr M;
   - ρ_R=R(ρ)/2 is a real density;
   - Tr(ρ_RR(E))=Tr(ρE) for every Hermitian E.

   Realification of a tensor product is not the product of the realifications.

Proof.

Item 1. Write H=[[a,b^*],[b,B]] with a>0, and put S=B−bb^*/a. Completing the square gives

$$(x,y)^* H(x,y) = a\,|x + b^* y/a|^2 + y^* S y,$$

so H≥0 iff S≥0, and this is one LDL step. Each case either reduces the dimension or gives a witness:

- A negative diagonal entry gives a basis-vector witness.
- A zero diagonal entry with a nonzero entry c beside it gives a witness. In [[0,c],[c^*,d]] with d≥0, take v=(−tc,1) with t=(d+1)/(2|c|^2). Then v^*Hv=−1.
- A zero row and column is deleted.

A witness y found for S lifts to (−b^*y/a,y) for H. Each step lowers the dimension, so the procedure terminates.

For the minors:

- *Necessity.* A principal submatrix of a PSD matrix is PSD, since it extends vectors by zero coordinates.
- *Sufficiency.* For every index set T of the remaining block, det(S_T)=det(H_{{1}∪T})/a≥0. Induct on the dimension. If no diagonal entry is positive, the two-by-two minors force H=0.

Item 2. Put Ω=Σ_ie_i⊗e_i.

- (⇒) Complete positivity gives (id⊗Φ)(ΩΩ^*)=J_Φ≥0.
- (⇐) Item 1 gives J_Φ=Σ_sd_sv_sv_s^*. Unvectorize with (K_s)_{ri}=(v_s)_{i·d_out+r}. Comparing entries gives Φ(X)=Σ_sd_sK_sXK_s^* on every matrix unit. Each term keeps every ancilla extension PSD.
- The (i,j) entry of Tr_outJ_Φ is Tr Φ(E_ij). So the partial-trace equation is trace preservation on a basis. By cyclicity of the trace it equals Σ_sd_sK_s^*K_s=I.

Item 3. This is the (⇐) step of item 2. For the unit-weight family, pq·(1/q^2)=p/q.

Item 4. The identities follow by block multiplication. For z=x+iy, z^*Hz=(x,y)$^\top$R(H)(x,y), so R preserves Hermiticity and positivity. Also Tr(ρ_RR(E))=Re Tr(ρE)=Tr(ρE).

Dimensions show the tensor statement: R(A⊗B) has size 2mn, while R(A)⊗R(B) has size 4mn. Already R(1)/2=I2/2, while the independently doubled product is I4/4.

Worked examples and controls.

- *Transpose.* On dimension two, J_T=[[1,0,0,0],[0,0,1,0],[0,1,0,0],[0,0,0,1]]. For v=(0,1,−1,0)$^\top$, v^*J_Tv=−2. So the transpose is positive and trace preserving but not completely positive.
- *Doubled identity.* X↦2X is completely positive but not trace preserving.

- *Leading-minor trap.* `diag(0,2,−1)` has every leading principal minor equal to `0` and trace one, but `e3` gives value `−1`.
- *Non-Hermitian.* `(1/2) [[1,1],[−1,1]]` has positive principal minors and trace one, but it is not Hermitian, so it is not a density.
- *Field boundary.* `ρ_y=(1/2) [[1,−i],[i,1]]` is a Gaussian-rational density. It has no real-rational form in this basis.
- *Realification controls.*
  - Complex rank one maps to real rank two.
  - In complex dimension one, `diag(1,0)` is a real density outside the image of `R`.
- *Hadamard-type channel.* Take `H=[[1,−1,−1,0],[1,1,0,1],[1,0,1,−1],[0,−1,1,1]]`. Then $H^\top H = 3I_4$, and $\Phi(X) = HXH^\top/3$ is trace preserving. Its Choi matrix is $\mathrm{vec}(H)\mathrm{vec}(H)^\top/3$, of rank one.

Correction on minimal Kraus counts. The audit wrote: "The coefficient data thus agree as channels, while their minimal unweighted presentations differ by field." **DP-8′ supersedes that sentence with the exact minima.** For the channel above, the least number of unit-weight Kraus operators is:

field	minimum	witness
`R`	1	`H/√3`
`Q(i)`	2	`c_1H`, `c_2H` with `c_1=(1+i)/3` and `c_2=1/3`
`Q`	3	three copies of `H/3`

Proof.

- *Every family is a multiple of `H`.* A rank-one Choi matrix forces every Kraus operator to be `K_s=c_sH`, with $\Sigma_s |c_s|^2 = 1/3$.
- *One operator.* This needs $|c|^2 = 1/3$. Write `c=(a+ib)/d` with integers of joint gcd one. Then $3(a^2+b^2) = d^2$. The squares modulo three are `0` and `1`, so $3 \mid d$. Then $3 \mid a^2+b^2$, so $3 \mid a$ and $3 \mid b$, which contradicts the gcd. So no single operator exists over `Q(i)`, and hence none over `Q`.
- *Two operators over `Q(i)`.* $|c_1|^2 + |c_2|^2 = 2/9 + 1/9 = 1/3$.
- *Two operators over `Q`.* This needs $a^2+b^2 = 1/3$ with rational `a,b`. Clearing denominators gives the same descent, so no such pair exists.
- *Three operators over `Q`.* $3 \cdot (1/9) = 1/3$.

So the field changes the minimal unit-weight count, `1`, `2` or `3`. It does not change the channel or its Choi matrix.

Evidence grade.

- *Written proof:* items 1–4, DP-8′ and the controls.
- *Exact finite replay:*
  - `dp8_identity_choi_psd`, `dp8_transpose_choi_matrix`, `dp8_transpose_choi_negative`, `dp8_transpose_refused_by_ldl`, `dp8_doubled_identity_cp_not_tp`;
  - `dp8_zero_diagonal_witness_minus_one`, `dp8_singular_leading_minor_trap`, `dp8_non_hermitian_positive_minors`, `dp8_rho_y_density`;
  - `dp8_HtH_3I`, `dp8_choi_rank_one_vecH`, `dp8_three_copy_completeness`, `dp8_two_copy_mutation_not_tp`, `dp8_mod3_descent_residues`;
  - `dp8_new_rational_two_squares_absent_d_le_60`, `dp8_new_rational_three_copies_minimum_witness`, `dp8_new_gaussian_two_kraus_completeness`, `dp8_new_gaussian_two_kraus_same_choi`;

  - dp8_realification_hom_trace_adjoint, dp8_realification_rank_one_to_rank_two, dp8_realification_tensor_dims, dp8_realification_unit_scaling.
- *Lean kernel:* the axiom report is lean4/component_quantum_axioms.txt.
  - WitnessedDifference.ComponentQuantum.choi_transpose_negative, WitnessedDifference.ComponentQuantum.choi_transpose_tp, WitnessedDifference.ComponentQuantum.leading_minor_trap;
  - WitnessedDifference.ComponentQuantum.kraus_h_orthogonal, WitnessedDifference.ComponentQuantum.three_copies_complete;
  - WitnessedDifference.ComponentQuantum.sq_mod_three, WitnessedDifference.ComponentQuantum.three_dvd_of_sq, WitnessedDifference.ComponentQuantum.mod_three_descent.

  mod_three_descent states, over natural numbers, that 3(a^2+b^2)=d^2 forces a=b=d=0. That supports both the one-operator absence over Q(i) and the two-operator absence over Q.
- *Native exact implementation:*
  - omega_quantum_physics::exact_hermitian decides PSD. When the matrix is not PSD it returns an exact negative vector with its value.
  - The channel controls replay the transpose Choi matrix with value −2. They also replay the absence of a single Gaussian-rational operator for the rank-one channel, and the two-operator family. Realification sizes replay as 2ab against 4ab.
  - The omega_wdp density and channel adapters re-check their premises. A shape mismatch is refused.
- *Unresolved residual:*
  - The omega_wdp carriers remain real symmetric over Q.
  - A budget refusal in unit-weight splitting is not a proof of nonexistence.
  - The equivalences of items 1 and 2 are not stated in Lean.
  - The rank-one step and realification are written proof with finite replay only.
  - The Q two-square absence is replayed exhaustively only for denominators up to 60. The general case rests on the descent.

P.19.c A derived short-time product-defect bound (DP-7, audited 2026-10-06) Hypotheses. Import the following finite-dimensional complex Hilbert-space operations: trace, partial trace, the trace norm $\|\cdot\|_1$, the operator norm $\|\cdot\|_\infty$, and constant-Hamiltonian evolution with ħ=1. They are premises; no quantum dynamics is constructed from finite scalar verification records.

- Let V be a finite set of sites, with positive local dimensions and a declared tensor order.
- Let $H=\Sigma_\alpha h_\alpha$ be a finite sum of Hermitian terms. Each h_α is time-independent, has support X_α with $1\le|X_\alpha|\le 2$, and acts as the identity off it.
- Put $g=\max_v \Sigma_{\alpha:v\in X_\alpha}\|h_\alpha\|_\infty$. One-site fields are included.
- Fix $\sigma\subseteq V$ with $k=|\sigma|$, the number of sites.
- Let $\rho(t)=e^{-itH}\rho(0)e^{itH}$, and write $\rho_S(t)$ for its reduction to a set of sites S.
- Put $\tau_\sigma(t)=\bigotimes_{i\in\sigma}\rho_i(t)$ and $F_\sigma(t)=\rho_\sigma(t)-\tau_\sigma(t)$.
- Put $D_\sigma(t)=\|F_\sigma(t)\|_1$, $n_\sigma=F_\sigma'(0)$ and $\nu_{act}=\|n_\sigma\|_1$.
- Assume $F_\sigma(0)=0$; a product initial state is sufficient.

Statement (DP-7′). For every real $t\ge 0$,

$$D_\sigma(t) \;\ge\; \nu_{\mathrm{act}} t - 4k(k+1)g^2t^2,$$

and for rational bounds $0 \le \nu \le \nu_{\text{act}}$ and $G \ge g$ the same holds with ν and G in place of ν_{act} and g.

When $k \ge 2$,

$$D_\sigma(t) \;\ge\; \nu t - 6k^2 G^2 t^2.$$

If $\nu > 0$ and $G > 0$, the weaker envelope has vertex $t_* = \nu/(12k^2G^2)$, and $D_\sigma(t) \ge \nu t/2$ for $0 \le t \le t_*$.

For the trace distance $T_\sigma = D_\sigma/2$, the valid weak form is $T_\sigma(t) \ge (\nu/2)t - 3k^2G^2t^2$.

Correction. The audit summary stated: "D is at least nu*t minus 4k(k+1)g^2t^2, hence the stated 6k^2* bound." **DP-7′ supersedes that sentence:** the step from $4k(k+1)$ to $6k^2$ needs $k \ge 2$, because $6k^2 - 4k(k+1) = 2k(k-2)$.

At $k = 1$ the difference is -2. There the relaxation runs the wrong way, and the $6k^2$ form is stronger than the inequality it claims to follow from. The bound itself is not violated at $k = 1$: the residual of a single site is $F \equiv 0$, so $\nu = 0$ and both sides are compatible. But "hence" is not a valid step there.

Proof.

Step 1 (support counting). For any finite matrices, $\|[A,B]\|_\infty \le 2\|A\|_\infty \|B\|_\infty$. Let O_S be supported on s sites. Only terms meeting S fail to commute with it, so

$$\|[H, O_S]\|_\infty \le 2sg\|O_S\|_\infty.$$

A contributing $[h_\alpha, O_S]$ is supported on $S \cup X_\alpha$, which has at most $s+1$ sites, because $|X_\alpha| \le 2$ and X_α meets S. Hence

$$\|[H, [H, O_S]]\|_\infty \le 4s(s+1)g^2\|O_S\|_\infty.$$

Step 2 (reduced derivatives). $\rho'' = -[H, [H, \rho]]$. For Hermitian A, $\|A\|_1 = \max\{\mathrm{Tr}(WA) : W = W^*, -I \preceq W \preceq I\}$. Also, for a density ρ, $|\mathrm{Tr}(\rho B)| \le \|B\|_\infty$. Then partial trace and cyclicity give, for every t:

- $\|\rho_S''(t)\|_1 \le 4s(s+1)g^2$;
- $\|\rho_i'(t)\|_1 \le 2g$;
- $\|\rho_i''(t)\|_1 \le 8g^2$.

None of these assumes the evolved state stays a product.

Step 3 (product of marginals). Apply the product rule:

$$\tau_\sigma'' = \sum_i \rho_i'' \otimes \bigotimes_{j \ne i} \rho_j + 2 \sum_{i<j} \rho_i' \otimes \rho_j' \otimes \bigotimes_{l \ne i,j} \rho_l.$$

The trace norm is multiplicative on tensor products of Hermitian matrices, and $\|\rho_i\|_1 = 1$. So

$$\|\tau_\sigma''\|_1 \le 8kg^2 + 4k(k-1)g^2 = 4k(k+1)g^2.$$

With step 2, this gives $\|F_\sigma''(t)\|_1 \le 8k(k+1)g^2$.

Step 4 (normal slope and remainder). For trace-zero Hermitian Z, put P_σZ=Σ_i(Tr_{σ\i}Z)⊗(⊗_{j≠i}ρ_j(0)).

- P_σZ has the same one-site marginals as Z. So P_σ is idempotent, and n_σ=(I−P_σ)ρ_σ′(0).
- Taylor's formula with integral remainder gives F_σ(t)=tn_σ+∫_0^t(t−u)F_σ″(u)du.
- The remainder has norm at most 8k(k+1)g²·t²/2=4k(k+1)g²t².
- The reverse triangle inequality gives the first bound.
- When k≥2, 2k(k−2)≥0 gives the second.

Step 5 (window). Put c=6k²G². Then f(t)=νt−ct²=ν²/(4c)−c(t−ν/(2c))², so the vertex is t*=ν/(2c). Also f(t)−νt/2=t(ν/2−ct)≥0 on [0,t*]. The remainder estimate holds at every t≥0; t* only bounds the half-slope claim.

Exact models.

- *Two sites.* Take H=X⊗X on two qubits, starting at |00⟩⟨00|. Then g=G=1.
  - n=i|00⟩⟨11|−i|11⟩⟨00|. The dual witness W=n has Tr(Wn)=2, so ν=2.
  - At k=2, both coefficients equal 24, and t*=1/24.
  - At t=1/100, the bound is 2/100−24/10000=11/625.
  - The exact defect is $D(t) = |\sin(2t)| + \frac{1}{2}\sin^2(2t)$. From x−x³/6≤sin x, sin(1/50)≥14999/750000, which exceeds 11/625=13200/750000.
- *Three sites.* Add a spectator qubit in |0⟩⟨0|, with H=(X⊗X)⊗I and k=3.
  - t*=2/108=1/54.
  - At t=1/200 the weak bound is 1/100−54/40000=173/20000.
  - The sharper bound is 1/100−48/40000=11/1250.

Controls (each drops one hypothesis).

1. *Fields omitted from g.* Take H=X⊗X+100(Z⊗I), so H²=10001I and g=101. At t0=π/√10001, which satisfies 0<t0<1/25<1/24, the defect is D(t0)=0. But the pair-only value g=1 would claim 2t0−24t0²>0.
2. *F_σ(0)≠0.* Take ψ=c|00⟩+is|11⟩ with c=9999/10001 and s=200/10001; then 9999²+200²=10001². At θ=2arctan(1/100)<1/50<1/48, the state is |00⟩, so D(θ)=0. Yet ν=1 would claim θ−24θ²>0.
3. *Time dependence.* Take H(t)=(1−2t/ε)(X⊗X) with ε=1/100. It returns to |00⟩ at t=ε, so D(ε)=0. Yet the bound gives 2ε−24ε²=11/625>0.
4. *Higher-body terms.* With N=27 sites, H=X^{⊗27}+Σ_{j=2}^{27}Z_j and O=Z_1, we have g=2. The double commutator has expectation 104 on the GHZ-minus ray, which exceeds the two-local bound 96.
5. *Half-trace mixing.* In the two-site model, T(1/100)≤101/10000. That is less than 176/10000, the raw bound misapplied to T.
6. *Malformed hosts.* These are refused:
   - a non-Hermitian term [[0,1],[0,0]];
   - diag(2,−1), which is not positive;
   - 2|0⟩⟨0|, which has trace two.
7. *Degenerate parameters.*
   - At k=1, F≡0.
   - G=0 forces every term to be zero.
   - ν=0 gives no positive window.
8. *Raw commutator.* For H=X⊗I from |00⟩⟨00|, the raw derivative is nonzero. But F_σ(t)=0 for all t, and the normal slope is 0.

Evidence grade.

- *Written proof:* steps 1–5, DP-7′ and the controls.
- *Exact finite replay:*
  - `dp7_support_growth_constant`, `dp7_P_sigma_idempotent`, `dp7_normal_derivative_n`, `dp7_n_survives_P_sigma`, `dp7_local_tangent_projected_to_zero`;
  - `dp7_relaxation_gap_is_2k_k_minus_2`, `dp7_relaxation_fails_exactly_at_k1`;
  - `dp7_tstar_half_slope_k2`, `dp7_tstar_half_slope_k3`, `dp7_k2_value_t_1_100`, `dp7_k3_values_t_1_200`, `dp7_sin_enclosure`, `dp7_inert_spectator_keeps_nu_2`, `dp7_I_pm_H_square_identity`;
  - `dp7_field_anticommute_H2`, `dp7_understated_field_enclosure`, `dp7_correlated_pythagorean`, `dp7_time_dependent_return`;
  - `dp7_higher_body_double_commutator`, `dp7_higher_body_ghz_expectation`;
  - `dp7_half_trace_control`, `dp7_half_trace_upper_value`.
- *Lean kernel:* scalar algebra only. The axiom report is `lean4/component_quantum_axioms.txt`.
  - `WitnessedDifference.ComponentQuantum.coefficient_relax`, `WitnessedDifference.ComponentQuantum.coefficient_relax_iff`, `WitnessedDifference.ComponentQuantum.coefficient_relax_fails_at_one`;
  - `WitnessedDifference.ComponentQuantum.half_slope_window`, `WitnessedDifference.ComponentQuantum.envelope_monotone`, `WitnessedDifference.ComponentQuantum.vertex_value`, `WitnessedDifference.ComponentQuantum.envelope_relax`;
  - `WitnessedDifference.ComponentQuantum.replay_k2`, `WitnessedDifference.ComponentQuantum.replay_k3`, `WitnessedDifference.ComponentQuantum.sine_enclosure_comparison`;
  - `WitnessedDifference.ComponentQuantum.field_control_square`, `WitnessedDifference.ComponentQuantum.correlated_control_normalization`, `WitnessedDifference.ComponentQuantum.time_dependent_control`.
- *Native exact implementation:*
  - `omega_quantum_physics::product_defect_bound` checks premises only. It certifies the premises, the lower envelope and the half-slope endpoint.
  - It accepts the weak $6k^2$ coefficient only for $k \geq 2$.
  - It refuses an understated field norm and a correlated initial state.
  - It names the analytic remainder as an explicit residual and does not discharge it.
- *Unresolved residual:*
  - The CODA k-cell evaluator is not yet bound to this premise checker.
  - Steps 1–4 use imported analytic facts and are a written proof: trace-norm duality, multiplicativity of the trace norm, the matrix exponential and Taylor's formula. They are not stated in Lean.
  - The existing scalar evaluator does not bind its inputs to a state, a Hamiltonian, norm witnesses or the host hypotheses.
  - Bounds on π and on `arctan` in controls 1 and 2 are written rational enclosures.

P.20 Coherence inside guarded components

P.18 left one question open: inside a connected component of the legal-swap graph Γ, do legal backtracks, distant squares and braid hexagons connect every pair of parallel legal swap proofs? This section gives a sufficient guard condition with a proof, shows that every native five-edit family with compatibly ordered precedences satisfies it, and records a bounded native search for the case where it fails.

The complex. Let $\mathcal{L}$ be the legal complete linear extensions of P.18. Let $X_{\mathcal{L}}$ be the 2-complex

with vertex set $\mathcal{L}$, the edges of Γ, a square for each pair of disjoint adjacent swaps whose four orders are all legal, and a hexagon for each window of three positions whose six orders are all legal. A square or hexagon in $X_{\mathcal{L}}$ is exactly a P.17 local relation whose intermediate orders are all legal. A combinatorial homotopy of edge paths inserts or deletes a backtrack, or replaces one arc of a face boundary by the complementary arc. Such a replacement decomposes into backtrack insertions followed by a P.17 square or braid move, with every intermediate order a vertex of the same face. Two parallel legal swap proofs are therefore joined by legal local 3-witnesses exactly when they are homotopic relative to their endpoints in $X_{\mathcal{L}}$. Coherence inside a component is simple connectivity of that component.

Definition (star-shaped legality). Fix $e_0 \in \mathcal{L}$. For an order h, let $\mathrm{Inv}(h)$ be the set of event pairs that h orders opposite to e_0. A *lower cover* of h is $h \cdot s_i$ where the adjacent pair at positions $i, i+1$ lies in $\mathrm{Inv}(h)$. The legal set is *star-shaped from* e_0 when every lower cover of every legal order is legal. Descending by lower covers shows that this is equivalent to the down-set property: every order whose inversion set is contained in that of a legal order is legal.

GEN-WDP-69 (guarded coherence under star-shaped legality). If $\mathcal{L}$ is star-shaped from some e_0, then Γ is connected and $X_{\mathcal{L}}$ is 1-connected ($\pi_1 = 0$). Any two legal swap proofs with the same framed source and target histories are joined by a finite sequence of legal backtracks, distant squares and braid hexagons, and every intermediate order is legal.

Proof. (i) *Faces below a legal order.* Let $v \in \mathcal{L}$ have inverted positions $i < j$. If $j > i+1$, the square spanned by s_i and s_j at v has its other three orders below v: their inversion sets are contained in $\mathrm{Inv}(v)$, so they are legal. If $j = i+1$, the window holds events A, B, C with (A, B) and (B, C) inverted. Then e_0 orders C before B before A, so (A, C) is inverted as well, and the six orders of the window are below v and legal. In both cases each swapped pair is ordered differently by e_0 and by v, both of which satisfy the declared precedences, so the pair is incomparable and the cell is a legal P.17 relation. (ii) *Connected lower links.* The lower covers of v are pairwise joined by the faces of (i), whose remaining vertices have smaller inversion count. The lower link of v is therefore a complete graph. It is nonempty unless $v = e_0$, because a nonidentity order relative to e_0 has an inverted adjacent pair. (iii) *Attachment.* An adjacent swap changes $|\,\mathrm{Inv}\,|$ by exactly one, so every edge joins different ranks, and every face is a weak-order interval with a unique top vertex. Build $X_{\mathcal{L}}$ from the point e_0 by attaching the legal orders in increasing rank. Each attachment adds the vertex, its edges to its lower covers and the faces with that vertex as top. This is a cone on its lower link. A cone on a nonempty connected graph, glued along that graph, preserves connectedness, and by van Kampen it preserves simple connectivity: the new loops through v are generated by pairs of lower covers, and each pair is killed by a face. (iv) The first paragraph converts homotopy into local moves. □

The proof is constructive. Let $D(h)$ be the canonical descent that repeatedly swaps the leftmost inverted pair until it reaches e_0. For an inverted position i of x, induction on rank turns the word $[i]\,D(xs_i)$ into $D(x)$. If i is the leftmost inversion this is the definition. Otherwise both descents pass through the bottom of the face of (i), one square or braid move exchanges them, and the inductive hypothesis handles the two lower paths. Rewriting a word w from h to h' edge by edge then gives $w \simeq D(h)D(h')^{-1}$. Applying this to two parallel words yields an explicit list of local replacements. The native owner emits that list, and an independent checker replays every intermediate order against the native guards.

P.17 is the special case $\mathcal{L}$ = all linear extensions of the declared poset. That set is a down-set from each of its elements, so the theorem recovers P.17's coherence without its all-order premise. P.18's

family is not star-shaped from either legal history: the condition fails exactly where connectivity fails.

Native corollary (compatibly ordered precedence). Rank the five native edit kinds as AddTerm 0, AddAssertion and Fill 1, Identify 2, Remove 3. Suppose the transitive closure of the declared precedences never places a higher-ranked event before a lower-ranked one. Then any kind-sorted linear extension e_0 is a valid base, and every five-edit model family of that shape satisfies GEN-WDP-69. (The bounded search below tests only the lexicographically least kind-sorted extension.) It suffices to check that an adjacent pair that a legal history orders against e_0 can be swapped toward e_0 with the run staying legal and ending in the same framed state. Such a pair has the lower-ranked event second, or has equal ranks.

- A Remove followed by an AddAssertion on the same count key: adding first only raises the count at the intermediate prefix, and both orders end with the same count.
- A Remove followed by an Identify: identification pushes counts forward by summation, so the pooled entry seen by the later Remove is at least the entry it saw before. Both orders end in the same pooled table.
- An Identify followed by a Fill: if the fill is legal after the identification it is nondegenerate and new there. Its preimage is then nondegenerate and new before, and both orders give the same filled set.
- An Identify followed by an AddAssertion: both are always enabled, and the pushed-forward counts agree.
- Events on different current keys, Fills against counts, and AddTerm against any event not using the term it creates commute, up to the token-to-native alignment of P.14.
- Two events of equal rank (two Removes, AddAssertions, Fills, Identifies or AddTerms) are legal in both orders whenever they are legal in one. Two identifications that would merge the same pair of classes fail in either order, so no legal history contains both.

The Lean module checks the abstract swap step (`helper_swap_preserves_run`), its count-key instance (`count_kernel_swap`), and the fact that the converse swap fails at count zero. It also checks that a cochain closed on the declared faces is invariant under backtrack and face moves (`moves_preserve_total`). The ten declarations in its axiom report use only `propext` and `Quot.sound`. The case analysis for Identify and Fill is a written proof checked on bounded families by the native search below.

When the condition fails (bounded native search). The native module `omega_wdp::genesis_guarded_history` enumerated 132,278 declared families in five tiers. The tiers use one to three initial terms; count, identification, fill and term-creation events; up to six events in three tiers and five in the other two; and up to three extra precedence edges. The menu is fixed in the search binary. Every one of the 38,158 kind-compatible families with a legal history had a star-shaped legal set, one component and zero rational first homology. The 65,253 constructive witness lists it generated all replayed. Outside the condition, 126 families have disconnected legal sets. The smallest has four events: count one, and two removals, each followed by its own addition. Exactly one family has a connected component with positive rational first homology.

Native counterexample to unconditional coherence. Take one term with $R(0,0) = 2$, three removals x_1, x_2, x_3 of that assertion and three additions a_1, a_2, a_3, with the declared precedences $x_i < a_i$. Let coordinate i count the events of pair i already performed. Each schedule is then a monotone lattice path from $(0,0,0)$ to $(2,2,2)$ in $\{0,1,2\}^3$, and an adjacent swap of events from different pairs flips one corner of the path. The count after a prefix is 2 minus the number of pairs at coordinate 1. A schedule is therefore legal exactly when its path avoids the centre $(1,1,1)$. Of

the 90 paths, 36 pass through the centre, so 54 are legal. They form one swap component with 84 legal swaps and 30 legal squares. There is no legal hexagon: a hexagon fills a unit cube of the grid, every unit cube has the centre as a corner, and some of its six paths pass through it. Exact elimination gives rational first homology of rank one. The retained certificate supplies two legal swap proofs from $x_1a_1x_2a_2x_3a_3$ to $x_3x_2a_3x_1a_1a_2$ with the same framed boundaries. It also supplies a 1-cochain supported on four edges, closed on all 30 legal squares, whose values on the two proofs are -1 and 0. By `moves_preserve_total`, no sequence of legal backtracks, squares and braids joins them. Coherence inside a guarded component therefore fails without the guard condition, already for the count kernel. Dropping any one of the three precedences removes the hole. With at most two extra precedences, the search found no component with positive rational first homology. The counterexample is not star-shaped from any legal history, in agreement with GEN-WDP-69.

The bounded search supports, and does not prove, the native corollary for Identify and Fill. Rational first homology zero would not by itself give simple connectivity; the coherence statements rest on the written proof.

Scope. GEN-WDP-69 is a written theorem with a constructive reduction. Its Lean counterpart covers only the local swap and invariance steps. Native evidence covers declared bounded families. The theorem does not give 4-cells, identify distinct 3-certificates, define restrictions over arbitrary source covers, or grant authority. GEN-WDP-O03 remains open beyond the components it covers.

P.21 Minimum retained information for two native edits

Definition. Fix the literal raw states with at most three terms, nonnegative integer count tables R, S of combined total at most four, and every subset of the available oriented nondegenerate filled triples. Use the 42 literal five-edit descriptors with addresses in 0..2, and at most two successful edits. Protect the joint result mark, the full literal balance record, native guard outcomes, carrier-bound admission, native addition-price pairs, and successful occurrences. A native-enabled edit excluded by the carrier bound retains its native price but is not an executed edge. A false native guard is a distinct outcome. Neither refusal consumes a successful-edit slot.

At a terminal prefix, one profile observes only the present mark and balance record; the second additionally permits nonexecuting status probes. At positive layers both observe every offer and all admitted continuations. These two profiles are declared separately, before computing their quotients.

Finite proposition (GEN-WDP-73). For either profile at horizon two, the coarsest adequate report partition is raw-state equality: 29,771 classes in total and $|M^{-1}(m)|$ classes in each result-mark fiber. An attaining report is

$$H(s) = (M(s), F(s), \min(R, 2), \min(S, 2)).$$

This is a minimum number of information classes, not a minimum number of fields or bytes.

Written proof. Equal protected futures determine whether each count is positive and whether it remains positive after one removal, hence every count capped at two. Removals stay inside the bound. Summing the root-normal table cancels its potentials, so the joint mark determines $\sigma = \sum R - \sum S$. Under total bound four, either no cap is two (every count is known), two caps are two (both counts are exactly two and all others zero), or exactly one cap is two. In the last case its fixed relation sign and σ recover its count after subtracting the other known signed counts. The mark supplies the term count and the literal balance keys supply the filled set. Thus equal

futures force raw equality; raw equality also preserves all five native operations. The displayed report decodes by the same argument. Counting the carriers gives $1+15+495+4\binom{22}{4}=29771$. ■

The complete derivation constructs the layer signatures, forward projections and backward path lifts from every supplied raw starting state. It gives identity and composition laws with the remaining horizon retained, whole-field deletion witnesses, and explicit AddTerm and Identify controls. The reciprocal-count pair $(R_{01}, R_{10}) = (1, 2)$ versus $(2, 1)$ has equal one-step reports only under the present-only terminal profile: after one removal, a terminal guard probe separates it under the second profile. There is no convention-independent least-horizon claim.

Seven native tests, independent replay of all 29,771 states and 1,250,382 offers against the native and pinned original implementations, and 21 selected formal declarations support different parts of this result. The selected formal model is not a verification of either concrete program. The source-specific finite result does not replace the broader source-refinement and higher-coherence obligations GEN-WDP-O02 and GEN-WDP-O03. Larger count bounds, transported action names, unbounded histories, observation instruments and authorization remain outside this statement.

P.22 Exact topology and coherence thresholds for paired count guards

P.20's count-kernel counterexample extends to an unbounded family with an exact classification. Fix $n \geq 2$ and $1 \leq c < n$. On one valid Source-25 count key, start with count c and take distinct event tags r_i, a_i, $1 \leq i \leq n$. Each r_i is a `Remove`, each a_i an `AddAssertion`, and every tag occurs once. Declare the task precedences $r_i < a_i$. These tags and precedences are additional presentation data; the native count update does not supply them. Other counts, terms and filled faces stay fixed.

For any pair-respecting prefix u, its active tasks are those whose removal has occurred and whose addition has not. Native replay succeeds exactly when all prefixes have at most c active tasks, and its count then equals $c - |\operatorname{Act}(u)|$. This follows by induction: removal opens one task and requires positive count; addition closes its already opened task. Every complete legal history returns to count c.

Let $K(n, c)$ be the full subcomplex of the event permutohedron consisting of ordered partitions $B = (B_1|\cdots|B_k)$ whose every total refinement is legal. Its vertices are complete tagged histories, and a face has dimension $2n - k$. Its two-dimensional faces are exactly P.20's legal squares and braid hexagons. Higher faces are explicitly included by the same refinement rule. A finite face test is

$$b(r_i) < b(a_i) \quad \text{for all } i, \qquad \max_t \#\{i : b(r_i) \leq t \leq b(a_i)\} \leq c,$$

where $b(e)$ is the block containing e. To prove the test, put every removal in a block before every addition there: this attains its largest possible occupancy. Both interval endpoints are included because an opening task can overlap a closing task within that refinement.

Define $T(n, c)$ as the task permutohedron subcomplex consisting of ordered partitions of $[n]$ whose blocks have size at most c. Also put

$$M(n, c+1) = \mathbb{R}^n \setminus \bigcup_{\substack{J \subseteq [n] \\ |J| = c+1}} \{x : x_i = x_j \text{ for all } i, j \in J\}.$$

This is the classical real no-$(c+1)$-equal arrangement complement.

Theorem (guarded schedule classification). For every such n, c,

$$|K(n, c)| \simeq |T(n, c)| \simeq M(n, c+1).$$

For $c = 1$, there are exactly $n!$ contractible components. For $2 \le c < n$, the complex is $(c-2)$-connected and is not $(c-1)$-connected; its first nonzero reduced integral homology occurs in degree $c-1$. In particular, legal swap proofs have square-and-braid coherence exactly when $c \ge 3$, within the connected range. Moreover,

$$K(n, n-1) \simeq S^{n-2}.$$

Proof of the classification. Index three finite covers by all strict task posets P of width at most c. Let $L(Q)$ denote the full linear-extension complex of a finite poset Q. It is contractible in every dimension: the proved contractibility lemma attaches the decreasing-run faces in increasing inversion rank; each face meets the preceding complex in a contractible polyhedral antistar. For each P, take

$$U_P = L(Q_P), \qquad T_P = L(P), \qquad V_P = \{x : x_i < x_j \text{ whenever } i <_P j\},$$

where Q_P is generated by $r_i < a_i$ and $a_i < r_j$ for $i <_P j$. Active tasks in U_P form a P-antichain, so $U_P \subseteq K(n, c)$. Conversely, a legal face defines intervals $[b(r_i), b(a_i)]$ and the interval order $i <_P j$ iff $b(a_i) < b(r_j)$. Pairwise intersecting intervals on a line share a point; therefore its width is the maximum occupancy in the face test, at most c. This shows that the U_P cover.

Every block of a face in $L(P)$ is a P-antichain, so $T_P \subseteq T(n, c)$. Conversely, the strict order between the blocks of any face of $T(n, c)$ is a poset of width at most c covering that face. Similarly, equal-coordinate blocks of $x \in M(n, c+1)$, ordered by their values, give a width-at-most-c poset with $x \in V_P$. A point of V_P cannot have $c+1$ equal coordinates, since they would form an antichain. Thus all three are covers of their stated spaces.

A finite intersection in any cover is nonempty exactly when the union of its task orders is acyclic. In the event cover, a task cycle gives an event cycle through $r_i < a_i$, while an event cycle contracts to a task cycle. When acyclic, the two polyhedral intersections are linear-extension complexes of the respective transitive closures and hence contractible. The Euclidean intersection is a nonempty open convex cone. The covers therefore have the same nerve and contractible nonempty intersections. The finite-subcomplex and open-good-cover forms of the nerve theorem give both equivalences.

Proof of the consequences. At capacity one, $T(n, 1)$ consists of the $n!$ task orders. At capacity $c \ge 2$, it contains the full $(c-1)$-skeleton of the task permutohedron: a d-face has no block larger than $d+1$. Adjoining the omitted cells, all of dimension at least c, cannot change π_j for $j \le c-2$, and produces a contractible polytope. This establishes $(c-2)$-connectivity. Projection from $M(n, c+1)$ to its first $c+1$ coordinates has a continuous section: place the remaining coordinates successively at one, two, and so on above the maximum retained coordinate. Its retract $M(c+1, c+1)$ is $\mathbb{R}^{c+1}$ minus its diagonal line and has homotopy type S^{c-1}. Hence π_{c-1} is nonzero, and H_{c-1} has $\mathbb{Z}$ as a direct summand. Finally, $T(n, n-1)$ is exactly the boundary of the $(n-1)$-dimensional task permutohedron, proving the sphere statement. Fundamental groups depend only on the two-skeleton; its path homotopies are precisely legal backtracks, squares and braid replacements. This demonstrates the coherence assertion. □

For $(n, c) = (3, 2)$, this recovers P.20's circle obstruction. At $(4, 2)$, T has 24 vertices, 36 edges and six square faces. Each square has a free edge belonging to no other retained face. Collapsing the six squares along such edges leaves a connected graph with 24 vertices and 30 edges, hence

$$H_1(K(4, 2); \mathbb{Z}) \cong \mathbb{Z}^7.$$

At $(4, 3)$, K's cell counts by dimension are $(1944, 5400, 5340, 2184, 302)$, totaling 15,170. The smaller model T has $(24, 36, 14)$, totaling 74; it is a two-sphere. This size reduction preserves the homotopy

type and its obstructions. It is not an execution quotient: T's vertices are task orders, not complete event histories. No preservation of every native answer, cost, permission or future continuation follows.

The full derivation supplies the written proof and relates it to established arrangement-complement and semaphore execution-space theory. The contribution here is the exact binding to the declared native guard and its legal-cell convention. The Lean module addresses prefix replay, the all-refinements capacity criterion, tagged refinements and selected order/cycle laws. These are separate from the written topological proof; they do not formalize the nerve theorem or sphere classification. The finite replay checks declared instances, native guard correspondence and cell/homology certificates; finite checks do not prove the theorem's unbounded quantifiers. Fresh run results and formal axiom dependencies belong in the verification receipt.

The full complex is essential for higher homology: discarding its higher cells can introduce additional classes. The theorem covers one count key and the specified paired events. Additional precedences, multiple keys, repeated loops, identification and filling require new analysis. It advances O03 for this family without asserting a classification of every guarded presentation.

P.23 Sparse integral transport and polynomial bounds at fixed degree

The complete signed construction and bounds make P.22's reduction effective on supplied chains. Fix the same tagged events, legal-cell convention and integral cellular differential. There are explicit augmentation-preserving chain maps and a degree-one operator

$$\begin{aligned} p &: C_*(K(n,c)) \longrightarrow C_*(T(n,c)), \\ i &: C_*(T(n,c)) \longrightarrow C_*(K(n,c)), \\ pi &= 1, \quad ip - 1 = \partial H + H\partial. \end{aligned}$$

The image of a requested cell is computed from its own carrier and recursive boundaries. No list of all execution cells, all task cells, or all linear extensions is needed.

Theorem (minimum-sweep sparse transport). Use the minimum-sweep contraction specified below, and fix a largest queried degree d. Put

$$R = \binom{2n}{2}, \qquad B = 2^{d+2} - 2, \qquad C = \max(2, RB).$$

For $0 \le q \le d$, a degree-q basis cell has projection and inclusion images with total absolute coefficient sum at most C^q, and homotopy image with sum at most $2R(q+1)C^{2q}$. For an input with s nonzero coefficients, these bounds multiplied by s bound the output supports. If the coefficients have at most b magnitude bits and M is the applicable unit-column bound, the output uses

$$O(sM[n\log(n+1) + b + \log(sM+1)])$$

bits. Evaluation has polynomial bit complexity in n, s, b for each fixed d. Checking a supplied certificate is polynomial in n and that certificate's own sparse encoding, with explicit bounds in the supplement. The exponent depends on d; no polynomial bound uniform in degree is asserted.

Construction and proof. A native cell A determines the interval order P_A, in which a task precedes another when its finish block is strictly before the other's start block. P.22 establishes its width is at most c. A task cell B determines the order P_B between its blocks. Write Q_P for the event order generated by pair precedences and these task precedences. These carriers shrink under faces.

For a finite order Q, choose its lexicographically first extension. Move its first, minimal label left across the blocks before it, assigning each merged cell the incidence sign of the face that puts the label after that block. The signed sum s_x satisfies $\partial s_x + s_x \partial = 1 - I_x D_x$, where D_x deletes a singleton x and is zero on a nonsingleton containing it, and I_x prepends the singleton. The supplement verifies all cancellation signs. Recursing on the remaining labels constructs h_Q with $\partial h_Q + h_Q \partial = 1 - r\epsilon$. It produces at most $\binom{m}{2}$ distinct unit-coefficient cells on m labels, using $O(m^3)$ label operations. Every crossed label is incomparable with the minimal one, so the construction remains inside $L(Q)$.

Map vertices to the corresponding roots and sequential paired words. For positive degrees, define

$$p(A) = h_{P_A} p(\partial A), \qquad i(B) = h_{Q_{P_B}} i(\partial B), \qquad H(A) = h_{Q_{P_A}} (ip(A) - A - H(\partial A)).$$

The homotopy formula also applies to vertices with $H\partial = 0$. Nested carriers and the contraction identity prove both chain laws and the homotopy identity by degree induction. The composite $pi(B)$ is carried by the closed face B, whose top chain group has only that generator. It fixes vertices; its boundary then forces $pi(B) = B$ in every degree. A degree-q cell has at most $2^{q+1} - 2$ facets. Combining this bound with the quadratic contraction bound gives the stated recurrences and estimates. All choices are capacity-independent, so p, i, H commute strictly with increasing-capacity inclusions. □

For a cycle z, the protected question is exactly whether it bounds over the integers. If $\partial b = p(z)$, then

$$\partial(i(b) - H(z)) = z.$$

The reconstructed filling is checked directly against z, using legality tests and boundaries of the supplied cells. The checker needs no transport producer or ambient matrix. Closed cochains pull back along p, and their pairings with included cycles are preserved by $pi = 1$. A pullback cochain may be represented by its evaluation rule; sparse input does not guarantee sparse support for its complete coordinate vector. If a finite cochain support is supplied, checking all legal adjacent-block merges of its support establishes global closure, rather than checking only visited cells.

The minimum-sweep maps are an explicit second version. The earlier first-descent contraction has different coordinates and only an exponential general bound from its inversion-descent proof. Both versions are retained and named in the native interface. Generic contractions and chain checking belong to `omega_homology`; WDP supplies the event carriers. The written all-size proof, selected Lean algebraic/recurrence proofs, and bounded Rust execution evidence remain distinct. These results concern the integral boundary question; they confer no authorization and do not identify arbitrary native histories or establish an empirical search advantage.

Appendix Q — Sound transfer of learned rejection rules and its accounted cost

The full derivation and correction record connect checked rejection to reuse without promoting a single failed candidate into an unproved general rule. The set-theoretic ingredients and cost arithmetic are standard. The WDP contribution here is their binding to declared tasks, adapters, candidate identity and independently replayed evidence. Supplemental local `ANT-*` results and GEN-WDP-48 leave the 154 canonical rows unchanged.

Q.1 One rejection versus every rejection

Declare candidate carriers `X,Y`, valid sets `G_S subset X,G_T subset Y`, and total adapter `alpha:Y->X`. A sound source rejection rule rejects `D subset X\G_S`. Then

$$\big[\forall D \subseteq X \setminus G_S :\ \alpha^{-1}(D) \cap G_T = \emptyset\big] \quad \Longleftrightarrow \quad \alpha(G_T) \subseteq G_S.$$

Proof. Inclusion makes a valid target's image disjoint from every such `D`. Conversely, if `y` is valid but `alpha(y)` is invalid, the singleton `D={alpha(y)}` is a sound source rejection that rejects `y`. For one fixed rejection rule only, the exact condition is disjointness of its preimage from `G_T`; this can hold when the full adapter audit fails. Image inclusion expresses one-way validity preservation, not general predicate factorization. A soundness certificate need not be a linear dual witness.

Q.2 Task parameters and partial adapters

For task families `G_S(d),G_T(e)` and task-data map `beta`, the overall criterion is `alpha(G_T(e)) subset G_S(beta(e))` for every admitted `e`. For derivative validation, the concrete condition on valid candidates is `partial(alpha(F))=beta(e)` whenever `partial(F)=e`. Identity maps satisfy this for the parameterized predicate `partial(F)!=e`. The source-bound predicate `partial(F)!=x` does not become safe for integrand `5` just because its code is reused: `F=5x` is then valid.

For a partial adapter with domain `A subset Y`, declare that out-of-domain inputs pass through unfiltered. The overall condition becomes `alpha(G_T intersect A) subset G_S`. If out-of-domain instead means reject, one additionally needs `G_T subset A`. A missing-input error is neither policy by default. Presence or equality of metadata fields does not prove the semantic inclusion; the actual predicates and maps must be checked.

Q.3 Conservative accumulation

If every transferred rejection `R_j` is disjoint from `G_T`, then their union is disjoint from `G_T`, so every valid candidate survives. This follows elementwise and requires either a checked transfer for each rule or a complete adapter audit plus each source soundness certificate. Source soundness alone is insufficient. Per-rule checks against the full declared valid set compose; a general adapter audit is a reusable alternative. Checking only candidates left by a potentially unsound earlier filter cannot establish preservation of all valid candidates. Incremental checking is sound when the invariant that all valid candidates remain is itself certified and each new step preserves it.

For the linear repair task, an exact annihilator `y D=0` gives a reusable rejection predicate `y b!=0`, sound for `b in D(A)` for any declared repair set `A`. Capacity bounds additionally give quantitative weighted-norm lower bounds; they are unnecessary for qualitative nonmembership. If acceptance allows `Phi<=epsilon`, rejection requires a certified lower bound strictly greater than `epsilon`, not just a nonzero pairing. Changing `D`, the coefficient domain or the task requires a new justification. Retaining all valid candidates establishes neither completeness of the rejection family nor efficient search among the survivors.

Q.4 Amortization on a declared workload

For `N` candidate occurrences, setup cost `C_0>=0`, guard cost `g>=0`, full evaluation cost `v>0` and `h` sound early rejections, the declared constant-cost model gives

$$T_{bare} = Nv, \qquad T_{guard} = C_0 + Ng + (N-h)v, \qquad \text{saving} = hv - Ng - C_0.$$

Strict benefit occurs exactly when `h v > N g + C_0`. With `N>0` this is `h/N > g/v+C_0/(Nv)`. If `g>=v`, strict benefit is unrealizable; equality of costs can occur. If all costs vanish there is no strict benefit.

For `N=10000,v=1,g=1/1000,C_0=2`, at least `13` hits are required. This is a conditional calculation, not a measured hit rate or performance claim. A singleton source rejection can have many target preimages even on a pool of distinct candidates. Repeated occurrences also count repeatedly, so a singleton guard need not have at most one hit.

Comparing `C_alpha+k c` with `k c'` for reusable versus per-rule audit is also conditional arithmetic. Division by `c'-c` is valid only when that difference is positive; no theorem forces one implementation's check cost to exceed another's. Target hit counts require a named workload or a proved bound. There is no canonical uniform frequency on an infinite rational carrier, and source hit rates do not supply target hit rates.

Q.5 Evidence, reuse and remaining obligations

The finite checker and original submission retain separate source soundness, adapter, per-rule transfer, accumulation and cost scopes. Replaying a candidate's certificate never grants authorization. A finite table audit establishes its table instance; a symbolic law is needed to cover an infinite polynomial or cochain family. Porting a guard to a new implementation requires preservation of the actual predicate and task binding, with a new cost assessment.

Restricting the valid target set or enlarging the valid source set preserves an already proved adapter inclusion. Enlarging the valid source set can invalidate the source rejection rule, however, so the complete transfer bundle must retain both obligations. These results justify sound reuse under explicit premises. Demonstrating that accumulated knowledge makes new tasks cheaper requires measured or proved target-workload costs in addition to soundness; general recursive re-entry remains a separate obligation.

Q.6 A checked finite example across two representation changes

The full finite derivation separates sound answers, coverage of all original solutions and preservation of a cost bound. Let `D={-1,0,1}`, `X0=D`^{2}, and let a valid original answer satisfy `x+y=d`, where `d∈{-1,0,1,2}`. Define exact image carriers and maps

$$f_{01}(x,y)=(x+y,x-y),\quad X_1=f_{01}(X_0),\quad g_{10}(u,v)=((u+v)/2,(u-v)/2),$$
$$f_{12}(u,v)=(u,v+u),\quad X_2=f_{12}(X_1),\quad g_{21}(r,t)=(r,t-r).$$

The maps are inverse bijections on these carriers. The composed original-answer decoder is `H(r,t)=(t/2,r-t/2)`, inverse to `F=f12∘f01`. The valid sets are `G0(d)={x+y=d}` within `X0`, `G1(d)={u=d}` within `X1` and `G2(d)={r=d}` within `X2`; thus `F(G0(d))=G2(d)`. For `d=1`, the complete solution paths are `(0,1)→(1,-1)→(1,0)` and `(1,0)→(1,1)→(1,2)`, decoded to the respective original pairs. Reporting the first coordinate alone does not decode the pair.

For the *actual* emittable set `E⊆X2`, including menu and fuel restrictions,

$$H(E)\subseteq G_0(d)\iff E\subseteq G_2(d),\qquad G_0(d)\subseteq H(E)\iff G_2(d)\subseteq E.$$

The first equivalence follows from pointwise validity transport. For the second, every valid original answer has a unique preimage under the bijection `H`; hence all are emittable exactly when those valid preimages belong to `E`. Soundness can hold vacuously when no solution is reachable, so it does not imply coverage.

The parameterized rejection R_d(u,v)≡u≠d transfers to level 2 through g21, giving r≠d; its level-0 pullback uses f01, giving x+y≠d. Fixed R1 is not a cross-task rule: at d=2 it rejects the unique valid level-2 candidate (2,2). With a mandatory guard and no fallback the resulting finite arena has an initial-containing, target-disjoint loss trap. Restoring the original route restores solvability. Also (1,0)∉X1 but (1,0)∈X2: admitting candidates by field names would accept an out-of-domain level-1 decoder value (1/2,1/2).

With proposal/check/emission prices 1,2,1, the supplied menu permits naming any candidate. Its optimal resolution cost is exactly 4: select a valid pair, check and emit; every successful run must pay these three charges. Making a safe guard mandatory adds 1/2 per occurrence, giving operating cost 9/2 and cost 15/2 including setup 3. Coverage survives, but a protected total bound of 4 does not. Installation is therefore not necessary to solve this finite profile within budget 18, including task d=2.

A **different, restricted menu** removes named selection, methods and alternate routes and draws without replacement in adversarial order. With v valid candidates and s soundly rejected invalid ones among nine, stopping at the first valid emission gives

$$T_0 = 31 - 3v, \qquad T_R = 36 - \tfrac{7}{2}v - 2s.$$

These values count all invalid candidates first, an attainable worst case. Operating nonincrease requires 4s≥10-v; including setup on the guarded side requires 4s≥16-v. They are scan-family values, not global optima of the named-proposal menu. Even maximal rejection at d=2 costs 33/2 to operate and 39/2>18 with setup. A check-free guarded extension would cost 5/2 per successful candidate, not 2, and is not the supplied check-retained menu.

One successful shared-ledger run at d=1 pays 4 for the parent answer, 3 for learning/checking/installation, 1 for transport to level 2, and 9/2 for each of its two decoded child answers: total 17≤18. Re-entry carries spent fuel unchanged. A separate outer continuation checker validates all fifteen steps, receiving completed solve episodes without treating an arena's terminal emission as a free continuation. Ten hostile traces are refused for skipped checks/setup/transport, changed bindings, incorrect emissions, reset/undercharge or an over-budget repeated episode. No wrapper optimality claim is made. Resetting after two transports and installation would expose 5+18=23 against the original 18, not create a valid new budget.

Reuse after a change follows the affected mathematical dependencies. A checked finite verdict-equivalence table can preserve acceptance across a checker refactoring, but cost additionally depends on the menu, successors, reports, targets and prices. A changed digest is a custody mismatch, not itself a mathematical counterexample. Adding an unused expensive action can leave the optimum unchanged; optimality still needs its lower inequality checked. A checker that admits sums in {d-1,d} instead admits invalid original (0,0) at d=1.

The Python replay checks all nine carrier rows, 6,144 emission subsets, 6,144 rejection subsets, 46 independently replayed finite arena certificates and 22 controls. The Lean 4 formalization checks the image/inverse/validity laws, generic soundness and coverage criteria, safe union, guard coverage, scan arithmetic, fuel laws and finite witnesses. It uses standard Lean logical axioms, with no custom axioms or sorry. Its image-restricted integer decoder and the Python off-image rational control have separate stated scopes. These checks complete this finite example, not a general correspondence between recursive engine execution and external tasks.

Appendix R — A plain-language picture of observers and authority

This appendix is an interpretive companion for readers outside the mathematics. It states no new result; every claim in it points to a section where the corresponding construction is defined or proved.

The rule card. Three children each say their block tower is best. The dispute has no answer until someone puts a card on the table: today "best" means tallest, measured with this stick from the floor; a loose block may be fixed once; a tower that falls does not count; equal heights are a tie. The card is the observer of Chapter 10. It is not a child, an eye or a tower. It says what is compared, which fixes are allowed and which outcomes exist. A new card ("best" means the tower that survives a gentle push) can name a different winner with the same towers and children. Neither card is wrong; they evaluate different declared questions (§10.1).

The clock. Two clocks read 12:00 and 12:03. "Three minutes" is a defect only once a time standard, a unit, a tolerance and the permitted repair (recalibration or replacement) are declared. In a classroom the difference may be acceptable; in a trading system three milliseconds may not be; in a distributed database the relevant defect may be whether event order can still be reconstructed. The same disagreement yields different defects under different evaluators (§§7.5–7.6, CH10-N01).

Five questions to ask of any reported result. What was compared? What counted as a defect? Which repairs were allowed? What residual remained? Which declared rule assigned the final status? A report that cannot answer them is a proposal, not an evaluation (§§10.1–10.2, §15.6).

Gauge, without fog. A difference that an allowed redescription removes (units, offsets, naming, formatting) is not an obstruction. A difference that survives every allowed redescription is one (§4.12). Seeing a difference does not make it real, and calling everything a convention does not make it go away.

Evaluation is not permission. A passed check is evidence for the question the check declares. Whether anyone may act on it is decided by a separately declared authority regime: who may propose, who may grant, within what scope, with what appeal (Chapter 15, Appendix P.6, P.19). The useful division of labour is that a model or person proposes, an evaluator evaluates, and a regime may authorize. Collapsing any two of these hides the step at which a claim acquired its standing.

Appendix S — Certified progress in a finite belief arena

This is the dynamic counterpart of Appendix Q's static rejection-rule transfer laws. A restrictive transferred guard can raise cost or destroy winning; a transfer violation alone is not a losing certificate. The submitted appendix and supplied adversarial review are retained byte for byte in source archive manifest in `docs/source_archive/appendix_s_review/`. The reviewed propositions below keep their proposed WDP-CHS identifiers, outside the 154-row canonical ledger.

S.0 Evidence and contribution

D01–D10 are definitions; P01–P19 are finite mathematical propositions; C01–C04 are conditional interpretations; X01–X08 are retained counterexamples; N01–N08 are nonclaims; O01–O03 remain open. Written proofs, Lean mathematical models, and concrete Python replay are distinct evidence.

Finite reachability attractors, alternating bisimulation, partition refinement, and minimax are standard mathematics. Their task-bound certificate specifications, exact cost obligations, and separation of evaluation from authorization are the construction studied here. No parity-game theorem or general Tarski theorem is needed for these elementary finite proofs.

Implementation, certificate laws, and quotient laws have separate scope. The proof/evidence supplement maps each obligation to its actual evidence.

S.1 Declared inputs and interpretation

WDP-CHS-D01. Declare a nonempty finite world carrier S, finite actions A, enablement En(s), nonempty successor sets δ(s,a) for enabled pairs, post-state reports Λ(s',a), nonempty initial belief b_0, and a finite listed resolution vocabulary R with total validation tables. Successful targets T are sets of nonempty beliefs, supplied by the model, independently of certificate keys. Targets stop execution. Reachable enabled belief/action pairs have positive rational costs. Task, assumptions, and authorization requirements are inputs.

WDP-CHS-D02. A belief is goal-resolvable when one listed resolution passes the supplied check in every possible world. Validity does not itself enable an emission blocked by a guard. Every enabled valid emission must deterministically reach a declared successful terminal. Therefore

$$V(b) \leq \min_{r:\,\text{valid and emit}(r)\,\text{enabled at}\,b} c(b, \text{emit}(r)).$$

This is an upper bound. Another emission or a preliminary route can be cheaper. Equality requires accounting for the alternative routes in the specific model.

WDP-CHS-D03. Check successful target membership before reporting a dead end. A successful terminal with no actions is successful. Otherwise use:

Report	Required evidence
Refused	Controller actually performed a declared refusal action
Stalled	Non-target belief has no admissible actions
Exhausted	Checked optimal finite value exceeds available rational budget
Impossible	Checked initial-containing loss trap and an explicit task policy admitting this report
LosingInDeclaredArena	Checked loss without that reporting policy
UndeterminedBudget	Only a feasible upper bound exceeds the budget

An invalid certificate yields RefusedCertificate; it does not mean the controller executed a refusal. Chapter 20's Proved/Refuted/Refused/Undetermined meanings remain unchanged. Report labels are not new institutional authorizations.

WDP-CHS-N01. Reporting precedence never removes dead ends from a trap or turns a successful terminal into a loss.

S.1.1 Assumptions and conditional conclusions

Assumption	Exact scope
A1	Accepted resolutions decode to valid original answers
A2	Finite listed resolution vocabulary and terminating table checks
A3	Each supplied resolution is correctly classified in both directions
A4	Every listed post-state observation is adversarially realizable
A5	Infinite nonterminal play is losing; legal loops can persist indefinitely
A6	All reachable enabled prices are at least some rational $\varepsilon>0$

WDP-CHS-C01. A checked win supplies a genuine original answer if targets carry sound original-answer decoders and A1 holds. Proof: compose P01's target reachability with the decoder, as in §20.6's backward transport.

WDP-CHS-C02. A non-target with empty admissibility is losing. Proof: its singleton is a target-disjoint trap with vacuous closure.

WDP-CHS-C03. Checked optimal value above budget rules out guaranteed success within that budget. Proof: P16's lower bound applies to every successful plan. A finite success above budget is not an unrealizable task.

WDP-CHS-C04. An arena loss implies source unsatisfiability only with an explicit map taking every original solution to an admissible winning strategy, preserving task and requirements. Proof: contrapositive of that coverage map. A3 alone checks supplied resolutions and supplies no missing solution or enabled operation. This is §20.6's constructive forward coverage obligation.

WDP-CHS-N02. Loss is relative to the declared arena, moves, and target. **WDP-CHS-N03.** Effective checks do not prove A1 or A3.

S.2 Beliefs and dynamic updates

WDP-CHS-D04. A belief is a nonempty subset of S. Let B be the beliefs reachable from b_0 before stopping at targets. Finite exploration gives $|B|\leq 2^{\wedge}|S|-1$.

WDP-CHS-D05. Adm(b) consists of actions enabled at every $s\in b$.

WDP-CHS-D06. For admissible a,

$$\mathrm{Post}_a(b)=\bigcup_{s\in b}\delta(s,a), \qquad \mathrm{Obs}_a(b)=\{\Lambda(s',a):s'\in \mathrm{Post}_a(b)\},$$
$$b'_{a,o}=\mathrm{Post}_a(b)\cap\Lambda_a^{-1}(o), \qquad \mathrm{Succ}(b,a)=\{b'_{a,o}:o\in \mathrm{Obs}_a(b)\}.$$

Each successor is nonempty because its observation is realized in the post-set. For $b=\{0\}$, $\delta(0,a)=\{1\}$, and observation fibre $\{1,2\}$, the posterior is $\{1\}$; neither prior intersection nor the unrestricted fibre is correct.

WDP-CHS-D07. A resolution procedure returns a listed r with a bound supplied check. The finite implementation represents validation tables. Computation, checking, and emission remain separately declared operations and costs.

S.3 Sharp finite reachability

WDP-CHS-P01. Starting with W_0=T∩B, put

$$W_{n+1} = W_n \cup \{b \in B \setminus T : \exists a \in \mathrm{Adm}(b), \mathrm{Succ}(b,a) \subseteq W_n\}.$$

There are at most |B| strict rounds. The stable W is exactly the winning region. It admits a memoryless strategy on W\T and a natural rank decreasing on every selected successor, with rank zero at targets. Its complement L=B\W is a target-disjoint trap: every admissible action at an L-belief has an L-successor.

Proof. Every strict round adds a new belief. At each admitted nonterminal choose a witnessing action and its first positive layer as rank. All successors have smaller layers, so every adversarial play reaches a target within that rank. At stabilization, an action with every successor in W would have admitted its source. Hence each enabled action at an L-belief has an L-successor. Choosing these successors prevents target reachability against every controller, including a history-dependent one; play either continues in L or ends at a non-target dead end. This establishes both directions. ■

WDP-CHS-P02. An avoidably reachable trap does not establish initial loss. Proof: s has finish$\to_5$t and trap$\to_1\ell$ with ℓ looping. $\{\ell\}$ is a trap, but s wins. A sound loss witness must contain b_0 itself. ■

WDP-CHS-P03. Winning need not be upward closed under belief inclusion. Proof: p enables only a→t and q only b→t; both singletons win but {p,q} stalls. ■

WDP-CHS-P04. An initial-containing target-disjoint L with an actual L-successor for every L/enabled pair establishes loss without computing W. Proof: follow the supplied closure at every controller choice; targets remain excluded. Dead ends remain in L, where closure is vacuous. ■

WDP-CHS-P05. Finite-table rejection-rule transfer α(G_T)⊆G_S is decidable and a failure has a target-good/source-bad element. It need not imply modified-arena loss. Proof: remove a direct cost-one emission but retain a cost-one repair followed by cost-one emission. Value rises from one to two and remains finite. Loss still requires P04. ■

WDP-CHS-P06. P01 quantifies exactly over the listed successors. Changing adversarial realizability can change winning. Proof: a sole action to a target wins; adjoining an unlisted looping non-target successor makes it losing. Suppressing that bad successor reverses the effect. ■

WDP-CHS-P07. W and L partition B with explicit controller and adversary strategies on their respective sides. Proof: the constructions of P01 cover B; rank descent and target-disjoint closure cannot hold at the same initial belief. ■

WDP-CHS-N04. This reachability argument does not prove parity objectives.

S.4 Submitted certificates

WDP-CHS-D08. WIN_OPTIMAL supplies a complete disjoint W/L partition of input-derived B, $b_0 \in W$, $T \cap B \subseteq W$, finite nonnegative rational values and exact natural ranks exactly on W, strategy exactly on $W \setminus T$, and resolution witnesses exactly where a common supplied resolution exists. Terminal values and ranks are zero. WIN_FEASIBLE supplies the same sound partition/strategy but does not claim lower optimality inequalities.

WDP-CHS-D09. LOSS supplies $L \subseteq B$ containing b_0 and excluding all model targets. Closure rows cover exactly all L/enabled pairs and name actual successors in L. The stalled list is exactly those L-members with no admissible action. Empty win fields and a false optimality flag are required. A standalone L need not cover B.

Both formats bind schema, model digest, task, assumptions, authorization metadata, and initial belief. The checker reconstructs targets and B from original inputs, checks every stalled claim, every map domain, exact scalar/rank types, and each supplied resolution on every possible world. It never replaces L by $L \setminus$stall. It neither searches for a strategy nor substitutes a solver's answer.

WDP-CHS-N05. A checked packet records evidence, not operational authority.

S.5 Belief-arena quotient and constructive refinement

WDP-CHS-P08. Let Π partition B. Within every block require equal target truth, enabled action sets, action prices, and successor-block sets. The map $q: B \to B/\Pi$ preserves winning and every achievable finite rational budget, hence optimal values including losing infinity. These are sufficient conditions, not necessary conditions for accidental agreement of one answer.

Proof. Define quotient tables by a representative; the laws make them well-defined. Every fine successor maps to a quotient successor, and every quotient successor at q(b) has an actual fine successor at b. Induction on horizon establishes target reachability equivalence: at horizon zero target tests agree; at the next horizon actions and full successor tests agree by the two directions and induction. Subtracting equal action prices gives the same induction for rational remaining budgets. Quantify over horizons. ■

This is a partition of beliefs. Targets and belief costs cannot be tested on untyped concrete-world representatives.

WDP-CHS-P09. A new target separating a merged block cannot factor through q. Proof: equal q-images would require equal target truth. ■

WDP-CHS-P10. Accidental cost equality does not imply congruence. Proof: p and q both have cost-one $d \to t$; p additionally has $a \to_2 t$ and q has $b \to_2 t$. Both values are one but action sets differ. Retarget p as successful: $V(p)=0$, whereas q's only successors t are non-target dead ends and $V(q)=\infty$. ■

WDP-CHS-P11. Information compression can destroy singleton wins, as P03 shows. That merge is not a validated P08 game quotient. ■

WDP-CHS-D10. Starting at Π_0, split each block by the finite signature

$$(1_T(b),\ \{(a, c(b,a), \{q_k(y) : y \in \mathrm{Succ}(b,a)\}) : a \in \mathrm{Adm}(b)\}) .$$

Repeat until unchanged. Blocks are only split; new measurements are not invented.

WDP-CHS-P12. The procedure terminates in at most $|B|-|\Pi_0|$ strict rounds and yields the coarsest P08-congruence refining Π_0.

Proof. Every strict round increases the number of nonempty blocks by at least one, bounded by |B|. Stable signatures give all four congruence laws. Let Ξ be any congruence refining Π_0. If it refines Π_k, its common target/action/ price data agree inside every Ξ-block; equal successor Ξ-block sets imply equal successor Π_k-block sets. Thus the next signature split never separates a Ξ-block. Induct over rounds. Ξ refines the stable output, proving coarseness and uniqueness. ■

WDP-CHS-P13. A refinement verification record binds original model, input partition, stable partition, and strict-round count. Reconstructing signatures checks the result; a changed model needs fresh replay. Proof: all four signature components are finite original tables. A verification record does not implement an external sensor. ■

WDP-CHS-N06. Deploying q requires an actual observer realizing its distinctions and preserving future enabled operations and prices. Section 14.7 constructs and optimizes such observation procedures for declared finite transition/report tables; physical realization and a new engine's source correspondence still require their own evidence.

WDP-CHS-P14. Quotient strategies lift by $\sigma(b)=\bar{\sigma}(q(b))$; class-constant fine strategies descend with the same worst-case budgets. An arbitrary fine strategy need not literally descend. Optimal quotient strategies exist by P15 and lift as optimal class-constant fine ones. Proof: P08 matches paths in both directions with equal cost and target truth; class constancy makes $\bar{\sigma}$ well-defined. ■

S.6 Costs, rank, and independent optimality

WDP-CHS-P15. On a nonterminal winning arena, let ε be the minimum enabled reachable price, c_max its maximum, R the maximum attractor rank, M=c_max R, and $H=\lfloor M/\varepsilon \rfloor$. Use synchronous finite-horizon recursion, with $V_0=0$ on targets and infinity elsewhere:

$$V_{n+1}(b) = \begin{cases} 0, & b \in T, \\ \min_{a \in \mathrm{Adm}(b)} \left(c(b,a) + \max_{y \in \mathrm{Succ}(b,a)} V_n(y) \right), & b \notin T. \end{cases}$$

The empty minimum is infinity. V_H is the exact optimum on W; outside W it is infinity. If only a successful initial target is reachable, H=0 and value zero.

Proof. An attractor strategy has cost at most M at every winner. Every strategy of guaranteed cost at most M uses at most H actions on every branch, because n actions cost at least $n\varepsilon$. There are finitely many contingent plans of depth H; backward induction computes their minimum. Any better strategy would also have cost at most M, hence depth at most H. This establishes attainment and optimality. Decomposing an optimal plan at its first action establishes the Bellman equation; conversely optimal successor plans can be composed after observing that action. Every minimizing action has $V(y) \le V(b)-\varepsilon$ for all successors. Its graph is acyclic. In increasing value order set $\rho(t)=0$ and $\rho(b)=1+\max_y \rho(y)$. This is the strategy's rank, not necessarily its earlier attractor rank. ■

WDP-CHS-P16. S.4's checked partition, rank, and terminal boundaries, together with the selected-action equality and all-action lower inequalities,

$$V(b) = c(b,\sigma(b)) + \max_{y \in \mathrm{Succ}(b,\sigma(b))} V(y),$$

$$V(b) \le c(b,a) + \max_{y \in \mathrm{Succ}(b,a)} V(y) \quad (a \in \mathrm{Adm}(b)),$$

are sufficient for exact optimality. Infinity in the second line is permitted only for a successor in the checked losing trap. P15 supplies a certificate in this normal form for every winning instance.

Proof. Induct on submitted rank to obtain a terminating selected strategy with cost at most V(b); terminal cost is zero. For a successful contingent plan, induct on finite depth and choose a successor attaining the lower line's maximum. An L-successor contradicts guaranteed success by trap closure; a W-successor has subtree cost at least V(y) by induction. Thus worst-case total is at least c(b,a)+V(y)≥V(b). Finite branching implies any strategy all of whose maximal plays reach a target has bounded depth: an unbounded tree would have an infinite nonterminal branch. Hence the lower argument also covers history-dependent guaranteed strategies. Upper and lower bounds agree. ■

Terminal zero fixes a constant otherwise unconstrained by local Bellman equalities. These laws define this certificate normal form; differently formatted proofs of one initial optimum need not contain all the same tables.

WDP-CHS-P17. A certified rank-decreasing strategy costs at most c_max ρ(b). Proof: at most ρ(b) paid actions occur and each costs at most c_max. ■

WDP-CHS-P18. D03's reporting procedure returns one label under its evidence and precedence. Proof: successful targets are tested first, then non-target stalls; W and L are disjoint, and budget comparisons use winning finite values. Feasible upper cost above budget supplies no negative budget result. ■

WDP-CHS-N07. Losing infinity, finite budget exhaustion, and bounded computation left Undetermined are different claims.

S.7 Optional actions and restrictive guards

WDP-CHS-P19. (1) Adding optional actions while preserving all old targets, enablements, prices, and successor sets cannot remove a win or raise optimum. Proof: every old strategy remains available unchanged. Removing adversarial branches of an old action also cannot worsen its guaranteed budget. (2) A restrictive guard can raise finite cost or destroy winning; actual loss requires P04. P05 gives the finite-cost alternative; replacing a sole emission with a non-target loop gives a singleton trap. (3) Setup payback requires finite operating savings greater than C_learn+C_check+C_install. Positive savings alone do not suffice. For a declared deployment distribution or count, sum finite savings and compare the specified total setup cost. ■

WDP-CHS-N08. Deployment probabilities, empirical hit rates, and setup costs are external declarations, not consequences of the game theorem.

S.8 Retained examples and changed-task controls

X01/X02: P03 refutes the singleton-world hope argument; D06 refutes prior/fibre posterior substitutes. X03/X04: P10 separates accidental equality from structural congruence, and P11 demonstrates a destructive information merge.

X05: For $s \to_1 u \to_1 t$ and $s \to_5 t$, optimum is two. Budget 3/2 yields Exhausted, not Impossible. A cost-five feasible upper bound does not refute budget three. The selected cheap strategy has rank two, though the direct action admits s in attractor layer one.

X06: On the filled triangle with edges (01,12,02), $d_0 f=(f_1-f_0,f_2-f_1,f_2-f_0)$. For $\delta=(0,1,0),(1,1,0),(0,2,0)$, the unit L^1 quotient obstructions are 1,2,2. Primal f=0 attains these raw norms. Dual $y=(1,1,-1)$ annihilates d_0, has infinity norm one, and pairs to the same values. Thus all three are nonexact. Blind checks reporting nonexact classify exactness but do not identify the vector. An optional self-loop preserves the available classification emission win; the identification task with only blind loops is losing. R7 must name which task it tests.

X07: False stall u cannot mask X05's cheaper route. Terminal value 100 cannot shift the cost-one optimum to 101. Empty winning witnesses omitting b_0, certificate-defined target omissions, and successful-terminal loss claims are refused. Retaining all of L repairs both genuine dead-end loss and optimal winning games with alternative losing detours.

X08: F=5x is invalid for the antiderivative task f=x and valid for f=5. A source rejection rule rejecting it cannot be transferred unchanged to that target task. If the only valid emission is replaced by a non-target loop, its singleton is a trap. Another solving route prevents that inference. In a sole-emission cost-1/2 example the optimum is 1/2; this does not imply D02's old general equality.

S.9 Executable replay

The authoritative implementation is the separate producer, model validator, and checker under code/certified_progress. No embedded obsolete sketch is an alternative checker. The producer uses exact rational values and an internal missing-value marker for infinity; no float infinity is a serialized witness. Strict disk JSON admission refuses duplicate keys and nonfinite literals.

Replay retains original model, submitted certificate, verification record, mutation controls, and source hashes, with no model calls. The tests include all seven supplied review failures, avoidable traps, incorrect resolutions, coefficient/type refusals, changed bindings, target and partition controls, both triangle tasks, and independent policy enumeration over 4,096 small three-state/two-action games. These are concrete finite checks, separately from the all-finite written proof.

S.10 Acceptance and open boundaries

Obligation	Mathematical evidence	Concrete evidence
Initial membership and complete win partition	P01/P04/P16; Lean rank/trap laws	Forged initial/partition controls
Input targets and terminal zero	P04/P16	Target-as-loss and shifted-value refusals
Dead ends remain in L	P01/P04	Genuine dead-end and losing-detour replay
Exact optimum and selected rank	P15/P16; Lean upper/lower budget laws	Independent policies and cost-two witness
Dynamic posterior	D06	Changed-world and ambiguous-report controls
Congruence suffices	P08/P14; Lean two-sided transport	Quotient and retargeting tests
Constructive coarsest refinement	P12; Lean finite relation construction	Signature splitting and stability/bound checks
Guard failure is not automatically loss	P05/P19	Finite-cost and singleton-trap examples

Obligation	Mathematical evidence	Concrete evidence
Budget/report separation	C03/P18	Feasible versus optimal comparisons

WDP-CHS-O01. Full parity/alternating objectives need their own certificate laws and proofs.

WDP-CHS-O02. Original-world validation soundness, completeness, and constructive winning-strategy coverage remain profile-specific adapter duties. Declaring A1–A3 does not discharge them.

WDP-CHS-O03. Native Rust binding and verification of Python source remain open. The written theorem, Lean model, and concrete replay have separate scopes. Physical observer realization, general guarded-history coherence, and GEN-WDP-O01–O05 retain their existing open meanings.

Theorem map — local acceptance, projection obstruction, checked repair, and WDP

Four names meet on one finite question, and they do not do the same work.

- Local acceptance supplies local satisfaction and the actual compatibility defect. A vector is locally invisible when it lies in $K = \ker Q$. It is an actual source defect when it also lies in the realizable image $R = \operatorname{im} D_s$.
- Projection Obstruction identifies what a report hides. For an observer A, the hidden set is $\ker A$. A hidden realizable defect is a nonzero vector of $R \cap \ker A$.
- Checked repair supplies checked transport and declared repair equivalence. Two vectors represent the same repair class when their difference lies in $B = \operatorname{im} D_r$. A certificate is accepted only when an independent checker replays the submitted witness. That acceptance is not an authorization to perform a repair.
- WDP retains the original question and the realizability obligations: which added observations distinguish realizable defects from one another and distinguish repair classes, and which of those observations are obtainable from the declared operations. An algebraic identity $B = V$ makes the quotient trivial. It does not exhibit an admissible source operation.

The equal-image count in §10.6 is the special case $R = B$. Theorem WDP-DIM-1 does not change that statement. Imported linear algebra is named in Appendix T before it is used. The WDP construction is the separation of R from B, the joint row count, the restricted menus, and the separate operation witness.

Targets 2–5 of `docs/five_outstanding_derivations.md`, general guarded-history coherence, unrestricted source refinement, physical reconstruction, and joint continuum foundations stay open.

Appendix T — Sharp completion with distinct images (WDP-DIM-1)

Coefficient field. $\mathbb{Q}$. All spaces below are finite-dimensional rational vector spaces, given by explicit matrices. No completeness, topology, or estimate is used.

Data. Linear maps $D_s : U \to V$, $D_r : F \to V$, and $Q : V \to Y$. Write $R = \operatorname{im} D_s$ for the realizable defect space and $B = \operatorname{im} D_r$ for the algebraic repair space. Do not assume $R = B$. Let $K = \ker Q$, and let $E : V \to \mathbb{Q}^m$ be an additional row map. Set $A = (Q, E) : V \to Y \oplus \mathbb{Q}^m$.

Imported linear algebra, used as algebra and not as a WDP theorem. Rank-nullity. For subspaces X, Y, $\dim(X + Y) + \dim(X \cap Y) = \dim X + \dim Y$. A linear map is injective

exactly when its kernel is $\{0\}$. If $\pi : Y \oplus \mathbb{Q}^m \to (Y \oplus \mathbb{Q}^m)/A(B)$ is the quotient projection, then $\ker(\pi \circ A) = A^{-1}(A(B)) = \ker A + B$, so the induced map $V/B \to (Y \oplus \mathbb{Q}^m)/A(B)$ has kernel $(\ker A + B)/B \cong \ker A/(\ker A \cap B)$, and that induced map is injective exactly when $\ker A \subseteq B$.

WDP-DIM-1. Let $T = R \cap K \cap B$. The three sharp row counts are

$$\begin{aligned} m_{\text{raw}} &= \dim(R \cap K), \\ m_{\text{quot}} &= \dim K - \dim(K \cap B), \\ m_{\text{both}} &= \dim K - \dim(K \cap B) + \dim T. \end{aligned}$$

The joint minimum row count is $m_{\text{both}} = \dim K - \dim(K \cap B) + \dim T$. The sum of the separate minima exceeds the joint minimum by $\dim((R \cap K)/T)$. The two counts agree exactly when $R \cap K \subseteq B$, which includes the equal-image case $R = B$.

T.1 Raw sufficiency

Raw sufficiency means that A distinguishes realizable defects: $R \cap \ker A = \{0\}$. Since $\ker A = K \cap \ker E$, this is exactly $(R \cap K) \cap \ker E = \{0\}$. Thus E must be injective on $R \cap K$, and rank-nullity gives the lower bound $m \geq \dim(R \cap K)$.

The bound is attained. Choose a basis of $R \cap K$, extend it to a basis of V, and take the dual coordinate rows of that first block. Those m_{raw} rows are injective on $R \cap K$. A lower certificate is any nonzero vector of $R \cap K \cap \ker E$, or the dimension count when $m < m_{\text{raw}}$. An upper certificate is the constructed row matrix, replayed by checking that its rank on a basis of $R \cap K$ equals $\dim(R \cap K)$.

T.2 Quotient sufficiency

The induced map $V/B \to (Y \oplus \mathbb{Q}^m)/A(B)$ is injective if and only if $\ker A \subseteq B$. For a block observer $A = (Q, E)$ one has $\ker A = K \cap \ker E$, so the criterion is $K \cap \ker E \subseteq B$. The largest subspace of K that may remain in the kernel is $K \cap B$, of codimension m_{quot} in K. This count does not depend on R.

Attaining rows: extend a basis of $K \cap B$ to a basis of K and then to a basis of V, and take the dual coordinates of the complementary block inside K. Those rows vanish on $K \cap B$ and are injective on a complement, so $K \cap \ker E = K \cap B \subseteq B$.

The kernel of the induced map is $(\ker A + B)/B$. It is zero exactly under the criterion above. A lower certificate is a vector of $\ker A$ outside B. An upper certificate is the constructed row matrix, replayed by checking $K \cap \ker E \subseteq B$.

T.3 The joint optimum is not the sum

Both properties hold together exactly when

$$K \cap \ker E \subseteq B \qquad \text{and} \qquad (K \cap \ker E) \cap R = \{0\}.$$

The second condition then says $K \cap \ker E \subseteq K \cap B$ meets T only at 0. Inside $K \cap B$, the largest subspace meeting T only at 0 has dimension $\dim(K \cap B) - \dim T$. Hence

$$\dim(K \cap \ker E) \leq \dim(K \cap B) - \dim T$$

whenever both properties hold. Rank-nullity on K gives $\dim(K \cap \ker E) \geq \dim K - m$, so

$$m \geq \dim K - \dim(K \cap B) + \dim T = m_{\text{both}}.$$

If $m < m_{\text{both}}$ and $K \cap \ker E \not\subseteq B$, the quotient property fails. If $m < m_{\text{both}}$ and $K \cap \ker E \subseteq B$, the dimension count forces a nonzero intersection with T, so the raw property fails. This case split is the proof that no smaller joint row count exists. It does not add m_{raw} to m_{quot}.

The excess of that sum is

$$m_{\text{raw}} + m_{\text{quot}} - m_{\text{both}} = \dim(R \cap K) - \dim T = \dim((R \cap K)/T).$$

It is zero exactly when $R \cap K \subseteq B$.

The count is attained. Split $K \cap B = T \oplus S_0$ and $K = (K \cap B) \oplus C$, and set $W = T \oplus C$. Then $\dim W = m_{\text{both}}$ and $S_0 \cap R = \{0\}$. Extend a basis that begins with S_0 and continues with W to a basis of V. The dual rows of the W block satisfy $K \cap \ker E = S_0 \subseteq B$ and $S_0 \cap R = \{0\}$.

When $R = B$, one has $T = R \cap K = B \cap K$, so $m_{\text{raw}} = \dim(B \cap K)$, $m_{\text{quot}} = \dim K - \dim(B \cap K)$, and $m_{\text{both}} = \dim K$. The sum equals the joint count. That is the §10.6 specialization, recovered here and not redefined.

A submitted witness carries a separate field, `lower_columns`. The checker accepts that field only when its columns are an obstruction basis for the claimed target. Width and rank equal to the sharp count are required, and they are not sufficient.

- Raw: the columns are a basis of $R \cap K$.
- Quotient: the columns lie in K, meet B only at 0, and number m_{quot}.
- Joint: the columns lie in K, number m_{both}, and meet $K \cap B$ only inside T.

The empty matrix is that basis when the sharp count is 0. On the $B \subset R$ example of T.4, the produced joint basis is a basis of $K = \text{span}\{e_1, e_2\}$. The two columns e_2 and e_3 have width 2 and rank 2, and $e_3 \notin \ker Q$, so they are not a joint obstruction basis. The checker rejects them. This acceptance rule does not replace the kernel-vector lower bounds proved in T.1 and T.2. Those vectors show that a too-small row map fails. The field `lower_columns` shows that the sharp count is forced, independently of which attaining rows were submitted.

T.4 Three inclusions

All three examples use the last-coordinate observer, so K is the span of the earlier coordinates. The shipped checker recomputes the dimensions from the matrices.

$R \subset B$. $V = \mathbb{Q}^3$, D_s has the single column $(1, 0, 0)$, D_r has columns $(1, 0, 0)$ and $(0, 1, 0)$, and $Q = (0, 0, 1)$. Then $B = K$, $R = \text{span}\{e_1\}$, $T = R$, $m_{\text{raw}} = 1$, $m_{\text{quot}} = 0$, $m_{\text{both}} = 1$, and the separate sum is 1.

$B \subset R$. The same V and Q, with D_s now both of $(1, 0, 0)$ and $(0, 1, 0)$, and D_r the single column $(1, 0, 0)$. Then $R = K$, $B = \text{span}\{e_1\}$, $T = B$, $m_{\text{raw}} = 2$, $m_{\text{quot}} = 1$, $m_{\text{both}} = 2$, and the separate sum is 3. The joint optimum is strictly smaller.

Neither inclusion. $V = \mathbb{Q}^4$, $Q = (0, 0, 0, 1)$, D_s has columns $(0, 1, 0, 0)$ and $(0, 0, 1, 0)$, and D_r has columns $(1, 0, 0, 0)$ and $(0, 1, 0, 0)$. Then $K = \text{span}\{e_1, e_2, e_3\}$, $B = \text{span}\{e_1, e_2\}$, $R = \text{span}\{e_2, e_3\}$, and $T = \text{span}\{e_2\}$. Neither image contains the other. Here $m_{\text{raw}} = 2$, $m_{\text{quot}} = 1$, $m_{\text{both}} = 2$, and the separate sum is 3.

A raw completion need not be a quotient completion. On $V = \mathbb{Q}^3$ with $R = \operatorname{span}\{e_1\}$, $B = \operatorname{span}\{e_3\}$, and $Q = (0, 0, 1)$, the single raw row $(1, 0, 0)$ kills every realizable invisible defect, while e_2 remains in $\ker A$ and outside B. The converse gap is the quotient row on the $B \subset R$ example: $(0, 1, 0)$ forces $K \cap \ker E = \operatorname{span}\{e_1\} \subseteq B$, and e_1 is a nonzero realizable defect still in the kernel.

T.5 Restricted menus

Unrestricted algebraic rows are arbitrary elements of the dual of V. A restricted menu is a smaller declared set.

Rows required to annihilate B. Then $B \subseteq \ker E$, so $T \subseteq R \cap \ker A$. If $T \neq \{0\}$, every such row system fails raw sufficiency and fails the joint property. The obstruction certificate is any nonzero vector of T. The $R \subset B$ example is this impossibility: $T = \operatorname{span}\{e_1\}$. If $T = \{0\}$, put a basis of B first, then a complement of $K \cap B$ inside K, then extend to V. The dual rows of that complement annihilate B, and they attain m_{both}, which equals m_{quot} in this case. Quotient sufficiency alone remains feasible at m_{quot} whether or not T vanishes: the same annihilator rows may kill T, and the quotient property allows that.

A finite instrument shelf. The rows must be chosen from a supplied finite list, not from their linear span. Positive rational costs are part of the shelf. Adding a row preserves sufficiency, so a minimum-cost sufficient subset exists whenever any subset is sufficient, and it is found by exhaustive enumeration of the finitely many subsets in increasing cost. The producer returns one such subset. A separate checker enumerates again from the submitted index set and cost. It rejects an insufficient subset, a cost that does not match the submitted indices, and any sufficient subset of strictly smaller cost. If no subset works, the checker accepts impossibility only after the full measurement family itself fails the same sufficiency test.

T.6 A repair claim needs an operation

Choosing $B = V$ makes $K \cap B = K$, so $m_{\text{quot}} = 0$ and the induced quotient map is automatically injective. That fact does not solve the original problem. A claim that a repair solves the original defect is accepted only with a separate witness $(D_{\text{adm}}, u, r_\star, \delta)$ satisfying $D_{\text{adm}} u = r_\star$, $r_\star \in B$, and $r_\star = \delta$, where δ is the declared realizable defect. The checker replays those matrix identities. It does not invent an operation. Certificate acceptance remains separate from an authorization to use the operation.

T.7 What this theorem does not say

WDP-DIM-1 is a finite rational row-count theorem. It does not assert a continuum reconstruction, a quantum code, a physical instrument, or a repair that has not been given as an admissible map. The Chapter 23 corrections are a different carrier and are proved in Appendix U.

T.8 Repair-class separation restricted to realizable candidates

WDP-DIM-R (realizable-domain corollary). Retain the rational linear data of WDP-DIM-1, but restrict candidates to $r \in R$. The protected answer is $q(r) = r + B$, and the observed report is the actual value $A(r) = (Q(r), E(r))$. A decoder on $A(R)$ exists exactly when

$$R \cap \ker A \subseteq B.$$

Writing $L = R \cap K$ and $T = L \cap B$, the minimum number of additional rational linear observation rows is

$$m_{\mathrm{quot},R} = \dim L - \dim T = \dim(R \cap K) - \dim(R \cap K \cap B).$$

Proof. Equal reports for $r, r' \in R$ mean $r - r' \in R \cap \ker A$; equal protected answers mean $r - r' \in B$. Conversely every vector in $R \cap \ker A$ occurs as the difference of itself and zero, so the fiber criterion is exactly the displayed inclusion. Equivalently, $\ker(E|_L) \subseteq T$. Rank-nullity therefore gives $m \geq \operatorname{rank}(E|_L) \geq \dim L - \dim T$. For attainment, extend a rational basis of T to one of L, and then to one of V. Take the dual coordinate rows of the complementary block inside L. Their kernel on L is exactly T, and there are $\dim L - \dim T$ rows. This includes the zero-row case. □

For example, let $V = \mathbb{Q}^2$, $R = \operatorname{span}\{e_1\}$, $B = 0$ and $Q = 0$. The row $E = (1, 0)$ suffices on R, whereas the global repair-class question needs two rows: e_2 and zero still collide under this E. If both raw and repair-class separation are required only on R, raw injectivity already supplies both, so the minimum is $\dim(R \cap K)$. The joint count in T.3 instead includes the global quotient obligation.

The restricted quotient presentation is $R/(R \cap B) \to A(R)/A(R \cap B)$. It is injective under exactly the same criterion: if $A(r) = A(b)$ for $b \in R \cap B$, then $r - b \in R \cap \ker A$, and the converse follows by taking $b = 0$. Do not replace its report quotient by the ambient quotient by $A(B)$. For $R = \operatorname{span}\{e_1\}$, $B = \operatorname{span}\{e_2\}$ and $A = (1, 1)$, actual reports distinguish every candidate in R, but $A(B) = \mathbb{Q}$ erases the entire report. Repairs outside R change that observation contract.

T.9 Finite encodings and instrument-menu boundaries

The observation-scope supplement gives two direct finite consequences of the report-fiber criterion. An unrestricted auxiliary binary encoding needs $\lceil \log_2 \max_{a \in p(X)} |q(p^{-1}(a))| \rceil$ bits for a nonempty finite candidate set; this standard counting result does not price an instrument shelf. A four-state, five-query shelf has pairwise non-derivable binary queries and inclusion-minimal separating sets of sizes two and three. Those sets cannot be the bases of a matroid. The exact table, proof and canonical native replay are supplied there; no theorem is attributed to an unread comparison paper. Section 14.6's hitting-set criterion remains the general finite-menu result. The supplement also states the calibrated-noise limitation and the remaining primary-source comparison obligations.

Appendix U — Corrected finite toric instrument mathematics

The corrected finite toric instrument mathematics has been promoted to canonical Chapter 23 (§23.1–§23.9). The sections below preserve the detailed proof calculations, the $L = 3, t = 2$ collision regression analysis, and the mechanized boundary for reference. Lattice assumption throughout: the periodic carrier $\mathbb{Z}_L \times \mathbb{Z}_L$ with $L \geq 3$, one qubit on each edge of the grid. There are $n = 2L^2$ edges. Horizontal edge $h_{i,j}$ has index $2(iL + j)$, and vertical edge $v_{i,j}$ has index one greater. Face $f_{i,j}$ is incident to $h_{i,j}$, $h_{i+1,j}$, $v_{i,j}$, and $v_{i,j+1}$. Star $s_{i,j}$ is incident to $h_{i,j}$, $h_{i,j-1}$, $v_{i,j}$, and $v_{i-1,j}$. Indices are periodic. The Pauli space is $\mathbb{F}_2^n \oplus \mathbb{F}_2^n$. The weight convention is

$$\mathrm{wt}(e) = |\operatorname{supp} e_X| + |\operatorname{supp} e_Z|,$$

so a single Y contributes 2. Detectors in this appendix are passive queries: a row returns a bit. No post-state is declared, and a classical bit table is not a proof that a physical measurement leaves a state fixed.

U.1 The commutation kernel is $S^\perp$, not S

Let A be the $L^2 \times n$ face-incidence matrix and A_v the star-incidence matrix, both over $\mathbb{F}_2$.

Each edge lies in two faces, so the sum of all rows of A is zero. The same holds for A_v. The dual graph, with a vertex per face and an edge per qubit, is the connected torus grid. A vector in the left kernel of A is constant on the two faces of every edge, hence constant on a connected graph. The left kernel is the constants and $\operatorname{rank} A = L^2 - 1$. The same connectedness argument gives $\operatorname{rank} A_v = L^2 - 1$.

The full stabilizer commutation syndrome is the block map

$$Q(x \mid z) = (Ax,\ A_v z).$$

Thus $\operatorname{rank} Q = 2L^2 - 2$ and, on the Pauli space of dimension $4L^2$,

$$\dim \ker Q = 4L^2 - (2L^2 - 2) = 2L^2 + 2.$$

A star meets every face evenly. The four edges of $s_{i,j}$ are shared two at a time with the four faces $f_{i,j}$, $f_{i,j-1}$, $f_{i-1,j}$, and $f_{i-1,j-1}$, and every other face is disjoint from that star because those four edges have only eight face incidences. Therefore $AA_v^{\mathsf{T}} = 0$, and transposition gives $A_v A^{\mathsf{T}} = 0$.

The stabilizer space S is spanned by the vectors (row of $A_v \mid 0$) and ($0 \mid$ row of A). The commutation relations put $S \subseteq \ker Q$. The two blocks are independent, so $\dim S = 2L^2 - 2$. Hence

$$\dim(\ker Q / S) = (2L^2 + 2) - (2L^2 - 2) = 4.$$

The kernel of Q is the set of Pauli vectors that commute with every generator of S. That set is the symplectic orthogonal $S^\perp$. It properly contains S, because the quotient dimension is 4, not 0.

WDP-CH23-P01C. On this carrier, with this weight convention, the full commutation syndrome has kernel $S^\perp \neq S$, and $\dim(S^\perp/S) = 4$. The imported sentence that a full independent stabilizer measurement has kernel S is withdrawn. The arithmetic that subtracts $\operatorname{rank} Q$ from the edge count n, rather than from the Pauli dimension $2n$, is the source of that mistake. The imported identifier WDP-CH23-P01 keeps its imported wording, including the quotient display of dimension 4 already present there; P01C is the statement that is used from here on.

An exhibiting vector is the weight-L support Z_1, the horizontal edges $h_{i,0}$ for $i = 0, \ldots, L-1$, placed in the X block with the Z block zero. It lies in $\ker A$. The row-zero horizontal flux, the sum of $h_{0,0}, \ldots, h_{0,L-1}$, is zero on every star, because each star meets that row in zero or two horizontal edges, and it equals 1 on Z_1 for every $L \geq 3$: the row intersects that support in exactly $h_{0,0}$. Thus Z_1 is outside the star span, the Pauli vector is outside S, and its weight is L. At $L = 3$ the shipped binary reconstruction and the Lean encoding both check kernel membership, exclusion from S, and weight 3.

U.2 Distance under the declared weight

Let $x \in \ker A$. Write $\alpha_i(x)$ for the sum of the horizontal edges in lattice row i, and $\beta_j(x)$ for the sum of the vertical edges in lattice column j. The sum of the face rows in lattice row i equals the horizontal edges of row i plus those of row $i + 1$, because each vertical edge of that strip lies in two of those faces. Therefore $Ax = 0$ forces $\alpha_i = \alpha_{i+1}$. Summing the face rows in lattice column j cancels the horizontal edges and forces the vertical fluxes in columns j and $j+1$ to agree. Each star

has two horizontal edges in one row and two vertical edges in one column, so both fluxes vanish on the row span of A_v. The pair (α, β) therefore descends to $\ker A/\operatorname{rowspan}(A_v)$ and separates the four classes: it is an isomorphism onto $\mathbb{F}_2^2$, since the quotient is two-dimensional and both nontrivial values occur on the exhibited loops of weight L.

If $\alpha = 1$, every lattice row contributes at least one horizontal edge, so the weight is at least L. If $\beta = 1$, the weight is at least L. The same lower bound holds in the other sector. A class in $\ker Q \setminus S$ is nontrivial in at least one sector, so its weight is at least L. The exhibited loops have weight L, and that is the minimum.

A class that is nontrivial in both sectors has weight at least $2L$ under the declared weight, because the two supports are counted separately and a shared qubit contributes to both. The imported value $2L - 1$ for the diagonal class counts non-identity Paulis, with a shared qubit read as one Y. That is a different convention.

WDP-CH23-P07C. Under $\mathrm{wt}(e) = |\operatorname{supp} e_X| + |\operatorname{supp} e_Z|$, the diagonal class has distance $2L$, not $2L - 1$. However, this distance does not establish the fault-tolerance threshold $t \leq L - 1$ for the coarser question $B = S + \langle u + v \rangle$. The relevant separation distance is

$$d_B = \min\{\mathrm{wt}(z) : z \in \ker Q \setminus B\},$$

measuring forbidden differences. Because the diagonal class $u + v$ belongs to the allowed subspace B, its weight of $2L$ does not determine that distance. The forbidden difference set $\ker Q \setminus B$ still contains the pure single-sector logical classes u and v, each of weight L. Thus $d_B = L$, retaining the single-sector threshold $2t < L$. The inference that $2t < 2L$ tolerates $t \leq L - 1$ fails: at $L = 3, t = 2$, the errors $e_1 = h_{0,0}$ and $e_2 = \{h_{1,0}, h_{2,0}\}$ have weights one and two, identical full syndromes, and difference $u \notin B$, producing an undetectable collision. **WDP-CH23-P08C** withdraws the imported slogans that this halves the required lattice and that the coarser question is factorially cheaper. No cost identity and no lattice-size identity of that form is proved. The coarser question retains the threshold requirement $2t < L$.

U.3 The restricted fiber criterion

Let E_{adm} be a declared finite set of Pauli vectors, let Q_D be a declared detector map, and let q be a protected question with $B = \ker q$, here a subspace.

WDP-CH23-P03C. For every admitted pair e, e',

$$Q_D(e) = Q_D(e') \quad \text{implies} \quad q(e) = q(e')$$

if and only if a recovery witness exists: one admitted representative e_s may be chosen in each nonempty fiber, and the decoder returns that representative. Both selection and verification terminate by finite enumeration. Equivalently, when q is linear,

$$(E_{\mathrm{adm}} - E_{\mathrm{adm}}) \cap \ker Q_D \subseteq B.$$

The forward direction subtracts the two recovery representatives. The reverse direction defines the answer on a fiber by the common value of q. The imported wording of WDP-CH23-P03 that begins the forward direction by assuming the difference already lies in B is not this proof.

A global rank defect is not an admitted collision. The quotient $S^{\perp}/S$ has dimension 4 and its nontrivial classes have weight at least L. At $t = 0$ the admitted set is $\{0\}$, so its difference set is

$\{0\} \subseteq B$. The forbidden-difference set outside B is empty, and the inclusion holds. **WDP-CH23-P10C.** Deleting one independent row from a full-rank syndrome drops the rank by 1 and raises $\dim(\ker Q_D/S)$ from 4 to 5. The imported count $\dim\ker Q_D = n - r_D$, using the edge count in place of the Pauli dimension, is withdrawn, and so is the imported quotient dimension 1. Whether the enlarged kernel contains a low-weight admitted pair is settled by searching E_{adm}, not by quoting the dimension.

U.4 Full rank, the empty purchase at $t = 0$, and the matroid boundary

Let $\mathcal{E}_t$ be the set of Pauli vectors of weight at most t, and let $B = S$.

WDP-CH23-P06C. Let $1 \le t$ and $2t < L$, and let $B = S$. A passive stabilizer-check set of rank $2L^2 - 2$ separates $\mathcal{E}_t$ up to S. Sufficiency uses the distance proved in U.2: the kernel of the full syndrome is $S^{\perp}$, and every vector of weight at most $2t$ in $S^{\perp}$ already lies in S. The converse is withdrawn. Full rank is not necessary for that separation. The sentence that every proper row subspace kills a weight-one column is withdrawn with it.

The column sentence already fails on the face-incidence matrix at $L = 3$. That matrix has rank $L^2 - 1 = 8$. Its rows indexed by $\{0, 1, 3, 5, 7, 8\}$ have rank 6. No column of this submatrix is zero, so no weight-one edge lies in its kernel. The edge pairs $(0, 1)$, $(4, 13)$, and $(5, 8)$ have columns summing to zero, so the kernel contains those weight-two vectors. The pair $(0, 1)$ lies outside the star span. The same six rows, placed in the X block and accompanied by no star row, do not separate $\mathcal{E}_1$.

Full rank also fails to be necessary for the full commutation syndrome. At $L = 4$ that syndrome has shape $(32, 64)$ and rank $30 = 2 \cdot 4^2 - 2$. Delete rows 1, 10, and 16. The remaining 29 rows are independent and have rank 29. On $\mathcal{E}_1$, the admitted-collision search returns no pair, so this suite separates $\mathcal{E}_1$ up to S. Here $t = 1$ and $2t = 2 < 4$. A dropped row can enlarge the kernel by a vector heavier than $2t$, and the fiber criterion can still hold.

WDP-CH23-P11C and WDP-CH23-P13C. At $t = 0$ the empty detector set is sufficient and has cost 0. A positive purchase is not required. For $1 \le t$ and $2t < L$ with $B = S$, rank $2L^2 - 2$ is a sufficient independent count. It need not be the minimum independent count: the $L = 4$ suite above separates $\mathcal{E}_1$ with 29 independent checks. The uniform-cost minimum on a declared finite shelf is the cost returned by the enumeration of P12C.

WDP-CH23-P12C. Greedy selection in the stabilizer-check matroid returns a minimum-cost basis of the check space only after sufficiency has been proved equal to full check-space rank for the obligation in hand. That hypothesis is withdrawn for $\mathcal{E}_t$ in the regime $1 \le t$, $2t < L$, $B = S$, by the rank-29 suite of P06C. The matroid shortcut is not used for this obligation. For a general finite family E_{adm} the obligation is the hitting condition of P03C. The terminating exact procedure is exhaustive enumeration of the shelf subsets under positive rational costs. The optimality checker is a second enumeration. It does not call the producer.

U.5 Adaptivity, schedules, and noise

An adaptive policy is a function from finite transcripts to the next passive query. A transcript is a finite list of selected shelf indices and the bits they returned. The information state is the set of admitted errors consistent with that transcript.

WDP-CH23-P17C. The imported unrestricted claim that adaptivity buys nothing is false. Take the four hypotheses $\{w, x, y, z\}$ and three passive tests: Split of cost 2, partitioning $\{w, x\}$ from

$\{y, z\}$; Wx of cost 2, separating w from x and not y from z; Yz of cost 2, separating y from z. Any two tests leave a colliding pair, so the nonadaptive minimum uses all three and costs 6. The adaptive policy buys Split and then the one test named by that bit. Every branch costs 4. A claimed adaptive lower bound of 6 is false, and the indistinguishable nonadaptive pairs are the certificate. The shipped checker rejects that false bound.

The further sentence that, for $t \geq 1$, $2t < L$, and $B = S$, every proper stabilizer-check subspace hides a weight-one error, and that the all-zero branch must accumulate rank $2L^2 - 2$, is withdrawn. The $L = 4$ rank-29 suite of P06C separates $\mathcal{E}_1$, so a passive policy may stop on that suite. **WDP-CH23-P18C** withdraws every unrestricted detector-purchase bound, including that full-rank branch. The $t = 0$ family remains a counterexample to any rule that buys a detector on every admitted family.

WDP-CH23-P15C. The imported cone radius is the definition $\lceil v_{\mathrm{LR}} \Delta t \rceil$. Comparing it with $\lfloor L/2 \rfloor$ is a geometric comparison. It is not an admitted collision and not a proof that a schedule fails. A failure certificate is still a pair in $(E^R - E^R) \cap \ker Q_{D,R}$ outside B^R, at the decision time, inside the declared family.

WDP-CH23-P16C. If a finite joint family of data and readout errors is actually supplied, together with a map to the observed record and a protected subspace, the fiber criterion of P03C is necessary and sufficient for deterministic factorization on that family. The imported spacetime chain, the Bernoulli readout model, and the repetition threshold $R \geq L$ are not reconstructed from the bare toric incidence matrices. They remain conditional on those missing matrices. Repeated measurements at different times are not the same row unless a proved state-evolution map identifies them. The detectors used here are passive queries. Nothing in the binary table is a theorem of physical non-disturbance.

Current answers and later operations. A passive query can match the current protected bit while leaving a later declared operation disabled. A witness that sets the current report matched, the future operation disabled, and nevertheless claims that the future operation remains available, is rejected. Preserving the current answer is not that later operation.

U.6 Executable and mechanized boundary

The Python reconstruction in `code/toric_code_observer_completion.py` builds the incidence matrices, the commutation syndrome, the stabilizer rows, the weight-L logical, the fiber search, the recovery representative, and the separate minimum-cost checker. `ToricSyndromeCorrespondence.lean` establishes the concrete $L = 3$ encoding: face rank 8, star rank 8, the logical vector in the face kernel and outside the star span, its weight 3, the row flux, and the identity $(2L^2 + 2) - (2L^2 - 2) = 4$ for every integer $L \geq 1$. Those Lean facts are not a formalization of the flux-distance argument for every L, of the rank-29 separation counterexample, of adaptivity, or of any physical schedule. The general-L distance and fiber proofs above are written proofs. The $L = 3$ face-row counterexample and the $L = 4$ rank-29 suite are executable witnesses in `code/toric_code_observer_completion.py`, replayed by `code/tests/test_toric_code_observer_completion.py`.

Appendix V — Accepted locality proofs L1–L4

The text below is the complete accepted locality argument, copied into this manuscript. Exact finite observer completion, conditional Lieb–Robinson readout bounds, reconstruction conditioning, noise refusal, and CAR locality with gauge constraints stay separate inside that argument. A Lieb–Robinson upper bound is not a reconstruction-error floor. The Bell, gauge, star-exchange, and

upper-envelope counterexamples are part of the proof, not a citation. The live file `unified_lieb_robinson_observer_completion_derivation.md` remains in place because locality replay pins its path. This copy, not a link to that file, is the canonical proof.

Locality, sufficient observation, and faithful recovery

Corrected derivation, 2026-10-05. The submitted draft is preserved byte for byte in `docs/source_archive/locality_and_smearing/source_manifest.json` in `docs/source_archive/locality_and_smearing/`. Its statements are not inherited as proved propositions.

1. The precise bridge

There are two different restrictions: information discarded by an observer, and operations excluded by physical locality. They meet at a declared readout map; neither restriction supplies the missing physical adapter for the other. The theorem below gives exact finite recovery conditions, then bounds the conditioning of a physical readout under explicit locality hypotheses. It does not identify a positive error upper bound with an error floor.

Fix a nonempty finite set E of admitted errors in $V = F_2{}^n$, a repair subspace $B \leq V$, and a linear syndrome map $Q{:}V \to F_2{}^m$ with $B \leq \ker Q$. Repair success means $e+r \in B$. An error in B has zero quotient obstruction; an error outside B does not become zero by selecting another representative. These are algebraic declarations. An actual decoder operation realizing r must still be supplied to use this model for a device or a proposal-protocol repair.

Theorem L1 (sharp admitted-error recovery). A syndrome-only decoder $r{:}Q(E) \to V$ succeeds for every $e \in E$ if and only if

$$Qe = Qf \quad \Longrightarrow \quad e + f \in B \qquad (e, f \in E).$$

Proof. If r succeeds, add $e+r(Qe) \in B$ and $f+r(Qf) \in B$; the common repair cancels. Conversely select one admitted representative e_s from each nonempty fiber $Q^{-1}(s) \cap E$ and set $r(s)=e_s$. The hypothesis gives $e+e_s \in B$. Both selection and verification terminate by finite enumeration. A positive certificate lists every used syndrome and its repair. A negative certificate is a pair e,f in the same fiber with $e+f \notin B$. The checker replays the supplied repair or pair; it does not replace an incorrect decoder with a new one. ■

This is not a left inverse of Q, $RQ=id_V$. That equation is unrealizable for a noninjective Q. Recovery is only required on E, modulo the declared B. For a full Hamming ball $E=\{e{:}wt(e) \leq t\}$, define $d=\min\{wt(v){:}v \in \ker Q \setminus B\}$, with $d=\infty$ when this set is empty. Then L1 is equivalent to $2t<d$: a forbidden vector of weight at most 2t splits into two vectors of weight at most t, and the sum of any two admitted errors has weight at most 2t. For a restricted admitted family, L1 is the criterion; the distance inequality is sufficient but need not be necessary.

Theorem L2 (minimum linear observer completion). Additional binary rows must annihilate B. A row suite C permits recovery precisely when every forbidden difference

$$F = \{e + f : e, f \in E,\ Qe = Qf,\ e + f \notin B\}$$

has a nonzero value under at least one row of C. Thus its minimum size is the minimum finite hitting suite among the rows in B^{perp}. There is an exhaustive terminating procedure: enumerate row subsets by increasing size, discard rows that do not annihilate B, and test all forbidden differences. A lower certificate must cover every smaller admitted suite with a retained collision. Just displaying a successful suite is only an upper bound.

If E is itself a linear subspace containing B, the minimum is dim(ker(Q|E)/B). Necessity follows from injectivity of the added map on that quotient. Sufficiency follows by choosing dual coordinate functionals there, extending them to E/B and then V/B. This dimension formula is not asserted for an arbitrary restricted E.

The retained example has n=3, Q(x,y,z)=x+y, B={000,110}, E=$F_2{}^3$. The original syndrome collides on 000 and 001. One extra row z produces (x+y,z), with repair (0,x+y,z). Zero extra rows fail and one succeeds, so the exact minimum is one. `LocalObservationCompletion.lean` establishes this concrete decoder separately from its abstract fiber criterion.

2. Conditional physical locality theorem

Let a finite tensor-product quantum system have a declared region R. Let ρ,σ be density matrices. Let O be a Hermitian evolved readout and O_R a Hermitian operator supported on the **joint** region R. Assume

$$\mathrm{Tr}_{R^c}\,\rho = \mathrm{Tr}_{R^c}\,\sigma, \qquad \|O - O_R\|_\infty \le \beta.$$

Theorem L3 (local readout bound).

$$|\,\mathrm{Tr}((\rho-\sigma)O)| \le \|\rho-\sigma\|_1\,\beta \le 2\beta.$$

Proof. Equal joint marginals give Tr((ρ−σ)O_R)=0. Subtract this identity and apply trace/operator norm duality. The trace distance bound follows from positivity and unit trace: $\|\rho-\sigma\|_1 \le \|\rho\|_1+\|\sigma\|_1=2$. ■

With approximate marginals at trace-norm distance d_R, the bound is instead d_R‖O_R‖∞+‖ρ−σ‖$_1$β. No correlation-decay assumption is needed for equal joint marginals. Equal **single-site** marginals do not give equal joint marginals on a multi-site region.

Here is an explicit way to obtain O_R from finite commutator estimates. For a qubit y outside R let E_y be the average of conjugation by I,X,Y,Z on y. It is a norm contraction and removes dependence on that qubit. Composing these maps gives E_R, the conditional expectation onto operators supported on R. If ‖[O,P_y]‖∞≤η_y for each of the three nonidentity Paulis, then

$$\|O - E_R O\|_\infty \le \tfrac{3}{4}\sum_{y\notin R}\eta_y.$$

Indeed O−E_yO is the average of the three differences O−P_yOP_y; each has the commutator norm. A telescoping sum over y, with preceding conditional expectations contractive, establishes the claim. Higher local dimensions use a declared finite unitary twirl and its corresponding average.

An imported Lieb–Robinson bound can therefore provide a β of the form C_R exp[−μ(r−v|t|)] after the spatial tail has actually been summed. C_R includes the geometry, operator norm,

and interaction hypotheses. This derivation does not infer such a bound from the name of a Hamiltonian. The standard locality results are imported from Bravyi–Hastings–Verstraete and Nachtergaele–Sims–Young. The finite twirl and the readout deduction above are self-contained.

The equality $\mathrm{Tr}([\rho,O])=0$ holds for every pair of finite square matrices. It is **not** an expression for $\mathrm{Tr}((\rho-\sigma)O)$. Confusing these two expressions would wrongly erase every readout difference.

3. Conditioning and noisy refusal

Let $z=R(\rho)$ and $z'=R(\sigma)$ be readout vectors, in a declared norm. Suppose $\|z-z'\|\le\eta$, a decoder F satisfies $\|F(z)-F(z')\|\le\kappa\|z-z'\|$, and its endpoint errors against the desired targets a,b are at most α.

Theorem L4. If $\|a-b\|=\delta$, then

$$\delta \le 2\alpha + \kappa\eta.$$

For exact endpoint decoding and $\eta\le 2\beta$, κ must satisfy $\kappa\ge\delta/(2\beta)$ when $\beta>0$. If $\beta=0$ and $\delta>2\alpha$, no such decoder exists. This is an obstruction to stable reconstruction, not a lower bound on the physical error of the device. For multiple channels their common β and vector norm must be declared; one cannot silently use a scalar bound for an arbitrary-dimensional norm.

Proof. Insert $F(z),F(z')$ between a and b and apply the triangle inequality. Combining the locality bound uses $\kappa\ge 0$. ■

If two readouts have distance at most 2ε in a normed real space, their closed noise balls of radius ε intersect at the midpoint. At the common report, any reconstruction has error at least $\delta/2$ at one endpoint. Thus $\beta\le\varepsilon$ is a sufficient indistinguishability condition in the scalar bound of L3. It is not a necessary physical horizon, nor evidence that the system has intrinsic error ε. The Lean proof establishes the common-report contradiction for exact rational targets and errors.

4. Retained carrier counterexamples

Bell marginals. The rational density matrices of $(|00\rangle\pm|11\rangle)/\sqrt{2}$ have identical individual marginals $I/2$. The observable $X\otimes X$ has expectations $+1$ and -1. Keeping current single-site outputs loses the joint readout. Setting $O_R=Z\otimes I$ and $O=O_R+\varepsilon X\otimes X$ gives readout difference 2ε and tail norm ε: the bound in L3 is sharp on these finite data. The executable checker checks density positivity, exact marginal equality, the submitted norm certificates, and the actual readout separately.

Gauge cycle. For oriented edges $a\to b\to c\to a$ and $\alpha=(0,1,0)$, the gradient is $(1,-1,0)$ if $d\alpha(\mathrm{edge})=\alpha(\mathrm{head})-\alpha(\mathrm{tail})$, or $(-1,1,0)$ for the opposite convention. Its L^1 norm is two, and its circulation is zero. Every gauge gradient telescopes around a cycle. Translation of the admitted repair set therefore preserves the quotient obstruction and the infimal repair cost. It cannot extinguish a nonzero cycle class. Changing the carrier, weights, or admissible repairs is a different operation requiring a preservation map.

Star exchange. Four physical sites $0,1,2,3$ form a star centered at 0. The two-particle configuration loop is $\{1,2\}\to\{0,2\}\to\{2,3\}\to\{0,3\}\to\{1,3\}\to\{0,1\}\to\{1,2\}$. Its legal directed hops are $1\to0,0\to3,2\to0,0\to1,3\to0,0\to2$. In ascending CAR occupation order their signs are $1,-1,1,1,-1,-1$, with product -1. Six signs all equal to -1 would instead multiply to $+1$. The exchange loop lives

in the configuration graph, not the physical site tree. The CAR/exterior-power carrier construction must be declared; it is not an inclusion of physical site cycles into configuration cycles.

Upper envelopes. An inequality 0≤error≤positive_bound does not imply error>0. Infimizing an upper envelope does not change this fact. A lower obstruction requires a collision, a separated pair, or another independently checked witness of the missing capability.

5. Scope relative to WDP, checked repair and local acceptance

L1 separates local syndrome satisfaction from global admitted-error recovery, the local-acceptance distinction. L2 specifies what an adequate observer must retain. The quotient ker Q/B connects the raw defect to the declared repair equivalence. L3–L4 give a physical readout restriction **only when a real carrier/readout adapter and locality hypotheses are supplied**. WDP binds those source, observer, repair and evaluation maps, instead of letting a name substitute for them. Proposal, independent checking and authorization remain separate.

There is no claim here that MUB6 is resolved, that a Majorana device realizes these declarations, that every noise model is correctable, or that a physical error floor follows from a Lieb–Robinson upper bound. The standard quantum error-correction setting is imported separately from Knill–Laflamme; L1 is a finite binary admitted-error theorem and is not a proof of that full operator theorem.

6. Evidence boundary

The full written theorem uses ordinary finite linear algebra, trace norms, and conditional imported analytic locality bounds. Python replays finite binary recovery, exact rational matrices, the star exchange, and rejection controls. `LocalObservationCompletion.lean` establishes the fiber criterion, the concrete binary decoder, rational locality/conditioning consequences, and cycle/sign identities. It does not formalize the imported analytic Lieb–Robinson theorem or verify the Python program. The reference package and source-bound verification records are described in `derivation_evidence.md`.

Appendix W — Accepted Trotter proofs and the Trotter-response boundary

The text below is the complete accepted Trotter argument, copied into this manuscript: finite pairing, fixed-carrier time consistency, positivity, the strict margin, the rational schedule, and cochain descent. Exact finite rational certificates, the all-M statement for one compressed channel, full fixed-four-site time convergence, finite spectral observations at selected M, general all-M inertia, and infinite-volume or joint spacetime convergence stay separate. The live file `trotter_cochain_extinction_derivation.md` remains in place because locality replay pins its path. This copy, not a link to that file, is the canonical proof.

Trotter response, smooth observations, and cochain descent

Corrected derivation, 2026-10-05. Original submitted bytes are retained under `docs/source_archive/locality_and_smearing/source_manifest.json` in `docs/source_archive/locality_and_smearing/`. The theorem closes a fixed finite spatial carrier's time limit. It does not assert a joint infinite-volume/continuum limit or Gaussian field law.

1. The exact supplied finite carrier

Take L=4 sites and the full 16-dimensional CAR Fock space. In the ascending occupation basis, c_x removes site x with sign (-1) raised to the number of occupied sites preceding x. Define

$$n_x = c_x^* c_x, \quad a_x = (-1)^x (n_x - \tfrac{1}{2} I), \qquad P_x = \tfrac{1}{2}(n_x + n_{x+1} - 2n_x n_{x+1} + c_x^* c_{x+1} + c_{x+1}^* c_x).$$

Indices are periodic. Each P_x is an orthogonal projection. Disjoint even bonds commute, as do disjoint odd bonds. Let $0<\theta\leq 1$, $M\geq 1$, $r=\theta/M$, $N=2M$, $T_0=(I+rP_0)(I+rP_2)$, $T_1=(I+rP_1)(I+rP_3)$, and K_j=T_(j mod 2). The input partition is

$$Z_M(z) = (1+r)^{-M} \operatorname{Tr}(K_{N-1} e^{\sum_x z_{N-1,x} a_x/2} \cdots K_0 e^{\sum_x z_{0,x} a_x/2}), \qquad G_M = 4\, D_z^2 \log Z_M(0).$$

The scalar prefactor cancels from G. Its matrix dimension is LN=8M; this is a spacetime response matrix, not a one-particle Hilbert-space matrix. For sampled real profiles ψ_(j,x), the declared physical quadratic is

$$q_M(\psi) = \frac{1}{LN^2} \psi^T G_M \psi.$$

This normalization is used throughout. The original Trotter-response implementation is reconstructed exactly at M=1; the new reference extends the same insertion convention, using full traces rather than guessed spectral formulas.

2. Reconstructed finite spectra

Independent congruence witnesses establish these inertias at $\theta=1/10$. Counts below are (positive, negative, zero).

M	step r	normalized partition	staggered eigenvalue	inertia
1	1/10	$\frac{17661}{1100}$	$-\frac{1}{706440}$	$(4,1,3)$
2	1/20	$\frac{755374987}{47040000}$	$-\frac{35281}{90644998440}$	$(8,1,7)$
3	1/30	$\frac{1395062306051761}{86870556000000}$	$-\frac{12188547841}{55802492242070440}$	$(12,1,11)$
4	1/40	$\frac{2379246859363301568641}{148151258316800000000}$	$-\frac{14588472130734881}{95169874374532062745640}$	$(16,1,15)$

For M=1 the full spectrum is 10/21,21/40,420/841 (twice), 0 (three times), and $-1/706440$. At a symbolic one-period r,

$$\lambda_1(r) = \frac{-r^4}{4(r+2)(r^2+4r+8)}.$$

Thus $\lambda_1(1/192)=-1/87426958080$. The denominator is $64+64r+24r^2+4r^3$, giving $\lambda_1=-r^4/64+r^5/64-5r^6/512+O(r^7)$. At r=1/10 the quadratic factor is 841/100, not 281/100. The table verifies four finite matrices; it is not an all-M inertia theorem. The time-limit theorem below does not depend on that unproved extrapolation.

3. A sharp finite pairing theorem

Let v_(j,x)=(−1)^j, so $\|v\|^2$=LN, and put bar A=$(LN)^{-1}\Sigma$_(j,x)(−1)^jψ_(j,x). If |ψ_(2k,x)−ψ_(2k+1,x)|≤D/N, then

$$|\bar{A}| \le \frac{D}{2N} = \frac{D}{4M}.$$

Proof. Pair the N terms into M adjacent differences. The total absolute sum is at most LM D/N; divide by LN and use N=2M. A linear sampled profile ψ_j=j/N attains equality, so the finite pair bound is sharp. A C^1 profile with sup|∂_τψ|≤D supplies this premise by the mean-value theorem. Time-independent profiles have bar A=0 exactly. ■

For the rank-one compressed channel P_vG_MP_v, set λ_v=$v^\top$G_Mv/(LN). Its contribution is exactly

$$q_{v,M} = \frac{\lambda_v}{N}\bar{A}^2, \qquad |q_{v,M}| \le \frac{|\lambda_v| D^2}{4N^3} = \frac{|\lambda_v| D^2}{32M^3}.$$

No eigenvector hypothesis is needed for this compression identity. When v is the sole negative eigenvector, as in the four checked instances, it is the entire negative spectral contribution. Outside those instances this identification needs its own proof.

All-M extinction of this channel. Set E=e^(4θ). Products of the K_j have norm at most E. Tr((T_1T_0)^M)≥16 because T_0,T_1≥I and T_1T_0 is similar to $T_0^{(1/2)T_1T_0}$(1/2)≥I. First and second normalized insertion moments have magnitudes at most E/2 and E/4. Hence each response entry has magnitude at most (E+E^2)/4, and |λ_v|≤2M(E+E^2). Consequently

$$|q_{v,M}| \le \frac{(E+E^2)D^2}{16M^2} \longrightarrow 0.$$

This establishes extinction for the declared smooth observed channel without postulating bounded eigenvalues or extrapolating the finite inertia table. It does not by itself prove positivity of the full q_M.

4. The fixed-L time-continuum theorem

Let H=Σ_xP_x, and let ψ_x(τ) be real, periodic C^1 functions with |ψ_x|≤B and |ψ'_x|≤D. Set A(τ)=Σ_xψ_x(τ)a_x, Z=Tr e^(θH), and ρ=e^(θH)/Z. Define the continuous time-ordered covariance kernel

$$K_{xy}(s) = Z^{-1}\,\mathrm{Tr}(e^{(1-s)\theta H} a_y e^{s\theta H} a_x) - \mathrm{Tr}(\rho a_x)\,\mathrm{Tr}(\rho a_y), \quad 0 \le s \le 1.$$

For τ_i≤τ_j use s=τ_j−τ_i; reverse order transposes x,y. This produces a continuous periodic symmetric kernel on pairs: at the endpoints the trace is cyclic, and at coinciding times the a_x commute. Define

$$Q_\theta(\psi) = \frac{1}{L}\sum_{x,y} \int_0^1\!\int_0^1 \psi_x(\tau) K_{xy}(\sigma-\tau)\psi_y(\sigma)\, d\tau d\sigma.$$

Theorem T1 (full consistency, explicit envelope). For the supplied L=4 carrier and samples ψ_(j,x)=ψ_x(j/N),

$$|q_M - Q_\theta| \le \frac{A_\theta(B,D)}{M}, \qquad A_\theta = L\{30(\theta+\theta^2)e^{16\theta}B^2 + 2\theta e^{4\theta}B^2 + \tfrac{1}{2}e^{16\theta}BD\}.$$

Proof, with conservative constants. Write $H_0=P_0+P_2, H_1=P_1+P_3$. Their norms are at most two and $\|H\|\le 4$. Taylor's integral remainder gives $\|T_i - e^{(rH_i)\|\le r^2 e}(2r)$. Duhamel's double integral gives $\|e^{(rH_1)e}(rH_0) - e^{(rH)\|\le (r^2/2)\|[H_1,H_0]\|e}(4r) \le 4r^2 e\hat{}(4r)$. Thus a complete period has error at most $6r^2 e\hat{}(4r)$. The same bound holds in the other layer order. Telescoping $k\le M$ periods gives $6Mr^2 e\hat{}(4\theta)$. For an unmatched half-period, the identity expansion and $\|H_i\|, \|H/2\| \le 2$ give $\|T_- i - e^{(rH/2)\|\le 2r+r^2+(e}(2r)-1) \le 6re\hat{}(2r)$, since $r\le 1$. Multiplying the remaining segment and adding the complete-period error therefore bounds **every** ordered segment by $\varepsilon_M = 6(\theta+\theta^2)e\hat{}(8\theta)/M$ against its ideal $e\hat{}(\theta$ length $H/N)$. This covers both starting parities and cyclic arcs.

For each trace there are 16 dimensions and $\|a_x\| = 1/2$. Both discrete and ideal normalizing traces are at least 16; their difference is at most $16\varepsilon_M$. The quotient identity $u/d - u'/d' = (u-u')/d + u'(d'-d)/(dd')$ then bounds first-moment errors by $\varepsilon_M(1+E)$, where $E = e\hat{}(4\theta)$. For the two-arc second moment the product telescopes into two errors, giving $\varepsilon_M(2E+E^2)$. Products of first moments differ by at most $E\,\varepsilon_M(1+E)$. Thus each covariance-entry error is at most $5\varepsilon_M E^2 = 30(\theta+\theta^2)e\hat{}(16\theta)/M$. Multiplying by $(LN^2)^{-1}\Sigma|\psi_i\psi_j|$ yields the first term of A_θ/M.

For temporal quadrature, $|K_{xy}| \le (E+E^2)/4 \le e\hat{}(16\theta)/2$. Differentiating a time segment introduces $\theta[H, a_x]$; trace duality gives $|K'_{xy}| \le 2\theta E$. This bound also controls the kernel across its continuous coinciding-time seam. The integrand has Lipschitz bound $2\theta EB^2 + (e\hat{}(16\theta)/2)BD$ in each time coordinate. Each grid cell has total coordinate error at most $2/N$. Summing all x,y and dividing by L yields $L(2\theta EB^2 + (e\hat{}(16\theta)/2)BD)/M$. Add the two bounds. ■

No spatial quadrature is taken in T1: the x indices are the original four sites. A joint L,M limit needs uniform spatial estimates and an actual spatial reconstruction map. Sampling a spatially oscillating function on a fixed grid cannot supply those hypotheses.

5. Positivity, strict margins and exact rational schedules

Theorem T2. $Q_\theta(\psi) \ge 0$. Strict positivity requires a nonzero channel in the spectral formula; a nonzero profile alone need not suffice.

Proof. The supplied H and a_x are real symmetric. Choose a real orthogonal eigenbasis for H, writing its eigenvalues h_a and thermal weights $w_a = e\hat{}(\theta h_a)/Z$. Expand $A(\tau)$ in temporal Fourier modes. For each unequal-energy ordered pair a,b and frequency $\omega_p = 2\pi p$, the real covariance coefficient is

$$\frac{(w_a - w_b)\,\theta(h_a - h_b)}{\theta^2(h_a - h_b)^2 + \omega_p^2} \ge 0.$$

Its contribution is this coefficient times $|\hat{A}_{ab}(p)|^2/L$. The unsymmetrized coefficient is $(w_a - w_b)/(\theta(h_a - h_b) + i\omega_p)$. Since $A(\tau)$ is real symmetric in this eigenbasis, $\hat{A}_{ab}(p) = \hat{A}_{ba}(p)$; the a,b and b,a coefficients are complex conjugates at the same frequency, so their imaginary parts cancel. This real-source hypothesis is part of this theorem; the argument is not asserted for arbitrary

time-dependent complex matrix sources. At equal energies only p=0 remains. Their combined term is the thermal variance of the energy-block-diagonal part of the temporal average of A, which is nonnegative by the weighted square inequality. Subtracting the mean occurs in this variance term, not separately at every frequency. These terms exhaust the finite spectral trace and its Fourier expansion. C^1 profiles justify the summation by square summability. ■

This is the usual finite thermal spectral argument, derived here from the declared H. It contains neither an unexplained |k| Fourier weight nor a replacement of time-dependent Fourier coefficients with only their p=0 averages. Positivity of a covariance also does not establish Gaussianity.

Corollary T3 (eventual finite positivity). If an independently justified margin Q_θ(ψ)≥m>0 is supplied, q_M>0 for every

$$M \geq M_0 = \max(1, \lfloor A_\theta/m \rfloor + 1).$$

The inequality is strict. A ceiling alone fails when A_θ/m is an integer. The result uses a uniform error envelope for all later M, not a single positive sample or an assumption of monotonicity.

There is a concrete source-derived positive-margin example: ψ_x(τ)=1. Then A=Σa_x has trace zero, Tr A^2=16, and thermal mean zero. The last statement follows from the unitary cyclic site translation, which preserves H and sends A to −A. In the constant-frequency Kubo trace each spectral weight before division by Z is ∫e^((1-s)θh_a+sθh_b)ds≥1, since H≥0. Therefore Q_θ≥Tr A^2/(LZ)≥1/(4e^(4θ)).

For exact rational schedules replace e^(kθ) by E_k=(1−kθ/32)^(-32), k∈{4,8,16}, 0≤θ≤1. The inequality −log(1−u)≥u establishes E_k≥e^(kθ). The reference computes a rational A and m=1/(4E_4), then checks M_0m>A independently. The margin is bound to the constant source profile; it is not silently assigned to other fields.

An optional general kernel reconstruction has cell values L G_ij, with each cell measure 1/(LN), because its double integral must equal q_M. If its L^1 kernel error is δ and profile interpolation error η, then a separate declared absolute kernel bound κ yields a quadratic error at most $\delta B^2+\kappa(2B\eta+\eta^2)$. A convolution of K with the uniform grid converges to the **average** of K; it does not reproduce K. This alternative cannot substitute for T1 without its own kernel-binding evidence.

6. Exact cochain descent and its limits

On a declared periodic N-cycle let (Df)j=f(j+1)−f_j. Then im D={x:Σx_j=0}; a primitive is constructed by cumulative sums. The Euclidean quotient norm on C^1/im D is √N times |mean(x)|. Constants are harmonic, not exact cochains. The staggered vector is exact: D(−v/2)=v.

Theorem T4 (adapter descent). For a declared target vector space W (with scalar R as the one-dimensional case), a linear adapter A:C^1→W factors through C^1/B if and only if A(B)=0. An orthogonal low-pass projector Π_W:C^1→W alone need not satisfy this. The valid contractive adapter onto W is Π_W(I−P_B), with P_B the orthogonal projector onto the **declared** B. It can discard further quotient information; descent is not faithfulness.

Proof. Descent implies A(x+b)=A(x), hence Ab=0. Conversely this identity makes the class formula [x]↦Ax well-defined. Orthogonal projections are contractions, so their composition is contractive. ■

For the full cycle B=im D, the adapter keeps only the constant mode if 0∈W, and is zero otherwise. A nonzero low-frequency exact mode is a counterexample to the claim that low-pass filtering alone kills B. To retain low-frequency fluctuations while rejecting only the staggered mode, one must instead declare the smaller B=span(v), and must justify that repair space for the represented task. Neither choice is automatically the CAR physics' repair relation. In a two-dimensional cochain complex, the general Hodge decomposition also includes im d_1^*, the coexact component; only on cocycles does quotienting im d_0 leave precisely the harmonic component.

For integrated edge cochains, subdivision by k maps each edge x_j to k copies of x_j/k. Potentials interpolate linearly, so D'F_0=F_1D. Summing the k fragments is a left inverse, and periods are preserved. If fine edge weight is k, Σ_fine k(x_j/k)^2=Σ_coarse x_j^2, giving the declared weighted L^2 isometry and quotient-norm preservation. Unweighted L^2 is not invariant. This convention is separate from WDP's L^1 price norm.

7. Evidence and remaining scope

The written T1–T3 prove the fixed four-site time limit and its positive-margin case. The general rank-one smearing theorem does not require an all-M negative-eigenspace classification. Python retains four original-operator matrices and independent congruence certificates, pairing witnesses, cycle primitives, weighted subdivisions and mutation controls. Lean establishes the finite pairing inequality, normalization, square/energy bounds, strict-margin consequences, and quotient-adapter/cycle identities. The analytic exponential, spectral and C^1 arguments are written proofs; they are not advertised as Lean formalization of the physical CAR evolution. Python is also not formally verified by the Lean mathematical model.

General all-M inertia, uniform infinite-volume estimates, joint spatial reconstruction, cumulant control, Gaussian limits and native physical repair bindings remain separate obligations. This scope does not close the paper's full continuum or guarded-history registers.

W.8 Trotter-response, and the closures that stay open

The active Trotter-response checker and its historical verification record stay where the locality replay already hashes them: `docs/meron_field_limit_derivations/code/gate_e_exact_counterexample.py` and `docs/meron_field_limit_derivations/code/gate_e_exact_counterexample_receipt.json`. This section copies the verification record's mathematical content. It does not replace the verification record and it does not rerun the writer that would overwrite it.

The checked point is $L = 4$, $M = 1$, $r = 1/10$. The verification record gives eigenvalue $-1/706440$, test vector $(-1, -1, -1, -1, +1, +1, +1, +1)$, and quadratic value $v^{\mathsf{T}} G v = -1/88305$, with `is_positive_semidefinite` false. The point $r = 1/10$ lies outside the declared region $r \leq 1/192$. One finite point does not establish a unrestricted finite-M claim, and it does not establish an asymptotic formula. The four inertias in the Trotter section above are four finite observations. No finite list of inertias is an all-M inertia formula.

The compressed-channel extinction proved above is an all-M statement for that declared rank-one channel only. Theorem T1 is full fixed-four-site time consistency. Neither statement is a spatial continuum theorem. A cochain factorization, including Theorem T4, does not supply a realizable physical operation.

Independently supported static Gaussian and bounded time-labelled cumulant results remain under the hypotheses already stated for those results. They are not a joint spacetime field limit. A

finite signed generating function is not a probability characteristic function. Euclidean covariance positivity, including Theorem T2, is not a physical quantum reconstruction theorem.

Earlier model responses proposed closing the uniform-volume, joint-limit, and extinction/positivity obligations. Those responses remain proposals. Their missing obligations are the estimates those responses name and do not finish: a uniform infinite-volume bound, a joint spacetime limit under the source normalization and sign conventions, and an extinction or positivity claim that is not the one-point Trotter-response verification record. Those obligations stay open. This appendix does not enlarge the Meron programme and does not give those claims a new name in place of the missing estimate.

Appendix X — Calibrated reports, finite recovery, and observation cost

This appendix connects the calibration bounds of Chapter 13 and Appendix I.4 to the report-fiber criterion and the finite cost results of Sections 14.6–14.7 and Appendix S. The new step is an exact finite representation of continuous interval reports. Calibration, units, and possible report combinations are explicit premises. A checked answer does not authorize an operation or establish that a physical instrument satisfies those premises.

X.1 Reports and their error bounds

Let X be a nonempty finite candidate set, $p : X \to P$ a free observation, and $q : X \to Q$ the protected answer. There are finitely many instruments $i \in I$, each with price $c_i > 0$ in one declared cost unit. A query leaves $x \in X$ unchanged. A complete report vector is fixed for the experiment; a query reveals one coordinate. Strategies are deterministic functions of the reports already received. Each instrument may be queried at most once. Repeated fresh measurements, changing states, and statistical noise reduction need a different model; Section 14.7 treats supplied state changes.

Instrument i has a declared raw coordinate and a calibrated coordinate. For each admitted realization of candidate x, let t_i be its ideal raw indication and r_i its recorded raw report. Assume

$$|r_i - t_i| \le \eta_i(x), \qquad |a_i t_i + b_i - f_i(x)| \le \kappa_i(x),$$

where $a_i \neq 0$, $\eta_i(x), \kappa_i(x) \ge 0$. The parser applies the fixed nominal conversion $z_i = a_i r_i + b_i$. The ideal indication t_i may vary across admitted calibration realizations; it is not another observed input. The gain has calibrated-unit/raw-unit dimension; b_i, f_i, and κ_i have calibrated units. The first bound includes any rounding or quantization error in the raw indication. The second includes the declared calibration-model error. These are uniform bounds over the admitted realizations for x, not confidence levels.

Proposition X.1 (transported enclosure). Every such report belongs to

$$J_i(x) = [f_i(x) - \epsilon_i(x), f_i(x) + \epsilon_i(x)], \qquad \epsilon_i(x) = |a_i|\eta_i(x) + \kappa_i(x).$$

Proof. Subtract $f_i(x)$ from $a_i r_i + b_i$, insert $a_i t_i$, and apply the triangle inequality. Both terms have calibrated units. No independence assumption is used. □

The calibration chain in Appendix I.4 can provide an error budget: its radius update is $(|a| + \alpha)r + \alpha|u| + \beta + \eta$ at nominal input center u. To use such a budget here, its direction and units

must match the reported coordinate and it must hold uniformly over the candidate's admitted raw indications. A bound on a different quantity or only one calibration sample does not supply this premise. Tighter verified enclosures may replace $J_i(x)$ throughout the argument.

Let L_x be the set of physically or symbolically admitted complete report vectors for x. **Outer-bound validity** means

$$\emptyset \neq L_x \subseteq \prod_{i \in I} J_i(x).$$

This condition suffices for answer soundness and conservative upper cost bounds. **Full report realizability** is the stronger condition

$$L_x = \prod_{i \in I} J_i(x).$$

It is an adversarial product model, not a statement about stochastic independence. It permits every combination of the individual interval values for each fixed candidate. Error propagation alone does not establish it. Correlated calibration errors can violate it even when each individual interval is exact.

X.2 Certified answers and abstention

A history consists of the free report p_0 and calibrated readings z_i for a queried subset $H \subseteq I$. Define

$$C(h) = \{x \in X : p(x) = p_0,\ z_i \in J_i(x) \text{ for every } i \in H\}.$$

Proposition X.2 (answer certificate). If outer-bound validity holds and the true candidate supplied the history, it belongs to $C(h)$. A proposed answer v is therefore correct if the checker establishes

$$C(h) \neq \emptyset \quad \text{and} \quad (\forall x \in C(h))\ q(x) = v.$$

If $C(h)$ contains two different answers, this checker abstains or requests another instrument. If $C(h) = \emptyset$, it reports inconsistency with the model; it cannot accept by vacuous truth.

Proof. Each genuine reading satisfies its enclosure, and its free report equals $p(x)$. Thus the true candidate satisfies every defining constraint. Apply the checked answer equality to that candidate. □

Under full report realizability this criterion is also necessary for a uniformly correct answer at a compatible history. Indeed each candidate in $C(h)$ admits a complete report vector extending the history: choose any point in each unqueried nonempty interval. Two such candidates with different answers give the same observed history, so one answer cannot be correct for both. This is the existing report-fiber obstruction. With only outer bounds, some candidates in $C(h)$ may be spurious; abstention can then be conservative.

An explicit parser is part of this statement. The finite reference code accepts a dictionary with a model digest, a string-valued free observation, and a list of distinct named instrument readings.

Each reading contains the expected raw-unit label and a canonical rational string, such as `-1/2`. Unknown fields, wrong units, repeated instruments, floating-point inputs, and a digest for another model are rejected. Calibration coefficients, enclosures, prices, and the answer table come from the caller's separately supplied model. The model digest binds those data; it does not authenticate the source of a reading or a calibration standard. Soundness remains conditional on that source and on the bounds above.

X.3 Exact finite representation of continuous reports

For each instrument define its membership signature

$$S_i(z) = \{x \in X : z \in J_i(x)\}, \qquad \mathcal{S}_i = \{S_i(z) : z \in \mathbb{R},\ S_i(z) \neq \emptyset\}.$$

A signature retains exactly which candidates are compatible with one reading. This does not round the reading to a nearby center.

Theorem X.3 (finite report reduction). The interval product model has the same possible answer histories and the same optimal fixed and adaptive worst-case observation costs as the finite deterministic model

$$\widehat{X} = \{(x, s) : x \in X,\ s \in \textstyle\prod_i \mathcal{S}_i,\ x \in s_i \text{ for every } i\},$$

$$\hat{p}(x, s) = p(x), \qquad \hat{q}(x, s) = q(x), \qquad m_i(x, s) = s_i,$$

with the same instrument prices. A compatible original candidate is exactly the projection of a compatible lifted state. There are at most $4|X| - 1$ nonempty signatures for each instrument.

Proof. List the distinct interval endpoints in increasing order, say $e_1 < \cdots < e_k$, with $k \leq 2|X|$. Membership is constant on each singleton $\{e_j\}$ and each open interval (e_j, e_{j+1}). Both exterior intervals have empty membership. Keep the nonempty signatures and merge equal ones. There are at most $2k - 1 \leq 4|X| - 1$. In particular the singleton endpoints must be retained: two closed intervals may meet at just one point. Rational endpoints give rational representatives, using endpoints and midpoints. Consequently restriction to exact rational reports retains every signature and both costs for rational data, including a nonzero rational raw conversion.

A continuous vector admitted for x yields a lifted state by applying S_i to each coordinate. Conversely choose one representative $z_i(s_i)$ for each nonempty signature. If $x \in s_i$, that representative lies in $J_i(x)$. Full report realizability makes the vector of representatives an admitted vector for x.

For a partial history, $x \in C(h)$ precisely when it belongs to all the observed signatures and has the right free report. Each unqueried interval is nonempty, so select one of its signatures to complete a lifted state. The reverse implication follows from the observed coordinates. Hence the projection of every compatible lifted fiber is exactly $C(h)$.

A strategy on signatures works on continuous readings by applying S_i after each query. Every resulting history is admitted in the finite model, so correctness and worst-case cost bounds transfer. In the other direction, run any continuous-reading strategy on the selected representatives $z_i(s_i)$ when a signature is observed. Every such path is admissible for its underlying candidate, including after adaptive choices, because every partial product assignment extends to a full one. It follows

that the finite strategy has no greater worst-case cost. The two translations equate the infima. There are finitely many deterministic trees on the finite alphabet with depth at most $|I|$, so every feasible finite optimum is attained; its continuous translation attains the same value. Fixed suites are unchanged in both translations. This establishes equality of both optima. □

The theorem compares information and cost, not the literal numeric contents of histories. Without full report realizability, every actual history still maps into this finite model, so its certified strategies remain sound upper bounds. The converse translation, and consequently physical optimality or impossibility, need not hold.

X.4 Separation and robust cost bounds

Write F_{box} and A_{box} for the optimal fixed and adaptive costs of the interval product model. The fixed suite is chosen before the free report; an adaptive strategy receives that report before its first paid query. An infeasible optimum is $+\infty$.

Corollary X.4 (fixed robust cost). A fixed suite $J \subseteq I$ ensures recovery under declared assumptions exactly when every pair x, y with $p(x) = p(y)$ and $q(x) \neq q(y)$ has some $i \in J$ with

$$J_i(x) \cap J_i(y) = \emptyset, \quad \text{equivalently} \quad |f_i(x) - f_i(y)| > \epsilon_i(x) + \epsilon_i(y).$$

Thus F_{box} is the existing minimum weighted hitting-set cost on these separating-instrument sets.

Proof. A disjoint coordinate prevents the pair from sharing a report. If no queried coordinate is disjoint, choose a common value in each of its intersections. Complete the unqueried coordinates separately for x and y. The two resulting lifted states have the same observed report and different answers. The equivalence with the strict inequality is the closed-interval intersection criterion. Section 14.6 now applies to the finite lift. □

The strict inequality matters: equality leaves a common endpoint. Under outer-bound validity alone the separation condition is sufficient, but its failure is not proof of actual indistinguishability.

For completeness, the existing adaptive recursion becomes particularly simple on a pair (C, R) of a nonempty compatible candidate set and the remaining instruments. It is zero if q is constant on C. Otherwise,

$$W(C, R) = \min_{i \in R} \left[c_i + \max_{s \in \mathcal{S}_i:\ C \cap s \neq \emptyset} W(C \cap s, R \setminus \{i\}) \right],$$

where the minimum over an empty set is $+\infty$. The initial adaptive cost is the maximum of $W(p^{-1}(p_0), I)$ over nonempty free-report fibers. The recursion terminates because R decreases, including when a report leaves C unchanged. Tracking only C and assuming every branch strictly shrinks it would be incorrect. By Theorem X.3 these are the same minimax costs as Section 14.6 on the lift; its complete-table checker can be reused. The maximum lists all realizable adversarial outcomes, not selected examples.

Corollary X.5 (comparison and stability). Let F_0, A_0 be the costs for the nominal deterministic reports $f_i(x)$. Then

$$A_{\mathrm{box}} \leq F_{\mathrm{box}}, \qquad F_0 \leq F_{\mathrm{box}}, \qquad A_0 \leq A_{\mathrm{box}}.$$

Enlarging any intervals while retaining their centers cannot decrease either product-model optimum. Moreover, suppose each instrument separates all of its distinct nominal report values despite the errors:

$$f_i(x) \neq f_i(y) \implies |f_i(x) - f_i(y)| > \epsilon_i(x) + \epsilon_i(y) \qquad (i \in I,\ x, y \in X).$$

Then $F_{\text{box}} = F_0$ and $A_{\text{box}} = A_0$.

Proof. Querying a sufficient fixed suite is an adaptive strategy with no greater worst-case cost. A nominal vector is an admitted product-model vector, so restriction of any robust strategy to nominal reports establishes the other two lower bounds. The same restriction argument establishes monotonicity for nested supports. Under the separation hypothesis, any compatible reading determines its unique nominal-value group. Decode that group and follow an optimal nominal strategy or fixed suite. The true candidate remains in the decoded group, so this strategy is correct at its nominal cost. Combine this upper bound with the lower bounds. □

For an actual correlated model $L_x \subseteq \prod_i J_i(x)$, only $F_{\text{actual}} \leq F_{\text{box}}$ and $A_{\text{actual}} \leq A_{\text{box}}$ follow without further assumptions. The product-model lower bounds must not be reported as physical lower bounds. A finite upper certificate ensures recovery under declared assumptions within its budget; under full realizability, a budget below the exact optimum cannot guarantee an answer on every admitted history. A sound procedure must then sometimes abstain, request a larger budget, or leave the stated model.

X.5 Changes of units

Proposition X.6 (coordinate invariance). For any invertible affine change of each calibrated coordinate,

$$z_i' = \lambda_i z_i + \tau_i, \quad f_i'(x) = \lambda_i f_i(x) + \tau_i, \quad \epsilon_i'(x) = |\lambda_i| \epsilon_i(x), \qquad \lambda_i \neq 0,$$

candidate compatibility, separation, and both observation costs are unchanged, provided the instrument prices retain their original cost unit. For ordinary changes of physical units $\lambda_i > 0$; a negative value also covers a reversal of coordinate orientation.

Proof. The image interval has the stated center and radius. Invertibility gives $z_i \in J_i(x)$ if and only if $z_i' \in J_i'(x)$. Thus all signatures and their allowed candidate memberships are unchanged. Both sides of the separation inequality are multiplied by $|\lambda_i|$. Apply Theorem X.3 to the identical finite observation table. Scaling the centers but leaving radii or unit tags unchanged is not this coordinate change. □

X.6 Exact examples and a necessary counterexample

Take four candidates with distinct protected answers and one constant free report. Their nominal calibrated reports and prices are

Instrument	00	01	10	11	Price
A	0	0	10	10	1
B	0	10	0	0	3

Instrument	00	01	10	11	Price
C	0	0	0	10	3

Give every report radius ϵ in the same calibrated unit. For $0 \le \epsilon < 5$, Corollary X.5 applies. The fixed cost is exactly 7: the pairs $(00, 01)$, $(00, 10)$, and $(10, 11)$ force B, A, and C, respectively. All three suffice. The adaptive cost is exactly 4: query A, then B on its low branch or C on its high branch. A budget below 4 cannot query A and a second distinct instrument; querying B or C alone also leaves different answers unresolved. At $\epsilon = 5$, the vector $(5, 5, 5)$ is compatible with all four candidates, so both optima are infinite in the product model. Larger radii retain this obstruction. The reference uses the raw conversion $z = 2r + 1$; changing coordinate units preserves the finite costs when the whole error interval is transported.

Full report realizability cannot be dropped from the lower-bound argument. Let $x \in \{0, 1\}$, with constant free observation and protected answer x, and let the same unknown $e \in [-1, 1]$ affect two unit-price instruments:

$$z_1 = x + e, \qquad z_2 = -x + e.$$

Their individual intervals are $[-1, 1]$ for $x = 0$, and $[0, 2]$, $[-2, 0]$ for $x = 1$. Each instrument alone has overlapping supports. The interval product model admits the report $(0, 0)$ for both candidates and therefore has infinite robust recovery cost. But the actual correlated model obeys $(z_1 - z_2)/2 = x$ for every admitted e. Two queries recover exactly, and one cannot guarantee recovery because of the individual overlaps. Its actual fixed and adaptive optima are both 2. This discrepancy persists even though every single-instrument interval is exact.

X.7 What is checked

The reference checker constructs the finite lift and checks answer proposals against a separately supplied model. It reuses `finite_observation.solver` for cost proposals and for complete-table verification; the verification path never calls the solver to replace a bad certificate. The reference has exact rational endpoints and refuses a lift above the existing ten-state limit. A resource refusal is not mathematical infeasibility. The four-candidate endpoint collision is checked directly, since its complete lift exceeds that limit. No new native evaluator or production observation-cost solver is claimed.

The cost certificate explicitly names `ideal_calibrated_interval_product`. It certifies the full interval product before raw-encoding restrictions. The raw-report API separately requires canonical rational strings of at most 160 characters and numerators/denominators of at most 256 bits, before calibration. Model admission does not guarantee closure under inverse calibration within those bounds. For example, gain 2^{-255}, offset 0, and point supports $\{0\}$ and $\{2^{255}\}$ admit a two-state model with ideal fixed and adaptive cost 1. The high state's only raw reading is 2^{510} (511 bits), refused as `INVALID_REPORT`; the low state's raw reading 0 verifies. This is a bounded API refusal, not ideal-model infeasibility or acceptance of a wrong answer. Applying the ideal costs to this API additionally requires proof that the needed reports have admitted raw encodings; this certificate does not provide that proof.

The six transfer lemmas compile using Lean core alone. Its six declarations establish partial-product extension, the exact projection of compatible lifted states, answer transfer, preservation

under outer supports, answer soundness, and the shared-report obstruction. The first three use classical choice. These declarations do not formalize real interval arithmetic, physical calibration, or the cost comparison. Those proofs are written above and use the finite cost results already stated in this paper. The Python controls test endpoint collisions, exact cost certificates, unit changes, invalid reports, false answers, substituted models, and correlated-noise limits. They are finite deterministic tests, not a device experiment or statistical validation.

Appendix Y — Exact recovery costs from joint report supports

Appendix X makes exact cost claims for products of intervals. This appendix retains the correlations between reports. It first characterizes the costs for arbitrary declared joint supports. It then gives a finite construction for a specified class of shared affine errors, including the example in which the product enclosure has infinite cost but two actual queries suffice. The costs concern deterministic, exact, worst-case recovery of a supplied answer. They do not price a statistical confidence level or authorize a query.

Y.1 Joint supports and compatible histories

Let X be nonempty and finite, with free report $p : X \to P$ and protected answer $q : X \to Q$. Let I be a finite instrument shelf with positive rational prices c_i, all in one declared cost unit. For each x, supply a nonempty set

$$L_x \subseteq \mathbb{R}^I.$$

A world is a pair (x, z) with $z \in L_x$; write Ω for their set. The protected answer of this world is $q(x)$. One complete vector is fixed throughout the experiment. Querying i reveals z_i and leaves the world unchanged. Each instrument is queried at most once. As in Appendix X, strategies are deterministic functions of the exact reports received. The general statements below impose no computability, measurability, or probability model on those functions.

For a history $h = (p_0, H, r_H)$, define

$$B(h) = \{(x, z) \in \Omega : p(x) = p_0,\ z_i = r_i \text{ for every } i \in H\},$$
$$C(h) = \{x : \exists z,\ (x, z) \in B(h)\}.$$

Equivalently, $x \in C(h)$ precisely when $p(x) = p_0$ and $r_H \in \pi_H(L_x)$, where π_H is coordinate projection. The same complete vector must extend every reading in the history. Checking each coordinate against its marginal support does not establish this condition.

A proposed answer v is certified when $C(h) \neq \emptyset$ and $q(x) = v$ for every $x \in C(h)$. If the true world belongs to the declared support, this answer is correct. If two compatible candidates have different answers, their admitted complete vectors realize the same history; a decoder of that history cannot be correct for both. Empty compatibility means inconsistency with the model, not acceptance. This is the report-fiber criterion applied to actual joint histories.

These distinctions also govern physical use. If the actual support is only known to be a nonempty subset of L_x, certification remains sound, but abstention can be conservative. Exact physical lower costs require that the declared joint supports are realizable. A bound on each coordinate does not establish that premise.

Y.2 Fixed suites and projected intersections

Proposition Y.1 (exact fixed cost). A fixed suite $J \subseteq I$, chosen before receiving the free report, is sufficient exactly when

$$\pi_J(L_x) \cap \pi_J(L_y) = \emptyset \quad \text{whenever} \quad p(x) = p(y),\ q(x) \neq q(y).$$

Consequently its optimum is

$$F_\Omega = \min \left\{ \sum_{i \in J} c_i : J \subseteq I \text{ satisfies this condition} \right\},$$

with value $+\infty$ when no suite is sufficient.

Proof. A point in a projected intersection is supplied by complete worlds with the same free and queried reports but different answers. Conversely, failure of answer constancy on a report fiber supplies just such an intersection. There are finitely many suites, so a finite minimum is attained. □

This remains a hitting-set problem, but the obligations are now pairs of complete worlds. For $p(x) = p(y)$ and $q(x) \neq q(y)$, put

$$E_{(x,z),(y,w)} = \{i \in I : z_i \neq w_i\}, \qquad z \in L_x,\ w \in L_y.$$

A suite is sufficient precisely when it meets every such set. Although there can be infinitely many world pairs, there are at most $2^{|I|}$ distinct sets of this form. Identifying which sets are realized can require a support-membership or existence oracle; the finite count is not an algorithm for an arbitrary supplied set.

In particular, recovery using the whole shelf is feasible exactly when the full supports L_x, L_y are disjoint for every cross-answer pair with the same free report. A shared complete vector obstructs every adaptive strategy too. When there is no such vector, querying the whole shelf and checking its fiber supplies a correct mathematical decoder. Marginal overlap by itself is not this obstruction.

Y.3 Adaptive cost depends on the residual joint support

Let B be a nonempty set of worlds contained in one already observed free-report fiber, and let R be the remaining instruments. No further free observation is available. Write $V(B, R)$ for the least worst-case additional cost of a strategy that returns the correct answer on every world in B. If no such strategy exists, set $V(B, R) = +\infty$. A strategy that sometimes abstains does not establish a finite value of this guaranteed-recovery objective. For convenience only, set $V(\emptyset, R) = 0$ on unreachable branches.

For $i \in R$, let

$$B_{i,t} = \{(x, z) \in B : z_i = t\}, \qquad T_i(B) = \{z_i : (x, z) \in B\}.$$

Theorem Y.2 (joint-history recursion). If q is constant on B, then $V(B, R) = 0$. Otherwise,

$$V(B,R)=\min_{i\in R}\left[c_i+\max_{t\in T_i(B)}V(B_{i,t},R\setminus\{i\})\right].$$

An empty minimum is $+\infty$. Every displayed maximum is attained, even if $T_i(B)$ is infinite. A finite value is attained by a strategy. The initial adaptive optimum is

$$A_\Omega=\max_{p_0\in p(X)}V(\{(x,z)\in\Omega:p(x)=p_0\},I).$$

Proof. Induct on $|R|$. With no remaining query, a nonconstant answer cannot be recovered. For the induction step, every finite child value belongs to the finite set of subset sums

$$\mathcal{K}_{R\setminus\{i\}}=\left\{\sum_{j\in S}c_j:S\subseteq R\setminus\{i\}\right\}.$$

Thus the nonempty set of child values lies in $\mathcal{K}_{R\setminus\{i\}}\cup\{+\infty\}$ and has an attained maximum. Any correct strategy with first query i has worst-case cost at least the corresponding displayed option. For a finite minimum, choose a minimizing i and use an optimal continuation on each realized branch. One can define these choices at all histories by fixing an order on I and always taking the first minimizing instrument; recursion decreases $|R|$. This supplies a strategy attaining the value. Its value is again a subset sum, since the maximum selects one child value and adds a previously unqueried price. The same argument handles infeasible branches with the extended value $+\infty$. The finite free-report image supplies the last maximum. □

Restriction to a smaller set of worlds cannot increase either cost. Also $A_\Omega\le F_\Omega$, by asking a sufficient fixed suite in order. Neither fact requires independent errors.

It is essential to retain $B(h)$, or an exact representation of its continuations. The pair consisting only of $C(h)$ and the remaining instruments is generally insufficient. For example, let the free report be constant, the answer be $x\in\{0,1\}$, and

$$L_0=\{(e,e):0\le e\le 1\},\qquad L_1=\{(e,2e-\tfrac{1}{3}):0\le e\le 1\}.$$

After a first reading of 0 or 1/3, the same candidates $\{0,1\}$ remain. At 0, a second unit-price reading distinguishes them. At 1/3, both coordinates agree and no remaining query can distinguish them. Their remaining costs are 1 and $+\infty$, respectively.

Y.4 Finite cost witnesses exist, but are not report encoders

The preceding recursion gives an existence result for arbitrary joint supports, without claiming an effective way to inspect them.

Theorem Y.3 (finite cost witness). There is a finite $W\subseteq\Omega$, meeting each candidate state, whose fixed and adaptive optima equal F_Ω and A_Ω. If $m=|I|$, one bound is

$$|W|\le|X|+2^{m+1}+\tau_m,\qquad \tau_0=2,\quad \tau_r=2+r\tau_{r-1}.$$

Proof. First consider any nonempty B contained in one fixed free-report fiber, as in Y.3, and an instrument set R of size r. We show that a nonempty finite $S \subseteq B$ preserves $V(B, R)$ and has at most τ_r worlds. If the answer is constant, one world suffices. For $r = 0$ and a nonconstant answer, choose two worlds with different answers. For $r > 0$ and a nonconstant answer, choose two such worlds and, for every $i \in R$, choose a reading t_i attaining the maximum in Theorem Y.2. By induction take $S_i \subseteq B_{i,t_i}$ preserving that branch's remaining value, with $|S_i| \le \tau_{r-1}$. Let S contain the chosen pair and every S_i.

Restriction gives $V(S, R) \le V(B, R)$. Since the chosen pair has different answers, any correct strategy on S must start with a query, say i. Its t_i branch contains S_i. Monotonicity therefore gives a remaining cost at least $V(S_i, R \setminus \{i\})$, which is the worst child value for query i on B. Minimizing over possible first queries gives the reverse inequality. This argument also applies when a value is infinite. The cardinality is at most $2 + r\tau_{r-1}$.

Choose a free-report fiber attaining A_Ω and retain the preceding finite witness for that fiber. For every insufficient fixed suite, choose two admitted worlds with the same reports on that suite and different answers. There are at most 2^m suites. Finally choose one admitted world per candidate state. Let W be the union of these selections. Every insufficient fixed suite still fails on its retained pair, while every sufficient suite stays sufficient after restriction. The retained worst free-report fiber gives $A_W \ge A_\Omega$, and restriction gives the opposite inequality. Counting the selections yields the stated bound. □

The selections use full support information and exact branch optima. This is an existence proof, not an algorithm that constructs a witness from arbitrary set descriptions. A guessed finite sample alone supplies lower costs, which can be strict. Its finite certificate becomes an exact certificate for the declared model only with a valid completeness argument. Nor does W represent every literal report or supply a decoder for every unsampled history.

There need not even be a sufficient finite encoding of each coordinate separately. For two unit-price instruments, take

$$L_0 = \{(u, u) : u \in [0, 1]\}, \qquad L_1 = \{(u, v) : u, v \in [0, 1],\ u \ne v\},$$

with constant free report and answer 0 or 1. The exact fixed and adaptive costs are both 2. For any finite-valued first-coordinate encoding, some distinct a, b have the same code. Then $(b, b) \in L_0$ and $(a, b) \in L_1$ have identical encoded reports, regardless of the second-coordinate encoding. The encoded problem cannot recover the answer. This example uses arbitrary, not necessarily closed supports; it makes no assertion about finite-precision instrument performance. It separates a finite witness of a cost from a finite encoding of all the reports needed for recovery.

Y.5 A constructive reduction for shared affine error

A useful explicit class does admit a finite cost construction. Index a finite nonempty family of segments by $k \in K$. Each has an owner $x(k) \in X$, with every candidate owning at least one segment, and

$$z_i = a_{ki}e + b_{ki}, \qquad e \in [\ell_k, u_k], \quad \ell_k \le u_k, \quad a_{ki} \ne 0.$$

All endpoints and coefficients are rational. The joint support L_x is exactly the union of the segments owned by x. Different components may overlap. A component label and its parameter are hidden variables, not free observations. Degenerate parameter intervals are permitted. The nonzero-slope condition is imposed for every component and instrument. The case $I = \emptyset$ is handled by the free report alone.

Fix a free report p_0 and a possible first query i. On component k with $p(x(k)) = p_0$, its value t lies in the closed interval D_{ki}, the image of $[\ell_k, u_k]$ under $e \mapsto a_{ki}e + b_{ki}$. That reading fixes the component's parameter and predicts every later coordinate as

$$g^i_{kj}(t) = \frac{a_{kj}}{a_{ki}}\, t + b_{kj} - \frac{a_{kj}}{a_{ki}}\, b_{ki}.$$

Make a finite set of critical values from:

1. every endpoint of every D_{ki} in this free-report fiber;
2. every isolated equality point $g^i_{kj}(t) = g^i_{lj}(t)$, for every component pair and every remaining coordinate j, that lies in $D_{ki} \cap D_{li}$.

Identical affine functions agree throughout their common domain; parallel unequal functions never agree. Neither contributes an isolated point. Partition the range at these critical values, retaining singleton points as well as open intervals between consecutive points. Use the point itself or the interval midpoint as a representative. Exterior intervals contain no admitted first reading.

On each cell, the active component indices are constant, as are all equalities between their predicted values for every remaining instrument. Hence the finite remaining observation problems at any two readings in the same cell differ only by a relabeling of outcomes. Their protected answers and costs agree. Overlapping components that describe the same candidate and full vector may be identified; duplicating such a world does not create an observable distinction.

Construct W by retaining one endpoint world from each segment and, for every p_0, i and cell whose compatible answers are not all equal, every world in the complete first-reading fiber at its representative. A first reading determines at most one world per component, so every such fiber is finite. Cells with one answer need no further observation.

Theorem Y.4 (affine critical-value reduction). This construction is finite and has

$$F_W = F_\Omega, \qquad A_W = A_\Omega.$$

It therefore supplies an instance for the existing finite solver and certificate checker. For rational input, every retained world is rational.

Proof. Activity can change only at an image endpoint. An equality of two predicted affine values can change only at an isolated root, unless the functions are identical. This establishes the cell statement above, including equality at a single critical point.

For fixed suites, $W \subseteq \Omega$ gives one direction. Suppose a nonempty suite J fails on Ω. Choose its conflicting admitted worlds and some $i \in J$. Their common i value belongs to a cell. The same two components are active at its representative and continue to agree on every coordinate in J, while their answers differ. This representative fiber was included in W, so J fails there too. For $J = \emptyset$,

the endpoint worlds preserve the candidate answers in every free-report fiber. Thus precisely the same fixed suites suffice.

For the adaptive result, first consider a nonconstant free-report fiber. For any proposed first query i, every nonconstant continuation problem is equivalent to a representative problem included in W. Moreover the fiber of W at that representative is the complete fiber of Ω: it contains all its worlds, and every sampled world is admitted in Ω. Constant-answer continuations have value zero. The remaining values have only finitely many cell types, so the largest continuation value is represented whenever it is positive or infinite. If all continuations have value zero, any nonempty restricted fiber has that value too. Thus no strategy on W with first query i can have a lower worst-case cost than the corresponding option on Ω. Theorem Y.2 and minimization over first queries give $A_W \geq A_\Omega$. Restriction gives the reverse inequality. Constant-answer free-report fibers have zero cost in both models. □

There are finitely many endpoints and pairwise affine roots; enumerating them requires only exact scalar rational arithmetic. With $n = |K|$ and $m = |I| \geq 1$, a coarse bound on the number of sampled worlds is

$$n + mn\left(4n + 2(m-1)\binom{n}{2} - 1\right).$$

Indeed each free-report fiber has at most $2n + (m-1)\binom{n}{2}$ critical values. It contributes at most twice this number minus one cells, each with at most its number of components. Summing the component counts over free-report fibers gives n. This bounds the finite construction, not the subsequent exponential cost-certificate computation.

A complete-vector collision with the same free report and different protected answers gives an earlier exit: retain the two admitted conflicting worlds. Both costs are then infinite by Proposition Y.1. For affine segments this collision can be checked by requiring all predicted-coordinate equalities at one common first-coordinate value, within both parameter-image intervals. This shortcut is separate from truncating an oversized sample.

The reduction also explains how a finite optimal policy can be used on continuous reports. If the free report already determines the answer, return it without querying an instrument. Otherwise take the finite policy's first query for that free report. If the returned value leaves one answer, stop. Otherwise retain that exact value t, find its cell representative s, and use the complete finite fiber at s. For a later coordinate, map the actual observed value to $g^i_{kj}(s)$ using any initially active component whose prediction at t equals that observation. The map is well-defined because predicted-value equalities are constant on the cell. Filter by the accumulated history and follow the finite policy. This preserves answers and never exceeds the certified worst-case cost. It is an encoding that uses the first reading and the history; it is not a fixed finite code for each coordinate in isolation.

Y.6 Exact examples and a misleading sampling baseline

For the shared-error supports

$$L_x = \{(x + e, -x + e) : -1 \leq e \leq 1\}, \qquad x \in \{0, 1\},$$

with constant free report, protected answer x, and unit prices, neither coordinate alone suffices, but $(z_1 - z_2)/2 = x$. Both exact costs are 2. The reference construction retains ten worlds and its finite certificate has the same costs. The product of the marginal intervals instead admits $(0, 0)$ for both answers and has infinite cost, as shown in Appendix X. At the actual joint report $(0, 0)$, only $x = 0$ is compatible.

Multiple components can encode correlation too. Let

$$L_0 = \{(0, 0), (1, 1)\}, \qquad L_1 = \{(0, 1), (1, 0)\}.$$

These are degenerate affine segments. Their marginal supports are identical, but their joint supports are disjoint. Both unit-price queries are necessary and sufficient. Matching the first reading to one component and the second reading to another component of the same candidate would give a false compatibility result.

The crossing example in Y.3 has the shared complete report $(1/3, 1/3)$. Both costs are infinite. A grid that happens to omit this point does not establish feasibility.

An isolated critical value can also change a finite optimum. Take three instruments A, B, C, with prices $1, 1, 3$, and

$$L_0 = \{(e, e, e) : -1 \le e \le 1\}, \qquad L_1 = \{(e, -e, e + 10) : -1 \le e \le 1\}.$$

The free report is constant and the protected answer is the support index. Instrument C alone separates its disjoint ranges, at cost 3. Any budget below 3 can use only A and B. Their joint report $(0, 0)$ is compatible with both answers, so no such budget suffices. The fixed and adaptive optima are therefore both 3.

Sampling only the parameter endpoints yields four worlds on which A and B suffice at cost 2. This finite answer is correct for that sample and false for the declared continuous model. The critical-value construction retains the interior collision of A and B, as well as the differing C values, and returns 3. This comparison tests the completeness of the reduction, not a statistical estimate of noise.

Y.7 Units, calibration and additional error

The calibrated coordinate z_i and its intercept b_{ki} have the instrument's declared output unit. The parameter e, ℓ_k, u_k have the component's declared parameter unit; a_{ki} has their unit ratio. Raw reports use the declared nonzero conversion $z_i = \gamma_i r_i + \delta_i$, as in Appendix X. All admitted effects of calibration must already be covered by the supplied joint support. The unit labels and a model digest cannot establish that physical premise.

For a coordinate change $z'_i = \lambda_i z_i + \tau_i$, with $\lambda_i \neq 0$, transport every component by

$$a'_{ki} = \lambda_i a_{ki}, \qquad b'_{ki} = \lambda_i b_{ki} + \tau_i,$$

and transport the raw conversion in the same way. This is a bijection of full and partial report spaces, preserving their compatible answers. It conjugates any query strategy without changing

the prices. Hence both costs are unchanged. Negative λ_i reverses orientation. Changing a unit label or only a nominal center is not this transformation.

A segment's marginal interval has center $a_{ki}(\ell_k + u_k)/2 + b_{ki}$ and radius $|a_{ki}|(u_k - \ell_k)/2$, with the corresponding output unit. Their product is generally larger than the segment. It remains an outer bound, with the limitations already stated in Appendix X.

The affine-segment algorithm does not cover an unmodeled independent rounding or calibration residual added to each coordinate. Such a residual thickens the support. After one reading, the latent parameter need not be uniquely determined, so the finite-fiber argument in Y.5 does not apply as written. The general joint-support criterion and recursion still apply to a correctly enlarged support; an effective reduction for that larger class is a separate obligation.

Y.8 Reference implementation and proof scope

The joint-support reference validates exact rational segment data, checks reports against a single common parameter in one component, constructs the critical finite witness, and reuses the existing finite observation solver and complete-table checker. Cost verification reconstructs the witness from the separately supplied model and checks the submitted certificate; it never invokes the proposal solver to replace a rejected certificate.

The model digest binds all components, unit labels, raw conversions, free reports, answer labels, prices and their common cost unit. It is not source authentication. The report parser reuses the bounded canonical-rational parser from Appendix X and rejects wrong units, repeated instruments, unknown fields and substituted models. A checked answer supplies no permission to act.

The reference admits at most ten segments and twelve instruments, with 256-bit rational input numerators and denominators. It refuses a constructed witness above the existing ten-world solver limit rather than deleting worlds or reporting infeasibility. The collision shortcut uses a genuine two-world obstruction. Duplicate descriptions of the same world are removed before the cap is applied.

The stored examples and targeted controls cover shared error, unions of components, isolated collisions, a missed interior cost peak, free observations, unit changes, certificate tampering and resource refusals. Degenerate supports are compared with a direct finite-checker baseline. These are exact deterministic examples, not instrument trials or evidence that a physical error model is valid.

The results in this appendix have written proofs. They are not new Lean formalizations, and no general joint-support decision procedure is implemented. The earlier six core Lean transfer lemmas retain their stated scope. The reference certifies costs for its admitted affine models; it does not implement a deployed controller for the continuous policy transformation described above.

Limitations & Explicit Nonclaims

This paper makes narrow, scoped claims under declared assumptions. Operational authority is null. Computational demonstrations and finite checks do not constitute unmechanized carrier-scoped results or institutional authorization.